

Notes on Numerical Analysis
and Numerical Methods for
Differential Equations
数值分析+ 微分方程数值解

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关于数学学习的几点建议

- A. 深入理解每一个知识点：证明或推导的每一步从哪里来的？争取做到“无一处无出处”，这有助于培养缜密的逻辑思维能力。
- B. 寻找新内容和已知内容或其他数学分支之间的联系。我们学习数值分析已经用到的其它分支包括分析基础和线性代数等。学习的本质是把新内容和已经牢固掌握的知识联系起来！
- C. 深入思考每一个知识点：一个定义捕捉到了什么？一个定理是否可以弱化条件？如果不能的话这些条件在证明中是在哪里出现的？作用是什么？一个定理的结论是否可以再加强？如果不能原因是什么？一个数学方法的适用范围是什么？局限性在哪里？
- D. 精准识记核心的定义定理，再以一定的逻辑关系把相关知识点串成一个故事，这些关系可以包括继承、组合、蕴含、特例等；构建这样一个脉络的目的是使自己知识体系的熵（混乱度）最小。
- E. 在完成知识体系构建的基础上尽可能地多做习题，但是构建知识体系永远比做题本身重要。
- F. 将新知识以一种和已有知识相容的方式纳入自己的知识体系。学数学的过程是盖一座大楼不是在一个平面上搭很多帐篷；一座大楼的高度取决于基础以及每一层的坚固度。
- G. 任何一门数学都包括内容和形式，两者相互依赖，互为补充。
- H. “骐骥一跃，不能十步；驽马十驾，功在不舍。锲而舍之，朽木不折；锲而不舍，金石可镂。”
One baby step at a time!
Do the simplest thing that could possibly work, then keep asking more and refining your answers.
- I. “一阴一阳之谓道，继之者善也，成之者性也。仁者见之谓之仁，知者见之谓之知，百姓日用不知，故君子之道鲜矣！”——《易经系辞上》
- J. “Think globally, act locally.”
- K. “重剑无锋，大巧不工”——《神雕侠侣》

Preface

This book comes out of my teaching two courses in the past decade at the school of mathematical sciences in Zhejiang University. The first six chapters correspond to the course “Numerical Analysis” (formerly “Numerical Approximation”) while the others to “Numerical Solutions of Differential Equations.” Both courses are on the core curriculum for the major of computational mathematics in Zhejiang University, so this book can be used as a textbook in upper-level undergraduate or first-year graduate courses. Nonetheless, any non-math major with a good grasp on linear algebra and calculus should be able to follow the contents.

If computational mathematics were a lady, math would be her soul, algorithms and programming would be her body, and the applications would be her appearance. Thus, although this book is mostly about the math part, I have frequently emphasized the interplay between theorems and algorithms, the domination of computer hardware over software, and the influence of algorithmic complexity upon mathematical analysis. Overall, the selected topics in this book converge to the theme of numerically solving differential equations with finite difference and finite volume methods.

Inside mathematics, numerical analysis is based on other more pure branches such as linear algebra, Fourier analysis, complex analysis, real analysis, and functional analysis. Thus it is recommended that a student already have a solid ground on linear algebra and calculus before starting her journey on numerical analysis. On the one hand, the first crucial step in learning anything new is to *link* the new concept to things that are already firmly established, which could be everyday experiences or certain preliminary knowledge. On the other hand, the study of numerical analysis could reinforce the memory or deepen the understanding of prerequisites. It is through these bidirectional processes that a student comprehends the new material, extends her network of knowledge, and even gain insights and power. To facilitate these learning processes, I have collected a sufficient amount of contents in the appendices so that the footprint of numerical analysis upon the more pure mathematical ground is clear; see Figure 1 for the structure of this ground in terms of a partial order of various spaces.

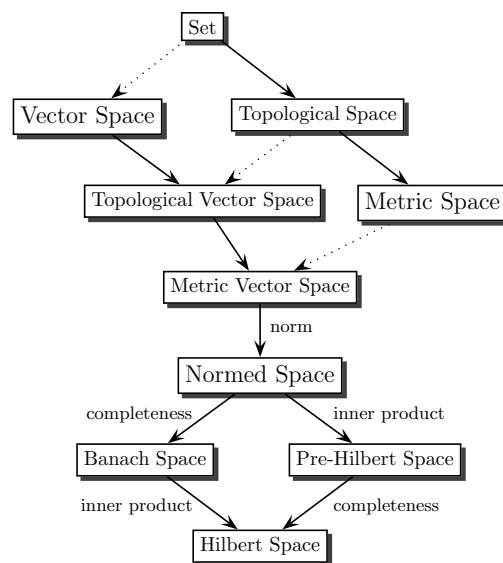


Figure 1: The inheritance diagram of various spaces upon which this book is based. Each arrow from A to B indicates the “is-a” relation, i.e., B is a special kind of A ; see Definition A.59 and Example A.63. In particular, an arrow with a dotted line indicates adding structures of the vector space in Definition C.2. Sets and logics are touched on in Appendix A, vector spaces in Appendix C, metric spaces in Appendices D, and Banach spaces in Appendices E

David Hilbert once wrote: “Math is a game played according to certain simple rules with meaningless marks on paper.” In order to make the “meaningless marks” meaningful, I have tried to convey the simple rules by drawing 76 figures and by explaining the central ideas behind the packaging.

Typically, a chapter is a coherent story that begins with its goal as finding solutions for a ubiquitous problem; then we progress towards that goal, under the guide of analysis and the simple ideas, until the problem is solved satisfactorily. The storyline consists of various environments: a **Definition** introduces a new concept, an **Example** usually connects the new concept to established facts, and a **Lemma**, a **Theorem** or a **Corollary** advances the storyline. In comparison, a **Remark** is more subjective as it gives motivations, compares concepts, offers an alternate proof, or explain the simple rules from the author’s viewpoint. Different from the above environments, a **Notation** is numbered within the entire book instead of within a chapter.

For each chapter and appendix, a condensed storyline is given as follows.

Chapter 4 concerns solving nonlinear equations by classical iterative algorithms including the bisection method, Newton’s method, the secant method, and fixed-point methods. Apart from the convergence of these algorithms, their speed of convergence is also quantified and analyzed. In particular, I am proud of the original proof of the convergence rate of the secant method. This opening chapter sets the stage for other chapters that an algorithm can be considered as a *contract* between the user and the developer: the user must ensure the preconditions of the input while the developer has to fulfill the postconditions of the output.

Also applying to theorems, this “contract” divides the responsibilities between the developer and the user and relates mathematical proofs to algorithmic implementations. However, rarely does a mathematics major spontaneously apply this contract idea at the task of designing an algorithm; I hope the contract idea could help!

Chapter 6 is devoted to interpolation by univariate polynomials. The unique interpolating polynomial can be constructed by the Lagrange formula or Newton’s formula. As an advantage of Newton’s formula, a newly added interpolation condition can be easily accommodated by appending an extra term to the original polynomial, increasing its degree by one. The implementation through the table of divided differences also works for Hermite interpolations. Unfortunately, polynomial interpolation of very high degree on uniform grids suffers from wide oscillations, cf. the Runge phenomenon; this is alleviated by Chebyshev polynomials that minimize the interpolation error through a wise choice of locations of interpolation nodes.

Chapter 7 focuses on splines, another way to circumvent the Runge phenomenon via interpolating the given function on a large number of intervals, with a polynomial of a fixed low degree on the interior of each interval and a lowered continuity at the junction of adjacent intervals. The piecewise-polynomial form of a spline can be straightforwardly constructed from this description, but for a different degree the process must be repeated. In contrast, the B-spline of degree $n + 1$ is defined recursively from that of degree n , leading to a better efficiency, flexibility, and reusability. This is analogous to the advantage of Newton’s formula over the Lagrange interpolation formula. In both cases, it is the divided difference that enables us to climb the ladder of polynomial degrees; as the key difference, it is regarded in Chapter 6 as a generalized derivative $f[x_0, \dots, x_k]$ of f whereas here as a linear functional $[x_0, \dots, x_k]f$ that acts on f .

Chapter 1 presents computer arithmetic and conditioning as an interplay between computer and analysis. The hardware of a floating-point number system is characterized by a quadruple (β, p, L, U) , from which we derive software concepts such as the machine precision. Then we analyze the generation and accumulation of rounding errors caused by the finite precision to obtain the model of machine arithmetic in Theorem 1.42 as a central result, which is rare in books on numerical analysis. In the last section, we elaborate on stability and conditioning of computer arithmetic.

Chapter 8 deals with the best approximation problem in the 2-norm. In an inner-product space, we construct an orthonormal basis by the Gram-Schmidt process, obtain the best approximation from Fourier expansions, and apply this abstract technique to continuous and discrete least squares. We also augment the technique to the solutions of ill-posed problems via spectral cutoff and Tikhonov regularization.

Chapter 9 further illustrates the power of orthogonal systems on numerical quadrature: the degree of exactness of a Gauss formula is almost twice of a corresponding Newton-Cotes formula. Formulas of numerical differentiation are also analyzed in the two cases of exact input and perturbed input.

Chapter 10 opens the second half of this book on numeri-

cally solving differential equations by an exposition on finite difference methods for boundary value problems. With a variety of techniques such as Green’s function and the discrete maximum principle, we show convergences in the 2-norm and the max-norm, in one dimension and two dimensions, and on regular and irregular domains with a number of different boundary conditions. Starting from this chapter, exercises are spreaded within the main text instead of agglomerating at the end of the chapter.

Chapter 3 briefly reviews classical iterative methods to smooth the way of later chapters.

Chapter 11 contains the bare necessities for a minimum story on multigrid methods; we mainly follow the excellent book by Briggs et al. [2000].

Chapters 12 and 13 cover classical numerical methods for solving ordinary differential equations. After laying down the mathematical foundations in Section 12.1, we study four basic methods in Section 12.2 to bring our attention to consistency, convergence, zero stability, and absolute stability. These essential concepts are revisited in the context of linear multistep methods in Section 12.4, with their analysis summarized as approximating the complex logarithm function by rational polynomials. In Section 12.5, we explain the concept of stiffness with physical examples and introduce the A-stability as its antidote. Section 13.1 shifts gears to one-step methods, transitioning to Runge-Kutta methods in Section 13.2, where we examine explicit and implicit Runge-Kutta methods, in particular the collocation methods, and emphasize various notions of stability such as the I-stability, the L-stability, the B-stability, and the algebraic stability.

Chapter 14 combines the theory and algorithms in Chapters 10, 12, and 13 to form the method of lines (MOL). In Section 14.1, we prove the famous Lax equivalence theorem, once again demonstrating the central theme of “consistency + stability = convergence” in numerical analysis. It is often desired that numerical methods preserve key properties of partial differential equations; this is illustrated by a discrete maximum principle in solving the heat equation. In Section 14.2, we introduce some legacy methods for the advection equation such as the unconditionally unstable FTCS method, the second-order Lax-Wendroff method, and the lop-sided upwind method, approaching the CFL condition from the viewpoint of domains of dependence. The behaviors of various methods, with respect to dichotomies of dissipation versus dispersion and group velocity versus phase velocity, are fully explained by the modified equation analysis. The von Neumann stability is also introduced as an alternative approach for analyzing MOL.

Chapter 15 pertains to high-order finite-volume methods for solving the Poisson equation and the advection-diffusion equation. Based on Chapters 10, 13, and 14, we start with discrete operators that act on face- and cell-averaged values and finish at sophisticated MOLs with fourth-order accuracy both in time and in space.

Appendix A lists the very basics of first-order logic and ordered sets.

Appendix C contains prerequisites in linear algebra. I give not only major theorems referenced in the main text but also intermediate lemmas that lead to these major the-

orem, covering contents from vector spaces to operators on inner-product spaces.

Appendix B reviews some major results in basic analysis, sprinkled with examples and exercises that illustrate key ideas of numerical analysis.

Appendix D transitions from basic analysis to functional analysis.

Appendix E is a sequence of lemmas and theorems that lead to the principle of uniform boundedness, the key in proving the convergence of numerical quadrature formulas in Chapter 9 and the Lax equivalence theorem in Chapter 14.

Appendix F helps to understand the behavior of various finite difference methods in solving linear partial differential equations.

Although the subject of numerical analysis is mature and well established, whenever appropriate I have tried to bring the contents up to date. For examples, the IEEE-754 standard for FPN arithmetic in Chapter 1 is the most recent version of 2019; the treatment of ill-posed problems in Section 8.4 could bridge gaps between the classical numerical analysis and the contemporary machine learning.

During my preparation of this book, many students and colleagues pointed out typos, raised questions, and suggested improvements; I am grateful to all of them. In particular, I would like to acknowledge the helps from Prof. Heyu Wang and from my students Difei Hu (who plotted many figures), Huiteng Li (who wrote the proof of the Lax equivalence theorem), Yang Li, and Junxiang Pan.

Chapter 1

Computer Arithmetic

Motivation 1.1. If scientific computing were a lady, then math would be her soul, computer science her body, and various applications her appearance. In light of this analogy, the first important principle in scientific computing is that *hardware affects software*.

A notable example is parallel computing. In the 20th century, computers with multiple cores were rare and most scientific computing programs were written for execution on a single thread. In contrast, nowadays parallel computing has become a standard feature of modern personal computers, smartphones, and tablets. During the past decades, the hardware development of supercomputers and multi-core chips has shifted the focus of algorithm design from one set of sequential steps to a multidimensional coordination of concurrent tasks. The performance metric has also been shifted from minimizing the total number of operations to maximizing concurrency and minimizing communication overhead. Although parallel algorithms will not be discussed in this book, the importance to understand computer hardware should be clear.

Fortunately, knowledge on computer hardware relevant to numerical analysis is only a handful of sand and mostly about the underlying floating-point number (FPN) system.

Roadmap 1.2. In Section 1.1, we describe FPNs and their representation by the IEEE standard 754 published in 2019. In an FPN system, there are only a finite number of FPNs and each FPN has a finite number of accuracy digits. Accordingly, the rounding errors incurred in approximating the real numbers with the FPNs and the accumulation of these rounding errors in floating-point operations must be examined to ensure the meaningfulness of FPN calculations; this is done in Sections 1.2, 1.3, and 1.4. In particular, Section 1.5 concerns the phenomenon of the subtraction of two nearly equal FPNs yielding a catastrophic loss of precision; how to minimize this precision loss is also discussed.

1.1 Floating-point number (FPN) systems

Definition 1.3. The *base or radix of a positional numeral system* is the number of unique symbols used to represent numbers.

Example 1.4. The *binary numeral system* consists of two digits: “0” and “1,” and thus its base is 2. The *decimal system* consists of ten digits: “0” – “9,” and thus its base is 10.

Definition 1.5. A *bit* is the basic unit of information in computing; it can have only one of two values 0 and 1.

Definition 1.6. A *byte* is a unit of information in computing that commonly consists of 8 bits; it is the smallest addressable unit of memory in many computers.

Definition 1.7. A *word* is a group of bits with fixed size that are handled as a unit by the instruction set architecture (ISA) and/or hardware of the processor. The *word size/width/length* is the number of bits in a word and is an important characteristic of processor or computer architecture.

Example 1.8. 32-bit and 64-bit computers are mostly common these days. A 32-bit register can store 2^{32} values, hence a processor with 32-bit memory address can directly access 4GB byte-addressable memory.

Definition 1.9 (FPN). A *floating-point number* (FPN) is a number of the form

$$x = \pm m \times \beta^e, \quad (1.1)$$

where β is the base or radix, $e \in [L, U]$, and the *significand* (or *mantissa*) m is a number of the form

$$m = \left(d_0 + \frac{d_1}{\beta} + \cdots + \frac{d_{p-1}}{\beta^{p-1}} \right), \quad (1.2)$$

where the integer d_i satisfies $\forall i \in [0, p-1], d_i \in [0, \beta-1]$. d_0 and d_{p-1} are called the *most significant digit* and the *least significant digit*, respectively. The *string of digits* of m is the string $d_0.d_1d_2 \cdots d_{p-1}$, of which the portion $.d_1d_2 \cdots d_{p-1}$ is called the *fraction* of m .

Algorithm 1.10. A decimal integer can be converted to a binary number via the following method:

- divide by 2 and record the remainder,
- repeat until you reach 0,
- concatenate the remainder backwards.

A decimal fraction can be converted to a binary number via the following method:

- multiply by 2 and check whether the integer part is no less than 1: if so record 1; otherwise record 0,
- repeat until you reach 0,
- concatenate the recorded bits forward.

Combine the above two methods and we can convert any decimal number to its binary counterpart.

Example 1.11. Convert 156 to binary number:

$$156 = (10011100)_2.$$

Definition 1.12 (FPN systems). A *floating-point number system* $\mathcal{F}$ is a proper subset of the rational numbers $\mathbb{Q}$, and it is characterized by a 4-tuple (β, p, L, U) with

- the *base* (or radix) β ;
- the *precision* (or significand digits) p ;
- the *exponent range* $[L, U]$.

Definition 1.13. An FPN is *normalized* if its mantissa satisfies $1 \leq m < \beta$.

Example 1.14. What is the normalized binary form of $\frac{2}{3}$?

$$\begin{aligned} \frac{2}{3} &= (0.a_1a_2a_3\cdots)_2 = (0.1010\cdots)_2 \\ &= (1.0101010\cdots)_2 \times 2^{-1}. \end{aligned}$$

Definition 1.15. The *subnormal* or *denormalized* numbers are FPNs of the form (1.1) with $e = L$ and $m \in (0, 1)$. A normalized FPN system can be *extended* by including the subnormal numbers.

Definition 1.16 (IEEE standard 754-2019). The *single precision* and *double precision* FPNs of current IEEE (Institute of Electrical and Electronics Engineers) standard 754 published in 2019 are normalized FPN systems with three binary formats (32, 64, and 128 bits) and two decimal formats (64 and 128 bits).

$$\beta = 2, p = 23 + 1, e \in [-126, 127]; \quad (1.3a)$$

$$\beta = 2, p = 52 + 1, e \in [-1022, 1023]; \quad (1.3b)$$

$$\beta = 2, p = 112 + 1, e \in [-16382, 16383]; \quad (1.3c)$$

$$\beta = 10, p = 16, e \in [-1022, 1023]; \quad (1.3d)$$

$$\beta = 10, p = 34, e \in [-6143, 6144]. \quad (1.3e)$$

±	exponent (e)	normalized significand (m)
• implicit radix point		

Figure 1.1: Representing floating-point numbers in the IEEE 754 standard.

Example 1.17. In the IEEE 754 standard, there are some further details on the representation specifications of FPNs.

Referring to Figure 1.1, some major representation specifications of the 32-bit FPNs are as follows.

- Out of the 32 bits, 1 is reserved for the sign, 8 for the exponents, 23 for the significand (see the plot above for the locations and the implicit radix point).
- The precision is 24 because we can choose $d_0 = 1$ for normalized binary floating-point numbers and get away with never storing d_0 .
- The exponent has $2^8 = 256$ possibilities. If we assign $1, 2, \dots, 256$ to these possibilities, it would not be possible to represent numbers whose magnitudes are smaller than one. Hence we subtract $1, 2, \dots, 256$ by 128 to shift the exponents to $-127, -126, \dots, 0, \dots, 127, 128$. Out of these numbers, $\pm m \times \beta^{-127}$ is reserved for ± 0 and subnormal numbers while $\pm m \times \beta^{128}$ is reserved for $\pm \infty$ and NaNs including **qNaN** (quiet) and **sNaN** (signaling).

Definition 1.18. The *machine precision* of a normalized FPN system $\mathcal{F}$ is the distance between 1.0 and the next larger FPN in $\mathcal{F}$,

$$\epsilon_M := \beta^{1-p}. \quad (1.4)$$

Definition 1.19. The underflow limit (UFL) and the overflow limit (OFL) of a normalized FPN system $\mathcal{F}$ are respectively

$$\text{UFL}(\mathcal{F}) := \min |\mathcal{F} \setminus \{0\}| = \beta^L, \quad (1.5)$$

$$\text{OFL}(\mathcal{F}) := \max |\mathcal{F}| = \beta^U (\beta - \beta^{1-p}). \quad (1.6)$$

Example 1.20. By default Matlab adopts IEEE 754 double precision arithmetic. Three characterizing constants are

- **eps** is the machine precision

$$\epsilon_M = \beta^{1-p} = 2^{1-(52+1)} = 2^{-52} \approx 2.22 \times 10^{-16},$$

- **realmin** is $\text{UFL}(\mathcal{F})$

$$\min |\mathcal{F} \setminus \{0\}| = \beta^L = 2^{-1022} \approx 2.22 \times 10^{-308},$$

- **realmax** is $\text{OFL}(\mathcal{F})$

$$\max |\mathcal{F}| = \beta^U (\beta - \beta^{1-p}) \approx 1.80 \times 10^{308}.$$

In C/C++, these constants are defined in `<float>` and `float.h` by macros `DBL_EPSILON`, `DBL_MIN`, and `DBL_MAX`.

Corollary 1.21. The cardinality of a normalized binary FPN system $\mathcal{F}$ is

$$\#\mathcal{F} = 2^p(U - L + 1) + 1. \quad (1.7)$$

Proof. The cardinality can be proved by Axiom A.33. The factor 2^p comes from the sign bit and the mantissa. By Example 1.17, $U - L + 1$ is the number of exponents represented in $\mathcal{F}$. The trailing “+1” in (1.7) accounts for the number 0. $\square$

Definition 1.22. The *range* of a normalized FPN system is a subset of $\mathbb{R}$ that consists of two intervals,

$$\mathcal{R}(\mathcal{F}) := \{x : x \in \mathbb{R}, \text{UFL}(\mathcal{F}) \leq |x| \leq \text{OFL}(\mathcal{F})\}. \quad (1.8)$$

Example 1.23. Consider a normalized FPN system with the characterization $\beta = 2, p = 3, L = -1, U = +1$.



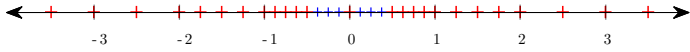
The four FPNs

$$1.00 \times 2^0, 1.01 \times 2^0, 1.10 \times 2^0, 1.11 \times 2^0$$

correspond to the four ticks in the plot starting at 1 while

$$1.00 \times 2^1, 1.01 \times 2^1, 1.10 \times 2^1, 1.11 \times 2^1$$

correspond to the four ticks starting at 2. Add subnormal FPNs and we have



Definition 1.24. Two normalized FPNs a, b are *adjacent* to each other in $\mathcal{F}$ iff

$$\forall c \in \mathcal{F} \setminus \{a, b\}, \quad |a - b| < |a - c| + |c - b|. \quad (1.9)$$

Lemma 1.25. Let a, b be two adjacent normalized FPNs satisfying $|a| < |b|$ and $ab > 0$. Then

$$\beta^{-1} \epsilon_M |a| < |a - b| \leq \epsilon_M |a|. \quad (1.10)$$

Proof. Consider $a > 0$, then $\Delta a := b - a > 0$. By Definitions 1.9 and 1.13, $a = m \times \beta^e$ with $1.0 \leq m < \beta$. a and b only differ from each other at the least significant digit, hence $\Delta a = \epsilon_M \beta^e$. Since $\frac{\epsilon_M}{\beta} < \frac{\epsilon_M}{m} \leq \epsilon_M$, we have $\frac{\Delta a}{a} \in (\beta^{-1} \epsilon_M, \epsilon_M]$. The other case is similar. $\square$

1.2 Rounding a single number

Definition 1.26 (Rounding). *Rounding* is a map $\text{fl} : \mathbb{R} \rightarrow \mathcal{F} \cup \{+\infty, -\infty, \text{NaN}\}$. The default rounding mode is *round to nearest*, i.e. $\text{fl}(x)$ is chosen to minimize $|\text{fl}(x) - x|$ for $x \in \mathcal{R}(\mathcal{F})$. In the case of a tie, $\text{fl}(x)$ is chosen by *round to even*, i.e. $\text{fl}(x)$ is the one with an even last digit d_{p-1} .

Definition 1.27. A rounded number $\text{fl}(x)$ *overflows* if $|x| > \text{OFL}(\mathcal{F})$, in which case $\text{fl}(x) = \text{NaN}$, or *underflows* if $0 < |x| < \text{UFL}(\mathcal{F})$, in which case $\text{fl}(x) = 0$. An underflow of an extended FPN system is called a *gradual underflow*.

Definition 1.28. The *unit roundoff* of $\mathcal{F}$ is the number

$$\epsilon_u := \frac{1}{2} \epsilon_M = \frac{1}{2} \beta^{1-p}. \quad (1.11)$$

Theorem 1.29 (Range of round-off errors). For $x \in \mathcal{R}(\mathcal{F})$ as in (1.8), we have

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| < \epsilon_u. \quad (1.12)$$

Proof. By Definition A.45, $\mathcal{R}(\mathcal{F})$ is a subset of $\mathbb{R}$ and is thus a chain. Therefore $\forall x \in \mathcal{R}(\mathcal{F}), \exists x_L, x_R \in \mathcal{F}$ s.t.

- x_L and x_R are adjacent,
- $x_L \leq x \leq x_R$.

If $x = x_L$ or x_R , then $\text{fl}(x) - x = 0$ and (1.12) clearly holds. Otherwise $x_L < x < x_R$. Then Lemma 1.25 and Definitions 1.24 and 1.26 yield

$$|\text{fl}(x) - x| \leq \frac{1}{2} |x_R - x_L| \leq \epsilon_u \min(|x_L|, |x_R|) < \epsilon_u |x|. \quad (1.13)$$

Hence $-\epsilon_u |x| < \text{fl}(x) - x < \epsilon_u |x|$, which yields (1.12). $\square$

Theorem 1.30. For $x \in \mathcal{R}(\mathcal{F})$, we have

$$\text{fl}(x) = \frac{x}{1 + \delta} \text{ where } |\delta| \leq \epsilon_u. \quad (1.14)$$

Proof. The proof is the same as that of Theorem 1.29, except that we replace the last inequality “ $< \epsilon_u |x|$ ” in (1.13) by “ $\leq \epsilon_u |\text{fl}(x)|$.” Consequently, the equality in (1.14) holds when $x = \frac{1}{2}(x_L + x_R)$ and $|\text{fl}(x)| = \min(|x_L|, |x_R|)$ has its significand as $m = 1.0$. $\square$

Example 1.31. Find x_L, x_R of $x = \frac{2}{3}$ in normalized single-precision IEEE 754 standard, which of them is $\text{fl}(x)$?

By Example 1.14, we have

$$\begin{aligned} \frac{2}{3} &= (0.1010 \dots)_2 = (1.0101010 \dots)_2 \times 2^{-1}. \\ x_L &= (1.010 \dots 10)_2 \times 2^{-1}; \\ x_R &= (1.010 \dots 11)_2 \times 2^{-1}, \end{aligned}$$

where the last bit of x_L must be 0 because the IEEE 754 standard states that 23 bits are reserved for the mantissa. It follows that

$$\begin{aligned} x - x_L &= \frac{2}{3} \times 2^{-24}, \\ x_R - x_L &= 2^{-24}, \\ x_R - x &= (x_R - x_L) - (x - x_L) = \frac{1}{3} \times 2^{-24}. \end{aligned}$$

Thus Definition 1.26 implies $\text{fl}(x) = x_R$.

1.3 Floating-point operations

Roadmap 1.32. Definitions 1.33, 1.37, and 1.40 in this section model the binary floating-point operations of computer arithmetic. The steps in these definitions may be different in varying hardware architectures. But the main point is to illustrate the interaction of software operations and hardware format in Lemmas 1.36, 1.39, 1.41 and Theorem 1.42. In other words, it suffices to scrutinize the model of machine arithmetic on only one specific hardware implementation.

Definition 1.33 (Addition/subtraction of two FPNs). Express $a, b \in \mathcal{F}$ as $a = M_a \times \beta^{e_a}$ and $b = M_b \times \beta^{e_b}$ where $M_a = \pm m_a$ and $M_b = \pm m_b$. With the assumption $|a| \geq |b|$, the sum $c := \text{fl}(a + b) \in \mathcal{F}$ is calculated in a register of precision at least $2p$ as follows.

(i) Exponent comparison:

- If $e_a - e_b > p + 1$, set $c = a$ and return c ;
- otherwise set $e_c \leftarrow e_a$ and $M_b \leftarrow M_b / \beta^{e_a - e_b}$.

(ii) Perform the addition $M_c \leftarrow M_a + M_b$ in the register with rounding to nearest.

(iii) Normalization:

- If $|M_c| = 0$, return 0.
- If $|M_c| \in (\beta, \beta^2)$, set $M_c \leftarrow M_c/\beta$ and $e_c \leftarrow e_c + 1$.
- If $|M_c| \in (0, \beta - \epsilon_u(p))$, repeat $M_c \leftarrow M_c\beta$, $e_c \leftarrow e_c - 1$ until $|M_c| \in [1, \beta)$.
- If $|M_c| \in [\beta - \epsilon_u(p), \beta]$, set $|M_c| \leftarrow 1.0$ and $e_c \leftarrow e_c + 1$.

(iv) Check range:

- return NaN if e_c overflows,
- return 0 if e_c underflows.

(v) Round M_c (to nearest) to precision p .

(vi) Set $c \leftarrow M_c \times \beta^{e_c}$.

Here $\epsilon_u(p)$ in step (iii) is the unit round-off for FPNs with precision p , cf. Definition 1.28.

Example 1.34. Consider the calculation of $c := \text{fl}(a + b)$ with $a = 1.234 \times 10^4$ and $b = 5.678 \times 10^0$ in an FPN system $\mathcal{F} : (10, 4, -7, 8)$.

- (i) $b \leftarrow 0.0005678 \times 10^4$; $e_c \leftarrow 4$.
- (ii) $m_c \leftarrow 1.2345678$.
- (iii) do nothing.
- (iv) do nothing.
- (v) $m_c \leftarrow 1.235$.
- (vi) $c = 1.235 \times 10^4$.

For $b = 5.678 \times 10^{-2}$, $c = a$ would be returned in step (i).

Example 1.35. Consider the calculation of $c := \text{fl}(a + b)$ with $a = 1.000 \times 10^0$ and $b = -9.000 \times 10^{-5}$ in an FPN system $\mathcal{F} : (10, 4, -7, 8)$.

- (i) $b \leftarrow -0.0000900 \times 10^0$; $e_c \leftarrow 0$.
- (ii) $m_c \leftarrow 0.9999100$.
- (iii) $e_c \leftarrow e_c - 1$; $m_c \leftarrow 9.9991000$.
- (iv) do nothing.
- (v) $m_c \leftarrow 9.999$.
- (vi) $c = 9.999 \times 10^{-1}$.

For $b = -9.000 \times 10^{-6}$, $c = a$ would be returned in step (i).

Lemma 1.36. For $a, b \in \mathcal{F}$, $a + b \in \mathcal{R}(\mathcal{F})$ implies

$$\text{fl}(a + b) = (a + b)(1 + \delta), \quad |\delta| < \epsilon_u. \quad (1.15)$$

Proof. The round-off error in step (v) always dominates those in step (i) and step (ii), both of which, because of the $2p$ precision, are nonzero only in the case of $e_a - e_b = p + 1$. Then (1.15) follows from Theorem 1.29. $\square$

Definition 1.37 (Multiplication of two FPNs). Express $a, b \in \mathcal{F}$ as $a = M_a \times \beta^{e_a}$ and $b = M_b \times \beta^{e_b}$ where $M_a = \pm m_a$ and $M_b = \pm m_b$. The product $c := \text{fl}(ab) \in \mathcal{F}$ is calculated in a register of precision at least $2p$ as follows.

(i) Exponent sum: $e_c \leftarrow e_a + e_b$.

(ii) Perform the multiplication $M_c \leftarrow M_a M_b$ in the register.

(iii) Normalization:

- If $|M_c| \in (\beta, \beta^2)$, set $M_c \leftarrow M_c/\beta$ and $e_c \leftarrow e_c + 1$.
- If $|M_c| \in [\beta - \epsilon_u(p), \beta]$, set $|M_c| \leftarrow 1.0$ and $e_c \leftarrow e_c + 1$.

(iv) Check range:

- return NaN if e_c overflows,
- return 0 if e_c underflows.

(v) Round M_c (to nearest) to precision p .

(vi) Set $c \leftarrow M_c \times \beta^{e_c}$.

Here $\epsilon_u(p)$ in step (iii) is the unit round-off for FPNs with precision p , cf. Definition 1.28.

Example 1.38. Consider the calculation of $c := \text{fl}(ab)$ with $a = 2.345 \times 10^4$ and $b = 6.789 \times 10^0$ in an FPN system $\mathcal{F} : (10, 4, -7, 8)$.

- (i) $e_c \leftarrow 4$.
- (ii) $M_c \leftarrow 15.920205$.
- (iii) $m_c \leftarrow 1.5920205$, $e_c \leftarrow 5$.
- (iv) do nothing.
- (v) $m_c \leftarrow 1.592$.
- (vi) $c = 1.592 \times 10^5$.

Lemma 1.39. For $a, b \in \mathcal{F}$, $|ab| \in \mathcal{R}(\mathcal{F})$ implies

$$\text{fl}(ab) = (ab)(1 + \delta), \quad |\delta| < \epsilon_u. \quad (1.16)$$

Proof. The error only comes from the round-off in steps (v). Then (1.16) follows from Theorem 1.29. $\square$

Definition 1.40 (Division of two FPNs). Express $a, b \in \mathcal{F}$ as $a = M_a \times \beta^{e_a}$ and $b = M_b \times \beta^{e_b}$ where $M_a = \pm m_a$ and $M_b = \pm m_b$. The quotient $c = \text{fl}\left(\frac{a}{b}\right) \in \mathcal{F}$ is calculated in a register of precision at least $2p + 1$ as follows.

- (i) If $m_b = 0$, return NaN; otherwise set $e_c \leftarrow e_a - e_b$.
- (ii) Perform the division $M_c \leftarrow M_a/M_b$ in the register with rounding to nearest.

(iii) Normalization:

- If $|M_c| < 1$, set $M_c \leftarrow M_c\beta$, $e_c \leftarrow e_c - 1$.

(iv) Check range:

- return NaN if e_c overflows,
- return 0 if e_c underflows.

(v) Round M_c (to nearest) to precision p .

(vi) Set $c \leftarrow M_c \times \beta^{e_c}$.

Lemma 1.41. For $a, b \in \mathcal{F}$, $\frac{a}{b} \in \mathcal{R}(\mathcal{F})$ implies

$$\text{fl}\left(\frac{a}{b}\right) = \frac{a}{b}(1 + \delta), \quad |\delta| < \epsilon_u. \quad (1.17)$$

Proof. In the case of $|M_a| = |M_b|$, there is no rounding error in Definition 1.40 and (1.17) clearly holds. Hereafter we denote by M_{c1} and M_{c2} the results of steps (ii) and (v) in Definition 1.40, respectively.

In the case of $|M_a| > |M_b|$, the condition $a, b \in \mathcal{F}$, Definition 1.18, and $|M_a|, |M_b| \in [1, \beta)$ imply

$$\left| \frac{M_a}{M_b} \right| \geq \frac{\beta - \epsilon_M}{\beta - 2\epsilon_M} > 1 + \beta^{-1}\epsilon_M, \quad (1.18)$$

which further implies that the normalization step (iii) in Definition 1.40 is not invoked. By Definitions 1.26, 1.18, and 1.28, the unit roundoff of a register with precision $p+k$ is

$$\frac{1}{2}\beta^{1-p-k} = \frac{1}{2}\beta^{1-p}\beta^{1-p}\beta^{p-1-k} = \beta^{p-1-k}\epsilon_u\epsilon_M,$$

and hence the unit roundoff of the register in Definition 1.40 is $\beta^{-2}\epsilon_u\epsilon_M$. Therefore we have

$$M_{c2} = M_{c1} + \delta_2 = \frac{M_a}{M_b} + \delta_1 + \delta_2 = \frac{M_a}{M_b}(1 + \delta),$$

where $|\delta_2| < \epsilon_u$, $|\delta_1| < \beta^{-2}\epsilon_u\epsilon_M$, and

$$|\delta| = \left| \frac{\delta_1 + \delta_2}{M_a/M_b} \right| < \frac{\epsilon_u(1 + \beta^{-2}\epsilon_M)}{1 + \beta^{-1}\epsilon_M} < \epsilon_u;$$

the first inequality follows from (1.18) and the triangle inequality.

Consider the last case $|M_a| < |M_b|$. It is impossible to have $|M_{c1}| = 1$ in step (ii) because

$$\left| \frac{M_a}{M_b} \right| \leq \frac{\beta - 2\epsilon_M}{\beta - \epsilon_M} = 1 - \frac{\epsilon_M}{\beta - \epsilon_M} < 1 - \beta^{-1}\epsilon_M$$

and the precision of the register is greater than $p+1$. Therefore $|M_{c1}| < 1$ must hold and in Definition 1.40 step (iii) is invoked to yield

$$\begin{aligned} M_{c1} &= \frac{M_a}{M_b} + \delta_1, & |\delta_1| &< \beta^{-2}\epsilon_u\epsilon_M; \\ M_{c2} &= \beta M_{c1} + \delta_2, & |\delta_2| &< \epsilon_u \\ &= \beta \frac{M_a}{M_b} \left(1 + \frac{\beta\delta_1 + \delta_2}{\beta M_a/M_b} \right), \end{aligned}$$

where the denominator in the parentheses satisfies

$$\beta \left| \frac{M_a}{M_b} \right| \geq \frac{\beta}{\beta - \epsilon_M} = 1 + \frac{\epsilon_M}{\beta - \epsilon_M} > 1 + \beta^{-1}\epsilon_M.$$

Hence we have

$$|\delta| = \left| \frac{\beta\delta_1 + \delta_2}{\beta M_a/M_b} \right| < \frac{\beta^{-1}\epsilon_u\epsilon_M + \epsilon_u}{1 + \beta^{-1}\epsilon_M} = \epsilon_u. \quad \square$$

Theorem 1.42 (Model of machine arithmetic). Denote by $\mathcal{F}$ a normalized FPN system with precision p . For each arithmetic operation $\odot = +, -, \times, /$, we have

$$\forall a, b \in \mathcal{F}, a \odot b \in \mathcal{R}(\mathcal{F}) \implies \text{fl}(a \odot b) = (a \odot b)(1 + \delta) \quad (1.19)$$

where $|\delta| < \epsilon_u$ if and only if these binary operations are performed in a register with precision $2p+1$.

Proof. This follows from Lemmas 1.36, 1.39, and 1.41. $\square$

1.4 The propagation of rounding errors

Theorem 1.43. If $\forall i = 0, 1, \dots, n$, $a_i \in \mathcal{F}$, $a_i > 0$, then

$$\text{fl} \left(\sum_{i=0}^n a_i \right) = (1 + \delta_n) \sum_{i=0}^n a_i, \quad (1.20)$$

where $|\delta_n| < (1 + \epsilon_u)^n - 1 \approx n\epsilon_u$.

Proof. Define $s_k := \sum_{i=0}^k a_i$,

$$\begin{cases} s_0 &:= a_0; \\ s_{k+1} &:= s_k + a_{k+1}, \end{cases} \quad \begin{cases} s_0^* &:= a_0; \\ s_{k+1}^* &:= \text{fl}(s_k^* + a_{k+1}), \end{cases}$$

$$\delta_k := \frac{s_k^* - s_k}{s_k}, \quad \epsilon_k := \frac{s_{k+1}^* - (s_k^* + a_{k+1})}{s_k^* + a_{k+1}}.$$

In other words, δ_k is the accumulated rounding error and ϵ_k is the rounding error at the k th step. Then we have

$$\begin{aligned} \delta_{k+1} &= \frac{s_{k+1}^* - s_{k+1}}{s_{k+1}} = \frac{(s_k^* + a_{k+1})(1 + \epsilon_k) - s_{k+1}}{s_{k+1}} \\ &= \frac{(s_k(1 + \delta_k) + a_{k+1})(1 + \epsilon_k) - s_k - a_{k+1}}{s_{k+1}} \\ &= \frac{(\epsilon_k + \delta_k + \epsilon_k \delta_k)s_k + \epsilon_k a_{k+1}}{s_{k+1}} \\ &= \frac{\epsilon_k s_{k+1} + \delta_k(1 + \epsilon_k)s_k}{s_{k+1}} = \epsilon_k + \delta_k(1 + \epsilon_k) \frac{s_k}{s_{k+1}}. \end{aligned}$$

The condition of a_i 's being positive implies $s_k < s_{k+1}$, and Theorem 1.29 states $|\epsilon_k| < \epsilon_u$. Hence we have

$$|\delta_{k+1}| < |\epsilon_k| + |\delta_k|(1 + \epsilon_u) < \epsilon_u + |\delta_k|(1 + \epsilon_u).$$

An easy induction then shows that

$$\begin{aligned} \forall k \in \mathbb{N}, |\delta_{k+1}| &< \epsilon_u \sum_{i=0}^k (1 + \epsilon_u)^i \\ &= \epsilon_u \frac{(1 + \epsilon_u)^{k+1} - 1}{1 + \epsilon_u - 1} = (1 + \epsilon_u)^{k+1} - 1, \end{aligned} \quad (1.21)$$

where the second step follows from the summation formula of geometric series. The proof is completed by the binomial theorem. $\square$

Theorem 1.44. For given $\mu \in \mathbb{R}^+$ and a positive integer $n \leq \lfloor \frac{\ln 2}{\mu} \rfloor$, suppose $|\delta_i| \leq \mu$ for each $i = 1, 2, \dots, n$. Then

$$1 - n\mu \leq \prod_{i=1}^n (1 + \delta_i) \leq 1 + n\mu + (n\mu)^2, \quad (1.22)$$

or equivalently, for $I_n := [-\frac{1}{1+n\mu}, 1]$,

$$\exists \theta \in I_n \text{ s.t. } \prod_{i=1}^n (1 + \delta_i) = 1 + \theta(n\mu + n^2\mu^2). \quad (1.23)$$

Proof. The condition $|\delta_i| \leq \mu$ implies

$$(1 - \mu)^n \leq \prod_{i=1}^n (1 + \delta_i) \leq (1 + \mu)^n.$$

Taylor expansion of $f(\mu) = (1 - \mu)^n$ at $\mu = 0$ with Lagrangian remainder yields

$$(1 - \mu)^n \geq 1 - n\mu,$$

which implies the first inequality in (1.22). On the other hand, the Taylor series of e^x for $x \in \mathbb{R}^+$ satisfies

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \\ &= 1 + x + \frac{x^2}{2!} \left(1 + \frac{x}{3} + \frac{2x^2}{4!} + \cdots \right) \\ &\leq 1 + x + \frac{x^2}{2} e^x. \end{aligned}$$

Set $x = n\mu$ in the above inequality, apply the condition $n\mu \leq \ln 2$, and we have

$$e^{n\mu} \leq 1 + n\mu + (n\mu)^2,$$

which, together with the inequality $(1 + \mu)^n \leq e^{n\mu}$, yields the second inequality in (1.22).

Finally, (1.22) implies that $\prod_{i=1}^n (1 + \delta_i)$ is in the range of the continuous function $f(\tau) = 1 + \tau(1 + n\mu)n\mu$ on I_n . The rest of the proof follows from the intermediate value theorem. $\square$

1.5 Avoiding catastrophic cancellation

Example 1.45 (Catastrophic cancellation). For two real numbers $x, y \in \mathcal{R}(\mathcal{F})$, Theorems 1.29 and 1.42 imply

$$\begin{aligned} \text{fl}(\text{fl}(x) \odot \text{fl}(y)) &= (\text{fl}(x) \odot \text{fl}(y))(1 + \delta_3) \\ &= (x(1 + \delta_1) \odot y(1 + \delta_2))(1 + \delta_3) \end{aligned}$$

where $|\delta_i| \leq \epsilon_u$. From Theorems 1.42 and 1.44, we know that *multiplication is accurate*:

$$\begin{aligned} \text{fl}(\text{fl}(x) \times \text{fl}(y)) &= xy(1 + \delta_1)(1 + \delta_2)(1 + \delta_3) \\ &= xy[1 + \theta(3\epsilon_u + 3\epsilon_u^2 + \epsilon_u^3)], \end{aligned}$$

where $\theta \in [-1, 1]$. Similarly, *division is also accurate*:

$$\begin{aligned} \text{fl}(\text{fl}(x)/\text{fl}(y)) &= \frac{x(1 + \delta_1)}{y(1 + \delta_2)}(1 + \delta_3) \\ &= \frac{x}{y}(1 + \delta_1)(1 - \delta_2 + \delta_2^2 - \cdots)(1 + \delta_3) \\ &\approx \frac{x}{y}(1 + \delta_1)(1 - \delta_2)(1 + \delta_3). \end{aligned}$$

However, *addition and subtraction might not be accurate*:

$$\begin{aligned} \text{fl}(\text{fl}(x) + \text{fl}(y)) &= (x(1 + \delta_1) + y(1 + \delta_2))(1 + \delta_3) \\ &= (x + y + x\delta_1 + y\delta_2)(1 + \delta_3) \\ &= (x + y) \left(1 + \delta_3 + \frac{x\delta_1 + y\delta_2}{x + y} + \delta_3 \frac{x\delta_1 + y\delta_2}{x + y} \right). \end{aligned}$$

In other words, the relative error of addition or subtraction can be arbitrarily large when $x + y \rightarrow 0$.

Theorem 1.46 (Loss of most significant digits). Suppose $x, y \in \mathcal{F}$, $x > y > 0$, and

$$\beta^{-t} \leq 1 - \frac{y}{x} \leq \beta^{-s}. \quad (1.24)$$

Then the number of most significant digits that are lost in the subtraction $x - y$ is at most t and at least s .

Proof. Rewrite $x = m_x \times \beta^n$ and $y = m_y \times \beta^m$ with $1 \leq m_x, m_y < \beta$. Definition 1.33 and the condition $x > y$ imply that m_y , the significand of y , is shifted so that y has the same exponent as x before $m_x - m_y$ is performed in the register. Then we have

$$\begin{aligned} y &= (m_y \times \beta^{m-n}) \times \beta^n \\ \Rightarrow x - y &= (m_x - m_y \times \beta^{m-n}) \times \beta^n \\ \Rightarrow m_{x-y} &= m_x \left(1 - \frac{m_y \times \beta^m}{m_x \times \beta^n} \right) = m_x \left(1 - \frac{y}{x} \right) \\ \Rightarrow \beta^{-t} &\leq m_{x-y} < \beta^{1-s}. \end{aligned}$$

To normalize m_{x-y} into the interval $[1, \beta)$, it should be multiplied by at least β^s and at most β^t . In other words, m_{x-y} should be shifted to the left for at least s times and at most t times. Therefore the conclusion on the number of lost significant digits follows. $\square$

Rule 1.47. Catastrophic cancellation should be avoided whenever possible.

Example 1.48. Calculate $y = f(x) = x - \sin x$ for $x \rightarrow 0$. When x is small, a straightforward calculation would result in a catastrophic cancellation because $x \approx \sin x$. The solution is to use the Taylor series

$$\begin{aligned} x - \sin x &= x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \right) \\ &= \frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots \end{aligned}$$

Example 1.49. To sum up a long sequence of n FPNs, it is a good idea to adopt *compensated summation*, where a compensation variable with a sufficiently high precision is utilized to sum up the FPNs so that the total error depends not on n but only on the floating-point precision of the result. For more details, see [Higham, 2002, §4].

Summary 1.50. By the model of machine arithmetic (Theorem 1.42), the relative round-off error is less than the unit roundoff. In addition, the accumulation of roundoff errors is at worst linear, thanks to the fact of the unit roundoff being much smaller than 1. These two facts, however, do not imply that the intended arithmetic can always be directly translated to floating-point operations. Whenever two nearly equal numbers are subtracted, the calculation needs to be rephrased in a mathematically equivalent way to avoid catastrophic cancellations.

1.6 Theoretical problems

- I. Convert the decimal integer 477 to a normalized FPN with $\beta = 2$.
- II. Convert the decimal fraction $3/5$ to a normalized FPN with $\beta = 2$.
- III. Let $x = \beta^e$, $e \in \mathbb{Z}$, $L < e < U$ be a normalized FPN in $\mathbb{F}$ and $x_L, x_R \in \mathbb{F}$ the two normalized FPNs adjacent to x such that $x_L < x < x_R$. Prove $x_R - x = \beta(x - x_L)$.

- | | |
|---|---|
| <p>IV. By reusing your result of II, find out the two normalized FPNs adjacent to $x = 3/5$ under the IEEE 754 single-precision protocol. What is $\text{fl}(x)$ and the relative roundoff error?</p> <p>V. If the IEEE 754 single-precision protocol did not round off numbers to the nearest, but simply dropped excess bits, what would the unit roundoff be?</p> <p>VI. Repeat Example 1.34 with $b = 8.769 \times 10^4$, $b = -5.678 \times 10^0$, and $b = -5.678 \times 10^3$.</p> <p>VII. Suppose the division of two FPNs is calculated in a register of precision $2p$. Give an example that contradicts the conclusion of the model of machine arithmetic.</p> <p>VIII. If we sort the positive numbers $a_i > 0$ according to their magnitudes and carry out the additions in this ascending order, we can minimize the rounding error term δ in Theorem 1.43. Can you give two examples?</p> <p>IX. Derive $\text{fl}(a_1b_1 + a_2b_2 + a_3b_3)$ for $a_i, b_i \in \mathcal{F}$ and comment on the corresponding derivation of $\text{fl}(\sum_i \prod_j a_{i,j})$.</p> | <p>X. How many bits of precision are lost in the subtraction $1 - \cos x$ when $x = \frac{1}{4}$?</p> <p>XI. Suggest at least two ways to compute $1 - \cos x$ to avoid catastrophic cancellation caused by subtraction.</p> |
|---|---|

1.7 Programming assignments

Consider a normalized FPN system $\mathbb{F}$ with the characterization $\beta = 2, p = 3, L = -1, U = +1$.

- compute $\text{UFL}(\mathbb{F})$ and $\text{OFL}(\mathbb{F})$ and output them as decimal numbers;
- enumerate all numbers in $\mathbb{F}$ and verify the corollary on the cardinality of $\mathbb{F}$ in the summary handout;
- plot $\mathbb{F}$ on the real axis;
- enumerate all the subnormal numbers of $\mathbb{F}$;
- plot the *extended* $\mathbb{F}$ on the real axis.

Chapter 2

Stability and Conditioning

Motivation 2.1. When navigating the ideal world of continuous mathematics, we often assume infinite precision, but as we have already seen in Chapter 1, digital computers must operate with discrete, finite-precision arithmetic. When we solve real-world problems on computers, a tiny measurement error or the rounding errors might explode into meaningless results. We do not want these to happen. Hence we must equip ourselves with the concepts of stability and conditioning, which allows you to predict whether a mathematical problem is inherently hopeless and, in the case of the problem being “well-conditioned,” whether an algorithm for solving the problem necessitates a higher precision or needs a complete reformulation. Knowledge in this short chapter is the backbone of robust technologies we rely on every day, from ensuring the structural safety of bridges and aircraft to maintaining the accuracy of medical imaging and weather forecasting. Mastering this chapter is the first step of moving beyond merely orchestrating an answer to gaining the expertise of guaranteeing the right answer in high-stakes computational environments.

Roadmap 2.2. This chapter concerns the accuracy and stability of an algorithm that solves a mathematical problem. After defining the accuracy of an algorithm in Section 2.1, we introduce the *qualitative* concepts of stability in Section 2.2. The most important concept of this chapter is the *quantitative* concept of stability, called the condition number, which measures how sensitive a problem or an algorithm is to small changes in input. In Sections 2.3, 2.4, and 2.6, we elaborate on condition numbers for a scalar function, a vector function, and an algorithm, respectively. Finally in Section 2.7, we combine main results to derive an error bound for computer solutions of mathematical problems.

2.1 Forward and backward errors

Definition 2.3. Let $\hat{x}$ be an approximation to $x \in \mathbb{R}$. The accuracy of $\hat{x}$ can be measured by its *absolute error*

$$E_{\text{abs}}(\hat{x}) = |\hat{x} - x| \quad (2.1)$$

and/or, in the case of $x \neq 0$, its *relative error*

$$E_{\text{rel}}(\hat{x}) = \frac{|\hat{x} - x|}{|x|}. \quad (2.2)$$

Definition 2.4. For an approximation $\hat{y}$ to $y = f(x)$ computed by $\hat{y} = \hat{f}(x)$, the *forward error* is the relative error of $\hat{y}$ in approximating y and the *backward error* is the smallest relative error in approximating x by an $\hat{x}$ that satisfies $f(\hat{x}) = \hat{y}$, assuming such an $\hat{x}$ exists.

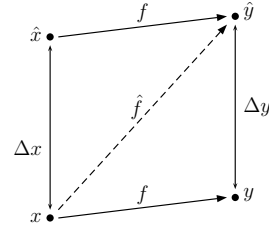


Figure 2.1: Forward and backward errors.

Definition 2.5 (Accuracy). An algorithm $\hat{y} = \hat{f}(x)$ for computing the function $y = f(x)$ is *accurate* if its forward error is small for all x , i.e. $\forall x \in \text{dom}(f)$, $E_{\text{rel}}(\hat{f}(x)) \leq c\epsilon_u$ where c is a small constant.

2.2 Backward stability and numerical stability

Definition 2.6 (Backward stability). An algorithm $\hat{f}(x)$ for computing $y = f(x)$ is *backward stable* if its backward error is small for all x , i.e.

$$\forall x \in \text{dom}(f), \exists \hat{x} \in \text{dom}(f), \text{ s.t.} \quad \hat{f}(x) = f(\hat{x}) \implies E_{\text{rel}}(\hat{x}) \leq c\epsilon_u, \quad (2.3)$$

where c is a small constant.

Definition 2.7. An algorithm $\hat{f}(x_1, x_2)$ for computing $y = f(x_1, x_2)$ is *backward stable* if

$$\forall (x_1, x_2) \in \text{dom}(f), \exists (\hat{x}_1, \hat{x}_2) \in \text{dom}(f) \text{ s.t.} \quad \hat{f}(x_1, x_2) = f(\hat{x}_1, \hat{x}_2) \implies \begin{cases} E_{\text{rel}}(\hat{x}_1) \leq c_1\epsilon_u, \\ E_{\text{rel}}(\hat{x}_2) \leq c_2\epsilon_u, \end{cases} \quad (2.4)$$

where c_1, c_2 are two small constants.

Lemma 2.8. For $f(x_1, x_2) = x_1 - x_2$, $x_1, x_2 \in \mathcal{R}(\mathcal{F})$, the algorithm $\hat{f}(x_1, x_2) = \text{fl}(x_1) - \text{fl}(x_2)$ is backward stable.

Proof. We have $\hat{f}(x_1, x_2) = (\text{fl}(x_1) - \text{fl}(x_2))(1 + \delta_3)$ from Theorem 1.42. Then Theorem 1.29 implies

$$\begin{aligned}\hat{f}(x_1, x_2) &= (x_1(1 + \delta_1) - x_2(1 + \delta_2))(1 + \delta_3) \\ &= x_1(1 + \delta_1 + \delta_3 + \delta_1\delta_3) - x_2(1 + \delta_2 + \delta_3 + \delta_2\delta_3).\end{aligned}$$

Take $\hat{x}_1$ and $\hat{x}_2$ to be the two terms in the above line and we have

$$\begin{aligned}E_{\text{rel}}(\hat{x}_1) &= |\delta_1 + \delta_3 + \delta_1\delta_3|, \\ E_{\text{rel}}(\hat{x}_2) &= |\delta_2 + \delta_3 + \delta_2\delta_3|.\end{aligned}$$

Then Definition 2.7 completes the proof. $\square$

Example 2.9. For $f(x) = 1 + x$, $x \in (0, \text{OFL})$, the algorithm $\hat{f}(x) = \text{fl}(1.0 + \text{fl}(x))$ is not backward stable.

We prove a stronger statement that implies the negation of (2.3). For each $x \in (0, \epsilon_u)$, Definition 1.26 yields $\hat{f}(x) = 1.0$. Then $\hat{f}(x) = f(\hat{x})$ implies $\hat{x} = 0$, which further implies $E_{\text{rel}}(\hat{x}) = 1$.

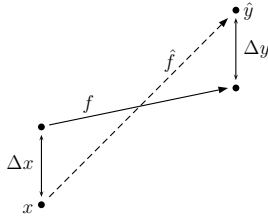


Figure 2.2: The concept of numerical stability. This plot is a shift of the dashed arrow of the upper triangle Figure 2.1. Note that we have $\hat{y} = \hat{f}(x)$ and $\hat{y} + \Delta y = f(x + \Delta x)$.

Definition 2.10. An algorithm $\hat{f}(x)$ for computing $y = f(x)$ is *stable* or *numerically stable* iff

$$\forall x \in \text{dom}(f), \exists \hat{x} \in \text{dom}(f) \text{ s.t. } \begin{cases} \left| \frac{\hat{f}(x) - f(\hat{x})}{f(\hat{x})} \right| \leq c_f \epsilon_u, \\ E_{\text{rel}}(\hat{x}) \leq c \epsilon_u, \end{cases} \quad (2.5)$$

where c_f, c are two small constants.

Lemma 2.11. If an algorithm is backward stable, then it is numerically stable.

Proof. By Definition 2.6, $f(\hat{x}) = \hat{f}(x)$, hence $c_f = 0$. The other condition also follows trivially. $\square$

Example 2.12. For $f(x) = 1 + x$, $x \in (0, \text{OFL})$, we show that the algorithm $\hat{f}(x) = \text{fl}(1.0 + \text{fl}(x))$ is stable.

If $|x| < \epsilon_u$, then $\hat{f}(x) = 1.0$. Choose $\hat{x} = x$, then $f(\hat{x}) - x = \hat{f}(x)$ and $\left| \frac{\hat{f}(x) - f(\hat{x})}{f(\hat{x})} \right| = \left| \frac{x}{1+x} \right| < 2\epsilon_u$.

Otherwise $|x| \geq \epsilon_u$. The definitions of the range and unit roundoff (Definitions 1.28 and 1.22) yield $x \in \mathcal{R}(\mathcal{F})$; here we have assumed $\epsilon_u > \text{UFL}$, which holds for all realworld FPN systems. By Theorem 1.29, $\hat{f}(x) = (1 + x(1 + \delta_1))(1 + \delta_2)$, i.e., $\hat{f}(x) = 1 + \delta_2 + x(1 + \delta_1 + \delta_2 + \delta_1\delta_2)$, where $|\delta_1|, |\delta_2| < \epsilon_u$.

Choose $\hat{x} = x(1 + \delta_1 + \delta_2 + \delta_1\delta_2)$ and we have

$$\begin{aligned}E_{\text{rel}}(\hat{x}) &= |\delta_1 + \delta_2 + \delta_1\delta_2| < 3\epsilon_u, \\ \Rightarrow \left| \frac{\hat{f}(x) - f(\hat{x})}{f(\hat{x})} \right| &= \left| \frac{\delta_2}{1 + x(1 + \delta_1 + \delta_2 + \delta_1\delta_2)} \right| \leq \epsilon_u,\end{aligned}$$

where the denominator is never close to zero since $x > 0$.

2.3 The condition number of a scalar function

Definition 2.13. The *absolute condition number* of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$\hat{C}_f(x) = \lim_{\delta \rightarrow 0} \sup_{|\Delta x| \leq \delta} \frac{|f(x + \Delta x) - f(x)|}{|\Delta x|}. \quad (2.6)$$

Definition 2.14. The *(relative) condition number* of a continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a measure of the relative change in the output for a small change in the input,

$$C_f(x) = \left| \frac{x f'(x)}{f(x)} \right|. \quad (2.7)$$

Definition 2.15. A problem with a low condition number is said to be *well-conditioned*. A problem with a high condition number is said to be *ill-conditioned*.

Example 2.16. Definition 2.14 yields

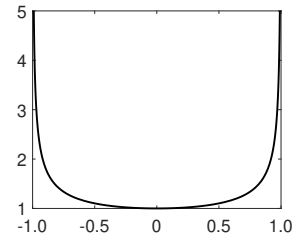
$$E_{\text{rel}}(\hat{y}) \lesssim C_f E_{\text{rel}}(\hat{x}). \quad (2.8)$$

The approximation mark “ $\approx$ ” refers to the fact that the quadratic term $(\Delta x)^2$ has been ignored. As one way to interpret (2.8) and to understand Definition 2.14, *the computed solution to an ill-conditioned problem may have a large forward error*.

Example 2.17. For the function $f(x) = \arcsin(x)$, its condition number, according to Definition 2.14, is

$$C_f(x) = \left| \frac{x f'(x)}{f(x)} \right| = \frac{x}{\sqrt{1-x^2} \arcsin x}.$$

Hence $C_f(x) \rightarrow +\infty$ as $x \rightarrow \pm 1$.



Lemma 2.18. Consider solving the equation $f(x) = 0$ near a simple root r , i.e. $f(r) = 0$ and $f'(r) \neq 0$. Suppose we perturb the function f to $F = f + \epsilon g$ where $f, g \in \mathcal{C}^2$, $g(r) \neq 0$, and $|\epsilon g'(r)| \ll |f'(r)|$. Then the root of F is $r + h$ where

$$h \approx -\epsilon \frac{g(r)}{f'(r)}. \quad (2.9)$$

Proof. Suppose $r + h$ is the new root, i.e. $F(r + h) = 0$, or,

$$f(r + h) + \epsilon g(r + h) = 0.$$

Taylor's expansion of $F(r + h)$ yields

$$f(r) + hf'(r) + \epsilon[g(r) + hg'(r)] = O(h^2)$$

and we have $h \approx -\epsilon \frac{g(r)}{f'(r) + \epsilon g'(r)} \approx -\epsilon \frac{g(r)}{f'(r)}$. $\square$

Example 2.19 (Wilkinson). Define

$$f(x) := \prod_{k=1}^p (x - k), \quad g(x) := x^p.$$

How is the root $x = p$ affected by perturbing f to $f + \epsilon g$?

By Lemma 2.18, the answer is

$$h \approx -\epsilon \frac{g(p)}{f'(p)} = -\epsilon \frac{p^p}{(p-1)!}.$$

For $p = 20, 30, 40$, the value of $\frac{p^p}{(p-1)!}$ is about 8.6×10^8 , 2.3×10^{13} , 5.9×10^{17} , respectively. Hence a small change of the coefficient in the monomial x^p would cause a large change of the root. Consequently, the problem of root finding for polynomials with very high degrees is hopeless.

2.4 The condition number of a vector function

Definition 2.20. The *condition number of a vector function* $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is

$$\text{cond}_{\mathbf{f}}(\mathbf{x}) = \frac{\|\mathbf{x}\| \|\nabla \mathbf{f}\|}{\|\mathbf{f}(\mathbf{x})\|}, \quad (2.10)$$

where $\|\cdot\|$ denotes a vector norm in Definition C.176 or a matrix norm in Definition C.281.

Example 2.21. In solving the linear system $A\mathbf{u} = \mathbf{b}$, the algorithm can be viewed as taking the input $\mathbf{b}$ and returning the output $A^{-1}\mathbf{b}$, i.e. $\mathbf{f}(\mathbf{b}) = A^{-1}\mathbf{b}$. Clearly $\nabla \mathbf{f} = A^{-1}$. Definition 2.20 yields

$$\text{cond}_{\mathbf{f}}(\mathbf{x}) = \frac{\|\mathbf{b}\| \|A^{-1}\|}{\|\mathbf{u}\|} = \frac{\|A\mathbf{u}\| \|A^{-1}\|}{\|\mathbf{u}\|}.$$

In practice the input $\mathbf{b}$ can take any value, hence we have

$$\max \text{cond}_{\mathbf{f}}(\mathbf{x}) = \max_{\mathbf{u} \neq 0} \frac{\|A\mathbf{u}\| \|A^{-1}\|}{\|\mathbf{u}\|} = \|A\| \|A^{-1}\|,$$

where we have used the common definition

$$\|A\| := \max_{\mathbf{u} \neq 0} \frac{\|A\mathbf{u}\|}{\|\mathbf{u}\|}. \quad (2.11)$$

This explains Definition 2.22.

Definition 2.22. The *condition number of a nonsingular square matrix* A is

$$\text{cond } A := \|A\| \|A^{-1}\|. \quad (2.12)$$

Lemma 2.23. Based on the 2-norm, the condition number of a nonsingular square matrix A is

$$\text{cond}_2 A := \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_{\max}}{\sigma_{\min}},$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are respectively the largest and the smallest singular values of A in Definition C.252. If A is also normal, we have

$$\text{cond}_2 A = \frac{|\lambda_{\max}|}{|\lambda_{\min}|},$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are eigenvalues of A with the largest and the smallest moduli, respectively. Furthermore, if A is unitary, we have $\text{cond}_2 A = 1$.

Example 2.24. Consider a linear system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{C}^{m \times n}$ that admits a solution. If we perturb $\mathbf{b}$ to $\hat{\mathbf{b}} = \mathbf{b} + \epsilon \mathbf{v}_j$ for some $\epsilon \in \mathbb{C}$, by Theorem C.255 the solution changes from $\mathbf{x}$ to $\hat{\mathbf{x}} = \mathbf{x} + \frac{\epsilon}{\sigma_j} \mathbf{u}_j$. Then the ratio of the change in the solution to that in $\mathbf{b}$ is

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|_2}{\|\hat{\mathbf{b}} - \mathbf{b}\|_2} = \frac{1}{\sigma_j}. \quad (2.13)$$

In the spirit of Definition 2.13, $\frac{1}{\sigma_{\min}}$ can be considered as the *absolute* condition number of $\hat{A}$.

Example 2.25. For the matrix

$$A = \begin{bmatrix} 1 & 1 - \delta \\ 1 & 1 + \delta \end{bmatrix} \quad (2.14)$$

and $\delta = 10^{-8}$, we have $\text{cond}_2 A = 199999999.137258$.

Theorem 2.26. Let A be an invertible matrix, $\hat{A}$ a perturbation of A satisfying

$$\|A^{-1}\| \|\hat{A} - A\| < 1, \quad (2.15)$$

and $\mathbf{x}$ and $\hat{\mathbf{x}}$ solutions to the linear systems

$$A\mathbf{x} = \mathbf{b}, \quad \hat{A}\hat{\mathbf{x}} = \hat{\mathbf{b}}. \quad (2.16)$$

Then the relative error of $\hat{\mathbf{x}}$ approximating $\mathbf{x}$ is bounded as

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond } A}{1 - (\text{cond } A) \frac{\|\hat{A} - A\|}{\|A\|}} \left(\frac{\|\hat{\mathbf{b}} - \mathbf{b}\|}{\|\mathbf{b}\|} + \frac{\|\hat{A} - A\|}{\|A\|} \right). \quad (2.17)$$

Proof. Write $\hat{A} = A[I + A^{-1}(\hat{A} - A)]$ and we have from Lemma C.304 and (2.15) that $\hat{A}$ is invertible and

$$\hat{A}^{-1} = [I + A^{-1}(\hat{A} - A)]^{-1} A^{-1}.$$

Then Theorem E.76 and the triangle inequality yield

$$\|\hat{A}^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\| \|\hat{A} - A\|}. \quad (2.18)$$

By (2.16) we can write $\hat{\mathbf{x}} - \mathbf{x} = \hat{A}^{-1}[\hat{\mathbf{b}} - \mathbf{b} - (\hat{A} - A)\mathbf{x}]$, which, together with (2.18) and the triangle inequality, yields

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond } A}{1 - \|\hat{A} - A\| \|A^{-1}\|} \left(\frac{\|\hat{\mathbf{b}} - \mathbf{b}\|}{\|\mathbf{b}\|} + \frac{\|\hat{A} - A\|}{\|A\|} \right).$$

Then (2.17) follows from $\|A\| \|\mathbf{x}\| \geq \|\mathbf{b}\|$. $\square$

Definition 2.27. The *componentwise condition number* of a vector function $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is

$$\text{cond}_{\mathbf{f}}(\mathbf{x}) = \|A(\mathbf{x})\|, \quad (2.19)$$

where the matrix $A(\mathbf{x}) = [a_{ij}(\mathbf{x})]$ and each component is

$$a_{ij}(\mathbf{x}) = \left| \frac{x_j \frac{\partial f_i}{\partial x_j}}{f_i(\mathbf{x})} \right|. \quad (2.20)$$

Example 2.28. For the vector function

$$\mathbf{f}(\mathbf{x}) := \begin{bmatrix} \frac{1}{x_1} + \frac{1}{x_2} \\ \frac{1}{x_1} - \frac{1}{x_2} \end{bmatrix},$$

its Jacobian matrix is

$$\nabla \mathbf{f} = -\frac{1}{x_1^2 x_2^2} \begin{bmatrix} x_2^2 & x_1^2 \\ x_2^2 & -x_1^2 \end{bmatrix}.$$

The condition number based on Definition 2.27 clearly captures the fact that $x_1 \pm x_2 \approx 0$ leads to ill-conditioning, $C_c = \|A_c\|$, where

$$A_c = \begin{bmatrix} \left| \frac{x_2}{x_1 + x_2} \right| & \left| \frac{x_1}{x_1 + x_2} \right| \\ \left| \frac{x_2}{x_1 - x_2} \right| & \left| \frac{x_1}{x_1 - x_2} \right| \end{bmatrix},$$

while that based on 1-norm of Definition 2.20 fails to capture the ill-conditioning,

$$C_1 = \frac{\|\mathbf{x}\|_1 \|\nabla \mathbf{f}\|_1}{\|\mathbf{f}\|_1} = \frac{|x_1| + |x_2|}{|x_1 x_2|} \frac{2 \max(x_1^2, x_2^2)}{|x_1 + x_2| + |x_1 - x_2|},$$

in that the condition $x_1 \pm x_2 \approx 0$ yields $C_1 \approx 2$. Note that we have used the well-known formula

$$\forall A \in \mathbb{R}^{n \times n}, \quad \|A\|_1 = \max_j \sum_i |a_{ij}|.$$

Example 2.29. If the parameters $t_1, t_2, \dots, t_n$ are equally spaced in $[-1, 1]$, the condition number of the Vandermonde matrices in Definition 6.3 based on the ∞ -norm is

$$\text{cond}_{\infty} V_n \sim \frac{1}{\pi} \exp \left[-\frac{\pi}{4} + \frac{n}{4} (\pi + 2 \ln 2) \right],$$

which is 9.86×10^8 for $n = 20$. See [Gautschi, 2012, p. 24] and Gautschi [1990] for more details.

Definition 2.30. The *Hilbert matrix* $H_n \in \mathbb{R}^{n \times n}$, named after Hilbert [1894], is

$$h_{i,j} = \frac{1}{i + j - 1}. \quad (2.21)$$

Example 2.31. The condition number of Hilbert matrices based on the 2-norm is

$$\text{cond}_2 H_n \sim \frac{(\sqrt{2} + 1)^{4n+4}}{2^{\frac{15}{4}} \sqrt{\pi n}},$$

which is 9.22×10^{14} for $n = 10$. See [Gautschi, 2012, p. 23] and Example 8.36 for more discussions.

2.5 The signature of an algorithm

Definition 2.32. An *algorithm* is a step-by-step procedure that takes some set of values as its *input* and produces some set of values as its *output*.

Definition 2.33. A *precondition* is a condition that holds for the input prior to the execution of an algorithm.

Definition 2.34. A *postcondition* is a condition that holds for the output after the execution of an algorithm.

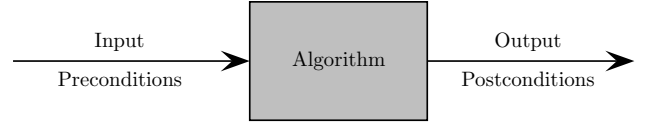


Figure 2.3: The signature of an algorithm.

Definition 2.35. The *signature of an algorithm* consists of its input, output, preconditions, postconditions, and how input parameters violating preconditions are handled.

2.6 The condition number of an algorithm

Definition 2.36. Consider approximating a function $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ with an algorithm $\mathbf{f}_A : \mathcal{F}^m \rightarrow \mathcal{F}^n$. Assume

$$\forall \mathbf{x} \in \mathcal{F}^m, \exists \mathbf{x}_A \in \mathbb{R}^m \text{ s.t. } \mathbf{f}_A(\mathbf{x}) = \mathbf{f}(\mathbf{x}_A), \quad (2.22)$$

the *condition number of the algorithm* $\mathbf{f}_A$ is given by

$$\text{cond}_A(\mathbf{x}) = \frac{1}{\epsilon_u} \inf_{\{\mathbf{x}_A\}} \frac{\|\mathbf{x}_A - \mathbf{x}\|}{\|\mathbf{x}\|}, \quad (2.23)$$

where ϵ_u is the unit roundoff of $\mathcal{F}$ in Definition 1.28. The algorithm $\mathbf{f}_A$ is *well conditioned* if its condition number is close to 1 and *ill conditioned* otherwise.

Example 2.37. Consider an algorithm A for calculating $y = \ln x$. Suppose that, for any positive number x , this program produces a y_A satisfying $y_A = (1 + \delta) \ln x$ where $|\delta| \leq 5\epsilon_u$. What is the condition number of the algorithm?

We clearly have

$$y_A = \ln x_A \text{ where } x_A = x^{1+\delta},$$

and consequently

$$\begin{aligned} E_{\text{rel}}(x_A) &= \left| \frac{x^{1+\delta} - x}{x} \right| = |x^\delta - 1| = |e^{\delta \ln x} - 1| \\ &\approx |\delta \ln x| \leq 5 |\ln x| \epsilon_u. \end{aligned}$$

Hence A is well conditioned except when $x \rightarrow 0^+$.

Theorem 2.38. Suppose a smooth function $f : \mathbb{R} \rightarrow \mathbb{R}$ is approximated by an algorithm $A : \mathcal{F} \rightarrow \mathcal{F}$, producing $f_A(x) = f(x)(1 + \delta(x))$ where $|\delta(x)| \leq \varphi(x)\epsilon_u$. If $\text{cond}_f(x)$ is bounded and nonzero, then we have

$$\forall x \in \mathcal{F}, \quad \text{cond}_A(x) \leq \frac{\varphi(x)}{\text{cond}_f(x)}. \quad (2.24)$$

Proof. Assume $\forall x, \exists x_A$ such that $f(x_A) = f_A(x)$. Write $x_A = x(1 + \epsilon_A)$ and we have

$$\begin{aligned} f(x)(1 + \delta) &= f(x_A) = f(x(1 + \epsilon_A)) = f(x + x\epsilon_A) \\ &= f(x) + x\epsilon_A f'(x) + O(\epsilon_A^2). \end{aligned}$$

Neglecting the quadratic term yields

$$x\epsilon_A f'(x) = f(x)\delta \implies \left| \frac{x_A - x}{x} \right| = |\epsilon_A| = \left| \frac{f(x)}{x f'(x)} \right| |\delta(x)|.$$

Dividing both sides by ϵ_u yields

$$\frac{1}{\epsilon_u} \left| \frac{x_A - x}{x} \right| = \frac{\delta(x)}{\epsilon_u \text{cond}_f(x)}.$$

Take inf with respect to all x_A 's, apply the condition $|\delta(x)| \leq \varphi(x)\epsilon_u$, and we have (2.24). $\square$

Example 2.39. Assume that $\sin x$ and $\cos x$ are computed with relative error within machine roundoff (this can be satisfied easily by truncating the Taylor series). Apply Theorem 2.38 to analyze the conditioning of the algorithm

$$f_A = \text{fl} \left[\frac{\text{fl}(1 - \text{fl}(\cos x))}{\text{fl}(\sin x)} \right] \quad (2.25)$$

that computes $f(x) = \frac{1 - \cos x}{\sin x}$ for $x \in (0, \pi/2)$.

By Definition 2.14, it is easy to compute that

$$\text{cond}_f(x) = \frac{x}{\sin x}.$$

Furthermore, by Theorem 1.42 and the assumptions on $\sin x$ and $\cos x$, we have

$$f_A(x) = \frac{(1 - (\cos x)(1 + \delta_1))(1 + \delta_2)}{(\sin x)(1 + \delta_3)}(1 + \delta_4),$$

where $|\delta_i| \leq \epsilon_u$ for $i = 1, 2, 3, 4$. Neglecting the quadratic terms of $O(\delta_i^2)$, the above equation is equivalent to

$$f_A(x) = \frac{1 - \cos x}{\sin x} \left\{ 1 + \delta_2 + \delta_4 - \delta_3 - \delta_1 \frac{\cos x}{1 - \cos x} \right\},$$

hence we have $\varphi(x) = 3 + \frac{\cos x}{1 - \cos x}$ and

$$\text{cond}_A(x) \leq \frac{\sin x}{x} \left(3 + \frac{\cos x}{1 - \cos x} \right).$$

Hence, $\text{cond}_A(x)$ may be unbounded as $x \rightarrow 0$. On the other hand, $\text{cond}_A(x)$ is controlled by $\frac{6}{\pi}$ as $x \rightarrow \frac{\pi}{2}$.

2.7 The overall error of a computer solution

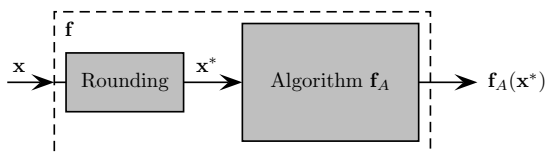


Figure 2.4: A scheme for considering the overall error of a computer solution.

Theorem 2.40. As shown in Figure 2.4, consider using normalized FPN arithmetic to solve a math problem

$$\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^n, \quad \mathbf{y} = \mathbf{f}(\mathbf{x}). \quad (2.26)$$

Denote the computer input and output as

$$\mathbf{x}^* \approx \mathbf{x}, \quad \mathbf{y}_A^* = \mathbf{f}_A(\mathbf{x}^*), \quad (2.27)$$

where $\mathbf{f}_A$ is the algorithm that approximates $\mathbf{f}$. The relative error of approximating $\mathbf{y}$ with $\mathbf{y}_A^*$ can be bounded as

$$E_{\text{rel}}(\mathbf{y}_A^*) \lesssim E_{\text{rel}}(\mathbf{x}^*) \text{cond}_{\mathbf{f}}(\mathbf{x}) + \epsilon_u \text{cond}_{\mathbf{f}}(\mathbf{x}^*) \text{cond}_A(\mathbf{x}^*), \quad (2.28)$$

where the relative error is defined in (2.2).

Proof. By the triangle inequality, we have

$$\frac{\|\mathbf{y}_A^* - \mathbf{y}\|}{\|\mathbf{y}\|} = \frac{\|\mathbf{f}_A(\mathbf{x}^*) - \mathbf{f}(\mathbf{x})\|}{\|\mathbf{f}(\mathbf{x})\|} \leq \frac{\|\mathbf{f}(\mathbf{x}^*) - \mathbf{f}(\mathbf{x})\|}{\|\mathbf{f}(\mathbf{x})\|} + \frac{\|\mathbf{f}_A(\mathbf{x}^*) - \mathbf{f}(\mathbf{x}^*)\|}{\|\mathbf{f}(\mathbf{x})\|}.$$

By (2.8), the first term satisfies

$$\frac{\|\mathbf{f}(\mathbf{x}^*) - \mathbf{f}(\mathbf{x})\|}{\|\mathbf{f}(\mathbf{x})\|} \lesssim \text{cond}_{\mathbf{f}}(\mathbf{x}) \frac{\|\mathbf{x}^* - \mathbf{x}\|}{\|\mathbf{x}\|} = E_{\text{rel}}(\mathbf{x}^*) \text{cond}_{\mathbf{f}}(\mathbf{x}).$$

By (2.8) and Definition 2.36, the second term is

$$\begin{aligned} \frac{\|\mathbf{f}_A(\mathbf{x}^*) - \mathbf{f}(\mathbf{x}^*)\|}{\|\mathbf{f}(\mathbf{x})\|} &= \frac{\|\mathbf{f}(\mathbf{x}_A^*) - \mathbf{f}(\mathbf{x}^*)\|}{\|\mathbf{f}(\mathbf{x})\|} \approx \frac{\|\mathbf{f}(\mathbf{x}_A^*) - \mathbf{f}(\mathbf{x}^*)\|}{\|\mathbf{f}(\mathbf{x}^*)\|} \\ &\leq \text{cond}_{\mathbf{f}}(\mathbf{x}^*) \frac{\|\mathbf{x}_A^* - \mathbf{x}^*\|}{\|\mathbf{x}^*\|} = \epsilon_u \text{cond}_A(\mathbf{x}^*) \text{cond}_{\mathbf{f}}(\mathbf{x}^*), \end{aligned}$$

where the last step follows from the fact that we only consider the $\mathbf{x}_A^*$ that is the least dangerous. $\square$

Summary 2.41. By Theorem 2.40, it is hopeless to approximate a function $\mathbf{f}$ whose condition number is too large: each summand on the RHS of (2.28) contains a factor $\text{cond}_{\mathbf{f}}(\mathbf{x}^*)$ or $\text{cond}_{\mathbf{f}}(\mathbf{x})$. On the other hand, for a function $\mathbf{f}$ with good conditioning, one still has to be careful in designing an approximating algorithm because otherwise the term $\text{cond}_A(\mathbf{x}^*)$ might make the RHS bound of (2.28) too large.

For more examples on the interaction between computer arithmetic and numerical stability, the reader is referred to the monograph by Higham [2002].

2.8 Theoretical problems

I. What are the condition numbers of the following functions? Where are they large?

- $(x - 1)^\alpha$,
- $\ln x$,
- e^x ,
- $\arccos x$.

II. Consider the function $f(x) = 1 - e^{-x}$ for $x \in [0, 1]$.

- Show that $\text{cond}_f(x) \leq 1$ for $x \in [0, 1]$.
- Let A be the algorithm that evaluates $f(x)$ for the machine number $x \in \mathbb{F}$. Assume that the exponential function is computed with relative error within machine roundoff. Estimate $\text{cond}_A(x)$ for $x \in [0, 1]$.

- Plot $\text{cond}_f(x)$ and the estimated upper bound of $\text{cond}_A(x)$ as a function of x on $[0, 1]$. Discuss your results.

III. Prove Lemma 2.23.

IV. Repeat Example 2.39 for $f(x) = \frac{\sin x}{1+\cos x}$ on the same interval.

V. The math problem of root finding for a polynomial

$$q(x) = \sum_{i=0}^n a_i x^i, \quad a_n = 1, a_0 \neq 0, a_i \in \mathbb{R} \quad (2.29)$$

can be considered as a vector function $f: \mathbb{R}^n \rightarrow \mathbb{C}$:

$$r = f(a_0, a_1, \dots, a_{n-1}).$$

Derive the componentwise condition number of f based on the 1-norm. For the Wilkinson example, compute your condition number, and compare your result with that in the Wilkinson Example. What does the comparison tell you?

2.9 Programming assignments

Print values of the functions in (2.30) at 101 equally spaced points covering the interval $[0.99, 1.01]$. Calculate each function in a straightforward way without rearranging or factoring. Note that the three functions are theoretically the same, but the computed values might be very different. Plot these functions near 1.0 using a magnified scale for the function values to see the variations involved. Discuss what you see. Which one is the most accurate? Why?

$$f(x) = x^8 - 8x^7 + 28x^6 - 56x^5 + 70x^4 - 56x^3 + 28x^2 - 8x + 1 \quad (2.30a)$$

$$g(x) = ((((((x - 8)x + 28)x - 56)x + 70)x - 56)x + 28)x - 8)x + 1 \quad (2.30b)$$

$$h(x) = (x - 1)^8 \quad (2.30c)$$

Chapter 3

Basic Methods for Solving Linear Systems

Motivation 3.1. Behind almost every scientific breakthrough and engineering marvel are numerical methods for solving linear systems. While basic algebra teaches us to solve two or three equations by hand, a linear system resulting from a real-world problem often contains millions or even billions of variables. From simulating the airflow over a Boeing aircraft wing to rendering the complex lighting in high-definition video games, the underlying problem is almost always the computer solution of a massive system of linear equations.

Roadmap 3.2. In the most general case of $A \in \mathbb{C}^{m \times n}$, the system $A\mathbf{x} = \mathbf{b}$ is always characterized by one of three possibilities: it has no solution, infinitely many solutions, or a unique solution. We will address the first two possibilities in Section 8.4.1, after learning the best approximation in the sense of the 2-norm. As for $A\mathbf{x} = \mathbf{b}$ having a unique solution, there is no loss of generality to assume that A is a nonsingular square real matrix. This chapter mainly concerns basic direct and iterative methods for solving systems of linear equations with this assumption.

At the foundation of direct methods is the solution of triangular systems discussed in Section 3.1. In Section 3.2, we study Gauss elimination as the most common direct method. In Section 3.2.4, we utilize the LU factorization to numerically estimate the condition number of a matrix. After introducing some special types of matrices in Section 3.3, we confine our exposition of basic iterative methods in Section 3.4 to linear fixed-point iterations, which will form an essential component of multigrid methods in Chapter 11.

Notation 1 (Colon). Let $A \in \mathbb{R}^{m \times n}$. We denote by $A(k, :)$ the k th row, i.e.,

$$A(k, :) := [a_{k1}, \dots, a_{kn}],$$

$A(:, k)$ the k th column, i.e.,

$$A(:, k) := [a_{1k}, \dots, a_{mk}]^\top,$$

and $A(i : j, k : l)$ a submatrix of A given by

$$A(i : j, k : l) := \begin{bmatrix} a_{ik} & \cdots & a_{il} \\ \vdots & \ddots & \vdots \\ a_{jk} & \cdots & a_{jl} \end{bmatrix}.$$

Furthermore, for a row vector $\mathbf{x} \in \mathbb{R}^n$, we denote by $\mathbf{x}(i : j)$ a vector in $\mathbb{R}^{j-i+1}$ given by

$$\mathbf{x}(i : j) := [x_i, \dots, x_j].$$

The column vector has similar notation.

Notation 2. Let $A \in \mathbb{R}^{m \times n}$. We denote by $|A|$ the matrix whose entries are absolute value of that of A , i.e.,

$$(|A|)_{ij} := |a_{ij}|.$$

For a row vector $\mathbf{x} \in \mathbb{R}^n$, we denote by $|\mathbf{x}|$ the vector whose components are absolute value of that of $\mathbf{x}$, i.e.,

$$|\mathbf{x}| := [|x_1|, \dots, |x_n|].$$

The column vector has similar notation.

3.1 Forward and back substitutions

Definition 3.3. Let $L \in \mathbb{R}^{n \times n}$ be a nonsingular lower triangular matrix. *Forward substitution* is the algorithm for solving a linear system $L\mathbf{x} = \mathbf{b}$, in which the solution $\mathbf{x}$ is obtained by solving for x_1 , then x_2 , up to x_n using the following formula:

$$\forall i = 1, 2, \dots, n, \quad x_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j=1}^{i-1} l_{ij}x_j \right). \quad (3.1)$$

Definition 3.4. Let $U \in \mathbb{R}^{n \times n}$ be a nonsingular upper triangular matrix. *Back substitution* is the algorithm for solving a linear system $U\mathbf{x} = \mathbf{b}$, in which the solution $\mathbf{x}$ is obtained by solving for x_n , then x_{n-1} , down to x_1 using the following formula:

$$\forall i = n, n-1, \dots, 1, \quad x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^n u_{ij}x_j \right). \quad (3.2)$$

Example 3.5. Consider the following lower triangular system:

$$\begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

If $l_{11}l_{22} \neq 0$, then the unknowns can be determined sequentially:

$$\begin{aligned} x_1 &= \frac{b_1}{l_{11}}, \\ x_2 &= \frac{b_2 - l_{21}x_1}{l_{22}}. \end{aligned}$$

Theorem 3.6 (backward error of triangular systems). The computed solutions $\hat{\mathbf{x}}$ obtained by forward substitution and back substitution satisfy,

$$(L + F)\hat{\mathbf{x}} = \mathbf{b}, \quad |F| \leq n\epsilon_u |L| + O(\epsilon_u^2), \quad (3.3)$$

and

$$(U + F)\hat{\mathbf{x}} = \mathbf{b}, \quad |F| \leq n\epsilon_u |U| + O(\epsilon_u^2), \quad (3.4)$$

respectively, where ϵ_u is the unit roundoff in Definition 1.28.

Proof. We only prove (3.3), while (3.4) is similar.

Let $\hat{s} = \text{fl}\left(b_i - \sum_{j=1}^{i-1} l_{ij}\hat{x}_j\right)$, Definition 3.3 and Theorem 1.42 yield

$$\exists \epsilon_j, \delta_k, \quad j, k = 1, 2, \dots, i-1, \quad |\epsilon_j|, |\delta_k| < \epsilon_u, \quad \text{s.t.}$$

$$\begin{aligned} \hat{s} &= \text{fl}\left(b_i - \sum_{j=1}^{i-1} \text{fl}(l_{ij}\hat{x}_j)\right) \\ &= \text{fl}(\dots \text{fl}(b_i - l_{i1}\hat{x}_1(1 + \epsilon_1)) - \dots - l_{i,i-1}\hat{x}_{i-1}(1 + \epsilon_{i-1})) \\ &= b_i \prod_{k=1}^{i-1} (1 + \delta_k) - \sum_{j=1}^{i-1} l_{ij}\hat{x}_j (1 + \epsilon_j) \prod_{k=j}^{i-1} (1 + \delta_k). \end{aligned}$$

Then,

$$\begin{aligned} \hat{x}_i &= \text{fl}\left(\frac{\hat{s}}{l_{ii}}\right) \\ &= \frac{\hat{s}}{l_{ii}}(1 + \delta_i) \\ &= \frac{b_i}{l_{ii}} \prod_{k=1}^i (1 + \delta_k) - \sum_{j=1}^{i-1} \frac{l_{ij}}{l_{ii}} \hat{x}_j (1 + \epsilon_j) \prod_{k=j}^i (1 + \delta_k) \\ &\Leftrightarrow b_i = l_{ii}\hat{x}_i \frac{1}{\prod_{k=1}^i (1 + \delta_k)} + \sum_{j=1}^{i-1} l_{ij}\hat{x}_j \frac{(1 + \epsilon_j)}{\prod_{k=1}^{j-1} (1 + \delta_k)} \quad (3.5) \end{aligned}$$

where $|\delta_i| < \epsilon_u$. And from proof of Theorem 1.30 and Theorem 1.43, we obtained that

$$\exists \mu_{ij}, \quad j = 1, 2, \dots, i, \quad |\mu_{ij}| < j\epsilon_u + O(\epsilon_u^2), \quad \text{s.t.}$$

$$(1 + \mu_{ij}) = \begin{cases} \frac{1}{\prod_{k=1}^i (1 + \delta_k)} & \text{if } j = i; \\ \frac{(1 + \epsilon_j)}{\prod_{k=1}^{j-1} (1 + \delta_k)} & \text{if } j \neq i. \end{cases}$$

Let $f_{ij} = l_{ij}\mu_{ij}$, then

$$\begin{aligned} \sum_{j=1}^i l_{ij} (1 + \mu_{ij}) \hat{x}_j &= \sum_{j=1}^i (l_{ij} + f_{ij}) \hat{x}_j = b_i \\ |f_{ij}| &= |\mu_{ij}| |l_{ij}| \leq j\epsilon_u |l_{ij}| + O(\epsilon_u^2) \end{aligned} \quad (3.6)$$

Then the proof is completed by rewriting (3.6) into matrix format and $j \leq n$. $\square$

Theorem 3.7 (Forward error of triangular systems). Let the triangular system $T\mathbf{x} = \mathbf{b}$, where $T \in \mathbb{R}^{n \times n}$ is nonsingular. The computed solutions $\hat{\mathbf{x}}$ obtained by substitution satisfy

$$\frac{\|\mathbf{x} - \hat{\mathbf{x}}\|_\infty}{\|\mathbf{x}\|_\infty} \leq \frac{\kappa(T, \mathbf{x})\gamma_n}{1 - \kappa(T)\gamma_n}, \quad (3.7)$$

where

$$\begin{aligned} \kappa(T, \mathbf{x}) &= \frac{\| |T^{-1}| |T| \|\mathbf{x}\|_\infty }{\|\mathbf{x}\|_\infty}, \\ \kappa(T) &= \| |T^{-1}| |T| \|_\infty, \\ \gamma_n &= \frac{n\epsilon_u}{1 - n\epsilon_u}. \end{aligned}$$

Example 3.8. $\kappa(T, x)$ can be arbitrarily large. This is illustrated by upper triangular matrix

$$U(\alpha) = (u_{ij}), \quad u_{ij} = \begin{cases} 1, & \text{if } i = j; \\ -\alpha, & \text{if } i < j, \end{cases}$$

for which

$$(U(\alpha)^{-1})_{ij} = \begin{cases} 1, & \text{if } i = j; \\ \alpha(1 + \alpha)^{j-i-1}, & \text{if } j > i. \end{cases}$$

Then we obtain that $\kappa(U(\alpha), \mathbf{e}) = \kappa(U(\alpha)) \sim 2\alpha^{n-1}$ as $\alpha \rightarrow \infty$. Therefore we cannot assert that all triangular systems are solved to high accuracy.

Lemma 3.9. Suppose the upper triangular matrix $U \in \mathbb{R}^{n \times n}$ satisfies

$$\forall j > i, \quad |u_{ii}| > |u_{ij}|. \quad (3.8)$$

Then the unit upper triangular matrix $W = |U^{-1}| |U|$ satisfies $w_{ij} \leq 2^{j-i}$ for all $j > i$, and hence $\kappa(U) \leq 2^n - 1$.

Proof. We can write $W = |V^{-1}| |V|$ where $V = D^{-1}U$ and $D = \text{diag}(u_{ii})$. The matrix V is the unit upper triangular with $v_{ij} \leq 1$, and it is easy to show that $|(V^{-1})_{ij}| \leq 2^{j-i-1}$ for $j > i$. Thus, for $j > i$,

$$w_{ij} = \sum_{k=i}^j |(V^{-1})_{ik}| v_{kj} \leq 1 + \sum_{k=i+1}^j 2^{k-i-1} \cdot 1 = 2^{j-i}.$$

$\square$

Theorem 3.10. Under the conditions of Lemma 3.9, the computed solution $\hat{\mathbf{x}}$ to $U\mathbf{x} = \mathbf{b}$ obtained by substitution satisfies

$$\forall i = 1, \dots, n, \quad |x_i - \hat{x}_i| \leq 2^{n-i+1} n\epsilon_u \max_{j \geq i} |\hat{x}_j| + O(\epsilon_u^2). \quad (3.9)$$

Proof. From Theorem 3.6 we have

$$|\mathbf{x} - \hat{\mathbf{x}}| = |U^{-1}F\hat{\mathbf{x}}| \leq n\epsilon_u |U^{-1}| |U| |\hat{\mathbf{x}}| + O(\epsilon_u^2).$$

Apply Lemma 3.9 we obtain that

$$\begin{aligned} |x_i - \hat{x}_i| &\leq n\epsilon_u \sum_{j=i}^n w_{ij} |\hat{x}_j| + O(\epsilon_u^2) \\ &\leq n\epsilon_u \max_{j \geq i} |\hat{x}_j| \sum_{j=i}^n 2^{j-i} + O(\epsilon_u^2) \\ &\leq 2^{n-i+1} n\epsilon_u \max_{j \geq i} |\hat{x}_j| + O(\epsilon_u^2). \end{aligned}$$

$\square$

Theorem 3.11. Analogues of Lemma 3.9 and Theorem 3.10 hold for lower triangular matrix L satisfies

$$\forall j < i, \quad |l_{ii}| \geq |l_{ij}|. \quad (3.10)$$

Exercise 3.12. Show that, if upper triangular matrix $U \in \mathbb{R}^{n \times n}$ is diagonally dominant (c.f. Definition 3.47) then $(|U^{-1}| |U|)_{ij} \leq i + j - 1$ and $\kappa(U) \leq 2n - 1$.

Example 3.13. The fact that the upper triangular matrix T satisfies (3.8) does not imply that $T^\top$ satisfies (3.10). Indeed, $\kappa(T^\top)$ can be arbitrarily large, as illustrated by the following example:

$$T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & \epsilon & \epsilon \\ 0 & 0 & 1 \end{bmatrix},$$

$$\kappa(T) = 5, \quad \kappa(T^\top) = 1 + \frac{2}{\epsilon}.$$

3.2 Gauss elimination

Notation 3. To write a (sparse) matrix, we almost always left off-diagonal zeros as blanks, except when the display of some zeros clearly indicate the structure of the matrix. An ellipsis, i.e., $\ddots$, or $\dots$, indicates a pattern that is evident from the two entries connected by the ellipsis.

3.2.1 Gauss transformations

Definition 3.14 (Gauss transformations). A *Gauss transformation* is a square matrix of the form

$$L_k := I_n - \ell_k \mathbf{e}_k^\top \quad (3.11)$$

$$= \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & -\ell_{k+1,k} & 1 & \\ & & \vdots & & \ddots \\ & & -\ell_{n,k} & & 1 \end{bmatrix},$$

where $I_n \in \mathbb{R}^{n \times n}$ is the identity matrix, the first k components of the *Gauss vector* $\ell_k \in \mathbb{R}^n$ are all zero, and $\mathbf{e}_k$ is the vector with 1 in position k and 0 elsewhere. The nonzero components of the Gauss vector are called the *multipliers*.

Theorem 3.15. Let $C \in \mathbb{R}^{n \times r}$ and $L_k = I_n - \ell_k \mathbf{e}_k^\top$ be a Gauss transformation. Then

$$L_k C = (I_n - \ell_k \mathbf{e}_k^\top) C = C - \ell_k C(k, :) \quad (3.12)$$

is an outer product update.

If $\hat{\ell}_k$ is the computed version of an exact Gauss vector ℓ_k , then

$$\hat{\ell}_k = \ell_k + \mathbf{e}, \quad |\mathbf{e}| \leq \epsilon_u |\ell_k|. \quad (3.13)$$

If $\hat{\ell}_k$ is used in a Gauss transformation update and $\text{fl}((I_n - \ell_k \mathbf{e}_k^\top) C)$ denotes the computed result, then

$$\text{fl}((I_n - \ell_k \mathbf{e}_k^\top) C) = (I_n - \ell_k \mathbf{e}_k^\top) C + E, \quad (3.14)$$

$$|E| \leq 3\epsilon_u (|C| + |\ell_k| C(k, :)) + O(\epsilon_u^2).$$

Definition 3.16. The *Gaussian elimination*, consisting of a sequence of row-wise operations, transforms matrix $A \in \mathbb{R}^{n \times m}$ into an upper triangular matrix U . At step k (for $k = 1, \dots, n - 1$), we eliminate the entries $a_{ik}^{(k)}$ for $i > k$ using the *pivot element* $a_{kk}^{(k)}$.

The transformation at step k is

$$A^{(k+1)} = L_k A^{(k)} = A^{(k)} - \ell_k A^{(k)}(k, :), \quad (3.15)$$

where L_k is Gauss transformation and the *multipliers* are given by

$$\forall i = k + 1, \dots, n, \quad l_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}. \quad (3.16)$$

$A^{(k)}$ is the matrix after the $(k - 1)$ th transformation and $A^{(1)} = A$. Additionally, we assume $a_{kk}^{(k)} \neq 0$.

Example 3.17. For a column vector $\alpha = [a_{1k}^{(k)}, \dots, a_{nk}^{(k)}]^\top$, Let $\ell_k = [0, \dots, 0, l_{k+1,k}, \dots, l_{nk}]^\top$, where l_{ik} are the same as (3.16) for each $i = k + 1, \dots, n$, then

$$L_k \alpha = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & -l_{k+1,k} & 1 & \\ & & \vdots & & \ddots \\ & & -l_{n,k} & & 1 \end{bmatrix} \begin{bmatrix} a_{1k}^{(k)} \\ \vdots \\ a_{kk}^{(k)} \\ a_{k+1,k}^{(k)} \\ \vdots \\ a_{nk}^{(k)} \end{bmatrix} = \begin{bmatrix} a_{1k}^{(k)} \\ \vdots \\ a_{kk}^{(k)} \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Example 3.18. Gaussian elimination applied to a matrix $A \in \mathbb{R}^{4 \times 4}$.

$$\begin{array}{ccc} \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{bmatrix} & \xrightarrow{L_1} & \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{bmatrix} \\ A^{(1)} & & A^{(2)} \\ \xrightarrow{L_2} \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{bmatrix} & \xrightarrow{L_3} & \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ 0 & \times & \times & \times \end{bmatrix} \\ A^{(3)} & & A^{(4)} \end{array}$$

3.2.2 The LU factorization

Lemma 3.19. Let $A \in \mathbb{R}^{n \times n}$ be nonsingular. The pivot elements $a_{kk}^{(k)}$ are all non-zero if and only if the i -th leading principal minor A_i of A is nonsingular for each $i = 1, 2, \dots, n$.

Theorem 3.20 (LU factorization). Let $A \in \mathbb{R}^{n \times n}$. If $\forall i = 1, 2, \dots, n$, the i th leading principal minor A_i of A is nonsingular, then there exists a unit lower triangular matrix L and an upper triangular matrix U such that

$$A = LU. \quad (3.17)$$

Furthermore, the entries of L below the diagonal are the multipliers, i.e., $(L)_{ik} = l_{ik}$ for all $i > k$, and $U = A^{(n)}$.

Proof. Directly from Theorem 3.19, we can apply Gaussian elimination on A . Then following (3.15), we obtain

$$U = A^{(n)} = L_{n-1} \dots L_2 L_1 A \\ \implies A = L_1^{-1} L_2^{-1} \dots L_{n-1}^{-1} U.$$

The sparsity pattern of ℓ_k implies $\mathbf{e}_i^\top \ell_k = 0$ for all $i \leq k$. Therefore $(I - \ell_k \mathbf{e}_k^\top)(I + \ell_k \mathbf{e}_k^\top) = I - \ell_k \mathbf{e}_k^\top \ell_k \mathbf{e}_k^\top = I$, i.e., $L_k^{-1} = I + \ell_k \mathbf{e}_k^\top$. Furthermore,

$$L_k^{-1} L_{k+1}^{-1} = (I + \ell_k \mathbf{e}_k^\top)(I + \ell_{k+1} \mathbf{e}_{k+1}^\top) = I + \ell_k \mathbf{e}_k^\top + \ell_{k+1} \mathbf{e}_{k+1}^\top.$$

Then we obtain that

$$L = L_1^{-1} L_2^{-1} \dots L_{n-1}^{-1} = I + \sum_{i=1}^{n-1} \ell_i \mathbf{e}_i^\top$$

is a unit lower triangular matrix satisfying $(L)_{ik} = \ell_{ik}$ for all $i > k$. $\square$

Algorithm 3.21. An outer product LU factorization algorithm.

Input: $A \in \mathbb{R}^{n \times n}$

Preconditions: A satisfies Theorem 3.20

Side effects: $A(i, i : n)$ is overwritten by $U(i, i : n)$ and $A(i + 1 : n, i)$ is overwritten by $L(i + 1 : n, i)$

```

1 for  $k = 1 : n - 1$  do
2    $\rho = k + 1 : n$ 
3    $A(\rho, k) = A(\rho, k) / A(k, k)$ 
4    $A(\rho, \rho) = A(\rho, \rho) - A(\rho, k) \cdot A(k, \rho)$ 
5 end
```

Example 3.22 (Viewpoint of outer product). Since the application of a Gauss transformation to a matrix involves an outer product, we can regard LU factorization as a sequence of outer product updates. Indeed, if

$$A = \begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{v} & B \end{bmatrix}$$

then the first step in Gaussian elimination results in the decomposition

$$\begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{v} & B \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \frac{\mathbf{v}}{\alpha} & I_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & B - \frac{\mathbf{v}\mathbf{w}^\top}{\alpha} \end{bmatrix} \begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{0} & I_{n-1} \end{bmatrix}.$$

Steps 2 through $n - 1$ compute the LU factorization

$$B - \frac{\mathbf{v}\mathbf{w}^\top}{\alpha} = L_1 U_1.$$

Then,

$$A = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \frac{\mathbf{v}}{\alpha} & I_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & L_1 U_1 \end{bmatrix} \begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{0} & I_{n-1} \end{bmatrix} \\ = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \frac{\mathbf{v}}{\alpha} & L_1 \end{bmatrix} \begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{0} & U_1 \end{bmatrix} \equiv LU.$$

Theorem 3.23. If no zero pivots are encountered during the execution of Algorithm 3.21, then the computed LU factors $\hat{L}$ and $\hat{U}$ of A satisfy

$$\hat{L}\hat{U} = A + H, \quad (3.18)$$

$$|H| \leq 2(n-1)\epsilon_u \left(|A| + |\hat{L}||\hat{U}| \right) + O(\epsilon_u^2). \quad (3.19)$$

Proof. The proof is by induction on n . The theorem obviously holds for $n = 1$. Suppose that $n \geq 2$ and that the theorem holds for case $n - 1$. If A is partitioned as follows

$$A = \begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{v} & B \end{bmatrix},$$

then the first step in Algorithm 3.21 is to compute

$$\hat{\mathbf{z}} = \text{fl}\left(\frac{\mathbf{v}}{\alpha}\right), \quad \hat{C} = \text{fl}(\hat{\mathbf{z}}\mathbf{w}^\top), \quad \hat{A}_1 = \text{fl}(B - \hat{C}),$$

from which we conclude that

$$\hat{\mathbf{z}} = \frac{\mathbf{v}}{\alpha} + \mathbf{f}, \quad (3.20)$$

$$|\mathbf{f}| \leq \epsilon_u \left| \frac{\mathbf{v}}{\alpha} \right|, \quad (3.21)$$

$$\hat{C} = \hat{\mathbf{z}}\mathbf{w}^\top + F_1, \quad (3.22)$$

$$|F_1| \leq \epsilon_u |\hat{\mathbf{z}}| |\mathbf{w}^\top|, \quad (3.23)$$

$$\hat{A}_1 = B - (\hat{\mathbf{z}}\mathbf{w}^\top + F_1) + F_2, \quad (3.24)$$

$$|F_2| \leq \epsilon_u (|B| + |\hat{\mathbf{z}}| |\mathbf{w}^\top|) + O(\epsilon_u^2), \quad (3.25)$$

$$|\hat{A}_1| \leq |B| + |\hat{\mathbf{z}}| |\mathbf{w}^\top| + O(\epsilon_u). \quad (3.26)$$

The algorithm proceeds to compute the LU factorization of $\hat{A}_1$. By induction, the computed factors $\hat{L}_1$ and $\hat{U}_1$ satisfy

$$\hat{L}_1 \hat{U}_1 = \hat{A}_1 + H_1 \quad (3.27)$$

where

$$|H_1| \leq 2(n-2)\epsilon_u \left(|\hat{A}_1| + |\hat{L}_1||\hat{U}_1| \right) + O(\epsilon_u^2). \quad (3.28)$$

Let

$$\hat{L} = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \hat{\mathbf{z}} & \hat{L}_1 \end{bmatrix}, \quad \hat{U} = \begin{bmatrix} \alpha & \mathbf{w}^\top \\ \mathbf{0} & \hat{U}_1 \end{bmatrix},$$

then

$$\hat{L}\hat{U} = A + H$$

where

$$H = \begin{bmatrix} 0 & \mathbf{0}^\top \\ \alpha \mathbf{f} & H_1 - F_1 + F_2 \end{bmatrix}. \quad (3.29)$$

To prove the theorem we must verify (3.19), i.e.,

$$|H| \leq 2(n-1)\epsilon_u \left[\begin{array}{cc} 2|\alpha| & 2|\mathbf{w}^\top| \\ |\mathbf{v}| + |\alpha||\mathbf{f}| & |B| + |\hat{L}_1||\hat{U}_1| + |\hat{\mathbf{z}}| |\mathbf{w}^\top| \end{array} \right] \\ + O(\epsilon_u^2).$$

Considering (3.29), this is obviously the case if

$$|H_1| + |F_1| + |F_2| \\ \leq 2(n-1)\epsilon_u \left(|B| + |\hat{\mathbf{z}}| |\mathbf{w}^\top| + |\hat{L}_1||\hat{U}_1| \right) + O(\epsilon_u^2). \quad (3.30)$$

Using (3.26) and (3.28) we have

$$|H_1| \leq 2(n-2)\epsilon_u \left(|B| + |\hat{\mathbf{z}}| |\mathbf{w}^\top| + |\hat{L}_1||\hat{U}_1| \right) + O(\epsilon_u^2),$$

while (3.23) and (3.25) imply

$$|F_1| + |F_2| \leq \epsilon_u (|B| + 2|\hat{\mathbf{z}}| |\mathbf{w}^\top|) + O(\epsilon_u^2).$$

These last two results establish (3.30). $\square$

Algorithm 3.24. Using LU factorization to solve a linear system $A\mathbf{x} = \mathbf{b}$ via steps as follows.

(LU-1) Compute LU factorization: $A = LU$,

(LU-2) Solve $L\mathbf{y} = \mathbf{b}$ by forward substitution,

(LU-3) Solve $U\mathbf{x} = \mathbf{y}$ by back substitution.

Theorem 3.25. Let $\hat{L}$ and $\hat{U}$ be the computed LU factors of A obtained by Algorithm 3.21. If $\hat{\mathbf{y}}$ and $\hat{\mathbf{x}}$ are the computed solutions to $\hat{L}\mathbf{y} = \mathbf{b}$ and $\hat{U}\mathbf{x} = \hat{\mathbf{y}}$, then $(A + E)\hat{\mathbf{x}} = \mathbf{b}$ with

$$|E| \leq n\epsilon_u \left(2|A| + 4|\hat{L}||\hat{U}| \right) + O(\epsilon_u^2). \quad (3.31)$$

Proof. From Theorem 3.6 we have

$$\begin{aligned} (\hat{L} + F)\hat{\mathbf{y}} &= \mathbf{b}, \quad |F| \leq n\epsilon_u|\hat{L}| + O(\epsilon_u^2), \\ (\hat{U} + G)\hat{\mathbf{x}} &= \hat{\mathbf{y}}, \quad |G| \leq n\epsilon_u|\hat{U}| + O(\epsilon_u^2), \end{aligned}$$

and thus

$$(\hat{L} + F)(\hat{U} + G)\hat{\mathbf{x}} = (\hat{L}\hat{U} + F\hat{U} + \hat{L}G + FG)\hat{\mathbf{x}} = \mathbf{b}.$$

It follows from Theorem 3.23 that $\hat{L}\hat{U} = A + H$ with

$$|H| \leq 2(n-1)\epsilon_u \left(|A| + |\hat{L}||\hat{U}| \right) + O(\epsilon_u^2),$$

and so by defining

$$E = H + F\hat{U} + \hat{L}G + FG$$

we find $(A + E)\hat{\mathbf{x}} = \mathbf{b}$. Moreover,

$$\begin{aligned} |E| &\leq |H| + |F||\hat{U}| + |\hat{L}||G| + O(\epsilon_u^2) \\ &\leq 2n\epsilon_u \left(|A| + |\hat{L}||\hat{U}| \right) + 2n\epsilon_u \left(|\hat{L}||\hat{U}| \right) + O(\epsilon_u^2), \end{aligned}$$

completing the proof of the theorem. $\square$

3.2.3 Pivoting

Example 3.26. Algorithm 3.24 is unstable, i.e., Gaussian elimination can give arbitrarily poor results, even for well-conditioned problems.

For example, suppose 3-digit floating point arithmetic is used to solve

$$A\mathbf{x} = \begin{bmatrix} .001 & 1.00 \\ 1.00 & 2.00 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1.00 \\ 3.00 \end{bmatrix}.$$

Applying Gaussian elimination we obtain

$$\hat{L} = \begin{bmatrix} 1 & 0 \\ 1000 & 1 \end{bmatrix}, \quad \hat{U} = \begin{bmatrix} .001 & 1 \\ 0 & -1000 \end{bmatrix},$$

then it can be shown that

$$\hat{L}\hat{U} = \begin{bmatrix} .001 & 1 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \equiv A + H.$$

If we solve problem using (LU-2) and (LU-3), we obtain a computed solution $\hat{\mathbf{x}} = [0, 1]^\top$, which is in contrast to the exact solution $\mathbf{x} = [1.002 \dots, .998 \dots]$.

Definition 3.27. A *permutation matrix* is a matrix whose columns are a permutation of the columns of the identity matrix.

Lemma 3.28. A permutation matrix P satisfies $P^\top P = I$.

Definition 3.29. *Partial pivoting* determines the permutation of rows at the k -th step such that the entry with the maximum magnitude in the k -th column on or below the diagonal is moved to the pivot position (k, k) . Specifically, we choose an index $p \in \{k, k+1, \dots, n\}$ such that

$$|a_{pk}^{(k)}| = \max_{k \leq i \leq n} |a_{ik}^{(k)}|. \quad (3.32)$$

We then swap row k and row p before proceeding with the elimination, which is equivalent to left multiplying a permutation matrix

$$P_k = \begin{cases} \begin{bmatrix} I_{k-1} & & & \\ & 0 & & 1 \\ & & I_{q-k-1} & \\ & 1 & & 0 \\ & & & & I_{n-q} \end{bmatrix}, & \text{if } k \neq q; \\ I_n, & \text{if } k = q. \end{cases}$$

Theorem 3.30. Let $A \in \mathbb{R}^{n \times n}$ be a nonsingular matrix. There exists a permutation matrix P , a unit lower triangular matrix L , and an upper triangular matrix U such that

$$PA = LU. \quad (3.33)$$

Proof. Applying partial pivoting Gaussian elimination to matrix A , we obtain that

$$U = A^{(n)} = L_{n-1}P_{n-1} \dots L_2P_2L_1P_1A.$$

Following from $P_k^{-1} = P_k = P_k^\top$, we obtain

$$\tilde{L}_{n-1} \dots \tilde{L}_2 \tilde{L}_1 PA = U,$$

where

$$P = P_{n-1} \dots P_2P_1$$

and

$$\tilde{L}_k = (P_{n-1} \dots P_{k+1}) L_k (P_{k+1} \dots P_{n-1}).$$

Moreover,

$$\tilde{L}_k = (P_{n-1} \dots P_{k+1}) (I_n - \ell_k \mathbf{e}_k^\top) (P_{k+1} \dots P_{n-1}) = I_n - \tilde{\ell}_k \mathbf{e}_k^\top$$

where $\tilde{\ell}_k = P_{n-1} \dots P_{k+1} \ell_k$, which shows that $\tilde{L}_k$ is a Gauss transformation. Then the proof is completed in the same way as Theorem 3.20. $\square$

Lemma 3.31. Partial pivoting ensures that all multipliers satisfy $|l_{ik}| \leq 1$.

Proof. This follows directly from (3.32). $\square$

Theorem 3.32. Let $\hat{P}, \hat{L}$ and $\hat{U}$ be the computed version of P, L , and U as produced by the partial pivoting Gaussian elimination. And $\hat{\mathbf{x}}$ is computed solution of $A\mathbf{x} = \mathbf{b}$ with processing similar to Algorithm 3.24, which satisfies $(A + E)\hat{\mathbf{x}} = \mathbf{b}$ where

$$\|E\|_\infty \leq n\epsilon_u \left(2\|A\|_\infty + 4n\|\hat{U}\|_\infty\right) + O(\epsilon_u^2). \quad (3.34)$$

Proof. From Lemma 3.31, it show that $\|\hat{L}\|_\infty \leq n$. Then the proof is completed by Theorem 3.25 and no rounding errors associated with permutation. $\square$

Definition 3.33. The *growth factor* ρ is defined by

$$\rho := \max_{i,j,k} \frac{|\hat{a}_{ij}^{(k)}|}{\|A\|_\infty}, \quad (3.35)$$

where $\hat{A}^{(k)}$ is the computed version of the matrix $A^{(k)} = L_k P_k \cdots L_1 P_1 A$.

Corollary 3.34. Let E be the same as that in Theorem 3.32, it has the bound

$$\|E\|_\infty \leq 6n^3 \rho \|A\|_\infty \epsilon_u + O(\epsilon_u^2). \quad (3.36)$$

It means that if the growth factor is modest then E satisfies

$$\|E\|_\infty \approx \epsilon_u \|A\|_\infty. \quad (3.37)$$

Example 3.35. Consider the matrix $A \in \mathbb{R}^{n \times n}$ defined by

$$a_{ij} = \begin{cases} 1 & \text{if } i = j \text{ or } j = n; \\ -1 & \text{if } i > j; \\ 0 & \text{otherwise.} \end{cases}$$

It can be shown that

$$A = \begin{bmatrix} 1 & & & \\ -1 & \ddots & & \\ \vdots & & 1 & \\ -1 & \cdots & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & 1 \\ & \ddots & & \vdots \\ & & 1 & 2^{n-2} \\ & & & 2^{n-1} \end{bmatrix},$$

and the growth factor of A is 2^{n-1} .

3.2.4 Condition estimation

Definition 3.36. The *residual* of a computed solution $\hat{\mathbf{x}}$ to the linear system $A\mathbf{x} = \mathbf{b}$ is the vector $\mathbf{b} - A\hat{\mathbf{x}}$.

Lemma 3.37. If the growth factor is modest, Gaussian elimination produces a solution $\hat{\mathbf{x}}$ with a relatively small residual.

Proof. From (3.37) we have

$$\|\mathbf{b} - A\hat{\mathbf{x}}\|_\infty \approx \epsilon_u \|A\|_\infty \|\hat{\mathbf{x}}\|_\infty. \quad (3.38) \quad \square$$

Theorem 3.38. If the growth factor is modest and the unit roundoff and condition satisfy $\epsilon_u \approx 10^{-d}$ and $\text{cond}_\infty(A) \approx 10^q$, then Gaussian elimination produces a solution $\hat{\mathbf{x}}$ that has about $d - q$ correct decimal digits.

Proof. From (3.38), we have

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|_\infty}{\|\mathbf{x}\|_\infty} \approx \epsilon_u \text{cond}_\infty(A). \quad (3.39)$$

$\square$

Example 3.39. Consider the system

$$\begin{bmatrix} .986 & .579 \\ .409 & .237 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} .235 \\ .107 \end{bmatrix}$$

in which $\text{cond}_\infty(A) \approx 700$ and $\mathbf{x} = [2, -3]^\top$. Here is what we find for various machine precisions:

ϵ_u	$\hat{x}_1$	$\hat{x}_2$	$\frac{\ \hat{\mathbf{x}} - \mathbf{x}\ _\infty}{\ \mathbf{x}\ _\infty}$	$\frac{\ \mathbf{b} - A\hat{\mathbf{x}}\ _\infty}{\ A\ _\infty \ \hat{\mathbf{x}}\ _\infty}$
10^{-3}	2.11	-3.17	$5 \cdot 10^{-2}$	$2.0 \cdot 10^{-3}$
10^{-4}	1.986	-2.975	$8 \cdot 10^{-3}$	$1.5 \cdot 10^{-4}$
10^{-5}	2.0019	-3.0032	$1 \cdot 10^{-3}$	$2.1 \cdot 10^{-6}$
10^{-6}	2.00025	-3.00094	$3 \cdot 10^{-4}$	$4.2 \cdot 10^{-7}$

Algorithm 3.40 (Condition estimator for triangular matrix). Let $T \in \mathbb{R}^{n \times n}$ be nonsingular upper triangular matrix. This algorithm computes unit ∞ -norm $\mathbf{y}$ and a scalar $\widehat{\text{cond}}(T)$ so $\|T\mathbf{y}\|_\infty \approx \frac{1}{\|T^{-1}\|_\infty}$ and $\widehat{\text{cond}}(T) \approx \text{cond}_\infty(T)$.

Input: $T \in \mathbb{R}^{n \times n}$
Preconditions: T is nonsingular
Output: The vector $\mathbf{y}$ and the scalar $\widehat{\text{cond}}(T)$

```

1  $\mathbf{p}(1:n) = 0$ 
2 for  $k = n$  to 1 do
3    $\mathbf{y}(k)^+ = (1 - \mathbf{p}(k)) / T(k, k)$ 
4    $\mathbf{y}(k)^- = (-1 - \mathbf{p}(k)) / T(k, k)$ 
5    $\mathbf{p}_k^+ = \mathbf{p}(1:k-1) + T(1:k-1, k) \mathbf{y}(k)^+$ 
6    $\mathbf{p}_k^- = \mathbf{p}(1:k-1) + T(1:k-1, k) \mathbf{y}(k)^-$ 
7   if  $|\mathbf{y}(k)^+| + \|\mathbf{p}_k^+\|_1 \geq |\mathbf{y}(k)^-| + \|\mathbf{p}_k^-\|_1$  then
8      $\mathbf{y}(k) = \mathbf{y}(k)^+$ 
9      $\mathbf{p}(1:k-1) = \mathbf{p}_k^+$ 
10  else
11     $\mathbf{y}(k) = \mathbf{y}(k)^-$ 
12     $\mathbf{p}(1:k-1) = \mathbf{p}_k^-$ 
13  end
14 end
15  $\widehat{\text{cond}}(T) = \|\mathbf{y}\|_\infty \|T\|_\infty$ 
16  $\mathbf{y} = \mathbf{y} / \|\mathbf{y}\|_\infty$ 
```

Algorithm 3.41. We estimate the condition of a square nonsingular matrix A whose partial pivoting LU factorization $PA = LU$ is available via steps as follows.

- (EC-1) Apply the lower triangular version of Algorithm 3.40 to $U^\top$ and obtain a large-norm solution to $U^\top \mathbf{y} = \mathbf{d}$,
- (EC-2) Solve the triangular systems $L^\top \mathbf{r} = \mathbf{y}$, $L\mathbf{w} = P\mathbf{r}$, and $U\mathbf{z} = \mathbf{w}$,
- (EC-3) Set $\widehat{\text{cond}}_\infty(A) = \frac{\|A\|_\infty \|\mathbf{z}\|_\infty}{\|\mathbf{r}\|_\infty}$.

3.3 Special types of matrices

3.3.1 Irreducible matrices

Definition 3.42. A matrix $A \in \mathbb{R}^{n \times n}$ is *reducible* if there exists a permutation matrix P such that

$$PAP^\top = \begin{bmatrix} A_{11} & \mathbf{0} \\ A_{21} & A_{22} \end{bmatrix}, \quad (3.40)$$

where $A_{11} \in \mathbb{R}^{r \times r}$, $A_{21} \in \mathbb{R}^{(n-r) \times r}$, and $A_{22} \in \mathbb{R}^{(n-r) \times (n-r)}$ are block matrices. An *irreducible* matrix is a square matrix that is not reducible.

Definition 3.43. A *partition* of a set S is a set of pairwise disjoint non-empty subsets $S_1, S_2, \dots, S_p$ whose union equals S .

Definition 3.44. A *decomposition* of a set S is a set of non-empty subsets $S_1, S_2, \dots, S_p$ whose union equals S .

Lemma 3.45. A matrix $A \in \mathbb{C}^{n \times n}$ is reducible if and only if there exists a partition $\{\mathcal{I}, \mathcal{J}\}$ of $\mathcal{W} = \{1, 2, \dots, n\}$ such that

$$\forall i \in \mathcal{I}, \forall j \in \mathcal{J}, \quad a_{i,j} = 0. \quad (3.41)$$

Exercise 3.46. Prove Lemma 3.45.

3.3.2 Diagonally dominant matrices

Definition 3.47. A matrix $A \in \mathbb{C}^{n \times n}$ is (*weakly*) *diagonally dominant* (by rows) iff

$$\forall i = 1, 2, \dots, n, \quad |a_{ii}| \geq \sum_{j \neq i, j=1}^n |a_{ij}|, \quad (3.42)$$

strictly diagonally dominant (by rows) iff

$$\forall i = 1, 2, \dots, n, \quad |a_{ii}| > \sum_{j \neq i, j=1}^n |a_{ij}|, \quad (3.43)$$

and *irreducibly diagonally dominant* iff it is irreducible and diagonally dominant with at least one strict inequality in (3.42).

Theorem 3.48 (Gershgorin). Any eigenvalue λ of a square matrix A is located in one of the closed discs of the complex plane centered at a_{ii} ,

$$\forall \lambda \in \sigma(A), \exists i \in [1, n] \text{ s.t. } |\lambda - a_{ii}| \leq \sum_{j \neq i, j=1}^n |a_{ij}|. \quad (3.44)$$

Proof. Choose the eigenvector associated with λ to be $\mathbf{u}$ such that $|u_i| = 1$ and $|u_j| \leq 1$ for each $j \neq i$. Then we have

$$\begin{aligned} A\mathbf{u} &= \lambda\mathbf{u} \implies \sum_{j=1}^n a_{ij}u_j = \lambda u_i \\ \implies (\lambda - a_{ii})u_i &= \sum_{j \neq i, j=1}^n a_{ij}u_j \\ \implies |\lambda - a_{ii}| &\leq \sum_{j \neq i, j=1}^n |a_{ij}||u_j| \leq \sum_{j \neq i, j=1}^n |a_{ij}|. \quad \square \end{aligned}$$

Theorem 3.49. A matrix A is nonsingular if it is strictly diagonally dominant or irreducibly diagonally dominant.

Proof. For a strictly diagonally dominant matrix A , the Gershgorin theorem 3.48 implies that 0 is not an eigenvalue of A and thus A is nonsingular. Suppose an irreducibly diagonally dominant matrix A is singular, i.e.,

$$\exists \mathbf{u} \in \mathbb{R}^n \text{ s.t. } \|\mathbf{u}\|_\infty = 1, \quad A\mathbf{u} = \mathbf{0}.$$

Then there exists a partition $\{\mathcal{I}, \mathcal{J}\}$ of $\mathcal{W} = \{1, 2, \dots, n\}$ such that

$$\mathcal{I} = \{i \in \mathcal{W} : |u_i| = 1\}, \quad \mathcal{J} = \{j \in \mathcal{W} : |u_j| < 1\}.$$

In addition, the irreducibly diagonal dominance of A implies $\mathcal{J} \neq \emptyset$. Then for $i \in \mathcal{I}$, the i th equation of $A\mathbf{u} = \mathbf{0}$ yields

$$\begin{aligned} a_{ii}u_i &= -\sum_{j \in \mathcal{I}; j \neq i} a_{ij}u_j - \sum_{j \in \mathcal{J}} a_{ij}u_j \\ \implies |a_{ii}| &\leq \sum_{j \in \mathcal{I}; j \neq i} |a_{ij}| + \sum_{j \in \mathcal{J}} |a_{ij}||u_j| < \sum_{j \neq i} |a_{ij}|, \end{aligned}$$

which contradicts (3.42). $\square$

Theorem 3.50. Suppose that A is strictly diagonally dominant by rows and write $\alpha := \min_i (|a_{ii}| - \sum_{j \neq i} |a_{ij}|)$. Then $\|A^{-1}\|_\infty \leq \frac{1}{\alpha}$.

Proof. By Theorem 3.49, A is invertible. Write $A\mathbf{x} = \mathbf{y}$ and we have, from Definition C.281,

$$\|A^{-1}\|_\infty = \sup_{\mathbf{y} \neq \mathbf{0}} \frac{\|A^{-1}\mathbf{y}\|_\infty}{\|\mathbf{y}\|_\infty} = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{x}\|_\infty}{\|A\mathbf{x}\|_\infty}.$$

Thus the reciprocal of $\|A^{-1}\|_\infty$ is the minimum contraction that A applies to any vector, i.e.,

$$\|A^{-1}\|_\infty^{-1} = \inf_{\mathbf{x} \neq \mathbf{0}} \frac{\|A\mathbf{x}\|_\infty}{\|\mathbf{x}\|_\infty}.$$

Hence it suffices to show that $\alpha\|\mathbf{x}\|_\infty \leq \|A\mathbf{x}\|_\infty$ holds for an arbitrary $\mathbf{x}$. Suppose $|x_k| = \|\mathbf{x}\|_\infty$. Then the condition on α gives $0 < \alpha \leq |a_{kk}| - \sum_{j \neq k} |a_{kj}|$ and

$$\begin{aligned} 0 < \alpha|x_k| &\leq |a_{kk}x_k| - \sum_{j \neq k} |a_{kj}||x_j| \\ &\leq |a_{kk}x_k| - \sum_{j \neq k} |a_{kj}x_j| \\ &\leq \sum_j |a_{kj}||x_j| \leq \max_k \left| \sum_j a_{kj}x_j \right| = \|A\mathbf{x}\|_\infty, \end{aligned}$$

where the first line follows from $|x_k| = \|\mathbf{x}\|_\infty$ and the second and the third from the triangle inequality. $\square$

Corollary 3.51. Suppose that A is strictly diagonally dominant by columns and write $\beta := \min_i (|a_{ii}| - \sum_{j \neq i} |a_{ji}|)$. Then $\|A^{-1}\|_1 \leq \frac{1}{\beta}$.

Proof. This follows from applying Theorem 3.50 and Lemma C.282 to A^* . $\square$

Corollary 3.52. Suppose that $A \in \mathbb{C}^{n \times n}$ is strictly diagonally dominant both by rows and by columns. Then

$$\|A^{-1}\|_2^{-1} = \sigma_n \geq \sqrt{\alpha\beta}, \quad (3.45)$$

where α, β are given in Theorem 3.50 and Corollary 3.51.

Proof. This follows from Theorem 3.50, Corollary 3.51, and a derivation similar to (C.115). $\square$

3.3.3 Positive-definite matrices

Lemma 3.53. The matrix of a positive-definite operator $T \in \mathcal{L}(V)$ (as in Definition C.250) has each of its diagonal entries positive.

Proof. This follows from setting $\mathbf{v}$ in Definition C.250 to be the i th column of the identity matrix. $\square$

Definition 3.54. A real positive-definite matrix $A \in \mathbb{R}^{n \times n}$ and a real positive semi-definite matrix $B \in \mathbb{R}^{n \times n}$ are matrices satisfying respectively

$$\forall \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}, \langle A\mathbf{x}, \mathbf{x} \rangle > 0, \quad (3.46)$$

$$\forall \mathbf{x} \in \mathbb{R}^n, \langle B\mathbf{x}, \mathbf{x} \rangle \geq 0. \quad (3.47)$$

Definition 3.55. A symmetric positive-definite (SPD) matrix is a real matrix $A \in \mathbb{R}^{n \times n}$ that is both symmetric and positive definite.

Lemma 3.56. Suppose $A \in \mathbb{C}^{n \times n}$ satisfies

$$\forall \mathbf{x} \in \mathbb{C}^n, \quad \text{Im} \langle A\mathbf{x}, \mathbf{x} \rangle = 0. \quad (3.48)$$

Then A is an Hermitian matrix.

Proof. $\langle A\mathbf{x}, \mathbf{x} \rangle$ being real for all $\mathbf{x} \in \mathbb{C}^n$ implies

$$\langle A\mathbf{x}, \mathbf{x} \rangle = \overline{\langle A\mathbf{x}, \mathbf{x} \rangle} = \langle \mathbf{x}, A\mathbf{x} \rangle = \langle A^* \mathbf{x}, \mathbf{x} \rangle.$$

It follows that $\forall \mathbf{x} \in \mathbb{C}^n, \langle (A^* - A)\mathbf{x}, \mathbf{x} \rangle = 0$. The proof is then completed by Lemma C.220. $\square$

Definition 3.57. A matrix $A \in \mathbb{C}^{n \times n}$ is Hermitian positive definite (HPD) iff

$$\forall \mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\}, \langle A\mathbf{x}, \mathbf{x} \rangle > 0. \quad (3.49)$$

Lemma 3.58. Any eigenvalue of an HPD matrix is positive.

Proof. By Theorem C.237, any HPD matrix A is unitarily similar to a diagonal matrix Λ , i.e., $A = Q\Lambda Q^*$ with $Q = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n]$ satisfying $Q^*Q = I$. Then each $(\lambda_i, \mathbf{q}_i)$ is an eigenpair of A . Consequently, any vector $\mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$ can be expressed as $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{q}_i$, for which we have

$$0 < \langle A\mathbf{x}, \mathbf{x} \rangle = \langle \sum_{i=1}^n x_i \lambda_i \mathbf{q}_i, \sum_{i=1}^n x_i \mathbf{q}_i \rangle = \sum_{i=1}^n |x_i|^2 \lambda_i.$$

The proof is completed by choosing $\mathbf{x}$ to be the standard basis vectors. $\square$

Lemma 3.59. The real part of any eigenvalue of a real positive-definite matrix A must be positive.

Proof. An eigenpair $(x + iy, \mathbf{u} + i\mathbf{v})$ of A satisfy $A(\mathbf{u} + i\mathbf{v}) = (x + iy)(\mathbf{u} + i\mathbf{v})$. Then

$$\begin{cases} A\mathbf{u} = x\mathbf{u} - y\mathbf{v} \\ A\mathbf{v} = x\mathbf{v} + y\mathbf{u} \end{cases} \implies \begin{cases} \mathbf{u}^\top A\mathbf{u} = x\|\mathbf{u}\|_2^2 - y\mathbf{u}^\top \mathbf{v} > 0 \\ \mathbf{v}^\top A\mathbf{v} = x\|\mathbf{v}\|_2^2 + y\mathbf{v}^\top \mathbf{u} > 0 \end{cases}$$

and $x > 0$ follows from $\mathbf{v}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{v}$. $\square$

Theorem 3.60. A real positive-definite matrix A satisfies

$$\exists \alpha \in \mathbb{R}^+ \text{ s.t. } \forall \mathbf{u} \in \mathbb{R}^n, \quad \langle A\mathbf{u}, \mathbf{u} \rangle \geq \alpha \|\mathbf{u}\|_2^2. \quad (3.50)$$

Proof. By Definition C.229, $H := \frac{1}{2}(A + A^*)$ is the Hermitian part of A . By Lemma C.230, $\langle A\mathbf{u}, \mathbf{u} \rangle = \langle H\mathbf{u}, \mathbf{u} \rangle$. Then the proof is completed by Lemma 3.58 and the Rayleigh theorem C.320 with $\alpha = \lambda_{\min}(H)$. $\square$

Lemma 3.61. If a symmetric matrix $A \in \mathbb{R}^{n \times n}$ is strictly diagonally dominant or irreducibly diagonally dominant and satisfies $a_{ii} > 0$ for each $i = 1, 2, \dots, n$, then A is SPD.

Proof. By Theorem 3.49, $A - \lambda I$ is nonsingular for any $\lambda \leq 0$. Hence all eigenvalues of A are positive real numbers. The symmetry of A implies that A is Hermitian and Corollary C.243 implies that A is unitarily similar to a diagonal matrix, whose diagonal entries are the positive eigenvalues of A . Hence (3.46) holds for A and the rest of the proof follows from Definition 3.55. $\square$

3.3.4 Nonnegative matrices

Definition 3.62. A nonnegative matrix is a matrix whose entries are nonnegative. A positive matrix is a matrix whose entries are positive.

Notation 4. For two matrices $A, B \in \mathbb{R}^{m \times n}$ with $m, n \in \mathbb{N}^+$, we write $A \leq B$ iff $a_{i,j} \leq b_{i,j}$ for all $1 \leq i \leq m$ and $1 \leq j \leq n$. In particular, $A \geq \mathbf{0}$ and $A > \mathbf{0}$ mean that A is a nonnegative and positive matrix, respectively.

Lemma 3.63. The binary relation “ $\leq$ ” given in Notation 4 is a partial order on $\mathbb{R}^{m \times n}$.

Proof. This follows immediately from Notation 4 and Definition A.59. $\square$

Lemma 3.64. Two nonnegative matrices A and B satisfy

- both AB and $A + B$ are nonnegative;
- A^n is nonnegative for $n \in \mathbb{N}^+$;
- $\mathbf{0} \leq A \leq B$ implies $\|A\|_1 \leq \|B\|_1$ and $\|A\|_\infty \leq \|B\|_\infty$.

Proof. This follows directly from Definition 3.62. $\square$

Theorem 3.65 (Perron-Frobenius). The spectral radius of a nonnegative irreducible matrix $A \in \mathbb{R}^{n \times n}$ is a simple eigenvalue of A , with which the associated eigenvector $\mathbf{u}$ can be chosen to be nonnegative.

Lemma 3.66. Nonnegative matrices A, B, C, D satisfy

$$A \leq B \implies AC \leq BC, \quad DA \leq DB. \quad (3.51)$$

Lemma 3.67. For nonnegative matrices A and B , we have

$$A \leq B \implies \forall n \in \mathbb{N}^+, A^n \leq B^n. \quad (3.52)$$

Exercise 3.68. Prove Lemmas 3.66 and 3.67.

Theorem 3.69. For square matrices A and B , we have

$$\mathbf{0} \leq A \leq B \implies \rho(A) \leq \rho(B). \quad (3.53)$$

Proof. This follows directly from Lemma 3.67 and Gelfand’s formula (Theorem C.307). $\square$

Theorem 3.70. A nonnegative square matrix A satisfies $\rho(A) < 1$ if and only if $I - A$ is invertible and $(I - A)^{-1} \geq \mathbf{0}$.

Proof. By $\rho(A) < 1$ and Lemma C.327, $I - A$ is invertible. Then Theorem C.329 and $A \geq \mathbf{0}$ give $(I - A)^{-1} \geq \mathbf{0}$.

As for the sufficiency, the Perron-Frobenius theorem 3.65 implies the existence of a nonnegative eigenvector $\mathbf{u}$ of A such that $A\mathbf{u} = \rho(A)\mathbf{u}$, i.e., $(I - A)^{-1}\mathbf{u} = \frac{1}{1-\rho(A)}\mathbf{u}$. Then $(I - A)^{-1} \geq \mathbf{0}$ implies $1 - \rho(A) > 0$. $\square$

3.3.5 M-matrices

Definition 3.71. An *M-matrix* A is a square matrix of size n that satisfies

(MMT-1) $\forall i = 1, 2, \dots, n, a_{ii} > 0$;

(MMT-2) $\forall i \neq j, i, j = 1, 2, \dots, n, a_{ij} < 0$;

(MMT-3) $\det(A) \neq 0$;

(MMT-4) $A^{-1} \geq \mathbf{0}$.

Theorem 3.72. Suppose a square matrix B satisfies (MMT-1,2). Then B is an M-matrix if and only if

$$\rho(I - D_B^{-1}B) < 1, \quad (3.54)$$

where D_B is the diagonal of B .

Proof. The conclusion follows from applying Theorem 3.70 to $A := I - D_B^{-1}B$. $\square$

Example 3.73. For the square matrix

$$B = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}, \quad (3.55)$$

elements of the matrix $A := I - D_B^{-1}B$ are given by

$$a_{i,j} = \begin{cases} \frac{1}{2} & \text{if } i - j = \pm 1; \\ 0 & \text{otherwise.} \end{cases}$$

Clearly $\|A\|_1 = 1$ and thus Lemma C.308 yields $\rho(A) \leq 1$.

Suppose $\rho(A) = 1$, then there exists an eigenvector $\mathbf{u}$ with $u_1 = 1$ and $A\mathbf{u} = \mathbf{u}$. The first $m - 1$ equations imply $u_i = i$, which contradicts the last equation $u_m = \frac{1}{2}u_{m-1}$. Hence $\rho(A) \neq 1$ and we have $\rho(A) < 1$. By Theorem 3.72, the matrix B in (3.55) is an M-matrix.

Theorem 3.74. If a square matrix A satisfies (MMT-2,3,4), then it satisfies (MMT-1) and (3.54).

Proof. Write $B = A^{-1}$. From $AB = I$ we have

$$\sum_{k=1}^n a_{ik}b_{ki} = 1 \implies a_{ii}b_{ii} = 1 - \sum_{k \neq i, k=1}^n a_{ik}b_{ki}.$$

(MMT-2) states that $a_{ik} \leq 0$ and (MMT-4) states that $b_{ki} \geq 0$. Hence the RHS of the last equation is no less than 1 and $b_{ii} \geq 0$ yields $a_{ii} > 0$. Hence A is an M-matrix and the proof is completed by Theorem 3.72. $\square$

Theorem 3.75. Suppose a square matrix B satisfies (MMT-2) and another square matrix A satisfies $A \leq B$. Then if A is an M-matrix, so is B .

Proof. By $A \leq B$ and (MMT-1), we have $D_B \geq D_A \geq \mathbf{0}$ and $D_A - A \geq D_B - B \geq \mathbf{0}$. It follows that

$$I - D_A^{-1}A \geq D_A^{-1}(D_B - B) \geq D_B^{-1}(D_B - B) = I - D_B^{-1}B \geq \mathbf{0},$$

where the second inequality follows from Lemma 3.66 and the fact that both D_A and D_B are nonnegative diagonal matrices. Then Theorems 3.69 and 3.72 yield

$$\rho(I - D_B^{-1}B) \leq \rho(I - D_A^{-1}A) < 1$$

and B being an M-matrix follows from Theorem 3.72. $\square$

3.4 Linear fixed-point iterations

Definition 3.76. For a nonsingular square matrix A , a *linear fixed-point iteration* for solving

$$A\mathbf{x} = \mathbf{b} \quad (3.56)$$

is an iteration of the form

$$\mathbf{x}^{(k+1)} = T\mathbf{x}^{(k)} + \mathbf{c}, \quad (3.57)$$

where $\mathbf{x}^{(k)}$ is the k th iterate that approximates $\mathbf{x}$ while the *iteration matrix* T and the vector $\mathbf{c}$ are functions of A and $\mathbf{b}$.

Roadmap 3.77. Four questions follow immediately from Definition 3.76.

- Does the iteration (3.57) converge?
- How to construct T and $\mathbf{c}$ from A and $\mathbf{b}$ so that the iteration (3.57) converges to the solution of (3.56)?
- If (3.57) converges, how fast is the convergence?
- Can we give some examples of fundamental linear fixed-point iterations?

A common way to construct T and $\mathbf{c}$ is given in Lemma 3.78, which partially answers question (b). Question (a) is (fully) answered by Theorem 3.79; see also Roadmap 3.80. Question (c) is addressed in Section 3.4.2 and question (d) by other subsections of Section 3.4.

Lemma 3.78. Suppose two matrices M and N satisfy $M - N = A$ and $\det(M) \neq 0$ and that the linear fixed-point iteration (3.57) given by

$$T = M^{-1}N, \quad \mathbf{c} = M^{-1}\mathbf{b} \quad (3.58)$$

converges. Then the limit of (3.57) is the solution of (3.56).

Proof. The fixed-point iteration (3.57) converges, say, to $\mathbf{x}^*$ and thus $(I - T)\mathbf{x}^* = \mathbf{c}$, which is equivalent to

$$M^{-1}A\mathbf{x}^* = M^{-1}\mathbf{b}.$$

Then the proof is completed by the nonsingularity of A . $\square$

3.4.1 General convergence criteria

Theorem 3.79. The fixed-point iteration in (3.57) converges for any $\mathbf{c}$ and $\mathbf{x}^{(0)}$ if and only if the spectral radius $\rho(T) < 1$.

Proof. If (3.57) converges to $\mathbf{x}$, then we have $\mathbf{x} = T\mathbf{x} + \mathbf{c}$, the subtraction of which from (3.57) gives

$$\mathbf{e}^{(k)} = \mathbf{x} - \mathbf{x}^{(k)} = T(\mathbf{x} - \mathbf{x}^{(k-1)}) = T^k(\mathbf{x} - \mathbf{x}^{(0)}) = T^k \mathbf{e}^{(0)}.$$

The convergence implies $\lim_{k \rightarrow \infty} T^k \mathbf{e}^{(0)} = \lim_{k \rightarrow \infty} \mathbf{e}^{(k)} = \mathbf{0}$. Then the arbitrariness of $\mathbf{e}^{(0)}$ yields $\lim_{k \rightarrow \infty} T^k = \mathbf{0}$ and Theorem C.326 gives $\rho(T) < 1$.

Conversely, $\rho(T) < 1$ and Lemma C.327 imply that $I - T$ is nonsingular. By Cramer's rule (Theorem C.279), there exists a unique solution $\mathbf{x}^*$ for the linear system $(I - T)\mathbf{x} = \mathbf{c}$. Then $\mathbf{x}^* = T\mathbf{x}^* + \mathbf{c}$ and (3.57) yield

$$\mathbf{x}^* - \mathbf{x}^{(k)} = T(\mathbf{x}^* - \mathbf{x}^{(k-1)}) = T^k(\mathbf{x}^* - \mathbf{x}^{(0)});$$

On the other hand, Theorem C.326 and $\rho(T) < 1$ yield $\lim_{k \rightarrow \infty} T^k = \mathbf{0}$, which implies $\lim_{k \rightarrow \infty} \mathbf{x}^{(k)} = \mathbf{x}^*$. $\square$

Roadmap 3.80. Theorem 3.79 is the most general result on the convergence of fixed-point iterations in Definition 3.76, without any assumptions on the iteration matrix T . In the rest of Section 3.4.1, we study the splitting $A = M - N$ in Lemma 3.78 and give two sufficient conditions for $\rho(T) < 1$.

This convergence criterion $\rho(T) < 1$ is analogous to the sufficient condition $\lambda = \max_{x \in [a, b]} |g'(x)| < 1$ in Theorem 4.42 for convergence of a nonlinear equation $x_{n+1} = g(x_n)$: a normal matrix T satisfies

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \quad \|T\mathbf{x} - T\mathbf{y}\|_2 \leq \|T\|_2 \|\mathbf{x} - \mathbf{y}\|_2 = \rho(T) \|\mathbf{x} - \mathbf{y}\|_2$$

and thus can be regarded as a contraction in the sense of Definition 4.38.

Theorem 3.81 (Householder–John criterion). For the splitting $A = M - N$ with $A, N \in \mathbb{R}^{n \times n}$, we have $\rho(M^{-1}N) < 1$ if both A and $A + N + N^\top$ are SPD.

Proof. Let $(\lambda, \mathbf{w}) \in \mathbb{C} \times \mathbb{C}^n$ be any eigenpair of the matrix $T := M^{-1}N = (A + N)^{-1}N$. Then we have

$$\begin{aligned} (A + N)^{-1}N\mathbf{w} &= \lambda\mathbf{w} \implies N\mathbf{w} = \frac{\lambda}{1-\lambda}A\mathbf{w} \\ \implies \mathbf{w}^*N\mathbf{w} &= \frac{\bar{\lambda}}{1-\bar{\lambda}}\mathbf{w}^*A\mathbf{w} \implies \mathbf{w}^*N^\top\mathbf{w} = \frac{\bar{\lambda}}{1-\bar{\lambda}}\mathbf{w}^*A\mathbf{w}, \end{aligned}$$

where $\bar{\lambda}$ is the complex conjugate of λ . Write $\mathbf{w} = \mathbf{u} + i\mathbf{v}$ where $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Then A being SPD implies

$$\mathbf{w}^*A\mathbf{w} = (\mathbf{u}^\top - i\mathbf{v}^\top)A(\mathbf{u} + i\mathbf{v}) = \mathbf{u}^\top A\mathbf{u} + \mathbf{v}^\top A\mathbf{v} > 0$$

Similarly, we have $\mathbf{w}^*(A + N + N^\top)\mathbf{w} > 0$, which implies

$$\left(1 + \frac{\lambda}{1-\lambda} + \frac{\bar{\lambda}}{1-\bar{\lambda}}\right) \mathbf{w}^*A\mathbf{w} = \frac{1-|\lambda|^2}{|1-\lambda|^2} \mathbf{w}^*A\mathbf{w} > 0,$$

where the first equality follows from complex arithmetic. Then the proof is completed by the last two inequalities. $\square$

Definition 3.82. A regular splitting of a square matrix A is a pair of matrices M, N satisfying

$$(RSM-1) \quad A = M - N,$$

$$(RSM-2) \quad \det(M) \neq 0,$$

$$(RSM-3) \quad M^{-1} \geq \mathbf{0},$$

$$(RSM-4) \quad N \geq \mathbf{0}.$$

Theorem 3.83. Let M, N be a regular splitting of A . Then $\rho(M^{-1}N) < 1$ if and only if A is invertible and $A^{-1} \geq \mathbf{0}$.

Proof. Write $T := M^{-1}N$ and we have $A = M(I - T)$. If $\rho(T) < 1$, Lemma C.327 implies that $I - T$ is nonsingular and (RSM-2) further yields that A is invertible. By Theorem 3.70, we have $(I - T)^{-1} \geq \mathbf{0}$, and (RSM-3) further yields $A^{-1} = (I - T)^{-1}M^{-1} \geq \mathbf{0}$.

As for the sufficiency, $\det(A) \neq 0$ and $\det(M) \neq 0$ imply that $I - T = M^{-1}A$ is invertible. By (RSM-3.4), we have $T \geq \mathbf{0}$. Then the Perron-Frobenius theorem 3.65 guarantees the existence of an eigenvector $\mathbf{x} \geq \mathbf{0}$ associated with the eigenvalue $\rho(T)$ such that $T\mathbf{x} = \rho(T)\mathbf{x}$. Then

$$\begin{aligned} A^{-1}N &= (M(I - M^{-1}N))^{-1}N = (I - T)^{-1}T \\ \implies A^{-1}N\mathbf{x} &= \frac{\rho(T)}{1-\rho(T)}\mathbf{x}. \end{aligned}$$

Furthermore, $A^{-1} \geq \mathbf{0}$ and (RSM-4) yield $\frac{\rho(T)}{1-\rho(T)} \geq 0$, which implies $\rho(T) \in [0, 1]$. $I - T$ being invertible implies $\rho(T) \neq 1$, which completes the proof. $\square$

3.4.2 Speed of convergence

Corollary 3.84. Suppose the matrix T in (3.57) satisfies $\|T\| < 1$ for some matrix norm induced from a vector norm. Then the fixed-point iteration (3.57) converges for any initial guess. Furthermore, the error norm of the k th iterate satisfies

$$\|\mathbf{e}^{(k)}\| \leq \frac{\|T\|^k}{1 - \|T\|} \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|; \quad (3.59)$$

$$\|\mathbf{e}^{(k)}\| \leq \frac{\|T\|}{1 - \|T\|} \|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|, \quad (3.60)$$

where $\mathbf{e}^{(k)} := \mathbf{x} - \mathbf{x}^{(k)}$.

Proof. The first statement follows directly from Theorem 3.79 and Lemma C.308. (3.60) follows from

$$\begin{aligned} \|\mathbf{x}^{(k)} - \mathbf{x}\| &= \|T(\mathbf{x}^{(k-1)} - \mathbf{x})\| \leq \|T\| \|\mathbf{x}^{(k-1)} - \mathbf{x}\| \\ &\leq \|T\| \|\mathbf{x}^{(k-1)} - \mathbf{x}^{(k)}\| + \|T\| \|\mathbf{x}^{(k)} - \mathbf{x}\|. \end{aligned}$$

By triangle inequalities, we have

$$\begin{aligned} \|\mathbf{x}^{(0)} - \mathbf{x}\| &\leq \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\| + \|T\| \|\mathbf{x}^{(0)} - \mathbf{x}\| \\ \implies \|\mathbf{x}^{(0)} - \mathbf{x}\| &\leq \frac{1}{1 - \|T\|} \|\mathbf{x}^{(0)} - \mathbf{x}^{(1)}\| \end{aligned}$$

which implies (3.59) since

$$\|\mathbf{x}^{(k)} - \mathbf{x}\| \leq \|T\|^k \|\mathbf{x}^{(0)} - \mathbf{x}\| \leq \frac{\|T\|^k}{1 - \|T\|} \|\mathbf{x}^{(0)} - \mathbf{x}^{(1)}\|. \quad \square$$

Definition 3.85. The *averaged convergence factor* of a fixed-point iteration in Definition 3.76 during k iterations is the averaged reduction ratio of error norms,

$$\psi_k(T) := \left(\sup_{\mathbf{e}^{(0)} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} \frac{\|\mathbf{e}^{(k)}\|}{\|\mathbf{e}^{(0)}\|} \right)^{\frac{1}{k}} \quad (3.61)$$

while the (*general*) *convergence factor* of the fixed-point iteration is the reduction ratio of error norms per iteration in the asymptotic range,

$$\phi(T) := \lim_{k \rightarrow +\infty} \psi_k(T). \quad (3.62)$$

The *convergence rate* of the fixed-point iteration is

$$\tau(T) := -\ln \phi(T). \quad (3.63)$$

Theorem 3.86. The general convergence factor is the spectral radius of the iteration matrix, i.e.,

$$\phi(T) = \rho(T). \quad (3.64)$$

Proof. Definition 3.85 yields

$$\begin{aligned} \phi(T) &= \lim_{k \rightarrow +\infty} \left(\max_{\mathbf{e}^{(0)} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} \frac{\|T^k \mathbf{e}^{(0)}\|}{\|\mathbf{e}^{(0)}\|} \right)^{\frac{1}{k}} \\ &= \lim_{k \rightarrow +\infty} \|T^k\|^{\frac{1}{k}} = \rho(T), \end{aligned}$$

where the second step follows from Definition C.281 and the last one from the Gelfand formula (Theorem C.307). $\square$

3.4.3 The Richardson iteration

Definition 3.87. The *approximate inverse* of a square matrix A is a matrix B that satisfies

$$\|I - BA\| < 1. \quad (3.65)$$

Definition 3.88. The *preconditioned system* of a linear system $A\mathbf{x} = \mathbf{b}$ is the linear system $M^{-1}A\bar{\mathbf{x}} = M^{-1}\mathbf{b}$ where the *preconditioning matrix* or the *preconditioner* M is non-singular so that $\bar{\mathbf{x}} = \mathbf{x}$.

Definition 3.89. The *Richardson iteration* is a fixed-point iteration in Definition 3.76 with $T = I - A$ and $\mathbf{c} = \mathbf{b}$.

3.4.4 The Jacobi iteration

Definition 3.90. The *Jacobi iteration* is an iterative method for solving (3.56) in which the next iterate $\mathbf{x}^{(k+1)}$ is determined by

$$\forall i = 1, 2, \dots, n, \quad a_{ii}x_i^{(k+1)} = - \sum_{j \neq i; j=1}^n a_{ij}x_j^{(k)} + b_i, \quad (3.66)$$

where $x_i^{(k)}$ is the i th component of the k th iterate $\mathbf{x}^{(k)}$.

Lemma 3.91. The Jacobi iteration is a fixed-point iteration for solving (3.56) with

$$\begin{cases} T_J = D^{-1}(L + U), \\ \mathbf{c} = D^{-1}\mathbf{b}, \end{cases} \quad (3.67)$$

where D , $-L$, and $-U$ are respectively the diagonal, lower triangular, and upper triangular part of A :

$$A = D - L - U. \quad (3.68)$$

Proof. This follows directly from (3.66). $\square$

Example 3.92. For the linear system $B\mathbf{x} = \mathbf{b}$ where B is given by (3.55), entries of the Jacobi iteration matrix $T_J = D^{-1}(L + U)$ are given by

$$t_{ij} = \begin{cases} \frac{1}{2} & \text{if } i - j = \pm 1; \\ 0 & \text{otherwise.} \end{cases}$$

Clearly $\|T_J\|_1 = 1$ and thus Lemma C.308 yields $\rho(T_J) \leq 1$. The arguments in Example 3.73 give $\rho(T_J) \neq 1$ and thus the Jacobi iteration converges.

Example 3.93. For the Jacobi iteration in Definition 3.90, $M_J = D$ and $N = L + U$ form a regular splitting of the matrix $B = D - L - U$ in (3.55), with M_J^{-1} being an approximate inverses of B . By Example 3.73, B is an M-matrix and thus (by Definition 3.71) is invertible and satisfies $B^{-1} \geq \mathbf{0}$. Then it follows from Lemma 3.91 and Theorem 3.83 that $\rho(T_J) < 1$ and therefore the Jacobi iteration converges.

Definition 3.94. The *weighted Jacobi iteration* is a fixed-point iteration of the form

$$\mathbf{x}_* = T_J \mathbf{x}^{(k)} + \mathbf{c} \quad (3.69a)$$

$$\mathbf{x}^{(k+1)} = (1 - \omega) \mathbf{x}^{(k)} + \omega \mathbf{x}_*, \quad (3.69b)$$

where T_J and $\mathbf{c}$ are given in (3.67) and ω is called the *relaxation parameter*.

Definition 3.95. The *optimal relaxation parameter* of the weighted Jacobi is

$$\omega_* := \arg \min_{\omega} \rho(T_{\omega}) = \arg \min_{\omega} \max\{|\omega\lambda + (1-\omega)| : \lambda \in \sigma(T_J)\}, \quad (3.70)$$

where $\sigma(T_J)$ is the spectrum of T_J .

Example 3.96. Consider the case that all eigenvalues of T_J are real and $\sigma(T_J) \subset [\alpha, \beta]$ holds with $-1 < \alpha < \beta < 1$. Since the function $f(\omega) = \omega\lambda + 1 - \omega$ is monotonically decreasing for any $\lambda \in (-1, 1)$, the optimal parameter must shift the interval $[\alpha, \beta]$ to center it around the origin. Hence

$$\omega_*\alpha + 1 - \omega_* = -(\omega_*\beta + 1 - \omega_*) \implies \omega_* = \frac{2}{2 - \alpha - \beta}.$$

3.4.5 The Gauss–Seidel iteration

Definition 3.97. The *Gauss–Seidel iteration* determines the next iterate $\mathbf{x}^{(k+1)}$ by annihilating the i th component of the residual in the order $i = 1, 2, \dots, n$ using new components of $\mathbf{x}^{(k+1)}$ whenever possible:

$$a_{ii}x_i^{(k+1)} = - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij}x_j^{(k)} + b_i. \quad (3.71)$$

Lemma 3.98. The Gauss–Seidel iteration is a fixed-point iteration for solving (3.56) with

$$\begin{cases} T_{GS} = (D - L)^{-1}U, \\ \mathbf{c} = (D - L)^{-1}\mathbf{b}, \end{cases} \quad (3.72)$$

where D , $-L$, and $-U$ are the same as in Lemma 3.91.

Proof. This follows directly from (3.71). $\square$

Definition 3.99. The *backward Gauss–Seidel iteration* is the same as the Gauss–Seidel iterate except that components of the next iterate $\mathbf{x}^{(k+1)}$ is updated in the reversed order $i = n, n-1, \dots, 1$:

$$a_{ii}x_i^{(k+1)} = -\sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^n a_{ij}x_j^{(k+1)} + b_i. \quad (3.73)$$

Lemma 3.100. The backward Gauss–Seidel iteration is a fixed-point iteration for solving (3.56) with

$$\begin{cases} T_{BGS} = (D - U)^{-1}L, \\ \mathbf{c} = (D - U)^{-1}\mathbf{b}, \end{cases} \quad (3.74)$$

where D , $-L$, and $-U$ are the same as in Lemma 3.91.

Proof. This follows directly from (3.73). $\square$

Theorem 3.101. For a linear system $A\mathbf{x} = \mathbf{b}$ with A being strictly diagonally dominant or irreducibly diagonally dominant, both Jacobi and Gauss–Seidel iterations converge.

Proof. By the diagonal dominance, the diagonal matrix D_A of A is nonsingular. Then we have

$$\forall |\lambda| \geq 1, \det(\lambda I - T_J) = \det(D^{-1}) \det(\lambda D - L - U) \neq 0,$$

because $|\lambda| \geq 1$, the diagonal dominance of $A = D - L - U$, and Theorem 3.49 imply $\det(\lambda D - L - U) \neq 0$. Similarly

$$\forall |\lambda| \geq 1, \det(\lambda I - T_{GS}) = \det((D - L)^{-1}) \det(\lambda D - \lambda L - U) \neq 0.$$

Therefore, we have $\rho(T_J) < 1$ and $\rho(T_{GS}) < 1$. $\square$

Theorem 3.102. For a linear system $A\mathbf{x} = \mathbf{b}$ with A being SPD, the Gauss–Seidel iteration converges.

Proof. Split $A = M - N$ with $M = D - L$ and $N = U$ and we have $A + N + N^\top = D$ from the symmetry of A . By Lemma 3.53, the diagonal matrix D is also SPD. Then Theorem 3.81 implies $\rho(T_{GS}) < 1$. $\square$

3.4.6 Successive over relaxation (SOR)

Definition 3.103. The *SOR iteration* determines the next iterate $\mathbf{x}^{(k+1)}$ by annihilating components of the residual in the order $i = 1, 2, \dots, n$ using a linear combination of the Gauss–Seidel iterate and the current one:

$$x_i^{(k+1)} = (1 - \omega)x_i^{(k)} + \omega x_i^{GS}, \quad (3.75)$$

where x_i^{GS} equals $x_i^{(k+1)}$ in (3.71).

Lemma 3.104. The SOR is a fixed-point iteration for solving (3.56) with

$$\begin{cases} T_{SOR} = (D - \omega L)^{-1}[\omega U + (1 - \omega)D], \\ \mathbf{c} = \omega(D - \omega L)^{-1}\mathbf{b}, \end{cases} \quad (3.76)$$

where D , $-L$, and $-U$ are the same as in Lemma 3.91.

Proof. This follows from (3.75) and (3.71). $\square$

Theorem 3.105. For a linear system $A\mathbf{x} = \mathbf{b}$ with A being strictly diagonally dominant or irreducibly diagonally dominant, the SOR iteration converges for any $\omega \in (0, 1)$.

Exercise 3.106. Prove Theorem 3.105.

Corollary 3.107. Convergence of the SOR iteration implies $\omega \in (0, 2)$.

Proof. Lemma 3.104 states that

$$T_{SOR} = (I - \omega D^{-1}L)^{-1}[(1 - \omega)I + \omega D^{-1}U].$$

Since $\det(I - \omega D^{-1}L) = 1$, we have

$$|\det(T_{SOR})| = |1 - \omega|^n = |\lambda_1 \lambda_2 \cdots \lambda_n|.$$

The convergence and Theorem 3.79 imply $\rho(T_{SOR}) < 1$. Then the above equation yields $\omega \in (0, 2)$. $\square$

Theorem 3.108. For a linear system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{n \times n}$ being SPD, the SOR iteration converges for any $\omega \in (0, 2)$.

Proof. For any eigenpair $(\lambda, \mathbf{u})$ of T_{SOR} in (3.76), we have

$$[(1 - \omega)D + \omega L^\top] \mathbf{u} = \lambda(D - \omega L) \mathbf{u},$$

where we have applied the symmetry of A . Then we have

$$\begin{aligned} \delta &:= \mathbf{u}^* D \mathbf{u}, \quad \alpha + \mathbf{i}\beta := \mathbf{u}^* L \mathbf{u} \\ \implies (1 - \omega)\delta + \omega(\alpha - \mathbf{i}\beta) &= \lambda[\delta - \omega(\alpha + \mathbf{i}\beta)] \\ \implies |\lambda|^2 &= \frac{[(1 - \omega)\delta + \omega\alpha]^2 + \omega^2\beta^2}{(\delta - \omega\alpha)^2 + \omega^2\beta^2}. \end{aligned}$$

The positive-definiteness of A , the splitting $A = D - L - U$, and the symmetry $L^\top = U$ imply $\delta - 2\alpha > 0$ while Lemma 3.53 yields $\delta > 0$. Then $\omega \in (0, 2)$ implies

$$\begin{aligned} &[(1 - \omega)\delta + \omega\alpha]^2 + \omega^2\beta^2 - (\delta - \omega\alpha)^2 - \omega^2\beta^2 \\ &= \omega\delta(\delta - 2\alpha)(\omega - 2) < 0, \end{aligned}$$

which yields $|\lambda| < 1$. $\square$

Definition 3.109. The *backward SOR* determines the next iterate $\mathbf{x}^{(k+1)}$ by annihilating components of the residual in the reversed order $i = n, n-1, \dots, 1$ using a linear combination of the backward Gauss–Seidel iterate and the current one:

$$x_i^{(k+1)} = (1 - \omega)x_i^{(k)} + \omega x_i^{BGS}, \quad (3.77)$$

where x_i^{BGS} equals $x_i^{(k+1)}$ in (3.73).

Lemma 3.110. The backward SOR is a fixed-point iteration for solving (3.56) with

$$\begin{cases} T_{BSOR} = (D - \omega U)^{-1}[\omega L + (1 - \omega)D], \\ \mathbf{c} = \omega(D - \omega U)^{-1}\mathbf{b}, \end{cases} \quad (3.78)$$

where D , $-L$, and $-U$ are the same as in Lemma 3.91.

Proof. This follows from (3.77) and (3.73). $\square$

Definition 3.111. Each step of the *symmetric SOR* (SSOR) consists of the SOR step followed by a backward SOR step,

$$(D - \omega L)\mathbf{x}_* = [(1 - \omega)D + \omega U]\mathbf{x}^{(k)} + \omega\mathbf{b}, \quad (3.79a)$$

$$(D - \omega U)\mathbf{x}^{(k+1)} = [(1 - \omega)D + \omega L]\mathbf{x}_* + \omega\mathbf{b}. \quad (3.79b)$$

Lemma 3.112. The SSOR is a fixed-point iteration for solving (3.56) with

$$\begin{aligned} T_{SSOR} &= \begin{pmatrix} (D - \omega U)^{-1}[(1 - \omega)D + \omega L] \\ (D - \omega L)^{-1}[(1 - \omega)D + \omega U] \end{pmatrix}, \\ \mathbf{c} &= \omega(2 - \omega)(D - \omega U)^{-1}D(D - \omega L)^{-1}\mathbf{b}, \end{aligned} \quad (3.80)$$

where D , $-L$, and $-U$ are the same as in Lemma 3.91.

Exercise 3.113. Prove Lemma 3.112.

Summary 3.114. Due to their complexity $O(n^3)$, direct methods are best for small general linear systems (typically $< 10^5$ unknowns). They are highly robust for well-posed problems and they yield highly accurate solutions without parameter tuning. In contrast, iterative methods are best for very large sparse systems that feature prominent structures such as SPD and diagonal dominance. They require low memory footprint because the matrix is never modified and even not assembled; usually the matrix is represented as a subroutine that operates on a long vector. When high accuracy of the solution is unnecessary, the iteration can be stopped after the approximate solution becomes good enough, which promotes efficiency.

3.5 Theoretical problems

- I. Show that, suppose a triangular system $T\mathbf{x} = \mathbf{b}$ of order n , where T is M-matrix and $\mathbf{b} > \mathbf{0}$. The computed solution $\hat{\mathbf{x}}$ obtained from substitution satisfies

$$|\mathbf{x} - \hat{\mathbf{x}}| \leq (n^2 + n + 1)\epsilon_u |\mathbf{x}| + O(\epsilon_u^2).$$

- II. Show that, according to Definition 3.88, the preconditioners of the Jacobi, Gauss-Seidel, SOR, and SSOR iterations are, respectively,

$$\begin{aligned} M_J &= D, \\ M_{GS} &= D - L, \\ M_{SOR} &= \frac{1}{\omega}(D - \omega L), \\ M_{SSOR} &= \frac{1}{\omega(2 - \omega)}(D - \omega L)D^{-1}(D - \omega U). \end{aligned}$$

- III. Show that for the Gauss-Seidel iteration in Lemma 3.98, $M_{GS} = D - L$ and $N = U$ form a regular splitting of the matrix B in (3.55) and M_{GS} is an approximate inverse of B . Therefore the Gauss-Seidel iteration converges by the same arguments in Example 3.93.

3.6 Programming assignments

Chapter 4

Solving Nonlinear Equations

Motivation 4.1. Nonlinear equations and systems of nonlinear equations arise ubiquitously in scientific computing, engineering design, economic forecasting, and many other fields. The vast majority of practical problems do not admit closed-form analytic solutions and must be approximated by numerical methods. This chapter illustrates the prevalence and importance of nonlinear root-finding problems through several concrete examples, and then provides rigorous mathematical definitions that lay the foundation for the subsequent study of the numerical methods.

Example 4.2. The motion of a planet around the sun is described by Kepler's equation

$$E - e \sin E = M,$$

where E is the eccentric anomaly, e the orbital eccentricity, and M the mean anomaly. For given e and M , solving for E is a transcendental equation that cannot be inverted in terms of elementary functions; numerical iterative methods (e.g., Newton's method) are indispensable.

Example 4.3. A two-link planar robot arm has end-effector coordinates (x, y) related to joint angles θ_1, θ_2 and given link lengths l_1, l_2 by

$$\begin{cases} x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2), \\ y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2). \end{cases}$$

Given a target position (x, y) , finding the joint angles constitutes a two-dimensional nonlinear system. Solutions may be non-unique and must respect joint limits, posing challenges to the robustness of numerical methods.

Example 4.4. In an economy with n commodities, the equilibrium price vector $\mathbf{p} = [p_1, \dots, p_n]^\top$ satisfies the zero-excess-demand conditions:

$$\forall i = 1, \dots, n, \quad Z_i(\mathbf{p}) = D_i(\mathbf{p}) - S_i(\mathbf{p}) = 0,$$

where D_i, S_i are often nonlinear functions (e.g., derived from CES utility functions). Solving large-scale nonlinear systems of this kind is a central task in computational economics and computable general equilibrium modeling.

Roadmap 4.5. This chapter is organized in a progression from scalar to multivariate problems and from robust to efficient methods. We begin with the *bisection method* to establish rigorous algorithm specifications, loop invariants, primary versus temporary variables, and techniques for proving correctness and simplifying algorithms; see Sections 4.1 and 4.2. Section 4.3 then formalizes the notion of *Q-order convergence* and establishes the linear convergence of the bisection method. Sections 4.4 and 4.5 present *Newton's method* and its derivative-free variant, the *secant method*. Both achieve fast (quadratic/superlinear) convergence but are sensitive to the initial guess; special attention is paid to the trade-off between convergence rate and computational cost. A unified framework of *fixed-point iteration* is developed in Section 4.6, revealing through contraction mapping theory the common structure shared by many iterative methods. Finally, Section 4.7 extends Newton's method and fixed-point iterations to *systems of nonlinear equations* and briefly introduces *Quasi-Newton methods*.

4.1 The bisection method

Definition 4.6. A *scalar nonlinear equation* is an equation of the form

$$f(x) = 0, \tag{4.1}$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a nonlinear function. The goal of numerical solution is to find $\alpha \in \mathbb{R}$, called a *root* or *solution*, such that $f(\alpha) = 0$ holds approximately within a prescribed tolerance.

Input: $f : [a, b] \rightarrow \mathbb{R}$, $a \in \mathbb{R}$, $b \in \mathbb{R}$,
 $M \in \mathbb{N}$, $\delta \in \mathbb{R}^+$, $\epsilon \in \mathbb{R}^+$
Preconditions : $f \in \mathcal{C}[a, b]$,
 $\text{sgn}(f(a)) \neq \text{sgn}(f(b))$
Output: c, h, k
Postconditions: $|f(c)| < \epsilon$ or $|h| < \delta$ or $k = M$

```

1  $u \leftarrow f(a)$ 
2  $v \leftarrow f(b)$ 
3 for  $k = 0 : M$  do
4    $h \leftarrow b - a$ 
5    $c \leftarrow a + h/2$ 
6   if  $|h| < \delta$  or  $k = M$  then break
7    $w \leftarrow f(c)$ 
8   if  $|w| < \epsilon$  then
9     break
10  else if  $\text{sgn}(w) \neq \text{sgn}(u)$  then
11     $b \leftarrow c$ 
12     $v \leftarrow w$ 
13  else
14     $a \leftarrow c$ 
15     $u \leftarrow w$ 
16  end
17 end

```

Figure 4.1: An algorithm of the bisection method.

Algorithm 4.7. For the root-finding problem of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$, the *bisection method* repeatedly reduce by half the given interval where the root must lie and returns the midpoint of the last interval as the solution; see Figure 4.1.

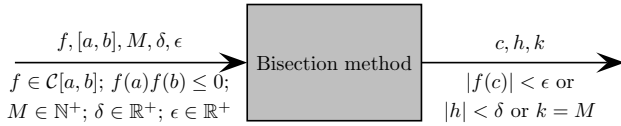


Figure 4.2: The signature of the bisection algorithm.

4.2 Proof of correctness and simplification of algorithms

Definition 4.8. An *invariant* is a condition that holds during the execution of an algorithm.

Definition 4.9. A variable is *temporary or derived* for a loop if it is initialized inside the loop. A variable is *persistent or primary* for a loop if it is initialized before the loop and its value changes across different iterations.

Exercise 4.10. What are the invariants in Algorithm 4.7? Which quantities do a, b, c, h, u, v, w represent? Which of them are primary? Which of these variables are temporary? Draw pictures to illustrate the life spans of these variables.

Algorithm 4.11. A simplified bisection algorithm.

Input: $f : [a, b] \rightarrow \mathbb{R}$, $a \in \mathbb{R}$, $b \in \mathbb{R}$,
 $M \in \mathbb{N}$, $\delta \in \mathbb{R}^+$, $\epsilon \in \mathbb{R}^+$
Preconditions : $f \in \mathcal{C}[a, b]$,
 $\text{sgn}(f(a)) \neq \text{sgn}(f(b))$
Output: c, h, k
Postconditions: $|f(c)| < \epsilon$ or $|h| < \delta$ or $k = M$

```

1  $h \leftarrow b - a$ 
2  $u \leftarrow f(a)$ 
3 for  $k = 0 : M$  do
4    $h \leftarrow h/2$ 
5    $c \leftarrow a + h$ 
6   if  $|h| < \delta$  or  $k = M$  then break
7    $w \leftarrow f(c)$ 
8   if  $|w| < \epsilon$  then
9     break
10  else if  $\text{sgn}(w) = \text{sgn}(u)$  then
11     $a \leftarrow c$ 
12  end
13 end

```

4.3 Q-order convergence

Definition 4.12 (Q-order convergence). A convergent sequence $\{x_n\}$ is said to *converge* to L with *Q-order* p ($p \geq 1$) if the limit $\lim_{n \rightarrow \infty} \frac{|x_{n+1} - L|}{|x_n - L|^p}$ exists as $n \rightarrow \infty$ and

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - L|}{|x_n - L|^p} = c > 0; \quad (4.2)$$

the constant c is called the *asymptotic factor*. In particular, $\{x_n\}$ has *Q-linear convergence* if $p = 1$ and *Q-quadratic convergence* if $p = 2$.

Example 4.13. Consider finding a root of $f(x) = \sin(x) = 0$ near $x \approx 3$. If the asymptotic factor $c = \frac{1}{10}$, then each iteration of a linearly convergent method will yield one more correct digit. If $c = \frac{1}{2}$, then each iteration of a linearly convergent method will yield one more correct bit. For a quadratically convergent method, each iteration will roughly double the number of correct digits/bits.

Definition 4.14. A sequence of iterates $\{x_n\}$ *converges linearly* to L if

$$\exists c \in (0, 1), \exists d > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, |x_n - L| \leq c^n d. \quad (4.3)$$

For a sequence $\{x_n\}$ that converges to L , its *order of convergence* is the maximum $p \in \mathbb{R}^+$ satisfying

$$\exists c > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, |x_{n+1} - L| \leq c|x_n - L|^p. \quad (4.4)$$

In particular, $\{x_n\}$ *converges quadratically* if $p = 2$.

Example 4.15. (4.3) also guarantees convergence while (4.4) does not: the sequence $\{1, -1, 1, -1, \dots\}$ satisfies (4.4) for $L = 0$, $p = 1$, and $c = 1$, but does not converge.

Theorem 4.16 (Monotonic sequence theorem). Every bounded monotonic sequence is convergent.

Theorem 4.17 (Convergence of bisection). For a continuous function $f : [a_0, b_0] \rightarrow \mathbb{R}$ satisfying $\text{sgn}(f(a_0)) \neq$

$\text{sgn}(f(b_0))$, the sequence of iterates in the bisection method converges linearly with asymptotic factor $\frac{1}{2}$,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} c_n = \alpha, \quad (4.5)$$

$$f(\alpha) = 0, \quad (4.6)$$

$$|c_n - \alpha| \leq 2^{-(n+1)}(b_0 - a_0), \quad (4.7)$$

where $[a_n, b_n]$ is the interval in the n th iteration and $c_n = \frac{1}{2}(a_n + b_n)$ is the solution returned by the bisection method.

Proof. It follows from the bisection method that

$$a_0 \leq a_1 \leq a_2 \leq \dots \leq b_0,$$

$$b_0 \geq b_1 \geq b_2 \geq \dots \geq a_0,$$

$$b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n).$$

In the rest of this proof, “lim” is a shorthand for “ $\lim_{n \rightarrow \infty}$.” By Theorem 4.16, both $\{a_n\}$ and $\{b_n\}$ converge. Also, $\lim(b_n - a_n) = \lim \frac{1}{2^n}(b_0 - a_0) = 0$, hence $\lim b_n = \lim a_n = \alpha$. By the given condition and the algorithm, the invariant $f(a_n)f(b_n) \leq 0$ always holds. Since f is continuous, $\lim f(a_n)f(b_n) = f(\lim a_n)f(\lim b_n)$, then $f^2(\alpha) \leq 0$ implies $f(\alpha) = 0$. (4.7) is another important invariant that can be proven by induction. Comparing (4.7) to (4.3) yields convergence of the bisection method. Also, the convergence is linear with asymptotic factor as $c = \frac{1}{2}$. $\square$

4.4 Newton's method

Input: $f : \mathbb{R} \rightarrow \mathbb{R}$, f' , $x_0 \in \mathbb{R}$, $M \in \mathbb{N}^+$, $\epsilon \in \mathbb{R}^+$

Preconditions : $f \in C^2$ and x_0 is sufficiently close to a root of f

Output: x, k

Postconditions: $|f(x)| < \epsilon$ or $k = M$

```

1  $x \leftarrow x_0$ 
2 for  $k = 0 : M$  do
3    $u \leftarrow f(x)$ 
4   if  $|u| < \epsilon$  then break
5    $x \leftarrow x - u/f'(x)$ 
6 end
```

Figure 4.3: An algorithm of Newton's method.

Algorithm 4.18. As shown in Figures 4.3 and 4.4, *Newton's method* finds the root of $f : \mathbb{R} \rightarrow \mathbb{R}$ near an initial guess x_0 by the iteration formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n \in \mathbb{N}. \quad (4.8)$$

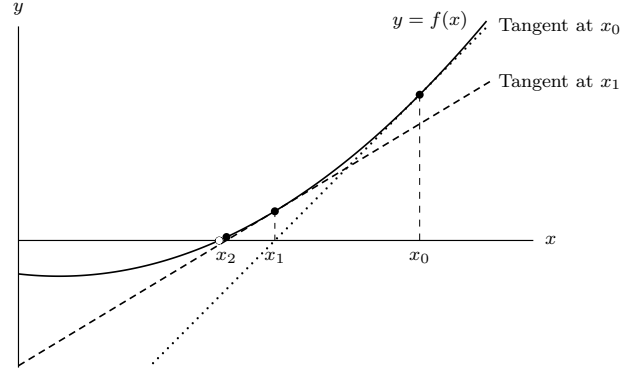


Figure 4.4: The geometric meaning of Newton's method.

Theorem 4.19 (Convergence of Newton's method). For a C^2 function $f : \mathcal{B} \rightarrow \mathbb{R}$ on $\mathcal{B} := [\alpha - \delta, \alpha + \delta]$ satisfying $f(\alpha) = 0$ and $f'(\alpha) \neq 0$, we have

$$\exists \delta_1 \in (0, \delta) \text{ s.t. } \forall x \in \mathcal{B}_1 := [\alpha - \delta_1, \alpha + \delta_1], f'(x) \neq 0.$$

Define $M := \frac{\max_{x \in \mathcal{B}_1} |f''(x)|}{2 \min_{x \in \mathcal{B}_1} |f'(x)|}$ and pick x_0 sufficiently close to α such that (i) $|x_0 - \alpha| = \delta_0 < \delta_1$ and (ii) $M\delta_0 < 1$ hold. Then the sequence of iterates $\{x_n\}$ in the Newton's method converges at least quadratically to the root α , i.e.

$$\lim_{n \rightarrow \infty} \frac{\alpha - x_{n+1}}{(\alpha - x_n)^2} = -\frac{f''(\alpha)}{2f'(\alpha)}. \quad (4.9)$$

Proof. Taylor's theorem (Theorem B.100) and the assumption $f \in C^2$ yield

$$f(\alpha) = f(x_n) + (\alpha - x_n)f'(x_n) + \frac{(\alpha - x_n)^2}{2}f''(\xi)$$

where ξ is between α and x_n . Then $f(\alpha) = 0$ gives

$$-\alpha = -x_n + \frac{f(x_n)}{f'(x_n)} + \frac{(\alpha - x_n)^2}{2} \frac{f''(\xi)}{f'(x_n)}.$$

By (4.8), we have

$$(*) : x_{n+1} - \alpha = x_n - \frac{f(x_n)}{f'(x_n)} - \alpha = (x_n - \alpha)^2 \frac{f''(\xi)}{2f'(x_n)}.$$

The definition of M and $(*)$ give $|x_{n+1} - \alpha| \leq M|x_n - \alpha|^2$. Comparing the above to (4.4) implies that if $\{x_n\}$ converges, then the order of convergence is 2. We must still show that (a) it converges and (b) it converges to α .

By (i) and (ii), we have $M|x_0 - \alpha| < 1$. Then an easy induction yields

$$|x_n - \alpha| \leq \frac{1}{M} (M|x_0 - \alpha|)^{2^n},$$

which shows both (a) and (b). Finally, the convergence rate could be higher than 2 in the case of $f''(\alpha) = 0$. $\square$

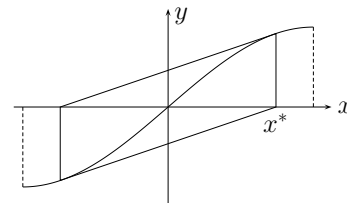


Figure 4.5: A pathological case for Newton's method that illustrates the necessity of condition (ii) in Theorem 4.19.

Theorem 4.20. A continuous function $f : [a, b] \rightarrow [c, d]$ is bijective if and only if it is strictly monotonic.

Theorem 4.21. If a $\mathcal{C}^2$ function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(\alpha) = 0$, $f' > 0$ and $f'' > 0$, then α is the only root of f and, $\forall x_0 \in \mathbb{R}$, the sequence of iterates $\{x_n\}$ in the Newton's method converges quadratically to α .

Proof. By Theorem 4.20, f is a bijection since f is continuous and strictly monotonic. With 0 in its range, f must have a unique root. When proving Theorem 4.19, we had

$$x_{n+1} - \alpha = (x_n - \alpha)^2 \frac{f''(\xi)}{2f'(x_n)}. \quad (4.10)$$

Then $f' > 0$ and $f'' > 0$ further imply $x_n > \alpha$ for all $n > 0$. f being strictly increasing implies that $f(x_n) > f(\alpha) = 0$ for all $n > 0$. By the definition of Newton's method, $x_{n+1} - \alpha = x_n - \alpha - \frac{f(x_n)}{f'(x_n)}$, hence the sequence $\{x_n - \alpha : n > 0\}$ is strictly monotonically decreasing with 0 as a lower bound. By Theorem 4.16 it converges.

Suppose $\lim_{n \rightarrow \infty} x_n = a$. Then take the limit of $n \rightarrow \infty$ on both sides of (4.8) and we have

$$a = a - \frac{f(a)}{f'(a)},$$

which implies $f(a) = 0$. By the uniqueness of the root of f , we have $a = \alpha$.

The quadratic convergence rate can be proved by an induction using (4.10), as in Theorem 4.19. $\square$

Definition 4.22. A subset $\mathcal{U}$ of a vector space $\mathcal{V}$ is a *convex set* iff

$$\forall x, y \in \mathcal{U}, \forall t \in (0, 1), \quad tx + (1 - t)y \in \mathcal{U}. \quad (4.11)$$

The *convex hull* of a subset $W \subset \mathcal{V}$ is the smallest convex set that contains W .

Definition 4.23. For a subset $\mathcal{U}$ of a real vector space, a function $f : \mathcal{U} \rightarrow \mathbb{R}$ is *convex* iff

$$\begin{aligned} &\forall t \in (0, 1), \forall x, y \in \mathcal{U}, \\ &x \neq y \implies f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y). \end{aligned} \quad (4.12)$$

In particular, f is *strictly convex* if we replace “ $\leq$ ” with “ $<$ ” in (4.12).

4.5 The secant method

Algorithm 4.24. As shown in Figure 4.6, the *secant method* finds a root of $f : \mathbb{R} \rightarrow \mathbb{R}$ near initial guesses x_0, x_1 by the iteration

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}, \quad n \in \mathbb{N}^+. \quad (4.13)$$

Input: $f : \mathbb{R} \rightarrow \mathbb{R}$, $x_0 \in \mathbb{R}$, $x_1 \in \mathbb{R}$,
 $M \in \mathbb{N}^+$, $\delta \in \mathbb{R}^+$, $\epsilon \in \mathbb{R}^+$
Preconditions : $f \in \mathcal{C}^2$; x_0, x_1 are sufficiently
close to a root of f
Output: x_n, x_{n-1}, k
Postconditions: $|f(x_n)| < \epsilon$ or $|x_n - x_{n-1}| < \delta$
or $k = M$

```

1  $x_n \leftarrow x_1$ 
2  $x_{n-1} \leftarrow x_0$ 
3  $u \leftarrow f(x_n)$ 
4  $v \leftarrow f(x_{n-1})$ 
5 for  $k = 2 : M$  do
6    $s \leftarrow \frac{x_n - x_{n-1}}{u - v}$ 
7    $x_{n-1} \leftarrow x_n$ 
8    $v \leftarrow u$ 
9    $x_n \leftarrow x_n - u \times s$ 
10  if  $|x_n - x_{n-1}| < \delta$  then break
11   $u \leftarrow f(x_n)$ 
12  if  $|u| < \epsilon$  then break
13 end
```

Figure 4.6: An algorithm of the secant method.

Definition 4.25. The sequence $\{F_n\}$ of *Fibonacci numbers* is defined as

$$F_0 = 0, F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}. \quad (4.14)$$

Theorem 4.26 (Binet's formula). Denote the golden ratio by $r_0 = \frac{1+\sqrt{5}}{2}$ and let $r_1 = 1 - r_0 = \frac{1-\sqrt{5}}{2}$, then

$$F_n = \frac{r_0^n - r_1^n}{\sqrt{5}}. \quad (4.15)$$

Proof. By Definition 4.25, we have

$$\mathbf{u}_k := \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = A \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix}, \quad \text{where } A := \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Hence we have $\mathbf{u}_n = A^n \mathbf{u}_0$. It follows from

$$\det(A - \lambda I) = \lambda^2 - \lambda - 1 = 0$$

that the two eigenvalues of A are $\lambda = r_0, r_1$, with their eigenvectors as $\mathbf{x}_0 = (r_0, 1)^\top$ and $\mathbf{x}_1 = (r_1, 1)^\top$, respectively. Indeed, the two eigenpairs stem from $\lambda^2 = \lambda + 1$, a nice relation between multiplication and addition by 1. Finally, we express $\mathbf{u}_0$ as a linear combination of $\mathbf{x}_0$ and $\mathbf{x}_1$,

$$\mathbf{u}_0 = \frac{1}{r_0 - r_1} (\mathbf{x}_0 - \mathbf{x}_1),$$

which, together with $\mathbf{u}_n = A^n \mathbf{u}_0$, yields

$$\mathbf{u}_n = \frac{1}{r_0 - r_1} (r_0^n \mathbf{x}_0 - r_1^n \mathbf{x}_1),$$

the second equation of which yields (4.15). $\square$

Corollary 4.27. The ratios r_0, r_1 in Theorem 4.26 satisfy

$$F_{n+1} = r_0 F_n + r_1^n. \quad (4.16)$$

Proof. This follows from (4.15) and values of r_0 and r_1 . $\square$

Lemma 4.28 (Error relation of the secant method). For the secant method (4.13), there exist ξ_n between x_{n-1} and x_n and ζ_n between $\min(x_{n-1}, x_n, \alpha)$ and $\max(x_{n-1}, x_n, \alpha)$ such that

$$x_{n+1} - \alpha = (x_n - \alpha)(x_{n-1} - \alpha) \frac{f''(\zeta_n)}{2f'(\xi_n)}. \quad (4.17)$$

Proof. Define a divided difference as

$$f[a, b] = \frac{f(a) - f(b)}{a - b}. \quad (4.18)$$

Then it takes some algebra to show that the formula (4.13) is equivalent to

$$x_{n+1} - \alpha = (x_n - \alpha)(x_{n-1} - \alpha) \frac{f[x_{n-1}, x_n] - f[x_n, \alpha]}{f[x_{n-1}, x_n]}. \quad (4.19)$$

By (4.18) and the mean value theorem (Theorem B.92), there exists ξ_n between x_{n-1} and x_n such that

$$f[x_{n-1}, x_n] = f'(\xi_n). \quad (4.20)$$

Define a function $g(x) := f[x, x_n]$, apply the mean value theorem to $g(x)$, and we have

$$\frac{f[x_{n-1}, x_n] - f[x_n, \alpha]}{x_{n-1} - \alpha} = g'(\beta) \quad (4.21)$$

for some β between x_{n-1} and α . Compute the derivative of $g'(\beta)$ from (4.18), use the Lagrangian remainder Theorem B.100, and we have

$$\frac{f[x_{n-1}, x_n] - f[x_n, \alpha]}{x_{n-1} - \alpha} = \frac{f''(\zeta_n)}{2} \quad (4.22)$$

for some ζ_n between $\min(x_{n-1}, x_n, \alpha)$ and $\max(x_{n-1}, x_n, \alpha)$. The proof is completed by substituting (4.20) and (4.22) into (4.19). $\square$

Theorem 4.29 (Convergence of the secant method). Consider a $\mathcal{C}^2$ function $f : \mathcal{B} \rightarrow \mathbb{R}$ on $\mathcal{B} = [\alpha - \delta, \alpha + \delta]$ satisfying $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. If both x_0 and x_1 are chosen sufficiently close to α and $f''(\alpha) \neq 0$, then the iterates $\{x_n\}$ in the secant method converges to the root α with order $p = \frac{1}{2}(1 + \sqrt{5}) \approx 1.618$.

Proof. The continuity of f' and the assumption $f'(\alpha) \neq 0$ yield

$$\exists \delta_1 \in (0, \delta) \text{ s.t. } \forall x \in \mathcal{B}_1, f'(x) \neq 0$$

where $\mathcal{B}_1 = [\alpha - \delta_1, \alpha + \delta_1]$. Define $E_i = |x_i - \alpha|$,

$$M = \frac{\max_{x \in \mathcal{B}_1} |f''(x)|}{2 \min_{x \in \mathcal{B}_1} |f'(x)|},$$

and we have from Lemma 4.28

$$ME_{n+1} \leq ME_n ME_{n-1}.$$

Pick x_0, x_1 such that

$$(i) \ E_0 < \delta, E_1 < \delta;$$

$$(ii) \ \max(ME_1, ME_0) = \eta < 1,$$

then an induction by the above equation shows that $E_n < \delta$, $ME_n < \eta$. To prove convergence, we write $ME_0 \leq \eta$, $ME_1 \leq \eta$, $ME_2 \leq ME_1 ME_0 \leq \eta^2$, $ME_3 \leq ME_2 ME_1 \leq \eta^3$, $\dots$, $ME_{n+1} \leq ME_n ME_{n-1} \leq \eta^{q_n + q_{n-1}} = \eta^{q_{n+1}}$, i.e.

$$E_n \leq B_n := \frac{1}{M} \eta^{q_n}.$$

$\{q_n\}$ is a Fibonacci sequence starting from $q_0 = 1, q_1 = 1$. By Theorem 4.26, as $n \rightarrow \infty$ we have $q_n \rightarrow \frac{1.618^{n+1}}{\sqrt{5}}$ since $|r_1| \approx 0.618 < 1$. Hence $\lim_{n \rightarrow \infty} E_n = 0$.

To guesstimate the convergence rate, we first examine the rate at which the upper bounds $\{B_n\}$ decrease:

$$\frac{B_{n+1}}{B_n^{r_0}} = \frac{\frac{1}{M} \eta^{q_{n+1}}}{\left(\frac{1}{M}\right)^{r_0} \eta^{r_0 q_n}} = M^{r_0-1} \eta^{q_{n+1}-r_0 q_n} \leq M^{r_0-1} \eta^{-1}$$

where $q_{n+1} - r_0 q_n = r_1^{n+1} > -1$.

To prove the convergence rate, we define

$$m_n := \left| \frac{f''(\zeta_n)}{2f'(\xi_n)} \right|, \quad m_\alpha := \left| \frac{f''(\alpha)}{2f'(\alpha)} \right|, \quad (4.23)$$

where ζ_n and ξ_n are the same as those in Lemma 4.28. By induction, we have

$$E_n = E_1^{F_n} E_0^{F_{n-1}} m_1^{F_{n-1}} \dots m_{n-1}^{F_1},$$

$$E_{n+1} = E_1^{F_{n+1}} E_0^{F_n} m_1^{F_n} \dots m_n^{F_2} m_n^{F_1},$$

where F_n is a Fibonacci number as in Definition 4.25. Then

$$\begin{aligned} \frac{E_{n+1}}{E_n^{r_0}} &= E_1^{F_{n+1}-r_0 F_n} E_0^{F_n-r_0 F_{n-1}} m_1^{F_n-r_0 F_{n-1}} m_2^{F_{n-1}-r_0 F_{n-2}} \\ &\quad \dots m_{n-2}^{F_3-r_0 F_2} m_{n-1}^{F_2-r_0 F_1} m_n^{F_1} \\ &= E_1^{r_1^n} E_0^{r_1^{n-1}} m_1^{r_1^{n-1}} m_2^{r_1^{n-2}} \dots m_{n-1}^{r_1^1} m_n^1, \end{aligned} \quad (4.24)$$

where the second step follows from Corollary 4.27. (4.23) and the convergence we just proved yield

$$\lim_{n \rightarrow +\infty} m_n = m_\alpha, \quad (4.25)$$

which means

$$\exists N \in \mathbb{N} \text{ s.t. } \forall n > N, m_n \in \left(\frac{1}{2} m_\alpha, 2m_\alpha \right). \quad (4.26)$$

We define

$$A := E_1^{r_1^n} \cdot E_0^{r_1^{n-1}} m_1^{r_1^{n-1}} \cdot m_2^{r_1^{n-2}} \dots m_{n-1}^{r_1^{n-N}},$$

$$B := m_{N+1}^{r_1^{n-N-1}} \cdot m_{N+2}^{r_1^{n-N-2}} \dots m_{n-1}^{r_1^1} \cdot m_n^1$$

so that $\frac{E_{n+1}}{E_n^{r_0}} = AB$. Since $|r_1| < 1$, we have $\lim_{n \rightarrow \infty} A = 1$. As for B , we have from (4.26)

$$\begin{aligned} B &\leq (2m_\alpha)^1 \left(\frac{1}{2} m_\alpha \right)^{r_1} \dots m_{N+1}^{r_1^{n-N-1}} \\ &\leq 2^{1-r_1+r_1^2-\dots+(-r_1)^{n-N-1}} \cdot m_\alpha^{1+r_1+r_1^2+\dots+r_1^{n-N-1}} \\ &\leq 2^{\sum_{i=0}^{n-N-1} (-r_1)^i} \cdot m_\alpha^{\sum_{i=0}^{n-N-1} r_1^i}, \end{aligned}$$

where the discrimination of the even-numbered factors in the first line comes from the fact of $r_1 < 0$. It follows that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{E_{n+1}}{E_n^{r_0}} &= \lim_{n \rightarrow \infty} A \lim_{n \rightarrow \infty} B \\ &= \lim_{n \rightarrow \infty} B \leq 2^{\frac{1}{1+r_1}} m_\alpha^{\frac{1}{r_0}}. \end{aligned}$$

The proof is then completed by Definition 4.12. $\square$

Corollary 4.30. Consider solving $f(x) = 0$ near a root α . Let m and sm be the time to evaluate $f(x)$ and $f'(x)$ respectively. The minimum time to obtain the desired absolute accuracy ϵ with Newton's method and the secant method are respectively

$$T_N = (1+s)m \lceil \log_2 K \rceil, \quad (4.27)$$

$$T_S = m \lceil \log_{r_0} K \rceil + m, \quad (4.28)$$

where $r_0 = \frac{1+\sqrt{5}}{2}$, $c = \left| \frac{f''(\alpha)}{2f'(\alpha)} \right|$,

$$K = \frac{\log c\epsilon}{\log c|x_0 - \alpha|}, \quad (4.29)$$

and $\lceil \cdot \rceil$ denotes the *ceiling function* or the *rounding-up operator* that rounds towards $+\infty$.

Proof. We showed $|x_n - \alpha| \leq \frac{1}{M} (M|x_0 - \alpha|)^{2^n}$ in proving Theorem 4.19. Denote $E_n = |x_n - \alpha|$, we have

$$ME_n \leq (ME_0)^{2^n}.$$

Let $i \in \mathbb{N}^+$ denote the smallest number of iterations such that the desired accuracy ϵ is satisfied, i.e. $(ME_0)^{2^i} \leq M\epsilon$. When ϵ is sufficiently small, $M \rightarrow c$. Hence we have

$$i = \lceil \log_2 K \rceil.$$

For each iteration, Newton's method incurs one function evaluation and one derivative evaluation, which cost time m and sm , respectively. Therefore (4.27) holds.

For the secant method, assume $ME_0 \geq ME_1$. By the proof of Theorem 4.29, we have

$$ME_n \approx (ME_0)^{\frac{\sqrt{5}}{5} r_0^{n+1}},$$

where we have ignored the term $-\frac{\sqrt{5}}{5} r_1^{n+1}$ in the exponent. Let $j \in \mathbb{N}^+$ be the smallest number of iterations such that the desired accuracy ϵ is satisfied, i.e. $r_0^j \leq \frac{\sqrt{5}}{r_0} K$. Hence

$$j = \left\lceil \log_{r_0} K + \log_{r_0} \frac{\sqrt{5}}{r_0} \right\rceil \leq \lceil \log_{r_0} K \rceil + 1.$$

Since the first two values x_0 and x_1 are given in the secant method, the least number of iterations is $\lceil \log_{r_0} K \rceil$ (compare to Newton's method!). In addition, only the function value $f(x_n)$ needs to be evaluated per iteration because $f(x_{n-1})$ has already been evaluated in the previous iteration. Finally, the extra m is due to the first step where the function value $f(x_0)$ has to be evaluated. $\square$

4.6 Fixed-point iterations

Definition 4.31. A *fixed point* of a function g is an independent parameter of g satisfying $g(\alpha) = \alpha$.

Example 4.32. A fixed point of $f(x) = x^2 - 3x + 4$ is $x = 2$.

Lemma 4.33. If $g : [a, b] \rightarrow [a, b]$ is continuous, then g has at least one fixed point in $[a, b]$.

Proof. The function $f(x) = g(x) - x$ satisfies $f(a) \geq 0$ and $f(b) \leq 0$. The proof is then completed by the intermediate value theorem (Theorem B.74). $\square$

Exercise 4.34. Let $A = [-1, 0) \cup (0, 1]$. Give an example of a continuous function $g : A \rightarrow A$ that does not have a fixed point. Give an example of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ that does not have a fixed point.

Definition 4.35. A *fixed-point iteration* is a method for finding a fixed point of g with a formula of the form

$$x_{n+1} = g(x_n), \quad n \in \mathbb{N}. \quad (4.30)$$

Example 4.36. Newton's method is a fixed-point iteration.

Exercise 4.37. To calculate the square root of some positive real number a , we can formulate the problem as finding the root of $f(x) = x^2 - a$. For $a = 1$, the initial guess of $x_0 = 2$, and the three choices of $g_1(x) := x^2 + x - a$, $g_2(x) := \frac{a}{x}$, and $g_3(x) := \frac{1}{2}(x + \frac{a}{x})$, verify that g_1 diverges, g_2 oscillates, g_3 converges. The theorems in this section will explain why.

Definition 4.38. A function $f : [a, b] \rightarrow [a, b]$ is a *contraction* or *contractive mapping* on $[a, b]$ if

$$\exists \lambda \in [0, 1) \text{ s.t. } \forall x, y \in [a, b], |f(x) - f(y)| \leq \lambda |x - y|. \quad (4.31)$$

Example 4.39. Any linear function $f(x) = \lambda x + c$ with $0 \leq \lambda < 1$ is a contraction.

Lemma 4.40. If $g(x)$ is a contraction on $[a, b]$ and it has a fixed point α in $[a, b]$, then for any choice $x_0 \in [a, b]$, the sequence $\{x_n\}$ generated by the fixed-point iteration (4.30) satisfies

$$\frac{1}{1+\lambda} |x_{n+1} - x_n| \leq |x_n - \alpha| \leq \frac{1}{1-\lambda} |x_{n+1} - x_n|. \quad (4.32)$$

Proof. By Definition 4.38, $x_{n+1} = g(x_n)$ implies that all x_n 's stay in $[a, b]$ and

$$|x_{n+1} - \alpha| = |g(x_n) - g(\alpha)| \leq \lambda |x_n - \alpha|.$$

Hence we have

$$|x_{n+1} - x_n| \leq |x_{n+1} - \alpha| + |x_n - \alpha| \leq (1+\lambda) |x_n - \alpha|,$$

which proves the first inequality in (4.32). As for the second inequality, it follows that

$$\begin{aligned} |x_n - \alpha| &\leq |x_{n+1} - x_n| + |x_{n+1} - \alpha| \\ &\leq |x_{n+1} - x_n| + \lambda |x_n - \alpha|, \end{aligned}$$

which implies $|x_n - \alpha| \leq \frac{1}{1-\lambda} |x_{n+1} - x_n|$. $\square$

Theorem 4.41 (Convergence of contractions). If $g(x)$ is a continuous contraction on $[a, b]$, then it has a unique fixed point α in $[a, b]$. Furthermore, the fixed-point iteration (4.30) converges to α for any choice $x_0 \in [a, b]$ and

$$|x_n - \alpha| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0|. \quad (4.33)$$

Proof. By Lemma 4.33, g has at least one fixed point in $[a, b]$. Suppose there are two distinct fixed points α and β , then $|\alpha - \beta| = |g(\alpha) - g(\beta)| \leq \lambda|\alpha - \beta|$, which implies $|\alpha - \beta| \leq 0$, i.e. the two fixed points are identical.

To prove convergence,

$$|x_{n+1} - x_n| = |g(x_n) - g(x_{n-1})| \leq \lambda|x_n - x_{n-1}|.$$

By induction, we have $|x_{n+1} - x_n| \leq \lambda^n |x_1 - x_0|$ and hence

$$|x_n - \alpha| \leq \frac{1}{1 - \lambda} |x_{n+1} - x_n| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0|,$$

where the first inequality follows from (4.32). $\square$

Theorem 4.42. Consider $g : [a, b] \rightarrow [a, b]$. If $g \in \mathcal{C}^1[a, b]$ and $\lambda = \max_{x \in [a, b]} |g'(x)| < 1$, then g has a unique fixed point α in $[a, b]$. Furthermore, the fixed-point iteration (4.30) converges to α for any choice $x_0 \in [a, b]$, the error bound (4.33) holds, and

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - \alpha}{x_n - \alpha} = g'(\alpha). \quad (4.34)$$

Proof. The mean value theorem (Theorem B.92) implies that, for all $x, y \in [a, b]$, $|g(x) - g(y)| \leq \lambda|x - y|$. Theorem 4.41 yields all the results except (4.34), which follows from

$$x_{n+1} - \alpha = g(x_n) - g(\alpha) = g'(\xi)(x_n - \alpha),$$

$\lim_{n \rightarrow \infty} x_n = \alpha$, and the fact that ξ is between x_n and α . $\square$

Corollary 4.43. Let α be a fixed point of $g : \mathbb{R} \rightarrow \mathbb{R}$ with $|g'(\alpha)| < 1$ and $g \in \mathcal{C}^1(\mathcal{B})$ on $\mathcal{B} = [\alpha - \delta, \alpha + \delta]$ with some $\delta > 0$. If x_0 is chosen sufficiently close to α , then the results of Theorem 4.41 hold.

Proof. Choose λ so that $|g'(\alpha)| < \lambda < 1$. Choose $\delta_0 \leq \delta$ so that $\max_{x \in \mathcal{B}_0} |g'(x)| \leq \lambda < 1$ on $\mathcal{B}_0 = [\alpha - \delta_0, \alpha + \delta_0]$. Then $g(\mathcal{B}_0) \subset \mathcal{B}_0$ and applying Theorem 4.42 completes the proof. $\square$

Corollary 4.44. Consider $g : [a, b] \rightarrow [a, b]$ with a fixed point $g(\alpha) = \alpha \in [a, b]$. The fixed-point iteration (4.30) converges to α with p th-order accuracy ($p > 1, p \in \mathbb{N}$) if

$$\begin{cases} g \in \mathcal{C}^p[a, b], \\ \forall k = 1, 2, \dots, p-1, g^{(k)}(\alpha) = 0, \\ g^{(p)}(\alpha) \neq 0 \end{cases} \quad (4.35)$$

and if x_0 is chosen sufficiently close to α .

Proof. By Corollary 4.43, $g'(\alpha) = 0$, and the continuity of g' , there exists $\delta > 0$ such that the fixed-point iteration converges uniquely to α inside $[\alpha - \delta, \alpha + \delta]$. By the Taylor expansion of g at α , we have

$$\begin{aligned} E_{\text{abs}}(x_{n+1}) &:= |x_{n+1} - \alpha| = |g(x_n) - g(\alpha)| \\ &= \left| \sum_{i=1}^{p-1} \frac{(x_n - \alpha)^i}{i!} g^{(i)}(\alpha) + \frac{(x_n - \alpha)^p}{p!} g^{(p)}(\xi) \right| \end{aligned}$$

for some $\xi \in [a, b]$. Since $g^{(p)}$ is continuous on $[a, b]$, Theorem B.71 implies that $g^{(p)}$ is bounded on $[a, b]$. Hence there exists a constant M such that $E_{\text{abs}}(x_{n+1}) \leq M E_{\text{abs}}^p(x_n)$. $\square$

Example 4.45. The following method has third-order convergence for computing $\sqrt{R}$:

$$x_{n+1} = \frac{x_n(x_n^2 + 3R)}{3x_n^2 + R}.$$

First, $\sqrt{R}$ is the fixed point of $F(x) = \frac{x(x^2 + 3R)}{3x^2 + R}$:

$$F(\sqrt{R}) = \frac{\sqrt{R}(R + 3R)}{3R + R} = \sqrt{R}.$$

Second, the derivatives of $F(x)$ are

n	$F^{(n)}(x)$	$F^{(n)}(\sqrt{R})$
1	$\frac{3(x^2 - R)^2}{(3x^2 + R)^2}$	0
2	$\frac{48Rx(x^2 - R)}{(3x^2 + R)^3}$	0
3	$\frac{-48R(9x^4 - 18Rx^2 + R^2)}{(3x^2 + R)^4}$	$\frac{-48R(-8R^2)}{(4R)^4} = \frac{3}{2R} \neq 0$

The rest follows from Corollary 4.44.

4.7 Systems of nonlinear equations

Definition 4.46. A system of nonlinear equations is a system of the form

$$\mathbf{f}(\mathbf{x}) = \mathbf{0}, \quad (4.36)$$

where $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a multivariate vector-valued function given by

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{bmatrix}, \quad (4.37)$$

each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a scalar function, at least one of f_i is nonlinear, and $\mathbf{x} = [x_1, x_2, \dots, x_n]^\top$. The numerical goal is to find $\boldsymbol{\alpha} \in \mathbb{R}^n$, called a *root* or *solution*, such that $\mathbf{f}(\boldsymbol{\alpha}) = \mathbf{0}$ holds approximately in a given norm.

Definition 4.47. A sequence of iterates $\{\mathbf{x}^{(k)}\} \subset \mathbb{R}^n$ converges linearly to $\mathbf{L} \in \mathbb{R}^n$ if

$$\exists c \in (0, 1), \exists d > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall k > N, \|\mathbf{x}^{(k)} - \mathbf{L}\|_\infty \leq c^k d. \quad (4.38)$$

For a sequence $\{\mathbf{x}^{(k)}\}$ that converges to $\mathbf{L}$, its *order of convergence* is the maximum $p \in \mathbb{R}^+$ satisfying

$$\exists c > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall k > N, \|\mathbf{x}^{(k+1)} - \mathbf{L}\|_\infty \leq c \|\mathbf{x}^{(k)} - \mathbf{L}\|_\infty^p. \quad (4.39)$$

In particular, $\{\mathbf{x}^{(k)}\}$ converges quadratically if $p = 2$.

4.7.1 Fixed-point iterations

Definition 4.48. A *fixed point* of a function $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an independent parameter of $\mathbf{g}$ satisfying $\mathbf{g}(\boldsymbol{\alpha}) = \boldsymbol{\alpha}$.

Example 4.49. In Definition 3.76, the solution of the linear system $A\mathbf{x} = \mathbf{b}$ is the fixed point of the function $\mathbf{g}(\mathbf{x}) = T\mathbf{x} + \mathbf{c}$.

Definition 4.50. A *fixed-point iteration* is a method for finding a fixed point of $\mathbf{g}$ with a formula of the form

$$\mathbf{x}^{(k+1)} = \mathbf{g}(\mathbf{x}^{(k)}), \quad k \in \mathbb{N}. \quad (4.40)$$

Theorem 4.51 (Brouwer's fixed point). With $\mathbb{D}^n := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_2 \leq 1\}$, any continuous function $\mathbf{f} : \mathbb{D}^n \rightarrow \mathbb{D}^n$ has a fixed point.

Proof. See Milnor [1978]. $\square$

Example 4.52. Take a map of your country C and place it on the ground of your room. Let $\mathbf{f}$ be the function assigning to each point in your country the point on the map corresponding to it. Then $\mathbf{f}$ can be considered as a continuous function $C \rightarrow C$. If C is homeomorphic to $\mathbb{D}^2$, then there must exist a point on the map that corresponds exactly to the point on the ground directly beneath it.

Exercise 4.53. Take two pieces of the same-sized paper and lay one on top of the other. Every point on the top sheet of paper is associated with some point right below it on the bottom sheet. Crumple the top sheet into a ball without ripping it. Place the crumpled ball on top of (and simultaneously within the realm of) the bottom sheet of paper. Use Theorem 4.51 to prove that there always exists some point in the crumpled ball that sits above the same point it sat above prior to crumpling.

Theorem 4.54. Let E be a closed, bounded and convex set in $\mathbb{R}^n$, cf. Definition 4.22. Assume that $\mathbf{g} : E \rightarrow E$ is a $\mathcal{C}^1$ function, i.e. $g_i \in \mathcal{C}^1(E)$ for each i , and further assume

$$\lambda = \max_{\mathbf{x} \in E} \|D\mathbf{g}(\mathbf{x})\|_\infty < 1, \quad (4.41)$$

where $D\mathbf{g}$ denotes the Jacobian matrix of $\mathbf{g}$, cf. Definition C.364. Then $\mathbf{g}$ has a unique fixed point $\boldsymbol{\alpha}$ in E , the fixed-point iteration (4.40) converges to $\boldsymbol{\alpha}$ for any choice $\mathbf{x}^{(0)} \in E$, and

$$\lim_{k \rightarrow \infty} \frac{\|\mathbf{x}^{(k+1)} - \boldsymbol{\alpha}\|_\infty}{\|\mathbf{x}^{(k)} - \boldsymbol{\alpha}\|_\infty} \leq \|D\mathbf{g}(\boldsymbol{\alpha})\|_\infty. \quad (4.42)$$

Furthermore, the error norm of the k th iterate satisfies

$$\|\mathbf{e}^{(k)}\|_\infty \leq \frac{\lambda^k}{1 - \lambda} \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_\infty; \quad (4.43)$$

$$\|\mathbf{e}^{(k)}\|_\infty \leq \frac{1}{1 - \lambda} \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_\infty, \quad (4.44)$$

where $\mathbf{e}^{(k)} := \boldsymbol{\alpha} - \mathbf{x}^{(k)}$.

Proof. For each $k \in \mathbb{N}^+$, define

$$\boldsymbol{\varphi}^{(k)} : [0, 1] \rightarrow E, \quad \boldsymbol{\varphi}^{(k)}(t) = (1 - t)\mathbf{x}^{(k-1)} + t\mathbf{x}^{(k)}.$$

It is well-defined because E is convex. Then

$$\begin{aligned} \mathbf{x}^{(k+1)} - \mathbf{x}^{(k)} &= \mathbf{g}(\mathbf{x}^{(k)}) - \mathbf{g}(\mathbf{x}^{(k-1)}) \\ &= \mathbf{g}(\boldsymbol{\varphi}^{(k)}(1)) - \mathbf{g}(\boldsymbol{\varphi}^{(k)}(0)) \\ &= D(\mathbf{g} \circ \boldsymbol{\varphi}^{(k)})(t_k) \\ &= D\mathbf{g}(\boldsymbol{\varphi}^{(k)}(t_k))(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}), \end{aligned}$$

where $t_k \in (0, 1)$, the third step follows from the vector mean value theorem C.370 and the last step from the chain rule (Theorem C.369). Hence $\boldsymbol{\varphi}^{(k)}(t_k) \in E$ and (4.41) yields

$$\begin{aligned} \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_\infty &\leq \|D\mathbf{g}(\boldsymbol{\varphi}^{(k)}(t_k))\|_\infty \|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty \\ &\leq \lambda \|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty. \end{aligned}$$

By induction,

$$(*) : \quad \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_\infty \leq \lambda^k \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_\infty,$$

which implies that for any $m \in \mathbb{N}^+$,

$$\begin{aligned} \|\mathbf{x}^{(k)} - \mathbf{x}^{(k+m)}\|_\infty &\leq \sum_{i=1}^m \|\mathbf{x}^{(k+i-1)} - \mathbf{x}^{(k+i)}\|_\infty \\ &\leq \sum_{i=1}^m \lambda^{k+i-1} \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_\infty \\ &< \frac{\lambda^k}{1 - \lambda} \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_\infty. \end{aligned}$$

Thus the sequence $\{\mathbf{x}^{(k)}\}$ is Cauchy and converges to some $\boldsymbol{\alpha} \in E$. Take limit of (4.40) and we have from the continuity of $\mathbf{g}$ that $\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\alpha})$.

As for uniqueness, suppose there are two distinct fixed points $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Then, as in the beginning of the proof, there exists $\boldsymbol{\xi} \in E$ on the line segment joining $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ such that

$$\boldsymbol{\alpha} - \boldsymbol{\beta} = \mathbf{g}(\boldsymbol{\alpha}) - \mathbf{g}(\boldsymbol{\beta}) = D\mathbf{g}(\boldsymbol{\xi})(\boldsymbol{\alpha} - \boldsymbol{\beta}).$$

Hence $\|\boldsymbol{\alpha} - \boldsymbol{\beta}\|_\infty \leq \lambda \|\boldsymbol{\alpha} - \boldsymbol{\beta}\|_\infty$, which yields $\|\boldsymbol{\alpha} - \boldsymbol{\beta}\|_\infty \leq 0$, i.e. the two fixed points are identical.

To prove convergence, for each k , there exists $\boldsymbol{\xi}^{(k)} \in E$ on the line segment joining $\mathbf{x}^{(k)}$ and $\boldsymbol{\alpha}$ such that

$$\mathbf{x}^{(k+1)} - \boldsymbol{\alpha} = \mathbf{g}(\mathbf{x}^{(k)}) - \mathbf{g}(\boldsymbol{\alpha}) = D\mathbf{g}(\boldsymbol{\xi}^{(k)})(\mathbf{x}^{(k)} - \boldsymbol{\alpha}).$$

It follows that

$$(**) : \quad \|\mathbf{x}^{(k+1)} - \boldsymbol{\alpha}\|_\infty \leq \|D\mathbf{g}(\boldsymbol{\xi}^{(k)})\|_\infty \|\mathbf{x}^{(k)} - \boldsymbol{\alpha}\|_\infty,$$

From (4.41), (**) implies

$$(\triangle) : \quad \|\mathbf{x}^{(k+1)} - \boldsymbol{\alpha}\|_\infty \leq \lambda \|\mathbf{x}^{(k)} - \boldsymbol{\alpha}\|_\infty.$$

By induction, we have $\|\mathbf{x}^{(k)} - \boldsymbol{\alpha}\|_\infty \leq \lambda^k \|\mathbf{x}^{(0)} - \boldsymbol{\alpha}\|_\infty$. With $\lambda < 1$, this shows $\mathbf{x}^{(k)} \rightarrow \boldsymbol{\alpha}$ as $k \rightarrow \infty$. Then (4.42) follows from (**), the continuity of $\|\cdot\|_\infty$, $\lim_{k \rightarrow \infty} \mathbf{x}^{(k)} = \boldsymbol{\alpha}$, and the fact that $\boldsymbol{\xi}^{(k)}$ is on the line segment joining $\mathbf{x}^{(k)}$ and $\boldsymbol{\alpha}$.

At last, the triangle inequality and $(\triangle)$ yield

$$\begin{aligned} \|\mathbf{x}^{(k)} - \boldsymbol{\alpha}\|_\infty &\leq \|\mathbf{x}^{(k)} - \mathbf{x}^{(k+1)}\|_\infty + \|\mathbf{x}^{(k+1)} - \boldsymbol{\alpha}\|_\infty \\ &\leq \|\mathbf{x}^{(k)} - \mathbf{x}^{(k+1)}\|_\infty + \lambda \|\mathbf{x}^{(k)} - \boldsymbol{\alpha}\|_\infty. \end{aligned}$$

Thus (4.44) holds, and together with (*), it implies (4.43). $\square$

Corollary 4.55. Let α be a fixed point of $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $\|D\mathbf{g}(\alpha)\|_\infty < 1$ and

$$\mathcal{B} = B(\alpha, \delta) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \alpha\|_\infty \leq \delta\} \quad (4.45)$$

with some $\delta > 0$. If $\mathbf{g} \in \mathcal{C}^1(\mathcal{B})$ and $\mathbf{x}^{(0)}$ is chosen sufficiently close to α , then the results of Theorem 4.54 hold.

Proof. Choose λ so that $\|D\mathbf{g}(\alpha)\|_\infty < \lambda < 1$. Choose $\delta_0 \leq \delta$ so that $\max_{\mathbf{x} \in \mathcal{B}_0} \|D\mathbf{g}(\mathbf{x})\|_\infty \leq \lambda < 1$ on $\mathcal{B}_0 = B(\alpha, \delta_0)$. Then $\mathbf{g}(\mathcal{B}_0) \subset \mathcal{B}_0$, and applying Theorem 4.54 completes the proof. $\square$

Example 4.56. Consider $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with

$$\begin{aligned} f_1([x_1, x_2]^\top) &= 3x_1^2 + 4x_2^2 - 1, \\ f_2([x_1, x_2]^\top) &= x_2^3 - 8x_1^3 - 1. \end{aligned}$$

We solve $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ iteratively with

$$\mathbf{x}^{(k+1)} = \mathbf{g}(\mathbf{x}^{(k)}) = \mathbf{x}^{(k)} - A\mathbf{f}(\mathbf{x}^{(k)}), \quad (4.46)$$

where

$$A = \begin{bmatrix} 0.016 & -0.17 \\ 0.52 & -0.26 \end{bmatrix}. \quad (4.47)$$

Given the initial value $\mathbf{x}^{(0)} = [-0.5, 0.25]^\top$, the numerical results of (4.46) are

k	$x_1^{(k)}$	$x_2^{(k)}$
0	-0.5	0.25
1	-0.497343750	0.254062500
2	-0.497254794	0.254077922
3	-0.497251343	0.254078566
4	-0.497251208	0.254078592

and the solution $\alpha \approx [-0.49725, 0.25408]^\top$. Then it is straightforward to compute

$$D\mathbf{g}(\alpha) \approx \begin{bmatrix} 0.0389 & 0.0004 \\ 0.0085 & -0.0066 \end{bmatrix}$$

and therefore $\|D\mathbf{g}(\alpha)\|_\infty \approx 0.0393 < 1$. Thus, from Corollary 4.55 and (4.42), it will be approximately true that

$$\frac{\|\mathbf{x}^{(k+1)} - \alpha\|_\infty}{\|\mathbf{x}^{(k)} - \alpha\|_\infty} \leq \|D\mathbf{g}(\alpha)\|_\infty \approx 0.0393$$

for all sufficiently large k .

4.7.2 Newton's method

Algorithm 4.57. *Newton's method* finds the root of $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ near an initial guess $\mathbf{x}^{(0)}$ by the iteration formula

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - [D\mathbf{f}(\mathbf{x}^{(k)})]^{-1}\mathbf{f}(\mathbf{x}^{(k)}), \quad k \in \mathbb{N}, \quad (4.48)$$

where $D\mathbf{f}$ is the Jacobian matrix of $\mathbf{f}$.

Theorem 4.58 (Convergence of Newton's method). For a $\mathcal{C}^2$ function $\mathbf{f} : \mathcal{B} \rightarrow \mathbb{R}^n$, where $\mathcal{B}$ is defined as in (4.45), satisfying $\mathbf{f}(\alpha) = \mathbf{0}$ and with $D\mathbf{f}(\alpha)$ nonsingular, there exists $\delta_1 \in (0, \delta)$ such that for all $\mathbf{x} \in \mathcal{B}_1 = B(\alpha, \delta_1)$, the Jacobian $D\mathbf{f}(\mathbf{x})$ is nonsingular. Define M_i as the maximum absolute

value over all second-order partial derivatives of f_i on $\mathcal{B}_1$, i.e.

$$M_i := \max_{1 \leq j, \ell \leq n} \max_{\mathbf{x} \in \mathcal{B}_1} \left| \frac{\partial^2 f_i}{\partial x_j \partial x_\ell}(\mathbf{x}) \right|.$$

Assume that $\mathbf{x}^{(0)}$ is sufficiently close to α such that (i) $\|\mathbf{x}^{(0)} - \alpha\|_\infty = \delta_0 < \delta_1$ and (ii) $M\delta_0 < 1$ hold, where

$$M := \frac{1}{2}n^2 \max_{\mathbf{x} \in \mathcal{B}_1} \|[D\mathbf{f}(\mathbf{x})]^{-1}\|_\infty \cdot \max_{1 \leq i \leq n} M_i.$$

Then the sequence of iterates $\{\mathbf{x}^{(k)}\}$ in (4.48) converges at least quadratically to the root α , i.e.

$$\|\alpha - \mathbf{x}^{(k+1)}\|_\infty \leq M\|\alpha - \mathbf{x}^{(k)}\|_\infty^2. \quad (4.49)$$

Proof. Taylor's theorem and $f_i \in \mathcal{C}^2$ yield

$$f_i(\alpha) = f_i(\mathbf{x}^{(k)}) + \sum_{j=1}^n (\alpha_j - x_j^{(k)}) \frac{f_i(\mathbf{x}^{(k)})}{\partial x_j} + \frac{1}{2}c_i^{(k)},$$

where $c_i^{(k)}$ is the i th element of $\mathbf{c}^{(k)}$,

$$c_i^{(k)} = \left[\sum_{j=1}^n (\alpha_j - x_j^{(k)}) \frac{\partial}{\partial x_j} \right]^2 f_i(\xi^{(k,i)}),$$

$\xi^{(k,i)}$ is on the line segment joining $\mathbf{x}^{(k)}$ and α . Then

$$\mathbf{0} = \mathbf{f}(\alpha) = \mathbf{f}(\mathbf{x}^{(k)}) + D\mathbf{f}(\mathbf{x}^{(k)})(\alpha - \mathbf{x}^{(k)}) + \frac{1}{2}\mathbf{c}^{(k)},$$

which, together with (4.48), implies

$$\begin{aligned} \mathbf{x}^{(k+1)} - \alpha &= \mathbf{x}^{(k)} - \alpha - [D\mathbf{f}(\mathbf{x}^{(k)})]^{-1}\mathbf{f}(\mathbf{x}^{(k)}) \\ &= \frac{1}{2}[D\mathbf{f}(\mathbf{x}^{(k)})]^{-1}\mathbf{c}^{(k)}. \end{aligned}$$

The definition of M_i yields

$$\begin{aligned} |c_i^{(k)}| &\leq \left[\sum_{j=1}^n |\alpha_j - x_j^{(k)}| \right]^2 M_i \\ &\leq \left(\sum_{j=1}^n \|\alpha - \mathbf{x}^{(k)}\|_\infty \right)^2 M_i = n^2 M_i \|\alpha - \mathbf{x}^{(k)}\|_\infty^2, \end{aligned}$$

which gives $\|\mathbf{c}^{(k)}\|_\infty \leq n^2 \max_{1 \leq i \leq n} M_i \|\alpha - \mathbf{x}^{(k)}\|_\infty^2$. Hence

$$\begin{aligned} \|\mathbf{x}^{(k+1)} - \alpha\|_\infty &\leq \frac{1}{2} \|[D\mathbf{f}(\mathbf{x}^{(k)})]^{-1}\|_\infty \|\mathbf{c}^{(k)}\|_\infty \\ &\leq \frac{1}{2} \max_{\mathbf{x} \in \mathcal{B}_1} \|[D\mathbf{f}(\mathbf{x})]^{-1}\|_\infty \cdot n^2 \max_{1 \leq i \leq n} M_i \|\alpha - \mathbf{x}^{(k)}\|_\infty^2 \\ &= M\|\alpha - \mathbf{x}^{(k)}\|_\infty^2. \end{aligned}$$

Comparing the above to (4.39) implies that if $\{\mathbf{x}^{(k)}\}$ converges, then the order of convergence is 2. We must still show that (a) it converges and (b) it converges to α .

By (i) and (ii), we have $M\|\mathbf{x}^{(0)} - \alpha\|_\infty < 1$. Then an easy induction yields

$$\|\mathbf{x}^{(k)} - \alpha\|_\infty \leq \frac{1}{M}(M\|\mathbf{x}^{(0)} - \alpha\|_\infty)^{2^k},$$

which shows both (a) and (b). Finally, the convergence rate could be higher than 2 in the case of $D^2\mathbf{f}(\boldsymbol{\alpha}) = \mathbf{0}$, i.e.

$$\forall i, j, \ell \in [1, n], \quad \frac{\partial^2 f_i}{\partial x_j \partial x_\ell}(\boldsymbol{\alpha}) = 0. \quad \square$$

Exercise 4.59. Let $\mathbf{g}(\mathbf{x}) = \mathbf{x} - [D\mathbf{f}(\mathbf{x})]^{-1}\mathbf{f}(\mathbf{x})$, where $\mathbf{f} \in \mathcal{C}^2$. Prove that $D\mathbf{g}(\boldsymbol{\alpha}) = O_{n \times n}$, provided $\boldsymbol{\alpha}$ is a root of $\mathbf{f}$ and $D\mathbf{f}(\boldsymbol{\alpha})$ is nonsingular.

4.7.3 Quasi-Newton Methods

Algorithm 4.60. For the *unconstrained optimization problem* $\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$, Newton's method can be applied by leveraging the first-order necessary condition $\nabla f(\mathbf{x}) = \mathbf{0}$. The iteration is given by

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - H_k^{-1} \mathbf{g}^{(k)}, \quad (4.50)$$

where $\mathbf{g}^{(k)} = \mathbf{g}(\mathbf{x}^{(k)}) = \nabla f(\mathbf{x}^{(k)})$ is the gradient vector of f evaluated at $\mathbf{x}^{(k)}$, and $H_k = H(\mathbf{x}^{(k)})$ is the Hessian matrix of f at $\mathbf{x}^{(k)}$, defined as $H = (h_{ij})_{n \times n}$, $h_{ij}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x})$.

Algorithm 4.61. *Quasi-Newton methods* aim to avoid the computation of second-order derivatives by using a symmetric positive-definite matrix to approximate either the Hessian matrix H_k or its inverse in (4.50), thereby optimizing the objective function f under this approximation. Different approximation schemes give rise to different variants of quasi-Newton methods.

Definition 4.62 (Quasi-Newton condition). For the Newton iteration (4.50), the *quasi-Newton condition* consists of one of the two equivalent relations

$$\mathbf{y}^{(k)} = H_k \boldsymbol{\delta}^{(k)}, \quad (4.51)$$

$$H_k^{-1} \mathbf{y}^{(k)} = \boldsymbol{\delta}^{(k)}, \quad (4.52)$$

where $\mathbf{y}^{(k)} = \mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}$ and $\boldsymbol{\delta}^{(k)} = \mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}$.

In quasi-Newton methods, one constructs a symmetric positive-definite matrix G_k (approximating H_k^{-1}) or B_k (approximating H_k), designed to satisfy (4.52) or (4.51).

Algorithm 4.63. The *DFP (Davidon–Fletcher–Powell) algorithm* is a quasi-Newton method that constructs and updates a matrix G_k to approximate the inverse Hessian matrix H_k^{-1} of the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Its iterative update formula is

$$G_{k+1} = G_k + \frac{\boldsymbol{\delta}^{(k)}(\boldsymbol{\delta}^{(k)})^\top}{(\boldsymbol{\delta}^{(k)})^\top \mathbf{y}^{(k)}} - \frac{G_k \mathbf{y}^{(k)}(\mathbf{y}^{(k)})^\top G_k}{(\mathbf{y}^{(k)})^\top G_k \mathbf{y}^{(k)}}, \quad (4.53)$$

where $\boldsymbol{\delta}^{(k)}$ and $\mathbf{y}^{(k)}$ are as defined in Definition 4.62. If the initial matrix G_0 is chosen as a positive-definite symmetric matrix (typically the identity matrix I_n), then every matrix G_k generated during the iteration remains positive-definite and symmetric.

Algorithm 4.64. The *BFGS (Broyden–Fletcher–Goldfarb–Shanno) algorithm* is a quasi-Newton method that constructs and updates a matrix B_k to approximate the Hessian

matrix H_k of the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Its iterative update formula is

$$B_{k+1} = B_k + \frac{\mathbf{y}^{(k)}(\mathbf{y}^{(k)})^\top}{(\mathbf{y}^{(k)})^\top \boldsymbol{\delta}^{(k)}} - \frac{B_k \boldsymbol{\delta}^{(k)}(\boldsymbol{\delta}^{(k)})^\top B_k}{(\boldsymbol{\delta}^{(k)})^\top B_k \boldsymbol{\delta}^{(k)}}, \quad (4.54)$$

where $\boldsymbol{\delta}^{(k)}$ and $\mathbf{y}^{(k)}$ are as defined in Definition 4.62. If the initial matrix B_0 is chosen as a positive-definite symmetric matrix (typically the identity matrix I_n), then every matrix B_k generated during the iteration remains positive-definite and symmetric.

Summary 4.65. This chapter systematically develops numerical methods for nonlinear equations and systems, including bisection, Newton, secant, fixed-point iteration, and multidimensional extensions (fixed-point iteration, Newton and Quasi-Newton). The theoretical core centers on convergence analysis (linear/quadratic/superlinear order, global/local criteria) and error estimation, complemented by quantitative efficiency comparisons (e.g., Newton vs. secant).

Distinctive features of these notes include: a rigorous proof of the secant method's convergence order — via error recursion and Fibonacci numbers, definitively establishing the golden ratio order; and cross-domain analogies linking nonlinear fixed-point iterations with linear system solvers — all unified under the contraction mapping principle. Theory and practice are tightly integrated: every method is accompanied by a complete signature and reinforced through programming assignments that enforce contract-based design.

4.8 Theoretical problems

I. Consider the bisection method starting with the initial interval $[1.5, 3.5]$. In the following questions “the interval” refers to the bisection interval whose width changes across different loops.

- What is the width of the interval at the n th step?
- What is the supremum of the distance between the root r and the midpoint of the interval?

II. In using the bisection algorithm with its initial interval as $[a_0, b_0]$ with $a_0 > 0$, we want to determine the root with its *relative* error no greater than ϵ . Prove that this goal of accuracy is guaranteed by the following choice of the number of steps,

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log a_0}{\log 2} - 1.$$

III. If the bisection method is used in single precision FPNs of IEEE 754 starting with the interval $[128, 129]$, can we compute the root with absolute accuracy $< 10^{-6}$? Why?

IV. Perform four iterations of Newton's method for the polynomial equation $p(x) = 4x^3 - 2x^2 + 3 = 0$ with the starting point $x_0 = -1$. Use a hand calculator and organize results of the iterations in a table.

- V. Consider a variation of Newton's method in which only the derivative at x_0 is used,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}.$$

Find C and s such that

$$e_{n+1} = Ce_n^s,$$

where e_n is the error of Newton's method at step n , s is a constant, and C may depend on x_n , the true solution α , and the derivative of the function f .

- VI. Within $(-\frac{\pi}{2}, \frac{\pi}{2})$, will the iteration $x_{n+1} = \tan^{-1} x_n$ converge?
- VII. Let $p > 1$. What is the value of the following continued fraction?

$$x = \frac{1}{p + \frac{1}{p + \frac{1}{p + \dots}}}$$

Prove that the sequence of values converges. (Hint: this can be interpreted as $x = \lim_{n \rightarrow \infty} x_n$, where $x_1 = \frac{1}{p}$, $x_2 = \frac{1}{p + \frac{1}{p}}$, $x_3 = \frac{1}{p + \frac{1}{p + \frac{1}{p}}}$, ..., and so forth.

Formulate x as a fixed point of some function.)

- VIII. What happens in problem II if $a_0 < 0 < b_0$? Derive an inequality of the number of steps similar to that in II. In this case, is the relative error still an appropriate measure?

- IX. Perform four iterations of Newton's method for the system

$$\begin{cases} x^2 + y^2 - 2x - 2y + 1 = 0, \\ x + y - 2xy = 0 \end{cases}$$

with the starting point $(x_0, y_0) = (2, 1)$. Use a hand calculator and organize results of the iterations in a table.

- X. (*) Consider solving $f(x) = 0$ ($f \in \mathcal{C}^{k+1}$) by Newton's method with the starting point x_0 close to a root of multiplicity k . Note that α is a zero of multiplicity k of the function f iff

$$f^{(k)}(\alpha) \neq 0; \quad \forall i < k, \quad f^{(i)}(\alpha) = 0.$$

- How can a multiple zero be detected by examining the behavior of the points $(x_n, f(x_n))$?
- Prove that if r is a zero of multiplicity k of the function f , then quadratic convergence in Newton's iteration will be restored by making this modification:

$$x_{n+1} = x_n - k \frac{f(x_n)}{f'(x_n)}.$$

4.9 Programming assignments

- A. Implement the bisection method, Newton's method, and the secant method in a C++ package. You should

- (a) design an abstract base class `EquationSolver` with a pure virtual method `solve`,
- (b) write a derived class of `EquationSolver` for each method to accommodate its particularities in the contract of solving nonlinear equations.

- B. Test your implementation of the bisection method on the following functions and intervals.

- $x^{-1} - \tan x$ on $[0, \frac{\pi}{2}]$,
- $x^{-1} - 2^x$ on $[0, 1]$,
- $2^{-x} + e^x + 2 \cos x - 6$ on $[1, 3]$,
- $(x^3 + 4x^2 + 3x + 5)/(2x^3 - 9x^2 + 18x - 2)$ on $[0, 4]$.

- C. Test your implementation of Newton's method by solving $x = \tan x$. Find the roots near 4.5 and 7.7.

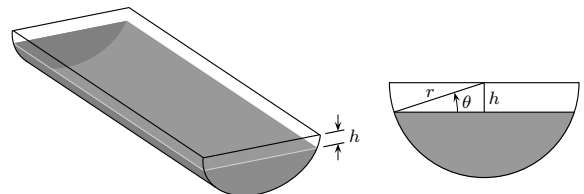
- D. Test your implementation of the secant method by the following functions and initial values.

- $\sin(x/2) - 1$ with $x_0 = 0, x_1 = \frac{\pi}{2}$,
- $e^x - \tan x$ with $x_0 = 1, x_1 = 1.4$,
- $x^3 - 12x^2 + 3x + 1$ with $x_0 = 0, x_1 = -0.5$.

You should play with other initial values and (if you get different results) think about the reasons.

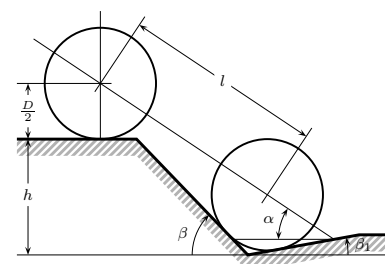
- E. As shown below, a trough of length L has a cross section in the shape of a semi-circle with radius r . When filled to within a distance h of the top, the water has the volume

$$V = L \left[0.5\pi r^2 - r^2 \arcsin \frac{h}{r} - h(r^2 - h^2)^{\frac{1}{2}} \right].$$



Suppose $L = 10\text{ft}$, $r = 1\text{ft}$, and $V = 12.4\text{ft}^3$. Find the depth of water in the trough to within 0.01ft by each of the three implementations in A.

- F. In the design of all-terrain vehicles, it is necessary to consider the failure of the vehicle when attempting to negotiate two types of obstacles. One type of failure is called *hang-up failure* and occurs when the vehicle attempts to cross an obstacle that causes the bottom of the vehicle to touch the ground. The other type of failure is called *nose-in failure* and occurs when the vehicle descends into a ditch and its nose touches the ground.



The above figure shows the components associated with the nose-in failure of a vehicle. The maximum angle α that can be negotiated by a vehicle when β is the maximum angle at which hang-up failure does not occur satisfies the equation

$$A \sin \alpha \cos \alpha + B \sin^2 \alpha - C \cos \alpha - E \sin \alpha = 0,$$

where

$$A = l \sin \beta_1, \quad B = l \cos \beta_1,$$

$$C = (h + 0.5D) \sin \beta_1 - 0.5D \tan \beta_1,$$

$$E = (h + 0.5D) \cos \beta_1 - 0.5D.$$

- (a) Use Newton's method to verify $\alpha \approx 33^\circ$ when $l = 89$ in., $h = 49$ in., $D = 55$ in. and $\beta_1 = 11.5^\circ$.
- (b) Use Newton's method to find α with the initial guess 33° for the situation when l, h, β_1 are the same as in part (a) but $D = 30$ in..

- (c) Use the secant method (with another initial value as far away as possible from 33°) to find α . Show that you get a different result if the initial value is too far away from 33° ; discuss the reasons.

G. Solve the system

$$\begin{cases} x^2 + xy^3 = 9, \\ 3x^2y - y^3 = 4 \end{cases}$$

using Newton's method for nonlinear systems. Use each of the initial guesses

$$(x_0, y_0) = (1.2, 2.5), (-2, 2.5), (-1.2, -2.5), (2, -2.5).$$

Observe which root to which the method converges, the number of iterates required, and the speed of convergence.

Chapter 5

Uniform Approximation

Definition 5.1. Given a normed vector space Y of functions and its subspace $X \subset Y$. A function $\hat{\varphi} \in X$ is called the *best approximation* to $f \in Y$ from X with respect to the norm $\|\cdot\|$ iff

$$\forall \varphi \in X, \quad \|f - \hat{\varphi}\| \leq \|f - \varphi\|. \quad (5.1)$$

Definition 5.2. The *fundamental problem of linear approximation* is to find the best approximation $\hat{\varphi} = \sum_{i=1}^n a_i u_i$ to $f \in Y$ from n elements $u_1, u_2, \dots, u_n \in X \subset Y$ that are linearly independent and given a priori.

Example 5.3. For $f(x) = e^x$ in $C^\infty[-1, 1]$, seeking its best approximation of the form $\hat{\varphi} = \sum_{i=1}^n a_i u_i$ in the subspace $X = \text{span}\{1, x, x^2, \dots\}$ is a problem of linear approximation, where n can be any positive integer and the norm can be the max-norm,

$$\|g\|_\infty := \max_{x \in [-1, 1]} |g(x)|, \quad (5.2)$$

the 1-norm

$$\|g\|_1 := \int_{-1}^{+1} |g(x)| dx, \quad (5.3)$$

or the 2-norm

$$\|g\|_2 := \left(\int_{-1}^{+1} |g(x)|^2 dx \right)^{\frac{1}{2}}. \quad (5.4)$$

The three different norms are motivated differently: the max-norm corresponds to the min-max error, the 1-norm is related to the area bounded between $g(x)$ and the x -axis, and the 2-norm is related to the Euclidean distance, cf. Section 8.3.

Definition 5.4. The *best uniform approximation* or *minimax approximation* is the best approximation with respect to the max-norm (5.2).

Example 5.5. For a simple closed curve $\gamma : [0, 1] \rightarrow \mathbb{R}^2$ and n points $\mathbf{x}_i \in \gamma$, consider a spline approximation $p : [0, 1] \rightarrow \mathbb{R}^2$ with its knots at $\mathbf{x}_i$'s and a scaled cumulative chordal length as in Definition 7.62. Denote by $\text{Int}(\gamma)$ as the complement of γ that always lies at the left of an observer who travels γ according to its parametrization. Then

the area difference between $\mathcal{S}_1 := \text{Int}(\gamma)$ and $\mathcal{S}_2 := \text{Int}(p)$ can be defined as

$$\|\mathcal{S}_1 \oplus \mathcal{S}_2\|_1 := \int_{\mathcal{S}_1 \oplus \mathcal{S}_2} d\mathbf{x},$$

where

$$\mathcal{S}_1 \oplus \mathcal{S}_2 := \mathcal{S}_1 \cup \mathcal{S}_2 \setminus (\mathcal{S}_1 \cap \mathcal{S}_2)$$

is the exclusive disjunction of $\mathcal{S}_1$ and $\mathcal{S}_2$.

The minimization of this area difference can be formulated by a best approximation problem based on the 1-norm.

Example 5.6. As in Notation 22, denote by $\mathbb{P}_n = \mathbb{P}_n(\mathbb{R})$ the set of all polynomials with coefficients in $\mathbb{R}$ and degree at most n . For $Y = \mathbb{P}_n$, and $X = \mathbb{P}_{n-1}$, the best uniform approximation to $f(x) = -x^n$ in Y from X is $\hat{\varphi} = \frac{T_n}{2^{n-1}} - x^n$, where T_n is the Chebyshev polynomial of degree n ; see theorem 5.68. Clearly $\hat{\varphi}$ satisfies (5.1).

5.1 Existence and uniqueness of the best approximation

Theorem 5.7. Suppose X is a finite-dimensional subspace of a normed vector space $(Y, \|\cdot\|)$. Then we have

$$\forall y \in Y, \exists \hat{\varphi} \in X \text{ s.t. } \forall \varphi \in X, \|\hat{\varphi} - y\| \leq \|\varphi - y\|. \quad (5.5)$$

Proof. For a given $y \in Y$, define a closed ball

$$B_y := \{x \in X : \|x\| \leq 2\|y\|\}.$$

Clearly $0 \in B_y$, and the distance from y to B_y is

$$\text{dist}(y, B_y) := \inf_{x \in B_y} \|y - x\| \leq \|y - 0\| = \|y\|.$$

By definition, any $z \in X$, $z \notin B_y$ must satisfy $\|z\| > 2\|y\|$, and thus

$$\|z - y\| \geq \|z\| - \|y\| > \|y\|.$$

Therefore, if a best approximation to y exists, it must be in B_y . As a subspace of X , B_y is finite dimensional, closed, and bounded, hence B_y is compact. The extreme value theorem states that a continuous scalar function attains its minimum and maximum on a compact set. A norm is a continuous function, hence the function $d : B_y \rightarrow \mathbb{R}^+ \cup \{0\}$ given by $d(x) = \|x - y\|$ must attain its minimum on B_y . $\square$

Theorem 5.8. Let Y be a normed linear space, X a subspace of Y , and $f \in Y$. Then the set U of best approximations to f from X is convex.

Proof. If U is empty, it is trivially convex. Suppose $\varphi_1, \varphi_2 \in U$, so that

$$\|f - \varphi_1\| = \|f - \varphi_2\| = \rho := \min_{\varphi \in X} \|f - \varphi\|.$$

For any $0 \leq t \leq 1$, we have

$$\begin{aligned} \rho &\leq \|f - (t\varphi_1 + (1-t)\varphi_2)\| \\ &= \|t(f - \varphi_1) + (1-t)(f - \varphi_2)\| \\ &\leq t\|f - \varphi_1\| + (1-t)\|f - \varphi_2\| \\ &= \rho. \end{aligned}$$

Thus $\|f - (t\varphi_1 + (1-t)\varphi_2)\| = \rho$ and $t\varphi_1 + (1-t)\varphi_2 \in U$, proving that U is convex. $\square$

Definition 5.9 (Strictly convex set). Let Y be a normed vector space. A subset $U \subset Y$ is said to be *strictly convex* if it is convex and for every pair of distinct points $x, y \in \partial U$, the open line segment $\{tx + (1-t)y : 0 < t < 1\}$ does not intersect ∂U .

Definition 5.10 (Strictly convex norm). A normed vector space $(Y, \|\cdot\|)$ is said to have a *strictly convex norm* if its closed unit ball $B = \{v \in Y : \|v\| \leq 1\}$ is strictly convex. A *strictly convex space* is a normed vector space with strictly convex norm.

Lemma 5.11. A normed vector space Y is strictly convex if and only if for every pair of distinct points $v_1, v_2 \in Y$ with $\|v_1\| = \|v_2\| = 1$, we have $\|tv_1 + (1-t)v_2\| < 1$ for all $0 < t < 1$.

Proof. This follows from Definition 5.9 and Definition 5.10. $\square$

Theorem 5.12 (Uniqueness of best approximation). Let Y be a strictly convex space and X be a subspace of Y . Then for any $f \in Y$, there exists at most one best approximation of f from X .

Proof. Suppose $\varphi_1, \varphi_2 \in X$ are two distinct best approximations of f from X and set

$$\rho := \|f - \varphi_1\| = \|f - \varphi_2\|.$$

Note that $\rho > 0$; otherwise $\varphi_1 = \varphi_2 = f$, contradicting their distinctness. Define

$$\phi_1 = \frac{f - \varphi_1}{\rho}, \quad \phi_2 = \frac{f - \varphi_2}{\rho}.$$

Then $\|\phi_1\| = \|\phi_2\| = 1$. By strict convexity of Y , we have

$$1 > \left\| \frac{\phi_1 + \phi_2}{2} \right\| = \frac{1}{\rho} \left\| f - \frac{\varphi_1 + \varphi_2}{2} \right\|.$$

Hence $\|f - \frac{\varphi_1 + \varphi_2}{2}\| < \rho$. However, by Theorem 5.8, $\frac{\varphi_1 + \varphi_2}{2}$ is also a best approximation of f , so its distance to f must equal to ρ , which is a contradiction. $\square$

Theorem 5.13. For $1 < p < \infty$, the p -norm given in Definition E.6 is strictly convex.

Proof. See [Rivlin, 1981, Chapter I]. $\square$

Example 5.14. The 1-norm is not strictly convex. For example, consider the space $(\mathcal{C}[-1, 1], \|\cdot\|_1)$ and define

$$f_1(x) = \frac{3}{2}x^2, \quad f_2(x) = \frac{3}{4}(1 - x^2).$$

A direct computation yields

$$\|f_1\|_1 = \|f_2\|_1 = \left\| \frac{f_1 + f_2}{2} \right\|_1 = 1,$$

which violates the strict convexity condition.

Exercise 5.15. Show that max-norm is not strictly convex.

5.2 Characterizing best uniform approximations

Theorem 5.16 (Existence of best uniform approximation). For any $f \in \mathcal{C}[a, b]$, there exists at least one best uniform approximation of f in $\mathbb{P}_n$.

Proof. Since $\mathbb{P}_n$ is a finite-dimensional subspace of $\mathcal{C}[a, b]$, this follows from Theorem 5.7. $\square$

Definition 5.17. A function $E(x)$ *equioscillates* on m points of $[a, b]$ if there exists $a \leq x_1 < x_2 < \cdots < x_m \leq b$ such that

$$|E(x_k)| = \|E\|_\infty = \max_{x \in [a, b]} |E(x)|, \quad 1 \leq k \leq m,$$

and

$$E(x_{k+1}) = -E(x_k), \quad 1 \leq k \leq m-1.$$

Example 5.18. Consider the function

$$E(x) = |x| - \frac{1}{8} - x^2$$

on $[-1, 1]$. It could be checked that

$$-E(-1) = E\left(\frac{1}{2}\right) = -E(0) = E\left(\frac{1}{2}\right) = -E(1) = \|E\|_\infty.$$

Therefore E equioscillates on $x = 0, \pm\frac{1}{2}, \pm 1$.

Example 5.19. The Chebyshev polynomial of degree n of the first kind $T_n(x) = \cos(n \arccos x)$ equioscillates on $n+1$ points $x'_k = \cos \frac{k}{n}\pi$ for $k = 0, 1, \dots, n$. See Section 5.7 for details.

Lemma 5.20. A best uniform approximation p_n^* of $f \in \mathcal{C}[a, b]$ in $\mathbb{P}_n$ equioscillates on at least 2 points.

Proof. Set

$$m = \min_{x \in [a, b]} (f - p_n^*), \quad M = \max_{x \in [a, b]} (f - p_n^*).$$

If $M = -m$, the extreme points of $f - p_n^*$ satisfy the property. Otherwise, $M + m \neq 0$. Define $c = \frac{M+m}{2}$ and consider $\tilde{p}_n^* = p_n^* + c$. Clearly $\tilde{p}_n^* \in \mathbb{P}_n$ and

$$\|f - \tilde{p}_n^*\| = \|f - p_n^*\| - |c| < \|f - p_n^*\|,$$

a contradiction. $\square$

Theorem 5.21 (Characterizing best uniform approximation). A polynomial p_n is a best uniform approximation of $f \in \mathcal{C}[a, b]$ in $\mathbb{P}_n$ if and only if $f - p_n$ equioscillates on at least $n + 2$ points in $[a, b]$.

Proof. Set $E(x) = f(x) - p_n(x)$. For sufficiency, assume E equioscillates on $n + 2$ points $a \leq x_1 < \dots < x_{n+2} \leq b$ but p_n is not a best approximation. By Theorem 5.16, there exists a best uniform approximation $p_n^* \in \mathbb{P}_n$. Then

$$\|f - p_n^*\|_\infty < \|f - p_n\|_\infty = |f - p_n(x_k)|, \quad 1 \leq k \leq n + 2.$$

Hence $p_n - p_n^*$ alternates in sign at $x_1 < \dots < x_{n+2}$. Consequently, it has at least $n + 1$ distinct zeros by intermediate value theorem. Being a polynomial of degree at most n , it must be identically zero, contradicting $p_n^* \neq p_n$. Thus p_n is a best approximation.

For necessity, suppose p_n is a best uniform approximation. If $E \equiv 0$, the conclusion is trivial. Otherwise, $\|E\|_\infty > 0$ and let $y_1 < y_2 < \dots < y_{k+1}$ be the points where E equioscillates. By Lemma 5.20, we have $k \geq 1$. We shall prove that $k \geq n + 1$. Assume to the contrary that $k \leq n$. Choose $x_i \in (y_i, y_{i+1})$ for $i = 1, \dots, k$. Then there is exactly one equioscillation point of E in each of the subintervals

$$[a, x_1), (x_1, x_2), \dots, (x_{k-1}, x_k), (x_k, b].$$

Define

$$q(x) = \pm c \prod_{i=1}^k (x - x_i), \quad c = \frac{1}{\left\| \prod_{i=1}^k (x - x_i) \right\|_\infty},$$

so $\|q\|_\infty = 1$. The sign is selected such that q is of same sign with E at y_1 . Since q also alternates signs at $y_1 < \dots < y_{k+1}$, it is of same sign with E at all y_i . Consider $p_n^* = p_n + \varepsilon q$ with $\varepsilon > 0$ left to be chosen. We claim that

$$\|f - p_n^*\|_\infty = \|E - \varepsilon q\|_\infty < \|E\|_\infty \quad (5.6)$$

for sufficiently small ε . Partition $[a, b]$ into

$$S^- = \{x \in [a, b] : q(x)E(x) \leq 0\},$$

$$S^+ = \{x \in [a, b] : q(x)E(x) > 0\}.$$

Since S^- is the preimage of a closed set under continuous mapping, it is closed and therefore compact by boundedness. Furthermore, S^+ includes all equioscillation points y_i by choice of sign of q . Hence for any $x \in S^-$, $|E(x)| < \|E\|_\infty$. By compactness, we have

$$d = \max_{x \in S^-} |E(x)| < \|E\|_\infty.$$

Now we could estimate $\|f - p_n^*\|_\infty = \|E - \varepsilon q\|_\infty$. There exists $\xi \in [a, b]$ such that

$$|E(\xi) - \varepsilon q(\xi)| = \|E - \varepsilon q\|_\infty.$$

If $\xi \in S^-$, then

$$\|E - \varepsilon q\|_\infty = |E(\xi)| + |\varepsilon q(\xi)| \leq d + \varepsilon \|q\|_\infty = d + \varepsilon.$$

If $\xi \in S^+$, by $q(\xi)E(\xi) > 0$, we have

$$\|E - \varepsilon q\|_\infty < \max\{|E(\xi)|, |\varepsilon q(\xi)|\} \leq \max\{\|E\|_\infty, \varepsilon\}.$$

Choosing $0 < \varepsilon < \|E\|_\infty - d$ yields (5.6), which contradicts the choice of p_n . Hence $k \geq n + 1$ and p_n equioscillates on at least $n + 2$. $\square$

Example 5.22. In Example 5.18, $E(x) = |x| - \frac{1}{8} - x^2$ equioscillates at 5 points. By Theorem 5.21, $p(x) = x^2 + \frac{1}{8}$ is the best uniform approximation of $f(x) = |x| \in \mathcal{C}[-1, 1]$ not only in $\mathbb{P}_2$, but also in $\mathbb{P}_3$. This shows that, if we seek a best uniform approximation $p_n \in \mathbb{P}_n$ for f , it may happen that $f - p_n$ equioscillates on more than $n + 2$ points. The following theorem gives a condition that ensures $f - p_n$ exactly equioscillates on $n + 2$ points.

Theorem 5.23. If $f \in \mathcal{C}^{n+1}[a, b]$ and $f^{(n+1)}$ has no zero in (a, b) , then $f - p_n$ equioscillates on $n + 2$ points and on no greater number of points, where p_n is the best uniform approximation of f in $\mathbb{P}_n$.

Proof. Clearly $f - p_n$ is not identically zero. Suppose $f - p_n$ equioscillates on $k \geq n + 3$ points. Then $f - p_n$ has at least $k - 1$ distinct zeros in (a, b) by intermediate value theorem. Applying Rolle's theorem $n + 1$ times shows that $f^{(n+1)}$ has at least $k - n - 2 \geq 1$ zeros in (a, b) , contradicting the hypothesis. Hence $f - p_n$ equioscillates on exactly $n + 2$ points. $\square$

Theorem 5.24 (Uniqueness of best uniform approximation). For any $f \in \mathcal{C}[a, b]$, there is a unique best uniform approximation of f in $\mathbb{P}_n$.

Proof. Suppose p_n and p_n^* are two distinct best uniform approximations of f and set $\rho = \|f - p_n\|_\infty = \|f - p_n^*\|_\infty$. By Theorem 5.8, $q = \frac{p_n + p_n^*}{2}$ is also a best approximation of f . By Theorem 5.21, $f - q$ equioscillates at $x_1 < \dots < x_{n+2}$. Then for some integer l , we have

$$\begin{aligned} \frac{f(x_j) - p_n(x_j)}{2} + \frac{f(x_j) - p_n^*(x_j)}{2} &= f(x_j) - q(x_j) \\ &= (-1)^{l+j} \rho, \quad 1 \leq j \leq n + 2. \end{aligned} \quad (5.7)$$

Since

$$\frac{|f(x_j) - p_n(x_j)|}{2} \leq \frac{\rho}{2}, \quad \frac{|f(x_j) - p_n^*(x_j)|}{2} \leq \frac{\rho}{2},$$

(5.7) can hold only if

$$f(x_j) - p_n(x_j) = f(x_j) - p_n^*(x_j) = (-1)^{l+j} \rho, \quad 1 \leq j \leq n + 2.$$

Thus we obtain

$$p_n(x_j) = p_n^*(x_j), \quad 1 \leq j \leq n + 2,$$

which implies that $p_n \equiv p_n^*$, a contradiction to previous assumption. Therefore the best uniform approximation in $\mathbb{P}_n$ is unique. $\square$

Definition 5.25. For $f \in \mathcal{C}[a, b]$, the *minimax polynomial* of degree n of f on $[a, b]$, denoted by p_n^* , is the best uniform approximation of f from $\mathbb{P}_n$ on $[a, b]$. We write $p_n^*(f; [a, b])$ when dependence on f and the interval needs emphasis.

Definition 5.26. For $f \in \mathcal{C}[a, b]$, the *minimax error* of degree n of f on $[a, b]$ is

$$E_n(f) = \|f - p_n^*\|_\infty, \quad (5.8)$$

and we write $E_n(f; [a, b])$ when the interval needs to be specified.

5.3 Bernstein polynomials

Definition 5.27. The *Bernstein basis polynomials* of degree $n \in \mathbb{N}^+$ relative to the unit interval $[0, 1]$ are

$$\forall k = 0, 1, \dots, n, \quad b_{n,k}(t) := \binom{n}{k} t^k (1-t)^{n-k} \quad (5.9)$$

and $b_{n,k}(t) = 0$ for all other values of the integer k .

Example 5.28. The Bernstein basis polynomials of degree 3 are t^3 , $3t^2(1-t)$, $3t(1-t)^2$, and $(1-t)^3$.

Definition 5.29. A *Bernstein polynomial* is a linear combination of Bernstein basis polynomials, i.e. $\sum_{k=0}^n \beta_k b_{n,k}$, where the coefficients are called *Bernstein coefficients* or *Bézier coefficients*.

Lemma 5.30. A Bernstein basis polynomial can always be expressed as a linear combination of Bernstein basis polynomials of higher degrees,

$$b_{n-1,k}(t) = \frac{n-k}{n} b_{n,k}(t) + \frac{k+1}{n} b_{n,k+1}(t). \quad (5.10)$$

Proof. This follows from Definition 5.27. $\square$

Lemma 5.31. The derivative of a Bernstein basis polynomial is a linear combination of two Bernstein basis polynomials of lower degrees, i.e.,

$$b'_{n,k}(t) = n b_{n-1,k-1}(t) - n b_{n-1,k}(t). \quad (5.11)$$

Proof. This follows from Definition 5.27. $\square$

Lemma 5.32. The integral of a Bernstein basis polynomial depends only on its degree, i.e.,

$$\forall k = 0, 1, \dots, n, \quad \int_0^1 b_{n,k}(t) dt = \frac{1}{n+1}. \quad (5.12)$$

Proof. This follows from (5.9) and integral by parts. $\square$

Lemma 5.33. The Bernstein basis polynomials of degree n form a basis of $\mathbb{P}_n$, the vector space of all polynomials with degree no more than n .

Proof. By Definition 5.27, we have $b_{n,k}(t) = \sum_{j=k}^n a_{k,j} t^j$ and thus

$$\begin{bmatrix} b_{n,0}(t) \\ b_{n,1}(t) \\ b_{n,2}(t) \\ \vdots \\ b_{n,n}(t) \end{bmatrix} = \begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & \cdots & a_{0,n} \\ & a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ & & a_{2,2} & \cdots & a_{2,n} \\ & & & \ddots & \vdots \\ & & & & a_{n,n} \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \\ \vdots \\ t^n \end{bmatrix},$$

where the matrix is uppertriangular with $a_{k,k} = \binom{n}{k} \neq 0$ on the diagonals. The rest of the proof follows from the list of monomials being a basis of $\mathbb{P}_n$. $\square$

Lemma 5.34. The Bernstein basis polynomials satisfy

$$\forall k = 0, 1, \dots, n, \quad \forall t \in (0, 1), \quad b_{n,k}(t) > 0 \quad (5.13a)$$

$$\sum_{k=0}^n b_{n,k}(t) = 1, \quad (5.13b)$$

$$\sum_{k=0}^n k b_{n,k}(t) = nt, \quad (5.13c)$$

$$\sum_{k=0}^n k^2 b_{n,k}(t) = n(n-1)t^2 + nt, \quad (5.13d)$$

$$\sum_{k=0}^n (k-nt)^2 b_{n,k}(t) = nt(1-t). \quad (5.13e)$$

Proof. (5.13a) follows from (5.9) and (5.13b) from the binomial theorem

$$\sum_{k=0}^n \binom{n}{k} t^k s^{n-k} = (t+s)^n. \quad (5.14)$$

Differentiate (5.14) with respect to t , multiply both sides with t , and we have

$$(*) : \quad \sum_{k=0}^n k \binom{n}{k} t^k s^{n-k} = nt(t+s)^{n-1}.$$

Then (5.13c) follows from setting $s = 1-t$ in $(*)$, (5.13d) from differentiating $(*)$ with respect to t and multiplying both sides with t , and (5.13e) from setting $s = 1-t$ in (5.13d) $-2nt(*) + n^2 t^2$ (5.14). $\square$

Definition 5.35. The n th *Bernstein polynomial of a function* $f \in \mathcal{C}[0, 1]$ is

$$(B_n f)(t) := \sum_{k=0}^n f\left(\frac{k}{n}\right) b_{n,k}(t), \quad (5.15)$$

where $b_{n,k}$ is a Bernstein basis polynomial in (5.9).

Definition 5.36. A linear operator $L : \mathcal{C}([a, b]) \rightarrow \mathcal{C}([c, d])$ is said to be *monotone* iff

$$\forall f, g \in \mathcal{C}([a, b]), \quad f \geq g \implies Lf \geq Lg, \quad (5.16)$$

where $f \geq g$ is a shorthand for $\forall x \in \text{dom}(f), f(x) \geq g(x)$.

Lemma 5.37. The *Bernstein operator* B_n defined by (5.15) is monotone.

Proof. This follows from (5.15), (5.16), and (5.13a). $\square$

5.4 The Weierstrass approximation theorem

Theorem 5.38 (Weierstrass approximation). Every continuous function $f : [a, b] \rightarrow \mathbb{R}$ can be uniformly approximated as closely as desired by a sequence of polynomials, i.e.,

$$\begin{aligned} \forall f \in \mathcal{C}[a, b], \forall \epsilon > 0, \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n > N, \\ \exists p_n \in \mathbb{P}_n \text{ s.t. } \forall x \in [a, b], |p_n(x) - f(x)| < \epsilon. \end{aligned} \quad (5.17)$$

Proof. Without loss of generality, we assume $a = 0, b = 1$. Set $p_n = B_n f$ in (5.15). For any $\epsilon > 0$, there exist $\delta > 0$ and $n \in \mathbb{N}^+$ such that

$$\begin{aligned} |(B_n f)(t) - f(t)| &= |(B_n f)(t) - f(t) \sum_{k=0}^n b_{n,k}(t)| \\ &\leq \sum_{k=0}^n |f(\frac{k}{n}) - f(t)| b_{n,k}(t) \\ &= \left(\sum_{k: |\frac{k}{n} - t| < \delta} + \sum_{k: |\frac{k}{n} - t| \geq \delta} \right) |f(\frac{k}{n}) - f(t)| b_{n,k}(t) \\ &\leq \sup_{|t-s| \leq \delta} |f(t) - f(s)| + \frac{\|f\|_\infty}{2n\delta^2} \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon, \end{aligned}$$

where the first inequality follows from (5.15) and the triangle inequality, the case $|k - nt| < n\delta$ in the second inequality follows from (5.13a) and (5.13b), the other case $|k - nt| \geq n\delta$ in the second inequality follows from (5.13e) and

$$\begin{aligned} \sum_{k: |\frac{k}{n} - t| \geq \delta} b_{n,k}(t) &\leq \sum_{k: |\frac{k}{n} - t| \geq \delta} b_{n,k}(t) \frac{(k-nt)^2}{\delta^2 n^2} \\ &\leq \sum_{k=0}^n b_{n,k}(t) \frac{(k-nt)^2}{\delta^2 n^2} = \frac{t(1-t)}{n\delta^2} \leq \frac{1}{4n\delta^2}, \end{aligned}$$

and the last inequality follows from the uniform continuity of f (cf. Theorem B.80) and the choice of $n > \frac{\|f\|_\infty}{\epsilon\delta^2}$. $\square$

Corollary 5.39. For $f \in \mathcal{C}[a, b]$, we have $E_n(f) \rightarrow 0$.

5.5 Modulus of continuity

Definition 5.40 (Modulus of continuity). The *modulus of continuity* of a function f on $[a, b]$ is a function $\omega_{f, [a, b]} : [0, \infty) \rightarrow [0, \infty]$ defined by

$$\omega_{f, [a, b]}(\delta) = \sup_{\substack{x_1, x_2 \in [a, b] \\ |x_1 - x_2| \leq \delta}} |f(x_1) - f(x_2)|. \quad (5.18)$$

When $[a, b]$ is clear from the context, we simply write $\omega_f(\delta)$ to emphasize the dependence on f ; when both f and $[a, b]$ is clear from the context or no particular function is intended, we write $\omega(\delta)$.

Definition 5.41. A function f on $[a, b]$ is said to be α -Hölder continuous or to satisfy a Hölder condition of order $\alpha > 0$ if there exists a constant $K > 0$ such that

$$|f(x_1) - f(x_2)| \leq K|x_1 - x_2|^\alpha, \quad \forall x_1, x_2 \in [a, b].$$

When $\alpha = 1$, this reduces to the definition of Lipschitz continuity.

Proposition 5.42. A function f on $[a, b]$ is α -Hölder continuous if and only if $\omega_f(\delta) \leq K\delta^\alpha$ for some constant $K > 0$.

Proof. This follows directly from Definition 5.40 and Definition 5.41. $\square$

Lemma 5.43. Suppose f is defined on $[a, b]$, then we have

$$|f(x_1) - f(x_2)| \leq \omega_f(|x_1 - x_2|), \quad \forall x_1, x_2 \in [a, b]. \quad (5.19)$$

Proof. This follows from Definition 5.40. $\square$

Lemma 5.44. The modulus of continuity ω is non-decreasing: if $0 \leq \delta_1 < \delta_2$, then $\omega(\delta_1) \leq \omega(\delta_2)$.

Lemma 5.45. A function f defined on $[a, b]$ is uniformly continuous if and only if

$$\lim_{\delta \rightarrow 0^+} \omega_f(\delta) = 0.$$

Exercise 5.46. Prove Lemma 5.44 and Lemma 5.45.

Lemma 5.47. For any $\lambda > 0$ and $\delta \geq 0$, the modulus of continuity satisfies

$$\omega(\lambda\delta) \leq (1 + \lambda)\omega(\delta).$$

Proof. Let n be the integer such that $n \leq \lambda < n + 1$. By Lemma 5.44, we have

$$\omega(\lambda\delta) \leq \omega((n + 1)\delta).$$

To estimate $\omega((n + 1)\delta)$, take any $x_1, x_2 \in [a, b]$ with $|x_1 - x_2| \leq (n + 1)\delta$. Without loss of generality, assume $x_1 < x_2$ and divide $[x_1, x_2]$ into $n + 1$ equal subintervals by the points

$$z_j = x_1 + j \frac{x_2 - x_1}{n + 1}, \quad j = 0, 1, \dots, n + 1.$$

Then $|z_j - z_{j+1}| \leq \delta$ for every $0 \leq j \leq n$. By (5.19),

$$|f(z_j) - f(z_{j+1})| \leq \omega(\delta), \quad j = 0, 1, \dots, n.$$

Consequently,

$$|f(x_1) - f(x_2)| \leq \sum_{j=0}^n |f(z_j) - f(z_{j+1})| \leq (n + 1)\omega(\delta).$$

Taking the supremum over all x_1, x_2 yields

$$\omega((n + 1)\delta) \leq (n + 1)\omega(\delta).$$

Combining the two estimates and using $n + 1 \leq \lambda + 1$, we obtain

$$\omega(\lambda\delta) \leq (n + 1)\omega(\delta) \leq (\lambda + 1)\omega(\delta),$$

which completes the proof. $\square$

Theorem 5.48. Suppose f is a bounded function on $[0, 1]$, then

$$\|f - B_n f\|_\infty \leq \frac{3}{2} \omega_f \left(\frac{1}{\sqrt{n}} \right). \quad (5.20)$$

Proof. By (5.9) and (5.19), we have

$$\begin{aligned} |f(t) - B_n f(t)| &= \left| \sum_{k=0}^n \left[f(t) - f\left(\frac{k}{n}\right) \right] b_{n,k}(t) \right| \\ &\leq \sum_{k=0}^n \left| f(t) - f\left(\frac{k}{n}\right) \right| b_{n,k}(t) \\ &\leq \sum_{k=0}^n \omega_f \left(\left| t - \frac{k}{n} \right| \right) b_{n,k}(t). \end{aligned}$$

Applying Lemma 5.47, we have

$$\begin{aligned} \omega_f \left(\left| t - \frac{k}{n} \right| \right) &= \omega_f \left(\sqrt{n} \left| t - \frac{k}{n} \right| \frac{1}{\sqrt{n}} \right) \\ &\leq \left(1 + \sqrt{n} \left| t - \frac{k}{n} \right| \right) \omega_f \left(\frac{1}{\sqrt{n}} \right). \end{aligned}$$

Therefore

$$\begin{aligned} |f(t) - B_n f(t)| &\leq \sum_{k=0}^n \left(1 + \sqrt{n} \left| t - \frac{k}{n} \right| \right) \omega_f \left(\frac{1}{\sqrt{n}} \right) b_{n,k}(t) \\ &= \omega_f \left(\frac{1}{\sqrt{n}} \right) \left[1 + \sqrt{n} \sum_{k=0}^n \left| t - \frac{k}{n} \right| b_{n,k}(t) \right]. \end{aligned}$$

Notice that

$$\begin{aligned} \sum_{k=0}^n \left| t - \frac{k}{n} \right| b_{n,k}(t) &= \sum_{k=0}^n \left(\left| t - \frac{k}{n} \right| \sqrt{b_{n,k}(t)} \right) \left(\sqrt{b_{n,k}(t)} \right) \\ &\leq \left[\sum_{k=0}^n \left(t - \frac{k}{n} \right)^2 b_{n,k}(t) \right]^{\frac{1}{2}} \left[\sum_{k=0}^n b_{n,k}(t) \right]^{\frac{1}{2}} \\ &= \left(\frac{t(1-t)}{n} \right)^{\frac{1}{2}} \leq \frac{1}{2\sqrt{n}}, \end{aligned}$$

where the second step follows from Cauchy–Schwarz inequality and the third step follows from (5.13b,e). Substituting this estimate into the previous inequality yields

$$|f(t) - B_n f(t)| \leq \frac{3}{2} \omega_f \left(\frac{1}{\sqrt{n}} \right).$$

The proof is completed by taking supremum over t on $[0, 1]$. $\square$

Corollary 5.49 (Weak estimate of minimax error). Suppose f is a bounded function on $[0, 1]$, then

$$E_n(f) \leq \frac{3}{2} \omega_f \left(\frac{1}{\sqrt{n}} \right). \quad (5.21)$$

Example 5.50. Suppose f is a bounded function on $[0, 1]$ and f is α -Hölder continuous, then the minimax error satisfies $E_n(f) = O(n^{-\alpha/2})$.

5.6 Jackson theorems

Definition 5.51 (Trigonometric system). The *trigonometric system* on $[-\pi, \pi]$ consists of the functions

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx, \dots$$

Theorem 5.52 (Orthogonality of the trigonometric system). For integers $m, n \geq 1$, the following orthogonality relations hold:

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = \pi \delta_{mn}, \quad (5.22a)$$

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \pi \delta_{mn}, \quad (5.22b)$$

$$\int_{-\pi}^{\pi} \cos mx \sin nx \, dx = 0, \quad (5.22c)$$

$$\int_{-\pi}^{\pi} 1 \cdot \cos mx \, dx = 0, \quad \int_{-\pi}^{\pi} 1 \cdot \sin mx \, dx = 0, \quad (5.22d)$$

$$\int_{-\pi}^{\pi} 1 \cdot 1 \, dx = 2\pi, \quad (5.22e)$$

where

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{if } m \neq n, \end{cases}$$

is the Kronecker delta function.

Definition 5.53 (Trigonometric polynomial). A *trigonometric polynomial* of degree n is a function of the form

$$T(\theta) = \frac{\alpha_0}{2} + \sum_{k=1}^n (\alpha_k \cos k\theta + \beta_k \sin k\theta),$$

where $\alpha_k, \beta_k \in \mathbb{R}$. It is called a *cosine polynomial* if all β_k vanish, and a *sine polynomial* if all α_k vanish.

Definition 5.54 (Fourier coefficients). For a 2π -periodic function $g \in \mathcal{C}[-\pi, \pi]$, its *Fourier coefficients* are defined by

$$\begin{aligned} a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(\phi) \cos k\phi \, d\phi, \quad k = 0, 1, 2, \dots, \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(\phi) \sin k\phi \, d\phi, \quad k = 1, 2, \dots \end{aligned} \quad (5.23)$$

The n -th *partial sum of the Fourier series* of g is the trigonometric polynomial

$$s_n(g; \theta) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos k\theta + b_k \sin k\theta).$$

Definition 5.55 (Modified partial sum). For each n , let $\{\rho_{k,n}\}_{k=1}^n$ be a sequence of real numbers. The *modified n -th partial sum* is defined by

$$q_n(g; \theta) = \frac{a_0}{2} + \sum_{k=1}^n \rho_{k,n} (a_k \cos k\theta + b_k \sin k\theta). \quad (5.24)$$

Lemma 5.56. If $g \in \mathcal{C}[-\pi, \pi]$ and has period 2π , then

$$q_n(g; \theta) = \frac{1}{\pi} \int_{-\pi}^{\pi} g(\phi + \theta) u_n(\phi) \, d\phi, \quad (5.25)$$

where

$$u_n(\phi) = \frac{1}{2} + \sum_{k=1}^n \rho_{k,n} \cos k\phi. \quad (5.26)$$

Proof. Substituting (5.23) into (5.24) and using the identity $\cos k\phi \cos k\theta + \sin k\phi \sin k\theta = \cos k(\phi - \theta)$, we have

$$\begin{aligned} q_n(g; \theta) &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(\phi) u_n(\phi - \theta) d\phi \\ &= \frac{1}{\pi} \int_{-\pi-\theta}^{\pi-\theta} g(\phi + \theta) u_n(\phi) d\phi. \end{aligned}$$

Since both g and u_n have period 2π ,

$$\frac{1}{\pi} \int_{-\pi-\theta}^{\pi-\theta} g(\phi + \theta) u_n(\phi) d\phi = \frac{1}{\pi} \int_{-\pi}^{\pi} g(\phi + \theta) u_n(\phi) d\phi,$$

and the proof is done. $\square$

Lemma 5.57.

$$|\theta| \leq \frac{\pi}{2} |\sin \theta|, \quad 0 \leq |\theta| \leq \frac{\pi}{2}.$$

Lemma 5.58.

$$\sin \theta \leq \theta, \quad \theta \geq 0.$$

Exercise 5.59. Prove Lemma 5.57 and Lemma 5.58.

Lemma 5.60. Suppose the coefficients $\rho_{1,n}, \dots, \rho_{n,n}$ are chosen so that u_n in (5.26) satisfies

$$u_n(\phi) \geq 0, \quad -\pi \leq \phi \leq \pi.$$

Then for $-\pi \leq \theta \leq \pi$,

$$|g(\theta) - q_n(g; \theta)| \leq \omega_g \left(\frac{1}{n} \right) \left[1 + \frac{n\pi}{\sqrt{2}} (1 - \rho_{1,n})^{\frac{1}{2}} \right], \quad (5.27)$$

where ω_g is short for $\omega_{g, [-\pi, \pi]}$.

Proof. By (5.22), we have $\frac{1}{\pi} \int_{-\pi}^{\pi} u_n(\phi) d\phi = 1$. Together with (5.25) and Definition 5.40, we have

$$\begin{aligned} |g(\theta) - q_n(g; \theta)| &= \left| \frac{1}{\pi} \int_{-\pi}^{\pi} [g(\theta) - g(\phi + \theta)] u_n(\phi) d\phi \right| \\ &\leq \frac{1}{\pi} \int_{-\pi}^{\pi} \omega_{g, [\theta-\pi, \theta+\pi]}(|\phi|) u_n(\phi) d\phi. \end{aligned}$$

Since g is 2π -periodic, it is not hard to prove that (Exercise 5.61)

$$\omega_{g, [\theta-\pi, \theta+\pi]}(|\phi|) = \omega_{g, [-\pi, \pi]}(|\phi|) := \omega_g(|\phi|).$$

Thus

$$|g(\theta) - q_n(g; \theta)| \leq \frac{1}{\pi} \int_{-\pi}^{\pi} \omega_g(|\phi|) u_n(\phi) d\phi.$$

By Lemma 5.47,

$$\omega_g(|\phi|) = \omega_g \left(n|\phi| \cdot \frac{1}{n} \right) \leq (1 + n|\phi|) \omega_g \left(\frac{1}{n} \right),$$

and hence

$$|g(\theta) - q_n(g; \theta)| \leq \omega_g \left(\frac{1}{n} \right) \left[1 + \frac{n}{\pi} \int_{-\pi}^{\pi} |\phi| u_n(\phi) d\phi \right]. \quad (5.28)$$

Now by Cauchy–Schwarz’s inequality for integral,

$$\begin{aligned} \frac{1}{\pi} \int_{-\pi}^{\pi} |\phi| u_n(\phi) d\phi &= \frac{1}{\pi} \int_{-\pi}^{\pi} (|\phi| [u_n(\phi)]^{\frac{1}{2}}) ([u_n(\phi)]^{\frac{1}{2}}) d\phi \\ &\leq \left[\frac{1}{\pi} \int_{-\pi}^{\pi} \phi^2 u_n(\phi) d\phi \right]^{\frac{1}{2}} \cdot \left[\frac{1}{\pi} \int_{-\pi}^{\pi} u_n(\phi) d\phi \right]^{\frac{1}{2}} \\ &= \left[\frac{1}{\pi} \int_{-\pi}^{\pi} \phi^2 u_n(\phi) d\phi \right]^{\frac{1}{2}}. \end{aligned} \quad (5.29)$$

According to Lemma 5.57, if $-\pi \leq \phi \leq \pi$,

$$\left(\frac{\phi}{2} \right)^2 \leq \frac{\pi^2}{4} \sin^2 \frac{\phi}{2} = \frac{\pi^2}{4} \frac{(1 - \cos \phi)}{2}.$$

Substituting this into (5.28) and using the identity in (5.22), we have

$$\begin{aligned} \frac{1}{\pi} \int_{-\pi}^{\pi} |\phi| u_n(\phi) d\phi &\leq \left[\frac{\pi}{2} \int_{-\pi}^{\pi} (1 - \cos \phi) u_n(\phi) d\phi \right]^{\frac{1}{2}} \\ &= \frac{\pi}{\sqrt{2}} (1 - \rho_{1,n})^{\frac{1}{2}}. \end{aligned} \quad (5.30)$$

Then the proof is done by substituting (5.30) into (5.28). $\square$

Exercise 5.61. Complete the proof of Lemma 5.60 by showing that if f has period T , then $\omega_{f, [a, a+T]}(\delta)$ is independent of a .

Lemma 5.62. Let $c_0, \dots, c_n$ be real numbers. Then the relation

$$u_n(\phi) = \left| \sum_{k=0}^n c_k e^{ik\phi} \right|^2$$

gives a nonnegative u_n and holds if

$$\begin{aligned} \sum_{k=0}^n c_k^2 &= \frac{1}{2}, \\ \rho_{l,n} &= 2 \sum_{k=0}^{n-l} c_k c_{k+l}, \quad l = 1, \dots, n. \end{aligned} \quad (5.31)$$

Proof. This follows from expansion of the expression and comparison of the corresponding terms. $\square$

Lemma 5.63. Choose

$$c_k = c \sin \frac{k+1}{n+2} \pi, \quad k = 0, 1, \dots, n,$$

where the constant c is determined by the normalization condition $\sum_{k=0}^n c_k^2 = \frac{1}{2}$, i.e.,

$$c^2 = \frac{1}{2 \sum_{k=0}^n \sin^2 \frac{k+1}{n+2} \pi}.$$

Then for the corresponding $\rho_{l,n}$ obtained from Lemma 5.62, we have

$$\rho_{1,n} = \cos \frac{\pi}{n+2}.$$

Proof. By Lemma 5.62, we have

$$\rho_{1,n} = 2c^2 \sum_{k=0}^{n-1} \sin \frac{k+1}{n+2} \pi \cdot \sin \frac{k+2}{n+2} \pi.$$

Notice that

$$\sum_{k=0}^{n-1} \sin \frac{k+1}{n+2} \pi \cdot \sin \frac{k+2}{n+2} \pi = \sum_{k=0}^n \sin \frac{k+1}{n+2} \pi \cdot \sin \frac{k+2}{n+2} \pi$$

and

$$\sum_{k=0}^{n-1} \sin \frac{k+1}{n+2} \pi \cdot \sin \frac{k+2}{n+2} \pi = \sum_{k=0}^n \sin \frac{k}{n+2} \pi \cdot \sin \frac{k+1}{n+2} \pi,$$

together we have

$$\begin{aligned} & \sum_{k=0}^{n-1} \sin \frac{k+1}{n+2} \pi \cdot \sin \frac{k+2}{n+2} \pi \\ &= \frac{1}{2} \sum_{k=0}^n \left(\sin \frac{k}{n+2} \pi + \sin \frac{k+2}{n+2} \pi \right) \cdot \sin \frac{k+1}{n+2} \pi \\ &= \sum_{k=0}^n \cos \frac{\pi}{n+2} \cdot \sin^2 \frac{k+1}{n+2} \pi \\ &= \frac{1}{2c^2} \cos \frac{\pi}{n+2}. \end{aligned}$$

Hence $\rho_{1,n} = \cos \frac{\pi}{n+2}$. $\square$

5.7 Chebyshev polynomials

Definition 5.64. The *Chebyshev polynomial* of degree $n \in \mathbb{N}$ of the first kind is a polynomial $T_n : [-1, 1] \rightarrow [-1, 1]$,

$$T_n(x) = \cos(n \arccos x). \quad (5.32)$$

Theorem 5.65. The Chebyshev polynomials of the first kind satisfy the following recursive relations,

$$\forall n \in \mathbb{N}^+, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x). \quad (5.33)$$

Proof. By trigonometric identities, we have

$$\begin{aligned} \cos(n+1)\theta &= \cos n\theta \cos \theta - \sin n\theta \sin \theta, \\ \cos(n-1)\theta &= \cos n\theta \cos \theta + \sin n\theta \sin \theta. \end{aligned}$$

The proof is then completed by adding up the two equations and setting $\cos \theta = x$. $\square$

Corollary 5.66. The coefficient of x^n in T_n is 2^{n-1} for each $n > 0$.

Proof. Use (5.33) and $T_1 = x$ in an induction. $\square$

Theorem 5.67. $T_n(x)$ has simple zeros at the n points

$$x_k = \cos \frac{2k-1}{2n} \pi, \quad (5.34)$$

where $k = 1, 2, \dots, n$. For $x \in [-1, 1]$ and $n \in \mathbb{N}^+$, $T_n(x)$ has extreme values at the $n+1$ points

$$x'_k = \cos \frac{k}{n} \pi, \quad k = 0, 1, \dots, n, \quad (5.35)$$

where it assumes the alternating values $(-1)^k$.

Proof. (5.32) and (5.34) yield

$$T_n(x_k) = \cos \left(n \arccos \left(\cos \frac{2k-1}{2n} \pi \right) \right) = \cos \left(\frac{2k-1}{2} \pi \right) = 0.$$

Differentiate (5.32) and we have

$$T'_n(x) = \frac{n}{\sqrt{1-x^2}} \sin(n \arccos x).$$

Then each x_k must be a simple zero since

$$T'_n(x_k) = \frac{n}{\sqrt{1-x_k^2}} \sin \left(\frac{2k-1}{2} \pi \right) \neq 0.$$

In contrast, $\forall k = 1, 2, \dots, n-1$,

$$T'_n(x'_k) = n \left(1 - \cos^2 \frac{k\pi}{n} \right)^{-\frac{1}{2}} \sin(k\pi) = 0;$$

$$T''_n(x) = \frac{n^2 \cos(n \arccos(x))}{x^2 - 1} + \frac{nx \sin(n \arccos(x))}{(1-x^2)^{3/2}};$$

$$T''_n(x'_k) \neq 0.$$

Hence a Taylor expansion of T_n yields

$$T_n(x'_k + \delta) = T_n(x'_k) + \frac{1}{2} T''_n(x'_k) \delta^2 + O(\delta^3),$$

and T_n must attain local extremes at each x'_k . For any $k = 0, 1, \dots, n$, $T_n(x)$ attains its extreme values at x'_k since $T_n(x'_0) = 1$, $T_n(x'_1) = -1$, ..., and by (5.32) we have $|T_n(x)| \leq 1$. Clearly these are the only extrema of $T_n(x)$ on $[-1, 1]$. $\square$

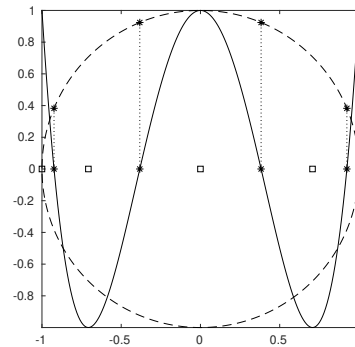


Figure 5.1: Zeros and extrema of Chebyshev polynomials with $n = 4$.

Theorem 5.68 (Chebyshev). Denote by $\tilde{\mathbb{P}}_n$ the class of all polynomials of degree $n \in \mathbb{N}^+$ with leading coefficient 1. Then

$$\forall p \in \tilde{\mathbb{P}}_n, \quad \max_{x \in [-1, 1]} \left| \frac{T_n(x)}{2^{n-1}} \right| \leq \max_{x \in [-1, 1]} |p(x)|. \quad (5.36)$$

Proof. By Theorem 5.67, $T_n(x)$ assumes its extrema $n + 1$ times at the points x'_k defined in (5.35). Suppose (5.36) does not hold. Then Theorem 5.67 implies that

$$\exists p \in \tilde{\mathbb{P}}_n \text{ s.t. } \max_{x \in [-1, 1]} |p(x)| < \frac{1}{2^{n-1}}. \quad (5.37)$$

Consider the polynomial $Q(x) = \frac{1}{2^{n-1}}T_n(x) - p(x)$.

$$Q(x'_k) = \frac{(-1)^k}{2^{n-1}} - p(x'_k), \quad k = 0, 1, \dots, n.$$

By (5.37), $Q(x)$ has alternating signs at these $n + 1$ points. Hence $Q(x)$ must have n zeros. However, by the construction of $Q(x)$, the degree of $Q(x)$ is at most $n - 1$. Therefore, $Q(x) \equiv 0$ and $p(x) = \frac{1}{2^{n-1}}T_n(x)$, which implies $\max |p(x)| = \frac{1}{2^{n-1}}$. This is a contradiction to (5.37). $\square$

Corollary 5.69. For $n \in \mathbb{N}^+$, we have

$$\max_{x \in [-1, 1]} |x^n + a_1 x^{n-1} + \dots + a_n| \geq \frac{1}{2^{n-1}}. \quad (5.38)$$

5.8 Theoretical problems

- I. Give a detailed proof of Lemma 5.30.

- II. Give a detailed proof of Lemma 5.32.

- III. A min-max problem. For $n \in \mathbb{N}^+$ and a fixed $a_0 \neq 0$, determine $\min \max_{x \in [a, b]} |a_0 x^n + a_1 x^{n-1} + \dots + a_n|$ where the minimum is taken over all $a_i \in \mathbb{R}$, $i = 1, 2, \dots, n$.

- IV. Imitate the proof of Chebyshev Theorem.

Express the Chebyshev polynomial of degree $n \in \mathbb{N}$ as a polynomial T_n and change its domain from $[-1, 1]$ to $\mathbb{R}$. For a fixed $a > 1$, define $\mathbb{P}_n^a := \{p \in \mathbb{P}_n : p(a) = 1\}$ and a polynomial $\hat{p}_n(x) \in \mathbb{P}_n^a$,

$$\hat{p}_n(x) := \frac{T_n(x)}{T_n(a)}.$$

For $\|f\|_\infty = \max_{x \in [-1, 1]} |f(x)|$, prove

$$\forall p \in \mathbb{P}_n^a, \quad \|\hat{p}_n\|_\infty \leq \|p\|_\infty.$$

5.9 Programming assignments

- A. Write a program to reproduce Figure 5.1.

Chapter 6

Polynomial Interpolation

Definition 6.1. *Interpolation* is the process of estimating unknown values that lie between given data points. Different from *extrapolation* that predicts values outside the known range, interpolation constructs a function, called an *interpolant*, that goes through a discrete set of known points to determine intermediate values.

Example 6.2. The interpolant may be a polynomial or a non-polynomial function such as a rational function and a piecewise-polynomial spline. Regardless of the form of the interpolant, we need to control the uncertainty of interpolation by determining the maximal difference between the original function and the interpolant.

6.1 The Vandermonde determinant

Definition 6.3. For n given parameters $t_1, t_2, \dots, t_n$, the associated n -by- n *Vandermonde matrix* V has its (i, j) th element as $v_{i,j} = t_j^{i-1}$; in matrix form, we have

$$V(t_1, t_2, \dots, t_n) = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ t_1 & t_2 & \cdots & t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_1^{n-1} & t_2^{n-1} & \cdots & t_n^{n-1} \end{bmatrix}. \quad (6.1)$$

Lemma 6.4. The determinant of a $(n+1)$ -by- $(n+1)$ Vandermonde matrix can be expressed as

$$\det V(x_0, x_1, \dots, x_n) = \prod_{i>j} (x_i - x_j). \quad (6.2)$$

Proof. Consider the function

$$U(x) = \det V(x_0, x_1, \dots, x_{n-1}, x) = \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ x_0 & x_1 & \cdots & x_{n-1} & x \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_0^n & x_1^n & \cdots & x_{n-1}^n & x^n \end{vmatrix}. \quad (6.3)$$

Clearly, $U(x) \in \mathbb{P}_n$ and it vanishes at $x_0, x_1, \dots, x_{n-1}$ since inserting these values in place of x yields two identical columns in the determinant. It follows that

$$U(x) = A \prod_{i=0}^{n-1} (x - x_i),$$

where A depends only on $x_0, x_1, \dots, x_{n-1}$. Meanwhile, the expansion of $U(x)$ in (6.3) by minors of its last column implies that the coefficient of x^n is $\det V(x_0, x_1, \dots, x_{n-1})$. Hence we have

$$U(x) = \det V(x_0, x_1, \dots, x_{n-1}) \prod_{i=0}^{n-1} (x - x_i),$$

and consequently the recursion

$$\det V(x_0, x_1, \dots, x_n) = \det V(x_0, x_1, \dots, x_{n-1}) \prod_{i=0}^{n-1} (x_n - x_i).$$

An induction based on $U(x_0, x_1) = x_1 - x_0$ yields (6.2). $\square$

Theorem 6.5 (Uniqueness of polynomial interpolation). Given distinct points $x_0, x_1, \dots, x_n \in \mathbb{C}$ and corresponding values $f_0, f_1, \dots, f_n \in \mathbb{C}$. Denote by $\mathbb{P}_n$ the class of polynomials of degree at most n . There exists a unique polynomial $p_n(x) \in \mathbb{P}_n$ such that

$$\forall i = 0, 1, \dots, n, \quad p_n(x_i) = f_i. \quad (6.4)$$

Proof. Set up a polynomial $\sum_{i=0}^n a_i x^i$ with $n+1$ undetermined coefficients a_i . The condition (6.4) leads to the system of $n+1$ equations:

$$a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n = f_i,$$

where $i = 0, 1, \dots, n$. By Lemma 6.4, the determinant of the system is $\prod_{i>j} (x_i - x_j)$. The proof is completed by the distinctness of the points and Cramer's rule (Theorem C.279). $\square$

6.2 The Cauchy remainder

Theorem 6.6 (Generalized Rolle). Let $n \geq 2$. Suppose that $f \in \mathcal{C}^{n-1}[a, b]$ and $f^{(n)}(x)$ exists at each point of (a, b) . Suppose that $f(x_0) = f(x_1) = \cdots = f(x_n) = 0$ for $a \leq x_0 < x_1 < \cdots < x_n \leq b$. Then there is a point $\xi \in (x_0, x_n)$ such that $f^{(n)}(\xi) = 0$.

Proof. Applying Rolle's theorem (Theorem B.91) on the n intervals (x_i, x_{i+1}) yields n points ζ_i where $f'(\zeta_i) = 0$. Consider $f', f'', \dots, f^{(n-1)}$ as new functions. Repeatedly applying the above arguments completes the proof. $\square$

Theorem 6.7 (Cauchy remainder of polynomial interpolation). Let $f \in \mathcal{C}^n[a, b]$ and suppose that $f^{(n+1)}(x)$ exists at each point of (a, b) . Let $p_n(f; x)$ denote the unique polynomial in $\mathbb{P}_n$ that coincides with f at $x_0, x_1, \dots, x_n$. Define

$$R_n(f; x) := f(x) - p_n(f; x) \quad (6.5)$$

as the *Cauchy remainder of the polynomial interpolation*. If $a \leq x_0 < x_1 < \dots < x_n \leq b$, then there exists some $\xi \in (a, b)$ such that

$$R_n(f; x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i) \quad (6.6)$$

where the value of ξ depends on $x, x_0, x_1, \dots, x_n$, and f .

Proof. Since $f(x_k) = p_n(f; x_k)$, the remainder $R_n(f; x)$ vanishes at x_k 's. Fix $x \neq x_0, x_1, \dots, x_n$ and define

$$K(x) = \frac{f(x) - p_n(f; x)}{\prod_{i=0}^n (x - x_i)}$$

and a function of t

$$W(t) = f(t) - p_n(f; t) - K(x) \prod_{i=0}^n (t - x_i).$$

The function $W(t)$ vanishes at $t = x_0, x_1, \dots, x_n$. In addition $W(x) = 0$. By Theorem 6.6, $W^{(n+1)}(\xi) = 0$ for some $\xi \in (a, b)$, i.e.

$$0 = W^{(n+1)}(\xi) = f^{(n+1)}(\xi) - (n+1)!K(x).$$

Hence $K(x) = f^{(n+1)}(\xi)/(n+1)!$ and (6.6) holds. $\square$

Corollary 6.8. Suppose $f(x) \in \mathcal{C}^{n+1}[a, b]$. Then

$$|R_n(f; x)| \leq \frac{M_{n+1}}{(n+1)!} \prod_{i=0}^n |x - x_i| \leq \frac{M_{n+1}}{(n+1)!} (b-a)^{n+1}, \quad (6.7)$$

where $M_{n+1} = \max_{x \in [a, b]} |f^{(n+1)}(x)|$.

Example 6.9. A value for $\arcsin(0.5335)$ is obtained by interpolating linearly between the values for $x = 0.5330$ and $x = 0.5340$. Estimate the error committed.

Let $f(x) = \arcsin(x)$. Then

$$f''(x) = x(1-x^2)^{-\frac{3}{2}}, \quad f'''(x) = (1+2x^2)(1-x^2)^{-\frac{5}{2}}.$$

Since the third derivative is positive over $[0.5330, 0.5340]$. The maximum value of f''' occurs at 0.5340. By Corollary 6.8 we have $|R_1| \leq 4.42 \times 10^{-7}$. The true error is about 1.10×10^{-7} .

6.3 The Lagrange formula

Definition 6.10. To interpolate given values $f_0, f_1, \dots, f_n$ at distinct points $x_0, x_1, \dots, x_n$, the *Lagrange formula* is

$$p_n(x) = \sum_{k=0}^n f_k \ell_k(x), \quad (6.8)$$

where the *fundamental polynomial for pointwise interpolation* (or *elementary Lagrange interpolation polynomial*) $\ell_k(x)$ is

$$\ell_k(x) = \prod_{i \neq k; i=0}^n \frac{x - x_i}{x_k - x_i}. \quad (6.9)$$

In particular, for $n = 0$, $\ell_0 = 1$.

Example 6.11. For $i = 0, 1, 2$, we are given $x_i = 1, 2, 4$ and $f(x_i) = 8, 1, 5$, respectively. The Lagrangian formula generates $p_2(x) = 3x^2 - 16x + 21$.

Lemma 6.12. For n distinct coordinates $x_0, x_1, \dots, x_{n-1}$, define a symmetric polynomial

$$\pi_n(x) = \begin{cases} 1, & n = 0; \\ \prod_{i=0}^{n-1} (x - x_i), & n > 0. \end{cases} \quad (6.10)$$

Then for $n > 0$ the fundamental polynomial for pointwise interpolation can be expressed as

$$\forall x \neq x_k, \quad \ell_k(x) = \frac{\pi_{n+1}(x)}{(x - x_k) \pi'_{n+1}(x_k)}. \quad (6.11)$$

Proof. By the chain rule, $\pi'_{n+1}(x)$ is the summation of $n+1$ terms, each of which is a product of n terms. When x is replaced with x_k , all of the $n+1$ terms vanish except one. $\square$

Lemma 6.13 (Cauchy relations). The fundamental polynomials $\ell_k(x)$ satisfy the Cauchy relations:

$$\sum_{k=0}^n \ell_k(x) \equiv 1; \quad (6.12)$$

$$\forall j = 1, \dots, n, \quad \sum_{k=0}^n (x_k - x)^j \ell_k(x) \equiv 0. \quad (6.13)$$

Proof. By Theorems 6.5 and 6.7, for each $q(x) \in \mathbb{P}_n$ we have $p_n(q; x) \equiv q(x)$. Interpolating the constant function $f(x) \equiv 1$ with the Lagrange formula yields (6.12).

Similarly, define $q_j(u) := (u - x)^j$ where x is a free parameter. Then we have, for any $j = 1, \dots, n$,

$$p_n(q_j; u) \equiv q_j(u) \Rightarrow \sum_{k=0}^n (x_k - x)^j \ell_k(u) \equiv (u - x)^j,$$

which yields (6.13) when we set $u = x$. $\square$

6.4 The Newton formula

Definition 6.14 (Divided difference and the Newton formula). The *Newton formula* for interpolating the values $f_0, f_1, \dots, f_n$ at distinct points $x_0, x_1, \dots, x_n$ is

$$p_n(x) = \sum_{k=0}^n a_k \pi_k(x), \quad (6.14)$$

where π_k is defined in (6.10) and the k th *divided difference* a_k is defined as the coefficient of x^k in $p_k(f; x)$ and is denoted by $f[x_0, x_1, \dots, x_k]$ or $[x_0, x_1, \dots, x_k]f$. In particular, $f[x_0] = f(x_0)$.

Corollary 6.15. Suppose $(i_0, i_1, i_2, \dots, i_k)$ is a permutation of $(0, 1, 2, \dots, k)$. Then

$$f[x_0, x_1, \dots, x_k] = f[x_{i_0}, x_{i_1}, \dots, x_{i_k}]. \quad (6.15)$$

Proof. The interpolating polynomial does not depend on the numbering of the interpolating nodes. The rest of the proof follows from the uniqueness of the interpolating polynomial in Theorem 6.5. $\square$

Corollary 6.16. The k th divided difference can be expressed as

$$f[x_0, x_1, \dots, x_k] = \sum_{i=0}^k \frac{f_i}{\prod_{j \neq i; j=0}^k (x_i - x_j)} = \sum_{i=0}^k \frac{f_i}{\pi'_{k+1}(x_i)}, \quad (6.16)$$

where $\pi_{k+1}(x)$ is defined in (6.10).

Proof. The uniqueness of interpolating polynomials in Theorem 6.5 implies that the two polynomials in (6.8) and (6.14) are the same. Then the first equality follows from (6.9) and Definition 6.14, while the second equality follows from Lemma 6.12. $\square$

Theorem 6.17. Divided differences satisfy the recursion

$$f[x_0, x_1, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}. \quad (6.17)$$

Proof. By Definition 6.14, $f[x_1, x_2, \dots, x_k]$ is the coefficient of x^{k-1} in a degree- $(k-1)$ interpolating polynomial, say, $P_2(x)$. Similarly, let $P_1(x)$ be the interpolating polynomial whose coefficient of x^{k-1} is $f[x_0, x_1, \dots, x_{k-1}]$. Construct a polynomial

$$P(x) = P_1(x) + \frac{x - x_0}{x_k - x_0} (P_2(x) - P_1(x)).$$

Clearly $P(x_0) = P_1(x_0)$. Furthermore, the interpolation condition implies $P_2(x_i) = P_1(x_i)$ for $i = 1, 2, \dots, k-1$. Hence $P(x_i) = P_1(x_i)$ for $i = 1, 2, \dots, k-1$. Lastly, $P(x_k) = P_2(x_k)$. Therefore, $P(x)$ as above is the interpolating polynomial for the given values at the $k+1$ points. In particular, the term $f[x_0, x_1, \dots, x_k]x^k$ in $P(x)$ is contained in $\frac{x}{x_k - x_0}(P_2(x) - P_1(x))$. The rest follows from Definition 6.14. $\square$

Definition 6.18. The k th divided difference ($k \in \mathbb{N}^+$) on the *table of divided differences*

$$\begin{array}{c|cccc} x_0 & f[x_0] & & & \\ x_1 & f[x_1] & f[x_0, x_1] & & \\ x_2 & f[x_2] & f[x_1, x_2] & f[x_0, x_1, x_2] & \\ x_3 & f[x_3] & f[x_2, x_3] & f[x_1, x_2, x_3] & f[x_0, x_1, x_2, x_3] \\ \dots & \dots & \dots & \dots & \dots \end{array}$$

is calculated as the difference of the entry immediately to the left and the one above it, divided by the difference of the x -value horizontal to the left and the one corresponding to the f -value found by going diagonally up.

Example 6.19. Derive the interpolating polynomial via the Newton formula for the function f with given values as follows. Then estimate $f(\frac{3}{2})$.

$$\begin{array}{c|cccc} x & 0 & 1 & 2 & 3 \\ \hline f(x) & 6 & -3 & -6 & 9 \end{array}$$

By Definition 6.18, we can construct the following table of divided difference,

$$\begin{array}{c|cccc} 0 & 6 & & & \\ 1 & -3 & -9 & & \\ 2 & -6 & -3 & 3 & \\ 3 & 9 & 15 & 9 & 2 \end{array} \quad (6.18)$$

By Definition 6.14, the interpolating polynomial is generated from the main diagonal and the first column of the above table as follows.

$$p_3 = 6 - 9x + 3x(x-1) + 2x(x-1)(x-2). \quad (6.19)$$

Hence $f(\frac{3}{2}) \approx p_3(\frac{3}{2}) = -6$.

Exercise 6.20. Redo Example 6.11 with the Newton formula.

Theorem 6.21. For distinct points $x_0, x_1, \dots, x_n$, and x , we have

$$\begin{aligned} f(x) &= f[x_0] + f[x_0, x_1](x - x_0) + \dots \\ &+ f[x_0, x_1, \dots, x_n] \prod_{i=0}^{n-1} (x - x_i) \\ &+ f[x_0, x_1, \dots, x_n, x] \prod_{i=0}^n (x - x_i). \end{aligned} \quad (6.20)$$

Proof. Take another point $z \neq x_i$. The Newton formula applied to $x_0, x_1, \dots, x_n, z$ yields an interpolating polynomial

$$\begin{aligned} Q(x) &= f[x_0] + f[x_0, x_1](x - x_0) + \dots \\ &+ f[x_0, x_1, \dots, x_n] \prod_{i=0}^{n-1} (x - x_i) \\ &+ f[x_0, x_1, \dots, x_n, z] \prod_{i=0}^n (x - x_i). \end{aligned}$$

The interpolation condition $Q(z) = f(z)$ yields

$$\begin{aligned} f(z) = Q(z) &= f[x_0] + f[x_0, x_1](z - x_0) + \dots \\ &+ f[x_0, x_1, \dots, x_n] \prod_{i=0}^{n-1} (z - x_i) \\ &+ f[x_0, x_1, \dots, x_n, z] \prod_{i=0}^n (z - x_i). \end{aligned}$$

Replacing the dummy variable z with x yields (6.20).

The above argument assumes $x \neq x_i$. We now consider the case of $x = x_j$ for some fixed j . Rewrite (6.20) as $f(x) = p_n(f; x) + R(x)$ where $R(x)$ is clearly the last term in (6.20). We need to show

$$\forall j = 0, 1, \dots, n, \quad p_n(f; x_j) + R(x_j) - f(x_j) = 0,$$

where $p_n(f; x_j)$ is the value of $p_n(f; x)$ at $x = x_j$; this clearly holds because $R(x_j) = 0$ and the interpolation condition at x_j dictates $p_n(f; x_j) = f(x_j)$. $\square$

Corollary 6.22. Suppose $f \in \mathcal{C}^n[a, b]$ and $f^{(n+1)}(x)$ exists at each point of (a, b) . If $a = x_0 < x_1 < \dots < x_n = b$ and $x \in [a, b]$, then there exists $\xi(x) \in (a, b)$ such that

$$f[x_0, x_1, \dots, x_n, x] = \frac{1}{(n+1)!} f^{(n+1)}(\xi(x)). \quad (6.21)$$

Proof. This follows from Theorems 6.21 and 6.7. $\square$

Corollary 6.23. If $x_0 < x_1 < \dots < x_n$ and $f \in \mathcal{C}^n[x_0, x_n]$, we have

$$\lim_{x_n \rightarrow x_0} f[x_0, x_1, \dots, x_n] = \frac{1}{n!} f^{(n)}(x_0). \quad (6.22)$$

Proof. Set $x = x_{n+1}$ in Corollary 6.22, replace $n+1$ by n , and we have $\xi \rightarrow x_0$ as $x_n \rightarrow x_0$. $\square$

Definition 6.24. A *bisequence* is a function $f: \mathbb{Z} \rightarrow \mathbb{R}$.

Definition 6.25. The *forward shift* E and the *backward shift* B are linear operators $V \mapsto V$ on the linear space V of bisequences given by

$$(Ef)(i) = f(i+1), \quad (Bf)(i) = f(i-1). \quad (6.23)$$

The *forward difference* Δ and the *backward difference* ∇ are linear operators $V \mapsto V$ given by

$$\Delta = E - I, \quad \nabla = I - B, \quad (6.24)$$

where I is the identity operator on V .

Example 6.26. With the notation $f_i := f(i)$ for a bisequence f , the n th forward difference and the n th backward difference are

$$\Delta^n f_i := (\Delta^n f)(i), \quad \nabla^n f_i := (\nabla^n f)(i). \quad (6.25)$$

In particular, for $n = 1$ we have

$$\Delta f_i = f_{i+1} - f_i, \quad \nabla f_i = f_i - f_{i-1}. \quad (6.26)$$

Theorem 6.27. The forward difference and backward difference satisfy

$$\forall n \in \mathbb{N}^+, \quad \Delta^n f_i = \nabla^n f_{i+n}. \quad (6.27)$$

Proof. An easy induction. $\square$

Theorem 6.28. The forward difference can be expressed explicitly as

$$\Delta^n f_i = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_{i+k}. \quad (6.28)$$

Proof. For $n = 1$, (6.28) reduces to $\Delta f_i = f_{i+1} - f_i$. The rest of the proof is an induction utilizing the identity

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}. \quad (6.29)$$

Suppose (6.28) holds. For the inductive step, we have

$$\begin{aligned} \Delta^{n+1} f_i &= \Delta \Delta^n f_i = \Delta \left(\sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_{i+k} \right) \\ &= \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_{i+k+1} - \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_{i+k} \\ &= \sum_{k=1}^{n+1} (-1)^{n+1-k} \binom{n}{k-1} f_{i+k} + f_{i+n+1} \\ &\quad + \sum_{k=1}^n (-1)^{n+1-k} \binom{n}{k} f_{i+k} + (-1)^{n+1} f_i \\ &= \sum_{k=0}^{n+1} (-1)^{n+1-k} \binom{n+1}{k} f_{i+k}, \end{aligned}$$

where the second line follows from (6.26), the third line from splitting one term out of each sum and replacing the dummy variable in the first sum, and the fourth line from (6.29) and the fact that $(-1)^{n+1} f_i$ and f_{i+n+1} contribute to the first and last terms, respectively. $\square$

Theorem 6.29. On a grid $x_i = x_0 + ih$ with uniform spacing h , the sequence of values $f_i = f(x_i)$ satisfies

$$\forall n \in \mathbb{N}^+, \quad f[x_0, x_1, \dots, x_n] = \frac{\Delta^n f_0}{n! h^n}. \quad (6.30)$$

Proof. Of course (6.30) can be proven by induction. Here we provide a more informative proof. For $\pi_{n+1}(x)$ defined in (6.10), we have $\pi'_{n+1}(x_k) = \prod_{i=0, i \neq k}^n (x_k - x_i)$. It follows from $x_k - x_i = (k-i)h$ that

$$\pi'_{n+1}(x_k) = \prod_{i=0, i \neq k}^n (k-i)h = h^n k! (n-k)! (-1)^{n-k}. \quad (6.31)$$

Then we have

$$\begin{aligned} f[x_0, x_1, \dots, x_n] &= \sum_{k=0}^n \frac{f_k}{\pi'_{n+1}(x_k)} = \sum_{k=0}^n \frac{(-1)^{n-k} f_k}{h^n k! (n-k)!} \\ &= \frac{1}{h^n n!} \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_k = \frac{\Delta^n f_0}{h^n n!}, \end{aligned}$$

where the first step follows from Corollary 6.16, the second from (6.31), and the last from Theorem 6.28. $\square$

Theorem 6.30 (Newton's forward difference formula). Suppose $p_n(f; x) \in \mathbb{P}_n$ interpolates $f(x)$ on a uniform grid $x_i = x_0 + ih$ at $x_0, x_1, \dots, x_n$ with $f_i = f(x_i)$. Then

$$\forall s \in \mathbb{R}, \quad p_n(f; x_0 + sh) = \sum_{k=0}^n \binom{s}{k} \Delta^k f_0, \quad (6.32)$$

where $\Delta^0 f_0 = f_0$ and

$$\binom{s}{k} = \frac{s(s-1) \cdots (s-k+1)}{k!}. \quad (6.33)$$

Proof. Set $f(x) = p_n(f; x)$ in Theorem 6.21, apply Theorem 6.29, and we have

$$p(x) = f_0 + \sum_{k=1}^n \frac{\Delta^k f_0}{k! h^k} \prod_{i=0}^{k-1} (x - x_i);$$

the remainder is zero because any $(n + 1)$ th divided difference applied to a degree n polynomial is zero. The proof is completed by $x = x_0 + sh$, $x_i = x_0 + ih$, and (6.33). $\square$

6.5 The Neville-Aitken algorithm

Theorem 6.31. Denote $p_0^{[i]} = f(x_i)$ for $i = 0, 1, \dots, n$. For all $k = 0, 1, \dots, n - 1$ and $i = 0, 1, \dots, n - k - 1$, define

$$p_{k+1}^{[i]}(x) = \frac{(x - x_i)p_k^{[i+1]}(x) - (x - x_{i+k+1})p_k^{[i]}(x)}{x_{i+k+1} - x_i}. \quad (6.34)$$

Then each $p_k^{[i]}$ is the interpolating polynomial for the function f at the points $x_i, x_{i+1}, \dots, x_{i+k}$. In particular, $p_n^{[0]}$ is the interpolating polynomial of degree n for the function f at the points $x_0, x_1, \dots, x_n$.

Proof. The induction basis clearly holds for $k = 0$ because of the definition $p_0^{[i]} = f(x_i)$. Suppose that $p_k^{[i]}$ is the interpolating polynomial of degree k for the function f at the points $x_i, x_{i+1}, \dots, x_{i+k}$. Then the interpolation conditions yield

$$\forall j = i + 1, i + 2, \dots, i + k, \quad p_k^{[i+1]}(x_j) = p_k^{[i]}(x_j) = f(x_j),$$

which, together with (6.34), implies

$$\forall j = i + 1, i + 2, \dots, i + k, \quad p_{k+1}^{[i]}(x_j) = f(x_j).$$

In addition, (6.34) and the induction hypothesis yield

$$\begin{aligned} p_{k+1}^{[i]}(x_i) &= p_k^{[i]}(x_i) = f(x_i), \\ p_{k+1}^{[i]}(x_{i+k+1}) &= p_k^{[i+1]}(x_{i+k+1}) = f(x_{i+k+1}). \end{aligned}$$

The proof is completed by the last three equations and the uniqueness of interpolating polynomials. $\square$

Example 6.32. To estimate $f(x)$ for $x = \frac{3}{2}$ directly from the table in Example 6.19, we construct a table by repeating (6.34) with $x_i = i$ for $i = 0, 1, 2, 3$.

x_i	$x - x_i$	$f(x_i)$	$p_1^{[i]}(x)$	$p_2^{[i]}(x)$	$p_3^{[i]}(x)$
0	$\frac{3}{2}$	6	$-\frac{15}{2}$	$-\frac{21}{4}$	-6
1	$\frac{1}{2}$	-3	$-\frac{9}{2}$	$-\frac{27}{4}$	
2	$-\frac{1}{2}$	-6	$-\frac{27}{2}$		
3	$-\frac{3}{2}$	9			

(6.35)

The result is the same as that in Example 6.19. In contrast, the calculation and layout of the two tables are distinct.

6.6 The Runge phenomenon

Example 6.33. The points $x_0, x_1, \dots, x_n$ in Theorem 6.5 are usually given *a priori*, e.g., as uniformly distributed over the interval $[x_0, x_n]$. As n increases, the degree of the interpolating polynomial also increases. Ideally we would like to have

$$\forall f \in \mathcal{C}[x_0, x_n], \forall x \in [x_0, x_n], \quad \lim_{n \rightarrow +\infty} p_n(f; x) = f(x). \quad (6.36)$$

However, this is not true for polynomial interpolation on equally spaced points. The famous Runge's example in Figure 6.1 illustrates the wide oscillations near the interval ends.

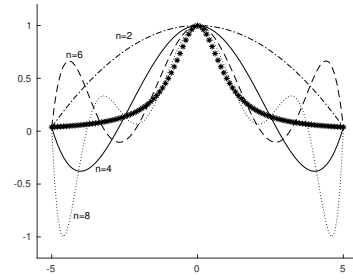


Figure 6.1: The Runge phenomenon created by interpolating $f(x) = \frac{1}{1+x^2}$ on $x_i = -5 + 10\frac{i}{n}$ with $i = 0, 1, \dots, n$ and $n = 2, 4, 6, 8$.

Example 6.34. Certain functions cannot be well approximated by polynomials. In Figure 6.2, a circle going through the origin has its radius as R and its center at $(-R \sin \theta, R \cos \theta)$. For $x \in (0, R - R \sin \theta)$, the circular arc can be expressed as the graph of a function

$$y = H(x) = R \cos \theta - \sqrt{R^2 - (x + R \sin \theta)^2}. \quad (6.37)$$

The values of factorials and derivatives of $H(x)$ are listed in the table in Figure 6.2.

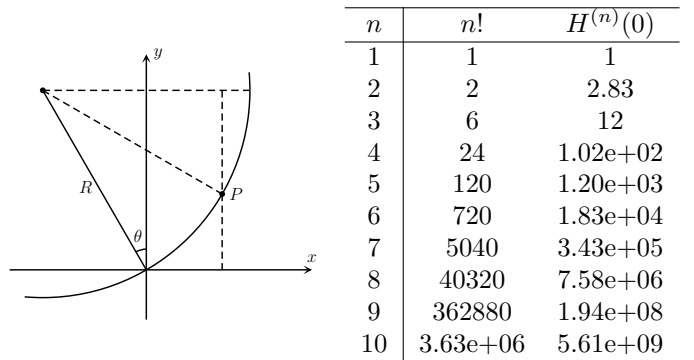


Figure 6.2: Representing a local circular arc with the function H in (6.37).

Corollary 6.35. Suppose polynomial interpolation is performed for f on the $n + 1$ zeros of $T_{n+1}(x)$ as in Theorem 5.67. The Cauchy remainder in Theorem 6.7 satisfies

$$|R_n(f; x)| \leq \frac{1}{2^n(n+1)!} \max_{x \in [-1, 1]} |f^{(n+1)}(x)|. \quad (6.38)$$

Proof. Theorem 6.7, Corollary 5.66, and Theorem 5.67 yield

$$|R_n(f; x)| = \frac{|f^{(n+1)}(\xi)|}{(n+1)!} \left| \prod_{i=0}^n (x - x_i) \right| = \frac{|f^{(n+1)}(\xi)|}{2^n(n+1)!} |T_{n+1}|.$$

Definition 5.64 completes the proof as $|T_{n+1}| \leq 1$. $\square$

6.7 The Lebesgue constant

Definition 6.36. The *polynomial interpolation operator* $\mathcal{L} : \mathcal{C}[a, b] \rightarrow \mathbb{P}_n$ for given nodes $a \leq x_0 < \dots < x_n \leq b$ is defined as

$$\forall f \in \mathcal{C}[a, b], \quad \mathcal{L}f := p_n(f, x), \quad (6.39)$$

where $p_n(f, x)$ is the polynomial in $\mathbb{P}_n$ that coincides with f at $x_0, \dots, x_n$.

Lemma 6.37. The polynomial interpolation operator $\mathcal{L}$ is linear and bounded with respect to the ∞ -norm.

Proof. The linearity of $\mathcal{L}$ follows directly from Theorem 6.5 and the Lagrange formula (6.8). For any $x \in [a, b]$, we have

$$(\mathcal{L}f)(x) = \left| \sum_{k=0}^n f_k \ell_k(x) \right| \leq \sum_{k=0}^n |f_k| |\ell_k(x)| \leq \sum_{k=0}^n |\ell_k(x)| \|f\|_\infty,$$

where ℓ_k is the elementary Lagrange interpolation polynomial in (6.9). Take the supremum over $[a, b]$ and we have

$$\|\mathcal{L}f\|_\infty \leq \|f\|_\infty \max_{x \in [a, b]} \sum_{k=0}^n |\ell_k(x)|. \quad (6.40)$$

Then the proof is completed by the boundedness of both f and ℓ_k on the compact interval $[a, b]$. $\square$

Definition 6.38 (Lebesgue constant). For given interpolation nodes $a \leq x_0 < \dots < x_n \leq b$, the *Lebesgue constant* Λ_n of the interpolation operator $\mathcal{L}$ is the ∞ -norm of $\mathcal{L}$, i.e.,

$$\Lambda_n := \|\mathcal{L}\|_\infty = \sup_{f \neq 0, f \in \mathcal{C}[a, b]} \frac{\|\mathcal{L}f\|_\infty}{\|f\|_\infty}. \quad (6.41)$$

Definition 6.39 (Lebesgue function). For given interpolation nodes $a \leq x_0 < \dots < x_n \leq b$, the *Lebesgue function* $\lambda : [a, b] \rightarrow \mathbb{R}$ of the interpolation operator $\mathcal{L}$ is

$$\lambda(x) = \lambda_n(x; x_0, \dots, x_n) := \sum_{k=0}^n |\ell_k(x)| \quad (6.42)$$

where ℓ_k is the elementary Lagrange interpolation polynomial in (6.9).

Lemma 6.40 (Evaluation of the Lebesgue constant). The Lebesgue constant and the Lebesgue function are related by

$$\Lambda_n = \max_{x \in [a, b]} \lambda(x). \quad (6.43)$$

Proof. (6.40) and (6.41) yield $\Lambda_n \leq \max_{x \in [a, b]} \lambda(x)$. Thus it suffices to show $\Lambda_n \geq \max_{x \in [a, b]} \lambda(x)$.

Suppose $\lambda(x_*) = \max_{x \in [a, b]} \lambda(x)$ for some $x_* \in [a, b]$. We construct a piecewise linear function $f^* \in \mathcal{C}[a, b]$ by (i) setting $f^*(x_k) = \text{sgn}(\ell_k(x_*))$ for each $k = 0, \dots, n$, where

“sgn” denotes the signum function, and (ii) extending f^* linearly between the nodes and constantly outside $[x_0, x_n]$. By construction, we have $\|f^*\|_\infty = 1$. In addition, evaluating $\mathcal{L}f^* = p_n(f^*, x)$ by the Lagrange formula (6.8) yields

$$\begin{aligned} p_n(f^*, x_*) &= \sum_{k=0}^n f^*(x_k) \ell_k(x_*) = \sum_{k=0}^n \text{sgn}(\ell_k(x_*)) \ell_k(x_*) \\ &= \sum_{k=0}^n |\ell_k(x_*)| = \lambda(x_*) = \max_{x \in [a, b]} \lambda(x). \end{aligned}$$

Now that $\frac{\|p_n(f^*)\|_\infty}{\|f^*\|_\infty} = \max_{x \in [a, b]} \lambda(x)$ already holds for the particular function $f^* \in \mathcal{C}[a, b]$, we know from (6.41) that $\Lambda_n \geq \max_{x \in [a, b]} \lambda(x)$. $\square$

Theorem 6.41. For a function $f \in \mathcal{C}[a, b]$, let $p^* \in \mathbb{P}_n[a, b]$ be the best polynomial approximation such that

$$\forall \varphi \in \mathbb{P}_n[a, b], \quad \|f - p^*\|_\infty \leq \|f - \varphi\|_\infty. \quad (6.44)$$

Let $p = \mathcal{L}f$ be an interpolating polynomial of f over the given nodes $a \leq x_0 < \dots < x_n \leq b$. Then we have

$$\|f - p\|_\infty \leq (\Lambda_n + 1) \|f - p^*\|_\infty. \quad (6.45)$$

Proof. Since $p - p^*$ interpolates $f - p^*$, Definition 6.38 yields $\|p - p^*\| \leq \Lambda_n \|f - p^*\|$. Then (6.45) follows from the triangle inequality $\|p - p^*\| \geq \|f - p\| - \|f - p^*\|$. $\square$

Theorem 6.42. For any $-1 \leq x_0 < x_1 < \dots < x_n \leq 1$, the Lebesgue constant satisfies

$$\Lambda_n \geq \frac{2}{\pi} \log(n+1) + \frac{2}{\pi} (\gamma + \log \frac{4}{\pi}), \quad (6.46)$$

where $\gamma := \lim_{n \rightarrow \infty} (\sum_{k=1}^n \frac{1}{k} - \log n) \approx 0.577$ is *Euler's constant* and $\frac{2}{\pi} (\gamma + \log \frac{4}{\pi}) \approx 0.52125$. For the Chebyshev extrema in (5.35), i.e., $x_k = \cos \frac{k}{n} \pi$, we have

$$\Lambda_n \leq \frac{2}{\pi} \log(n+1) + 1, \quad (6.47a)$$

$$\Lambda_n \sim \frac{2}{\pi} \log(n) \text{ as } n \rightarrow \infty. \quad (6.47b)$$

For nodes with uniform spacing, i.e. $x_k = -1 + 2\frac{k}{n}$, we have

$$\forall n \geq 1, \quad \Lambda_n > \frac{1}{n^2} 2^{n-2}, \quad (6.48a)$$

$$\Lambda_n \sim \frac{2^{n+1}}{en \log(n)} \text{ as } n \rightarrow \infty, \quad (6.48b)$$

where $e \approx 2.71828$ is *Euler's number*.

Proof. See [Trefethen, 2013, Theorem 15.2] $\square$

6.8 The Hermite interpolation

Definition 6.43. Given distinct points $x_0, x_1, \dots, x_k$ in $[a, b]$, non-negative integers $m_0, m_1, \dots, m_k$, and a function $f \in \mathcal{C}^M[a, b]$ where $M = \max_i m_i$, the *Hermite interpolation problem* seeks a polynomial p of the lowest degree such that

$$\forall i = 0, 1, \dots, k, \quad \forall \mu = 0, 1, \dots, m_i, \quad p^{(\mu)}(x_i) = f_i^{(\mu)}, \quad (6.49)$$

where $f_i^{(\mu)} = f^{(\mu)}(x_i)$ is the value of the μ th derivative of f at x_i ; in particular, $f_i^{(0)} = f(x_i)$.

Theorem 6.44. There exists a unique solution to the Hermite interpolation problem in Definition 6.43.

Proof. Write $N := k + \sum_{i=0}^k m_i$. The interpolation conditions in (6.49) lead to a linear system of $N + 1$ equations, where the unknowns are coefficients of the monomials in the polynomial $p(x) = \sum_i a_i x^i$. It suffices to show that the matrix M of this linear system is invertible, i.e., $\ker M = \{\mathbf{0}\}$. To this end, consider the linear system $M\mathbf{a} = \mathbf{0}$. Since each x_i in (6.49) is a zero of $p(x)$ with its multiplicity no less than $(m_i + 1)$, the product $P(x) = \prod_{i=0}^k (x - x_i)^{m_i+1}$ must be a factor of $p(x)$. However, the degree of P is $N + 1$ whereas that of p is N , thus we must have $p \equiv 0$. $\square$

Definition 6.45 (Generalized divided difference). Let $x_0, x_1, \dots, x_k$ be $k + 1$ pairwise distinct points with each x_i repeated $m_i + 1$ times; write $N := k + \sum_{i=0}^k m_i$. The N th divided difference associated with these points is the coefficient of the monomial x^N in the polynomial p that uniquely solves the Hermite interpolation problem in Definition 6.43.

Corollary 6.46. The n th divided difference at $n + 1$ “confluent” (i.e. identical) points is

$$f[x_0, \dots, x_0] = \frac{1}{n!} f^{(n)}(x_0), \quad (6.50)$$

where x_0 is repeated $n + 1$ times on the left-hand side.

Proof. For the Hermite interpolation problem with $k = 0$ and $m_0 = n$, the unique solution is a polynomial of degree n . By Definition 6.45, we have $p^{(n)} \equiv n!f[x_0, \dots, x_0]$, then the interpolation conditions yield

$$f^{(n)}(x_0) = p^{(n)}(x_0) = n!f[x_0, \dots, x_0]. \quad \square$$

Theorem 6.47. For the Hermite interpolation problem in Definition 6.43, denote $N = k + \sum_i m_i$. Denote by $p_N(f; x)$ the unique element of $\mathbb{P}_N$ for which (6.49) holds. Suppose $f^{(N+1)}(x)$ exists in (a, b) . Then there exists some $\xi \in (a, b)$ such that

$$f(x) - p_N(f; x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} \prod_{i=0}^k (x - x_i)^{m_i+1}. \quad (6.51)$$

Proof. The proof is similar to that of Theorem 6.7. Pay attention to the difference caused by the multiple roots of the polynomial $\prod_{i=0}^k (x - x_i)^{m_i+1}$. $\square$

Example 6.48. For the Hermite interpolation problem

$$p(x_0) = f_0, \quad p'(x_0) = f'_0, \quad p''(x_0) = f''_0,$$

Newton’s formula yields the interpolating polynomial as

$$p(x) = f_0 + f'_0(x - x_0) + \frac{1}{2}f''_0(x - x_0)^2,$$

which is exactly the Taylor polynomial of degree 2. Thus a Taylor polynomial is a special case of a Hermite interpolating polynomial. By Theorem 6.47, the Cauchy remainder of this interpolation is

$$R_2(f; x) = f(x) - p_2(f; x) = \frac{f^{(3)}(\xi)}{6}(x - x_0)^3,$$

which is Lagrange’s formula of the remainder term in Taylor’s formula; see Theorem B.100.

Example 6.49. For the Hermite interpolation problem

$$p(x_0) = f_0, \quad p(x_1) = f_1, \quad p'(x_1) = f'_1, \quad p(x_2) = f_2$$

shown in Figure 6.3, the table of divided differences has the form

$$\begin{array}{c|cccc} x_0 & f_0 & & & \\ x_1 & f_1 & f[x_0, x_1] & & \\ x_1 & f_1 & f'_1 & f[x_0, x_1, x_1] & \\ x_2 & f_2 & f[x_1, x_2] & f[x_1, x_1, x_2] & f[x_0, x_1, x_1, x_2] \end{array}$$

and the interpolating polynomial follows from Newton’s formula. By Theorem 6.47, the Cauchy remainder is

$$R_3(f; x) = \frac{f^{(4)}(\xi)}{4!}(x - x_0)(x - x_1)^2(x - x_2)$$

for some $\xi \in (x_0, x_2)$.

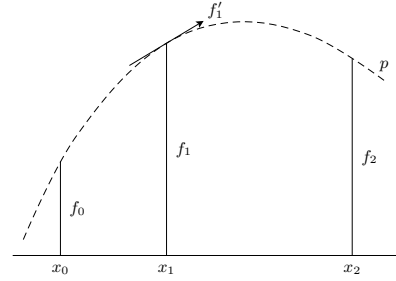


Figure 6.3: An example of Hermite interpolation.

6.9 Hermite splines

Definition 6.50. A *cubic Hermite spline* or *cubic Hermite interpolator* is a $\mathcal{C}^1$ function that consists of piecewise cubic polynomials and solves the Hermite interpolation problem in Definition 6.43 with $m_i \equiv 1$.

Lemma 6.51. Consider the Hermite interpolation problem where function values f_i ’s and derivative values f'_i ’s are specified at points x_i ’s where $i = 0, 1, \dots, n$ and $x_j < x_{j+1}$ for $j = 0, 1, \dots, n - 1$. The corresponding cubic Hermite spline p_f belongs to $\mathcal{C}^1([x_0, x_n])$ and is given by

$$p_f(x)|_{[x_j, x_{j+1}]} =: p_j(x) = \sum_{k=0}^3 \beta_k b_{3,k} \left(\frac{x - x_j}{x_{j+1} - x_j} \right), \quad (6.52a)$$

$$\beta_0 = f_j, \quad \beta_1 = f_j + \frac{1}{3}f'_j(x_{j+1} - x_j); \quad (6.52b)$$

$$\beta_3 = f_{j+1}, \quad \beta_2 = f_{j+1} - \frac{1}{3}f'_{j+1}(x_{j+1} - x_j), \quad (6.52c)$$

where $b_{3,k}$ is the k th Bernstein base polynomial of degree 3 in Definition 5.27.

Proof. It is straightforward to verify from (6.52a) that $p_j(x)$ satisfies the interpolation conditions

$$\begin{aligned} p_j(x_j) &= f_j, \quad p_j(x_{j+1}) = f_{j+1}; \\ p'_j(x_j) &= f'_j, \quad p'_j(x_{j+1}) = f'_{j+1} \end{aligned}$$

in the interval $[x_j, x_{j+1}]$. By Definitions 6.43 and 6.50, p_f is indeed a cubic Hermite spline and is $\mathcal{C}^1$. On the other hand, the four interpolation conditions on each interval uniquely determines p_j because Lemma 5.33 states that $\{b_{3,k} : k = 0, 1, 2, 3\}$ is a basis of $\mathbb{P}_3$ and the coordinate transformation $x \mapsto \frac{x-x_j}{x_{j+1}-x_j}$ is a bijection. $\square$

6.10 Bézier curves

Definition 6.52. The Bézier curve of $n+1$ distinct points $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_n \in \mathbb{R}^D$ is the closed curve $\mathbf{B} : [0, 1] \rightarrow \mathbb{R}^D$,

$$\mathbf{B}(t) := \sum_{k=0}^n \mathbf{q}_k b_{n,k}(t), \quad (6.53)$$

where $n \in \mathbb{N}^+$, $b_{n,k}$ is the k th Bernstein base polynomial of degree n in Definition 5.27, and the points $\mathbf{q}_k$'s are called the *control points* of the Bézier curve.

Example 6.53. A linear Bézier curve is a line segment, i.e., a Bézier curve of two distinct points, a quadratic Bézier curve is defined by three distinct points, and a cubic Bézier curve by four distinct points.

Lemma 6.54. For distinct points $\mathbf{p}_i \in W \subset \mathbb{R}^n$ where $i = 1, \dots, m$ and $m \geq 2$, any point $\sum_{i=1}^m \lambda_i \mathbf{p}_i$ with $\sum_{i=1}^m \lambda_i = 1$ and $\lambda_i \in [0, 1]$ belongs to the convex hull of W .

Proof. The induction basis $m = 2$ holds directly from Definition 4.22. As for the induction step, we have

$$\sum_{i=1}^m \lambda_i \mathbf{p}_i = \lambda_m \mathbf{p}_m + (1 - \lambda_m) \mathbf{q}_m,$$

where $\mathbf{q}_m := \sum_{i=1}^{m-1} \frac{\lambda_i}{1 - \lambda_m} \mathbf{p}_i$, by the induction hypothesis, belongs to the convex hull of W . Then the proof is completed by the induction basis. $\square$

Lemma 6.55. The Bézier curve of $n+1$ distinct points is contained in the convex hull of these points.

Proof. This follows from Lemma 6.54, (5.13a,b), and Definitions 4.22 and 6.52. $\square$

Lemma 6.56. A Bézier curve of degree n satisfies

$$B(0) = \mathbf{q}_0, \quad B'(0) = n(\mathbf{q}_1 - \mathbf{q}_0); \quad (6.54a)$$

$$B(1) = \mathbf{q}_n, \quad B'(1) = n(\mathbf{q}_n - \mathbf{q}_{n-1}). \quad (6.54b)$$

Proof. Rewrite (6.53) as

$$\mathbf{B}(t) = \mathbf{q}_n t^n + \mathbf{q}_0(1-t)^n + \sum_{k=1}^{n-1} \mathbf{q}_k \binom{n}{k} t^k (1-t)^{n-k},$$

which yields the first equations in (6.54) and

$$\begin{aligned} \mathbf{B}'(t) &= n\mathbf{q}_n t^{n-1} - n\mathbf{q}_0(1-t)^{n-1} \\ &\quad + \sum_{k=1}^{n-1} \mathbf{q}_k \binom{n}{k} [kt^{k-1}(1-t)^{n-k} - (n-k)t^k(1-t)^{n-k-1}]. \end{aligned}$$

Then the second equations follow. $\square$

Algorithm 6.57 (Curve fitting via Cubic Bézier curves). A curve $\gamma : (0, 1) \rightarrow \mathbb{R}^D$ with given tangent vector γ' can be approximated by cubic Bézier curves as follows.

- Select $m+1$ consecutive marker points $\mathbf{p}_0, \dots, \mathbf{p}_m$ on γ and calculate tangent vector at these markers.
- For each $j = 0, \dots, m-1$, calculate $\mathbf{q}_0 := \mathbf{p}_j$, $\mathbf{q}_1 := \frac{1}{3}\gamma'(\mathbf{p}_j) + \mathbf{p}_j$, $\mathbf{q}_2 := \mathbf{p}_{j+1} - \frac{1}{3}\gamma'(\mathbf{p}_j)$, $\mathbf{q}_3 := \mathbf{p}_{j+1}$ and determine a cubic Bézier curve (6.53) from $\mathbf{q}_j$'s.
- Connect the m cubic Bézier curves.

6.11 Theoretical problems

- For $f \in \mathcal{C}^2[x_0, x_1]$ and $x \in (x_0, x_1)$, linear interpolation of f at x_0 and x_1 yields

$$f(x) - p_1(f; x) = \frac{f''(\xi(x))}{2}(x-x_0)(x-x_1).$$

Consider the case $f(x) = \frac{1}{x}$, $x_0 = 1$, $x_1 = 2$.

- Determine $\xi(x)$ explicitly.
- Extend the domain of ξ continuously from (x_0, x_1) to $[x_0, x_1]$. Find $\max \xi(x)$, $\min \xi(x)$, and $\max f''(\xi(x))$.

- Let $\mathbb{P}_m^+$ be the set of all polynomials of degree $\leq m$ that are non-negative on the real line,

$$\mathbb{P}_m^+ = \{p : p \in \mathbb{P}_m, \forall x \in \mathbb{R}, p(x) \geq 0\}.$$

Find $p \in \mathbb{P}_{2n}^+$ such that $p(x_i) = f_i$ for $i = 0, 1, \dots, n$ where $f_i \geq 0$ and x_i are distinct points on $\mathbb{R}$.

- Consider $f(x) = e^x$.

- Prove by induction that

$$\forall t \in \mathbb{R}, \quad f[t, t+1, \dots, t+n] = \frac{(e-1)^n}{n!} e^t.$$

- From Corollary 6.22 we know

$$\exists \xi \in (0, n) \text{ s.t. } f[0, 1, \dots, n] = \frac{1}{n!} f^{(n)}(\xi).$$

Determine ξ from the above two equations. Is ξ located to the left or to the right of the midpoint $n/2$?

- Consider $f(0) = 5$, $f(1) = 3$, $f(3) = 5$, $f(4) = 12$.

- Use the Newton formula to obtain $p_3(f; x)$;
- The data suggest that f has a minimum in $x \in (1, 3)$. Find an approximate value for the location $x_{\min}$ of the minimum.

- Consider $f(x) = x^7$.

- Compute $f[0, 1, 1, 1, 2, 2]$.
- We know that this divided difference is expressible in terms of the 5th derivative of f evaluated at some $\xi \in (0, 2)$. Determine ξ .

- f is a function on $[0, 3]$ for which one knows that

$$f(0) = 1, f(1) = 2, f'(1) = -1, f(3) = f'(3) = 0.$$

- Estimate $f(2)$ using Hermite interpolation.

- Estimate the maximum possible error of the above answer if one knows, in addition, that $f \in \mathcal{C}^5[0, 3]$ and $|f^{(5)}(x)| \leq M$ on $[0, 3]$. Express the answer in terms of M .

VII. Define forward difference by

$$\Delta f(x) = f(x+h) - f(x),$$

$$\Delta^{k+1} f(x) = \Delta \Delta^k f(x) = \Delta^k f(x+h) - \Delta^k f(x)$$

and backward difference by

$$\nabla f(x) = f(x) - f(x-h),$$

$$\nabla^{k+1} f(x) = \nabla \nabla^k f(x) = \nabla^k f(x) - \nabla^k f(x-h).$$

For $x_j = x + jh$, prove

$$\Delta^k f(x) = k!h^k f[x_0, x_1, \dots, x_k],$$

$$\nabla^k f(x) = k!h^k f[x_0, x_{-1}, \dots, x_{-k}].$$

VIII. Assume f is differentiable at x_0 . Prove

$$\frac{\partial}{\partial x_0} f[x_0, x_1, \dots, x_n] = f[x_0, x_0, x_1, \dots, x_n].$$

What about the partial derivative with respect to one of the other variables?

6.12 Programming assignments

A. Implement the Newton formula in a subroutine that produces the value of the interpolation polynomial $p_n(f; x_0, x_1, \dots, x_n; x)$ at any real x , where $n \in \mathbb{N}^+$, x_i 's are distinct, and f is a function assumed to be available in the form of a subroutine.

B. Run your routine on the function

$$f(x) = \frac{1}{1+x^2}$$

for $x \in [-5, 5]$ using $x_i = -5 + 10\frac{i}{n}$, $i = 0, 1, \dots, n$, and $n = 2, 4, 6, 8$. Plot the polynomials against the exact function to reproduce the plot in the notes that illustrate the Runge phenomenon.

C. Reuse your subroutine of Newton interpolation to perform Chebyshev interpolation for the function

$$f(x) = \frac{1}{1+25x^2}$$

for $x \in [-1, 1]$ on the zeros of Chebyshev polynomials T_n with $n = 5, 10, 15, 20$. Clearly the Runge function

$f(x)$ is a scaled version of the function in B. Plot the interpolating polynomials against the exact function to observe that the Chebyshev interpolation is free of the wide oscillations in the previous assignment.

D. A car traveling along a straight road is clocked at a number of points. The data from the observations are given in the following table, where the time is in seconds, the displacement is in feet, and the velocity is in feet per second.

Time	0	3	5	8	13
displacement	0	225	383	623	993
velocity	75	77	80	74	72

- Use a Hermite polynomial to predict the position of the car and its speed for $t = 10$ seconds.
- Use the derivative of the Hermite polynomial to determine whether the car ever exceeds the speed limit of 55 miles per hour, i.e., 81 feet per second.

E. It is suspected that the high amounts of tannin in mature oak leaves inhibit the growth of the winter moth larvae that extensively damage these trees in certain years. The following table lists the average weight of two samples of larvae at times in the first 28 days after birth. The first sample was reared on young oak leaves, whereas the second sample was reared on mature leaves from the same tree.

Day	0	6	10	13	17	20	28
Sp1	6.67	17.3	42.7	37.3	30.1	29.3	28.7
Sp2	6.67	16.1	18.9	15.0	10.6	9.44	8.89

- Use Newton's formula to approximate the average weight curve for each sample.
- Predict whether the two samples of larvae will die after another 15 days.

F. The roots of the following equation constitute a closed planar curve in the shape of a heart:

$$x^2 + \left(\frac{3}{2}y - \sqrt{|x|}\right)^2 = 3. \quad (6.55)$$

Write a program to approximate the heart by cubic Bézier curves and plot your approximant, i.e. the piecewise cubic *polynomials*. Choose $m = 10, 40, 160$ in Algorithm 6.57 and produce three plots of the heart function. Your knots should include the characteristic points.

Chapter 7

Univariate Polynomial Splines



Figure 7.1: An example spline. The L-shapes brackets and circles represent fixtures and the shaded thick curve a wooden bar that bends under the constraints of the fixtures.

Definition 7.1. Given nonnegative integers n , k , and a strictly increasing sequence $(x_i)_{i=1}^N$ that partitions $[a, b]$,

$$a = x_1 < x_2 < \cdots < x_N = b, \quad (7.1)$$

the set of *spline functions of degree n and smoothness class k* relative to the partitioned interval is

$$\mathbb{S}_n^k = \{s : s \in \mathcal{C}^k[a, b]; \forall i \in [1, N-1], s|_{[x_i, x_{i+1}]} \in \mathbb{P}_n\}. \quad (7.2)$$

The x_i 's are called *knots* of the spline.

Notation 5. In Chapter 6 and this chapter, the polynomial degree is denoted by n for all methods. Here we use N to denote the number of knots for a spline.

Example 7.2. As an extreme, $\mathbb{S}_n^n = \mathbb{P}_n$, i.e. all the pieces of $s \in \mathbb{S}_n^n$ belong to a single polynomial. On the other end, $\mathbb{S}_1^0$ is the class of piecewise linear interpolating functions. The most popular splines are the cubic splines in $\mathbb{S}_3^2$, on which we focus in this chapter.

7.1 PP-form splines

7.1.1 Constructing PP-form splines

Lemma 7.3. Let $m_i := s'(f; x_i)$ denote the derivative of $s \in \mathbb{S}_3^2$ at x_i . Then, for each $i = 2, 3, \dots, N-1$, we have

$$\lambda_i m_{i-1} + 2m_i + \mu_i m_{i+1} = 3\mu_i f[x_i, x_{i+1}] + 3\lambda_i f[x_{i-1}, x_i], \quad (7.3)$$

where

$$\mu_i = \frac{x_i - x_{i-1}}{x_{i+1} - x_{i-1}}, \quad \lambda_i = \frac{x_{i+1} - x_i}{x_{i+1} - x_{i-1}}. \quad (7.4)$$

Proof. Denote $p_i(x) = s|_{[x_i, x_{i+1}]}$ and $K_i = f[x_i, x_{i+1}]$. The table of divided difference for the Hermite interpolation problem $p_i(x_i) = f_i$, $p_i(x_{i+1}) = f_{i+1}$, $p_i'(x_i) = m_i$, $p_i'(x_{i+1}) = m_{i+1}$ is

x_i	f_i				
x_i	f_i	m_i			
x_{i+1}	f_{i+1}	K_i	$\frac{K_i - m_i}{x_{i+1} - x_i}$		
x_{i+1}	f_{i+1}	m_{i+1}	$\frac{m_{i+1} - K_i}{x_{i+1} - x_i}$	$\frac{m_i + m_{i+1} - 2K_i}{(x_{i+1} - x_i)^2}$	

Then the Newton formula yields

$$p_i(x) = f_i + (x - x_i)m_i + (x - x_i)^2 \frac{K_i - m_i}{x_{i+1} - x_i} + (x - x_i)^2 (x - x_{i+1}) \frac{m_i + m_{i+1} - 2K_i}{(x_{i+1} - x_i)^2}, \quad (7.5)$$

or equivalently

$$\begin{cases} p_i(x) &= c_{i,0} + c_{i,1}(x - x_i) + c_{i,2}(x - x_i)^2 + c_{i,3}(x - x_i)^3, \\ c_{i,0} &= f_i, \\ c_{i,1} &= m_i, \\ c_{i,2} &= \frac{3K_i - 2m_i - m_{i+1}}{x_{i+1} - x_i}, \\ c_{i,3} &= \frac{m_i + m_{i+1} - 2K_i}{(x_{i+1} - x_i)^2}. \end{cases} \quad (7.6)$$

$s \in \mathcal{C}^2$ implies that $p_{i-1}''(x_i) = p_i''(x_i)$, i.e.

$$3c_{i-1,3}(x_i - x_{i-1}) = c_{i,2} - c_{i-1,2}.$$

The substitution of the coefficients $c_{i,j}$ into the above equation yields (7.3). $\square$

Lemma 7.4. Denote $M_i = s''(f; x_i)$ for $s \in \mathbb{S}_3^2$. Then, for each $i = 2, 3, \dots, N-1$, we have

$$\mu_i M_{i-1} + 2M_i + \lambda_i M_{i+1} = 6f[x_{i-1}, x_i, x_{i+1}] \quad (7.7)$$

where μ_i and λ_i are the same as those in (7.4).

Proof. Taylor expansion of $s(x)$ at x_i yields

$$s(x) = f_i + s'(x_i)(x - x_i) + \frac{M_i}{2}(x - x_i)^2 + \frac{s'''(x_i)}{6}(x - x_i)^3. \quad (7.8)$$

For $x \in (x_i, x_{i+1}]$, differentiate (7.8) twice, set $x = x_{i+1}$, and we have

$$s'''(x_i) = \frac{M_{i+1} - M_i}{x_{i+1} - x_i}. \quad (7.9)$$

Substitute (7.9) into (7.8), set $x = x_{i+1}$, and we have

$$s'(x_i) = f[x_i, x_{i+1}] - \frac{1}{6}(M_{i+1} + 2M_i)(x_{i+1} - x_i). \quad (7.10)$$

Similarly, for $x \in [x_{i-1}, x_i]$, differentiate (7.8) twice, set $x = x_{i-1}$, and we have $s'''(x_i) = \frac{M_{i-1} - M_i}{x_{i-1} - x_i}$, the substitution of which into (7.8) yields

$$s'(x_i) = f[x_{i-1}, x_i] - \frac{1}{6}(M_{i-1} + 2M_i)(x_{i-1} - x_i). \quad (7.11)$$

The subtraction of (7.10) from (7.11) yields (7.7). $\square$

Definition 7.5 (Boundary conditions of cubic splines). Common cubic splines include the following.

- A *complete cubic spline* $s \in \mathbb{S}_3^2$ satisfies boundary conditions $s'(f; a) = f'(a)$ and $s'(f; b) = f'(b)$.
- A *cubic spline with specified second derivatives at its end points*: $s''(f; a) = f''(a)$ and $s''(f; b) = f''(b)$.
- A *natural cubic spline* $s \in \mathbb{S}_3^2$ satisfies boundary conditions $s''(f; a) = 0$ and $s''(f; b) = 0$.
- A *not-a-knot cubic spline* $s \in \mathbb{S}_3^2$ satisfies that $s'''(f; x)$ exists at $x = x_2$ and $x = x_{N-1}$.
- A *periodic cubic spline* $s \in \mathbb{S}_3^2$ is obtained by replacing $s(f; b) = f(b)$ with $s(f; b) = s(f; a)$, $s'(f; b) = s'(f; a)$, and $s''(f; b) = s''(f; a)$.

Lemma 7.6. For a complete cubic spline $s \in \mathbb{S}_3^2$, denote $M_i = s''(f; x_i)$ and we have

$$2M_1 + M_2 = 6f[x_1, x_1, x_2], \quad (7.12)$$

$$M_{N-1} + 2M_N = 6f[x_{N-1}, x_N, x_N]. \quad (7.13)$$

Proof. As for (7.12), the cubic polynomial on $[x_1, x_2]$ can be written as

$$s_1(x) = f[x_1] + f[x_1, x_1](x - x_1) + \frac{M_1}{2}(x - x_1)^2 + \frac{s_1'''(x_1)}{6}(x - x_1)^3.$$

Differentiate the above equation twice, replace x with x_2 , and we have $s_1'''(x_1) = \frac{M_2 - M_1}{x_2 - x_1}$, which implies

$$s_1(x) = f[x_1] + f[x_1, x_1](x - x_1) + \frac{M_1}{2}(x - x_1)^2 + \frac{M_2 - M_1}{6(x_2 - x_1)}(x - x_1)^3. \quad (7.14)$$

Set $x = x_2$, divide both sides by $x_2 - x_1$, and we have

$$f[x_1, x_2] = f[x_1, x_1] + \left(\frac{M_1}{2} + \frac{M_2 - M_1}{6} \right) (x_2 - x_1),$$

which yields (7.12). (7.13) can be proven similarly. $\square$

Theorem 7.7. For any continuous function $f : [a, b] \rightarrow \mathbb{R}$, there exists a unique complete/natural/periodic cubic spline $s(f; x) \in \mathbb{S}_3^2$ that interpolates f .

Proof. We only prove the case of complete cubic splines since the other cases are similar.

By the proof of Lemma 7.3, s is uniquely determined if all the m_i 's are uniquely determined on all intervals. For a complete cubic spline we already have $m_1 = f'(a)$ and $m_N = f'(b)$. Assemble (7.3) into a linear system $\mathbf{A}\mathbf{m} = \mathbf{b}$,

$$\begin{bmatrix} 2 & \mu_2 & & & \\ \lambda_3 & 2 & \mu_3 & & \\ & \ddots & \ddots & \ddots & \\ & & \lambda_i & 2 & \mu_i \\ & & & \ddots & \ddots & \ddots \\ & & & & \lambda_{N-2} & 2 & \mu_{N-2} \\ & & & & & \lambda_{N-1} & 2 \end{bmatrix} \begin{bmatrix} m_2 \\ m_3 \\ \vdots \\ m_i \\ \vdots \\ m_{N-2} \\ m_{N-1} \end{bmatrix} = \mathbf{b}, \quad (7.15)$$

where the vector $\mathbf{b}$ is determined from the known information. (7.4) implies that the matrix in the above equation is strictly diagonally dominant as in Definition 3.47. Then Theorem 3.49 implies that the matrix be nonsingular and Cramer's rule (Theorem C.279) dictates that the m_i 's be uniquely determined.

Alternatively, a complete cubic spline can be uniquely determined from Lemmas 7.4 and 7.6, following arguments similar to the above. $\square$

Theorem 7.8. The max-norm-based condition number of the linear algorithm that determines the unique complete/natural/periodic cubic spline in Theorem 7.7 is no greater than 3, independent of the size N of the linear system $\mathbf{A}\mathbf{m} = \mathbf{b}$.

Proof. (7.4) implies $\mu_i + \lambda_i = 1$ for any $i = 2, \dots, N-1$. Hence Lemma C.282 gives $\|A\|_\infty = 3$. Also, Theorem 3.50 yields $\|A^{-1}\|_\infty \leq 1$. The proof is then completed by Definition 2.22. $\square$

Example 7.9. Construct a complete cubic spline $s(x)$ on points $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 6$ from the function values of $f(x) = \ln(x)$ and its derivatives at x_1 and x_5 . Approximate $\ln(5)$ by $s(5)$.

From the given conditions, we set up the table of divided differences as follows.

x_i	$f[x_i]$		
1	0		
1	0	1	
2	0.6931	0.6931	-0.3069
3	1.0986	0.4055	-0.1438
4	1.3863	0.2877	-0.05889
6	1.7918	0.2027	-0.02831
6	1.7918	0.1667	-0.01803

All values of λ_i and μ_i are $\frac{1}{2}$ except that

$$\lambda_4 = \frac{2}{3}, \quad \mu_4 = \frac{1}{3}.$$

Then Lemma 7.4 and Lemma 7.6 yield a linear system

$$\begin{bmatrix} 2 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & 4 & 1 & \\ & & 1 & 6 & 2 \\ & & & 1 & 2 \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ M_3 \\ M_4 \\ M_5 \end{bmatrix} \approx \begin{bmatrix} -1.84112 \\ -1.72610 \\ -0.70670 \\ -0.50967 \\ -0.10820 \end{bmatrix},$$

where elements in the right-hand side (RHS) vector are obtained from the last column of the table of divided differences by multiplying 6, 12, 12, 18, and 6. Why? Solve the linear system and we have all the M_i 's. Then we derive an expression of the spline on the last interval following the procedures similar to those for (7.14). After this expression is obtained, we then evaluate it and obtain $s(5) \approx 1.60977$. In comparison, $\ln(5) \approx 1.60944$.

7.1.2 The minimum properties

Theorem 7.10 (Minimum bending energy). For any function $g \in \mathcal{C}^2[a, b]$ that satisfies $g'(a) = f'(a)$, $g'(b) = f'(b)$, and $g(x_i) = f(x_i)$ for each $i = 1, 2, \dots, N$, the complete cubic spline $s = s(f; x)$ satisfies

$$\int_a^b [s''(x)]^2 dx \leq \int_a^b [g''(x)]^2 dx, \quad (7.16)$$

where the equality holds only when $g(x) = s(f; x)$.

Proof. Define $\eta(x) = g(x) - s(x)$. From the given conditions we have $\eta \in \mathcal{C}^2[a, b]$, $\eta'(a) = \eta'(b) = 0$, and $\forall i = 1, 2, \dots, N$, $\eta(x_i) = 0$. Then

$$\begin{aligned} \int_a^b [g''(x)]^2 dx &= \int_a^b [s''(x) + \eta''(x)]^2 dx \\ &= \int_a^b [s''(x)]^2 dx + \int_a^b [\eta''(x)]^2 dx + 2 \int_a^b s''(x) \eta''(x) dx. \end{aligned}$$

From

$$\begin{aligned} \int_a^b s''(x) \eta''(x) dx &= \sum_{i=1}^{N-1} \int_{x_i}^{x_{i+1}} s''(x) d\eta' \\ &= \sum_{i=1}^{N-1} s''(x) \eta'(x) \Big|_{x_i}^{x_{i+1}} - \sum_{i=1}^{N-1} \int_{x_i}^{x_{i+1}} \eta'(x) s'''(x) dx \\ &= s''(b) \eta'(b) - s''(a) \eta'(a) - \sum_{i=1}^{N-1} \int_{x_i}^{x_{i+1}} \eta'(x) s'''(x) dx \\ &= - \sum_{i=1}^{N-1} s'''(x) \eta(x) \Big|_{x_i}^{x_{i+1}} + \sum_{i=1}^{N-1} \int_{x_i}^{x_{i+1}} \eta(x) s^{(4)}(x) dx \\ &= 0, \end{aligned}$$

we have

$$\int_a^b [g''(x)]^2 dx = \int_a^b [s''(x)]^2 dx + \int_a^b [\eta''(x)]^2 dx,$$

which completes the proof. $\square$

Theorem 7.11 (Minimum bending energy). For any function $g \in \mathcal{C}^2[a, b]$ with $g(x_i) = f(x_i)$ for each $i = 1, 2, \dots, N$, the natural cubic spline $s = s(f; x)$ satisfies

$$\int_a^b [s''(x)]^2 dx \leq \int_a^b [g''(x)]^2 dx, \quad (7.17)$$

where the equality holds only when $g(x) = s(f; x)$.

Proof. The proof is similar to that of Theorem 7.10. Although $\eta'(a) = \eta'(b) = 0$ does not hold, we do have $s''(a) = s''(b) = 0$. $\square$

Lemma 7.12. Suppose a $\mathcal{C}^2$ function $f : [a, b] \rightarrow \mathbb{R}$ is interpolated by a complete cubic spline or a cubic spline with specified second derivatives at its end points. Then

$$\forall x \in [a, b], \quad |s''(x)| \leq 3 \max_{x \in [a, b]} |f''(x)|. \quad (7.18)$$

Proof. Since $s''(x)$ is linear on $[x_i, x_{i+1}]$, $|s''(x)|$ attains its maximum at x_j for some j . If $j = 2, \dots, N-1$, it follows from Lemma 7.4 and Corollary 6.22 that

$$\begin{aligned} 2M_j &= 6f[x_{j-1}, x_j, x_{j+1}] - \mu_j M_{j-1} - \lambda_j M_{j+1} \\ \Rightarrow 2|M_j| &\leq 6|f[x_{j-1}, x_j, x_{j+1}]| + (\mu_j + \lambda_j)|M_j| \\ \Rightarrow \exists \xi \in (x_{j-1}, x_{j+1}) \text{ s.t. } |M_j| &\leq 3|f''(\xi)| \\ \Rightarrow |s''(x)| &\leq 3 \max_{x \in [a, b]} |f''(x)|. \end{aligned} \quad (7.19)$$

If $|s''(x)|$ attains its maximum at x_1 or x_N , (7.19) clearly holds at these end points for a cubic spline with specified second derivatives. After all, $s''(a) = f''(a)$ and $s''(b) = f''(b)$. As for the complete cubic spline, it suffices to prove (7.19) when $|s''(x)|$ attains its maximum at x_1 . Since the first derivative $f'(a) = f[x_1, x_1]$ is specified, $f[x_1, x_1, x_2]$ is a constant. By (7.12), we have

$$2|M_1| \leq 6|f[x_1, x_1, x_2]| + |M_2| \leq 6|f[x_1, x_1, x_2]| + |M_1|$$

which, together with Corollary 6.22, implies

$$\exists \xi \in (x_1, x_2) \text{ s.t. } |M_1| \leq 3|f''(\xi)|.$$

This completes the proof. $\square$

7.1.3 Error analysis

Theorem 7.13. Denote by $X_b := (x_i)_{i=1}^N$ the sequence of breakpoints in (7.1) and suppose that a scalar function $f \in \mathcal{C}^2([a, b]) \cap \mathcal{C}^4([a, b] \setminus X_b)$ is interpolated by a complete cubic spline or a cubic spline with specified second derivatives at its end points. Then we have

$$\begin{aligned} \forall x \in [a, b], \quad \forall j = 0, 1, 2, \\ \left| f^{(j)}(x) - s^{(j)}(x) \right| &\leq c_j h^{4-j} \max_{\xi \in [a, b] \setminus X_b} |f^{(4)}(\xi)|, \end{aligned} \quad (7.20)$$

where $c_0 = \frac{1}{16}$, $c_1 = c_2 = \frac{1}{2}$, and $h = \max_{i=1}^{N-1} |x_{i+1} - x_i|$.

Proof. Our plan is to first prove the case of $j = 2$, then utilize the conclusion to validate other cases.

Consider an auxiliary function $\hat{s} \in \mathcal{C}^2[a, b]$ that satisfies

$$\forall i = 1, 2, \dots, N-1, \quad \begin{cases} \hat{s}|_{[x_i, x_{i+1}]} \in \mathbb{P}_3; \\ \hat{s}''(x_i) = f''(x_i) \end{cases}$$

and $\hat{s}''(x_N) = f''(x_N)$. We can obtain such an $\hat{s}$ by interpolating $f''(x)$ with some $\tilde{s} \in \mathbb{S}_1^0$ and integrating $\tilde{s}$ twice. Then the theorem of Cauchy remainder (Theorem 6.7) implies

$$\begin{aligned} \exists \quad \xi_i \in (x_i, x_{i+1}), \quad \text{s.t. } \forall x \in [x_i, x_{i+1}], \\ |f''(x) - \tilde{s}(x)| \leq \frac{1}{2} |f^{(4)}(\xi_i)| |(x - x_i)(x - x_{i+1})|, \end{aligned}$$

hence we have

$$|f''(x) - \hat{s}''(x)|_{x \in [x_i, x_{i+1}]} \leq \frac{1}{8} \max_{\xi \in (x_i, x_{i+1})} |f^{(4)}(\xi)| (x_{i+1} - x_i)^2,$$

which further yields

$$\forall x \in [a, b], \quad |f''(x) - \hat{s}''(x)| \leq \frac{1}{8} h^2 \max_{\xi \in [a, b] \setminus X_b} |f^{(4)}(\xi)|. \quad (7.21)$$

Now interpolate $f - \hat{s}$ with a cubic spline. Since $\hat{s} \in \mathbb{S}_3^2$, the interpolant must be $s - \hat{s}$. Then Lemma 7.12 yields

$$\forall x \in [a, b], \quad |s''(x) - \hat{s}''(x)| \leq 3 \max_{\xi \in [a, b]} |f''(\xi) - \hat{s}''(\xi)|,$$

which, together with (7.21), leads to (7.20) for $j = 2$:

$$\begin{aligned} |f''(x) - s''(x)| &\leq |f''(x) - \hat{s}''(x)| + |\hat{s}''(x) - s''(x)| \\ &\leq 4 \max_{\xi \in [a, b]} |f''(\xi) - \hat{s}''(\xi)| \\ &\leq \frac{1}{2} h^2 \max_{\xi \in [a, b] \setminus X_b} |f^{(4)}(\xi)|. \end{aligned}$$

For $j = 1$, the interpolation conditions dictate $f(x) - s(x) = 0$ for $x = x_i, x_{i+1}$ and Rolle's theorem B.91 implies $f'(\xi_i) - s'(\xi_i) = 0$ for some $\xi_i \in (x_i, x_{i+1})$. Then the second fundamental theorem of calculus (Theorem B.128) yields

$$\forall x \in [x_i, x_{i+1}], \quad f'(x) - s'(x) = \int_{\xi_i}^x (f''(t) - s''(t)) dt,$$

which, together with the integral mean value theorem B.124 and (7.1.3), gives

$$\begin{aligned} |f'(x) - s'(x)|_{x \in [x_i, x_{i+1}]} &= |x - \xi_i| |f''(\eta_i) - s''(\eta_i)| \\ &\leq \frac{1}{2} h^3 \max_{\xi \in [a, b] \setminus X_b} |f^{(4)}(\xi)|. \end{aligned}$$

Finally, for $j = 0$, interpolate $f - s$ with some linear spline $\bar{s} \in \mathbb{S}_1^0$. The interpolation conditions dictate $\forall x \in [a, b]$, $\bar{s}(x) \equiv 0$. Hence

$$\begin{aligned} |f(x) - s(x)|_{x \in [x_i, x_{i+1}]} &= |f(x) - s(x) - \bar{s}|_{x \in [x_i, x_{i+1}]} \\ &\leq \frac{1}{8} (x_{i+1} - x_i)^2 \max_{\xi \in [x_i, x_{i+1}]} |f''(\xi) - s''(\xi)| \\ &\leq \frac{1}{16} h^4 \max_{\xi \in [a, b] \setminus X_b} |f^{(4)}(\xi)|, \end{aligned}$$

where the second step follows from Theorem 6.7 and the third step from (7.1.3). $\square$

Exercise 7.14. Verify Theorem 7.13 using the results in Example 7.9.

7.2 B-form splines

Notation 6. In the notation $\mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$, t_i 's in the parentheses represent knots of a spline. When there is no danger of ambiguity, we also use the shorthand notation $\mathbb{S}_{n,N}^{n-1} := \mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$ or simply $\mathbb{S}_{n,N}^{n-1}$.

Theorem 7.15. The set of splines $\mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$ is a linear space with dimension $n + N - 1$.

Proof. It is easy to verify from (7.2) and Definition C.2 that $\mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$ is indeed a linear space. Note that the additive identity is the zero function not the number zero. One polynomial of degree n is determined by $n + 1$ coefficients. The $N - 1$ intervals lead to $(N - 1)(n + 1)$ coefficients. At each of the $N - 2$ interval knots, the smoothness condition requires that the 0th, 1st, $\dots$, $(n - 1)$ th derivatives of adjacent polynomials match. Hence the dimension is $(N - 1)(n + 1) - n(N - 2) = n + N - 1$. $\square$

Example 7.16. The cubic splines in Definition 7.5, have $n = 3$ and hence the dimension of $\mathbb{S}_3^2$ is $N + 2$. Apart from the N interpolation conditions at the knots, we need to impose two other conditions at the ends of the interpolating interval to obtain a unique spline, this leads to different types of cubic splines in Definition 7.5.

7.2.1 Truncated power functions

Definition 7.17. The *truncated power function* with exponent n is defined as

$$x_+^n = \begin{cases} x^n & \text{if } x \geq 0, \\ 0 & \text{if } x < 0; \end{cases} \quad (7.22)$$

see Figure 7.2 for an illustration.

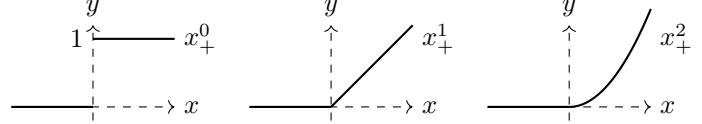


Figure 7.2: Truncated power functions (7.22) for $n = 0, 1, 2$.

Example 7.18. According to Definition 7.17, we have

$$\forall t \in [a, b], \quad \int_a^b (t - x)_+^n dx = \int_a^t (t - x)^n dx = \frac{(t - a)^{n+1}}{n + 1}. \quad (7.23)$$

Lemma 7.19. A basis of $\mathbb{S}_n^{n-1}(t_1, \dots, t_N)$ is

$$1, x, x^2, \dots, x^n, (x - t_2)_+^n, (x - t_3)_+^n, \dots, (x - t_{N-1})_+^n. \quad (7.24)$$

Proof. $\forall i = 2, 3, \dots, N - 1$, $(x - t_i)_+^n \in \mathbb{S}_{n,N}^{n-1}$. Also, $\forall i = 0, 1, \dots, n$, $x^i \in \mathbb{S}_{n,N}^{n-1}$. Suppose

$$\sum_{i=0}^n a_i x^i + \sum_{j=2}^{N-1} a_{n+j-1} (x - t_j)_+^n = \mathbf{0}(x). \quad (7.25)$$

To satisfy (7.25) for all $x < t_2$, a_i must be 0 for each $i = 0, 1, \dots, n$. To satisfy (7.25) for all $x \in (t_2, t_3)$, a_{n+1} must be 0. Similarly, all a_{n+j} 's must be zero. By Definition C.26, the functions in (7.24) are linearly independent. The proof is completed by Theorem 7.15, Lemma C.42, and the fact that there are $n + N - 1$ functions in (7.24). $\square$

Corollary 7.20. Any $s \in \mathbb{S}_{n,N}^{n-1}$ can be expressed as

$$\forall x \in [t_1, t_N], \quad s(x) = \sum_{i=0}^n a_i (x - t_1)^i + \sum_{j=1}^{N-2} a_{n+j} (x - t_{j+1})_+^n. \quad (7.26)$$

Proof. By Lemma 7.19, it suffices to point out that

$$\text{span}\{1, x, \dots, x^n\} = \text{span}\{1, (x - t_1), \dots, (x - t_1)^n\}. \quad \square$$

Example 7.21. Consider the piecewise linear interpolation using the truncated power basis, i.e., (7.26) with $n = 1$. Imagine a plastic rod that is initially straight. Place one of its end at (t_1, f_1) and let it go through (t_2, f_2) . In general (t_3, f_3) will be off the rod, but we can bend the rod at (t_2, f_2) so that it goes through (t_3, f_3) . Similarly, we bend the rod at (t_3, f_3) so that it goes through (t_4, f_4) . This bending action is repeated until the rod is finally bent at (t_{N-1}, f_{N-1}) . To sum up, the truncated power functions in the last summation of (7.26) realize the above “bending” process and yields C^0 continuity of the interpolant. For $n > 1$, this analogy of bending also holds and lead to a C^{n-1} continuity of the interpolant.

Example 7.22 (Piecewise linear interpolation with the truncated power basis could be ill-conditioned due to the non-uniformness of knots). Consider once again the piecewise linear interpolation in $\mathbb{S}(t_1, t_2, \dots, t_6)$ using the truncated power basis with $t_i = i$ except that $t_4 = 5 - h$. By Corollary 7.20, the interpolation conditions $(t_i, f_i)_{i=1}^6$ yield a linear system

$$M\mathbf{a} = \mathbf{f} := [f_1, f_2, \dots, f_6]^\top \quad (7.27)$$

where $\mathbf{a} = [a_0, a_1, \dots, a_5]^\top$ and $n = 1$ results in the matrix M being lower triangular,

$$M = \begin{bmatrix} 1 & & & & & \\ 1 & 1 & & & & \\ 1 & 2 & 1 & & & \\ 1 & 4-h & 3-h & 2-h & & \\ 1 & 4 & 3 & 2 & h & \\ 1 & 5 & 4 & 3 & 1+h & 1 \end{bmatrix}.$$

When h is small, M becomes almost singular because the fourth and fifth rows of M are almost linearly dependent. Indeed, for $h < 1$, the condition number of M based on the 2-norm, according to Lemma 2.23, is $\text{cond}_2 M = \frac{2-h}{h}$, which is unbounded as $h \rightarrow 0$.

7.2.2 The local support of B-splines

Definition 7.23. The *hat function* at t_i is

$$\hat{B}_i(x) = \begin{cases} \frac{x-t_{i-1}}{t_i-t_{i-1}} & x \in (t_{i-1}, t_i], \\ \frac{t_{i+1}-x}{t_{i+1}-t_i} & x \in (t_i, t_{i+1}], \\ 0 & \text{otherwise.} \end{cases} \quad (7.28)$$

Theorem 7.24. The hat functions form a basis of $\mathbb{S}_1^0$.

Proof. By Definition 7.23, we have

$$\hat{B}_i(t_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \quad (7.29)$$

Suppose $\sum_{i=1}^N c_i \hat{B}_i(x) = \mathbf{0}(x)$. Then we have $c_i = 0$ for each $i = 1, 2, \dots, N$ by setting $x = t_j$ and applying (7.29). Hence by Definition C.26 these N hat functions are linearly independent. The proof is completed by Theorem 7.15 and Lemma C.42. $\square$

Example 7.25 (Piecewise linear interpolation with the hat function basis is perfectly conditioned). By Example 7.22, piecewise interpolation with the truncated power basis could be ill-conditioned if the knots are non-uniform. In comparison, piecewise interpolation with the hat function basis is insensitive to knot positions. By Theorem 7.24, a function $f \in \mathbb{S}_1^0(t_1, t_2, \dots, t_6)$ can be expressed as

$$f(x) = \sum_{i=1}^6 a_{i-1} \hat{B}_i(x), \quad (7.30)$$

where the coefficients in $\mathbf{a} = [a_0, a_1, \dots, a_5]^\top$ are determined by solving the linear system $M\mathbf{a} = \mathbf{f}$ with the same $\mathbf{f}$ as that in (7.27). However, here (7.29) implies that $M = [\hat{B}_i(t_j)]$

is the identity matrix and thus $a_i = f(t_{i+1})$. Substitute this into (7.30) and we see from (7.28) that $f|_{[t_i, t_{i+1}]}$ is obtained by simply connecting the two conditions (t_i, f_i) and (t_{i+1}, f_{i+1}) by a straight line segment.

Definition 7.26. *B-splines* are defined recursively by

$$B_i^{n+1}(x) = \frac{x-t_{i-1}}{t_{i+n}-t_{i-1}} B_i^n(x) + \frac{t_{i+n+1}-x}{t_{i+n+1}-t_i} B_{i+1}^n(x). \quad (7.31)$$

The recursion base is the B-spline of degree zero,

$$B_i^0(x) = \begin{cases} 1 & \text{if } x \in (t_{i-1}, t_i], \\ 0 & \text{otherwise.} \end{cases} \quad (7.32)$$

Example 7.27. The hat functions in Definition 7.23 are clearly the B-splines of degree one:

$$B_i^1 = \hat{B}_i. \quad (7.33)$$

In (7.31), B-splines of higher degrees are defined by generalizing the idea of hat functions.

Example 7.28. The quadratic B-splines $B_i^2(x) =$

$$\begin{cases} \frac{(x-t_{i-1})^2}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}, & x \in (t_{i-1}, t_i]; \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(t_{i+2}-x)(x-t_i)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}]; \\ \frac{(t_{i+2}-x)^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}]; \\ 0, & \text{otherwise.} \end{cases} \quad (7.34)$$

Definition 7.29. The *support* of a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is

$$\text{supp}(f) = \text{closure}\{x \in \mathbb{R}^n : f(x) \neq 0\}; \quad (7.35)$$

where the closure of a set in the context of metric spaces is given in Definition D.67.

Lemma 7.30 (Support of B-splines). For $n \in \mathbb{N}^+$, the interval of support of B_i^n is $[t_{i-1}, t_{i+n}]$ where

$$\forall x \in (t_{i-1}, t_{i+n}), B_i^n(x) > 0. \quad (7.36)$$

Proof. This is an easy induction by (7.32) and (7.31). $\square$

Definition 7.31. Let X be a vector space. For each $x \in X$ we associate a unique real (or complex) number $L(x)$. If $\forall x, y \in X$ and $\forall \alpha, \beta \in \mathbb{R}$ (or $\mathbb{C}$), we have

$$L(\alpha x + \beta y) = \alpha L(x) + \beta L(y), \quad (7.37)$$

then L is called a *linear functional* over X .

Example 7.32. $X = \mathcal{C}[a, b]$, then the elements of X are functions continuous over $[a, b]$.

$$L(f) = \int_a^b f(x) dx, \quad L(f) = \int_a^b x^2 f(x) dx$$

are both linear functionals over X .

Notation 7. We have used the notation $f[x_0, \dots, x_k]$ for the k th divided difference of f , inline with considering $f[x_0, \dots, x_k]$ as a generalization of the Taylor expansion. Hereafter, for analyzing B-splines, it is both semantically and syntactically better to use the notation $[x_0, \dots, x_k]f$, inline with considering the *procedures* of a divided difference as a linear functional over $\mathcal{C}[x_0, x_k]$.

Theorem 7.33 (Leibniz formula). For $k \in \mathbb{N}$, the k th divided difference of a product of two functions satisfies

$$[x_0, \dots, x_k]fg = \sum_{i=0}^k [x_0, \dots, x_i]f \cdot [x_i, \dots, x_k]g. \quad (7.38)$$

Proof. The induction basis $k = 0$ holds because (7.38) reduces to $[x_0]fg = f(x_0)g(x_0)$. Now suppose (7.38) holds. For the induction step, we have from Theorem 6.17 that

$$[x_0, \dots, x_{k+1}]fg = \frac{[x_1, \dots, x_{k+1}]fg - [x_0, \dots, x_k]fg}{x_{k+1} - x_0}.$$

By the induction hypothesis, we have

$$\begin{aligned} [x_1, \dots, x_{k+1}]fg &= \sum_{i=0}^k [x_1, \dots, x_{i+1}]f \cdot [x_{i+1}, \dots, x_{k+1}]g \\ &= S_1 + \sum_{i=0}^k [x_0, \dots, x_i]f \cdot [x_{i+1}, \dots, x_{k+1}]g, \text{ where} \\ S_1 &= \sum_{i=0}^k (x_{i+1} - x_0) \cdot [x_0, \dots, x_{i+1}]f \cdot [x_{i+1}, \dots, x_{k+1}]g \\ &= \sum_{i=1}^{k+1} (x_i - x_0) \cdot [x_0, \dots, x_i]f \cdot [x_i, \dots, x_{k+1}]g. \\ [x_0, \dots, x_k]fg &= \sum_{i=0}^k [x_0, \dots, x_i]f \cdot [x_i, \dots, x_k]g \\ &= -S_2 + \sum_{i=0}^k [x_0, \dots, x_i]f \cdot [x_{i+1}, \dots, x_{k+1}]g, \text{ where} \\ S_2 &= \sum_{i=0}^k [x_0, \dots, x_i]f \cdot (x_{k+1} - x_i) \cdot [x_i, \dots, x_{k+1}]g. \end{aligned}$$

In the above derivation, we have applied Theorem 6.17 to go from the k th divided difference to the $(k+1)$ th. Then

$$\begin{aligned} [x_0, \dots, x_{k+1}]fg &= \frac{S_1 + S_2}{x_{k+1} - x_0} \\ &= \sum_{i=0}^{k+1} [x_0, \dots, x_i]f \cdot [x_i, \dots, x_{k+1}]g, \end{aligned}$$

which completes the inductive proof. $\square$

Example 7.34. There exists a relation between B-splines and truncated power functions, e.g.,

$$\begin{aligned} &(t_{i+1} - t_{i-1})[t_{i-1}, t_i, t_{i+1}](t-x)_+ \\ &= [t_i, t_{i+1}](t-x)_+ - [t_{i-1}, t_i](t-x)_+ \\ &= \frac{(t_{i+1} - x)_+ - (t_i - x)_+}{t_{i+1} - t_i} - \frac{(t_i - x)_+ - (t_{i-1} - x)_+}{t_i - t_{i-1}} \\ &= B_i^1 = \begin{cases} \frac{x - t_{i-1}}{t_i - t_{i-1}} & x \in (t_{i-1}, t_i], \\ \frac{t_{i+1} - x}{t_{i+1} - t_i} & x \in (t_i, t_{i+1}], \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

The algebra is illustrated in Figure 7.3.

The significance is that, by applying divided difference to truncated power functions we can “cure” their drawback of non-local support. This idea is made precise in the next Theorem.

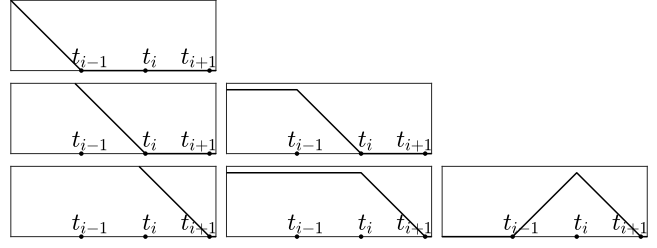


Figure 7.3: Divided difference of truncated power functions.

Theorem 7.35 (B-splines as divided difference of truncated power functions). For any $n \in \mathbb{N}$, we have

$$B_i^n(x) = (t_{i+n} - t_{i-1}) \cdot [t_{i-1}, \dots, t_{i+n}](t-x)_+^n. \quad (7.39)$$

Proof. For $n = 0$, (7.39) reduces to

$$\begin{aligned} B_i^0(x) &= (t_i - t_{i-1}) \cdot [t_{i-1}, t_i](t-x)_+^0 \\ &= (t_i - x)_+^0 - (t_{i-1} - x)_+^0 \\ &= \begin{cases} 0 & \text{if } x \in (-\infty, t_{i-1}], \\ 1 & \text{if } x \in (t_{i-1}, t_i], \\ 0 & \text{if } x \in (t_i, +\infty), \end{cases} \end{aligned}$$

which is the same as (7.32). Hence the induction basis holds. Now assume the induction hypothesis (7.39) hold.

By Definition 7.17, $(t-x)_+^{n+1} = (t-x)(t-x)_+^n$. Then the application of the Leibniz formula (Theorem 7.33) with $f = (t-x)$ and $g = (t-x)_+^n$ yields

$$\begin{aligned} &[t_{i-1}, \dots, t_{i+n}](t-x)_+^{n+1} \\ &= (t_{i-1} - x) \cdot [t_{i-1}, \dots, t_{i+n}](t-x)_+^n \\ &\quad + [t_i, \dots, t_{i+n}](t-x)_+^n. \end{aligned} \quad (7.40)$$

Definition 7.26 and the induction hypothesis yield

$$B_i^{n+1}(x) = \beta(x) + \gamma(x), \text{ with}$$

$$\begin{aligned} \beta(x) &= \frac{x - t_{i-1}}{t_{i+n} - t_{i-1}} B_i^n(x) \\ &= (x - t_{i-1}) \cdot [t_{i-1}, \dots, t_{i+n}](t-x)_+^n \\ &= [t_i, \dots, t_{i+n}](t-x)_+^n - [t_{i-1}, \dots, t_{i+n}](t-x)_+^{n+1}, \end{aligned}$$

where the last step follows from (7.40). Similarly,

$$\begin{aligned} \gamma(x) &= \frac{t_{i+n+1} - x}{t_{i+n+1} - t_i} B_{i+1}^n(x) \\ &= (t_{i+n+1} - x) \cdot [t_i, \dots, t_{i+n+1}](t-x)_+^n \\ &= (t_{i+n+1} - t_i) \cdot [t_i, \dots, t_{i+n+1}](t-x)_+^n \\ &\quad + (t_i - x) \cdot [t_i, \dots, t_{i+n+1}](t-x)_+^n \\ &= [t_{i+1}, \dots, t_{i+n+1}](t-x)_+^n - [t_i, \dots, t_{i+n}](t-x)_+^n \\ &\quad + [t_i, \dots, t_{i+n+1}](t-x)_+^{n+1} \\ &\quad - [t_{i+1}, \dots, t_{i+n+1}](t-x)_+^n \\ &= [t_i, \dots, t_{i+n+1}](t-x)_+^{n+1} - [t_i, \dots, t_{i+n}](t-x)_+^n, \end{aligned}$$

where the second last step follows from Theorem 6.17 and (7.40). The above arguments yield

$$\begin{aligned} B_i^{n+1}(x) &= [t_i, \dots, t_{i+n+1}](t-x)_+^{n+1} \\ &\quad - [t_{i-1}, \dots, t_{i+n}](t-x)_+^{n+1} \\ &= (t_{i+n+1} - t_{i-1}) \cdot [t_{i-1}, \dots, t_{i+n+1}](t-x)_+^{n+1}, \end{aligned}$$

which completes the inductive proof. $\square$

7.2.3 Integrals and derivatives

Corollary 7.36 (Integrals of B-splines). The average of a B-spline over its support only depends on its degree,

$$\frac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} B_i^n(x) dx = \frac{1}{n+1}. \quad (7.41)$$

Proof. The left-hand side (LHS) of (7.41) is

$$\begin{aligned} & \frac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} B_i^n(x) dx \\ &= \int_{t_{i-1}}^{t_{i+n}} [t_{i-1}, \dots, t_{i+n}](t-x)_+^n dx \\ &= [t_{i-1}, \dots, t_{i+n}] \int_{t_{i-1}}^{t_{i+n}} (t-x)_+^n dx \\ &= [t_{i-1}, \dots, t_{i+n}] \frac{(t-t_{i-1})^{n+1}}{n+1} \\ &= \frac{1}{n+1}, \end{aligned}$$

where the first step follows from Theorem 7.35, the second step from the commutativity of integration and taking divided difference, the third step from (7.23), and the last step from Corollary 6.22. $\square$

Theorem 7.37 (Derivatives of B-splines). For $n \geq 2$, we have, $\forall x \in \mathbb{R}$,

$$\frac{d}{dx} B_i^n(x) = \frac{n B_i^{n-1}(x)}{t_{i+n-1} - t_{i-1}} - \frac{n B_{i+1}^{n-1}(x)}{t_{i+n} - t_i}. \quad (7.42)$$

For $n = 1$, (7.42) holds for all x except at the three knots t_{i-1} , t_i , and t_{i+1} , where the derivative of B_i^1 is not defined.

Proof. We first show that (7.42) holds for all x except at the knots t_j . By (7.33), (7.28), and (7.32), we have

$$\forall x \in \mathbb{R} \setminus \{t_{i-1}, t_i, t_{i+1}\}, \quad \frac{d}{dx} B_i^1(x) = \frac{1}{t_i - t_{i-1}} B_i^0(x) - \frac{1}{t_{i+1} - t_i} B_{i+1}^0(x).$$

Hence the induction basis holds. Now suppose (7.42) holds $\forall x \in \mathbb{R} \setminus \{t_{i-1}, \dots, t_{i+n}\}$. Differentiate (7.31), apply the induction hypothesis (7.42), and we have

$$\frac{d}{dx} B_i^{n+1}(x) = \frac{B_i^n(x)}{t_{i+n} - t_{i-1}} - \frac{B_{i+1}^n(x)}{t_{i+n+1} - t_i} + nC(x), \quad (7.43)$$

where $C(x)$ is

$$\begin{aligned} & \frac{x-t_{i-1}}{t_{i+n}-t_{i-1}} \left[\frac{B_i^{n-1}(x)}{t_{i+n-1}-t_{i-1}} - \frac{B_{i+1}^{n-1}(x)}{t_{i+n}-t_i} \right] \\ &+ \frac{t_{i+n+1}-x}{t_{i+n+1}-t_i} \left[\frac{B_{i+1}^{n-1}(x)}{t_{i+n}-t_i} - \frac{B_{i+2}^{n-1}(x)}{t_{i+n+1}-t_{i+1}} \right] \\ &= \frac{1}{t_{i+n}-t_{i-1}} \left[\frac{(x-t_{i-1})B_i^{n-1}(x)}{t_{i+n-1}-t_{i-1}} + \frac{(t_{i+n}-x)B_{i+1}^{n-1}(x)}{t_{i+n}-t_i} \right] \\ &- \frac{1}{t_{i+n+1}-t_i} \left[\frac{(x-t_i)B_{i+1}^{n-1}(x)}{t_{i+n}-t_i} + \frac{(t_{i+n+1}-x)B_{i+2}^{n-1}(x)}{t_{i+n+1}-t_{i+1}} \right] \\ &= \frac{B_i^n(x)}{t_{i+n}-t_{i-1}} - \frac{B_{i+1}^n(x)}{t_{i+n+1}-t_i}, \end{aligned}$$

where the last step follows from (7.31). Then (7.43) can be written as

$$\frac{d}{dx} B_i^{n+1}(x) = \frac{(n+1)B_i^n(x)}{t_{i+n}-t_{i-1}} - \frac{(n+1)B_{i+1}^n(x)}{t_{i+n+1}-t_i},$$

which completes the inductive proof of (7.42) except at the knots. Since $B_i^1 = \hat{B}_i$ is continuous, an easy induction with (7.31) shows that B_i^n is continuous for all $n \geq 1$. Hence the RHS of (7.42) is continuous for all $n \geq 2$. Therefore, if $n \geq 2$, $\frac{d}{dx} B_i^n(x)$ exists for all $x \in \mathbb{R}$. This completes the proof. $\square$

Corollary 7.38 (Smoothness of B-splines). $B_i^n \in \mathbb{S}_n^{n-1}$.

Proof. For $n = 1$, the induction basis $B_i^1(x) \in \mathbb{S}_1^0$ holds because of (7.33). The rest of the proof follows from (7.31) and Theorem 7.37 via an easy induction. $\square$

7.2.4 Marsden's identity

Theorem 7.39 (Marsden's identity). For any $n \in \mathbb{N}$,

$$(t-x)^n = \sum_{i=-\infty}^{+\infty} (t-t_i) \cdots (t-t_{i+n-1}) B_i^n(x), \quad (7.44)$$

where the product $(t-t_i) \cdots (t-t_{i+n-1})$ is defined as 1 for $n = 0$.

Proof. For $n = 0$, (7.44) follows from Definition 7.26. Now suppose (7.44) holds. A linear interpolation of the linear function $f(t) = t - x$ is the function itself,

$$t-x = \frac{t-t_{i+n}}{t_{i-1}-t_{i+n}}(t_{i-1}-x) + \frac{t-t_{i-1}}{t_{i+n}-t_{i-1}}(t_{i+n}-x). \quad (7.45)$$

Hence for the inductive step we have

$$\begin{aligned} (t-x)^{n+1} &= (t-x) \sum_{i=-\infty}^{+\infty} (t-t_i) \cdots (t-t_{i+n-1}) B_i^n(x) \\ &= \sum_{i=-\infty}^{+\infty} (t-t_i) \cdots (t-t_{i+n}) \frac{t_{i-1}-x}{t_{i-1}-t_{i+n}} B_i^n(x) \\ &\quad + \sum_{i=-\infty}^{+\infty} (t-t_{i-1}) \cdots (t-t_{i+n-1}) \frac{t_{i+n}-x}{t_{i+n}-t_{i-1}} B_i^n(x) \\ &= \sum_{i=-\infty}^{+\infty} (t-t_i) \cdots (t-t_{i+n}) \frac{x-t_{i-1}}{t_{i+n}-t_{i-1}} B_i^n(x) \\ &\quad + \sum_{i=-\infty}^{+\infty} (t-t_i) \cdots (t-t_{i+n}) \frac{t_{i+n+1}-x}{t_{i+n+1}-t_i} B_{i+1}^n(x) \\ &= \sum_{i=-\infty}^{+\infty} (t-t_i) \cdots (t-t_{i+n}) B_i^{n+1}(x), \end{aligned}$$

where the first step follows from the induction hypothesis, the second step from (7.45), the third step from replacing i with $i+1$ in the second summation, and the last step from (7.31). $\square$

Corollary 7.40 (Truncated power functions as linear combinations of B-splines). For any $j \in \mathbb{Z}$ and $n \in \mathbb{N}$,

$$(t_j - x)_+^n = \sum_{i=-\infty}^{j-n} (t_j - t_i) \cdots (t_j - t_{i+n-1}) B_i^n(x). \quad (7.46)$$

Proof. We need to show that the RHS is $(t_j - x)^n$ if $x \leq t_j$ and 0 otherwise. Set $t = t_j$ in (7.44) and we have

$$(t_j - x)^n = \sum_{i=-\infty}^{+\infty} (t_j - t_i) \cdots (t_j - t_{i+n-1}) B_i^n(x).$$

For each $i = j - n + 1, \dots, j$, the corresponding term in the summation is zero regardless of x ; for each $i \geq j + 1$, Lemma 7.30 implies that $B_i^n(x) = 0$ for all $x \leq t_j$. Hence

$$x \leq t_j \implies \sum_{i=-\infty}^{j-n} (t_j - t_i) \cdots (t_j - t_{i+n-1}) B_i^n(x) = (t_j - x)^n.$$

Otherwise $x > t_j$, then Lemma 7.30 implies $B_i^n(x) = 0$ for each $i \leq j - n$. This completes the proof. $\square$

7.2.5 Symmetric polynomials

Definition 7.41. The *elementary symmetric polynomial* of degree k in n variables is the sum of all products of k distinct variables chosen from the n variables,

$$\sigma_k(x_1, \dots, x_n) = \sum_{1 \leq i_1 < \dots < i_k \leq n} x_{i_1} x_{i_2} \cdots x_{i_k}. \quad (7.47)$$

In particular, $\sigma_0(x_1, \dots, x_n) = 1$ and

$$\forall k > n, \quad \sigma_k(x_1, \dots, x_n) = 0.$$

If the distinctiveness condition is dropped, we have the *complete symmetric polynomial* of degree k in n variables,

$$\tau_k(x_1, \dots, x_n) = \sum_{1 \leq i_1 \leq \dots \leq i_k \leq n} x_{i_1} x_{i_2} \cdots x_{i_k}. \quad (7.48)$$

Example 7.42. $\sigma_2(x_1, x_2, x_3) = x_1 x_2 + x_1 x_3 + x_2 x_3$. In comparison, $\tau_2(x_1, x_2, x_3) = \sigma_2(x_1, x_2, x_3) + x_1^2 + x_2^2 + x_3^2$.

Lemma 7.43. For $k \leq n$, the elementary symmetric polynomials satisfy a recursion,

$$\begin{aligned} & \sigma_{k+1}(x_1, \dots, x_n, x_{n+1}) \\ &= \sigma_{k+1}(x_1, \dots, x_n) + x_{n+1} \sigma_k(x_1, \dots, x_n). \end{aligned} \quad (7.49)$$

Proof. The terms in $\sigma_{k+1}(x_1, \dots, x_n, x_{n+1})$ can be assorted into two groups: (a) those that contain the factor x_{n+1} and (b) those that do not. By the symmetry in (7.47), group (a) must be $x_{n+1} \sigma_k(x_1, \dots, x_n)$ and group (b) must be $\sigma_{k+1}(x_1, \dots, x_n)$. $\square$

Example 7.44. $\sigma_2(x_1, x_2, x_3) = x_1 x_2 + x_3(x_1 + x_2)$.

Definition 7.45. The *generating function for the elementary symmetric polynomials* is

$$g_{\sigma,n}(z) = \prod_{i=1}^n (1 + x_i z) = (1 + x_1 z) \cdots (1 + x_n z) \quad (7.50)$$

while that for the complete symmetric polynomials is

$$g_{\tau,n}(z) = \prod_{i=1}^n \frac{1}{1 - x_i z} = \frac{1}{1 - x_1 z} \cdots \frac{1}{1 - x_n z}. \quad (7.51)$$

Lemma 7.46 (Generating elementary and complete symmetric polynomials). The elementary and complete symmetric polynomials are related to their generating functions as

$$g_{\sigma,n}(z) = \sum_{k=0}^n \sigma_k(x_1, \dots, x_n) z^k. \quad (7.52)$$

$$g_{\tau,n}(z) = \sum_{k=0}^{+\infty} \tau_k(x_1, \dots, x_n) z^k. \quad (7.53)$$

Proof. With Lemma 7.43, we can prove (7.52) by an easy induction. For (7.53), (7.51) and the identity

$$\frac{1}{1-x} = \sum_{k=0}^{+\infty} x^k \quad (7.54)$$

yield

$$\begin{aligned} g_{\tau,n}(z) &= \prod_{i=1}^n \sum_{k=0}^{+\infty} x_i^k z^k \\ &= (1 + x_1 z + x_1^2 z^2 + \cdots)(1 + x_2 z + x_2^2 z^2 + \cdots) \\ &\quad \cdots (1 + x_n z + x_n^2 z^2 + \cdots). \end{aligned}$$

The coefficient of the monomial z^k , is the sum of all possible products of k variables from $x_1, x_2, \dots, x_n$. Definition 7.41 then completes the proof. $\square$

Example 7.47.

$$\begin{aligned} & (1 + x_1 z)(1 + x_2 z)(1 + x_3 z) \\ &= 1 + (x_1 + x_2 + x_3)z \\ &\quad + (x_1 x_2 + x_1 x_3 + x_2 x_3)z^2 + x_1 x_2 x_3 z^3. \end{aligned}$$

Lemma 7.48 (Recursive relations of complete symmetric polynomials). The complete symmetric polynomials satisfy a recursion,

$$\begin{aligned} & \tau_{k+1}(x_1, \dots, x_n, x_{n+1}) \\ &= \tau_{k+1}(x_1, \dots, x_n) + x_{n+1} \tau_k(x_1, \dots, x_n, x_{n+1}). \end{aligned} \quad (7.55)$$

Proof. (7.51) implies

$$g_{\tau,n+1} = g_{\tau,n} + x_{n+1} z g_{\tau,n+1}. \quad (7.56)$$

The proof is completed by requiring that the coefficient of z^{k+1} on the LHS equal that of z^{k+1} on the RHS. $\square$

Theorem 7.49 (Complete symmetric polynomials as divided difference of monomials). The complete symmetric polynomial of degree $m - n$ in $n + 1$ variables is the n th divided difference of the monomial x^m , i.e.

$$\begin{aligned} & \forall m \in \mathbb{N}^+, \forall i \in \mathbb{N}, \forall n = 0, 1, \dots, m, \\ & \tau_{m-n}(x_i, \dots, x_{i+n}) = [x_i, \dots, x_{i+n}] x^m. \end{aligned} \quad (7.57)$$

Proof. By Lemma 7.48, we have

$$\begin{aligned} & (x_{n+1} - x_1) \tau_k(x_1, \dots, x_n, x_{n+1}) \\ &= \tau_{k+1}(x_1, \dots, x_n, x_{n+1}) - \tau_{k+1}(x_1, \dots, x_n) \\ &\quad - x_1 \tau_k(x_1, \dots, x_n, x_{n+1}) \\ &= \tau_{k+1}(x_2, \dots, x_n, x_{n+1}) + x_1 \tau_k(x_1, \dots, x_n, x_{n+1}) \\ &\quad - \tau_{k+1}(x_1, \dots, x_n) - x_1 \tau_k(x_1, \dots, x_n, x_{n+1}) \\ &= \tau_{k+1}(x_2, \dots, x_n, x_{n+1}) - \tau_{k+1}(x_1, \dots, x_n). \end{aligned} \quad (7.58)$$

The rest of the proof is an induction on n . For $n = 0$, (7.57) reduces to

$$\tau_m(x_i) = [x_i] x^m,$$

which is trivially true. Now suppose (7.57) holds for a non-negative integer $n < m$. Then (7.58) and the induction hypothesis yield

$$\begin{aligned} & (x_{i+n+1} - x_i) \tau_{m-n-1}(x_i, \dots, x_{i+n+1}) \\ &= \tau_{m-n}(x_{i+1}, \dots, x_{i+n+1}) - \tau_{m-n}(x_i, \dots, x_{i+n}) \\ &= [x_{i+1}, \dots, x_{i+n+1}] x^m - [x_i, \dots, x_{i+n}] x^m \\ &= (x_{i+n+1} - x_i) [x_i, \dots, x_{i+n+1}] x^m, \end{aligned}$$

which completes the proof. $\square$

7.2.6 B-splines indeed form a basis

Theorem 7.50. Given any $k \in \mathbb{N}$, the monomial x^k can be expressed as a linear combination of B-splines for any fixed $n \geq k$, in the form

$$\binom{n}{k} x^k = \sum_{i=-\infty}^{+\infty} \sigma_k(t_i, \dots, t_{i+n-1}) B_i^n(x), \quad (7.59)$$

where $\sigma_k(t_i, \dots, t_{i+n-1})$ is the elementary symmetric polynomial of degree k in the n variables $t_i, \dots, t_{i+n-1}$.

Proof. Lemma 7.46 yields

$$(1 + t_i x) \cdots (1 + t_{i+n-1} x) = \sum_{k=0}^n \sigma_k(t_i, \dots, t_{i+n-1}) x^k.$$

Replace x with $-\frac{1}{t}$, multiply the above equation with t^n , and we have

$$(t - t_i) \cdots (t - t_{i+n-1}) = \sum_{k=0}^n \sigma_k(t_i, \dots, t_{i+n-1}) (-1)^k t^{n-k}.$$

Substituting the above into (7.44) yields

$$\begin{aligned} (t - x)^n &= \sum_{i=-\infty}^{+\infty} \sum_{k=0}^n \sigma_k(t_i, \dots, t_{i+n-1}) (-1)^k t^{n-k} B_i^n(x) \\ &= \sum_{k=0}^n \left\{ t^{n-k} (-1)^k \sum_{i=-\infty}^{+\infty} \sigma_k(t_i, \dots, t_{i+n-1}) B_i^n(x) \right\}. \end{aligned}$$

On the other hand, the binomial theorem states that

$$(t - x)^n = \sum_{k=0}^n \binom{n}{k} t^{n-k} (-x)^k = \sum_{k=0}^n t^{n-k} (-1)^k \binom{n}{k} x^k.$$

Comparing the last two equations completes the proof. $\square$

Corollary 7.51 (Partition of Unity).

$$\forall n \in \mathbb{N}, \quad \sum_{i=-\infty}^{+\infty} B_i^n = 1. \quad (7.60)$$

Proof. Setting $k = 0$ in Theorem 7.50 yields (7.60). $\square$

Theorem 7.52. The following list of B-splines is a basis of $\mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$,

$$B_{2-n}^n(x), B_{3-n}^n(x), \dots, B_N^n(x). \quad (7.61)$$

Proof. It is easy to verify that

$$\forall t_i \in \mathbb{R}, \quad (x - t_i)_+^n = (x - t_i)^n - (-1)^n (t_i - x)_+^n. \quad (7.62)$$

Then it follows from Theorem 7.39 and Corollary 7.40 that each truncated power function $(x - t_i)_+^n$ can be expressed as a linear combination of B-splines. By Lemma 7.19, each element in $\mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$ can be expressed as a linear combination of

$$1, x, x^2, \dots, x^n, (x - t_2)_+^n, (x - t_3)_+^n, \dots, (x - t_{N-1})_+^n.$$

Theorem 7.50 states that each monomial x^j can also be expressed as a linear combination of B-splines. Since the domain is restricted to $[t_1, t_N]$, we know from Lemma 7.30 that only those B-splines in the list of (7.61) appear in the linear combination. Therefore, these B-splines form a spanning list of $\mathbb{S}_n^{n-1}(t_1, t_2, \dots, t_N)$. The proof is completed by Lemma C.41, Theorem 7.15, and the fact that the length of the list (7.61) is also $n + N - 1$. $\square$

7.2.7 Cardinal B-splines

Definition 7.53. The *cardinal B-spline* of degree n , denoted by $B_{i,\mathbb{Z}}^n$, is the B-spline in Definition 7.26 on the knot set $\mathbb{Z}$.

Corollary 7.54. Cardinal B-splines of the same degree are translates of one another, i.e.

$$\forall x \in \mathbb{R}, \quad B_{i,\mathbb{Z}}^n(x) = B_{i+1,\mathbb{Z}}^n(x+1). \quad (7.63)$$

Proof. The recurrence relation (7.31) reduces to

$$B_{i,\mathbb{Z}}^{n+1}(x) = \frac{x-i+1}{n+1} B_{i,\mathbb{Z}}^n(x) + \frac{i+n+1-x}{n+1} B_{i+1,\mathbb{Z}}^n(x). \quad (7.64)$$

The rest of the proof is an easy induction on n . $\square$

Corollary 7.55. A cardinal B-spline is symmetric about the center of its interval of support, i.e.

$$\forall n > 0, \forall x \in \mathbb{R}, \quad B_{i,\mathbb{Z}}^n(x) = B_{i,\mathbb{Z}}^n(2i+n-1-x). \quad (7.65)$$

Proof. The proof is similar with that of Corollary 7.54. $\square$

Example 7.56. For $t_i = i$, the quadratic B-spline in Example 7.28 simplifies to

$$B_{i,\mathbb{Z}}^2(x) = \begin{cases} \frac{(x-i+1)^2}{2} & \text{if } x \in (i-1, i]; \\ \frac{3}{4} - \left(x - (i + \frac{1}{2})\right)^2 & \text{if } x \in (i, i+1]; \\ \frac{(i+2-x)^2}{2} & \text{if } x \in (i+1, i+2]; \\ 0 & \text{otherwise.} \end{cases} \quad (7.66)$$

It is straightforward to verify Corollaries 7.54 and 7.55. It also follows from (7.66) that

$$B_{i,\mathbb{Z}}^2(j) = \begin{cases} \frac{1}{2} & \text{if } j \in \{i, i+1\}; \\ 0 & \text{if } j \in \mathbb{Z} \setminus \{i, i+1\}. \end{cases} \quad (7.67)$$

Example 7.57. For $t_i = i$, the cubic cardinal B-spline is

$$B_{i,\mathbb{Z}}^3(x) = \begin{cases} \frac{(x-i+1)^3}{6} & \text{if } x \in (i-1, i]; \\ \frac{2}{3} - \frac{1}{2}(x-i+1)(i+1-x)^2 & \text{if } x \in (i, i+1]; \\ B_{i,\mathbb{Z}}^3(2i+2-x) & \text{if } x \in (i+1, i+3); \\ 0 & \text{otherwise.} \end{cases} \quad (7.68)$$

It follows that

$$B_{i,\mathbb{Z}}^3(j) = \begin{cases} \frac{1}{6} & \text{if } j \in \{i, i+2\}; \\ \frac{2}{3} & \text{if } j = i+1; \\ 0 & \text{if } j \in \mathbb{Z} \setminus \{i, i+1, i+2\}. \end{cases} \quad (7.69)$$

This illustrates Corollary 7.54 that cardinal B-splines have the same shape, i.e., they are invariant under integer translations.

Theorem 7.58. The cardinal B-spline of degree n can be explicitly expressed as

$$B_{i,\mathbb{Z}}^n(x) = \frac{1}{n!} \sum_{k=-1}^n (-1)^{n-k} \binom{n+1}{k+1} (k+i-x)_+^n. \quad (7.70)$$

Proof. Theorems 7.35, 6.29, and 6.28 yield

$$\begin{aligned} B_{i,\mathbb{Z}}^n(x) &= (n+1)[i-1, \dots, i+n](t-x)_+^n \\ &= \frac{n+1}{(n+1)!} \Delta^{n+1}(i-1-x)_+^n \\ &= \frac{1}{n!} \sum_{k=0}^{n+1} (-1)^{n+1-k} \binom{n+1}{k} (i-1+k-x)_+^n. \end{aligned}$$

Replacing k with $k+1$ and accordingly changing the summation bounds complete the proof. $\square$

Corollary 7.59. The value of a cardinal B-spline at an integer j is

$$B_{i,\mathbb{Z}}^n(j) = \frac{1}{n!} \sum_{k=j-i+1}^n (-1)^{n-k} \binom{n+1}{k+1} (k+i-j)^n \quad (7.71)$$

for $j \in [i, n+i]$ and is zero otherwise.

Proof. This follows directly from Theorem 7.58 and Definition 7.17. $\square$

Theorem 7.60 (Unique interpolation by complete cubic cardinal B-splines). There is a unique B-spline $S(x) \in \mathbb{S}_3^2$ that interpolates $f(x)$ at $1, 2, \dots, N$ with $S'(1) = f'(1)$ and $S'(N) = f'(N)$. Furthermore, this B-spline is

$$S(x) = \sum_{i=-1}^N a_i B_{i,\mathbb{Z}}^3(x), \quad (7.72)$$

where

$$a_{-1} = a_1 - 2f'(1), \quad a_N = a_{N-2} + 2f'(N), \quad (7.73)$$

and $\mathbf{a}^\top = [a_0, \dots, a_{N-1}]$ is the solution of the linear system $M\mathbf{a} = \mathbf{b}$ with

$$\begin{aligned} \mathbf{b}^\top &= [3f(1) + f'(1), 6f(2), \\ &\quad \dots, 6f(N-1), 3f(N) - f'(N)], \\ M &= \begin{bmatrix} 2 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 2 \end{bmatrix}. \end{aligned}$$

Proof. By Theorem 7.52 and Lemma 7.30, we have, at each interpolation site $i = 1, 2, \dots, N$,

$$f(i) = a_{i-2} B_{i-2,\mathbb{Z}}^3(i) + a_{i-1} B_{i-1,\mathbb{Z}}^3(i) + a_i B_{i,\mathbb{Z}}^3(i).$$

Then (7.69) yields

$$\forall i = 1, 2, \dots, N, \quad a_{i-2} + 4a_{i-1} + a_i = 6f(i), \quad (7.74)$$

which proves the middle $N-2$ equations of $M\mathbf{a} = \mathbf{b}$. By Theorem 7.37, we have

$$\frac{d}{dx} B_{i,\mathbb{Z}}^n(x) = B_{i,\mathbb{Z}}^{n-1}(x) - B_{i+1,\mathbb{Z}}^{n-1}(x). \quad (7.75)$$

Differentiate (7.72), apply (7.75), set $x = 1$, apply (7.67) and we have the first identity in (7.73), which, together with (7.74), yields

$$2a_0 + a_1 = f'(1) + 3f(1);$$

this proves the first equation of $M\mathbf{a} = \mathbf{b}$. The last equation of $M\mathbf{a} = \mathbf{b}$ and the second identity in (7.73) can be shown similarly. The strictly diagonal dominance of M implies a nonzero determinant of M and therefore $\mathbf{a}$ is uniquely determined. The uniqueness of $S(x)$ then follows from (7.73). $\square$

Theorem 7.61. There is a unique B-spline $S(x) \in \mathbb{S}_2^1$ that interpolates $f(x)$ at $t_i = i + \frac{1}{2}$ for each $i = 1, 2, \dots, N-1$ with end conditions $S(1) = f(1)$ and $S(N) = f(N)$. Furthermore, this B-spline is

$$S(x) = \sum_{i=0}^N a_i B_{i,\mathbb{Z}}^2(x), \quad (7.76)$$

where

$$a_0 = 2f(1) - a_1, \quad a_N = 2f(N) - a_{N-1}, \quad (7.77)$$

and $\mathbf{a}^\top = [a_1, \dots, a_{N-1}]$ is the solution of the linear system $M\mathbf{a} = \mathbf{b}$ with

$$\begin{aligned} \mathbf{b}^\top &= \left[8f\left(\frac{3}{2}\right) - 2f(1), 8f\left(\frac{5}{2}\right), \right. \\ &\quad \left. \dots, 8f\left(N - \frac{3}{2}\right), 8f\left(N - \frac{1}{2}\right) - 2f(N) \right], \\ M &= \begin{bmatrix} 5 & 1 & & & \\ 1 & 6 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 6 & 1 \\ & & & 1 & 5 \end{bmatrix}. \end{aligned}$$

Proof. It follows from Lemma 7.30 and Definition 7.53 that there are three quadratic cardinal B-splines, namely $B_{i-1,\mathbb{Z}}^2$, $B_{i,\mathbb{Z}}^2$, and $B_{i+1,\mathbb{Z}}^2$, that have nonzero values at each interpolation site $t_i = i + \frac{1}{2}$. Hence we have

$$f(t_i) = a_{i-1} B_{i-1,\mathbb{Z}}^2(t_i) + a_i B_{i,\mathbb{Z}}^2(t_i) + a_{i+1} B_{i+1,\mathbb{Z}}^2(t_i). \quad (7.78)$$

Hence the dimension of the space of relevant cardinal B-splines is $N-1+2 = N+1$, which is different from that in the proof of Theorem 7.52! By Theorem 7.58, we can calculate the values of B-splines as:

$$\begin{aligned} B_{0,\mathbb{Z}}^2(x) &= \frac{1}{2} \sum_{k=-1}^2 (-1)^{2-k} \binom{3}{k+1} (k-x)_+^2, \\ B_{0,\mathbb{Z}}^2\left(\frac{1}{2}\right) &= \frac{3}{4}, \\ B_{0,\mathbb{Z}}^2\left(-\frac{1}{2}\right) &= B_{0,\mathbb{Z}}^2\left(\frac{3}{2}\right) = \frac{1}{8}, \end{aligned}$$

where for $B_{0,\mathbb{Z}}^2\left(-\frac{1}{2}\right)$ we have used Corollary 7.55. Then Corollary 7.54 and (7.78) yield

$$a_{i-1} + 6a_i + a_{i+1} = 8f(t_i), \quad (7.79)$$

which proves the middle $N - 3$ equations in $M\mathbf{a} = \mathbf{b}$. At the end point $x = 1$, only two quadratic cardinal B-splines, $B_{0,\mathbb{Z}}^2$ and $B_{1,\mathbb{Z}}^2$, are nonzero. Then Example 7.28 yields

$$\frac{1}{2}a_0 + \frac{1}{2}a_1 = f(1)$$

and this proves the first identity in (7.77). Also, the above equation and (7.79) with $i = 1$ yield

$$5a_1 + a_2 = 8f\left(\frac{3}{2}\right) - 2f(1),$$

which proves the first equation in $M\mathbf{a} = \mathbf{b}$. The last equation in $M\mathbf{a} = \mathbf{b}$ can be proven similarly. $\square$

7.3 Curve fitting

Definition 7.62. The *cumulative chordal lengths* of a point sequence $(\mathbf{x}_i)_{i=1}^N$ in $\mathbb{R}^D$ form a sequence of N real numbers,

$$t_i = \begin{cases} 0, & \text{if } i = 1; \\ t_{i-1} + \|\mathbf{x}_i - \mathbf{x}_{i-1}\|_2, & \text{otherwise,} \end{cases} \quad (7.80)$$

where $\|\cdot\|_2$ denotes the 2-norm.

Algorithm 7.63. A curve γ is approximated by fitting D spline functions from N characteristic points $(\mathbf{x}_i)_{i=1}^N$ on γ :

- (a) compute the cumulative chordal length for each $\mathbf{x}_i$;
- (b) fit a spline function for each coordinate of γ with the independent parameter as the cumulative chordal length.

Theorem 7.64. An $O(\epsilon)$ perturbation to the position of each characteristic point in Algorithm 7.63 causes an $O(\epsilon)$ error at each point of the fitted spline. In other words, Algorithm 7.63 is well conditioned.

Proof. We only consider the case of a *periodic* cubic spline S with N characteristic points $(\mathbf{x}_{N+1} = \mathbf{x}_1)$ since other cases can be proved similarly. By Lemma 7.3, the PP form of S can be constructed via the linear system

$$A\mathbf{m} := \begin{bmatrix} 2 & \mu_1 & & & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \lambda_{N-1} & 2 & \mu_{N-1} \\ \mu_N & & & \lambda_N & 2 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \\ m_N \end{bmatrix} = \mathbf{b}, \quad (7.81)$$

where $t_1, t_2, \dots, t_{N+1}$ are the cumulative chordal lengths of $(\mathbf{x}_i)_{i=1}^{N+1}$, the first derivative $m_i := S'(t_i)$ is at the i th breakpoint, the off-diagonal elements of A are

$$\mu_1 = \frac{t_{N+1} - t_N}{t_{N+1} - t_N + t_2 - t_1}, \quad \lambda_1 = \frac{t_2 - t_1}{t_{N+1} - t_N + t_2 - t_1},$$

$$\forall i = 2, 3, \dots, N, \quad \begin{cases} \mu_i = \frac{t_i - t_{i-1}}{t_{i+1} - t_{i-1}}; \\ \lambda_i = \frac{t_{i+1} - t_i}{t_{i+1} - t_{i-1}}, \end{cases}$$

and the elements of $\mathbf{b}$ are $b_i = 3\mu_i K_i + 3\lambda_i K_{i-1}$ with $K_i := S[t_i, t_{i+1}] = \frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{t_{i+1} - t_i}$.

All μ_i 's and λ_i 's are positive and satisfy $\mu_i + \lambda_i = 1$, thus A in (7.81) is strictly diagonally dominant, invertible, and well conditioned. Then, for each $i = 1, \dots, N$, we have

$$S|_{[t_i, t_{i+1}]}(t) = S(t_i) + (t - t_i)m_i + (t - t_i)^2 \frac{K_i - m_i}{t_{i+1} - t_i} + (t - t_i)^2 (t - t_{i+1}) \frac{m_i + m_{i+1} - 2K_i}{(t_{i+1} - t_i)^2}. \quad (7.82)$$

An $O(\epsilon)$ perturbation to $(\mathbf{x}_i)_{i=1}^N$ yields a new sequence $(\hat{\mathbf{x}}_i)_{i=1}^N$ of breakpoints satisfying

$$\hat{\mathbf{x}}_i - \mathbf{x}_i = O(\epsilon); \quad \Delta \hat{t}_i - \Delta t_i = O(\epsilon), \quad (7.83)$$

where $\Delta t_i := t_{i+1} - t_i$ and $\hat{t}_i$'s are cumulative chordal lengths of $\hat{\mathbf{x}}_i$'s.

A perturbed spline $\hat{S}$ can be constructed from $\hat{\mathbf{x}}_i$'s via

$$\hat{A}\hat{\mathbf{m}} = \hat{\mathbf{b}},$$

where $\hat{A}$, $\hat{\mathbf{m}}$, and $\hat{b}_i$ are similar to those in (7.81). For each $i = 1, \dots, N$, the form of the perturbed spline is

$$\begin{aligned} \hat{S}|_{[\hat{t}_i, \hat{t}_{i+1}]}(\hat{t}) &= \hat{S}(\hat{t}_i) + (\hat{t} - \hat{t}_i)\hat{m}_i + (\hat{t} - \hat{t}_i)^2 \frac{\hat{K}_i - \hat{m}_i}{\hat{t}_{i+1} - \hat{t}_i} \\ &\quad + (\hat{t} - \hat{t}_i)^2 (\hat{t} - \hat{t}_{i+1}) \frac{\hat{m}_i + \hat{m}_{i+1} - 2\hat{K}_i}{(\hat{t}_{i+1} - \hat{t}_i)^2}. \end{aligned} \quad (7.84)$$

The one-to-one correspondence of $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ furnishes a bijection $v : [0, t_{N+1}] \rightarrow [0, \hat{t}_{N+1}]$,

$$\hat{t}|_{[\hat{t}_i, \hat{t}_{i+1}]} = v|_{[t_i, t_{i+1}]}(t) = \frac{\Delta \hat{t}_i}{\Delta t_i} (t - t_i) + \hat{t}_i,$$

Then (7.83) yields

$$\begin{aligned} &(\hat{t}|_{[\hat{t}_i, \hat{t}_{i+1}]} - \hat{t}_i) - (t|_{[t_i, t_{i+1}]} - t_i) \\ &= [v|_{[t_i, t_{i+1}]}(t) - \hat{t}_i] - (t - t_i) \\ &= \frac{\Delta \hat{t}_i - \Delta t_i}{\Delta t_i} (t|_{[t_i, t_{i+1}]} - t_i) \leq |\Delta \hat{t}_i - \Delta t_i| = O(\epsilon). \end{aligned} \quad (7.85)$$

It follows from (7.83) that

$$\hat{\mu}_i - \mu_i = -(\hat{\lambda}_i - \lambda_i) = O(\epsilon), \quad \hat{K}_i - K_i = O(\epsilon), \quad \hat{b}_i - b_i = O(\epsilon), \quad (7.86)$$

which, together with (7.81), yields

$$\begin{aligned} \hat{\mathbf{m}} - \mathbf{m} &= \hat{A}^{-1}\hat{\mathbf{b}} - A^{-1}\mathbf{b} \\ &= \hat{A}^{-1}(\hat{\mathbf{b}} - \mathbf{b}) - \hat{A}^{-1}(\hat{A} - A)A^{-1}\mathbf{b} \\ &= O(\epsilon). \end{aligned} \quad (7.87)$$

Subtract (7.82) from (7.84), apply (7.83), (7.85), (7.86), and (7.87), and we arrive at the conclusion, i.e.,

$$\forall t \in [0, t_{N+1}], |\hat{S}(v(t)) - S(t)| = O(\epsilon). \quad \square$$

7.4 Theoretical problems

- I. Consider $s \in \mathbb{S}_3^2$ on $[0, 2]$:

$$s(x) = \begin{cases} p(x) & \text{if } x \in [0, 1], \\ (2 - x)^3 & \text{if } x \in [1, 2]. \end{cases}$$

Determine $p \in \mathbb{P}_3$ such that $s(0) = 0$. Is $s(x)$ a natural cubic spline?

- II. Given $f_i = f(x_i)$ of some scalar function at points $a = x_1 < x_2 < \dots < x_n = b$, we consider interpolating f on $[a, b]$ with a quadratic spline $s \in \mathbb{S}_2^1$.

- (a) Why is an additional condition needed to determine s uniquely?
- (b) Define $m_i = s'(x_i)$ and $p_i = s|_{[x_i, x_{i+1}]}$. Determine p_i in terms of f_i, f_{i+1} , and m_i for $i = 1, 2, \dots, n-1$.
- (c) Suppose $m_1 = f'(a)$ is given. Show how $m_2, m_3, \dots, m_{n-1}$ can be computed.
- III. Let $s_1(x) = 1 + c(x+1)^3$ where $x \in [-1, 0]$ and $c \in \mathbb{R}$. Determine $s_2(x)$ on $[0, 1]$ such that

$$s(x) = \begin{cases} s_1(x) & \text{if } x \in [-1, 0], \\ s_2(x) & \text{if } x \in [0, 1] \end{cases}$$

is a natural cubic spline on $[-1, 1]$ with knots $-1, 0, 1$. How must c be chosen if one wants $s(1) = -1$?

- IV. Consider $f(x) = \cos(\frac{\pi}{2}x)$ with $x \in [-1, 1]$.
- (a) Determine the natural cubic spline interpolant to f on knots $-1, 0, 1$.
- (b) As discussed in the class, natural cubic splines have the minimal total bending energy. Verify this by taking $g(x)$ be (i) the quadratic polynomial that interpolates f at $-1, 0, 1$, and (ii) $f(x)$.
- V. The quadratic B-spline $B_i^2(x)$.
- (a) Derive the same explicit expression of $B_i^2(x)$ as that in the notes from the recursive definition of B-splines and the hat function.
- (b) Verify that $\frac{d}{dx}B_i^2(x)$ is continuous at t_i and t_{i+1} .
- (c) Show that only one $x^* \in (t_{i-1}, t_{i+1})$ satisfies $\frac{d}{dx}B_i^2(x^*) = 0$. Express x^* in terms of the knots within the interval of support.
- (d) Consequently, show $B_i^2(x) \in [0, 1]$.
- (e) Plot $B_i^2(x)$ for $t_i = i$.
- VI. Verify Theorem 7.35 algebraically for the case of $n = 2$, i.e.

$$(t_{i+2} - t_{i-1})[t_{i-1}, t_i, t_{i+1}, t_{i+2}](t - x)_+^2 = B_i^2.$$

- VII. Scaled integral of B-splines.
Deduce from the Theorem on derivatives of B-splines that the scaled integral of a B-spline $B_i^n(x)$ over its support is independent of its index i even if the spacing of the knots is not uniform.
- VIII. Symmetric Polynomials.
We have a theorem on expressing complete symmetric polynomials as divided difference of monomials.
- (a) Verify this theorem for $m = 4$ and $n = 2$ by working out the table of divided difference and comparing the result to the definition of complete symmetric polynomials.
- (b) Prove this theorem by the lemma on the recursive relation on complete symmetric polynomials.

- IX. In fitting a curve by cubic splines, one gets inaccurate results when the distance between two adjacent points is much smaller than those of other adjacent pairs. Use the condition number of a matrix to explain this phenomenon.

7.5 Programming assignments

- A. Write a program for cubic-spline interpolation of the function

$$f(x) = \frac{1}{1 + 25x^2}$$

on evenly spaced nodes within the interval $[-1, 1]$ with $N = 6, 11, 21, 41, 81$. Compute for each N the max-norm of the interpolation error vector at mid-points of the subintervals and report the errors and convergence rates with respect to the number of subintervals.

Your algorithm should follow the example of interpolating the natural logarithm in the notes and your program must use an implementation of **lapack**.

Plot the interpolating spline against the exact function to observe that spline interpolation is free of the wide oscillations in the Runge phenomenon.

- B. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a given function. Implement two subroutines to interpolate f by the quadratic and cubic cardinal B-splines, which corresponds to Theorems 7.61 and 7.60, respectively.

- C. Run your subroutines on the function

$$f(x) = \frac{1}{1 + x^2}, \quad x \in [-5, 5],$$

using $t_i = -6 + i$, $i = 1, \dots, 11$ for Theorem 7.60 and $t_i = i - \frac{11}{2}$, $i = 1, \dots, 10$ for Theorem 7.61, respectively. Plot the polynomials against the exact function.

- D. Define $E_S(x) = |S(x) - f(x)|$ as the interpolation error. For the two cardinal B-spline interpolants, output values of $E_S(x)$ at the sites

$$x = -3.5, -3, -0.5, 0, 0.5, 3, 3.5.$$

Output these values by a program. Why are some of the errors close to machine precision? Which of the two B-splines is more accurate?

- E. Approximate the heart curve in (6.55) by fitting a cubic spline whose parameter is the cumulative chordal length in (7.80). What boundary conditions are appropriate here? Choose $N = 10, 40, 160$, plot your results, and compare them to those of cubic Bézier curves in Section 6.12. What can you say about the differences between these two curve-fitting methods?
- F. (*) Write a program to illustrate (7.39) by plotting the truncated power functions for $n = 1, 2$ and build a table of divided difference where the entries are figures instead of numbers. The pictures you generated for $n = 1$ should be the same as those in Example 7.34.

Chapter 8

Least-squares Approximation

Theorem 8.1. The set $\mathcal{C}[a, b]$ of continuous functions over $[a, b]$ is an inner-product space over $\mathbb{C}$ with its inner product as

$$\langle u, v \rangle := \int_a^b \rho(t) u(t) \overline{v(t)} dt, \quad (8.1)$$

where $\overline{v(t)}$ is the complex conjugate of $v(t)$ and the *weight function* $\rho(x) \in \mathcal{L}[a, b]$ satisfies $\rho(x) > 0$ for all $x \in (a, b)$. In addition, $\mathcal{C}[a, b]$ with

$$\|u\|_2 := \left(\int_a^b \rho(t) |u(t)|^2 dt \right)^{\frac{1}{2}} \quad (8.2)$$

is a normed vector space over $\mathbb{C}$.

Proof. This follows from Definitions C.2, C.170, and C.175. $\square$

Definition 8.2. The *least-squares approximation* on $\mathcal{C}[a, b]$ is a best approximation problem with the norm in (5.1) set to that in (8.2).

8.1 Orthogonal systems

Definition 8.3. A subset S of an inner product space X is *orthogonal* iff

$$\forall u, v \in S, \quad \langle u, v \rangle = \begin{cases} 0 & \text{if } u \neq v, \\ > 0 & \text{if } u = v. \end{cases} \quad (8.3)$$

Theorem 8.4. Any finite set of nonzero orthogonal elements $u_1, u_2, \dots, u_n$ is linearly independent.

Proof. This is proved by contradiction using Definitions C.26 and 8.3. $\square$

Definition 8.5. A subset S of an inner product space X is *orthonormal* if it is both orthogonal as in Definition 8.3 and satisfies $\forall u \in S, \langle u, u \rangle = 1$.

Example 8.6. The standard basis vectors in $\mathbb{R}^n$ are orthonormal.

Example 8.7. The Chebyshev polynomials of the first kind as in Definition 5.64 are orthogonal with respect to (8.1) where $a = -1, b = 1, \rho = \frac{1}{\sqrt{1-x^2}}$. However, they are not orthonormal.

8.1.1 Orthogonal polynomials

Definition 8.8. A set of *orthogonal polynomials* is a set P of polynomials satisfying

$$\forall p_i, p_j \in P, \quad i \neq j \implies \langle p_i, p_j \rangle = 0, \quad (8.4)$$

where p_i is a polynomial of degree i .

Example 8.9. In Chapter 9, the inner product in (8.4) for real polynomials p_i and p_j is taken to be

$$\langle p_i, p_j \rangle := \int_a^b \rho(x) p_i(x) p_j(x) dx, \quad (8.5)$$

which is a special case of (8.1).

Definition 8.10. The *monic shifted Legendre polynomials* are

$$\left\{ \tilde{P}_s(t) = \frac{s!}{(2s)!} \frac{d^s}{dt^s} \phi_s(t) : s \in \mathbb{N}; \phi_s(t) = t^s(t-1)^s \right\}, \quad (8.6)$$

where $\tilde{P}_0(t) \equiv 1$ for $s = 0$.

Lemma 8.11. The monic shifted Legendre polynomials are orthogonal with respect to (8.5) with $a = 0, b = 1$, and $\rho \equiv 1$.

Proof. It suffices to verify that

$$\forall Q \in \mathbb{P}_{s-1}, \quad \int_0^1 \tilde{P}_s(t) Q(t) dt = 0. \quad (8.7)$$

Definition 8.10 implies $\phi_s^{(k)}(0) = \phi_s^{(k)}(1) = 0$ for each $k = 0, 1, \dots, s-1$ and we have

$$\begin{aligned} \int_0^1 \tilde{P}_s(t) Q(t) dt &= \int_0^1 \frac{s!}{(2s)!} \phi^{(s)}(t) Q(t) dt \\ &= - \int_0^1 \frac{s!}{(2s)!} \phi^{(s-1)}(t) Q'(t) dt \\ &= \dots = (-1)^s \int_0^1 \frac{s!}{(2s)!} \phi(t) Q^{(s)}(t) dt = 0, \end{aligned}$$

where the second step follows from integration by parts and the last from $Q^{(s)}(t) = 0$. $\square$

Theorem 8.12. For a set of real orthogonal polynomials over $[a, b]$ with respect to (8.5), each zero of any such polynomials is real, simple, and inside (a, b) .

Proof. For fixed $n \geq 1$, suppose $p_n(x)$ does not change sign in $[a, b]$. Then

$$\exists c \in \mathbb{R}^+ \text{ s.t. } \int_a^b \rho(x) p_n(x) dx = c \langle p_n, p_0 \rangle \neq 0,$$

which contradicts the orthogonality of p_n and p_0 . Hence there exists $x_1 \in (a, b)$ satisfying $p_n(x_1) = 0$.

Suppose there were a zero at x_1 which is multiple. Then $\frac{p_n(x)}{(x-x_1)^2}$ would be a polynomial of degree $n-2$. Hence $0 = \langle p_n(x), \frac{p_n(x)}{(x-x_1)^2} \rangle = \langle 1, \frac{p_n^2(x)}{(x-x_1)^2} \rangle > 0$, which is false. Therefore every zero is simple.

Suppose that only $j < n$ zeros of p_n , say $x_1, x_2, \dots, x_j$, are inside (a, b) and all other zeros are out of (a, b) . Let $v_j(x) = \prod_{i=1}^j (x - x_i) \in \mathbb{P}_j$. Then $p_n v_j = P_{n-j} v_j^2$ where P_{n-j} is a polynomial of degree $n-j$ that does not change sign on $[a, b]$. Hence $|\langle P_{n-j}, v_j^2 \rangle| > 0$, which contradicts the orthogonality of $p_n(x)$ and $v_j(x)$. $\square$

8.1.2 The Gram–Schmidt process

Definition 8.13. The *Gram–Schmidt process* takes in a finite or infinite independent list $(u_1, u_2, \dots)$ and output two other lists $(v_1, v_2, \dots)$ and $(u_1^*, u_2^*, \dots)$ by

$$v_{n+1} = u_{n+1} - \sum_{k=1}^n \langle u_{n+1}, u_k^* \rangle u_k^*, \quad (8.8a)$$

$$u_{n+1}^* = v_{n+1} / \|v_{n+1}\|, \quad (8.8b)$$

with the recursion basis as $v_1 = u_1$, $u_1^* = v_1 / \|v_1\|$.

Theorem 8.14. For a finite or infinite independent list $(u_1, u_2, \dots)$, the Gram–Schmidt process yields constants

$$\begin{array}{ccc} a_{11} & & \\ a_{21} & a_{22} & \\ a_{31} & a_{32} & a_{33} \\ \vdots & & \end{array}$$

such that $a_{kk} = \frac{1}{\|v_k\|} > 0$ and the elements $u_1^*, u_2^*, \dots$

$$\begin{array}{l} u_1^* = a_{11} u_1 \\ u_2^* = a_{21} u_1 + a_{22} u_2 \\ u_3^* = a_{31} u_1 + a_{32} u_2 + a_{33} u_3 \\ \vdots \end{array} \quad (8.9)$$

are orthonormal.

Proof. Definition 8.13 implies that the formulae (8.8) can be rewritten in the form of (8.9); this can be proven by induction. The induction basis is the recursion basis $u_1^* = u_1 / \|u_1\|$ and the inductive step follows from (8.8) as

$$u_n^* = \frac{1}{\|v_n\|} \left(u_n - \sum_{k=1}^{n-1} \langle u_n, u_k^* \rangle u_k^* \right),$$

where each u_k^* is a linear combination of $u_1, u_2, \dots, u_k$ and a_{nn} , the coefficient of u_n in (8.9), is clearly $\frac{1}{\|v_n\|}$. By (8.8b),

u_{n+1}^* is normal. We show by induction that u_{n+1}^* is orthogonal to $u_n^*, u_{n-1}^*, \dots, u_1^*$. The induction base holds because

$$\begin{aligned} \langle v_2, u_1^* \rangle &= \langle u_2 - \langle u_2, u_1^* \rangle u_1^*, u_1^* \rangle \\ &= \langle u_2, u_1^* \rangle - \langle u_2, u_1^* \rangle \langle u_1^*, u_1^* \rangle = 0, \end{aligned}$$

where the second step follows from (IP-3) in Definition C.170 and the third step from u_1^* being normal. The inductive step also holds because for any $j < n+1$ we have

$$\begin{aligned} \langle v_{n+1}, u_j^* \rangle &= \langle u_{n+1} - \sum_{k=1}^n \langle u_{n+1}, u_k^* \rangle u_k^*, u_j^* \rangle \\ &= \langle u_{n+1}, u_j^* \rangle - \sum_{k=1}^n \langle u_{n+1}, u_k^* \rangle \langle u_k^*, u_j^* \rangle \\ &= \langle u_{n+1}, u_j^* \rangle - \langle u_{n+1}, u_j^* \rangle = 0, \end{aligned} \quad (8.10)$$

where the third step follows from the induction hypothesis and (8.8b), i.e.,

$$\langle u_k^*, u_j^* \rangle = \begin{cases} 1 & \text{if } k = j; \\ 0 & \text{otherwise.} \end{cases} \quad (8.11) \quad \square$$

Exercise 8.15. Prove $a_{kk} = \frac{1}{\|v_k\|}$ by using $\|u_n^*\| = 1$.

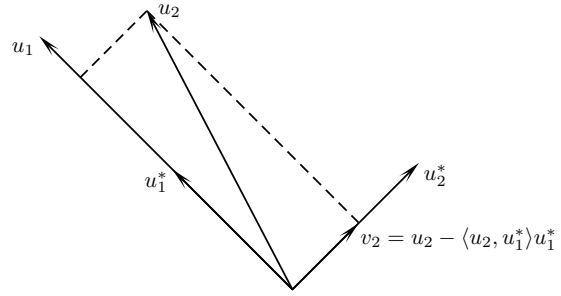


Figure 8.1: The geometric meaning of the Gram–Schmidt process (8.8). Here v_{n+1} is the error residual of u_{n+1} after projecting u_{n+1} onto the space spanned by the orthonormal bases $(u_1^*, u_2^*, \dots, u_n^*)$ while u_{n+1}^* is the normalization of v_{n+1} . Also see Definition 8.20 and Example 8.21.

Corollary 8.16. For a finite or infinite independent list $(u_1, u_2, \dots)$, we can find constants

$$\begin{array}{ccc} b_{11} & & \\ b_{21} & b_{22} & \\ b_{31} & b_{32} & b_{33} \\ \vdots & & \end{array}$$

and an orthonormal list $(u_1^*, u_2^*, \dots)$ such that $b_{ii} > 0$ and

$$\begin{array}{l} u_1 = b_{11} u_1^* \\ u_2 = b_{21} u_1^* + b_{22} u_2^* \\ u_3 = b_{31} u_1^* + b_{32} u_2^* + b_{33} u_3^* \\ \vdots \end{array} \quad (8.12)$$

Proof. This follows from (8.9) and that a lower-triangular matrix with positive diagonal elements is invertible. $\square$

Corollary 8.17. In Theorem 8.14, we have $\langle u_n^*, u_i \rangle = 0$ for each $i = 1, 2, \dots, n-1$.

Proof. By Corollary 8.16, each u_i can be expressed as

$$u_i = \sum_{k=1}^i b_{ik} u_k^*.$$

Inner product the above equation with u_n^* , apply the orthogonal conditions, and we reach the conclusion. $\square$

Definition 8.18. Using the Gram–Schmidt orthonormalizing process with the inner product (8.1), we obtain from the independent list of monomials $(1, x, x^2 \dots)$ the following *classic orthonormal polynomials*:

	a	b	$\rho(x)$
Chebyshev polynomials of the first kind	-1	1	$\frac{1}{\sqrt{1-x^2}}$
Chebyshev polynomials of the second kind	-1	1	$\sqrt{1-x^2}$
Legendre polynomials	-1	1	1
Jacobi polynomials	-1	1	$(1-x)^\alpha(1+x)^\beta$
Laguerre polynomials	0	$+\infty$	$x^\alpha e^{-x}$
Hermite polynomials	$-\infty$	$+\infty$	e^{-x^2}

where $\alpha, \beta > -1$ for Jacobi polynomials and $\alpha > -1$ for Laguerre polynomials.

Example 8.19. We compute the first 3 Legendre polynomials using the Gram–Schmidt process.

$$\begin{aligned} u_1 &= 1, \quad v_1 = 1, \quad \|v_1\|^2 = \int_{-1}^{+1} dx = 2, \quad u_1^* = \frac{1}{\sqrt{2}}. \\ u_2 &= x, \quad v_2 = x - \left\langle x, \frac{1}{\sqrt{2}} \right\rangle \frac{1}{\sqrt{2}} = x, \quad \|v_2\|^2 = \frac{2}{3}, \\ u_2^* &= \sqrt{\frac{3}{2}} x. \\ v_3 &= x^2 - \left\langle x^2, \sqrt{\frac{3}{2}} x \right\rangle \sqrt{\frac{3}{2}} x - \left\langle x^2, \frac{1}{\sqrt{2}} \right\rangle \frac{1}{\sqrt{2}} = x^2 - \frac{1}{3}, \\ \|v_3\|^2 &= \int_{-1}^{+1} \left(x^2 - \frac{1}{3} \right)^2 dx = \frac{8}{45}, \\ u_3^* &= \frac{3}{4} \sqrt{10} \left(x^2 - \frac{1}{3} \right). \end{aligned}$$

8.1.3 Fourier expansions

Definition 8.20. Let $(u_1^*, u_2^*, \dots)$ be a finite or infinite orthonormal list. The *orthogonal expansion* or *Fourier expansion* for an arbitrary w is the series

$$\sum_{n=1}^m \langle w, u_n^* \rangle u_n^*, \quad (8.13)$$

where the constants $\langle w, u_n^* \rangle$ are known as the *Fourier coefficients* of w and the term $\langle w, u_n^* \rangle u_n^*$ the *projection* of w on u_n^* . The *error of the Fourier expansion* of w with respect to $(u_1^*, u_2^*, \dots)$ is simply $\sum_n \langle w, u_n^* \rangle u_n^* - w$.

Example 8.21. With the Euclidean inner product in Definition C.174, we select orthonormal vectors in $\mathbb{R}^3$ as

$$u_1^* = (1, 0, 0)^\top, \quad u_2^* = (0, 1, 0)^\top, \quad u_3^* = (0, 0, 1)^\top.$$

For the vector $w = (a, b, c)^\top$, the Fourier coefficients are

$$\langle w, u_1^* \rangle = a, \quad \langle w, u_2^* \rangle = b, \quad \langle w, u_3^* \rangle = c,$$

and the sum of projections of w onto u_1^* , u_2^* , and u_3^* gives the Fourier expansion of w as

$$w = \langle w, u_1^* \rangle u_1^* + \langle w, u_2^* \rangle u_2^* + \langle w, u_3^* \rangle u_3^*.$$

Exercise 8.22. For the orthonormal list in $L_{\rho=1}^2[-\pi, \pi]$,

$$\frac{1}{\sqrt{2\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\cos x}{\sqrt{\pi}}, \dots, \frac{\sin(nx)}{\sqrt{\pi}}, \frac{\cos(nx)}{\sqrt{\pi}}, \dots, \quad (8.14)$$

derive the *Fourier series* of a function $f(x)$ as

$$f(x) \sim \frac{a_0}{2} + \sum_{k=1}^{+\infty} (a_k \cos kx + b_k \sin kx), \quad (8.15)$$

where the coefficients are

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx, \quad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx dx.$$

Theorem 8.23. Let $u_1, u_2, \dots, u_n$ be linearly independent and let u_i^* be the u_i 's orthonormalized by the Gram–Schmidt process. If $w = \sum_{i=1}^n a_i u_i$, then

$$w = \sum_{i=1}^n \langle w, u_i^* \rangle u_i^*, \quad (8.16)$$

i.e. w is equal to its Fourier expansion.

Proof. By the condition $w = \sum_{i=1}^n a_i u_i$ and Corollary 8.16, we can express w as a linear combination of u_i^* 's, i.e., $w = \sum_{i=1}^n c_i u_i^*$. Then the orthonormality of u_i^* 's implies

$$\forall k = 1, 2, \dots, n, \quad \langle w, u_k^* \rangle = c_k,$$

which completes the proof. $\square$

Theorem 8.24 (Minimum properties of Fourier expansions). Let $u_1^*, u_2^*, \dots$ be an orthonormal system and let w be arbitrary. Then

$$\left\| w - \sum_{i=1}^N \langle w, u_i^* \rangle u_i^* \right\| \leq \left\| w - \sum_{i=1}^N a_i u_i^* \right\|, \quad (8.17)$$

for any selection of constants $a_1, a_2, \dots, a_N$.

Proof. With the shorthand notation $\sum_i = \sum_{i=1}^N$, we deduce

from the definition and properties of inner products

$$\begin{aligned}
\left\| w - \sum_i a_i u_i^* \right\|^2 &= \left\langle w - \sum_i a_i u_i^*, w - \sum_i a_i u_i^* \right\rangle \\
&= \langle w, w \rangle - \left\langle w, \sum_i a_i u_i^* \right\rangle - \left\langle \sum_i a_i u_i^*, w \right\rangle \\
&\quad + \left\langle \sum_i a_i u_i^*, \sum_i a_i u_i^* \right\rangle \\
&= \langle w, w \rangle - \sum_i \bar{a}_i \langle w, u_i^* \rangle - \sum_i a_i \langle u_i^*, w \rangle \\
&\quad + \sum_i \sum_j a_i \bar{a}_j \langle u_i^*, u_j^* \rangle \\
&= \langle w, w \rangle - \sum_i \bar{a}_i \langle w, u_i^* \rangle - \sum_i a_i \langle u_i^*, w \rangle + \sum_i |a_i|^2 \\
&\quad - \sum_i \langle u_i^*, w \rangle \langle w, u_i^* \rangle + \sum_i \langle u_i^*, w \rangle \langle w, u_i^* \rangle \\
&= \|w\|^2 - \sum_i |\langle w, u_i^* \rangle|^2 + \sum_i |a_i - \langle w, u_i^* \rangle|^2, \quad (8.18)
\end{aligned}$$

where “ $|\cdot|$ ” denotes the modulus of a complex number. The first two terms are independent of a_i . Therefore $\|w - \sum_i a_i u_i^*\|^2$ is minimized only when $a_i = \langle w, u_i^* \rangle$. $\square$

Corollary 8.25. Let $(u_1, u_2, \dots, u_n)$ be an independent list. The fundamental problem of linearly approximating an arbitrary vector w is solved by the best approximation $\hat{\varphi} = \sum_k \langle w, u_k^* \rangle u_k^*$ where u_k^* 's are the u_k 's orthonormalized by the Gram–Schmidt process. The error norm is

$$\|w - \hat{\varphi}\|^2 := \min_{a_k} \left\| w - \sum_{k=1}^n a_k u_k \right\|^2 = \|w\|^2 - \sum_{k=1}^n |\langle w, u_k^* \rangle|^2. \quad (8.19)$$

Proof. This follows directly from (8.18). $\square$

Corollary 8.26 (Bessel inequality). If $u_1^*, u_2^*, \dots, u_N^*$ are orthonormal, then, for an arbitrary w ,

$$\sum_{i=1}^N |\langle w, u_i^* \rangle|^2 \leq \|w\|^2. \quad (8.20)$$

Proof. This follows directly from Corollary 8.25 and the real positivity of a norm. $\square$

Corollary 8.27. The Gram–Schmidt process in Definition 8.13 satisfies

$$\forall n \in \mathbb{N}^+, \quad \|v_{n+1}\|^2 = \|u_{n+1}\|^2 - \sum_{k=1}^n |\langle u_{n+1}, u_k^* \rangle|^2. \quad (8.21)$$

Proof. By (8.8a), each v_{n+1} can be regarded as the error of Fourier expansion of u_{n+1} with respect to the orthonormal list $(u_1^*, u_2^*, \dots, u_n^*)$. In Corollary 8.25, identifying w with u_{n+1} completes the proof. $\square$

Example 8.28. Consider the problem in Example 5.3 in the sense of least square approximation with the weight function $\rho = 1$. It is equivalent to

$$\min_{a_i} \int_{-1}^{+1} \left(e^x - \sum_{i=0}^n a_i x^i \right)^2 dx. \quad (8.22)$$

For $n = 1, 2$, use the Legendre polynomials derived in Example 8.19:

$$u_1^* = \frac{1}{\sqrt{2}}, \quad u_2^* = \sqrt{\frac{3}{2}}x, \quad u_3^* = \frac{1}{4}\sqrt{10}(3x^2 - 1),$$

and we have the Fourier coefficients of e^x as

$$\begin{aligned}
b_1 &= \int_{-1}^{+1} \frac{1}{\sqrt{2}} e^x dx = \frac{1}{\sqrt{2}} \left(e - \frac{1}{e} \right), \\
b_2 &= \int_{-1}^{+1} \sqrt{\frac{3}{2}} x e^x dx = \sqrt{6} e^{-1}, \\
b_3 &= \int_{-1}^{+1} \frac{1}{4} \sqrt{10} (3x^2 - 1) e^x dx = \frac{\sqrt{10}}{2} \left(e - \frac{7}{e} \right).
\end{aligned}$$

The minimizing polynomials are thus

$$\hat{\varphi}_n = \begin{cases} \frac{1}{2e}(e^2 - 1) + \frac{3}{e}x & \text{if } n = 1; \\ \hat{\varphi}_1 + \frac{5}{4e}(e^2 - 7)(3x^2 - 1) & \text{if } n = 2. \end{cases} \quad (8.23)$$

8.2 The normal equations

Theorem 8.29. Let $u_1, u_2, \dots, u_n \in X$ be linearly independent and let u_i^* be the u_i 's orthonormalized by the Gram–Schmidt process. Then, for any element w ,

$$\forall j = 1, 2, \dots, n, \quad \left(w - \sum_{k=1}^n \langle w, u_k^* \rangle u_k^* \right) \perp u_j^*, \quad (8.24)$$

where “ $\perp$ ” denotes orthogonality.

Proof. If $w \in X$, we have $w - \sum_{k=1}^n \langle w, u_k^* \rangle u_k^* = \mathbf{0}$ and thus (8.24) holds trivially. For the other case of $w \notin X$, set $w = u_{n+1}$, apply (8.11), and we have (8.24). $\square$

Corollary 8.30. Let $u_1, u_2, \dots, u_n \in X$ be linearly independent. If $\hat{\varphi} = \sum_{k=1}^n a_k u_k$ is the best linear approximant to w , then

$$\forall j = 1, 2, \dots, n, \quad (w - \hat{\varphi}) \perp u_j. \quad (8.25)$$

Proof. Since $\hat{\varphi} = \sum_{k=1}^n a_k u_k$ is the best linear approximant to w , Theorem 8.24 implies that

$$\sum_{k=1}^n a_k u_k = \sum_{k=1}^n \langle w, u_k^* \rangle u_k^*.$$

Corollary 8.16 and Theorem 8.29 complete the proof. $\square$

Definition 8.31. Let $u_1, u_2, \dots, u_n$ be a sequence of elements in an inner product space. The $n \times n$ matrix

$$G = G(u_1, u_2, \dots, u_n) = (\langle u_i, u_j \rangle) = \begin{bmatrix} \langle u_1, u_1 \rangle & \langle u_1, u_2 \rangle & \dots & \langle u_1, u_n \rangle \\ \langle u_2, u_1 \rangle & \langle u_2, u_2 \rangle & \dots & \langle u_2, u_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle u_n, u_1 \rangle & \langle u_n, u_2 \rangle & \dots & \langle u_n, u_n \rangle \end{bmatrix} \quad (8.26)$$

is the *Gram matrix* of $u_1, u_2, \dots, u_n$. Its determinant

$$g = g(u_1, u_2, \dots, u_n) = \det(\langle u_i, u_j \rangle) \quad (8.27)$$

is the *Gram determinant*.

Lemma 8.32. Let $w_i = \sum_{j=1}^n a_{ij} u_j$ for $i = 1, 2, \dots, n$. Let $A = (a_{ij})$ and its conjugate transpose $A^H = (\overline{a_{ji}})$. Then we have

$$G(w_1, w_2, \dots, w_n) = AG(u_1, u_2, \dots, u_n)A^H \quad (8.28)$$

and

$$g(w_1, w_2, \dots, w_n) = |\det A|^2 g(u_1, u_2, \dots, u_n). \quad (8.29)$$

Proof. The inner product of u_i and w_j yields

$$\begin{aligned} & \begin{bmatrix} \langle u_1, w_1 \rangle & \langle u_1, w_2 \rangle & \dots & \langle u_1, w_n \rangle \\ \langle u_2, w_1 \rangle & \langle u_2, w_2 \rangle & \dots & \langle u_2, w_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle u_n, w_1 \rangle & \langle u_n, w_2 \rangle & \dots & \langle u_n, w_n \rangle \end{bmatrix} \\ &= \begin{bmatrix} \langle u_1, u_1 \rangle & \langle u_1, u_2 \rangle & \dots & \langle u_1, u_n \rangle \\ \langle u_2, u_1 \rangle & \langle u_2, u_2 \rangle & \dots & \langle u_2, u_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle u_n, u_1 \rangle & \langle u_n, u_2 \rangle & \dots & \langle u_n, u_n \rangle \end{bmatrix} \begin{bmatrix} \overline{a_{11}} & \dots & \overline{a_{n1}} \\ \overline{a_{12}} & \dots & \overline{a_{n2}} \\ \vdots & \ddots & \vdots \\ \overline{a_{1n}} & \dots & \overline{a_{nn}} \end{bmatrix} \\ &= G(u_1, u_2, \dots, u_n)A^H. \end{aligned}$$

Therefore (8.28) holds since

$$\begin{aligned} G(w_1, w_2, \dots, w_n) &= \begin{bmatrix} \langle w_1, w_1 \rangle & \langle w_1, w_2 \rangle & \dots & \langle w_1, w_n \rangle \\ \langle w_2, w_1 \rangle & \langle w_2, w_2 \rangle & \dots & \langle w_2, w_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle w_n, w_1 \rangle & \langle w_n, w_2 \rangle & \dots & \langle w_n, w_n \rangle \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} \langle u_1, w_1 \rangle & \langle u_1, w_2 \rangle & \dots & \langle u_1, w_n \rangle \\ \langle u_2, w_1 \rangle & \langle u_2, w_2 \rangle & \dots & \langle u_2, w_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle u_n, w_1 \rangle & \langle u_n, w_2 \rangle & \dots & \langle u_n, w_n \rangle \end{bmatrix} \\ &= AG(u_1, u_2, \dots, u_n)A^H. \end{aligned}$$

The following properties of complex conjugate are well known:

$$\overline{z + w} = \overline{z} + \overline{w}, \quad \overline{zw} = \overline{z} \overline{w}.$$

Then the identity $\det(A) = \det(A^\top)$ and the Leibniz formula of determinants (Definition C.269) yield

$$\overline{\det A} = \det A^\top = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n \overline{a_{i, \sigma_i}} = \det A^H.$$

Finally, (8.29) follows from the determinant of (8.28) and the identity $\det(AB) = \det(A) \det(B)$. $\square$

Theorem 8.33. For nonzero elements $u_1, u_2, \dots, u_n \in X$, we have

$$0 \leq g(u_1, u_2, \dots, u_n) \leq \prod_{k=1}^n \|u_k\|^2, \quad (8.30)$$

where the lower equality holds if and only if $u_1, u_2, \dots, u_n$ are linearly dependent and the upper equality holds if and only if they are orthogonal.

Proof. Suppose $u_1, u_2, \dots, u_n$ are linearly dependent. Then we can find constants $c_1, c_2, \dots, c_n$ satisfying $\sum_{i=1}^n c_i u_i = \mathbf{0}$ with at least one c_j being nonzero. Construct vectors

$$w_k = \begin{cases} \sum_{i=1}^n c_i u_i = \mathbf{0}, & k = j; \\ u_k, & k \neq j. \end{cases}$$

We have $g(w_1, w_2, \dots, w_n) = 0$ because $\langle w_j, w_k \rangle = 0$ for each k . By the Laplace theorem (Theorem C.275), we can expand the determinant of $C = (c_{ij})$ according to minors of its j th row:

$$\begin{aligned} \det(C) &= \det \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ c_1 & \dots & c_{j-1} & c_j & c_{j+1} & \dots & c_n \\ & & & 1 & & & \\ & & & & \ddots & & \\ & & & & & & 1 \end{bmatrix} \\ &= 0 + \dots + 0 + c_j + 0 + \dots + 0 = c_j \neq 0, \end{aligned}$$

where the determinant of each minor matrix M_i of c_i with $i \neq j$ is zero because the i_0 th row of each M_i is a row of all zeros, with $i_0 = i$ if $i < j$ and $i_0 = i - 1$ if $i > j$. Then Lemma 8.32 yields $g(u_1, u_2, \dots, u_n) = 0$.

Now suppose $u_1, u_2, \dots, u_n$ are linearly independent. Theorem 8.14 yields constants a_{ij} such that $a_{kk} > 0$ and the following vectors are orthonormal:

$$u_k^* = \sum_{i=1}^k a_{ki} u_i.$$

Then Definition 8.31 implies $g(u_1^*, u_2^*, \dots, u_n^*) = 1$. Also, we have $\det(a_{ij}) = \prod_{k=1}^n a_{kk}$ because the matrix (a_{ij}) is triangular. It then follows from Lemma 8.32 that

$$g(u_1, u_2, \dots, u_n) = \prod_{k=1}^n \frac{1}{a_{kk}^2} > 0. \quad (8.31)$$

Since the list of vectors $(u_1, u_2, \dots, u_n)$ is either dependent or independent, the arguments so far show that $g(u_1, u_2, \dots, u_n) = 0$ if and only if $u_1, u_2, \dots, u_n$ are linearly dependent.

Suppose $u_1, u_2, \dots, u_n$ are orthogonal. By Definition 8.31, $G(u_1, u_2, \dots, u_n)$ is a diagonal matrix with $\|u_k\|^2$ on the diagonals. Hence the orthogonality of u_k 's implies

$$g(u_1, u_2, \dots, u_n) = \prod_{k=1}^n \|u_k\|^2. \quad (8.32)$$

For the converse statement, suppose (8.32) holds. Then $u_1, u_2, \dots, u_n$ must be independent because otherwise it would contradict the lower equality of (8.30) proved as above. Apply the Gram–Schmidt process to $(u_1, u_2, \dots, u_n)$ and we know from Theorem 8.14 that $\frac{1}{a_{kk}} = \|v_k\|$. Set the length of the list in Theorem 8.14 to $1, 2, \dots, n$ and we know from (8.31) and (8.32) that

$$\forall k = 1, 2, \dots, n, \quad \|u_k\|^2 = \|v_k\|^2. \quad (8.33)$$

Then Corollary 8.27 and (8.33) imply

$$\forall k = 1, 2, \dots, n, \quad \sum_{j=1}^{k-1} |\langle u_k, u_j^* \rangle|^2 = 0,$$

which further implies

$$\forall k = 1, 2, \dots, n, \quad \forall j = 1, 2, \dots, k-1, \quad \langle u_k, u_j^* \rangle = 0,$$

which, together with Corollary 8.16, implies the orthogonality of u_k 's. Finally, we remark that the maximum of $g(u_1, u_2, \dots, u_n)$ is indeed $\prod_{k=1}^n \|u_k\|^2$ because of (8.31), $\frac{1}{a_{kk}} = \|v_k\|$, and Corollary 8.27. $\square$

Theorem 8.34. Let $\hat{\varphi} = \sum_{i=1}^n a_i u_i$ be the best approximation to w constructed from the list of independent vectors $(u_1, u_2, \dots, u_n)$. Then the coefficients

$$\mathbf{a} = [a_1, a_2, \dots, a_n]^\top$$

are uniquely determined from the linear system of *normal equations*,

$$G(u_1, u_2, \dots, u_n)^\top \mathbf{a} = \mathbf{c}, \quad (8.34)$$

where $\mathbf{c} = [\langle w, u_1 \rangle, \langle w, u_2 \rangle, \dots, \langle w, u_n \rangle]^\top$.

Proof. Take inner product of $\hat{\varphi} = \sum_{i=1}^n a_i u_i$ with u_j , apply Corollary 8.30, and we have

$$\langle w, u_j \rangle = \sum_{k=1}^n a_k \langle u_k, u_j \rangle,$$

which is simply the j th equation of (8.34). The uniqueness of the coefficients follows from Theorem 8.33 and Cramer's rule (Theorem C.279). $\square$

Example 8.35. Solve Example 8.28 by normal equations.

To find the best approximation $\hat{\varphi} = a_0 + a_1 x + a_2 x^2$ to e^x from the linearly independent list $(1, x, x^2)$, we first construct the Gram matrix from (8.26), (8.1), and $\rho = 1$:

$$G(1, x, x^2) = \begin{bmatrix} \langle 1, 1 \rangle & \langle 1, x \rangle & \langle 1, x^2 \rangle \\ \langle x, 1 \rangle & \langle x, x \rangle & \langle x, x^2 \rangle \\ \langle x^2, 1 \rangle & \langle x^2, x \rangle & \langle x^2, x^2 \rangle \end{bmatrix} = \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix}.$$

We then calculate the vector

$$\mathbf{c} = \begin{bmatrix} \langle e^x, 1 \rangle \\ \langle e^x, x \rangle \\ \langle e^x, x^2 \rangle \end{bmatrix} = \begin{bmatrix} e - \frac{1}{e} \\ \frac{2}{3} \\ e - \frac{5}{e} \end{bmatrix}.$$

The normal equations then yields

$$a_0 = \frac{3(11 - e^2)}{4e}, \quad a_1 = \frac{3}{e}, \quad a_2 = \frac{15(e^2 - 7)}{4e}.$$

With these values, it is easily verified that the best approximation $\hat{\varphi} = a_0 + a_1 x + a_2 x^2$ equals that in (8.23).

Example 8.36 (The Hilbert matrix as the Gram matrix). Consider the least-square approximation of a given function $f \in \mathcal{C}[0, 1]$ by a polynomial $p(x) = \sum_{k=0}^n \alpha_k x^k$. If the monomials are used as the basis of the subspace, their Gram matrix is the Hilbert matrix as in Definition 2.30 because

$$\langle x^j, x^k \rangle = \int_0^1 x^j x^k dx = \frac{1}{j+k+1}.$$

The coefficients α_k 's are determined by the normal equations

$$\forall j = 0, 1, \dots, n, \quad \sum_{k=0}^n \frac{\alpha_k}{j+k+1} = \int_0^1 f(x) x^j dx. \quad (8.35)$$

For the special case of $f(x) = \frac{1}{1+x}$, the RHS

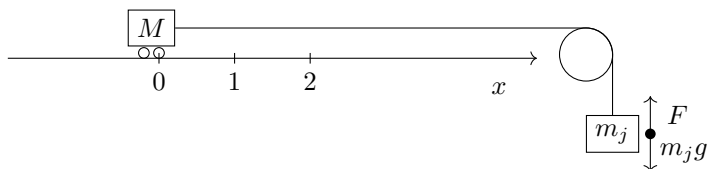
$$r_j := \int_0^1 \frac{x^j}{1+x} dx$$

can be derived as

$$\forall j = 0, 1, \dots, n, \quad r_j = (-1)^j \ln 2 + \sum_{i=1}^j (-1)^{i+j} \frac{1}{i}. \quad (8.36)$$

8.3 Discrete least squares (DLS)

Example 8.37 (Processing experimental data on Newton's second law using discrete least squares). A cart with mass M is pulled along a horizontal track by a cable attached to a weight of mass m_j through a pulley.



Neglect the friction of the track and the pulley system, assume Newton's second law holds and we have

$$m_j g = (m_j + M) a_j = (m_j + M) \frac{d^2 x}{dt^2}.$$

The following experiments verify Newton's second law.

- (i) For fixed M and m_j , we measure a number of data points (t_i, x_i) by recording the position of the cart with a high-speed camera.
- (ii) Fit a quadratic polynomial $p(t) = c_0 + c_1 t + c_2 t^2$ by minimizing the total length squared,

$$\min_i \sum (x_i - p(t_i))^2.$$

- (iii) Take $a_j = 2c_2$ as the experimental result of acceleration for the force $F_j = m_j(g - a_j)$.

- (iv) Change the weight m_j and repeat steps (i)-(iii) a number of times to get data points (a_j, F_j) .
- (v) Fit a linear polynomial $f(x) = c_0 + c_1x$ by minimizing the total length squared,

$$\min \sum_j (F_j - f(a_j))^2,$$

where the expressions in steps (ii) and (v) justify the name “least squares.”

One verifies Newton’s second law by showing that the data fitting result c_1 is very close to M . This verification process have neither derived nor proved Newton’s second law. But we claim that we did not find any self-inconsistency of Newton’s second law, nor did we find any discrepancy between the experimental results and Newton’s second law.

8.3.1 Gaussian and Dirac delta functions

Definition 8.38. A *Gaussian function*, or a *Gaussian*, is a function of the form

$$f(x) = a \exp\left(-\frac{(x-b)^2}{2c^2}\right), \quad (8.37)$$

where $a \in \mathbb{R}^+$ is the height of the curve’s peak, $b \in \mathbb{R}$ is the position of the center of the peak and $c \in \mathbb{R}^+$ is the standard deviation or the *Gaussian RMS (root mean square) width*.

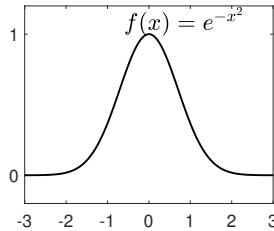


Figure 8.2: A Gaussian has a characteristic symmetric *bell curve* shape; its standard deviation controls the width of the bell

Lemma 8.39. The integral of a Gaussian is

$$\int_{-\infty}^{+\infty} a e^{-\frac{(x-b)^2}{2c^2}} dx = ac\sqrt{2\pi}. \quad (8.38)$$

Proof. By the trick of combining two one-dimensional Gaussians and the Polar coordinate transformation, we have

$$\begin{aligned} \int_{-\infty}^{+\infty} e^{-x^2} dx &= \sqrt{\left(\int_{-\infty}^{+\infty} e^{-x^2} dx\right) \left(\int_{-\infty}^{+\infty} e^{-y^2} dy\right)} \\ &= \sqrt{\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)} dx dy} \\ &= \sqrt{\int_0^{2\pi} \int_0^{+\infty} e^{-r^2} r dr d\theta} \\ &= \sqrt{2\pi \cdot \left(-\frac{1}{2} e^{-r^2}\right) \Big|_0^{+\infty}} = \sqrt{\pi}, \end{aligned}$$

and hence

$$\int_{-\infty}^{+\infty} a e^{-\frac{(x-b)^2}{2c^2}} dx = \sqrt{2\pi} ac \int_{-\infty}^{+\infty} e^{-y^2} dy = ac\sqrt{2\pi},$$

where it follows from the transformation of $x = b + \sqrt{2}cy$. $\square$

Definition 8.40. A *normal distribution* or *Gaussian distribution* is a continuous probability distribution of the form

$$f_{\mu, \sigma} = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad (8.39)$$

where μ is the *mean* or *expectation* and σ is the *standard deviation*.

Definition 8.41. The *Dirac delta function* $\delta(x-\bar{x})$ centered at $\bar{x}$ is

$$\delta(x-\bar{x}) = \lim_{\epsilon \rightarrow 0} \phi_\epsilon(x-\bar{x}) \quad (8.40)$$

where $\phi_\epsilon(x-\bar{x}) = f_{\bar{x}, \epsilon}$ is a normal distribution with its mean at $\bar{x}$ and its standard deviation as ϵ .

Lemma 8.42. The Dirac delta function satisfies

$$\delta(x-\bar{x}) = \begin{cases} +\infty, & x = \bar{x}, \\ 0, & x \neq \bar{x}; \end{cases} \quad (8.41a)$$

$$\int_{-\infty}^{+\infty} \delta(x-\bar{x}) dx = 1. \quad (8.41b)$$

Proof. These follow directly from Definitions 8.40 and 8.41 and Lemma 8.39. $\square$

Lemma 8.43 (Sifting property of δ). If $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then

$$\int_{-\infty}^{+\infty} \delta(x-\bar{x}) f(x) dx = f(\bar{x}). \quad (8.42)$$

Proof. Since $I_\epsilon := [\bar{x} - \epsilon, \bar{x} + \epsilon]$ is a compact interval and $f(x)$ is continuous over I_ϵ , $f(x)$ is bounded over I_ϵ , say, $f(x) \in [m, M]$. The nonnegativeness of ϕ_ϵ and the integral mean value theorem B.124 imply that

$$(*) : \int_{-\infty}^{+\infty} \delta(x-\bar{x}) f(x) dx = \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x-\bar{x}) f(x) dx$$

is bounded within the interval

$$\left[m \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x-\bar{x}) dx, M \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x-\bar{x}) dx \right].$$

It follows that

$$\lim_{\epsilon \rightarrow 0} \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x-\bar{x}) f(x) dx = f(\bar{x}) \lim_{\epsilon \rightarrow 0} \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x-\bar{x}) dx = f(\bar{x}).$$

Apply $\lim_{\epsilon \rightarrow 0}$ to $(*)$ and we have (8.42). $\square$

Definition 8.44. The *Heaviside function* or *step function* is

$$H(x) := \begin{cases} 0 & \text{if } x < 0; \\ 1 & \text{if } x \geq 0. \end{cases} \quad (8.43)$$

Lemma 8.45. The Dirac delta function and the Heaviside function are related as

$$\int_{-\infty}^x \delta(t) dt = H(x). \quad (8.44)$$

Proof. This follows from Definitions 8.41 and 8.44 and Lemma 8.42. $\square$

8.3.2 Reusing the formalism

Definition 8.46. Define a function $\lambda : \mathbb{R} \rightarrow \mathbb{R}$

$$\lambda(t) = \begin{cases} 0 & \text{if } t \in (-\infty, a), \\ \int_a^t \rho(\tau) d\tau & \text{if } t \in [a, b], \\ \int_a^b \rho(\tau) d\tau & \text{if } t \in (b, +\infty). \end{cases} \quad (8.45)$$

Then a corresponding *continuous measure* $d\lambda$ can be defined as

$$d\lambda = \begin{cases} \rho(t)dt & \text{if } t \in [a, b], \\ 0 & \text{otherwise,} \end{cases} \quad (8.46)$$

where the *support of the continuous measure* $d\lambda$ is the interval $[a, b]$.

Definition 8.47. The *discrete measure* or the *Dirac measure* associated with the point set $\{t_1, t_2, \dots, t_N\}$ is a measure $d\lambda$ that is nonzero only at the points t_i and has the value ρ_i there. The *support of the discrete measure* is the set $\{t_1, t_2, \dots, t_N\}$.

Lemma 8.48. For a function $u : \mathbb{R} \rightarrow \mathbb{R}$, define

$$\lambda(t) = \sum_{i=1}^N \rho_i H(t - t_i), \quad (8.47)$$

and we have

$$\int_{\mathbb{R}} u(t) d\lambda = \sum_{i=1}^N \rho_i u(t_i). \quad (8.48)$$

Proof. By (8.47) and Lemmas 8.43 and 8.45, we have

$$\int_{\mathbb{R}} u(t) d\lambda = \int_{\mathbb{R}} \sum_{i=1}^N \rho_i \delta(t - t_i) u(t) dt = \sum_{i=1}^N \rho_i u(t_i). \quad \square$$

8.3.3 DLS via normal equations

Example 8.49. Consider the sales record in Figure 8.3.

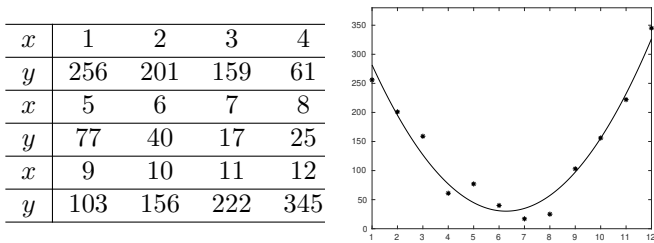


Figure 8.3: A sales record as an example of DLS.

From the plot of the discrete data, it appears that a quadratic polynomial would be a good fit. Hence we formulate the least square problem as finding the coefficients of a quadratic polynomial to minimize

$$\sum_{i=1}^{12} \left(y_i - \sum_{j=0}^2 a_j x_i^j \right)^2.$$

Reusing the procedures in Example 8.35, we have

$$\begin{aligned} G(1, x, x^2) &= \begin{bmatrix} \langle 1, 1 \rangle & \langle 1, x \rangle & \langle 1, x^2 \rangle \\ \langle x, 1 \rangle & \langle x, x \rangle & \langle x, x^2 \rangle \\ \langle x^2, 1 \rangle & \langle x^2, x \rangle & \langle x^2, x^2 \rangle \end{bmatrix} \\ &= \begin{bmatrix} 12 & 78 & 650 \\ 78 & 650 & 6084 \\ 650 & 6084 & 60710 \end{bmatrix}, \\ \mathbf{c} &= \begin{bmatrix} \langle y, 1 \rangle \\ \langle y, x \rangle \\ \langle y, x^2 \rangle \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{12} y_i \\ \sum_{i=1}^{12} y_i x_i \\ \sum_{i=1}^{12} y_i x_i^2 \end{bmatrix} = \begin{bmatrix} 1662 \\ 11392 \\ 109750 \end{bmatrix}. \end{aligned}$$

Then the normal equations yield

$$\mathbf{a} = G^{-1} \mathbf{c} = \begin{bmatrix} 386.00 \\ -113.43 \\ 9.04 \end{bmatrix}.$$

The corresponding polynomial is shown in the above plot.

8.3.4 DLS via QR decomposition

Definition 8.50. A matrix $A \in \mathbb{R}^{n \times n}$ is *orthogonal* iff $A^T A = I$.

Theorem 8.51 (QR factorization). For any $A \in \mathbb{R}^{m \times n}$, there exists an orthogonal matrix $Q \in \mathbb{R}^{m \times m}$ and an upper-triangular matrix $R \in \mathbb{R}^{m \times n}$ so that $A = QR$.

Proof. Rewrite $A = [\xi_1, \xi_2, \dots, \xi_n] \in \mathbb{R}^{m \times n}$ and denote by r the column rank of A . Construct a rank- r matrix

$$A_r = [u_1, u_2, \dots, u_r]$$

by the following steps.

- (S-1) Set $u_1 = \xi_{k_1}$ where k_1 satisfies $\xi_{k_1} \neq \mathbf{0}$ and $\forall \ell < k_1$, $\xi_\ell = \mathbf{0}$.
- (S-2) For each $j = 2, \dots, r$, set $u_j = \xi_{k_j}$ where k_j satisfies the condition that $K_j = (\xi_{k_1}, \dots, \xi_{k_j})$ is a list of independent column vectors and, $\forall \ell \in R_j := \{k_{j-1} + 1, \dots, k_j - 1\}$, ξ_ℓ can be expressed as a linear combination of the column vectors in K_{j-1} .

In plain words, (S-1) means that we jump over the lead zero vectors and (S-2) states that, starting from u_{j-1} , we pick the first vector as u_j that is not in $\text{span}(u_1, u_2, \dots, u_{j-1})$.

By Corollary 8.16, the Gram-Schmidt process determines a unique matrix $A_r^* = [u_1^*, u_2^*, \dots, u_r^*] \in \mathbb{R}^{m \times r}$ and a unique upper-triangular matrix such that

$$A_r = A_r^* \begin{bmatrix} b_{11} & b_{21} & \dots & b_{r1} \\ & b_{22} & \dots & b_{r2} \\ & & \ddots & \vdots \\ & & & b_{rr} \end{bmatrix}, \quad (8.49)$$

where $u_1^*, u_2^*, \dots, u_r^*$ are orthogonal. By definition of the column rank of a matrix, we have $r \leq m$.

To continue, we insert each column vector in

$$X = \{\xi_1, \xi_2, \dots, \xi_n\} \setminus \{u_1, u_2, \dots, u_r\}$$

back into (8.49) and show that the QR form of (8.49) is maintained. For those zero column vectors in (S-1), we have

$$A_\xi = [\xi_1 \ \dots \ \xi_{k_1-1} \ u_1 \ u_2 \ \dots \ u_r] \quad (8.50)$$

$$= A_r^* \begin{bmatrix} 0 & \dots & 0 & b_{11} & b_{21} & \dots & b_{r1} \\ 0 & \dots & 0 & & b_{22} & \dots & b_{r2} \\ \vdots & \ddots & \vdots & & & \ddots & \vdots \\ 0 & \dots & 0 & & & & b_{rr} \end{bmatrix}.$$

For each ξ_ℓ with $\ell \in R_j$ in (S-2), we have

$$[u_1, u_2, \dots, u_{j-1}, \xi_\ell] \quad (8.51)$$

$$= [u_1^*, u_2^*, \dots, u_{j-1}^*] \begin{bmatrix} b_{11} & \dots & b_{j-1,1} & c_{\ell,1} \\ & \ddots & \vdots & \vdots \\ & & b_{j-1,j-1} & c_{\ell,j-1} \end{bmatrix},$$

where $\xi_\ell = c_{\ell,1}u_1^* + \dots + c_{\ell,j-1}u_{j-1}^*$. With (8.50) as the induction basis and (8.51) as the inductive step, it is straightforward to prove by induction that we have $A = A_r^* R$ where R is an upper-triangular matrix.

If $r = m$, Definitions 8.50 and 8.5 complete the proof. Otherwise $r < m$ and the proof is completed by the well-known fact in linear algebra that a list of orthonormal vectors can be extended to an orthonormal basis. $\square$

Lemma 8.52. An orthogonal matrix preserves the 2-norm of the vectors it acts on.

Proof. Definition 8.50 yields

$$\forall \mathbf{x} \in \text{dom}(Q), \quad \|Q\mathbf{x}\|_2^2 = \mathbf{x}^\top Q^\top Q \mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|_2^2. \quad \square$$

Theorem 8.53. Consider an over-determined linear system $A\mathbf{x} = \mathbf{b}$ where $A \in \mathbb{R}^{m \times n}$ and $m \geq n$. The discrete linear least square problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|_2^2$$

is solved by $\mathbf{x}^*$ satisfying

$$R_1 \mathbf{x}^* = \mathbf{c}, \quad (8.52)$$

where $R_1 \in \mathbb{R}^{n \times n}$ and $\mathbf{c} \in \mathbb{R}^n$ result from the QR factorization of A :

$$Q^\top A = R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}, \quad Q^\top \mathbf{b} = \begin{bmatrix} \mathbf{c} \\ \mathbf{r} \end{bmatrix}. \quad (8.53)$$

Furthermore, the minimum is $\|\mathbf{r}\|_2^2$.

Proof. For any $\mathbf{x} \in \mathbb{R}^n$, we have

$$\|A\mathbf{x} - \mathbf{b}\|_2^2 = \|Q^\top A\mathbf{x} - Q^\top \mathbf{b}\|_2^2 = \|R_1 \mathbf{x} - \mathbf{c}\|_2^2 + \|\mathbf{r}\|_2^2,$$

where the first step follows from Lemma 8.52. $\square$

8.4 Solving ill-posed linear systems

Definition 8.54. A problem in mathematical physics is *well-posed* iff (i) it admits a solution, (ii) the solution is unique, and (iii) the problem has a small condition number; otherwise it is *ill-posed*.

8.4.1 No solutions or multiple solutions

Theorem 8.55. Suppose the linear system $A\mathbf{x} = \mathbf{b}$ in Theorem C.255 has no solutions. Then $\mathbf{x}^\oplus$ in (C.93) solves the problem of minimizing the 2-norm of the residual, i.e.,

$$\mathbf{x}^\oplus = \arg \min_{\mathbf{x} \in \mathbb{C}^n} \|A\mathbf{x} - \mathbf{b}\|_2. \quad (8.54)$$

Proof. The Fourier expansion of $\mathbf{b}$ is

$$\mathbf{b} = \sum_{i=1}^m \langle \mathbf{b}, \mathbf{v}_i \rangle \mathbf{v}_i.$$

It follows from (C.93) that

$$\forall \mathbf{b} \notin \text{range } A, \quad A\mathbf{x}^\oplus - \mathbf{b} \in \text{span}(\mathbf{v}_{r+1}, \mathbf{v}_{r+2}, \dots, \mathbf{v}_m).$$

On the other hand, the rank of A being r implies

$$\forall \mathbf{x} \in \mathbb{C}^n, \quad A\mathbf{x} - A\mathbf{x}^\oplus \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r).$$

Hence $\langle A\mathbf{x}^\oplus - \mathbf{b}, A\mathbf{x} - A\mathbf{x}^\oplus \rangle = 0$ and the Pythagorean theorem C.196 yields

$$\|A\mathbf{x} - \mathbf{b}\|_2^2 = \|A\mathbf{x} - A\mathbf{x}^\oplus\|_2^2 + \|A\mathbf{x}^\oplus - \mathbf{b}\|_2^2,$$

which completes the proof. $\square$

Theorem 8.56. Suppose a linear system $A\mathbf{x} = \mathbf{b}$ has multiple solutions. Then its general solution is $\mathbf{x}^\oplus + \mathbf{y}$ where $\mathbf{x}^\oplus$ is given by (C.93) and $\mathbf{y}$ is any vector in the null space of A . Furthermore, $\mathbf{x}^\oplus$ is the solution with the minimum 2-norm.

Proof. By Theorem C.255, we have

$$\begin{cases} \mathbf{x}^\oplus \in \text{span}(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r), \\ \text{null } A = \text{span}(\mathbf{u}_{r+1}, \mathbf{u}_{r+2}, \dots, \mathbf{u}_n). \end{cases}$$

Hence $\forall \mathbf{y} \in \text{null } A$, $\langle \mathbf{x}^\oplus, \mathbf{y} \rangle = 0$ and the Pythagorean theorem C.196 yields

$$\|\mathbf{x}^\oplus + \mathbf{y}\|_2 = \|\mathbf{x}^\oplus\|_2 + \|\mathbf{y}\|_2.$$

The proof is completed by the fact that all solutions of the linear system are of the form $\mathbf{x}^\oplus + \mathbf{y}$. $\square$

8.4.2 The Moore–Penrose inverse

Definition 8.57. The *Moore–Penrose inverse* or *pseudo-inverse* or *generalized inverse* of a matrix $A \in \mathbb{F}^{m \times n}$ where $\mathbb{F} = \mathbb{R}, \mathbb{C}$ is the matrix $A^+ \in \mathbb{F}^{n \times m}$ given by

$$\forall \mathbf{y} \in \mathbb{F}^m, \quad A^+ \mathbf{y} = \sum_{j=1}^r \frac{1}{\sigma_j} \langle \mathbf{y}, \mathbf{v}_j \rangle \mathbf{u}_j, \quad (8.55)$$

where r is the rank of A and $(\sigma_j, \mathbf{u}_j, \mathbf{v}_j)$ is the singular system of A , cf. Definition C.254.

Theorem 8.58. The pseudo-inverse in Definition 8.57 has the following properties.

(PDI-1) AA^+ maps all columns of A to themselves:
 $AA^+A = A$;

(PDI-2) A^+ acts like a weak inverse:

$$A^+AA^+ = A^+;$$

(PDI-3) Both AA^+ and A^+A are Hermitian:

$$(AA^+)^* = AA^+, (A^+A)^* = A^+A.$$

Lemma 8.59. If A has linearly independent columns, then A^*A is invertible and we have

$$A^+ = (A^*A)^{-1}A^*, \quad (8.56)$$

which is then a *left inverse*, i.e. $A^+A = I$. Similarly, if A has linearly independent rows, then AA^* is invertible and

$$A^+ = A^*(AA^*)^{-1}, \quad (8.57)$$

which is then a *right inverse*, i.e. $AA^+ = I$.

8.4.3 Spectral cutoff for ill-conditioning

Example 8.60. The condition number of solving the linear system $A\mathbf{x} = \mathbf{b}$ is dominated by the small singular values. If we perturb $\mathbf{b}$ to $\hat{\mathbf{b}} = \mathbf{b} + \epsilon \mathbf{v}_j$ for some $\epsilon \in \mathbb{C}$, by Theorem C.255 the solution would change from $\mathbf{x}$ to $\hat{\mathbf{x}} = \mathbf{x} + \frac{\epsilon}{\sigma_j} \mathbf{u}_j$. Hence the condition number based on *absolute* errors is

$$\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|_2}{\|\hat{\mathbf{b}} - \mathbf{b}\|_2} = \frac{1}{\sigma_j}.$$

Thus a solvable linear system is ill-conditioned if its matrix has small singular values.

Definition 8.61. The *spectral cutoff* is a method to stabilize an ill-conditioned linear system by neglecting the terms with small singular values in the solution (C.93).

Definition 8.62. Given a perturbed RHS $\hat{\mathbf{b}}$ to a linear system $A\mathbf{x} = \mathbf{b}$, the *discrepancy principle* is a strategy to come up with an approximate solution $\hat{\mathbf{x}}$ so that

$$\|A\hat{\mathbf{x}} - \hat{\mathbf{b}}\|_2 = C\|\mathbf{b} - \hat{\mathbf{b}}\|_2, \quad (8.58)$$

where $C \geq 1$ is some suitable constant.

Example 8.63. By the discrepancy principle for the spectral cutoff, we approximate the solution (C.93) by

$$\mathbf{x}_p := \begin{cases} \mathbf{0} & \text{if } p = 0; \\ \sum_{j=1}^p \frac{1}{\sigma_j} \langle \hat{\mathbf{b}}, \mathbf{v}_j \rangle \mathbf{u}_j & \text{if } p = 1, \dots, r, \end{cases} \quad (8.59)$$

where p is determined by the perturbation error $\|\hat{\mathbf{b}} - \mathbf{b}\|_2$ so that (8.58) is satisfied. A common choice is $C = 1$.

Theorem 8.64. Suppose the linear system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{C}^{m \times n}$ is solvable and a perturbation $\hat{\mathbf{b}}$ to $\mathbf{b}$ satisfies

$$\|\hat{\mathbf{b}} - \mathbf{b}\|_2 \leq \epsilon \leq \|\hat{\mathbf{b}}\|_2. \quad (8.60)$$

Then there exists a smallest integer $p = p(\epsilon)$ such that

$$\|A\mathbf{x}_p - \hat{\mathbf{b}}\|_2 \leq \epsilon, \quad (8.61)$$

where $\mathbf{x}_p$ is defined in (8.59). The discrepancy principle for the spectral cutoff is *convergent* in the sense that

$$\lim_{\epsilon \rightarrow 0} \mathbf{x}_p = A^+\mathbf{b}. \quad (8.62)$$

Proof. The function $f : \{0, 1, \dots, r\} \rightarrow \mathbb{R}$ given by

$$f(p) := \|A\mathbf{x}_p - \hat{\mathbf{b}}\|_2^2 - \epsilon^2$$

can be expressed as

$$f(p) = \sum_{j=p+1}^m \left| \langle \hat{\mathbf{b}}, \mathbf{v}_j \rangle \right|^2 - \epsilon^2, \quad (8.63)$$

where $\mathbf{v}_j$ is a left singular vector of A . Then (8.60) yields $f(0) = \|\hat{\mathbf{b}}\|_2^2 - \epsilon^2 \geq 0$ and $f(r) = -\epsilon^2 < 0$ if the rank r of A equals m . If $r < m$, the solvability condition of the linear system and Theorem C.255 yield

$$\forall j = r+1, \dots, m, \quad \langle \mathbf{b}, \mathbf{v}_j \rangle = 0.$$

Then we have

$$f(r) = \sum_{j=r+1}^m \left| \langle (\hat{\mathbf{b}} - \mathbf{b}), \mathbf{v}_j \rangle \right|^2 - \epsilon^2 \leq \|\hat{\mathbf{b}} - \mathbf{b}\|_2^2 - \epsilon^2 \leq 0$$

where the last step follows from (8.60). By (8.63), f is monotonically decreasing, and the first conclusion concerning (8.61) follows from $f(0) \geq 0$ and $f(r) \leq 0$.

As for the second conclusion, (8.61) yields

$$\|A\mathbf{x}_p - \mathbf{b}\|_2 \leq \|A\mathbf{x}_p - \hat{\mathbf{b}}\|_2 + \|\hat{\mathbf{b}} - \mathbf{b}\|_2 \leq 2\epsilon.$$

Then (8.62) follows from

$$\forall \mathbf{u} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r), \quad A^+A\mathbf{u} = \mathbf{u},$$

which is a consequence of (C.91a) and (8.55). $\square$

8.4.4 Tikhonov regularization

Definition 8.65. *Tikhonov regularization* is a method to approximate the solution of an ill-conditioned linear system $A\mathbf{x} = \mathbf{b}$ by that of a regularized linear system

$$\alpha\mathbf{x} + A^*A\mathbf{x} = A^*\mathbf{b}, \quad (8.64)$$

where $\alpha > 0$ is called the *regularization parameter*.

Theorem 8.66. Suppose a matrix $A \in \mathbb{C}^{m \times n}$ has rank r . Then for any $\mathbf{b} \in \mathbb{C}^m$ the Tikhonov regularized linear system (8.64) admits the unique solution

$$\mathbf{x}_\alpha := \sum_{j=1}^r \frac{\sigma_j}{\alpha + \sigma_j^2} \langle \mathbf{b}, \mathbf{v}_j \rangle \mathbf{u}_j, \quad (8.65)$$

where $(\sigma_j, \mathbf{u}_j, \mathbf{v}_j)$ is the singular system of A , cf. Definition C.254. Furthermore, the Tikhonov regularization satisfies

$$\lim_{\alpha \rightarrow 0} (\alpha I + A^*A)^{-1} A^*\mathbf{b} = A^+\mathbf{b}. \quad (8.66)$$

Proof. The matrix $B := \alpha I + A^*A$ is strictly positive definite and thus nonsingular. Furthermore, $B\mathbf{u}_j = (\alpha + \sigma_j^2)\mathbf{u}_j$ and $B^* = B$ imply that $(\alpha + \sigma_j^2, \mathbf{u}_j, \mathbf{u}_j)$ is a singular system of B . Apply Theorem C.255 to (8.64) and we have

$$\mathbf{x} = \sum_{j=1}^r \frac{1}{\alpha + \sigma_j^2} \langle A^*\mathbf{b}, \mathbf{u}_j \rangle \mathbf{u}_j = \sum_{j=1}^r \frac{1}{\alpha + \sigma_j^2} \langle \mathbf{b}, A\mathbf{u}_j \rangle \mathbf{u}_j = \mathbf{x}_\alpha,$$

where the last step follows from (C.91a) and (8.65). Finally, (8.66) follows directly from (8.55) and (8.65). $\square$

Theorem 8.67. For any $\mathbf{b} \in \mathbb{C}^m$, the solution $\mathbf{x}_\alpha$ in (8.65) to the Tikhonov regularized linear system (8.64) is equivalent to the unique minimizer of $\|\mathbf{Ax} - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}\|_2^2$, i.e.,

$$\|\mathbf{Ax}_\alpha - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}_\alpha\|_2^2 = \inf_{\mathbf{x} \in \mathbb{C}^n} \{\|\mathbf{Ax} - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}\|_2^2\}. \quad (8.67)$$

Proof. Define $\mathbf{z}_\alpha := \alpha\mathbf{x}_\alpha + A^*\mathbf{Ax}_\alpha - A^*\mathbf{b}$. The law of cosines (Theorem C.181) implies

$$\begin{aligned} \|\mathbf{Ax} - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}\|_2^2 &= 2\operatorname{Re} \langle \mathbf{x} - \mathbf{x}_\alpha, \mathbf{z}_\alpha \rangle \\ &+ \|\mathbf{Ax}_\alpha - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}_\alpha\|_2^2 + \|\mathbf{Ax} - \mathbf{Ax}_\alpha\|_2^2 + \alpha\|\mathbf{x} - \mathbf{x}_\alpha\|_2^2. \end{aligned} \quad (8.68)$$

Hence the solution $\mathbf{x}_\alpha$ of (8.64) satisfies $\mathbf{z}_\alpha = \mathbf{0}$ and consequently (8.67) holds.

Conversely, let $\mathbf{x}_\alpha$ be the solution of (8.67) and suppose $\mathbf{z}_\alpha \neq \mathbf{0}$. Then for $\mathbf{x} := \mathbf{x}_\alpha - \epsilon\mathbf{z}_\alpha$ with $\epsilon \in \mathbb{R}$, (8.68) yields

$$\|\mathbf{Ax} - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}\|_2^2 = \|\mathbf{Ax}_\alpha - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}_\alpha\|_2^2 - 2\epsilon a + \epsilon^2 b,$$

where $a := \|\mathbf{z}_\alpha\|_2^2$ and $b := \|\mathbf{Az}_\alpha\|_2^2 + \alpha\|\mathbf{z}_\alpha\|_2^2$. The choice of $\epsilon = \frac{a}{b}$ gives

$$\|\mathbf{Ax} - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}\|_2^2 < \|\mathbf{Ax}_\alpha - \mathbf{b}\|_2^2 + \alpha\|\mathbf{x}_\alpha\|_2^2,$$

which contradicts the starting point. $\square$

Example 8.68. In practical applications, the RHS side of a linear system is often perturbed with some uncertainties, say, $\|\hat{\mathbf{b}} - \mathbf{b}\|_2 < \epsilon$. Then the Tikhonov regularization approximates the solution $\mathbf{x} = A^+\mathbf{b}$ by $\mathbf{x}_\alpha$ that satisfies

$$B\mathbf{x}_\alpha = A^*\hat{\mathbf{b}},$$

where $B := \alpha I + A^*A$. Hence the total absolute error is

$$\mathbf{x}_\alpha - \mathbf{x} = B^{-1}A^*(\hat{\mathbf{b}} - \mathbf{b}) + (B^{-1}A^*\mathbf{b} - A^+\mathbf{b}),$$

where the first RHS term is the error caused by the RHS data and the second is the error caused by the regularization. On the one hand, α should be small to keep the regularization error small. On the other hand, Lemma 2.23 implies

$$\operatorname{cond}_2 B = \frac{\alpha + \sigma_1^2}{\alpha + \sigma_n^2}.$$

Thus α should be large to make the data error small; otherwise the conditioning of the linear system $B\mathbf{x} = A^*(\hat{\mathbf{b}} - \mathbf{b})$ would be large. Note that this conclusion remains valid even if the condition number is based on absolute errors as in Example 8.60. To sum up, the choice of α have to be made through a compromise between accuracy and stability.

Theorem 8.69. Suppose the linear system $\mathbf{Ax} = \mathbf{b}$ with $A \in \mathbb{C}^{m \times n}$ is solvable and a perturbation $\hat{\mathbf{b}}$ to $\mathbf{b}$ satisfies

$$\|\hat{\mathbf{b}} - \mathbf{b}\|_2 \leq \epsilon < \|\hat{\mathbf{b}}\|_2. \quad (8.69)$$

Then there exists a unique $\alpha = \alpha(\epsilon)$ such that

$$\|\mathbf{Ax}_\alpha - \hat{\mathbf{b}}\|_2 = \epsilon, \quad (8.70)$$

where $\mathbf{x}_\alpha$ is the unique solution of

$$(\alpha I + A^*A)\mathbf{x} = A^*\hat{\mathbf{b}}. \quad (8.71)$$

In addition, the discrepancy principle for Tikhonov regularization is *convergent*, i.e.,

$$\lim_{\epsilon \rightarrow 0} \mathbf{x}_\alpha = A^+\mathbf{b}. \quad (8.72)$$

Proof. By (8.65), the function $f : (0, +\infty) \rightarrow \mathbb{R}$ given by

$$f(\alpha) := \|\mathbf{Ax}_\alpha - \hat{\mathbf{b}}\|_2^2 - \epsilon^2$$

can be expressed as

$$f(\alpha) = \sum_{j=1}^r \left| \frac{\alpha}{\alpha + \sigma_j^2} \langle \hat{\mathbf{b}}, \mathbf{v}_j \rangle \right|^2 + \sum_{j=r+1}^m \left| \langle \hat{\mathbf{b}}, \mathbf{v}_j \rangle \right|^2 - \epsilon^2, \quad (8.73)$$

where $\mathbf{v}_j$ is a left singular vector of A . Being continuous and strictly monotonically increasing, f also satisfies

$$\begin{aligned} \lim_{\alpha \rightarrow 0} f(\alpha) &= \sum_{j=r+1}^m \left| \langle \hat{\mathbf{b}}, \mathbf{v}_j \rangle \right|^2 - \epsilon^2 = \sum_{j=r+1}^m \left| \langle \hat{\mathbf{b}} - \mathbf{b}, \mathbf{v}_j \rangle \right|^2 - \epsilon^2 \\ &\leq \|\hat{\mathbf{b}} - \mathbf{b}\|_2^2 - \epsilon^2 \leq 0, \end{aligned}$$

$$\lim_{\alpha \rightarrow +\infty} f(\alpha) = \sum_{j=1}^m \left| \langle \hat{\mathbf{b}}, \mathbf{v}_j \rangle \right|^2 - \epsilon^2 = \|\hat{\mathbf{b}}\|_2^2 - \epsilon^2 > 0.$$

hence f must have exactly one root α that satisfies (8.70).

Then (8.70) and the triangle inequality yield

$$\|\hat{\mathbf{b}}\|_2 - \epsilon = \|\hat{\mathbf{b}}\|_2 - \|\mathbf{Ax}_\alpha - \hat{\mathbf{b}}\|_2 \leq \|\mathbf{Ax}_\alpha\|_2.$$

On the other hand, $\mathbf{x}_\alpha$ being a solution of (8.71) gives

$$\alpha\|\mathbf{Ax}_\alpha\|_2 = \|AA^*(\hat{\mathbf{b}} - \mathbf{Ax}_\alpha)\|_2 \leq \|AA^*\|_2 \epsilon.$$

By (8.69), we have $\|\hat{\mathbf{b}}\|_2 \geq \|\mathbf{b}\|_2 - \epsilon$, which, together with the above two inequalities, gives

$$\alpha \leq \frac{\|AA^*\|_2 \epsilon}{\|\mathbf{b}\|_2 - 2\epsilon}.$$

Hence $\lim_{\epsilon \rightarrow 0} \alpha = 0$. Then (8.72) follows from (8.69), (8.55), and (8.65). $\square$

8.5 Theoretical problems

- I. Give a detailed proof of Theorem 8.1.
- II. Consider the Chebyshev polynomials of the first kind.
 - (a) Show that they are orthogonal on $[-1, 1]$ with respect to the inner product in Theorem 8.1 with the weight function $\rho(x) = \frac{1}{\sqrt{1-x^2}}$.
 - (b) Normalize the first three Chebyshev polynomials to arrive at an orthonormal system.
- III. Least-square approximation of a continuous function. Approximate the circular arc given by the equation $y(x) = \sqrt{1-x^2}$ for $x \in [-1, 1]$ by a quadratic polynomial with respect to the inner product in Theorem 8.1.
 - (a) $\rho(x) = \frac{1}{\sqrt{1-x^2}}$ with Fourier expansion,

(b) $\rho(x) = \frac{1}{\sqrt{1-x^2}}$ with normal equations.

IV. Discrete least square via orthonormal polynomials. Consider the example on the table of sales record in Example 8.49.

(a) Starting from the independent list $(1, x, x^2)$, construct orthonormal polynomials by the Gram-Schmidt process using

$$\langle u(t), v(t) \rangle = \sum_{i=1}^N \rho(t_i) u(t_i) v(t_i) \quad (8.74)$$

as the inner product with $N = 12$ and $\rho(x) = 1$.

(b) Find the best approximation $\hat{\varphi} = \sum_{i=0}^2 a_i x^i$ such that $\|y - \hat{\varphi}\| \leq \|y - \sum_{i=0}^2 b_i x^i\|$ for all $b_i \in \mathbb{R}$. Verify that $\hat{\varphi}$ is the same as that of the example on the table of sales record in the notes.

(c) Suppose there are other tables of sales record in the same format as that in the example. Values of N and x_i 's are the same, but the values of y_i 's are different. Which of the above calculations can be reused? Which cannot be reused? What advantage of orthonormal polynomials over normal equations does this reuse imply?

V. Prove Theorem 8.58 and Lemma 8.59.

8.6 Programming assignments

A. Write a program to perform discrete least square via normal equations. Your subroutine should take two arrays x and y as the input and output three coefficients a_0, a_1, a_2 that determines a quadratic polynomial as the best fitting polynomial in the sense of least squares with the weight function $\rho = 1$.

Run your subroutine on the following data.

x	0.0	0.5	1.0	1.5	2.0	2.5	3.0
y	2.9	2.7	4.8	5.3	7.1	7.6	7.7
x	3.5	4.0	4.5	5.0	5.5	6.0	6.5
y	7.6	9.4	9.0	9.6	10.0	10.2	9.7
x	7.0	7.5	8.0	8.5	9.0	9.5	10.0
y	8.3	8.4	9.0	8.3	6.6	6.7	4.1

B. Write a program to solve the previous discrete least square problem via QR factorization. Report the condition number based on the 2-norm of the matrix G in the normal-equation approach and that of the matrix R_1 in the QR-factorization approach, verifying that the former is much larger than the latter.

C. First, prove that the solution of (8.35) with $f(x) = \frac{1}{1+x}$ is of the form

$$\forall j = 0, 1, \dots, n, \quad \alpha_j = \beta_j \ln 2 + \gamma_j,$$

where β_j and γ_j are rational numbers. Hint: derive (8.36) from $r_0 = \ln 2$ and the geometric series

$$\sum_{i=1}^j (-1)^{i-1} x^{i-1} = \frac{1 - (-1)^j x^j}{1+x}.$$

Second, write a program to perform Gauss elimination on the Hilbert matrix in terms of rational numbers to obtain β_j and γ_j .

Third, for $n = 1, 2, \dots, 6$, compute α_j 's by approximating $\ln 2$ to machine precision and present your results in a table to show convergence of $\alpha_j = (-1)^j$ to the Taylor series of $f(x) = \frac{1}{1+x}$.

Fourth, compute the RHS r_j 's in (8.36) by floating-point arithmetic with $\ln 2 \approx 0.69315$ (the relative error is 10^{-5}) and then solve (8.35) to obtain α_j 's. Can you still obtain convergence? Why? Your explanation should contain at least the condition numbers of the Hilbert matrix for $n = 2, 3, \dots, 6$.

Finally, use Tikhonov regularization with $\alpha = 10^{-10}$ to solve (8.35); show that you can recover convergence at least for α_0 and α_1 .

Chapter 9

Numerical Integration and Differentiation

Definition 9.1. A *weighted quadrature formula* $I_n(f)$ is a linear functional

$$I_n(f) := \sum_{k=1}^n w_k f(x_k) \quad (9.1)$$

that approximates the integral of a function $f \in \mathcal{C}[a, b]$,

$$I(f) := \int_a^b f(x) \rho(x) dx, \quad (9.2)$$

where the weight function $\rho \in \mathcal{L}[a, b]$ satisfies $\forall x \in (a, b)$, $\rho(x) > 0$. The points x_k 's at which the integrand f is evaluated are called *nodes* or *abscissas*, and the multiplier w_k 's are called *weights* or *coefficients*.

Example 9.2. If a and/or b are infinite, $I(f)$ and $I_n(f)$ in (9.1) may still be well defined if the *moment of weight function*

$$\mu_j := \int_a^b x^j \rho(x) dx \quad (9.3)$$

exists and is finite for all $j \in \mathbb{N}$.

9.1 Accuracy and convergence

Definition 9.3. The *remainder*, or *error*, of $I_n(f)$ is

$$E_n(f) := I(f) - I_n(f). \quad (9.4)$$

$I_n(f)$ is said to be *convergent* for $\mathcal{C}[a, b]$ iff

$$\forall f \in \mathcal{C}[a, b], \quad \lim_{n \rightarrow +\infty} I_n(f) = I(f). \quad (9.5)$$

Definition 9.4. A subset $\mathbb{V} \subset \mathcal{C}[a, b]$ is *dense* in $\mathcal{C}[a, b]$ iff

$$\forall f \in \mathcal{C}[a, b], \forall \epsilon > 0, \exists f_\epsilon \in \mathbb{V}, \text{ s.t. } \max_{x \in [a, b]} |f(x) - f_\epsilon(x)| \leq \epsilon. \quad (9.6)$$

Theorem 9.5. Let $\{I_n(f) : n \in \mathbb{N}^+\}$ be a sequence of quadrature formulas that approximate $I(f)$, where I_n and $I(f)$ are defined in (9.1) and (9.2). Let $\mathbb{V}$ be a dense subset of $\mathcal{C}[a, b]$. $I_n(f)$ is convergent for $\mathcal{C}[a, b]$ if and only if

$$(a) \quad \forall f \in \mathbb{V}, \lim_{n \rightarrow +\infty} I_n(f) = I(f),$$

$$(b) \quad \exists B \in \mathbb{R} \text{ s.t. } \forall n \in \mathbb{N}^+, W_n := \sum_{k=1}^n |w_k| < B.$$

Proof. For sufficiency, we need to prove that for any given f we have $\lim_{n \rightarrow +\infty} I_n(f) = I(f)$. To this end, we find $f_\epsilon \in \mathbb{V}$ such that (9.6) holds, define $K := \max_{x \in [a, b]} |f(x) - f_\epsilon(x)|$. Then we have

$$\begin{aligned} |E_n(f)| &\leq |I(f) - I(f_\epsilon)| + |I(f_\epsilon) - I_n(f_\epsilon)| + |I_n(f_\epsilon) - I_n(f)| \\ &= \left| \int_a^b [f(x) - f_\epsilon(x)] \rho(x) dx \right| \\ &\quad + |I(f_\epsilon) - I_n(f_\epsilon)| + \left| \sum_{k=1}^n w_k [f(x_k) - f_\epsilon(x_k)] \right| \\ &\leq K \left[\int_a^b \rho(x) dx + \sum_{k=1}^n |w_k| \right] + |I(f_\epsilon) - I_n(f_\epsilon)|, \end{aligned}$$

where the first step follows from the triangle inequality, the second from Definition 9.1, and the third from the definition of K . The terms inside the brackets is bounded because of $\rho \in \mathcal{C}[a, b]$ and condition (b). By condition (a), $|I(f_\epsilon) - I_n(f_\epsilon)|$ can be made arbitrarily small. Since K can also be arbitrarily small, we have (9.5).

For necessity, (9.5) trivially yields (a) while it is non-trivial to deduce (b) from (9.5) and the principle of uniform boundedness (Theorem E.83). The numerical quadrature formula $I_n : \mathcal{C}[a, b] \rightarrow \mathbb{R}$ is a linear functional and is continuous at $f = \mathbf{0}$ in the sense of Definition D.82 because

$$\begin{aligned} \forall \epsilon > 0, \exists \delta = \frac{\epsilon}{2 \sum_k |w_k|}, \text{ s.t. } \forall f \in \mathcal{C}[a, b], \\ \|f - \mathbf{0}\|_\infty < \delta \implies |I_n(f) - I_n(\mathbf{0})| = \left| \sum_k w_k f(x_k) \right| \\ \leq \sum_k |w_k| |f(x_k)| \leq \delta \sum_k |w_k| < \epsilon. \end{aligned}$$

By Theorem E.41, I_n is continuous, i.e., $\forall n \in \mathbb{N}^+$, $I_n \in \mathcal{CL}(\mathcal{C}[a, b], \mathbb{R})$. The convergence (9.5) implies $\forall f \in \mathcal{C}[a, b]$, $\sup_{n \in \mathbb{N}^+} |I_n(f)| < +\infty$. Then Theorems E.27 and E.83 yield (b), where W_n , by Lemma E.56, equals the operator norm of I_n . $\square$

Definition 9.6. A weighted quadrature formula (9.1) has (polynomial) *degree of exactness* d_E iff

$$\begin{cases} \forall f \in \mathbb{P}_{d_E}, & E_n(f) = 0, \\ \exists g \in \mathbb{P}_{d_E+1}, \text{ s.t. } & E_n(g) \neq 0, \end{cases} \quad (9.7)$$

where $\mathbb{P}_d$ denotes the set of polynomials with degree no more than d .

Example 9.7. By Definition 9.6, $d_E \geq 0$ implies that $\sum_k w_k$ is bounded since $I_n(c) = c \int_a^b \rho(x) dx$ holds for any constant $c \in \mathbb{R}$.

Lemma 9.8. For a quadrature formula $I_n(f)$ in (9.1) with $d_E \geq n - 1$, its weights can be deduced from its distinct nodes $x_1, \dots, x_n$ as

$$\forall k = 1, \dots, n, \quad w_k = \int_a^b \rho(x) \ell_k(x) dx, \quad (9.8)$$

where $\ell_k(x)$ is the fundamental polynomial in (6.9) applied to the given nodes, i.e.,

$$\ell_k(x) := \prod_{i \neq k; i=1}^n \frac{x - x_i}{x_k - x_i}. \quad (9.9)$$

Proof. Let $p_{n-1}(f; x)$ be the unique polynomial that interpolates f at the distinct nodes, as in the theorem on the uniqueness of polynomial interpolation (Theorem 6.5). Then we have

$$\begin{aligned} \sum_{k=1}^n w_k p_{n-1}(x_k) &= \int_a^b p_{n-1}(f; x) \rho(x) dx \\ &= \int_a^b \sum_{k=1}^n \{\ell_k(x) f(x_k)\} \rho(x) dx = \sum_{k=1}^n w_k f(x_k), \end{aligned}$$

where the first step follows from $d_E \geq n - 1$ and the second step from the interpolation conditions (6.4), the Lagrange formula, and the uniqueness of $p_{n-1}(f; x)$. The proof is completed by setting f to be the hat function $\hat{B}_k(x)$ (see Definition 7.23) for each x_k . $\square$

9.2 Newton–Cotes formulas

Definition 9.9. A *Newton–Cotes formula* is a formula (9.1) based on approximating $f(x)$ by interpolating it on uniformly spaced nodes $x_1, \dots, x_n \in [a, b]$.

Definition 9.10. The *trapezoidal rule* is a formula (9.1) based on approximating $f(x)$ by the straight line that connects the points $(a, f(a))^\top$ and $(b, f(b))^\top$. In particular, for $\rho(x) \equiv 1$, it is simply

$$I^T(f) = \frac{b-a}{2} [f(a) + f(b)]. \quad (9.10)$$

Example 9.11. Derive the trapezoidal rule for the weight function $\rho(x) = x^{-1/2}$ on the interval $[0, 1]$. Note that one cannot apply (9.10) to $\rho(x)f(x)$ because $\rho(0) = \infty$. (9.8) yields

$$\begin{aligned} w_1 &= \int_0^1 x^{-1/2} (1-x) dx = \frac{4}{3}, \\ w_2 &= \int_0^1 x^{-1/2} x dx = \frac{2}{3}. \end{aligned}$$

Hence the formula is

$$I^T(f) = \frac{2}{3} [2f(0) + f(1)]. \quad (9.11)$$

Theorem 9.12. For $f \in \mathcal{C}^2[a, b]$ with weight function $\rho(x) \equiv 1$, the remainder of the trapezoidal rule satisfies

$$\exists \zeta \in (a, b) \text{ s.t. } E^T(f) = -\frac{(b-a)^3}{12} f''(\zeta). \quad (9.12)$$

Proof. By Theorem 6.5, the interpolating polynomial $p_1(f; x)$ is unique. Then we have

$$\begin{aligned} E^T(f) &= - \int_a^b \frac{f''(\xi(x))}{2} (x-a)(b-x) dx \\ &= -\frac{f''(\zeta)}{2} \int_a^b (x-a)(b-x) dx = -\frac{(b-a)^3}{12} f''(\zeta), \end{aligned}$$

where the first step follows from Theorem 6.7 and the second step from the integral mean value theorem (Theorem B.124). Here we can apply Theorem B.124 because

$$w(x) = (x-a)(b-x)$$

is always positive on (a, b) . Also note that ξ is a function of x while ζ is a constant depending only on f , a , and b . $\square$

Definition 9.13. The *midpoint rule* is a formula (9.1) based on approximating $f(x)$ by the constant $f(\frac{a+b}{2})$. In particular, for $\rho(x) \equiv 1$, it is simply

$$I^M(f) = (b-a) f\left(\frac{b+a}{2}\right). \quad (9.13)$$

Definition 9.14. *Simpson's rule* is a formula (9.1) based on approximating $f(x)$ by a quadratic polynomial that goes through the points $(a, f(a))^\top$, $(b, f(b))^\top$, and $(\frac{a+b}{2}, f(\frac{a+b}{2}))^\top$. For $\rho(x) \equiv 1$, it is simply

$$I^S(f) = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]. \quad (9.14)$$

Theorem 9.15. For $f \in \mathcal{C}^4[a, b]$ with weight $\rho(x) \equiv 1$, the remainder of Simpson's rule satisfies

$$\exists \zeta \in (a, b) \text{ s.t. } E^S(f) = -\frac{(b-a)^5}{2880} f^{(4)}(\zeta). \quad (9.15)$$

Proof. It is difficult to imitate the proof of Theorem 9.12, since $(x-a)(x-b)(x-\frac{a+b}{2})$ changes sign over $[a, b]$ and the integral mean value theorem is not applicable. To overcome this difficulty, we can formulate the interpolation via a Hermite problem so that Theorem B.124 can be applied. See problem I in Section 9.6 for the main steps. $\square$

Example 9.16. Consider the integral

$$I = \int_{-4}^4 \frac{dx}{1+x^2} = 2 \tan^{-1}(4) = 2.6516 \dots \quad (9.16)$$

Let $n-1$ be the number of sub-intervals that partition $[a, b]$ in Definition 9.9. As shown below, the Newton–Cotes formula appears to be non-convergent.

$n-1$	2	4	6	8	10
I_{n-1}	5.4902	2.2776	3.3288	1.9411	3.5956

For equally spaced nodes, the interpolating polynomials have wilder and wilder oscillations as the degree increases. Consequently, condition (b) of Theorem 9.5 does not hold. Hence Newton–Cotes formulas are not convergent even for well-behaved functions in $\mathcal{C}[a, b]$. In practice, Newton–Cotes formula with $n > 8$ is seldom used.

9.3 Composite formulas

Definition 9.17. The *composite trapezoidal rule* for approximating $I(f)$ in (9.2) with $\rho(x) \equiv 1$ is

$$I_n^T(f) = \frac{h}{2}f(x_0) + h \sum_{k=1}^{n-1} f(x_k) + \frac{h}{2}f(x_n), \quad (9.17)$$

where $h = \frac{b-a}{n}$ and $x_k = a + kh$.

Theorem 9.18. For $f \in C^2[a, b]$, the remainder of the composite trapezoidal rule satisfies

$$\exists \xi \in (a, b) \text{ s.t. } E_n^T(f) = -\frac{b-a}{12}h^2 f''(\xi). \quad (9.18)$$

Proof. Apply Theorem 9.12 to the subintervals, sum up the errors, and we have

$$E_n^T(f) = -\frac{b-a}{12}h^2 \left[\frac{1}{n} \sum_{k=0}^{n-1} f''(\xi_k) \right]. \quad (9.19)$$

The proof is completed by (9.19), the intermediate value Theorem B.74, and the fact $f \in C^2[a, b] \implies f'' \in C[a, b]$. $\square$

Definition 9.19. The *composite Simpson's rule* for approximating $I(f)$ in (9.2) with $\rho(x) \equiv 1$ is

$$I_n^S(f) = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \cdots + 4f(x_{n-1}) + f(x_n)], \quad (9.20)$$

where $h = \frac{b-a}{n}$, $x_k = a + kh$, and n is even.

Theorem 9.20. For $f \in C^4[a, b]$ and $n \in 2\mathbb{N}^+$, the remainder of the composite Simpson's rule satisfies

$$\exists \xi \in (a, b) \text{ s.t. } E_n^S(f) = -\frac{b-a}{180}h^4 f^{(4)}(\xi). \quad (9.21)$$

Proof. Exercise. $\square$

Lemma 9.21. The trapezoidal rule satisfies

$$\forall f \in \mathbb{T}_{n-1}[0, 2\pi], \quad E_n^T(f) = 0, \quad (9.22)$$

where $\mathbb{T}_n[0, 2\pi]$ is the class of trigonometric polynomials of degree at most n ,

$$\mathbb{T}_n[0, 2\pi] := \text{span}\{1, \cos x, \sin x, \dots, \cos(nx), \sin(nx)\}.$$

Proof. It suffices to verify that (9.22) holds for the complex exponential $e_m(x) := e^{imx} = \cos mx + i \sin mx$, $m \in \mathbb{N}$, i.e.

$$\begin{aligned} E_n^T(e_m) &= \int_0^{2\pi} e_m(x) dx \\ &\quad - \frac{2\pi}{n} \left[\frac{e_m(0) + e_m(2\pi)}{2} + \sum_{k=1}^{n-1} e_m\left(\frac{2k\pi}{n}\right) \right] \\ &= \int_0^{2\pi} e^{imx} dx - \frac{2\pi}{n} \sum_{k=0}^{n-1} e^{imk \cdot 2\pi/n}. \end{aligned}$$

Since $\int_0^{2\pi} e^{imx} dx = (im)^{-1} \cdot e^{imx} \Big|_0^{2\pi} = 0$ for $m > 0$, the geometric series yields

$$E_n^T(e_m) = \begin{cases} 0 & \text{if } m = 0; \\ -2\pi & \text{if } m = 0 \pmod{n}, m > 0; \\ -\frac{2\pi}{n} \frac{1 - e^{imn \cdot 2\pi/n}}{1 - e^{im \cdot 2\pi/n}} = 0 & \text{if } m \neq 0 \pmod{n}. \end{cases} \quad (9.23)$$

Hence (9.22) holds as $E_n^T(e_m) = 0$ for $m = 0, \dots, n-1$. $\square$

9.4 Gauss formulas

Lemma 9.22. Let $n, m \in \mathbb{N}^+$ and $m \leq n$. Given polynomials $p = \sum_{i=0}^{n+m} p_i x^i \in \mathbb{P}_{n+m}$ and $s = \sum_{i=0}^n s_i x^i \in \mathbb{P}_n$ satisfying $p_{n+m} \neq 0$ and $s_n \neq 0$, there exist unique polynomials $q \in \mathbb{P}_m$ and $r \in \mathbb{P}_{n-1}$ such that

$$p = qs + r. \quad (9.24)$$

Proof. Rewrite (9.24) as

$$\sum_{i=0}^{n+m} p_i x^i = \left(\sum_{i=0}^m q_i x^i \right) \left(\sum_{i=0}^n s_i x^i \right) + \sum_{i=0}^{n-1} r_i x^i. \quad (9.25)$$

Since monomials are linearly independent, (9.25) consists of $n+m+1$ equations, the last $m+1$ of which are

$$\begin{aligned} p_{n+m} &= q_m s_n, \\ p_{n+m-1} &= q_m s_{n-1} + q_{m-1} s_n, \\ &\vdots \\ p_n &= q_m s_{n-m} + \cdots + q_0 s_n, \end{aligned}$$

which can be written as $S\mathbf{q} = \mathbf{p}$ with S being a lower triangular matrix whose diagonal entries are $s_n \neq 0$. The coefficient vector $\mathbf{q}$ can be determined uniquely from coefficients of p and s . Then r can be determined uniquely by $p - qs$ from (9.25). $\square$

Definition 9.23. The *node polynomial* associated with the nodes x_k 's of a weighted quadrature formula is

$$v_n(x) = \prod_{k=1}^n (x - x_k). \quad (9.26)$$

Theorem 9.24. An *interpolatory formula*, i.e. a quadrature formula (9.1) with $d_E \geq n-1$, can be improved to have $d_E \geq n+j-1$ where $j = 1, 2, \dots, n$ by and only by imposing the additional conditions on its node polynomial and weight function:

$$\forall p \in \mathbb{P}_{j-1}, \quad \int_a^b v_n(x) p(x) \rho(x) dx = 0. \quad (9.27)$$

Proof. For the necessity, we have

$$\int_a^b v_n(x) p(x) \rho(x) dx = \sum_{k=1}^n w_k v_n(x_k) p(x_k) = 0,$$

where the first step follows from the facts $d_E \geq n+j-1$ and $v_n(x)p(x) \in \mathbb{P}_{n+j-1}$, and the second step from (9.26).

To prove the sufficiency, we must show that $E_n(p) = 0$ for any $p \in \mathbb{P}_{n+j-1}$. Lemma 9.22 yields

$$\forall p \in \mathbb{P}_{n+j-1}, \exists! q \in \mathbb{P}_{j-1}, \exists! r \in \mathbb{P}_{n-1}, \text{ s.t. } p = qv_n + r. \quad (9.28)$$

Consequently, we have

$$\begin{aligned} \int_a^b p(x)\rho(x)dx &= \int_a^b q(x)v_n(x)\rho(x)dx + \int_a^b r(x)\rho(x)dx \\ &= \int_a^b r(x)\rho(x)dx = \sum_{k=1}^n w_k r(x_k) \\ &= \sum_{k=1}^n w_k [p(x_k) - q(x_k)v_n(x_k)] = \sum_{k=1}^n w_k p(x_k), \end{aligned}$$

where the first step follows from (9.28), the second from (9.27), the third from the condition of $d_E \geq n-1$, the fourth from (9.28), and the last from (9.26). $\square$

Definition 9.25. A *Gaussian quadrature formula* (or simply a *Gauss formula*) is an interpolatory formula that satisfies (9.27) for $j = n$ and some node polynomial v_n in (9.26); the nodes of the formula are exactly the roots of $v_n(x)$.

Corollary 9.26. A Gauss formula has $d_E = 2n-1$.

Proof. The index j in (9.27) cannot be $n+1$ because the node polynomial $v_n(x) \in \mathbb{P}_n$ cannot be orthogonal to itself. Therefore we know that $j = n$ in Theorem 9.24 is optimal: the formula (9.1) achieves the highest degree of exactness $2n-1$. From an algebraic viewpoint, the $2n$ degrees of freedom of nodes and weights in (9.1) determine a polynomial of degree at most $2n-1$. The proof is completed by Theorem 9.24 and Definition 9.6. $\square$

Corollary 9.27. Weights of a Gauss formula $I_n(f)$ are

$$\forall k = 1, \dots, n, \quad w_k = \int_a^b \frac{v_n(x)}{(x-x_k)v'_n(x_k)} \rho(x)dx, \quad (9.29)$$

where $v_n(x)$ is the node polynomial that defines $I_n(f)$.

Proof. This follows from Lemma 9.8; also see (6.11). $\square$

Example 9.28. Derive the Gauss formula of $n = 2$ for the weight function $\rho(x) = x^{-1/2}$ on the interval $[0, 1]$.

We first construct an orthogonal polynomial

$$\pi(x) = c_0 - c_1x + x^2$$

such that

$$\forall p \in \mathbb{P}_1, \quad \langle p(x), \pi(x) \rangle := \int_0^1 p(x)\pi(x)\rho(x)dx = 0,$$

which is equivalent to $\langle 1, \pi(x) \rangle = 0$ and $\langle x, \pi(x) \rangle = 0$ because $\mathbb{P}_1 = \text{span}\{1, x\}$. These two conditions yield

$$\begin{aligned} \int_0^1 (c_0 - c_1x + x^2)x^{-1/2}dx &= \frac{2}{5} + 2c_0 - \frac{2}{3}c_1 = 0, \\ \int_0^1 x(c_0 - c_1x + x^2)x^{-1/2}dx &= \frac{2}{7} + \frac{2}{3}c_0 - \frac{2}{5}c_1 = 0. \end{aligned}$$

Hence $c_1 = \frac{6}{7}$, $c_0 = \frac{3}{35}$, and the orthogonal polynomial is

$$\pi(x) = \frac{3}{35} - \frac{6}{7}x + x^2$$

with its zeros at

$$x_1 = \frac{1}{7} \left(3 - 2\sqrt{\frac{6}{5}} \right), \quad x_2 = \frac{1}{7} \left(3 + 2\sqrt{\frac{6}{5}} \right).$$

To calculate w_1 and w_2 , we could again use (9.8), but it is simpler to set up a linear system of equations by exploiting Corollary 9.26, i.e. Gauss quadrature is exact for all constants and linear polynomials,

$$\begin{aligned} w_1 + w_2 &= \int_0^1 x^{-1/2}dx = 2, \\ x_1w_1 + x_2w_2 &= \int_0^1 xx^{-1/2}dx = \frac{2}{3}, \end{aligned}$$

which yields

$$\begin{aligned} w_1 &= \frac{-2x_2 + \frac{2}{3}}{x_1 - x_2} = 1 + \frac{1}{3}\sqrt{\frac{5}{6}}, \\ w_2 &= \frac{2x_1 - \frac{2}{3}}{x_1 - x_2} = 1 - \frac{1}{3}\sqrt{\frac{5}{6}}. \end{aligned}$$

The desired two-point Gauss formula is thus

$$\begin{aligned} I_2^G(f) &= \left(1 + \frac{1}{3}\sqrt{\frac{5}{6}} \right) f \left(\frac{3}{7} - \frac{2}{7}\sqrt{\frac{6}{5}} \right) \\ &\quad + \left(1 - \frac{1}{3}\sqrt{\frac{5}{6}} \right) f \left(\frac{3}{7} + \frac{2}{7}\sqrt{\frac{6}{5}} \right). \end{aligned} \quad (9.30)$$

The degree of exactness of the trapezoidal rule is 1 while that of the two-point Gauss formula is 3. Hence we expect that the Gauss formula be much more accurate. Indeed, calculate errors of the two formulas (9.11) and (9.30) for $f(x) = \cos(\frac{1}{2}\pi x)$ and we have

$$\begin{aligned} E^T &= 0.226453\dots, \\ E_2^G &= 0.002197\dots, \end{aligned}$$

which can be verified by simple calculations.

Corollary 9.29. All nodes of a Gauss formula are real, distinct, and contained in (a, b) .

Proof. This follows directly from Definition 9.25 and Theorem 8.12. $\square$

Lemma 9.30. Gauss formulas have positive weights.

Proof. For each $k = 1, 2, \dots, n$, the definition of $\ell_k(x)$ in (9.9) implies $\ell_k^2 \in \mathbb{P}_{2n-2}$, then we have

$$w_k = \sum_{j=1}^n w_j \ell_k^2(x_j) = \int_a^b \rho(x) \ell_k^2(x) dx > 0,$$

where the first step follows from (9.9), second step from $d_E = 2n-1$ and the last step from the conditions on ρ . $\square$

Lemma 9.31. A Gauss formula satisfies

$$\sum_{k=1}^n w_k = \mu_0 \in (0, +\infty).$$

Proof. This follows from setting $j = 0$ in (9.3) and applying the condition on ρ in Definition 9.1. $\square$

Theorem 9.32. Gauss formulas are convergent for $\mathcal{C}[a, b]$.

Proof. Denote by $\mathbb{P}$ the set of real polynomials. Theorem 5.38 states that $\mathbb{P}$ is dense in $\mathcal{C}[a, b]$, i.e. condition (a) in Theorem 9.5 holds. Condition (b) also holds because of Lemmas 9.31 and 9.30. Then the proof is completed by Theorem 9.5. $\square$

Theorem 9.33. For $f \in \mathcal{C}^{2n}[a, b]$, the remainder of a Gauss formula $I_n(f)$ satisfies

$$\exists \xi \in (a, b) \text{ s.t. } E_n^G(f) = \frac{f^{(2n)}(\xi)}{(2n)!} \int_a^b \rho(x) v_n^2(x) dx, \quad (9.31)$$

where v_n is the node polynomial that defines I_n .

Proof. One proof is suggested in 1885 by Markov, a student of Chebyshev and famous for his work in probability theory on certain random processes now known as Markov chains. See exercise IV in Section 9.6. $\square$

9.5 Numerical differentiation

Formula 9.34 (The method of undetermined coefficients). A general method to derive FD formulas that approximate $u^{(k)}(\bar{x})$ is based on an arbitrary stencil of $n > k$ distinct points $x_1, x_2, \dots, x_n$. Taylor expansions of u at each point x_i in the stencil about $u(\bar{x})$ yield

$$u(x_i) = u(\bar{x}) + (x_i - \bar{x})u'(\bar{x}) + \dots + \frac{1}{k!}(x_i - \bar{x})^k u^{(k)}(\bar{x}) + \dots$$

for $i = 1, 2, \dots, n$. This leads to a linear combination of point values that approximates $u^{(k)}(\bar{x})$,

$$u^{(k)}(\bar{x}) = c_1 u(x_1) + c_2 u(x_2) + \dots + c_n u(x_n) + O(h^p),$$

where the c_j 's are chosen to make p as large as possible:

$$\forall i = 0, \dots, p-1, \quad \frac{1}{i!} \sum_{j=1}^n c_j (x_j - \bar{x})^i = \begin{cases} 1 & \text{if } i = k, \\ 0 & \text{otherwise.} \end{cases} \quad (9.32)$$

Example 9.35. To approximate $u'(\bar{x})$ with an FD formula

$$D_2 u(\bar{x}) = a u(\bar{x}) + b u(\bar{x} - h) + c u(\bar{x} - 2h), \quad (9.33)$$

we determine the coefficients a , b , and c to give the best possible accuracy. Taylor expansions at $\bar{x}$ yield

$$\begin{aligned} D_2 u(\bar{x}) &= (a + b + c)u(\bar{x}) - (b + 2c)h u'(\bar{x}) \\ &\quad + \frac{1}{2}(b + 4c)h^2 u''(\bar{x}) - \frac{1}{6}(b + 8c)h^3 u'''(\bar{x}) \\ &\quad + O(h^4). \end{aligned}$$

Set $a + b + c = 0$, $b + 2c = -\frac{1}{h}$, and $b + 4c = 0$, solve

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{h} \\ 0 \end{bmatrix}, \quad (9.34)$$

and we get

$$a = \frac{3}{2h}, \quad b = -\frac{2}{h}, \quad c = \frac{1}{2h}. \quad (9.35)$$

Therefore the FD formula is determined as

$$D_2 u(\bar{x}) = \frac{1}{2h} [3u(\bar{x}) - 4u(\bar{x} - h) + u(\bar{x} - 2h)]. \quad (9.36)$$

Definition 9.36. In approximating the derivative of a smooth function, an FD formula is p -th order accurate if its error E has the form

$$E(h) = \Theta(h^p), \text{ as } h \rightarrow 0, \quad (9.37)$$

where h is the maximum distance of adjacent points in the stencil.

Example 9.37. Consider approximating $u'(x)$ at a point $\bar{x}$ using the nearby function values $u(\bar{x} \pm h)$. Three commonly used formulas are

$$D_+ u(\bar{x}) := \frac{u(\bar{x} + h) - u(\bar{x})}{h}, \quad (9.38)$$

$$D_- u(\bar{x}) := \frac{u(\bar{x}) - u(\bar{x} - h)}{h}, \quad (9.39)$$

$$D_0 u(\bar{x}) := \frac{u(\bar{x} + h) - u(\bar{x} - h)}{2h} = \frac{1}{2}(D_+ + D_-)u(\bar{x}). \quad (9.40)$$

For $u(x) = \sin(x)$ and $\bar{x} = 1$, we calculate the errors of the above three formulas in approximating $u'(1) = \cos(1) \approx 0.5403023$ with $h = 0.01$ and 0.005 .

Define the error as $E := Du(\bar{x}) - u'(\bar{x})$ and the following table shows the values of E for the three formulas.

h	$D_+ u(\bar{x})$	$D_- u(\bar{x})$	$D_0 u(\bar{x})$
$h_1 = 1.0\text{e-}2$	-4.2e-3	-4.2e-3	-9.00e-6
$h_2 = 5.0\text{e-}3$	-2.1e-3	-2.1e-3	-2.25e-6
$p = \log_2 \frac{E(Du(\bar{x}, h_1))}{E(Du(\bar{x}, h_2))}$	1	1	2

The last row of the table shows that the error behaves like $E(h) \approx Ch^p$, which means reduction of h by a factor of r leads to error reduction by a factor of r^p . See Figure 9.1.

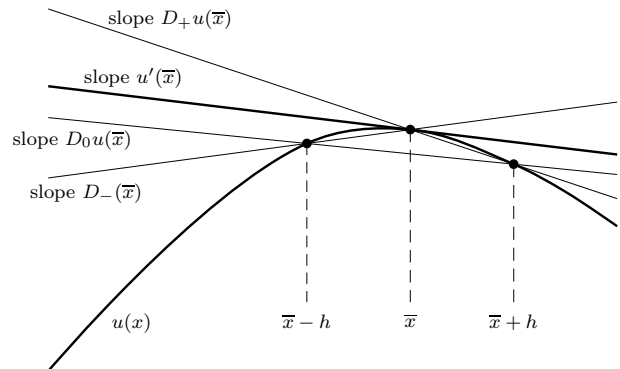


Figure 9.1: Geometric illustrations of low-order FD formulas.

Exercise 9.38. Show that the FD formulas $D_+u(\bar{x})$ and $D_-u(\bar{x})$ are first-order accurate while $D_0u(\bar{x})$ is second-order accurate.

Exercise 9.39. Construct a table of divided difference (as in Definition 6.18) to derive a quadratic polynomial that agrees with $u(x)$ at $\bar{x}$, $\bar{x} - h$, and $\bar{x} - 2h$. Then take derivative of this polynomial to obtain the FD formula (9.36).

Lemma 9.40. In approximating the second derivative of $u \in C^4(\mathbb{R})$, the formula

$$D^2u(\bar{x}) = \frac{u(\bar{x} - h) - 2u(\bar{x}) + u(\bar{x} + h)}{h^2} \quad (9.41)$$

is second-order accurate. Furthermore, if the input function values $u(\bar{x} - h)$, $u(\bar{x})$, and $u(\bar{x} + h)$ are perturbed with random errors $\epsilon \in [-E, E]$, then there exists $\xi \in (\bar{x} - h, \bar{x} + h)$ such that

$$|u''(\bar{x}) - D^2u(\bar{x})| \leq \frac{h^2}{12}|u^{(4)}(\xi)| + \frac{4E}{h^2}. \quad (9.42)$$

9.6 Theoretical problems

I. Simpson's rule.

- (a) Show that Simpson's rule on $[-1, 1]$ with $\rho = 1$ can be obtained by

$$\int_{-1}^1 y(t)dt = \int_{-1}^1 p_3(y; -1, 0, 0, 1; t)dt + E^S(y),$$

where $y \in C^4[-1, 1]$ and $p_3(y; -1, 0, 0, 1; t)$ is the interpolation polynomial of y that satisfies $p_3(-1) = y(-1)$, $p_3(0) = y(0)$, $p_3'(0) = y'(0)$, and $p_3(1) = y(1)$.

- (b) Derive $E^S(y)$.
 (c) Using (a), (b), and a change of variable, derive the composite Simpson's rule and prove the theorem on its error estimation.

II. Estimate the number of subintervals required to approximate $\int_0^1 e^{-x^2} dx$ to six correct decimal places, i.e. the absolute error is less than 0.5×10^{-6} ,

- (a) by the composite trapezoidal rule,
 (b) by the composite Simpson's rule.

III. Gauss-Laguerre quadrature formula.

- (a) Construct a polynomial $\pi_2(t) = t^2 + at + b$ that is orthogonal to $\mathbb{P}_1$ with respect to the weight function $\rho(t) = e^{-t}$, i.e.

$$\forall p \in \mathbb{P}_1, \quad \int_0^{+\infty} p(t)\pi_2(t)\rho(t)dt = 0.$$

(hint: $\int_0^{+\infty} t^m e^{-t} dt = m!$)

- (b) Derive the two-point Gauss-Laguerre quadrature formula

$$\int_0^{+\infty} f(t)e^{-t}dt = w_1f(t_1) + w_2f(t_2) + E_2(f)$$

and express $E_2(f)$ in terms of $f^{(4)}(\tau)$ for some $\tau > 0$.

- (c) Apply the formula in (b) to approximate

$$I = \int_0^{+\infty} \frac{1}{1+t} e^{-t} dt.$$

Use the remainder to estimate the error and compare your estimate with the true error. With the true error, identify the unknown quantity τ contained in $E_2(f)$.

(hint: use the exact value $I = 0.596347361 \dots$)

IV. Remainder of Gauss formulas. Consider the Hermite interpolation problem: find $p \in \mathbb{P}_{2n-1}$ such that

$$\forall m = 1, 2, \dots, n, \quad p(x_m) = f_m, \quad p'(x_m) = f'_m. \quad (9.43)$$

There are *elementary Hermite interpolation polynomials* h_m, q_m such that the solution of (9.43) can be expressed in the form

$$p(t) = \sum_{m=1}^n [h_m(t)f_m + q_m(t)f'_m],$$

analogous to the Lagrange interpolation formula.

- (a) Seek h_m and q_m in the form

$$h_m(t) = (a_m + b_mt)\ell_m^2(t), \quad q_m(t) = (c_m + d_mt)\ell_m^2(t)$$

where ℓ_m is the elementary Lagrange polynomial in (6.9). Determine the constants a_m, b_m, c_m, d_m .

- (b) Obtain the quadrature rule

$$I_n(f) = \sum_{k=1}^n [w_k f(x_k) + \mu_k f'(x_k)]$$

that satisfies $E_n(p) = 0$ for all $p \in \mathbb{P}_{2n-1}$.

- (c) What conditions on the node polynomial or on the nodes x_k must be imposed so that $\mu_k = 0$ for each $k = 1, 2, \dots, n$?

V. Prove Lemma 9.40. How do you choose h to minimize the error bound in (9.42)? Design a fourth-order accurate formula based on a symmetric stencil, derive its error bound, and minimize the error bound. What do you observe in comparing the second-order case and the fourth-order case?

9.7 Programming assignments

Chapter 10

Finite Difference Methods for Boundary Value Problems (BVPs)

Motivation 10.1. In boundary value problems (BVPs), the steady state of the interior – such as the temperature of a metal plate or the stress on a bridge beam – is dictated by conditions at the boundaries. BVPs are the backbone of engineering and physics. They allow us to predict how sound waves travel through a room, how heat flows through an engine block, or how electromagnetic fields distribute around an antenna. Without the ability to define and solve these problems, it would be impossible to design safe infrastructure, efficient electronics, or advanced medical imaging tech. They turn abstract physical laws into predictive tools that ensure our designs can withstand the specific pressures and constraints of the real world.

Definition 10.2. A *partial differential equation* (PDE) is an equation involving an unknown function of two or more variables and some of its partial derivatives.

Definition 10.3. *Laplace equation* is a second-order PDE of the form

$$\Delta u(\mathbf{x}) = 0, \quad (10.1)$$

where the unknown is a function $u : \bar{\Omega} \rightarrow \mathbb{R}$, $\bar{\Omega}$ is the closure of an open set $\Omega \subset \mathbb{R}^n$, and the *Laplacian operator* $\Delta : \mathcal{C}^2(\Omega) \rightarrow \mathcal{C}(\Omega)$ is

$$\Delta := \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}. \quad (10.2)$$

Example 10.4. *Potential flow* is a special type of flow where the velocity field $\mathbf{u}$ can be expressed as the gradient of a scalar function:

$$\mathbf{u} = \nabla \varphi,$$

where φ is called the *velocity potential*. For incompressible fluids with $\nabla \cdot \mathbf{u} = 0$, the velocity potential satisfies a Laplace equation $\Delta \varphi = 0$.

Definition 10.5. *Poisson's equation* is a second-order PDE of the form

$$-\Delta u(\mathbf{x}) = f(\mathbf{x}), \quad (10.3)$$

where the unknown is a function $u : \bar{\Omega} \rightarrow \mathbb{R}$, $\bar{\Omega}$ is the closure of an open set $\Omega \subset \mathbb{R}^n$, and the RHS function $f : \Omega \rightarrow \mathbb{R}$ is given a priori.

Definition 10.6. A *boundary value problem* (BVP) is a differential equation together with a set of additional constraints, called the *boundary conditions*, that hold only on the domain boundary.

Definition 10.7. Common types of boundary conditions for a one-dimensional interval $\Omega = (a, b)$ are

- *Dirichlet conditions*: $u(a) = \alpha$ and $u(b) = \beta$;
- *Mixed conditions*: $u(a) = \alpha$ and $\frac{\partial u}{\partial x} \Big|_b = \beta$;
- *Neumann conditions*: $\frac{\partial u}{\partial x} \Big|_a = \alpha$ and $\frac{\partial u}{\partial x} \Big|_b = \beta$.

Theorem 10.8. Suppose f and g are two sufficiently smooth functions. Then there exists a unique solution (up to an additive constant) for the Neumann BVP

$$\Delta \phi = f \quad \text{in } \Omega; \quad (10.4a)$$

$$\mathbf{n} \cdot \nabla \phi = g \quad \text{on } \partial\Omega \quad (10.4b)$$

if and only if

$$\int_{\Omega} f \, dV = \int_{\partial\Omega} g \, dA. \quad (10.5)$$

Proof. See [Taylor, 2011, page 409]. $\square$

Example 10.9. The fundamental theorem of vector calculus states that a continuously differentiable vector field $\mathbf{v}^*$ can be uniquely decomposed into a divergence-free part and a curl-free part:

$$\begin{cases} \mathbf{v}^* &= \mathbf{v} + \nabla \phi, \\ \nabla \cdot \mathbf{v} &= 0, \quad \nabla \times \nabla \phi = \mathbf{0}. \end{cases} \quad (10.6)$$

When a given boundary condition of $\mathbf{v}$ satisfies $\oint_{\partial\Omega} \mathbf{v} \cdot \mathbf{n} = 0$, the decomposition is realized by solving the Neumann BVP

$$\Delta \phi = \nabla \cdot \mathbf{v}^* \quad \text{in } \Omega, \quad (10.7a)$$

$$\mathbf{n} \cdot \nabla \phi = \mathbf{n} \cdot (\mathbf{v}^* - \mathbf{v}) \quad \text{on } \partial\Omega, \quad (10.7b)$$

for which the existence of the unique solution is guaranteed by Theorem 10.8 and

$$\int_{\Omega} \nabla \cdot \mathbf{v}^* \, dV = \int_{\Omega} \nabla \cdot (\mathbf{v}^* - \mathbf{v}) \, dV = \int_{\partial\Omega} \mathbf{n} \cdot (\mathbf{v}^* - \mathbf{v}) \, dA.$$

Roadmap 10.10. In this chapter, we focus on finite difference methods for solving BVPs. We present the algorithmic steps for solving one-dimensional BVPs with Dirichlet boundary conditions in Section 10.1 and define in Section 10.2 elementary concepts such as grid functions, local truncation errors, and global solution errors. In Section 10.3, we show that the method presented in Example 10.12 is second-order convergent both in the 2-norm and in the ∞ -norm, illustrating the fundamental pattern that convergence is equivalent to consistency and stability. The method and its analysis are generalized to Neumann and mixed conditions in Section 10.4 and to two-dimensional regular and irregular domains in Section 10.5.

10.1 The finite difference (FD) discretization

Formula 10.11. In solving a linear BVP, the general procedures of an FD method are as follows.

- (FD-1) Discretize the problem domain by a grid.
- (FD-2) Approximate each spatial derivative in the PDE with some finite difference formula at every grid point to get a system of linear equations $A\mathbf{U} = \mathbf{F}$ where the vector $\mathbf{U}$ approximates the unknown variable on the grid while the vector $\mathbf{F}$ contains given conditions of the BVP such as boundary conditions and derivatives of the unknown function.
- (FD-3) Solve the system of algebraic equations.

Example 10.12 (An FD method for Poisson's equation in a unit interval). Consider the one-dimensional BVP

$$-u''(x) = f(x) \text{ in } \Omega := (0, 1) \quad (10.8)$$

with Dirichlet boundary conditions

$$u(0) = \alpha, \quad u(1) = \beta. \quad (10.9)$$

The general procedures of an FD method based on the central difference are as follows.

- (a) Discretize Ω by a Cartesian grid with uniform spacing,

$$x_j = jh, \quad h = \frac{1}{m+1}, \quad j = 0, 1, \dots, m+1.$$

Set $U_0 = \alpha$, $U_{m+1} = \beta$ and we will compute m values $U_1, \dots, U_m$ where each U_j approximates $u(x_j)$.

- (b) Approximate the second derivative u'' with a centered difference

$$u''(x_j) = \frac{u(x_{j-1}) - 2u(x_j) + u(x_{j+1}))}{h^2} + O(h^2) \quad (10.10)$$

and we get the following system of linear equations:

$$\begin{aligned} -\frac{\alpha - 2U_1 + U_2}{h^2} &= f(x_1), \\ -\frac{U_{j-1} - 2U_j + U_{j+1}}{h^2} &= f(x_j), \quad j = 2, \dots, m-1, \\ -\frac{U_{m-1} - 2U_m + \beta}{h^2} &= f(x_m). \end{aligned}$$

These equations are written in the form

$$A\mathbf{U} = \mathbf{F}, \quad (10.11)$$

where

$$\mathbf{U} = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ \vdots \\ U_{m-1} \\ U_m \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} f(x_1) + \frac{\alpha}{h^2} \\ f(x_2) \\ f(x_3) \\ \vdots \\ f(x_{m-1}) \\ f(x_m) + \frac{\beta}{h^2} \end{bmatrix}, \quad (10.12)$$

$$A = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}. \quad (10.13)$$

- (c) Solve the linear system (10.11).

10.2 Errors and consistency

Definition 10.13. The *global error* or *solution error* of an FD method in Formula 10.11 is

$$\mathbf{E} = \mathbf{U} - \hat{\mathbf{U}}, \quad (10.14)$$

where $\hat{\mathbf{U}} = [u(x_1), u(x_2), \dots, u(x_m)]^\top$ is the vector of true values and $\mathbf{U}$ the computed solution.

Definition 10.14. A *grid function* is a function $\mathbf{g} : \mathbf{X} \rightarrow \mathbb{R}$ on a discrete grid $\mathbf{X}$ that contains a finite number of points.

Definition 10.15. The L_q -norm of a grid function $\mathbf{g}$ on a one-dimensional grid $\mathbf{X} := \{x_1, x_2, \dots, x_N\}$ is

$$\|\mathbf{g}\|_{L_q} = \left(h \sum_{i=1}^N |g_i|^q \right)^{\frac{1}{q}}, \quad (10.15)$$

where $\mathbf{g} = (g_1, g_2, \dots, g_N)$. In particular, the L_1 -norm is

$$\|\mathbf{g}\|_{L_1} = h \sum_{i=1}^N |g_i| \quad (10.16)$$

and the L_∞ -norm is

$$\|\mathbf{g}\|_{L_\infty} = \max_{1 \leq i \leq N} |g_i|. \quad (10.17)$$

When there is no danger of ambiguity, we also write $\|\mathbf{g}\|_q = \|\mathbf{g}\|_{L_q}$.

Exercise 10.16. Suppose a grid function $\mathbf{g} : \mathbf{X} \rightarrow \mathbb{R}$ has $\mathbf{X} := \{x_1, x_2, \dots, x_N\}$, $g_1 = O(h)$, $g_N = O(h)$, and $g_j = O(h^2)$ for all $j = 2, \dots, N-1$. Show that

$$\|\mathbf{g}\|_{L_\infty} = O(h), \quad \|\mathbf{g}\|_{L_1} = O(h^2), \quad \|\mathbf{g}\|_{L_2} = O(h^{\frac{3}{2}}). \quad (10.18)$$

As the main point of this exercise, the differences in the max-norm, 1-norm, and 2-norm of a grid function often reveal the percentage of components with large magnitude.

Definition 10.17. The *local truncation error* (LTE) of an FD method in Formula 10.11 is the error caused by replacing a continuous derivative with an FD formula.

Example 10.18. When we approximate Δu with

$$D^2 u(x_j) := \frac{u(x_{j-1}) - 2u(x_j) + u(x_{j+1}))}{h^2}, \quad (10.19)$$

the LTE of the FD method in Example 10.12 is

$$\tau_j = -D^2 u(x_j) - (-u''(x_j)) = -\frac{h^2}{12} u''''(x_j) + O(h^4).$$

Lemma 10.19. Let $A\mathbf{U} = \mathbf{F}$ be the linear system obtained by applying Formula 10.11 to a linear BVP $\mathcal{L}u = f(x)$ in $(0, 1)$ with Dirichlet conditions. Then the LTE of this FD method is the error of calculating the RHS function $\mathbf{F}$ by replacing U_j with the exact solution $u(x_j)$, i.e.,

$$\boldsymbol{\tau} = A\hat{\mathbf{U}} - \mathbf{F}, \quad (10.20)$$

where $\hat{\mathbf{U}}$ is the vector of true solution values. In particular, this holds for the FD method in Example 10.12.

Proof. We only prove the case in Example 10.12 since other linear BVPs can be proven by similar arguments.

By (10.8) and (10.11), we have, $\forall j = 2, \dots, m-1$,

$$(A\hat{\mathbf{U}} - \mathbf{F})_j = -\frac{u(x_{j-1}) - 2u(x_j) + u(x_{j+1}))}{h^2} + \Delta u(x_j),$$

which also holds for $j = 1$ and $j = m$ since the boundary conditions yield $u(x_0) = u(0) = \alpha$ and $u(x_{m+1}) = u(1) = \beta$. Then the proof is completed by Definition 10.17. $\square$

Lemma 10.20. The LTE and the global error are related as

$$A\mathbf{E} = -\boldsymbol{\tau}. \quad (10.21)$$

Proof. $A\mathbf{E} = A(\mathbf{U} - \hat{\mathbf{U}}) = \mathbf{F} - (\mathbf{F} + \boldsymbol{\tau}) = -\boldsymbol{\tau}$. $\square$

Definition 10.21. An FD method in Formula 10.11 is said to be *consistent* with the BVP if

$$\lim_{h \rightarrow 0} \|\boldsymbol{\tau}^h\|_{L_q} = 0, \quad (10.22)$$

where $\boldsymbol{\tau}^h$ is the LTE.

10.3 Stability and convergence

Definition 10.22. An FD method is *convergent* if

$$\lim_{h \rightarrow 0} \|\mathbf{E}^h\|_{L_q} = 0, \quad (10.23)$$

where $\mathbf{E}^h$ is the solution error in Definition 10.13.

Definition 10.23. An FD method in Formula 10.11 is *stable* if

- (a) $\exists h_0 \in \mathbb{R}^+$ s.t. $\forall h \in (0, h_0)$, $\det(A) \neq 0$, where A is the matrix of the linear system for the grid size h ;
- (b) $\lim_{h \rightarrow 0} \|A^{-1}\| = O(1)$,

where $\|A^{-1}\|$ is the operator norm of A^{-1} .

Theorem 10.24. A consistent and stable FD method is convergent.

Proof. By Definition 10.15, the L_q -norm $\|\cdot\|_{L_q}$ and the corresponding vector norm $\|\cdot\|_q$ are related by $\|\mathbf{E}\|_{L_q} = h^{\frac{1}{q}} \|\mathbf{E}\|_q$. By Lemma 10.20, we have

$$\begin{aligned} \|\mathbf{E}^h\|_{L_q} &= h^{\frac{1}{q}} \|\mathbf{E}^h\|_q = h^{\frac{1}{q}} \|(A^h)^{-1} \boldsymbol{\tau}^h\|_q \\ &\leq \|(A^h)^{-1}\| h^{\frac{1}{q}} \|\boldsymbol{\tau}^h\|_q = \|(A^h)^{-1}\| \|\boldsymbol{\tau}^h\|_{L_q}. \end{aligned}$$

Then (10.23) follows from Definitions 10.23 and 10.21. $\square$

10.3.1 Convergence in the 2-norm

Lemma 10.25. The eigenvalues λ_k and eigenvectors $\mathbf{w}_k$ of the matrix A in (10.13) are

$$\lambda_k(A) = \frac{4}{h^2} \sin^2 \frac{k\pi}{2(m+1)}, \quad (10.24)$$

$$w_{k,j} = \sin \frac{jk\pi}{m+1}, \quad (10.25)$$

where $j, k = 1, 2, \dots, m$.

Proof. It is straightforward to verify the conclusions using the trigonometric identity

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}. \quad \square$$

Exercise 10.26. Show that the set of eigenvectors (10.25) of A in (10.13) is orthogonal, i.e.,

$$\langle \mathbf{w}_i, \mathbf{w}_k \rangle = \begin{cases} 0 & \text{if } i \neq k; \\ \frac{m+1}{2} & \text{if } i = k, \end{cases}$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product.

Theorem 10.27. The FD method in Example 10.12 is second-order convergent in the 2-norm.

Proof. The symmetry of A gives $\|A\|_2 = \rho(A)$. Then Lemma 10.25 implies the stability, i.e., condition (b) in Definition 10.23:

$$\lim_{h \rightarrow 0} \|A^{-1}\|_2 = \lim_{h \rightarrow 0} \frac{1}{\min |\lambda_k(A)|} = \lim_{h \rightarrow 0} \frac{h^2}{4 \sin^2 \frac{\pi h}{2}} = \frac{1}{\pi^2}. \quad (10.26)$$

The rest of the proof follows from Example 10.18, Definition 10.21, and Theorem 10.24. $\square$

10.3.2 Green's function

Definition 10.28. For any fixed $\bar{x} \in [0, 1]$, the *Green's function* $G(x; \bar{x})$ is the function of x that solves the BVP

$$\begin{cases} u''(x) = \delta(x - \bar{x}); \\ u(0) = u(1) = 0, \end{cases} \quad (10.27)$$

where $\delta(x - \bar{x})$ is the Dirac delta function in Definition 8.41.

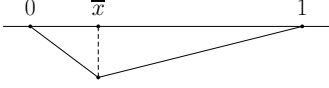


Figure 10.1: The Green function that solves the BVP (10.27).

Lemma 10.29. The Green's function $G(x; \bar{x})$ that solves (10.27) is

$$G(x; \bar{x}) = \begin{cases} (\bar{x} - 1)x, & x \in [0, \bar{x}], \\ \bar{x}(x - 1), & x \in [\bar{x}, 1]. \end{cases} \quad (10.28)$$

Proof. For any fixed ϵ , we have from (10.27) and (8.41b) that

$$\begin{aligned} \int_{x_0 - \epsilon}^{x_0 + \epsilon} G''(x) dx &= \int_{x_0 - \epsilon}^{x_0 + \epsilon} \delta(x - \bar{x}) dx \\ &= \begin{cases} 0, & \bar{x} \notin (x_0 - \epsilon, x_0 + \epsilon), \\ 1, & \bar{x} \in (x_0 - \epsilon, x_0 + \epsilon). \end{cases} \end{aligned}$$

Take limit $\epsilon \rightarrow 0$ of the above equation and we deduce from the second fundamental theorem of calculus (Theorem B.128) that

$$\begin{aligned} &\lim_{\epsilon \rightarrow 0} G'(x_0 + \epsilon) - \lim_{\epsilon \rightarrow 0} G'(x_0 - \epsilon) \\ &= \begin{cases} 0 & \text{if } x_0 \in (0, \bar{x}) \cup (\bar{x}, 1), \\ 1 & \text{if } x_0 = \bar{x}. \end{cases} \end{aligned} \quad (10.29)$$

Substitute

$$G(x; \bar{x}) = \begin{cases} ax + b, & x \in [0, \bar{x}], \\ cx + d, & x \in [\bar{x}, 1] \end{cases}$$

into (10.29) and (10.27) and the continuity of $G(x; \bar{x})$ yields

$$\begin{cases} c = a + 1 \\ b = 0 \\ c + d = 0 \\ a\bar{x} + b = c\bar{x} + d \end{cases} \Rightarrow \begin{cases} a = \bar{x} - 1 \\ b = 0 \\ c = \bar{x} \\ d = -\bar{x} \end{cases},$$

which completes the proof. $\square$

Corollary 10.30. The solution to the linear BVP

$$\begin{cases} u''(x) = c\delta(x - \bar{x}), \\ u(0) = u(1) = 0 \end{cases}$$

is

$$u(x) = cG(x; \bar{x}).$$

Proof. This follows directly from Lemma 10.29. $\square$

10.3.3 Convergence in the max-norm

Lemma 10.31. For the matrix A in (10.13), any element of its inverse $B = A^{-1}$ is

$$b_{ij} = -hG(x_i; x_j) = \begin{cases} -hx_j(x_i - 1), & i \leq j, \\ -hx_j(x_i - 1), & i \geq j. \end{cases} \quad (10.30)$$

More explicitly, the matrix B is

$$B = -h \begin{bmatrix} x_1(x_1 - 1) & x_1(x_2 - 1) & \cdots & x_1(x_m - 1) \\ x_1(x_2 - 1) & x_2(x_2 - 1) & \cdots & x_2(x_m - 1) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(x_m - 1) & x_2(x_m - 1) & \cdots & x_m(x_m - 1) \end{bmatrix}.$$

Proof. To verify that B is indeed the inverse of A , it suffices to multiply the i th row of $h^2 A$ and the j th column of $-\frac{1}{h}B$,

$$\begin{bmatrix} 0, \dots, 0, -1, 2, -1, 0, \dots, 0 \end{bmatrix} \begin{bmatrix} x_1(x_j - 1) \\ \vdots \\ x_{j-1}(x_j - 1) \\ x_j(x_j - 1) \\ x_j(x_{j+1} - 1) \\ \vdots \\ x_j(x_m - 1) \end{bmatrix},$$

the only nonzero case is when $i = j$:

$$2x_j(x_j - 1) - x_{j-1}(x_j - 1) - x_j(x_{j+1} - 1) = -h. \quad \square$$

Theorem 10.32. The max-norm of $B = A^{-1}$ satisfies

$$\|B\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^m |b_{ij}| \leq 1. \quad (10.31)$$

Proof. Lemma 10.31 yields

$$\begin{aligned} \sum_{j=1}^m |b_{ij}| &= \sum_{j=1}^i hx_j |x_i - 1| + \sum_{j=i+1}^m hx_j |x_j - 1| \\ &\leq \sum_{j=1}^i h \left(\frac{m}{m+1} \right)^2 + \sum_{j=i+1}^m h \left(\frac{m}{m+1} \right)^2 \\ &= mh \left(\frac{m}{m+1} \right)^2 = \left(\frac{m}{m+1} \right)^3 \leq 1. \end{aligned} \quad \square$$

Corollary 10.33. The condition number $\text{cond}_\infty(A)$ of the matrix A in (10.13) based on the ∞ -norm is 4.

Proof. This follows directly from Theorem 10.32, Lemma C.282, and Definition 2.22. $\square$

10.3.4 A solution via Green's function

Lemma 10.34. Suppose $\mathcal{L}$ is an invertible linear differential operator that satisfies

$$\mathcal{L}u(x) = f(x). \quad (10.32)$$

Then we have

$$u(x) = \int G(x; \bar{x}) f(\bar{x}) d\bar{x}, \quad (10.33)$$

where G is Green's function satisfying

$$\mathcal{L}G(x; \bar{x}) = \delta(x - \bar{x}). \quad (10.34)$$

Proof. Multiply (10.34) by $f(\bar{x})$, integrate w.r.t. $\bar{x}$, and we have

$$\int \mathcal{L}G(x; \bar{x}) f(\bar{x}) d\bar{x} = \int \delta(x - \bar{x}) f(\bar{x}) d\bar{x} = f(x),$$

where the second equality follows from the sifting property of the Dirac delta function (Lemma 8.43). Therefore

$$\mathcal{L} \int G(x; \bar{x}) f(\bar{x}) d\bar{x} = f(x),$$

which further implies (10.33) since $\mathcal{L}$ is invertible. $\square$

Theorem 10.35. The Dirichlet BVP

$$\begin{cases} u''(x) = f(x) & \text{in } (0, 1), \\ u(0) = \alpha, \quad u(1) = \beta \end{cases} \quad (10.35)$$

is solved by

$$u(x) = \alpha G_0(x) + \beta G_1(x) + \hat{U}(x), \quad (10.36)$$

where $G_0(x)$, $G_1(x)$ and $\hat{U}(x)$ are defined by BVPs as follows

$$\begin{cases} G_0''(x) = 0, \\ G_0(0) = 1, G_0(1) = 0 \end{cases} \Rightarrow G_0(x) = 1 - x, \quad (10.37a)$$

$$\begin{cases} G_1''(x) = 0, \\ G_1(0) = 0, G_1(1) = 1 \end{cases} \Rightarrow G_1(x) = x, \quad (10.37b)$$

$$\begin{cases} \hat{U}''(x) = f(x), \\ \hat{U}(0) = 0, \hat{U}(1) = 0 \end{cases} \Rightarrow \hat{U}(x) = \int_0^1 f(\bar{x}) G(x; \bar{x}) d\bar{x}. \quad (10.37c)$$

Proof. This follows from the linearity of the BVP (10.35). $\square$

Lemma 10.36. In solving the BVP problem (10.35) with the FD method in Example 10.12, an $O(\epsilon)$ perturbation to the RHS function f at any x_j where $j = 1, 2, \dots, m$ results in an $O(\epsilon h)$ error in the solution while an $O(\epsilon)$ perturbation to a Dirichlet condition results in an $O(\epsilon)$ error in the solution.

Proof. We discretize the Dirichlet BVP (10.35) into

$$A_g \mathbf{U}_g = \mathbf{F}_g$$

where $\mathbf{U}$ is the same as that in (10.12) and

$$A_g = \frac{1}{h^2} \begin{bmatrix} h^2 & 0 & & & \\ 1 & -2 & 1 & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & -2 & 1 \\ & & & & 1 & -2 & 1 \\ & & & & & 0 & h^2 \end{bmatrix},$$

$$\mathbf{U}_g = \begin{bmatrix} U_0 \\ \mathbf{U} \\ U_{m+1} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{m-1}) \\ f(x_m) \end{bmatrix}, \quad \mathbf{F}_g = \begin{bmatrix} \alpha \\ \mathbf{F} \\ \beta \end{bmatrix}.$$

To figure out $B_g = A_g^{-1}$ from $B = A^{-1}$, we write

$$B_g = \begin{bmatrix} & \mathbf{0}^\top & \\ \mathbf{b}_{g,0} & B & \mathbf{b}_{g,m+1} \\ & \mathbf{0}^\top & \end{bmatrix}$$

where $\mathbf{b}_{g,0}$ and $\mathbf{b}_{g,m+1}$ are respectively the first and last columns of B_g . Then we have

$$\mathbf{U}_g = \begin{bmatrix} & \mathbf{0}^\top & \\ \mathbf{b}_{g,0} & B & \mathbf{b}_{g,m+1} \\ & \mathbf{0}^\top & \end{bmatrix} \begin{bmatrix} \alpha \\ \mathbf{F} \\ \beta \end{bmatrix} = \alpha \mathbf{b}_{g,0} + \beta \mathbf{b}_{g,m+1} + \begin{bmatrix} 0 \\ B\mathbf{F} \\ 0 \end{bmatrix}.$$

By Theorem 10.35, $\mathbf{b}_{g,0}$ and $\mathbf{b}_{g,m+1}$ come from evaluating G_0 and G_1 on the grid $[x_0, x_1, \dots, x_{m+1}]$. Then the proof is completed by the splitting of $\mathbf{U}_g$ in the above equation and the fact that entries of $\mathbf{b}_{g,0}$ and $\mathbf{b}_{g,m+1}$ are $O(1)$ while those of B are $O(h)$. $\square$

10.4 Other boundary conditions

Example 10.37. Consider the second-order BVP

$$u''(x) = f(x) \quad \text{in } (0, 1) \quad (10.38)$$

with mixed boundary conditions

$$u'(0) = \sigma, \quad u(1) = \beta. \quad (10.39)$$

As the crucial difference between this BVP and the BVP with pure Dirichlet conditions in Example 10.12, the value of $u(x)$ at $x = 0$ becomes an unknown to be solved for.

The first approach is to use a one-sided expression

$$\frac{U_1 - U_0}{h} = \sigma \quad (10.40)$$

to arrive at

$$A_E \mathbf{U}_E = \mathbf{F}_E \quad (10.41)$$

where $\mathbf{U}_E = [U_0, U_1, \dots, U_m, U_{m+1}]^\top$, $\mathbf{F}_E = [\sigma, f(x_1), \dots, f(x_m), \beta]^\top$, and

$$A_E = \frac{1}{h^2} \begin{bmatrix} -h & h & & & \\ 1 & -2 & 1 & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & -2 & 1 \\ & & & & 1 & -2 & 1 \\ & & & & & 0 & h^2 \end{bmatrix}.$$

The LTE at $x_0 = 0$ is

$$\begin{aligned}\tau_0 &= \frac{1}{h^2}(hu(x_1) - hu(x_0)) - \sigma \\ &= u'(x_0) + \frac{1}{2}hu''(x_0) + O(h^2) - \sigma \\ &= \frac{1}{2}hu''(x_0) + O(h^2),\end{aligned}$$

which is only first order accurate.

The second approach is to extend the domain with a *ghost cell* $x_{-1} = -h$ and use a central difference to obtain

$$\frac{U_1 - U_{-1}}{2h} = \sigma \quad (10.42)$$

that is second-order accurate for the LTE. We do not have any information for U_{-1} , so we want to eliminate it by

$$\frac{1}{h^2}(U_{-1} - 2U_0 + U_1) = f(x_0). \quad (10.43)$$

(10.42) and (10.43) yield

$$\frac{1}{h}(-U_0 + U_1) = \sigma + \frac{h}{2}f(x_0). \quad (10.44)$$

The resulting matrix is the same as that of the first approach in (10.41) except that the first component of $\mathbf{F}_E$ has an additional term $\frac{h}{2}f(x_0)$.

The third approach is to approximate $u'(0)$ with U_0, U_1 , and U_2 so that the corresponding FD formula is second-order accurate, cf. Example 9.35,

$$-\frac{1}{h}\left(\frac{3}{2}U_0 - 2U_1 + \frac{1}{2}U_2\right) = \sigma + O(h^2). \quad (10.45)$$

This results in the linear system

$$A_F \mathbf{U}_E = \mathbf{F}_E \quad (10.46)$$

where

$$A_F = \frac{1}{h^2} \begin{bmatrix} -\frac{3}{2}h & 2h & -\frac{1}{2}h & & & \\ 1 & -2 & 1 & & & \\ & 1 & -2 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & -2 & 1 \\ & & & & 1 & -2 & 1 \\ & & & & & 0 & h^2 \end{bmatrix}.$$

Exercise 10.38. Show that all elements of the first column of $B_E = A_E^{-1}$ are $O(1)$.

Example 10.39. Consider the second-order BVP

$$u''(x) = f(x) \text{ in } (0, 1), \quad (10.47)$$

with pure Neumann conditions

$$u'(0) = \sigma_0, \quad u'(1) = \sigma_1. \quad (10.48)$$

To ensure the existence of a solution, the following compatibility condition on $f(x)$, σ_0 , and σ_1 must be satisfied:

$$\int_0^1 f(x)dx = \int_0^1 u''(x)dx = u'(1) - u'(0) = \sigma_1 - \sigma_0. \quad (10.49)$$

In fact, if (10.49) holds, there are an infinite number of solutions: if v is a solution of (10.47), $v + \mathbb{R}$ are also solutions.

Using procedures similar to those in Example 10.37, we can discretize (10.47) and (10.48) as

$$A_F \mathbf{U}_E = \mathbf{F}_F, \quad (10.50)$$

where

$$A_F = \frac{1}{h^2} \begin{bmatrix} -h & h & & & & \\ 1 & -2 & 1 & & & \\ & 1 & -2 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & -2 & 1 \\ & & & & 1 & -2 & 1 \\ & & & & & h & -h \end{bmatrix}, \quad (10.51)$$

$$\mathbf{F}_F = \begin{bmatrix} \sigma_0 + \frac{h}{2}f(x_0) \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{m-1}) \\ f(x_m) \\ -\sigma_1 + \frac{h}{2}f(x_{m+1}) \end{bmatrix}.$$

Lemma 10.40. The matrix A_F in (10.51) satisfies

$$\dim \mathcal{N}(A_F) = 1. \quad (10.52)$$

Proof. Clearly, $\mathbf{e} = (1, 1, \dots, 1)^\top$ is in $\mathcal{N}(A_F)$, the null space of A_F . The rest follows from the well-posedness of the BVP with pure Neumann conditions. $\square$

Theorem 10.41 (Solvability condition). The linear system (10.50) has a solution if and only if

$$\frac{h}{2}f(x_0) + h \sum_{i=1}^m f(x_i) + \frac{h}{2}f(x_{m+1}) = \sigma_1 - \sigma_0. \quad (10.53)$$

Proof. The fundamental theorem of linear algebra (Theorem C.92) implies

$$\mathbb{R}^{m+2} = \mathcal{R}(A_F) \oplus \mathcal{N}(A_F^\top) \quad (10.54)$$

and $\dim \mathcal{N}(A_F^\top) = \dim \mathcal{N}(A_F)$. Lemma 10.40 further yields $\dim \mathcal{N}(A_F^\top) = 1$. Then it is readily verified that

$$\mathcal{N}(A_F^\top) = \text{span} \{(1, h, h, \dots, h, 1)^\top\}.$$

For sufficiency, the above equation and (10.53) imply that $\mathbf{F}_F$ is orthogonal to $\mathcal{N}(A_F^\top)$ and thus (10.54) yields $\mathbf{F}_F \in \mathcal{R}(A_F)$. By Theorem C.255, (10.50) must have a solution. As for necessity, the existence of a solution of (10.50) implies $\mathbf{F}_F \in \mathcal{R}(A_F)$, which, together with the above two equations, implies (10.53). $\square$

10.5 BVPs in two dimensions

Example 10.42 (An FD method for Poisson's equation in a unit square). Consider the two-dimensional BVP

$$-\frac{\partial^2}{\partial x^2}u(x, y) - \frac{\partial^2}{\partial y^2}u(x, y) = f(x, y) \quad (10.55)$$

in $\Omega := (0, 1)^2$ with homogeneous Dirichlet conditions

$$u(x, y)|_{\partial\Omega} = 0. \quad (10.56)$$

A uniform Cartesian grid can be generated with

$$x_i = ih, \quad y_j = jh, \quad i, j = 0, 1, \dots, m, m+1, \quad (10.57)$$

where $h = \Delta x = \Delta y = \frac{1}{m+1}$ is the uniform grid size.

Approximate $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$ separately and we have, $\forall i, j = 1, 2, \dots, m$,

$$-\frac{U_{i-1,j} - 2U_{ij} + U_{i+1,j}}{h^2} - \frac{U_{i,j-1} - 2U_{ij} + U_{i,j+1}}{h^2} = f_{ij}. \quad (10.58)$$

These $m \times m$ equations organize into a single system

$$A_{2D}\mathbf{U} = \mathbf{F}. \quad (10.59)$$

Exercise 10.43. Show that the LTE of the FD method in Example 10.42 is

$$\tau_{i,j} = -\frac{1}{12}h^2 \left(\frac{\partial^4 u}{\partial x^4} + \frac{\partial^4 u}{\partial y^4} \right) \Big|_{(x_i, y_j)} + O(h^4). \quad (10.60)$$

	7	8	9
	4	5	6
	1	2	3

Figure 10.2: The row ordering for two-dimensional structured grids.

Example 10.44. For $m = 3$ with the row ordering in Fig-

ure 10.2, we have

$$A_{2D} = \frac{1}{h^2} \begin{bmatrix} +4 & -1 & & -1 & & & \\ -1 & +4 & -1 & & -1 & & \\ & -1 & +4 & & & -1 & \\ -1 & & & +4 & -1 & & -1 \\ & -1 & & -1 & +4 & -1 & \\ & & -1 & & -1 & +4 & -1 \\ & & & -1 & & -1 & +4 \end{bmatrix}$$

$$= \frac{1}{h^2} \begin{bmatrix} T & -I & \\ -I & T & -I \\ & -I & T \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} U_{11} \\ U_{21} \\ U_{31} \\ U_{12} \\ U_{22} \\ U_{32} \\ U_{13} \\ U_{23} \\ U_{33} \end{bmatrix},$$

where $\mathbf{U}$ is obtained by stacking the columns on top of each other and

$$T = \begin{bmatrix} +4 & -1 & 0 \\ -1 & +4 & -1 \\ 0 & -1 & +4 \end{bmatrix}. \quad (10.61)$$

Lemma 10.45. Let $\mathbf{E} = \mathbf{U} - \hat{\mathbf{U}}$ denote the global error of the linear system (10.59). Then the LTE (10.60) satisfies

$$A_{2D}\mathbf{E} = -\boldsymbol{\tau}. \quad (10.62)$$

Proof. The proof is the same as that of Lemma 10.20. $\square$

Motivation 10.46. We are not satisfied with the results of A_{2D} in Example 10.42 because

- the connection of A_{2D} to the 1D second-order discrete Laplacian A is not clear,
- it does not generalize to higher-dimensional spaces,
- it is difficult to analyze the stability and convergence of the FD method in Example 10.42.

These difficulties can be overcome by relating A_{2D} to the 1D second-order discrete Laplacian A and by reusing the analysis of A .

Roadmap 10.47. In the rest of Section 10.5 we *completely* solve the BVP in 2D. Section 10.5.1 contains just enough materials to express A_{2D} in terms of A while in Sections 10.5.2 and 10.5.3 we show the second-order convergence of the FD method in Example 10.42 in the 2-norm and in the max-norm, respectively. The tools taken in these two sections are different: the former is *global* while the latter is *local*. Finally, the maximum principle in Section 10.5.3 is generalized to irregular domains in Section 10.5.4.

10.5.1 Kronecker product

Definition 10.48. The *Kronecker product* of two matrices $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$ is another matrix $A \otimes B \in \mathbb{C}^{mp \times nq}$

given by

$$A \otimes B := \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix}, \quad (10.63)$$

where a_{ij} is the (i, j) th element of A .

Example 10.49.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \otimes \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ 0 & 15 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{bmatrix}.$$

Definition 10.50. For $X \in \mathbb{C}^{m \times n}$, $\text{vec}(X)$ is defined to be a column vector of size mn made of the columns of X stacked on top of one another from left to right.

Lemma 10.51. Any $A \in \mathbb{C}^{m \times m}$, $B \in \mathbb{C}^{n \times n}$, and $X \in \mathbb{C}^{m \times n}$ satisfy

$$\text{vec}(AX) = (I_n \otimes A)\text{vec}(X), \quad (10.64)$$

$$\text{vec}(XB) = (B^\top \otimes I_m)\text{vec}(X). \quad (10.65)$$

Proof. We have

$$\begin{aligned} \text{vec}(AX) &= \text{vec}([AX_1, AX_2, \dots, AX_n]) \\ &= \begin{bmatrix} AX_1 \\ AX_2 \\ \vdots \\ AX_n \end{bmatrix} = \begin{bmatrix} A & & \\ & A & \\ & & \ddots \\ & & & A \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = (I_n \otimes A) \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}. \end{aligned}$$

Let $Y = XB$, then

$$Y_j = Xb_j \Rightarrow y_{kj} = \sum_{i=1}^n x_{ki}b_{ij}. \quad (10.66)$$

Let $C = B^\top \otimes I_m$, then the (i, j) -th sub-block of C is

$$C_{ij} = b_{ji}I_m. \quad (10.67)$$

Let $D = C\text{vec}(X)$, then the j -th block of D is

$$D_j = \sum_{i=1}^n C_{ji}X_i = \sum_{i=1}^n b_{ij}I_m X_i = \sum_{i=1}^n b_{ij}X_i, \quad (10.68)$$

and the (k, j) -th entry of D is (Here we also use (k, j) to denote the scalar index corresponding to the multi-index (k, j) .)

$$d_{(k,j)} = \sum_{i=1}^n b_{ij}x_{ki} = \sum_{i=1}^n x_{ki}b_{ij}. \quad (10.69)$$

Combining (10.66) and (10.69) yields (10.65). $\square$

10.5.2 Convergence in the 2-norm

Lemma 10.52. The linear system (10.59) is equivalent to

$$AU_{m \times m} + U_{m \times m}A = F_{m \times m}, \quad (10.70)$$

where the (i, j) th element of $U_{m \times m}$ is the computed solution at the (i, j) th grid point, the (i, j) th element of $F_{m \times m}$ is

$$(F_{m \times m})_{ij} = f(ih, jh),$$

and A is the 1D discrete Laplacian in (10.13).

Proof. A direct computation gives

$$\begin{cases} (AU_{m \times m})_{ij} = \frac{1}{h^2}(-U_{i-1,j} + 2U_{ij} - U_{i+1,j}), \\ (U_{m \times m}A)_{ij} = \frac{1}{h^2}(-U_{i,j-1} + 2U_{ij} - U_{i,j+1}), \end{cases} \quad (10.71)$$

and the *homogeneous* Dirichlet condition yields

$$AU_{m \times m} + U_{m \times m}A = F_{m \times m}. \quad \square$$

Lemma 10.53. The 1D discrete Laplacian A in (10.13) satisfies

$$\text{vec}(AU_{m \times m} + U_{m \times m}A) = (I_m \otimes A + A \otimes I_m)\text{vec}(U_{m \times m}).$$

Proof. Lemma 10.51 yields

$$\begin{aligned} \text{vec}(AU_{m \times m}) &= (I_m \otimes A)\text{vec}(U_{m \times m}); \\ \text{vec}(U_{m \times m}A) &= (A^\top \otimes I_m)\text{vec}(U_{m \times m}) \\ &= (A \otimes I_m)\text{vec}(U_{m \times m}), \end{aligned}$$

where the last step follows from the symmetry of A . Adding these two equations gives the desired result. $\square$

Theorem 10.54. With matrix ordering, the linear system (10.59) can be written as

$$A_{2D} = I_m \otimes A + A \otimes I_m, \quad \mathbf{U} = \text{vec}(U_{m \times m}), \quad \mathbf{F} = \text{vec}(F_{m \times m}).$$

Proof. This follows from Lemma 10.52 and Lemma 10.53. $\square$

Definition 10.55. The *discrete Laplacian* in n -dimensional space analogous to the 1D discrete Laplacian (10.13) is

$$A_{nD} = \sum_{j=0}^{n-1} \underbrace{I_m \otimes \cdots \otimes I_m}_{\#I_m=j} \otimes A \otimes \underbrace{I_m \otimes \cdots \otimes I_m}_{\#I_m=n-j-1}. \quad (10.72)$$

Example 10.56. For $n = 3$, we have

$$A_{3D} = A \otimes I_m \otimes I_m + I_m \otimes A \otimes I_m + I_m \otimes I_m \otimes A.$$

Theorem 10.57. The eigenpairs of A_{2D} are

$$\lambda_{ij} = \lambda_i + \lambda_j, \quad \mathbf{W}_{ij} = \text{vec}(\mathbf{w}_i \mathbf{w}_j^\top), \quad (10.73)$$

where $i, j = 1, 2, \dots, m$ and $(\lambda_i, \mathbf{w}_i)$ is an eigenpair of A in Lemma 10.25.

Proof. Lemma 10.25 yields

$$\begin{aligned} A\mathbf{w}_i \mathbf{w}_j^\top + \mathbf{w}_i \mathbf{w}_j^\top A &= \lambda_i \mathbf{w}_i \mathbf{w}_j^\top + \mathbf{w}_i \mathbf{w}_j^\top A^\top \\ &= \lambda_i \mathbf{w}_i \mathbf{w}_j^\top + \mathbf{w}_i (A\mathbf{w}_j)^\top = \lambda_i \mathbf{w}_i \mathbf{w}_j^\top + \lambda_j \mathbf{w}_i \mathbf{w}_j^\top = (\lambda_i + \lambda_j) \mathbf{w}_i \mathbf{w}_j^\top. \end{aligned}$$

Then Theorem 10.54 and Lemma 10.53 yield

$$\begin{aligned} A_{2D} \text{vec}(\mathbf{w}_i \mathbf{w}_j^\top) &= (I_m \otimes A + A \otimes I_m) \text{vec}(\mathbf{w}_i \mathbf{w}_j^\top) \\ &= \text{vec}(A(\mathbf{w}_i \mathbf{w}_j^\top) + (\mathbf{w}_i \mathbf{w}_j^\top)A) = (\lambda_i + \lambda_j) \text{vec}(\mathbf{w}_i \mathbf{w}_j^\top), \end{aligned}$$

and hence $\lambda_i + \lambda_j$ is an eigenvalue of A_{2D} with corresponding eigenvector $\mathbf{W}_{ij}$.

It remains to show that the above eigenpairs are *all* eigenpairs of A_{2D} . This indeed holds because the set of eigenvectors $\{\mathbf{W}_{ij}\}$ is orthogonal:

$$\begin{aligned} \mathbf{W}_{ij}^\top \mathbf{W}_{kl} &= \text{vec}(\mathbf{w}_i \mathbf{w}_j^\top)^\top \text{vec}(\mathbf{w}_k \mathbf{w}_\ell^\top) \\ &= \text{Trace} \left((\mathbf{w}_i \mathbf{w}_j^\top)^\top \mathbf{w}_k \mathbf{w}_\ell^\top \right) = \text{Trace}(\mathbf{w}_j (\mathbf{w}_i^\top \mathbf{w}_k) \mathbf{w}_\ell^\top) \\ &= \begin{cases} 0 & \text{if } i \neq k; \\ \frac{m+1}{2} \mathbf{w}_\ell^\top \mathbf{w}_j & \text{if } i = k \end{cases} = \begin{cases} 0 & \text{if } i \neq k \text{ or } \ell \neq j; \\ \frac{(m+1)^2}{4} & \text{otherwise,} \end{cases} \end{aligned}$$

where the last two steps follow from Exercise 10.26. $\square$

Theorem 10.58. The FD method in Example 10.42 is second-order convergent in the 2-norm.

Proof. We have $\|A_{2D}\|_2 = \rho(A_{2D})$ since A_{2D} is symmetric. Then Theorem 10.57 yields

$$\lim_{h \rightarrow 0} \|A_{2D}^{-1}\|_2 = \lim_{h \rightarrow 0} \frac{1}{\min |\lambda_{ij}|} = \lim_{h \rightarrow 0} \frac{h^2}{8 \sin^2 \frac{\pi h}{2}} = \frac{1}{2\pi^2} = O(1).$$

By Definition 10.23, the method is stable. The proof is completed by (10.60), Definition 10.21, Theorem 10.24, and Lemma 10.45. $\square$

10.5.3 Convergence in the max-norm via a discrete maximum principle

Theorem 10.59. The FD method in Example 10.42 is second-order convergent in the max-norm.

Proof. Let $\mathbf{X}_I$ be the grid obtained by removing from $\mathbf{X}$ in (10.57) those grid points satisfying $i = 0, m+1$ or $j = 0, m+1$. For the linear map $\hat{A}_{2D} : \{\mathbf{X} \rightarrow \mathbb{R}\} \rightarrow \{\mathbf{X}_I \rightarrow \mathbb{R}\}$ given by

$$\begin{aligned} \hat{A}_{2D} U_{i,j} &:= (\hat{A}_{2D} \mathbf{U})_{i,j} \\ &:= \frac{1}{h^2} (4U_{i,j} - U_{i+1,j} - U_{i-1,j} - U_{i,j+1} - U_{i,j-1}), \end{aligned} \quad (10.74)$$

the matrix of $\hat{A}_{2D}$ is different from A_{2D} in Example 10.42: the latter is a square matrix acting only on equation-discretization points while the former is a rectangular matrix on both equation-discretization and Dirichlet points. Nonetheless, we can prove a result similar to Lemma 10.20,

$$\hat{A}_{2D} E_{i,j} = \hat{A}_{2D} (U_{i,j} - u_{i,j}) = f_{i,j} - \hat{A}_{2D} u_{i,j} = -\tau_{i,j}, \quad (10.75)$$

where the second step follows from the definition of $U_{i,j}$ and the last from (10.55) and Definition 10.17.

Define a *comparison function* $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$\phi(x, y) := \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 \quad (10.76)$$

and write $\phi_{i,j} := \phi(ih, jh)$. (10.74) and (10.76) yield

$$\hat{A}_{2D} \phi_{i,j} = -4. \quad (10.77)$$

Denote by $E : \mathbf{X} \rightarrow \mathbb{R}$ and $\tau : \mathbf{X}_I \rightarrow \mathbb{R}$ the solution error and the LTE, respectively, and write $\tau_m := \max_{i,j} |\tau_{i,j}|$. Construct a grid function

$$\psi_{i,j} := E_{i,j} + \frac{1}{4} \tau_m \phi_{i,j} \quad (10.78)$$

and we have, from (10.75) and (10.77),

$$\hat{A}_{2D} \psi_{i,j} = -\tau_{i,j} - \tau_m \leq 0.$$

By (10.74), $\hat{A}_{2D} \psi_{i,j} \leq 0$ dictates that $\psi_{i,j}$ be no greater than at least one of its neighbors $\psi_{i+1,j}$, $\psi_{i-1,j}$, $\psi_{i,j+1}$, and $\psi_{i,j-1}$. Therefore, *the maximum value of ψ must occur at a boundary point*, i.e., a point with $i = 0, m+1$ or $j = 0, m+1$. Consequently, there exists some constant $C > 0$ such that

$$E_{i,j} \leq \psi_{i,j} \leq \frac{1}{8} \tau_m < Ch^2,$$

where the first step follows from (10.78) and $\tau_m \phi_{i,j} \geq 0$, the second step from $E_{i,j}$ being zero at all Dirichlet points and the fact that the maximum of ϕ is $\frac{1}{2}$ at the domain corners, and the last step from (10.60).

By similar arguments, the grid function

$$\chi_{i,j} := -E_{i,j} + \frac{1}{4} \tau_m \phi_{i,j} \quad (10.79)$$

satisfies $\hat{A}_{2D} \chi_{i,j} \leq 0$ for all grid points and thus

$$-E_{i,j} \leq \chi_{i,j} \leq \frac{1}{8} \tau_m < Ch^2.$$

To sum up, we have $|E_{i,j}| = O(h^2)$ for all grid points. $\square$

Definition 10.60. For a BVP $\mathcal{L}u = f$ on a domain Ω , the set $\mathbf{X}_\Omega$ of *equation-discretization points* is the set of grid points where values of the unknown function u are sought while the set $\mathbf{X}_{\partial\Omega} \subset \partial\Omega$ of *Dirichlet points* is the set of grid points where each point $Q \in \mathbf{X}_{\partial\Omega}$ satisfies

- $u(Q)$ is prescribed by a Dirichlet condition,
- this Dirichlet condition at Q is involved in discretizing the BVP at some nearby point $P \in \mathbf{X}_\Omega$.

We write $\mathbf{X}_\mathcal{L} := \mathbf{X}_\Omega \cup \mathbf{X}_{\partial\Omega}$.

Lemma 10.61 (Discrete maximum principle). Suppose that a linear BVP $\mathcal{L}u = f$ on a domain Ω discretizes to

$$\forall P \in \mathbf{X}_\Omega, \quad L_h U_P + g_P = f_P, \quad (10.80)$$

where $f_P = f(P)$, g_P corresponds to all boundary data except Dirichlet conditions, and L_h and $\mathbf{X}_\Omega$ satisfy

(DMP-1) for each equation-discretization point $P \in \mathbf{X}_\Omega$, L_h is of the form

$$L_h U_P = c_P U_P - \sum_{Q \in Q_P} c_Q U_Q, \quad (10.81)$$

where $Q_P \subset \mathbf{X}_\mathcal{L}$, $c_P > 0$ and each $c_Q > 0$. The set $\{P\} \cup Q_P$ in (10.81) is called the *P-stencil* or the *stencil of L_h at P* ;

(DMP-2) $\forall P \in \mathbf{X}_\Omega$, $c_P \geq \sum_{Q \in Q_P} c_Q$;

(DMP-3) $\mathbf{X}_\Omega$ is *connected* in the sense that

$$\begin{aligned} &\forall P_0, P_m \in \mathbf{X}_\Omega, \exists P_1, P_2, \dots, P_{m-1} \text{ s.t.} \\ &\forall r = 1, 2, \dots, m, P_r \text{ is in the } P_{r-1}\text{-stencil;} \end{aligned} \quad (10.82)$$

(DMP-4) at least one equation (10.81) involves a boundary point $Q \in \partial\Omega$ that is also a Dirichlet point.

Then for any grid function $\psi : \mathbf{X}_\mathcal{L} \rightarrow \mathbb{R}$ satisfying

$$\begin{cases} \max_{P \in \mathbf{X}_\mathcal{L}} \psi_P \geq 0, \\ \forall P \in \mathbf{X}_\Omega, \quad L_h \psi_P \leq 0, \end{cases} \quad (10.83)$$

we have

$$\max_{P \in \mathbf{X}_\Omega} \psi_P \leq \max_{Q \in \mathbf{X}_{\partial\Omega}} \psi_Q. \quad (10.84)$$

Proof. Suppose (10.84) is false, i.e., ψ attains its maximum at some $P \in \mathbf{X}_\Omega$:

$$\psi(P) = M_\Omega := \max_{Q \in \mathbf{X}_\Omega} \psi_Q > M_{\partial\Omega} := \max_{Q \in \mathbf{X}_{\partial\Omega}} \psi_Q.$$

Then (DMP-1) and (10.83) yield

$$(*) : \quad M_\Omega = \psi_P \leq \frac{1}{c_P} \sum_{Q \in Q_P} c_Q \psi_Q \leq \frac{1}{c_P} \sum_{Q \in Q_P} c_Q M_\Omega \leq M_\Omega,$$

where the first inequality follows from (10.83) and (10.81), the second from the definition of M_Ω , and the third from (DMP-2) and $M_\Omega \geq 0$. For the second “ $\leq$ ” in (*) to be “ $=$,” we must have $\psi_Q = \psi_P$ for each Q . By (DMP-3), ψ takes the same value M_Ω on all equation-discretization points. Then (DMP-4) implies $M_\Omega = M_{\partial\Omega}$, which contradicts the starting point $M_\Omega > M_{\partial\Omega}$. $\square$

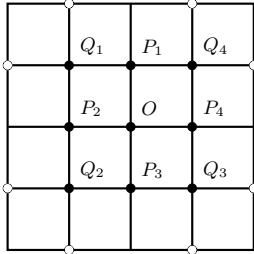


Figure 10.3: The importance of precisely defining connectedness for the discrete maximum principle; see Example 10.62.

Example 10.62. Suppose the concept of connectedness in (DMP-3) were defined as

$$\begin{aligned} &\forall P_0, P_m \in \mathbf{X}_\Omega, \exists P_1, P_2, \dots, P_{m-1} \text{ s.t. } \forall r = 1, 2, \dots, m, \\ &\text{both } U_{P_r} \text{ and } U_{P_{r-1}} \text{ appear in some equation (10.80).} \end{aligned} \quad (10.85)$$

Then the conclusion (10.84) of Lemma 10.61 would not hold. For the counter-example in Figure 10.3, let L_h be

$$\begin{aligned} L_h \psi_{P_1} &= 3\psi_{P_1} - \psi_{P_2} - \psi_{P_3} - \psi_{P_4}, \\ L_h \psi_{P_2} &= 3\psi_{P_2} - \psi_{P_1} - \psi_{P_3} - \psi_{P_4}, \\ L_h \psi_{P_3} &= 3\psi_{P_3} - \psi_{P_1} - \psi_{P_2} - \psi_{P_4}, \\ L_h \psi_{P_4} &= 3\psi_{P_4} - \psi_{P_1} - \psi_{P_2} - \psi_{P_3}, \\ L_h \psi_O &= 4\psi_O - \psi_{P_1} - \psi_{P_2} - \psi_{P_3} - \psi_{P_4}, \end{aligned}$$

and the expressions of L_h at Q_i 's be similar with that of L_h at O such that (DMP-1,2,4) hold. Thus (10.85) holds. The following distribution of the grid function ψ ,

$$\psi_{\mathbf{x}} = \begin{cases} 10 & \text{if } \mathbf{x} = P_1, P_2, P_3, P_4; \\ 1 & \text{otherwise,} \end{cases}$$

satisfies (10.83), but the conclusion (10.84) does not hold.

Theorem 10.63. Suppose an FD discretization of a BVP $\mathcal{L}u = f$ satisfies the conditions (DMP-1,2,3,4) in Lemma 10.61. Then the solution error $E_P := U_P - u(P)$ of the FD method (10.80) is bounded by

$$\forall P \in \mathbf{X}_\mathcal{L}, \quad |E_P| \leq T_{\max} \max_{Q \in \mathbf{X}_{\partial\Omega}} \phi(Q), \quad (10.86)$$

where $T_{\max} = \max_{P \in \mathbf{X}_\Omega} |T_P|$, T_P is the LTE at P and $\phi : \mathbf{X}_\mathcal{L} \rightarrow \mathbb{R}$ is a nonnegative grid function satisfying

$$\forall P \in \mathbf{X}_\Omega, \quad L_h \phi_P \leq -1. \quad (10.87)$$

Proof. By (10.75), we have $L_h E_P = -T_P$. Define

$$\psi_P := E_P + T_{\max} \phi_P$$

and we know from (10.87) that

$$L_h \psi_P \leq -T_P - T_{\max} \leq 0.$$

Furthermore, we have $\max_{P \in \mathbf{X}} \psi_P \geq 0$ because $\phi_P \geq 0$ and

$$(*) : \quad \forall Q \in \mathbf{X}_{\partial\Omega}, \quad E_Q = 0,$$

cf. Definition 10.60. Then we have

$$\begin{aligned} E_P &\leq \max_{P \in \mathbf{X}_\mathcal{L}} (E_P + T_{\max} \phi_P) \\ &\leq \max_{Q \in \mathbf{X}_{\partial\Omega}} (E_Q + T_{\max} \phi(Q)) = T_{\max} \max_{Q \in \mathbf{X}_{\partial\Omega}} (\phi(Q)), \end{aligned}$$

where the first step follows from $T_{\max} \phi_P \geq 0$, the second from Lemma 10.61, and the third from (*).

Repeat the above arguments with $\psi_P = -E_P + T_{\max} \phi_P$ and we have

$$-E_P \leq T_{\max} \max_{Q \in \mathbf{X}_{\partial\Omega}} (\phi_Q). \quad \square$$

10.5.4 Convergence on irregular domains

Example 10.64 (An FD method for Poisson's equation in 2D irregular domains). Consider the BVP

$$-\frac{\partial^2}{\partial x^2} u(x, y) - \frac{\partial^2}{\partial y^2} u(x, y) = f(x, y) \quad (10.88)$$

in a 2D irregular domain Ω with Dirichlet conditions.

An equation-discretization point is *regular* if the standard 5-point stencil is applicable; otherwise it is *irregular*. For an irregular point, we modify the FD discretization in Example 10.42 to incorporate the info of local geometry and Dirichlet conditions. For example, in Figure 10.4, the discrete operator becomes

$$L_h U_P := \frac{(1 + \theta)U_P - U_A - \theta U_W}{\frac{1}{2}\theta(1 + \theta)h^2} + \frac{(1 + \alpha)U_P - U_B - \alpha U_S}{\frac{1}{2}\alpha(1 + \alpha)h^2}. \quad (10.89)$$

In the resulting linear system $L_h U_P - f_P = 0$, the form of $L_h U_P$ as in (10.89) is different from that in (10.58) at a regular equation-discretization point. Consequently, a global analysis of this linear system is difficult.

Use a direct method such as the LU factorization to solve the linear system, with your favorite implementation of a BLAS or LAPACK.

II. For the function

$$u(x, y) = \exp(y + \sin(x)), \quad (10.94)$$

derive the corresponding $f(x, y)$ and the boundary conditions. Test your solver for all combinations of (a, b) in I on grids with $n = 8, 16, 32, 64$ along each dimension, report the 1-, 2-, and ∞ -norms of the errors and the corresponding convergence rates on the four grids. You should also design at least two of your own test functions and carry out the same process.

III. User-specified parameters must be clearly listed in an input file with a key-value syntax such as json (<https://www.json.org/json-en.html>), but

you may use or write your own parser. The main program is supposed to read the input file, create objects from the input, call your BVP solver, generate test results, and report errors and convergence rates.

IV. The generation of pictures on solution and errors can be performed outside your program using another tool of your choice.

V. Write a GNU **makefile** under your root directory so that the command “**make run**” would trigger the compilation of your source code, the production of the executable, and the running of your tests.

VI. Write a report to summarize the main points of your numerical experiments, which should be designed to verify the analytic results in the notes. To some extent, your grade depends on the number of key conclusions in the notes confirmed by your numerical experiments.

Chapter 11

Multigrid Methods for Poisson's Equations

Motivation 11.1. Optimal complexity is the main reason for adopting a multigrid method to solve linear systems that results from discretizing BVPs. The convergence rates of classical iterative method depend on the grid spacing, or problem size. In contrast, convergence rates of multigrid methods do not. Let $N = n^3$ be the total number of unknowns in a 3D cube where the solution of a Poisson equation is sought. To solve the linear system resulting from the FD discretization of the Poisson equation, methods based on Cholesky factorization have best complexity $O(N^2)$, methods based on Krylov space techniques such as the conjugate gradient method have best complexity $O(N^{1.17})$, and methods based on geometric multigrid have the optimal complexity $O(N)$.

Roadmap 11.2. In this chapter, we tell a story that only consists of the bare necessities of multigrid methods; our exposition mostly follows the book by Briggs et al. [2000]. Instead of the original linear system, we solve the residual equation discussed in Section 11.1, which facilitates the iterative and recursive implementation of multigrid algorithms and makes it general to only consider the one-dimensional Poisson equations with homogeneous Dirichlet conditions in Section 11.2. The algorithmic components in Section 11.3 and the convergence analysis in Section 11.4 generalizes to multiple dimensions and other types of boundary conditions in a straightforward manner.

11.1 The residual equation

Definition 11.3. In solving a linear system $A\mathbf{x} = \mathbf{b}$, the error of an approximate solution $\tilde{\mathbf{x}}$ is

$$\mathbf{e}(\tilde{\mathbf{x}}) := \mathbf{x} - \tilde{\mathbf{x}} \quad (11.1)$$

and the residual of $\tilde{\mathbf{x}}$ is

$$\mathbf{r}(\tilde{\mathbf{x}}) := \mathbf{b} - A\tilde{\mathbf{x}}. \quad (11.2)$$

Lemma 11.4. The error and the residual of an approximate solution $\tilde{\mathbf{x}}$ satisfy the residual equation

$$A\mathbf{e} = \mathbf{r}. \quad (11.3)$$

Proof. This follows from Definition 11.3 in the same way that Lemma 10.20 follows from Lemma 10.19. $\square$

Theorem 11.5. The relative error of an approximate solution is bounded by its relative residual.

$$\frac{1}{\text{cond}(A)} \frac{\|\mathbf{r}\|_2}{\|\mathbf{b}\|_2} \leq \frac{\|\mathbf{e}\|_2}{\|\mathbf{x}\|_2} \leq \text{cond}(A) \frac{\|\mathbf{r}\|_2}{\|\mathbf{b}\|_2}, \quad (11.4)$$

where $\text{cond}(A)$ is the condition number of A as in Definition 2.22.

Exercise 11.6. Prove Theorem 11.5.

11.2 The model problem

Definition 11.7. The *model problem* for our exposition of multigrid methods is the one-dimensional Poisson equation with homogeneous boundary condition

$$\begin{cases} -\Delta u = f & \text{in } \Omega := (0, 1); \\ u = 0 & \text{on } \partial\Omega. \end{cases} \quad (11.5)$$

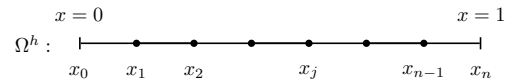


Figure 11.1: FD discretization of $\Omega = [0, 1]$ for multigrid methods.

Example 11.8. As a special case of Example 10.12 with $\alpha = 0$, $\beta = 0$, and $m = n - 1$, our discretization of (11.5) as shown in Figure 11.1 yields a linear system

$$A\mathbf{u} = \mathbf{f}, \quad (11.6)$$

where the unit interval Ω is discretized by a uniform grid size $h = \frac{1}{n}$ into n cells with cell boundaries at $x_j = jh = \frac{j}{n}$ for $j = 0, 1, \dots, n$, the knowns $f_j = f(x_j)$ and the unknowns u_j are located at the internal nodes x_j with $j = 1, 2, \dots, n - 1$. The matrix $A \in \mathbb{R}^{(n-1) \times (n-1)}$ is the same as that in (10.13), i.e., a Toeplitz matrix given by

$$a_{ij} = \begin{cases} \frac{2}{h^2} & \text{if } i = j; \\ -\frac{1}{h^2} & \text{if } i - j = \pm 1; \\ 0 & \text{otherwise.} \end{cases} \quad (11.7)$$

By Lemma 10.25, the eigenvalues and eigenvectors of A are

$$\lambda_k(A) = \frac{4}{h^2} \sin^2 \frac{k\pi}{2n} = \frac{4}{h^2} \sin^2 \frac{kh\pi}{2}, \quad (11.8)$$

$$w_{k,j} = \sin \frac{jk\pi}{n} = \sin(x_j k\pi), \quad (11.9)$$

where $j, k = 1, 2, \dots, n-1$.

11.3 Algorithmic components

Notation 8. Ω^h denotes the uniform grid of n intervals that discretizes the problem domain Ω . Occasionally we also abuse the notation to mean the corresponding vector space of grid functions $\{\Omega^h \rightarrow \mathbb{R}\}$.

11.3.1 Fourier modes on Ω^h

Definition 11.9. The *wavelength* of a sinusoidal function is the distance of one sinusoidal period. The *wavenumber* k of a sinusoidal function is the number of half sinusoidal waves in unit length.

Lemma 11.10. The k th Fourier mode with its j th component as $w_{k,j} = \sin(x_j k\pi)$ has wavelength $L = \frac{2}{k}$.

Proof. By Definition 11.9, $\sin(x_j k\pi) = -\sin(x_j + \frac{L}{2})k\pi$ implies $x_j k\pi = (x_j + \frac{L}{2})k\pi - \pi$. Hence $k = \frac{2}{L}$. $\square$

Exercise 11.11. For $\Omega = (0, 1)$, plot to show that the maximum wavenumber that is representable on Ω^h is $k_{\max} = \frac{1}{h}$. What if we require that the Fourier mode be 0 at the boundary points?

Lemma 11.12 (Aliasing). For $k \in (n, 2n)$ on Ω^h , the Fourier mode $\mathbf{w}_k$ of which the j th component is $w_{k,j} = \sin(x_j k\pi)$ is actually represented as the additive inverse of the mode $\mathbf{w}_{k'}$ where $k' = 2n - k$.

Proof. It is readily verified that

$$\begin{aligned} \sin(x_j k\pi) &= -\sin(2j\pi - x_j k\pi) = -\sin(x_j (2n - k)\pi) \\ &= -\sin(x_j k'\pi) = -w_{k',j}. \end{aligned} \quad \square$$

Example 11.13. According to Lemma 11.12, the mode with $k = \frac{3}{2}n$ is represented by $k = \frac{1}{2}n$.

Exercise 11.14. Plot the case of $n = 6$ for Example 11.13.

Definition 11.15. On Ω^h , the Fourier modes with wavenumbers $k \in [1, \frac{n}{2})$ are called the *low-frequency* (LF) or *smooth* modes, those with $k \in [\frac{n}{2}, n)$ the *high-frequency* (HF) or *oscillatory* modes.

11.3.2 Relaxation

Lemma 11.16. For the linear system (11.6), the weighted Jacobi in Definition 3.94 has the iteration matrix

$$T_\omega = (1 - \omega)I + \omega D^{-1}(L + U) = I - \frac{\omega h^2}{2} A, \quad (11.10)$$

whose eigenvectors are the same as those of A , with the corresponding eigenvalues as

$$\lambda_k(T_\omega) = 1 - 2\omega \sin^2 \frac{k\pi}{2n}, \quad (11.11)$$

where $k = 1, 2, \dots, n-1$.

Exercise 11.17. Prove Lemma 11.16.

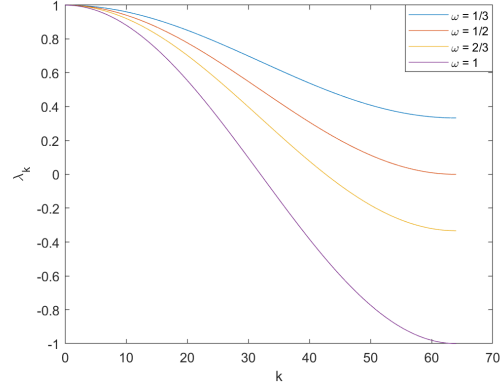


Figure 11.2: Eigenvalues of the iteration matrix T_ω for different wave numbers.

Exercise 11.18. Write a program to reproduce Figure 11.2. For $n = 64$, $\omega \in [0, 1]$, verify $\rho(T_\omega) \geq 0.9986$ and hence slow convergence.

Definition 11.19. The *smoothing factor* μ of an iterative method is the best damping factor for HF modes,

$$\mu := \min_{\omega \in (0, 1]} \max_{k \in [\frac{n}{2}, n)} |\lambda_k(T_\omega)|, \quad (11.12)$$

where T_ω is the iteration matrix of the iterative method and $\lambda_k(T_\omega)$ is the k th eigenvalue of T_ω . An iterative method is said to have the *smoothing property* if $\mu < 1$ and μ is independent of the grid size.

Example 11.20. The smoothing factor of the weighted Jacobi is determined by the optimization problem (11.12). For any fixed k , the function $\lambda_k(T_\omega)$ in (11.11) is monotonically decreasing in terms of ω . For any fixed ω , $\lambda_k(T_\omega)$ is also monotonically decreasing in terms of k . In particular, $\lambda_k(T_0) \equiv 1$ and $\lambda_k(T_1) = \cos \frac{k\pi}{n}$. To minimize $\max_{k \in [\frac{n}{2}, n)} |\lambda_k(T_\omega)|$ over $\omega \in [0, 1]$, we must have

$$1 - \omega = \lambda_{\frac{n}{2}}(T_\omega) = -\lambda_n(T_\omega) = -1 + 2\omega,$$

which implies $\omega = \frac{2}{3}$. Consequently we have $|\lambda_k| \leq \mu = \frac{1}{3}$.

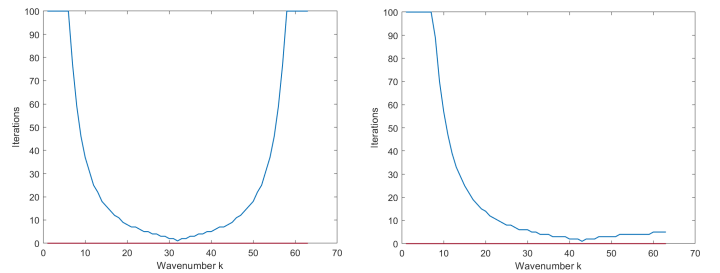


Figure 11.3: The number of iterations required by the weighted Jacobi method to reduce the norm of the initial error by a factor of 100 for each eigenvector $\mathbf{w}_k$; left: $\omega = 1$, right: $\omega = \frac{2}{3}$.

Exercise 11.21. Write a program to reproduce Figure 11.3 to verify that regular Jacobi is only good for damping modes $16 \leq k \leq 48$. In contrast, for $\omega = \frac{2}{3}$, the modes $16 \leq k < 64$ are all damped out quickly.

11.3.3 Restriction and prolongation

Lemma 11.22. The k th LF mode on Ω^h becomes the k th mode (LF or HF) on Ω^{2h} :

$$w_{k,2j}^h = w_{k,j}^{2h}. \quad (11.13)$$

LF modes $k \in [\frac{n}{4}, \frac{n}{2})$ of Ω^h will become HF modes on Ω^{2h} .

Proof. It is readily verified that

$$w_{k,2j}^h = \sin \frac{2jk\pi}{n} = \sin \frac{jk\pi}{\frac{n}{2}} = w_{k,j}^{2h}, \quad (11.14)$$

where $k \in [1, \frac{n}{2})$. Because of the smaller range of k on Ω^{2h} , the modes with $k \in [\frac{n}{4}, \frac{n}{2})$ are HF by definition since the highest wavenumber is $\frac{n}{2}$ on Ω^{2h} . $\square$

Definition 11.23. The *restriction* operator

$$I_h^{2h} : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^{\frac{n}{2}-1}$$

maps a vector on the fine grid Ω^h to its counterpart on the coarse grid Ω^{2h} :

$$I_h^{2h} \mathbf{v}^h = \mathbf{v}^{2h}. \quad (11.15)$$

Definition 11.24. The *injection* operator is a restriction operator given by

$$v_j^{2h} = v_{2j}^h, \quad (11.16)$$

where $j = 1, 2, \dots, \frac{n}{2} - 1$.

Definition 11.25. The *full-weighting* operator is a restriction operator given by

$$v_j^{2h} = \frac{1}{4} (v_{2j-1}^h + 2v_{2j}^h + v_{2j+1}^h), \quad (11.17)$$

where $j = 1, 2, \dots, \frac{n}{2} - 1$.

Example 11.26. For $n = 8$, the full-weighting operator is

$$I_h^{2h} = \frac{1}{4} \begin{bmatrix} 1 & 2 & 1 & & & \\ & & 1 & 2 & 1 & \\ & & & & 1 & 2 & 1 \end{bmatrix}. \quad (11.18)$$

Definition 11.27. The *prolongation or interpolation* operator

$$I_{2h}^h : \mathbb{R}^{\frac{n}{2}-1} \rightarrow \mathbb{R}^{n-1}$$

maps a vector on the coarse grid Ω^{2h} to its counterpart on the fine grid Ω^h :

$$I_{2h}^h \mathbf{v}^{2h} = \mathbf{v}^h. \quad (11.19)$$

Definition 11.28. The *linear interpolation* operator is a prolongation operator given by

$$\begin{aligned} v_{2j}^h &= v_j^{2h}, \\ v_{2j+1}^h &= \frac{1}{2}(v_j^{2h} + v_{j+1}^{2h}). \end{aligned} \quad (11.20)$$

Example 11.29. For $n = 8$, the linear interpolation operator is

$$I_{2h}^h = \frac{1}{2} \begin{bmatrix} 1 & & & & & \\ 2 & & & & & \\ 1 & 1 & & & & \\ & 2 & & & & \\ & 1 & 1 & & & \\ & & 2 & & & \\ & & 1 & & & \end{bmatrix}. \quad (11.21)$$

11.3.4 Two-grid correction

Definition 11.30. The *two-grid correction scheme*

$$\mathbf{v}^h \leftarrow \text{TG}(\mathbf{v}^h, \mathbf{f}^h, \nu_1, \nu_2) \quad (11.22)$$

solves $\mathbf{A}\mathbf{u} = \mathbf{f}$ in (11.6) via steps as follows.

- (TG-1) Relax $A^h \mathbf{u}^h = \mathbf{f}^h$ for ν_1 times on Ω^h with initial guess $\mathbf{v}^h$: $\mathbf{v}^h \leftarrow T_{\omega}^{\nu_1} \mathbf{v}^h + \mathbf{c}_1(f)$,
- (TG-2) Compute the fine-grid residual $\mathbf{r}^h = \mathbf{f}^h - A^h \mathbf{v}^h$ and restrict it to the coarse grid by $\mathbf{r}^{2h} = I_h^{2h} \mathbf{r}^h$: $\mathbf{r}^{2h} \leftarrow I_h^{2h}(\mathbf{f}^h - A^h \mathbf{v}^h)$,
- (TG-3) Solve $A^{2h} \mathbf{e}^{2h} = \mathbf{r}^{2h}$ on Ω^{2h} : $\mathbf{e}^{2h} \leftarrow (A^{2h})^{-1} \mathbf{r}^{2h}$,
- (TG-4) Interpolate the coarse-grid error to the fine grid by $\mathbf{e}^h = I_{2h}^h \mathbf{e}^{2h}$ and correct the fine-grid approximation: $\mathbf{v}^h \leftarrow \mathbf{v}^h + I_{2h}^h \mathbf{e}^{2h}$,
- (TG-5) Relax $A^h \mathbf{u}^h = \mathbf{f}^h$ for ν_2 times on Ω^h with initial guess $\mathbf{v}^h$: $\mathbf{v}^h \leftarrow T_{\omega}^{\nu_2} \mathbf{v}^h + \mathbf{c}_2(f)$.

Lemma 11.31. Acting on the error vector, the iteration matrix of the two-grid correction scheme (11.22) is

$$TG = T_{\omega}^{\nu_2} [I - I_{2h}^h (A^{2h})^{-1} I_h^{2h} A^h] T_{\omega}^{\nu_1}. \quad (11.23)$$

Proof. By Definition 11.30, the residual on the fine grid is

$$\mathbf{r}^h(\mathbf{v}^h) = \mathbf{f}^h - A^h (T_{\omega}^{\nu_1} \mathbf{v}^h + \mathbf{c}_1(f)).$$

The two-grid correction scheme with $\nu_2 = 0$ replaces the initial guess with

$$\mathbf{v}^h \leftarrow T_{\omega}^{\nu_1} \mathbf{v}^h + \mathbf{c}_1(f) + I_{2h}^h (A^{2h})^{-1} I_h^{2h} \mathbf{r}^h(\mathbf{v}^h)$$

which also holds for the exact solution $\mathbf{u}^h$

$$\mathbf{u}^h \leftarrow T_{\omega}^{\nu_1} \mathbf{u}^h + \mathbf{c}_1(f) + I_{2h}^h (A^{2h})^{-1} I_h^{2h} \mathbf{r}^h(\mathbf{u}^h).$$

Subtracting the two equations yields

$$\mathbf{e}^h \leftarrow T_{\omega}^{\nu_1} \mathbf{e}^h - I_{2h}^h (A^{2h})^{-1} I_h^{2h} A^h T_{\omega}^{\nu_1} \mathbf{e}^h.$$

Similar arguments apply to step (TG-5) yield (11.23). $\square$

11.3.5 Multigrid cycles

Definition 11.32. As shown in Figure 11.4, the *V-cycle scheme*

$$\mathbf{v}^h \leftarrow \mathbf{VC}^h(\mathbf{v}^h, \mathbf{f}^h, \nu_1, \nu_2) \quad (11.24)$$

solves $\mathbf{A}\mathbf{u} = \mathbf{f}$ in (11.6) via steps as follows.

(VC-1) Relax ν_1 times on $A^h \mathbf{u}^h = \mathbf{f}^h$ with initial guess $\mathbf{v}^h$.

(VC-2) If Ω^h is the coarsest grid, solve the linear system and go to (VC-4), otherwise

$$\begin{aligned} \mathbf{f}^{2h} &\leftarrow I_h^{2h}(\mathbf{f}^h - A^h \mathbf{v}^h), \\ \mathbf{v}^{2h} &\leftarrow \mathbf{0}, \\ \mathbf{v}^{2h} &\leftarrow \mathbf{VC}^{2h}(\mathbf{v}^{2h}, \mathbf{f}^{2h}, \nu_1, \nu_2). \end{aligned}$$

(VC-3) Interpolate error back and correct the solution:

$$\mathbf{v}^h \leftarrow \mathbf{v}^h + I_{2h}^h \mathbf{v}^{2h}.$$

(VC-4) Relax ν_2 times on $A^h \mathbf{u}^h = \mathbf{f}^h$ with initial guess $\mathbf{v}^h$.

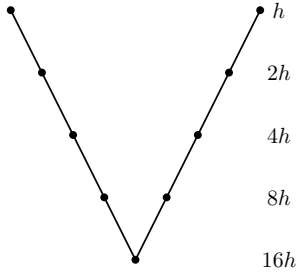


Figure 11.4: The V-cycle scheme with four recursions.

Lemma 11.33. In a D -dimensional domain with $n = 2^m$ cells ($m \in \mathbb{N}^+$) along each dimension, the storage cost of V-cycles is

$$2n^D (1 + 2^{-D} + 2^{-2D} + \dots + 2^{-mD}) < \frac{2n^D}{1 - 2^{-D}}. \quad (11.25)$$

Let WU denote the computational cost of performing one relaxation sweep on the finest grid. After neglecting the intergrid transfer, the computational cost of a single V-cycle with $\nu_1 = \nu_2 = 1$ is

$$2\text{WU} (1 + 2^{-D} + 2^{-2D} + \dots + 2^{-mD}) < \frac{2}{1 - 2^{-D}} \text{WU}. \quad (11.26)$$

Proof. On each grid, both vectors of errors and residuals must be stored, and this justifies the factor of 2 in (11.25); the rest of (11.25) follows from Definition 11.32. A similar argument yields (11.26). $\square$

Definition 11.34. As shown in Figure 11.5, the *full multigrid V-cycle*

$$\mathbf{v}^h \leftarrow \mathbf{FMG}^h(\mathbf{f}^h, \nu_1, \nu_2) \quad (11.27)$$

solves $\mathbf{A}\mathbf{u} = \mathbf{f}$ in (11.6) via steps as follows.

(FMG-1) If Ω^h is the coarsest grid, set $\mathbf{v}^h \leftarrow \mathbf{0}$ and go to (FMG-3), otherwise

$$\begin{aligned} \mathbf{f}^{2h} &\leftarrow I_h^{2h} \mathbf{f}^h, \\ \mathbf{v}^{2h} &\leftarrow \mathbf{FMG}^{2h}(\mathbf{f}^{2h}, \nu_1, \nu_2). \end{aligned}$$

(FMG-2) Correct $\mathbf{v}^h \leftarrow I_{2h}^h \mathbf{v}^{2h}$.

(FMG-3) Perform a V-cycle with the initial guess as $\mathbf{v}^h$:

$$\mathbf{v}^h \leftarrow \mathbf{VC}^h(\mathbf{v}^h, \mathbf{f}^h, \nu_1, \nu_2).$$

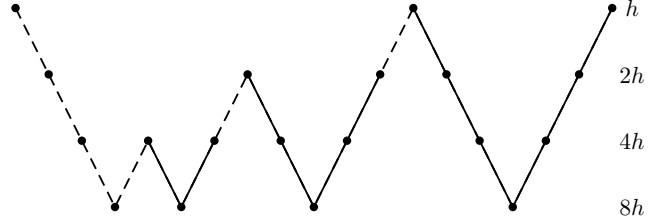


Figure 11.5: The full multigrid V-cycle with three levels of coarsening.

Exercise 11.35. Show that, for $\nu_1 = \nu_2 = 1$, the computational cost of an FMG cycle is less than $\frac{2}{(1-2^{-D})^2}$ WU. Give upper bounds as tight as possible for computational costs of an FMG cycle for $D = 1, 2, 3$.

11.4 Convergence analysis

11.4.1 The spectral picture

Definition 11.36. A Fourier mode $\mathbf{w}_k^h$ with $k \in [1, \frac{n}{2})$ and the mode $\mathbf{w}_{k'}^h$ with $k' = n - k$ are called *complementary modes* on Ω^h .

Lemma 11.37. A pair of complementary modes satisfy

$$w_{k',j}^h = (-1)^{j+1} w_{k,j}^h. \quad (11.28)$$

Proof. This follows from

$$w_{k',j}^h = \sin \frac{(n-k)j\pi}{n} = \sin \left(j\pi - \frac{kj\pi}{n} \right) = (-1)^{j+1} w_{k,j}^h. \quad \square$$

Lemma 11.38. The action of the full-weighting operator on a pair of complementary modes on Ω^h is

$$I_h^{2h} \mathbf{w}_k^h = c_k \mathbf{w}_k^{2h} := \cos^2 \frac{k\pi}{2n} \mathbf{w}_k^{2h}, \quad (11.29a)$$

$$I_h^{2h} \mathbf{w}_{k'}^h = -s_k \mathbf{w}_k^{2h} := -\sin^2 \frac{k\pi}{2n} \mathbf{w}_k^{2h}, \quad (11.29b)$$

where $k \in [1, \frac{n}{2})$, $k' = n - k$. In addition, $I_h^{2h} \mathbf{w}_{\frac{n}{2}}^h = \mathbf{0}$.

Proof. For the smooth mode k , we have

$$\begin{aligned} &(I_h^{2h} \mathbf{w}_k^h)_j \\ &= \frac{1}{4} \sin \frac{(2j-1)k\pi}{n} + \frac{1}{2} \sin \frac{2jk\pi}{n} + \frac{1}{4} \sin \frac{(2j+1)k\pi}{n} \\ &= \frac{1}{2} \left(1 + \cos \frac{k\pi}{n} \right) \sin \frac{2jk\pi}{n} = \cos^2 \frac{k\pi}{2n} w_{k,j}^{2h}, \end{aligned}$$

where the last step follows from Lemma 11.22. (11.29b) can be proved by similar steps by replacing k with $n - k$. $\square$

Lemma 11.39. The action of the linear interpolation operator on Ω^{2h} is

$$I_{2h}^h \mathbf{w}_k^{2h} = c_k \mathbf{w}_k^h - s_k \mathbf{w}_{k'}^h, \quad (11.30)$$

where $k' = n - k$.

Proof. Lemma 11.37 and trigonometric identities yield

$$\begin{aligned} c_k w_{k,j}^h - s_k w_{k',j}^h &= \left(\cos^2 \frac{k\pi}{2n} + (-1)^j \sin^2 \frac{k\pi}{2n} \right) w_{k,j}^h \\ &= \begin{cases} w_{k,j}^h & \text{if } j \text{ is even;} \\ \cos \frac{k\pi}{n} w_{k,j}^h & \text{if } j \text{ is odd.} \end{cases} \end{aligned}$$

On the other hand, by Definition 11.28, we have

$$(I_{2h}^h \mathbf{w}_k^{2h})_j = \begin{cases} w_{k,j}^h, & \text{if } j \text{ is even,} \\ \frac{1}{2} \sin \frac{k\pi(j-1)}{n} + \frac{1}{2} \sin \frac{k\pi(j+1)}{n} & \text{if } j \text{ is odd,} \end{cases}$$

where last expression simplifies to $\cos \frac{k\pi}{n} w_{k,j}^h$. $\square$

Theorem 11.40. The two-grid correction operator is invariant on the subspace $W_k^h = \text{span}\{\mathbf{w}_k^h, \mathbf{w}_{k'}^h\}$.

$$TG \mathbf{w}_k = \lambda_k^{\nu_1 + \nu_2} s_k \mathbf{w}_k + \lambda_k^{\nu_1} \lambda_{k'}^{\nu_2} s_k \mathbf{w}_{k'} \quad (11.31a)$$

$$TG \mathbf{w}_{k'} = \lambda_{k'}^{\nu_1} \lambda_k^{\nu_2} c_k \mathbf{w}_k + \lambda_{k'}^{\nu_1 + \nu_2} c_k \mathbf{w}_{k'}, \quad (11.31b)$$

where λ_k is the eigenvalue of T_ω .

Proof. Recall from (11.8) that $\frac{4}{h^2} s_k$ is the eigenvalue of A^h . Consider first the case of $\nu_1 = \nu_2 = 0$.

$$A^h \mathbf{w}_k^h = \frac{4s_k}{h^2} \mathbf{w}_k^h \quad (11.32a)$$

$$\Rightarrow I_h^{2h} A^h \mathbf{w}_k^h = \frac{4c_k s_k}{h^2} \mathbf{w}_k^{2h} \quad (11.32b)$$

$$\Rightarrow (A^{2h})^{-1} I_h^{2h} A^h \mathbf{w}_k^h = \frac{4c_k s_k}{h^2} \frac{(2h)^2}{4 \sin^2 \frac{k\pi}{n}} \mathbf{w}_k^{2h} = \mathbf{w}_k^{2h} \quad (11.32c)$$

$$\Rightarrow -I_{2h}^h (A^{2h})^{-1} I_h^{2h} A^h \mathbf{w}_k^h = -c_k \mathbf{w}_k^h + s_k \mathbf{w}_{k'}^h \quad (11.32d)$$

$$\Rightarrow [I - I_{2h}^h (A^{2h})^{-1} I_h^{2h} A^h] \mathbf{w}_k^h = s_k \mathbf{w}_k^h + s_k \mathbf{w}_{k'}^h. \quad (11.32e)$$

Similarly, we have

$$A^h \mathbf{w}_{k'}^h = \frac{4s_{k'}}{h^2} \mathbf{w}_{k'}^h = \frac{4c_k}{h^2} \mathbf{w}_{k'}^h \quad (11.33a)$$

$$\Rightarrow I_h^{2h} A^h \mathbf{w}_{k'}^h = -\frac{4c_k s_k}{h^2} \mathbf{w}_k^{2h} \quad (11.33b)$$

$$\Rightarrow (A^{2h})^{-1} I_h^{2h} A^h \mathbf{w}_{k'}^h = -\frac{4c_k s_k}{h^2} \frac{(2h)^2}{4 \sin^2 \frac{k\pi}{n}} \mathbf{w}_k^{2h} = -\mathbf{w}_k^{2h} \quad (11.33c)$$

$$\Rightarrow -I_{2h}^h (A^{2h})^{-1} I_h^{2h} A^h \mathbf{w}_{k'}^h = c_k \mathbf{w}_k^h - s_k \mathbf{w}_{k'}^h \quad (11.33d)$$

$$\Rightarrow (I - I_{2h}^h (A^{2h})^{-1} I_h^{2h} A^h) \mathbf{w}_{k'}^h = c_k \mathbf{w}_k^h + c_k \mathbf{w}_{k'}^h, \quad (11.33e)$$

where $c_k = s_{k'}$ is applied in (11.33a).

Adding pre-smoothing incurs a scaling of $\lambda_k^{\nu_1}$ for (11.32e) and $\lambda_{k'}^{\nu_1}$ for (11.33e). In contrast, adding post-smoothing incurs a scaling of $\lambda_k^{\nu_2}$ for $\mathbf{w}_k^h$ and a scaling of $\lambda_{k'}^{\nu_2}$ for $\mathbf{w}_{k'}^h$ in both (11.32e) and (11.33e). Hence (11.31) holds. $\square$

Exercise 11.41. Rewrite (11.31) as

$$TG \begin{bmatrix} \mathbf{w}_k \\ \mathbf{w}_{k'} \end{bmatrix} = \begin{bmatrix} \lambda_k^{\nu_1 + \nu_2} s_k & \lambda_k^{\nu_1} \lambda_{k'}^{\nu_2} s_k \\ \lambda_{k'}^{\nu_1} \lambda_k^{\nu_2} c_k & \lambda_{k'}^{\nu_1 + \nu_2} c_k \end{bmatrix} \begin{bmatrix} \mathbf{w}_k \\ \mathbf{w}_{k'} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} \begin{bmatrix} \mathbf{w}_k \\ \mathbf{w}_{k'} \end{bmatrix}. \quad (11.34)$$

Deduce the main conclusion $\rho(TG) \approx 0.1$ by reproducing the plots in Figure 11.6 of the damping coefficients of two-grid correction with weighted Jacobi for $n = 64$ and $\omega = \frac{2}{3}$. The horizontal axis represents the wavenumber k . Repeat the plots for $n = 128$ to show the independence of $\rho(TG) \approx 0.1$ from the grid size.

Hint: It is tricky to plot the coefficients defined in (11.34). Since c_2, c_4 act on HF modes, one has to ensure that the components in the vectors s_k and c_k indeed correspond to those in $\mathbf{w}_{k'}$. If s_k and c_k are computed from an increasing order of the frequencies, then their components will have to be reversed for plotting. Physical intuition helps in this case: c_1 and c_4 should form one curve while c_2 and c_3 should form another.

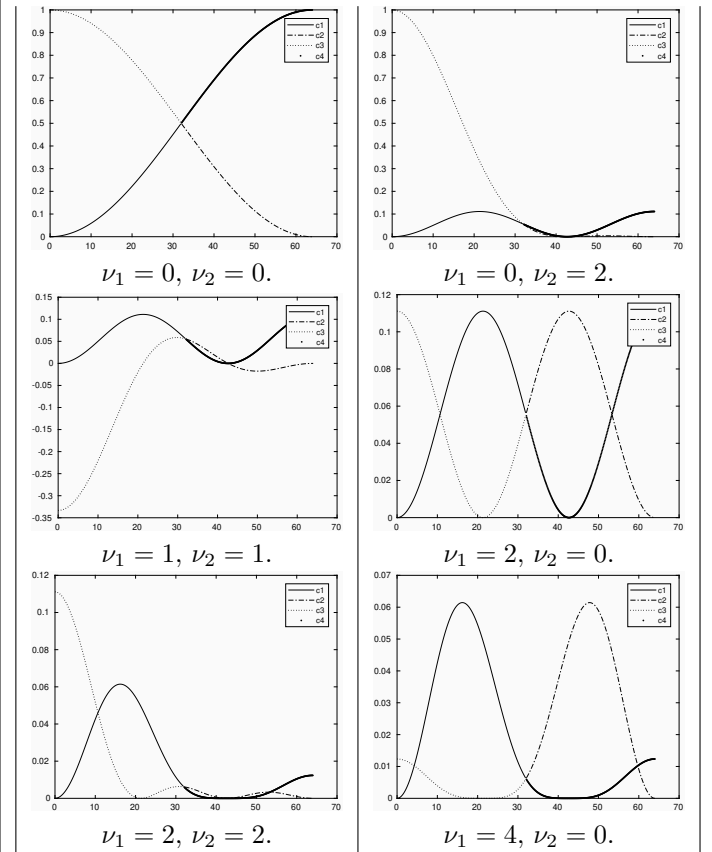


Figure 11.6: Damping coefficients of the two-grid correction operator in (11.34).

11.4.2 The algebraic picture

Lemma 11.42. The full-weighting operator and the linear interpolation operator satisfy the *variational properties*

$$I_{2h}^h = c(I_h^{2h})^\top, \quad (11.35a)$$

$$I_h^{2h} A^h I_{2h}^h = A^{2h}, \quad (11.35b)$$

where $c = 2$ and the property (11.35b) is also called the *Galerkin condition*.

Proof. The conclusions follow from (11.7) and Definitions 11.25 and 11.28. $\square$

Lemma 11.43. A basis for $\mathcal{R}(I_{2h}^h)$, the range of the linear interpolation operator, is given by its columns. The range and the null space of I_{2h}^h satisfy

$$\dim \mathcal{R}(I_{2h}^h) = \frac{n}{2} - 1, \quad \mathcal{N}(I_{2h}^h) = \{\mathbf{0}\}. \quad (11.36)$$

Proof. Since $\mathcal{R}(I_{2h}^h) = \{I_{2h}^h \mathbf{v}^{2h} : \mathbf{v}^{2h} \in \Omega^{2h}\}$, the maximum dimension of $\mathcal{R}(I_{2h}^h)$ is $\frac{n}{2} - 1$. Write $\mathbf{v}^{2h} = \sum v_j^{2h} \mathbf{e}_j^{2h}$ and we have $I_{2h}^h \mathbf{v}^{2h} = \sum v_j^{2h} I_{2h}^h \mathbf{e}_j^{2h}$, where $I_{2h}^h \mathbf{e}_j^{2h}$'s are columns of the matrix of I_{2h}^h and thus linearly independent; see Example 11.29 and Figure 11.7. $\square$

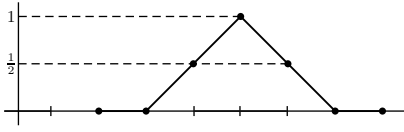


Figure 11.7: A typical basis vector of the range of linear interpolation.

Lemma 11.44. The full-weighting operator satisfies

$$\dim \mathcal{R}(I_h^{2h}) = \frac{n}{2} - 1, \quad \dim \mathcal{N}(I_h^{2h}) = \frac{n}{2}. \quad (11.37)$$

Exercise 11.45. Prove Lemma 11.44.

Lemma 11.46. The null space of the full-weighting operator is given by

$$\mathcal{N}(I_h^{2h}) = \text{span}\{A^h \mathbf{e}_j^h : j \text{ is odd}\}, \quad (11.38)$$

where $\mathbf{e}_j^h$ is the j th unit vector on Ω^h .

Proof. Consider $I_h^{2h} A^h$. The j th row of I_h^{2h} has $2(j-1)$ leading zeros and the next three nonzero entries are $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Since the bandwidth of A^h is 3, it suffices to consider only five columns of A^h for potentially non-zero dot-product $\sum_i (I_h^{2h})_{ji} (A^h)_{ik}$. For $2j \pm 1$, these dot products are zero; for $2j$, the dot product is $\frac{1}{2}$; for $2j \pm 2$, the dot product is $-\frac{1}{4}$. Hence for any odd j , we have $I_h^{2h} A^h \mathbf{e}_j^h = \mathbf{0}$. $\square$

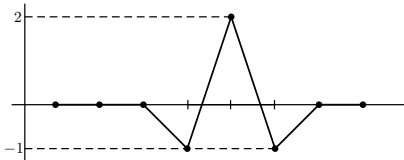


Figure 11.8: A typical basis vector of the null space of full weighting.

Theorem 11.47. The null space of the two-grid correction operator (without relaxation) is the range of linear interpolation:

$$\mathcal{N}(TG) = \mathcal{R}(I_{2h}^h). \quad (11.39)$$

Proof. If $\mathbf{s}^h \in \mathcal{R}(I_{2h}^h)$, then there exists some $\mathbf{q}^{2h} \in \mathbb{R}^{\frac{n}{2}-1}$ satisfying $\mathbf{s}^h = I_{2h}^h \mathbf{q}^{2h}$.

$$TGS^h = [I - I_{2h}^h (A^{2h})^{-1} I_{2h}^{2h} A^h] I_{2h}^h \mathbf{q}^{2h} = \mathbf{0},$$

where the last step comes from (11.35b). Hence we have $\mathcal{R}(I_{2h}^h) \subseteq \mathcal{N}(TG)$. Furthermore, $\mathbf{t}^h \in \mathcal{N}(I_h^{2h} A^h)$ implies

$$TG\mathbf{t}^h = [I - I_{2h}^h (A^{2h})^{-1} I_{2h}^{2h} A^h] \mathbf{t}^h = \mathbf{t}^h,$$

i.e., TG is the identity operator when acting on $\mathcal{N}(I_h^{2h} A^h)$. Apply the fundamental theorem of linear maps (Theorem C.59) to TG and we have

$$n - 1 = \dim \mathcal{N}(TG) + \dim \mathcal{R}(TG).$$

TG being an identity operator on $\mathcal{N}(I_h^{2h} A^h)$ implies

$$\dim \mathcal{R}(TG) \geq \dim \mathcal{N}(I_h^{2h} A^h) = \dim \mathcal{N}(I_h^{2h}) = \frac{n}{2},$$

where the second step follows from A^h having full rank on $\mathbb{R}^{n-1}$ and the last step from Lemma 11.44. The above two equations imply $\dim \mathcal{N}(TG) \leq \frac{n}{2} - 1$, which, together with Lemma 11.43 and $\mathcal{R}(I_{2h}^h) \subseteq \mathcal{N}(TG)$, yields (11.39). $\square$

Definition 11.48. Let A be an $n \times n$ symmetric positive definite matrix. The A -inner product or energy inner product of two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is defined as

$$\langle \mathbf{u}, \mathbf{v} \rangle_A := \langle A\mathbf{u}, \mathbf{v} \rangle, \quad (11.40)$$

where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product on $\mathbb{R}^n$. Naturally, the A -norm or energy norm is defined as

$$\|\mathbf{u}\|_A := \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle_A}. \quad (11.41)$$

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are A -orthogonal iff $\langle \mathbf{u}, \mathbf{v} \rangle_A = 0$.

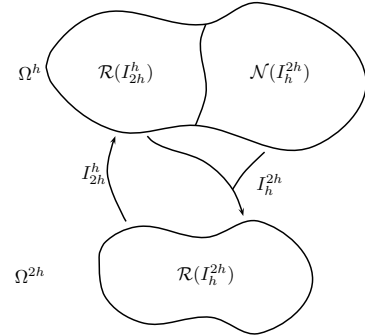


Figure 11.9: The subspace decomposition of the grid functions Ω^h and effects of intergrid operators.

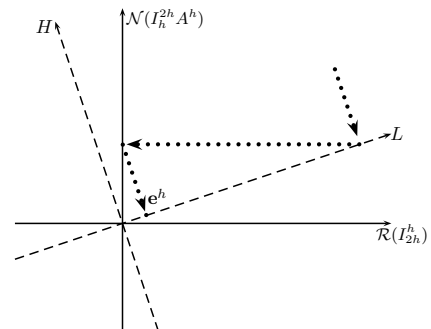


Figure 11.10: Interpret residue reduction via projections in multigrid methods.

11.4.3 The optimal complexity of FMG

Definition 11.49. The error of a computed value v_i^h from the corresponding exact solution $u(x_i)$ is

$$E_i^h := v_i^h - u(x_i) = v_i^h - u_i^h + u_i^h - u(x_i).$$

The *discretization error* is the error $u_i^h - u(x_i)$ incurred by truncating the Taylor series of exact values and the *algebraic error* is the error $v_i^h - u_i^h$ incurred by inexact solution of the linear system.

Lemma 11.50. When linearly interpolating errors from a coarse grid to the fine grid, we have

$$\sqrt{c} \|\mathbf{v}^{2h} - \mathbf{u}^{2h}\|_{A^{2h}} = \|I_{2h}^h \mathbf{v}^{2h} - I_{2h}^h \mathbf{u}^{2h}\|_{A^h}, \quad (11.42)$$

where $c \in \mathbb{R}^+$ is the constant in (11.35).

Proof. Definition 11.48 and Lemma 11.42 yield

$$\begin{aligned} \|\mathbf{v}^{2h} - \mathbf{u}^{2h}\|_{A^{2h}}^2 &= \langle A^{2h}(\mathbf{v}^{2h} - \mathbf{u}^{2h}), \mathbf{v}^{2h} - \mathbf{u}^{2h} \rangle \\ &= \langle I_h^{2h} A^h I_{2h}^h (\mathbf{v}^{2h} - \mathbf{u}^{2h}), \mathbf{v}^{2h} - \mathbf{u}^{2h} \rangle \\ &= \left\langle A^h I_{2h}^h (\mathbf{v}^{2h} - \mathbf{u}^{2h}), \frac{1}{c} I_{2h}^h (\mathbf{v}^{2h} - \mathbf{u}^{2h}) \right\rangle \\ &= \frac{1}{c} \|I_{2h}^h \mathbf{v}^{2h} - I_{2h}^h \mathbf{u}^{2h}\|_{A^h}^2, \end{aligned}$$

where the second step and the third step follow from Lemma 11.42. $\square$

Lemma 11.51. Suppose there exists a constant $K \in \mathbb{R}^+$ independent of the grid size h such that

$$\|I_{2h}^h \mathbf{u}^{2h} - \mathbf{u}^h\|_{A^h} \leq Kh^p. \quad (11.43)$$

Then a single FMG cycle in Definition 11.34 reduces the algebraic error from $O(1)$ to $O(h^p)$, i.e.,

$$\|\mathbf{e}^h\|_{A^h} \leq Kh^p, \quad (11.44)$$

where $p = 2$ is the order of accuracy of the discrete Laplacian (11.7).

Proof. We prove (11.44) by induction. On the coarsest grid, FMG is exact and thus (11.44) holds for the induction basis.

For the induction hypothesis, we assume that the linear system on Ω^{2h} has been solved to the level of discretization error so that

$$(*) : \quad \|\mathbf{e}^{2h}\|_{A^{2h}} \leq K(2h)^p.$$

We need to show that $(*)$ implies (11.44).

The initial algebraic error on Ω^h is

$$\mathbf{e}_0^h = I_{2h}^h \mathbf{v}^{2h} - \mathbf{u}^h,$$

which yields

$$\begin{aligned} \|\mathbf{e}_0^h\|_{A^h} &\leq \|I_{2h}^h \mathbf{v}^{2h} - I_{2h}^h \mathbf{u}^{2h}\|_{A^h} + \|I_{2h}^h \mathbf{u}^{2h} - \mathbf{u}^h\|_{A^h} \\ &= \sqrt{c} \|\mathbf{v}^{2h} - \mathbf{u}^{2h}\|_{A^{2h}} + \|I_{2h}^h \mathbf{u}^{2h} - \mathbf{u}^h\|_{A^h} \\ &\leq \sqrt{c} K(2h)^p + Kh^p = (1 + \sqrt{c} 2^p) Kh^p, \end{aligned}$$

where the second step follows from Lemma 11.50 and the third step from (11.43) and the induction hypothesis. Then we have $1 + \sqrt{c} 2^p < 7$ from $p = 2$ and $c = 2$. Exercise 11.41 states that $\rho(TG) \approx 0.1$ for a VC(2,1)-cycle and hence one V-cycle is enough to reduce $\|\mathbf{e}_0^h\|_{A^h}$ to less than Kh^p . $\square$

Theorem 11.52. For the FD discretization (in Example 11.8) of the model problem in Definition 11.7, a single FMG cycle is of the complexity $O(\frac{1}{h})$ and is sufficient to achieve second-order accuracy.

Proof. By Definition 11.49, we have

$$\|\mathbf{E}^h\| \leq \|\mathbf{u}^h - \mathbf{u}\| + \|\mathbf{u}^h - \mathbf{v}^h\|.$$

Taylor expansion yields $\|\mathbf{u}^h - \mathbf{u}\| = O(h^2)$. Then the proof is completed by Exercise 11.35 and Lemma 11.51. $\square$

Summary 11.53. Geometric multigrid methods are optimal for solving BVPs on regular domains with structured grids. For irregular domains, algebraic multigrid methods with unstructured grids are very well developed; see [Briggs et al., 2000, chapter 9] for an introduction. On the other hand, irregular domains can be alternatively handled by coupling geometric multigrid methods with cut-cell methods to efficiently solve the linear system of BVPs; see Liu et al. [2026].

11.5 Programming assignments

Write a C++ package to implement the one-dimensional multigrid method discussed in this chapter to solve the model problem in Definition 11.7.

I. Your package must give the user the following options:

- (a) boundary conditions: Dirichlet, Neumann, or mixed (partly Dirichlet and partly Neumann).
- (b) restriction operators: full weighting and injection;
- (c) interpolation operators: linear and quadratic;
- (d) cycles: V-cycle and FMG;
- (e) stopping criteria: the number of maximum iterations and the relative accuracy ϵ of the solution;
- (f) the initial guess.

As for the bottom solver, you can either implement a Gaussian elimination in your own package or use the one in BLAS or LAPACK.

II. For the function in (10.94), derive the corresponding $f(x, y)$ and the boundary conditions in two dimensions and set $y = x$ to obtain $f(x)$ in one dimension. For $\epsilon = 10^{-8}$ and the zero-vector initial guess, test your multigrid solver for all combinations of (b,c,d) in I on grids with $n = 32, 64, 128, 256$, report the residual and the reduction rate of the residuals for each V-cycle. Report the maximum norm of the error vector and the corresponding convergence rates on the four grids. You should also design at least two of your own test functions and carry out the same process.

III. Gradually reduce ϵ towards 2.2×10^{-16} , under which critical value of ϵ does your program fail to achieve the preset accuracy? Why?

The requirements III-VI in Section 10.6 should also be met in this assignment.

Chapter 12

Linear Multistep Methods (LMMs) for Initial Value Problems (IVPs)

Definition 12.1. A system of ordinary differential equations (ODEs) of dimension N is a set of differential equations of the form

$$\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}(t), t), \quad (12.1)$$

where t is time, $\mathbf{u} \in \mathbb{R}^N$ is the evolutionary variable, and the RHS function has the signature $\mathbf{f} : \mathbb{R}^N \times (0, +\infty) \rightarrow \mathbb{R}^N$. In particular, (12.1) is an ODE for $N = 1$.

Definition 12.2. A system of ODEs is *linear* if its RHS function can be expressed as $\mathbf{f}(\mathbf{u}, t) = A(t)\mathbf{u} + \beta(t)$, and *nonlinear* otherwise; it is *homogeneous* if it is linear and $\beta(t) = \mathbf{0}$; it is *autonomous* if $\mathbf{f}$ does not depend on t explicitly; and *nonautonomous* otherwise. A linear ODE system is further said to have *constant coefficient* if the matrix A is independent on time.

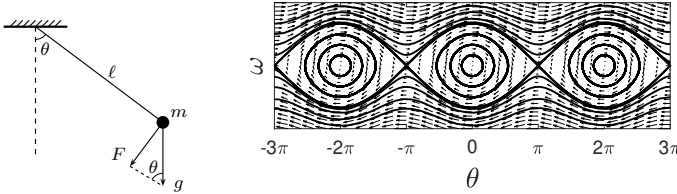


Figure 12.1: A simple pendulum, the RHS vector of the ODE that governs its dynamics, and the solution trajectories in its phase space.

Example 12.3. For the simple pendulum shown in the LHS of Figure 12.1, the moment of inertial and the torque are

$$I = m\ell^2, \quad \tau = -mg\ell \sin \theta,$$

and the equation of motion can be derived from Newton's second law for rotation $\tau = I\theta''(t)$ as

$$\theta''(t) = -\frac{g}{\ell} \sin \theta, \quad (12.2)$$

which admits a unique solution if we impose two initial conditions

$$\theta(0) = \theta_0, \quad \theta'(0) = \omega_0.$$

Alternatively, (12.2) can be derived from the fact that the total energy remains a constant with respect to time.

$$L = \frac{1}{2}m(\ell\theta')^2 + mg\ell(1 - \cos \theta);$$

$$\frac{dL}{dt} = 0 \Rightarrow m\ell^2\theta'\theta'' + mg\ell\theta' \sin \theta = 0.$$

The ODE in (12.2) is autonomous, nonlinear, and second-order; it can be reduced to a first-order ODE system,

$$\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}) := \begin{pmatrix} \omega \\ -\frac{g}{\ell} \sin \theta \end{pmatrix},$$

where $\omega = \theta'$ and $\mathbf{u} = (\theta, \omega)^\top$. Its solutions are illustrated in the RHS of Figure 12.1. Any point in the set

$$\{(\theta, \omega) : \omega = 0; \theta = 2n\pi, n \in \mathbb{Z}\}$$

is a *stable fixed point* to which a small perturbation changes the solution from the fixed point to a small circle close to the point. In contrast, any point in the set

$$\{(\theta, \omega) : \omega = 0; \theta = (2n + 1)\pi, n \in \mathbb{Z}\}$$

is an *unstable fixed point* to which a small perturbation changes the solution from the fixed point to another unstable fixed point far away from the original one.

If the initial kinetic energy is large enough, the pendulum will forever circulate around the fixture; accordingly, the solution will follow the wavy curve with θ increasing indefinitely. On the other hand, if the initial kinetic energy is not large enough, the pendulum will oscillate in a manner symmetric to the vertical line of $\theta = 0$; accordingly, the solution will follow the simple closed curves. An open curve that separates these two cases connects two unstable fixed points. For more on dynamical systems, see Guckenheimer and Holmes [1983] or Wiggins [2003].

Definition 12.4. Given $T > 0$, $\mathbf{f} : \mathbb{R}^N \times [0, T] \rightarrow \mathbb{R}^N$, and $\mathbf{u}_0 \in \mathbb{R}^N$, the *initial value problem* (IVP) is to find $\mathbf{u}(t) \in \mathcal{C}^1$ such that

$$\begin{cases} \mathbf{u}(0) = \mathbf{u}_0, \\ \forall t \in [0, T], \quad \mathbf{u}'(t) = \mathbf{f}(\mathbf{u}(t), t). \end{cases} \quad (12.3)$$

12.1 Mathematical foundation

12.1.1 Existence and uniqueness of solution

Definition 12.5. A function $\mathbf{f} : \mathbb{R}^N \times [0, +\infty) \rightarrow \mathbb{R}^N$ is *Lipschitz continuous* in its first variable over some domain

$$\mathcal{D} = \{(\mathbf{u}, t) : \|\mathbf{u} - \mathbf{u}_0\| \leq a, t \in [0, T]\} \quad (12.4)$$

iff

$$\exists L \geq 0 \text{ s.t. } \forall (\mathbf{u}, t), (\mathbf{v}, t) \in \mathcal{D}, \quad \|\mathbf{f}(\mathbf{u}, t) - \mathbf{f}(\mathbf{v}, t)\| \leq L \|\mathbf{u} - \mathbf{v}\|. \quad (12.5)$$

Example 12.6. If $\mathbf{f}(\mathbf{u}, t) = \mathbf{f}(t)$, then $L = 0$.

Example 12.7. If $\mathbf{f}(\mathbf{u}, t)$ is not continuous in $\mathbf{u}$, then $\mathbf{f}$ is not Lipschitz continuous in $\mathbf{u}$.

Theorem 12.8 (Fundamental theorem of ODEs). If $\mathbf{f}(\mathbf{u}(t), t)$ is Lipschitz continuous in $\mathbf{u}$ and continuous in t over some region $\mathcal{D} = \{(\mathbf{u}, t) : \|\mathbf{u} - \mathbf{u}_0\| \leq a, t \in [0, T]\}$, then there is a unique solution to the IVP in Definition 12.4 at least up to time

$$T^* = \min \left(T, \frac{a}{S} \right), \quad \text{where } S = \max_{(\mathbf{u}, t) \in \mathcal{D}} \|\mathbf{f}(\mathbf{u}, t)\|. \quad (12.6)$$

Proof. Referring to Section D.1.4, we denote by (M, ρ) the complete metric space of continuous functions $\mathbf{u} : [0, T] \rightarrow \mathbb{R}^N$ with $\mathbf{u}(0) = \mathbf{u}_0$ and with the metric

$$(*) : \quad \rho(\mathbf{u}, \mathbf{v}) = \sup_{t \in [0, T]} \exp(-Kt) \|\mathbf{u}(t) - \mathbf{v}(t)\|$$

where $K > L$.

For a given $\mathbf{u} \in M$, define $\phi(\mathbf{u})$ as the solution $\mathbf{U}$ on $[0, T]$ to the IVP in Definition 12.4, which is solvable by integration as

$$\phi(\mathbf{u})(t) = \mathbf{u}_0 + \int_0^t \mathbf{f}(\mathbf{u}(s), s) ds. \quad (12.7)$$

ϕ is a contractive mapping because $\forall \mathbf{u}, \mathbf{v} \in M$,

$$\begin{aligned} & \rho(\phi(\mathbf{u}), \phi(\mathbf{v})) \\ &= \sup_{t \in [0, T]} \exp(-Kt) \left\| \int_0^t (\mathbf{f}(\mathbf{u}(s), s) - \mathbf{f}(\mathbf{v}(s), s)) ds \right\| \\ &\leq \sup_{t \in [0, T]} \exp(-Kt) \int_0^t \|\mathbf{f}(\mathbf{u}(s), s) - \mathbf{f}(\mathbf{v}(s), s)\| ds \\ &\leq L \sup_{t \in [0, T]} \exp(-Kt) \int_0^t \|\mathbf{u}(s) - \mathbf{v}(s)\| ds \\ &= L \sup_{t \in [0, T]} \exp(-Kt) \int_0^t \exp(-Ks) \|\mathbf{u}(s) - \mathbf{v}(s)\| \exp(Ks) ds \\ &\leq L \rho(\mathbf{u}, \mathbf{v}) \sup_{t \in [0, T]} \exp(-Kt) \int_0^t \exp(Ks) ds \\ &\leq \frac{L}{K} \rho(\mathbf{u}, \mathbf{v}), \end{aligned}$$

where the third step follows from Definition 12.5 and the fifth from (*). Then Theorem D.120 completes the proof. $\square$

Example 12.9. Consider $N = 1$, $u'(t) = \sqrt{u(t)}$, $u(0) = 0$.

$$\lim_{u \rightarrow 0} f'(u) = \lim_{u \rightarrow 0} \frac{1}{2\sqrt{u}} = +\infty.$$

Hence $f(u)$ is not Lipschitz continuous near $u = 0$. However, $u(t) \equiv 0$ and $u(t) = \frac{1}{4}t^2$ are both solutions. Hence the Lipschitz condition is not necessary for the existence of solutions.

Example 12.10. Consider the IVP $u'(t) = u^2$ with $u(0) = \eta > 0$. The slope $f'(u) = 2u$ goes to $+\infty$ as $u \rightarrow +\infty$. So there is no guarantee of a unique solution on $[0, +\infty)$, but there might exist T^* such that unique solutions are guaranteed on $[0, T^*]$.

In fact, $u(t) = \frac{1}{\eta - 1 - t}$ is a solution, but blows up at $t = \frac{1}{\eta}$. According to Theorem 12.8, $f(u) = u^2$ and we would like to maximize $\frac{a}{S}$. Since $S = \max_{\mathcal{D}} |f(u)| = (\eta + a)^2$, it is equivalent to find $\min_{a > 0} \frac{1}{a}(\eta + a)^2$:

$$\frac{1}{a}(\eta + a)^2 = 2\eta + a + \frac{\eta^2}{a} \geq 2\eta + 2\sqrt{\eta^2} = 4\eta.$$

Hence $T^* = \frac{1}{4\eta}$. The estimation of T^* in Theorem 12.8 is thus quite conservative for this case.

Example 12.11. For the simple pendulum in Example 12.3, we have

$$|\sin \theta - \sin \theta^*| \leq |\theta - \theta^*| \leq \|\mathbf{u} - \mathbf{u}^*\|_\infty$$

since $\cos \theta^* \leq 1$. In addition, we have $|\omega - \omega^*| \leq \|\mathbf{u} - \mathbf{u}^*\|_\infty$.

$$\begin{aligned} \|\mathbf{f}(\mathbf{u}) - \mathbf{f}(\mathbf{u}^*)\|_\infty &= \max \left(|\omega - \omega^*|, \frac{g}{\ell} |\sin \theta - \sin \theta^*| \right) \\ &\leq \max \left(\frac{g}{\ell}, 1 \right) \|\mathbf{u} - \mathbf{u}^*\|_\infty. \end{aligned}$$

Therefore, $\mathbf{f}$ is Lipschitz continuous with $L = \max(\frac{g}{\ell}, 1)$.

12.1.2 Well-posedness

Definition 12.12. The IVP in Definition 12.4 is *well-posed* iff

- (i) it admits a unique solution for any $t \in [0, T]$,
- (ii) for the perturbed IVP

$$\mathbf{v}' = \mathbf{f}(\mathbf{v}, t) + \boldsymbol{\delta}(t), \quad \mathbf{v}(0) = \mathbf{u}_0 + \boldsymbol{\epsilon}_0, \quad (12.8)$$

there exists a constant $\hat{\epsilon} > 0$ such that for all $\epsilon < \hat{\epsilon}$, the inequalities $\|\boldsymbol{\epsilon}_0\| < \epsilon$ and $\|\boldsymbol{\delta}(t)\| < \epsilon$ imply that (12.8) admits a unique solution for any $t \in [0, T]$ and

$$\exists C > 0 \text{ s.t. } \forall t \in [0, T], \quad \|\mathbf{u}(t) - \mathbf{v}(t)\| \leq C\epsilon,$$

where C depends on $\mathbf{f}$ and $\mathbf{u}_0$ but not on ϵ .

Theorem 12.13. If $\mathbf{f}(\mathbf{u}, t)$ is Lipschitz continuous in $\mathbf{u}$ and continuous in t on $\mathcal{D} = \{(\mathbf{u}, t) : \mathbf{u} \in \mathbb{R}^N, t \in [0, T]\}$, then the IVP in Definition 12.4 is well-posed.

Proof. Theorem 12.8 has already established the existence and uniqueness of the solution of the IVP. It remains to prove that the solution is overly sensitive neither to the initial condition nor to the RHS $\mathbf{f}(\mathbf{u}, t)$. To this end, we consider two IVPs

$$\begin{cases} \mathbf{v}' = \mathbf{f}(\mathbf{v}, t) \\ \mathbf{v}(0) = \mathbf{v}_0 \end{cases} \quad \text{and} \quad \begin{cases} \mathbf{w}' = \mathbf{f}(\mathbf{w}, t) + \boldsymbol{\delta}(t) \\ \mathbf{w}(0) = \mathbf{w}_0 \end{cases}$$

where $\mathbf{v}_0$, $\mathbf{w}_0$ and $\boldsymbol{\delta}(t)$ satisfy the conditions

$$(*) : \begin{cases} \forall t \in [0, T], \quad \|\boldsymbol{\delta}(t)\| < \epsilon \\ \|\mathbf{v}_0 - \mathbf{w}_0\| < \epsilon \end{cases},$$

with ϵ being a small positive constant. Obviously, the two IVPs are respectively solved by

$$\begin{aligned} \mathbf{v}(t) &= \mathbf{v}_0 + \int_0^t \mathbf{f}(\mathbf{v}(s), s) ds, \\ \mathbf{w}(t) &= \mathbf{w}_0 + \int_0^t [\mathbf{f}(\mathbf{w}(s), s) + \boldsymbol{\delta}(s)] ds. \end{aligned}$$

Thus we have

$$\mathbf{v}(t) - \mathbf{w}(t) = \mathbf{v}_0 - \mathbf{w}_0 + \int_0^t [\mathbf{f}(\mathbf{v}(s), s) - \mathbf{f}(\mathbf{w}(s), s) - \boldsymbol{\delta}(s)] ds.$$

Take the 2-norm on both sides and we have

$$\begin{aligned} & \|\mathbf{v}(t) - \mathbf{w}(t)\| \\ &= \left\| \mathbf{v}_0 - \mathbf{w}_0 + \int_0^t [\mathbf{f}(\mathbf{v}(s), s) - \mathbf{f}(\mathbf{w}(s), s) - \boldsymbol{\delta}(s)] ds \right\| \\ &\leq \|\mathbf{v}_0 - \mathbf{w}_0\| + \left\| \int_0^t [\mathbf{f}(\mathbf{v}(s), s) - \mathbf{f}(\mathbf{w}(s), s) - \boldsymbol{\delta}(s)] ds \right\| \\ &\leq \|\mathbf{v}_0 - \mathbf{w}_0\| + \int_0^t \|\mathbf{f}(\mathbf{v}(s), s) - \mathbf{f}(\mathbf{w}(s), s) - \boldsymbol{\delta}(s)\| ds \\ &\leq \|\mathbf{v}_0 - \mathbf{w}_0\| + \int_0^t \|\mathbf{f}(\mathbf{v}(s), s) - \mathbf{f}(\mathbf{w}(s), s)\| + \|\boldsymbol{\delta}(s)\| ds \\ &\leq \epsilon + \int_0^t L \|\mathbf{v}(s) - \mathbf{w}(s)\| ds + \int_0^t \epsilon ds \\ &= (1+t)\epsilon + \int_0^t L \|\mathbf{v}(s) - \mathbf{w}(s)\| ds, \end{aligned}$$

where the fifth step follows from (*). To proceed with

$$(**) : \|\mathbf{v}(t) - \mathbf{w}(t)\| \leq (1+t)\epsilon + \int_0^t L \|\mathbf{v}(s) - \mathbf{w}(s)\| ds,$$

we define a function $h : [0, T] \rightarrow \mathbb{R}$,

$$h(s) = e^{-sL} \int_0^s L \|\mathbf{v}(r) - \mathbf{w}(r)\| dr,$$

of which the derivative is

$$h'(s) = Le^{-sL} \left(\|\mathbf{v}(s) - \mathbf{w}(s)\| - \int_0^s L \|\mathbf{v}(r) - \mathbf{w}(r)\| dr \right).$$

Consequently, we have

$$\begin{aligned} & e^{-tL} \int_0^t L \|\mathbf{v}(r) - \mathbf{w}(r)\| dr \\ &= h(t) = h(t) - h(0) = \int_0^t h'(s) ds \\ &= \int_0^t Le^{-sL} (\|\mathbf{v}(s) - \mathbf{w}(s)\| - \int_0^s L \|\mathbf{v}(r) - \mathbf{w}(r)\| dr) ds \\ &\leq \int_0^t Le^{-sL} (1+s)\epsilon ds \\ \Rightarrow & \int_0^t L \|\mathbf{v}(r) - \mathbf{w}(r)\| dr \leq \int_0^t Le^{L(t-s)} (1+s)\epsilon ds, \end{aligned}$$

where the first step follows from the definition of $h(t)$, the second from $h(0) = 0$ and the fifth from (**). Substitute the above inequality to (**) and we have

$$\begin{aligned} \|\mathbf{v}(t) - \mathbf{w}(t)\| &\leq (1+t)\epsilon + \int_0^t Le^{L(t-s)} (1+s)\epsilon ds \\ &= \epsilon + \epsilon(1 + \frac{1}{L})(e^{Lt} - 1). \end{aligned}$$

The proof is then completed by Definition 12.12 where the constant is chosen as $C := 1 + (1 + \frac{1}{L})(e^{LT} - 1)$. $\square$

Exercise 12.14. Show that the solution of IVP is not overly sensitive to the initial condition of problems that satisfy a Lipschitz condition. In other words, if both $\mathbf{v}$ and $\mathbf{w}$ satisfy the *same* IVP with initial condition $\mathbf{v}(a) = \mathbf{v}_0$ and $\mathbf{w}(a) = \mathbf{w}_0$, then we have

$$\|\mathbf{v}(t) - \mathbf{w}(t)\| \leq \|\mathbf{v}_0 - \mathbf{w}_0\| \exp(L(t-a)).$$

12.1.3 Constant-coefficient linear IVPs

Theorem 12.15 (Duhamel's principle). For a linear IVP

$$\mathbf{u}'(t) = A\mathbf{u} + \mathbf{g}(t) \quad (12.9)$$

with a time-independent matrix A , the solution is

$$\mathbf{u}(t) = e^{tA}\mathbf{u}_0 + \int_0^t e^{(t-\tau)A}\mathbf{g}(\tau) d\tau. \quad (12.10)$$

Proof. (12.10) is verified by the Leibniz formula

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, y) dy = \int_a^b \frac{\partial}{\partial x} f(x, y) dy - f(x, a) \frac{da}{dx} + f(x, b) \frac{db}{dx}$$

and Lemma C.339. $\square$

Example 12.16. Many linear problems are naturally formulated in the form of a single high-order ODE

$$v^{(m)}(t) - \sum_{j=1}^m c_j v^{(m-j)}(t) = \phi(t). \quad (12.11)$$

By setting $u_j(t) = v^{(j-1)}(t)$ and $\mathbf{u} = [u_1, u_2, \dots, u_m]^\top$, we can rewrite (12.11) as

$$\mathbf{u}'(t) = A\mathbf{u} + \mathbf{g}(t), \quad (12.12)$$

where $\mathbf{g}(t) = [0, \dots, 0, \phi(t)]^\top$ and

$$a_{ij} = \begin{cases} 1 & \text{if } i = j - 1 < m, \\ c_{m+1-j} & \text{if } i = m, \\ 0 & \text{otherwise.} \end{cases}$$

If A is diagonalizable, say $A = X^{-1}\Lambda X$, then we can define $\mathbf{v} = X\mathbf{u}$ and rewrite (12.9) as

$$\mathbf{v}'(t) = \Lambda\mathbf{v} + X\mathbf{g}(t),$$

which consists of a number of decoupled scalar IVPs whose solutions are given by Theorem 12.15.

Example 12.17. The matrix of the linear IVP system

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} \quad (12.13)$$

is not diagonalizable. For v we have $v(t) = v(0)e^{\lambda t}$. For u we consider the form of the solution

$$u = (E + Ft)e^{\lambda t};$$

then (12.13) yields $E = u(0)$ and $F = v(0)$, i.e.,

$$u(t) = u(0)e^{\lambda t} + tv(t).$$

12.2 Basic numerical methods

Notation 9. To numerically solve the IVP (12.3), we are given initial condition $\mathbf{U}^0 = \mathbf{u}_0$, and want to compute approximations $\mathbf{U}^1, \mathbf{U}^2, \dots$ such that

$$\mathbf{U}^n \approx \mathbf{u}(t_n),$$

where k is the uniform time-step size and $t_n = nk$.

Definition 12.18. The *(forward) Euler method* solves the IVP (12.3) by

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k\mathbf{f}(\mathbf{U}^n, t_n), \quad (12.14)$$

which is based on replacing $\mathbf{u}'(t_n)$ with the forward difference $(\mathbf{U}^{n+1} - \mathbf{U}^n)/k$ and $\mathbf{u}(t_n)$ with $\mathbf{U}^n$ in $\mathbf{f}(\mathbf{u}, t)$.

Definition 12.19. The *backward Euler method* solves the IVP (12.3) by

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k\mathbf{f}(\mathbf{U}^{n+1}, t_{n+1}), \quad (12.15)$$

which is based on replacing $\mathbf{u}'(t_{n+1})$ with the backward difference $(\mathbf{U}^{n+1} - \mathbf{U}^n)/k$ and $\mathbf{u}(t_{n+1})$ with $\mathbf{U}^{n+1}$ in $\mathbf{f}(\mathbf{u}, t)$.

Definition 12.20. The *trapezoidal method* is

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \frac{k}{2} (\mathbf{f}(\mathbf{U}^n, t_n) + \mathbf{f}(\mathbf{U}^{n+1}, t_{n+1})). \quad (12.16)$$

Definition 12.21. The *midpoint (or leapfrog) method* is

$$\mathbf{U}^{n+1} = \mathbf{U}^{n-1} + 2k\mathbf{f}(\mathbf{U}^n, t_n). \quad (12.17)$$

Example 12.22. For the IVP

$$u'(t) = u, \quad u(0) = 1, \quad t \in [0, 1],$$

the Euler method (12.14) with step size $k = 0.2$ yields the following results.

n	U^n	$k\mathbf{f}(U^n, t_n)$
0	1	0.2
1	1.2	$0.2 \times 1.2 = 0.24$
2	1.44	$0.2 \times 1.44 = 0.288$
3	1.728	$0.2 \times 1.728 = 0.3456$
4	2.0736	$0.2 \times 2.0736 = 0.41472$
5	2.48832	

12.2.1 Truncation and solution errors

Definition 12.23. The *local truncation error* (LTE) is the error caused by replacing continuous derivatives with finite difference formulas.

Example 12.24. The LTE of the leapfrog method is

$$\begin{aligned} \tau^n &= \frac{\mathbf{u}(t_{n+1}) - \mathbf{u}(t_{n-1})}{2k} - \mathbf{f}(\mathbf{u}(t_n), t_n) \\ &= \left[\mathbf{u}'(t_n) + \frac{1}{6}k^2\mathbf{u}'''(t_n) + O(k^4) \right] - \mathbf{u}'(t_n) \\ &= \frac{1}{6}k^2\mathbf{u}'''(t_n) + O(k^4), \end{aligned}$$

where $\mathbf{v} = O(k^4)$ for a vector function $\mathbf{v} : \mathbb{R} \rightarrow \mathbb{R}^n$ means that each component of $\mathbf{v}$ is $O(k^4)$, cf. Notation 19. Hereafter we use this asymptotic notation without further ado.

Definition 12.25. For a numerical method of the form

$$\mathbf{U}^{n+1} = \phi(\mathbf{U}^{n+1}, \mathbf{U}^n, \dots, \mathbf{U}^{n-m}), \quad (12.18)$$

the *one-step error* is defined by

$$\mathcal{L}^n := \mathbf{u}(t_{n+1}) - \phi(\mathbf{u}(t_{n+1}), \mathbf{u}(t_n), \dots, \mathbf{u}(t_{n-m})). \quad (12.19)$$

In other words, $\mathcal{L}^n$ is the error that would be introduced in one time step if the values $\mathbf{U}^{n+1}, \mathbf{U}^n, \mathbf{U}^{n-1}, \dots$ were all taken to be the exact values from $\mathbf{u}(t)$.

Example 12.26. The one-step error of the leapfrog method is

$$\begin{aligned} \mathcal{L}^n &= \mathbf{u}(t_{n+1}) - \mathbf{u}(t_{n-1}) - 2k\mathbf{f}(\mathbf{u}(t_n), t_n) = 2k\tau^n \\ &= \frac{1}{3}k^3\mathbf{u}'''(t_n) + O(k^5). \end{aligned}$$

Definition 12.27. The *solution error* of a numerical method for solving the IVP in Definition 12.4 is

$$\mathbf{E}^N := \mathbf{U}^{T/k} - \mathbf{u}(T); \quad \mathbf{E}^n = \mathbf{U}^n - \mathbf{u}(t_n). \quad (12.20)$$

12.2.2 Convergence of the Euler method

Definition 12.28. A numerical method of (12.18) is *convergent* iff its application to *any* IVP with $\mathbf{f}(\mathbf{u}, t)$ Lipschitz continuous in $\mathbf{u}$ and continuous in t yields

$$\forall T > 0, \quad \lim_{\substack{k \rightarrow 0 \\ Nk=T}} \mathbf{U}^N = \mathbf{u}(T). \quad (12.21)$$

Lemma 12.29. Consider a linear IVP of the form

$$\begin{cases} \mathbf{u}'(t) = \lambda\mathbf{u}(t) + \mathbf{g}(t), \\ \mathbf{u}(0) = \mathbf{u}_0, \end{cases} \quad (12.22)$$

where λ is either a scalar or a diagonal matrix. The solution error and the LTE of the Euler method satisfy

$$\mathbf{E}^{n+1} = (1 + k\lambda)\mathbf{E}^n - k\tau^n. \quad (12.23)$$

Proof. Definition 12.23 yields

$$\begin{aligned} \tau^n &= \frac{\mathbf{u}(t_{n+1}) - \mathbf{u}(t_n)}{k} - \mathbf{u}'(t_n) \\ &= \frac{\mathbf{u}(t_{n+1}) - \mathbf{u}(t_n)}{k} - (\lambda\mathbf{u}(t_n) + \mathbf{g}(t_n)), \end{aligned}$$

which gives $\mathbf{u}(t_{n+1}) = (1 + k\lambda)\mathbf{u}(t_n) + k\mathbf{g}(t_n) + k\boldsymbol{\tau}^n$. The Euler method applied to the linear IVP (12.22) reads

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k(\lambda\mathbf{U}^n + \mathbf{g}(t_n)) = (1 + k\lambda)\mathbf{U}^n + k\mathbf{g}(t_n).$$

Subtracting the above two equations yields (12.23). $\square$

Lemma 12.30. For the linear IVP (12.22), the solution errors and the LTEs of the Euler method satisfy

$$\mathbf{E}^n = (1 + k\lambda)^n \mathbf{E}^0 - k \sum_{m=1}^n (1 + k\lambda)^{n-m} \boldsymbol{\tau}^{m-1}. \quad (12.24)$$

Proof. We proceed by induction on n .

The induction basis holds because of (12.23). Suppose (12.24) holds for all integers no greater than n . Then for $n + 1$, we have

$$\begin{aligned} \mathbf{E}^{n+1} &= (1 + k\lambda)\mathbf{E}^n - k\boldsymbol{\tau}^n \\ &= (1 + k\lambda)^{n+1} \mathbf{E}^0 - k \sum_{m=1}^{n+1} (1 + k\lambda)^{n+1-m} \boldsymbol{\tau}^{m-1}, \end{aligned}$$

where the first equality follows from (12.23) and the second from the induction hypothesis. $\square$

Theorem 12.31. With $\|\mathbf{E}^0\| = O(k)$, the Euler method is convergent for solving the linear IVP (12.22).

Proof. $|1 + k\lambda| \leq 1 + |k\lambda| \leq e^{k|\lambda|}$ implies $m < n \leq \frac{T}{k}$ and

$$(1 + k\lambda)^{n-m} \leq e^{(n-m)k|\lambda|} \leq e^{nk|\lambda|} \leq e^{|\lambda|T}.$$

Then Lemma 12.30 yields

$$\begin{aligned} \|\mathbf{E}^n\| &\leq e^{|\lambda|T} (\|\mathbf{E}^0\| + k \sum_{m=1}^n \|\boldsymbol{\tau}^{m-1}\|) \\ &\leq e^{|\lambda|T} (\|\mathbf{E}^0\| + nk \max_{1 \leq m \leq n} \|\boldsymbol{\tau}^{m-1}\|). \end{aligned}$$

The LTE of the Euler method is

$$\boldsymbol{\tau}^n = \frac{\mathbf{u}(t_{n+1}) - \mathbf{u}(t_n)}{k} - \mathbf{u}'(t_n) = \frac{1}{2}k\mathbf{u}''(t_n) + O(k^2)$$

and hence $\|\mathbf{E}^N\| \leq e^{|\lambda|T} (\|\mathbf{E}^0\| + TO(k)) = O(k)$. $\square$

Lemma 12.32. Consider a nonlinear IVP of the form

$$\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}(t), t),$$

where $\mathbf{f}(\mathbf{u}, t)$ is continuous in t and is Lipschitz continuous in $\mathbf{u}$ with L as the Lipschitz constant. The Euler method satisfies

$$\|\mathbf{E}^{n+1}\| \leq (1 + kL)\|\mathbf{E}^n\| + k\|\boldsymbol{\tau}^n\|. \quad (12.25)$$

Proof. The definition of the LTE yields

$$\boldsymbol{\tau}^n = \frac{\mathbf{u}(t_{n+1}) - \mathbf{u}(t_n)}{k} - \mathbf{f}(\mathbf{u}(t_n), t_n),$$

and hence

$$\mathbf{u}(t_{n+1}) = \mathbf{u}(t_n) + k\mathbf{f}(\mathbf{u}(t_n), t_n) + k\boldsymbol{\tau}^n,$$

the Euler method is

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k\mathbf{f}(\mathbf{U}^n, t_n),$$

subtracting the above two equations gives

$$\mathbf{E}^{n+1} = \mathbf{E}^n + k(\mathbf{f}(\mathbf{U}^n, t_n) - \mathbf{f}(\mathbf{u}(t_n), t_n)) - k\boldsymbol{\tau}^n,$$

the triangle inequality and Lipschitz continuity of $\mathbf{f}$ yield

$$\begin{aligned} \|\mathbf{E}^{n+1}\| &\leq \|\mathbf{E}^n\| + k\|\mathbf{f}(\mathbf{U}^n, t_n) - \mathbf{f}(\mathbf{u}(t_n), t_n)\| + k\|\boldsymbol{\tau}^n\| \\ &\leq (1 + kL)\|\mathbf{E}^n\| + k\|\boldsymbol{\tau}^n\|. \end{aligned} \quad \square$$

Theorem 12.33. For the nonlinear IVP in Lemma 12.32, the Euler method is convergent.

Proof. Follow the procedures of linear IVPs to show that

$$\|\mathbf{E}^N\| \leq e^{LT} (\|\mathbf{E}^0\| + T\|\boldsymbol{\tau}\|) = O(k). \quad \square$$

12.2.3 Zero stability and absolute stability

Definition 12.34. A numerical method is *stable* or *zero-stable* iff its application to *any* IVP with $\mathbf{f}(\mathbf{u}, t)$ Lipschitz continuous in $\mathbf{u}$ and continuous in t yields

$$\forall T > 0, \quad \lim_{\substack{k \rightarrow 0 \\ Nk=T}} \|\mathbf{U}^N\| < \infty. \quad (12.26)$$

Example 12.35. Consider the scalar IVP

$$u'(t) = \lambda(u - \cos t) - \sin t,$$

with $\lambda = -2100$ and $u(0) = 1$. The exact solution is clearly

$$u(t) = \cos t.$$

The following table shows the error at time $T = 2$ when the Euler method is used with various values of k .

k	$E(T)$
2.00e-4	1.98e-8
4.00e-4	3.96e-8
8.00e-4	7.92e-8
9.50e-4	3.21e-7
9.76e-4	5.88e+35
1.00e-3	1.45e+76

The first three lines confirm the first-order accuracy of the Euler method, but something dramatic happens between $k = 9.76\text{e-}4$ and $k = 9.50\text{e-}4$. What's going on?

Definition 12.36. The Euler method

$$U^{n+1} = (1 + k\lambda)U^n$$

for solving the scalar IVP

$$u'(t) = \lambda u(t) \quad (12.27)$$

is *absolutely stable* or has *eigenvalue stability* iff

$$|1 + k\lambda| \leq 1 + O(k). \quad (12.28)$$

Definition 12.37. The *region of absolute stability* (RAS) for the forward Euler method applied to (12.27) is the set of all points

$$\{z \in \mathbb{C} : |1 + z| \leq 1 + O(k)\}, \quad (12.29)$$

where $z := k\lambda$ and λ is a function of k ; see Figure 12.2.

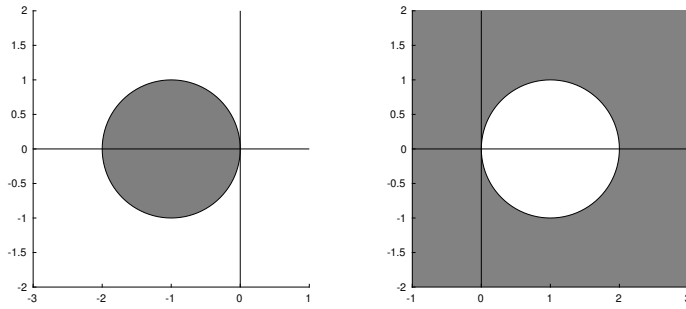


Figure 12.2: Regions of absolute stability (RAS) for the forward Euler (left) and the backward Euler (right).

Example 12.38. The backward Euler method for (12.27) reads

$$U^{n+1} = U^n + k\lambda U^{n+1} \Rightarrow U^{n+1} = \frac{1}{1 - k\lambda} U^n.$$

Hence the region of absolute stability for the backward Euler method is

$$\{z \in \mathbb{C} : |1 - z| \geq 1\}; \quad (12.30)$$

see the RHS of Figure 12.2.

Lemma 12.39. Consider an autonomous, homogeneous, and linear system of IVPs

$$\mathbf{u}'(t) = A\mathbf{u} \quad (12.31)$$

where $\mathbf{u} \in \mathbb{R}^N$, $N > 1$, and A is diagonalizable with eigenvalues as λ_i 's. The Euler method is absolutely stable if and only if each $z_i := k\lambda_i$ is within the stability region (12.29).

Proof. For the sufficiency, the Euler method for (12.31) reads

$$\mathbf{U}^{n+1} = \mathbf{U}^n + kA\mathbf{U}^n = (I + kA)\mathbf{U}^n.$$

Since A is diagonalizable, we have $AR = R\Lambda$ where R contains the eigenvectors of A that span $\mathbb{R}^N$. Then

$$R^{-1}\mathbf{U}^{n+1} = R^{-1}(I + kA)RR^{-1}\mathbf{U}^n.$$

Set $\mathbf{V} := R^{-1}\mathbf{U}$ and we have

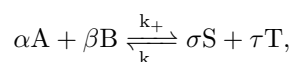
$$\mathbf{V}^{n+1} = (I + k\Lambda)\mathbf{V}^n.$$

After advancing $\mathbf{V}^0$ to $\mathbf{V}^n$, we use $\mathbf{U}^n = R\mathbf{V}^n$ to recover the solution of (12.31).

Since A is diagonalizable, the eigenvectors of A span the domain of A . Hence the necessity follows. $\square$

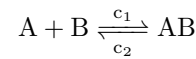
Definition 12.40. The *law of mass action* states that the rate of a chemical reaction is proportional to the product of the concentration of the reacting substances, with each concentration raised to a power equal to the coefficient that occurs in the reaction.

Example 12.41. For the reaction



the forward reaction rate is $k_+[A]^\alpha[B]^\beta$ and the backward reaction rate is $k_-[S]^\sigma[T]^\tau$.

Example 12.42. Consider



with $c_1, c_2 > 0$. Let

$$\mathbf{u} := \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} [A] \\ [B] \\ [AB] \end{bmatrix}.$$

By Example 12.41, we have

$$\begin{aligned} u_1' &= -c_1 u_1 u_2 + c_2 u_3; \\ u_2' &= -c_1 u_1 u_2 + c_2 u_3; \\ u_3' &= c_1 u_1 u_2 - c_2 u_3, \end{aligned}$$

which can be written more compactly as

$$\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}).$$

Let $\mathbf{v}(t) := \mathbf{u}(t) - \bar{\mathbf{u}}$ with $\bar{\mathbf{u}}$ independent of time. Then

$$\begin{aligned} \mathbf{v}'(t) &= \mathbf{u}'(t) = \mathbf{f}(\mathbf{u}(t)) = \mathbf{f}(\mathbf{v} + \bar{\mathbf{u}}) \\ &= \mathbf{f}(\bar{\mathbf{u}}) + \mathbf{f}'(\bar{\mathbf{u}})\mathbf{v}(t) + O(\|\mathbf{v}\|^2), \end{aligned}$$

and hence

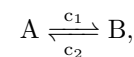
$$\mathbf{v}'(t) \approx A\mathbf{v}(t) + \mathbf{b},$$

where $A = \mathbf{f}'(\bar{\mathbf{u}})$ is the Jacobian matrix, i.e.,

$$A = \begin{bmatrix} -c_1 u_2 & -c_1 u_1 & c_2 \\ -c_1 u_2 & -c_1 u_1 & c_2 \\ c_1 u_2 & c_1 u_1 & -c_2 \end{bmatrix},$$

with eigenvalues $\lambda_1 = -c_1(u_1 + u_2) - c_2$ and $\lambda_2 = \lambda_3 = 0$. As the total concentration of species A and B , the time-dependent quantity $u_1 + u_2 \in \mathbb{R}^+$ is bounded by the constant $M_0 := u_1(0) + u_2(0) + 2u_3(0)$. Therefore, any choice of k that satisfies $k < \frac{2}{c_1 M_0 + c_2}$ guarantees absolute stability for the Euler method because $u_1(t), u_2(t), u_3(t) \geq 0$ and $u_1(t) + u_2(t) \leq M$ always hold.

Example 12.43. For the reaction



we obtain the linear IVPs

$$\begin{cases} u_1' = -c_1 u_1 + c_2 u_2; \\ u_2' = c_1 u_1 - c_2 u_2. \end{cases}$$

Formula 12.44. A general way of reducing an IVP

$$\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}, t)$$

to a collection of scalar, linear model problems of the form

$$w_i'(t) = \lambda_i w_i(t), \quad i = 1, 2, \dots, n$$

consists of steps as follows.

- (a) Linearization: at the neighborhood of a particular solution $\mathbf{u}^*(t)$, we write $\mathbf{u}(t) = \mathbf{u}^*(t) + (\delta\mathbf{u})(t)$ and apply Taylor expansion

$$\mathbf{f}(\mathbf{u}, t) = \mathbf{f}(\mathbf{u}^*, t) + J(t)\delta\mathbf{u} + O(\|\delta\mathbf{u}\|^2)$$

to obtain an ODE system on $\delta\mathbf{u}$:

$$(\delta\mathbf{u})'(t) \approx J(t)(\delta\mathbf{u}).$$

- (b) Freezing coefficients: set $A = J(t^*)$ where t^* is the particular time of interest.
- (c) Diagonalization: assume A is diagonalizable by V and we write $(\delta\mathbf{u})'(t) = V(V^{-1}AV)V^{-1}(\delta\mathbf{u})$. Define $\mathbf{w} := V^{-1}(\delta\mathbf{u})$ and we have a decoupled system of scalar IVPs,

$$\mathbf{w}'(t) = \Lambda\mathbf{w}(t),$$

where $\Lambda = V^{-1}AV$ is a diagonal matrix.

12.3 The generic form of LMMs

Definition 12.45. For solving the IVP (12.3), an s -step linear multistep method (LMM) has the form

$$\sum_{j=0}^s \alpha_j \mathbf{U}^{n+j} = k \sum_{j=0}^s \beta_j \mathbf{f}(\mathbf{U}^{n+j}, t_{n+j}), \quad (12.32)$$

where $\alpha_s = 1$ is assumed WLOG.

Adams-Bashforth		Adams-Moulton		Nyström		Generalized Milne-Simpson		Backward Differentiation	
α_j	β_j	α_j	β_j	α_j	β_j	α_j	β_j	α_j	β_j
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	○

Figure 12.3: Classic formulas of linear multistep methods.

Definition 12.46. An LMM (12.32) is *explicit* if $\beta_s = 0$ and is *implicit* otherwise.

Definition 12.47. An *Adams formula* is an LMM of the form

$$\mathbf{U}^{n+s} = \mathbf{U}^{n+s-1} + k \sum_{j=0}^s \beta_j \mathbf{f}(\mathbf{U}^{n+j}, t_{n+j}), \quad (12.33)$$

where β_j 's are chosen to maximize the order of accuracy.

Definition 12.48. An *Adams-Bashforth formula* is an Adams formula with $\beta_s = 0$. An *Adams-Moulton formula* is an Adams formula with $\beta_s \neq 0$.

Example 12.49. The Euler method is the 1-step Adams-Bashforth formula with

$$s = 1, \alpha_1 = 1, \alpha_0 = -1, \beta_1 = 0, \beta_0 = 1,$$

where we have repeated known values of Adams-Bashforth formula for the ease of computation in Section 12.4.1.

Example 12.50. The trapezoidal method is a 1-step Adams-Moulton formula with

$$s = 1, \alpha_1 = 1, \alpha_0 = -1, \beta_1 = \beta_0 = \frac{1}{2}.$$

Another 1-step Adams-Moulton formula is the backward Euler method.

Definition 12.51. A *Nyström formula* is an LMM of the form

$$\mathbf{U}^{n+s} = \mathbf{U}^{n+s-2} + k \sum_{j=0}^{s-1} \beta_j \mathbf{f}(\mathbf{U}^{n+j}, t_{n+j}), \quad (12.34)$$

where β_j 's are chosen to give order s .

Example 12.52. The midpoint method is the 2-step Nyström formula with

$$s = 2, \alpha_2 = 1, \alpha_1 = 0, \alpha_0 = -1, \beta_1 = 2, \beta_0 = \beta_2 = 0.$$

Definition 12.53. A *backward differentiation formula* (BDF) is an LMM of the form

$$\mathbf{U}^{n+s} + \sum_{j=0}^{s-1} \alpha_j \mathbf{U}^{n+j} = k \beta_s \mathbf{f}(\mathbf{U}^{n+s}, t_{n+s}), \quad (12.35)$$

where α_j 's and β_s are chosen to give order s .

Example 12.54. The backward Euler method is the 1-step BDF with

$$s = 1, \alpha_1 = \beta_1 = 1, \alpha_0 = -1.$$

12.4 Analysis of LMMs

12.4.1 Consistency and accuracy

Definition 12.55. The *characteristic polynomials* or *generating polynomials* for the LMM (12.32) are

$$\rho(\zeta) := \sum_{j=0}^s \alpha_j \zeta^j; \quad \sigma(\zeta) := \sum_{j=0}^s \beta_j \zeta^j. \quad (12.36)$$

Example 12.56. The forward Euler method (12.14) has

$$\rho(\zeta) = \zeta - 1, \quad \sigma(\zeta) = 1, \quad (12.37)$$

while the backward Euler method (12.15) has

$$\rho(\zeta) = \zeta - 1, \quad \sigma(\zeta) = \zeta. \quad (12.38)$$

Example 12.57. The trapezoidal method (12.16) has

$$\rho(\zeta) = \zeta - 1, \quad \sigma(\zeta) = \frac{1}{2}(\zeta + 1), \quad (12.39)$$

and the midpoint method (12.17) has

$$\rho(\zeta) = \zeta^2 - 1, \quad \sigma(\zeta) = 2\zeta. \quad (12.40)$$

Notation 10. Denote by Z a *time shift operator* that acts on both discrete functions according to

$$Z\mathbf{U}^n := \mathbf{U}^{n+1} \quad (12.41)$$

and on analytic functions according to

$$Z\mathbf{u}(t) := \mathbf{u}(t+k). \quad (12.42)$$

Definition 12.58. The *difference operator of an LMM* is an operator $\mathcal{L}$ on the linear space of continuously differentiable functions given by

$$\mathcal{L} := \rho(Z) - k\mathcal{D}\sigma(Z), \quad (12.43)$$

where $\mathcal{D}\mathbf{u}(t_n) = \mathbf{u}_t(t_n) := \frac{d\mathbf{u}}{dt}(t_n)$, Z is the time shift operator, and ρ, σ are characteristic polynomials for the LMM.

Lemma 12.59. The one-step error of the LMM (12.32) is

$$\mathcal{L}\mathbf{u}(t_n) = C_0\mathbf{u}(t_n) + C_1k\mathbf{u}_t(t_n) + C_2k^2\mathbf{u}_{tt}(t_n) + \cdots, \quad (12.44)$$

where

$$\begin{aligned} C_0 &= \sum_{j=0}^s \alpha_j \\ C_1 &= \sum_{j=0}^s (j\alpha_j - \beta_j) \\ C_2 &= \sum_{j=0}^s \left(\frac{1}{2}j^2\alpha_j - j\beta_j\right) \\ &\vdots \\ C_q &= \sum_{j=0}^s \left(\frac{1}{q!}j^q\alpha_j - \frac{1}{(q-1)!}j^{q-1}\beta_j\right). \end{aligned} \quad (12.45)$$

Proof. Definitions 12.25, 12.58, and 12.55 yield

$$\begin{aligned} \mathcal{L}\mathbf{u}(t_n) &= \sum_{j=0}^s \alpha_j \mathbf{u}(t_{n+j}) - k \sum_{j=0}^s \beta_j \mathbf{f}(\mathbf{u}(t_{n+j}), t_{n+j}) \\ &= \sum_{j=0}^s \alpha_j \mathbf{u}(t_{n+j}) - k \sum_{j=0}^s \beta_j \mathbf{u}'(t_{n+j}). \end{aligned}$$

Taylor expansions yield

$$\begin{aligned} \mathbf{u}(t_{n+j}) &= \mathbf{u}(t_n) + jk\mathbf{u}'(t_n) + \frac{1}{2}(jk)^2\mathbf{u}''(t_n) + \cdots \\ \mathbf{u}'(t_{n+j}) &= \mathbf{u}'(t_n) + jk\mathbf{u}''(t_n) + \frac{1}{2}(jk)^2\mathbf{u}'''(t_n) + \cdots \end{aligned}$$

Substitution of the above into $\mathcal{L}\mathbf{u}(t_n)$ yields (12.44). $\square$

Definition 12.60. An LMM has *order of accuracy* p iff

$$\forall \mathbf{u} \in \mathcal{C}^{p+1}, \quad \mathcal{L}\mathbf{u}(t_n) = \Theta(k^{p+1}), \quad (12.46)$$

i.e., if in (12.45) we have $C_0 = C_1 = \cdots = C_p = 0$ and $C_{p+1} \neq 0$. Then C_{p+1} is called the *error constant*.

Definition 12.61. An LMM is *preconsistent* if $\rho(1) = 0$, i.e., $\sum_{i=0}^s \alpha_i = 0$ or $\sum_{i=0}^{s-1} \alpha_i = -1$.

Definition 12.62. An LMM is *consistent* if it has order of accuracy $p \geq 1$.

Example 12.63. For the Euler method, the coefficients C_j 's in (12.45) can be computed directly from Example 12.49 as $C_0 = C_1 = 0, C_2 = \frac{1}{2}, C_3 = \frac{1}{6}$.

Exercise 12.64. Compute the first five coefficients C_j 's of the trapezoidal rule and the midpoint rule from Examples 12.50 and 12.52.

Example 12.65. A necessary condition for $\|\mathbf{E}^n\| = O(k)$ is $\|\mathcal{L}\mathbf{u}(t_n)\| = O(k^2)$ since there are $\frac{T}{k}$ time steps and hence C_0 and C_1 in (12.45) should be zero, i.e.,

$$\sum_{j=0}^s \alpha_j = 0, \quad \sum_{j=0}^s j\alpha_j = \sum_{j=0}^s \beta_j, \quad (12.47)$$

which is equivalent to

$$\rho(1) = 0 \quad \text{and} \quad \rho'(1) = \sigma(1). \quad (12.48)$$

In addition, if second-order accuracy is desired, we need

$$\frac{1}{2} \sum_{j=0}^s j^2 \alpha_j = \sum_{j=0}^s j \beta_j.$$

In general, p th-order accuracy requires (12.47) and

$$\forall q = 2, \dots, p, \quad \sum_{j=0}^s \frac{1}{q!} j^q \alpha_j = \sum_{j=0}^s \frac{1}{(q-1)!} j^{q-1} \beta_j. \quad (12.49)$$

Exercise 12.66. Express conditions of $\|\mathcal{L}\mathbf{u}(t_n)\| = O(k^3)$ using characteristic polynomials.

Exercise 12.67. Derive coefficients of LMMs shown below by the method of undetermined coefficients and a programming language with symbolic computation such as **Matlab**.

Adams-Bashforth formulas in Definition 12.48

s	p	β_s	β_{s-1}	β_{s-2}	β_{s-3}	β_{s-4}
1	1	0	1			
2	2	0	$\frac{3}{2}$	$-\frac{1}{2}$		
3	3	0	$\frac{23}{12}$	$-\frac{16}{12}$	$\frac{5}{12}$	
4	4	0	$\frac{55}{24}$	$-\frac{59}{24}$	$\frac{37}{24}$	$-\frac{9}{24}$

Adams-Moulton formulas in Definition 12.48

s	p	β_s	β_{s-1}	β_{s-2}	β_{s-3}	β_{s-4}
1	1	1				
1	2	$\frac{1}{2}$	$\frac{1}{2}$			
2	3	$\frac{5}{12}$	$\frac{8}{12}$	$-\frac{1}{12}$		
3	4	$\frac{9}{24}$	$\frac{19}{24}$	$-\frac{5}{24}$	$\frac{1}{24}$	
4	5	$\frac{251}{720}$	$\frac{646}{720}$	$-\frac{264}{720}$	$\frac{106}{720}$	$-\frac{19}{720}$

Backward differentiation formulas in Definition 12.53

s	p	α_s	α_{s-1}	α_{s-2}	α_{s-3}	α_{s-4}	β_s
1	1	1	-1				1
2	2	1	$-\frac{4}{3}$	$\frac{1}{3}$			$\frac{2}{3}$
3	3	1	$-\frac{18}{11}$	$\frac{9}{11}$	$-\frac{2}{11}$		$\frac{6}{11}$
4	4	1	$-\frac{48}{25}$	$\frac{36}{25}$	$-\frac{16}{25}$	$\frac{3}{25}$	$\frac{12}{25}$

Example 12.68. To derive coefficients of the 2nd-order Adams-Bashforth formula, we interpolate $\mathbf{f}(t)$ by a linear polynomial

$$q(t) = \mathbf{f}^{n+1} - \frac{t_{n+1} - t}{k}(\mathbf{f}^{n+1} - \mathbf{f}^n)$$

and then calculate

$$\mathbf{U}^{n+2} - \mathbf{U}^{n+1} = \int_{t_{n+1}}^{t_{n+2}} q(t) dt = \frac{3}{2} k \mathbf{f}^{n+1} - \frac{1}{2} k \mathbf{f}^n.$$

Lemma 12.69. An LMM with $\sigma(1) \neq 0$ has order of accuracy p if and only if

$$\frac{\rho(e^\kappa)}{\sigma(e^\kappa)} = \kappa + \Theta(\kappa^{p+1}), \quad (12.50)$$

where $\kappa := k\mathcal{D}$, the linear operator $\mathcal{D}$ represents the (well-posed) IVP, and the asymptotic symbol Θ in Notation 19 is generalized to linear operators by the operator norm in Definition C.288 in the asymptotic range of $\kappa \rightarrow \mathbf{0}$.

Proof. By Taylor's theorem,

$$\mathbf{u}(t_{n+1}) = \mathbf{u}(t_n) + k\mathbf{u}_t(t_n) + \frac{1}{2}k^2\mathbf{u}_{tt}(t_n) + \dots$$

By Notation 10, we also have $\mathbf{u}(t_{n+1}) = Z\mathbf{u}(t_n)$. A comparison of the two equalities yields

$$Z = 1 + (k\mathcal{D}) + \frac{1}{2!}(k\mathcal{D})^2 + \dots + \frac{1}{n!}(k\mathcal{D})^n + \dots = e^{k\mathcal{D}},$$

where the last step follows from Definition C.330. Set $\kappa := k\mathcal{D}$ and we have

$$\mathcal{L} = \rho(e^\kappa) - \kappa\sigma(e^\kappa) = \Theta(k^{p+1}) = \Theta(\kappa^{p+1}),$$

where the first step follows from Definition 12.58 and the second from Definition 12.60 and the LMM having order of accuracy p . Since $\sigma(e^\kappa)$ is an analytic function of k and is nonzero at $k = 0$, we divide by it the second and the last terms of the above equation to get (12.50). $\square$

Theorem 12.70. An LMM with $\sigma(1) \neq 0$ has order of accuracy p if and only if

$$\begin{aligned} \frac{\rho(z)}{\sigma(z)} &= \log z + \frac{C_{p+1}}{\sigma(1)}(z-1)^{p+1} + O((z-1)^{p+2}) \\ &= \left[(z-1) - \frac{1}{2}(z-1)^2 + \frac{1}{3}(z-1)^3 - \dots \right] \\ &\quad + \Theta((z-1)^{p+1}). \end{aligned} \quad (12.51)$$

as $z \rightarrow 1$, where C_{p+1} is defined in (12.45).

Proof. We only prove the case of scalar IVPs. To get from (12.50) to the first equality, we make the change of variables $z = e^\kappa$, $\kappa = \log z$, noting that $\Theta(\kappa^{p+1})$ as $\kappa \rightarrow \mathbf{0}$ has the same meaning as $\Theta((z-1)^{p+1})$ as $z \rightarrow 1$ since $e^\kappa = 1$ and $d(e^\kappa)/d\kappa \neq 0$ at $\kappa = \mathbf{0}$. The second equality is just the usual Taylor series for $\log z$ at 1. $\square$

Example 12.71. The trapezoidal rule has $\rho(z) = z-1$ and $\sigma(z) = \frac{1}{2}(z+1)$. A comparison of (12.51) with the expansion

$$\frac{\rho(z)}{\sigma(z)} = \frac{z-1}{\frac{1}{2}(z+1)} = (z-1) \left[1 - \frac{z-1}{2} + \frac{(z-1)^2}{4} - \dots \right]$$

confirms that the trapezoidal rule has order 2 with error constant $-\frac{1}{12}$.

Exercise 12.72. For the third-order BDF in Definition 12.53 and Exercise 12.67, derive its characteristic polynomials and apply Theorem 12.70 to verify that the order of accuracy is indeed 3.

Exercise 12.73. Prove that an s -step LMM has order of accuracy p if and only if, when applied to an ODE $u_t = q(t)$, it gives exact results whenever q is a polynomial of degree $< p$, but not whenever q is a polynomial of degree p . Assume arbitrary continuous initial condition u_0 and exact numerical initial data $v^0, \dots, v^{s-1}$.

12.4.2 Zero stability

Example 12.74 (A consistent but unstable LMM). The LMM

$$\mathbf{U}^{n+2} - 3\mathbf{U}^{n+1} + 2\mathbf{U}^n = -k\mathbf{f}(\mathbf{U}^n, t_n) \quad (12.52)$$

has a one-step error given by

$$\begin{aligned} \mathcal{L}\mathbf{u}(t_n) &= \mathbf{u}(t_{n+2}) - 3\mathbf{u}(t_{n+1}) + 2\mathbf{u}(t_n) + k\mathbf{u}'(t_n) \\ &= \frac{1}{2}k^2\mathbf{u}''(t_n) + O(k^3), \end{aligned}$$

so the method is consistent with first-order accuracy. But the solution error may not exhibit first order accuracy, or even convergence. For the trivial IVP

$$u'(t) = 0, \quad u(0) = 0$$

with its solution as $u(t) \equiv 0$, the LMM (12.52) reads

$$U^{n+2} = 3U^{n+1} - 2U^n \Rightarrow U^{n+2} - U^{n+1} = 2(U^{n+1} - U^n),$$

and therefore $U^n = 2U^0 - U^1 + 2^n(U^1 - U^0)$. For $U^1 - U^0 = \epsilon$, we have, as $k \rightarrow 0$,

$$U^n = \epsilon(2^n - 1) = \epsilon(2^{T/k} - 1) \rightarrow +\infty.$$

Definition 12.75. An s -step LMM is *stable* or *zero-stable* if all solutions $\{\mathbf{U}^n\}$ of the recurrence

$$\rho(Z)\mathbf{U}^n = \sum_{j=0}^s \alpha_j \mathbf{U}^{n+j} = \mathbf{0} \quad (12.53)$$

are bounded as $n \rightarrow +\infty$.

Definition 12.76. The *companion matrix* of a monic polynomial $p(z) = z^s + \sum_{i=0}^{s-1} \alpha_i z^i$ (Definition C.147) is the s -by- s square matrix

$$M := \begin{bmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ -\alpha_0 & -\alpha_1 & \cdots & -\alpha_{s-2} & -\alpha_{s-1} \end{bmatrix}. \quad (12.54)$$

Lemma 12.77. Any monic polynomial $p(z) = \sum_{j=0}^s \alpha_j z^j$ with $\alpha_s = 1$ is the characteristic polynomial of its companion matrix, cf. Definition C.152 and Theorem C.280.

Proof. We proceed by induction on s . The induction basis clearly holds since for $s = 1$,

$$M = [-\alpha_0] \Rightarrow \det(zI - [-\alpha_0]) = z + \alpha_0.$$

Suppose the conclusion holds for all integers no greater than $s - 1$. Then for s , we have

$$\begin{aligned} \det(zI - M) &= \begin{vmatrix} z & -1 & & & \\ & z & -1 & & \\ & & \ddots & \ddots & \\ & & & z & -1 \\ \alpha_0 & \alpha_1 & \cdots & \alpha_{s-2} & z + \alpha_{s-1} \end{vmatrix} \\ &= z \begin{vmatrix} z & -1 & & & \\ & z & -1 & & \\ & & \ddots & \ddots & \\ & & & z & -1 \\ \alpha_1 & \alpha_2 & \cdots & \alpha_{s-2} & z + \alpha_{s-1} \end{vmatrix} \\ &\quad + (-1)^{s+1} \alpha_0 \begin{vmatrix} -1 & & & & \\ z & -1 & & & \\ & \ddots & \ddots & & \\ & & & z & -1 \end{vmatrix} \\ &= z \left(z^{s-1} + \sum_{j=1}^{s-1} \alpha_j z^{j-1} \right) + (-1)^{s+1} \alpha_0 (-1)^{s-1} \\ &= z^s + \sum_{j=0}^{s-1} \alpha_j z^j, \end{aligned}$$

where the second equality follows from Laplace's formula (Theorem C.275) and the third from the induction hypothesis. Then the proof is completed by Theorem C.280. $\square$

Theorem 12.78 (Dahlquist root condition). An LMM is zero-stable if and only if any simple root of $\rho(z)$ satisfies $|z| \leq 1$ and any multiple root satisfies $|z| < 1$.

Proof. WLOG, we only prove the case of scalar IVPs; see Hairer et al. [1993] for the vector case. For a scalar IVP, we write (12.53) as $U^{n+s} + \sum_{j=0}^{s-1} \alpha_j U^{n+j} = 0$; this s -step recurrence formula can be expressed in matrix form as

$$\mathbf{V}^{n+1} = M\mathbf{V}^n, \quad (12.55)$$

where M is the companion matrix (12.54) and

$$\mathbf{V}^n = (U^n, U^{n+1}, \dots, U^{n+s-1})^\top.$$

Hence we have

$$\mathbf{V}^n = M^n \mathbf{V}^0.$$

By Lemma 12.77, the characteristic polynomial of M is $\rho(z)$, i.e., $p_M(z) = \rho(z)$. By Definition C.152, the set of eigenvalues of M is the set of roots of ρ , and these eigenvalues determine how the powers M^n behave asymptotically as $n \rightarrow +\infty$. The scalar sequence $\{U^n\}_{n=0}^{+\infty}$ is bounded as $n \rightarrow +\infty$ if and only if the vector sequence $\{\mathbf{V}^n\}$ is bounded, and $\{\mathbf{V}^n\}$ is bounded if and only if all elements of M^n are bounded. Since $\|\mathbf{V}^n\| \leq \|M^n\| \|\mathbf{V}^0\|$, the zero-stability is now equivalent to the power-boundedness of M .

By Theorem C.169, we have $M = RJR^{-1}$ and thus

$$M^n = RJ^nR^{-1}.$$

Therefore the boundedness of M^n is determined by that of

$$J_i^n = \begin{bmatrix} \lambda_i^n & \binom{n}{1} \lambda_i^{n-1} & \binom{n}{2} \lambda_i^{n-2} & \cdots & \binom{n}{m_i-1} \lambda_i^{n-m_i+1} \\ & \lambda_i^n & \binom{n}{1} \lambda_i^{n-1} & \cdots & \binom{n}{m_i-2} \lambda_i^{n-m_i+2} \\ & & \ddots & \ddots & \vdots \\ & & & \lambda_i^n & \binom{n}{1} \lambda_i^{n-1} \\ & & & & \lambda_i^n \end{bmatrix},$$

which follows from $J_i^n = (\lambda_i I + \eta)^n$ where η is the nilpotent matrix with $\eta^{m_i} = \mathbf{0}$,

$$\eta_{ij} = \begin{cases} 1 & \text{if } j - i = 1; \\ 0 & \text{otherwise.} \end{cases} \quad (12.56)$$

By Definition C.114, the dimension of the eigenspace of the companion matrix M is 1 for each eigenvalue of M because the upper-right $(s-1) \times (s-1)$ block of $zI - M$ is nonsingular for any $z \in \mathbb{C}$. Hence the geometric multiplicity $m_g(\lambda)$ is 1 for any eigenvalue λ of M . By Theorem C.169, there is exactly one Jordan block for each eigenvalue of M .

As $n \rightarrow \infty$, the nonzero elements of J_i^n approach 0 if $|\lambda_i| < 1$ and ∞ if $|\lambda_i| > 1$. For $|\lambda_i| = 1$, they are bounded in the case of a 1×1 block, but unbounded if $m_i \geq 2$. $\square$

12.4.3 Absolute stability

Definition 12.79. The *stability polynomial* of an LMM is

$$\pi_\kappa(\zeta) := \rho(\zeta) - \kappa\sigma(\zeta) = \sum_{j=0}^s (\alpha_j - \kappa\beta_j) \zeta^j. \quad (12.57)$$

Definition 12.80. An LMM is *absolutely stable* for some κ if all solutions $\{\mathbf{U}^n\}$ of

$$\pi_\kappa(Z)\mathbf{U}^n = [\rho(Z) - \kappa\sigma(Z)]\mathbf{U}^n = \mathbf{0}$$

are bounded as $n \rightarrow +\infty$, where Z is the time-shift operator in Notation 10.

Theorem 12.81 (*Root condition* for absolute stability). An LMM is absolutely stable for $\kappa := k\lambda$ if and only if each root of $\pi_\kappa(\zeta)$ satisfy $|\zeta| \leq 1 + O(k)$, and any root satisfying $|\zeta| \geq 1$ is simple.

Proof. This proof is similar with that of Theorem 12.78. As the key difference, the stability polynomial is a function of k and thus eigenvalues of its companion matrix might as well depend on k . $\square$

Definition 12.82. The *region of absolute stability (RAS)* for an LMM is the set of all $\kappa \in \mathbb{C}$ for which the method is absolutely stable.

Example 12.83. For the Euler method (12.14),

$$\pi_\kappa(\zeta) = (\zeta - 1) - \kappa = \zeta - (1 + \kappa), \quad (12.58)$$

with the single root $\zeta_1 = 1 + \kappa$. Thus the RAS for the Euler method is the disk:

$$\mathcal{R} = \{\kappa : |1 + \kappa| \leq 1\}. \quad (12.59)$$

Example 12.84. For backward Euler method (12.15),

$$\pi_\kappa(\zeta) = (\zeta - 1) - \kappa\zeta = (1 - \kappa)\zeta - 1, \quad (12.60)$$

with root $\zeta_1 = (1 - \kappa)^{-1}$. Thus the RAS for backward Euler method is:

$$\mathcal{R} = \{\kappa : |(1 - \kappa)^{-1}| \leq 1\} = \{\kappa : |1 - \kappa| \geq 1\}. \quad (12.61)$$

Example 12.85. For the trapezoidal method (12.16),

$$\pi_\kappa(\zeta) = (\zeta - 1) - \frac{1}{2}\kappa(\zeta + 1) = \left(1 - \frac{1}{2}\kappa\right)\zeta - \left(1 + \frac{1}{2}\kappa\right). \quad (12.62)$$

Thus the RAS for the trapezoidal method is the left half-plane:

$$\begin{aligned} \mathcal{R} &= \left\{ \kappa \in \mathbb{C} : \left| \frac{2 + \kappa}{2 - \kappa} \right| \leq 1 \right\} \\ &= \{\kappa \in \mathbb{C} : \operatorname{Re} \kappa \leq 0\}. \end{aligned} \quad (12.63)$$

Example 12.86. For the midpoint method (12.17),

$$\pi_\kappa(\zeta) = \zeta^2 - 2\kappa\zeta - 1. \quad (12.64)$$

$\pi_\kappa(\zeta) = 0$ implies

$$2\kappa = \zeta - \frac{1}{\zeta}.$$

For $\zeta := ae^{i\theta}$, we have $\frac{1}{\zeta} = a^{-1}e^{-i\theta}$. Hence there are always one zero with $|\zeta_1| \leq 1$ and another zero with $|\zeta_2| \geq 1$, depending on the sign of κ . The only possibility for both roots to have a modulus no greater than one is $|\zeta_1| = |\zeta_2| = 1 = a$. So the stability region only consists of the open interval from $-i$ to i on the imaginary axis:

$$\mathcal{R} = \{\kappa \in \mathbb{C} : \kappa = i\alpha \text{ with } |\alpha| < 1\}. \quad (12.65)$$

Definition 12.87. The *boundary locus* method finds the RAS of an LMM (ρ, σ) with $\sigma(e^{i\theta}) \neq 0$ by steps as follows:

(a) compute the *root locus curve*

$$\gamma(\theta) = \frac{\rho(e^{i\theta})}{\sigma(e^{i\theta})}, \quad \theta \in [0, 2\pi]; \quad (12.66)$$

(b) the closed curve γ divides the complex plane $\mathbb{C}$ into a number of connected regions;

(c) for each connected region $S \subset \mathbb{C}$, choose a convenient interior point $\kappa_p \in S$ and solve the equation $\rho(\zeta) - \kappa_p\sigma(\zeta) = 0$: S is part of the RAS if all roots are in the unit disk; otherwise S is not. Note that one can exit the loop upon the first finding of such a path-connected region.

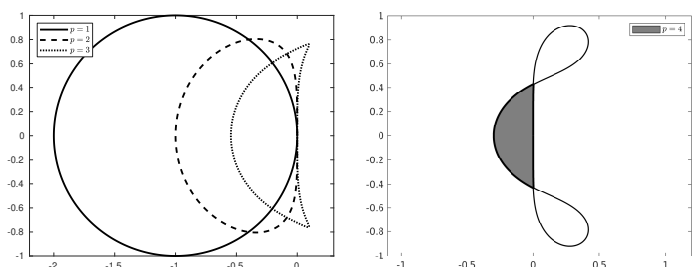


Figure 12.4: The bounded RAS of Adams-Bashforth formulas for $p = 1, 2, 3$ (left) and $p = 4$ (right).

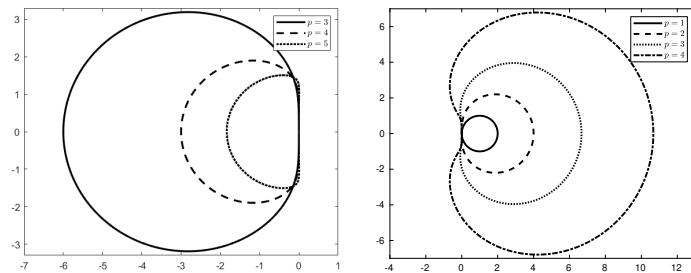


Figure 12.5: The bounded RASs of Adams-Moulton formulas with $p = 3, 4, 5$ (left) and the unbounded RASs of backward differentiation formulas with $p = 1, 2, 3, 4$ (right).

Example 12.88. RASs of some LMMs are shown in Figures 12.4 and 12.5.

Exercise 12.89. Write a program to reproduce the RAS plots in Figures 12.4 and 12.5.

12.4.4 Linear difference equations

Definition 12.90. A *system of linear difference equations* is a set of equations of the form

$$X_n = A_n X_{n-1} + \phi_n, \quad (12.67)$$

where $n, s \in \mathbb{N}^+$, $X_n \in \mathbb{C}^s$, $\phi_n \in \mathbb{C}^s$, and $A_n \in \mathbb{C}^{s \times s}$. With the initial vector X_0 specified, the system of linear difference equations becomes an initial value problem. The system is *homogeneous* if $\phi_n = \mathbf{0}$.

Example 12.91. A linear difference equation of the form

$$y_n = \alpha_{n1}y_{n-1} + \alpha_{n2}y_{n-2} + \cdots + \alpha_{ns}y_{n-s} + \psi_n$$

can be easily recast in the form (12.67) by writing

$$A_n = \begin{bmatrix} \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{n,s-1} & \alpha_{ns} \\ 1 & 0 & & & \\ & 1 & \ddots & & \\ & & \ddots & 0 & \\ & & & 1 & 0 \end{bmatrix}, X_{n-1} = \begin{bmatrix} y_{n-1} \\ y_{n-2} \\ \vdots \\ y_{n-s} \end{bmatrix},$$

$$\phi_n = [\psi_n \ 0 \ 0 \ \cdots \ 0]^\top.$$

Theorem 12.92. The problem (12.67) with initial value X_0 has the unique solution

$$\begin{aligned} X_n &= \left(\prod_{i=1}^n A_i \right) X_0 \\ &+ \left(\prod_{i=2}^n A_i \right) \phi_1 + \left(\prod_{i=3}^n A_i \right) \phi_2 + \cdots + A_n \phi_{n-1} + \phi_n, \end{aligned} \quad (12.68)$$

where

$$\prod_{i=m}^n A_i = \begin{cases} A_n A_{n-1} \cdots A_{m+1} A_m & \text{if } m \leq n; \\ \mathbf{0} & \text{otherwise.} \end{cases}$$

Proof. For $n = 1$, (12.68) reduces to (12.67). The rest of the proof is a straightforward induction. $\square$

Theorem 12.93. Let θ_n be the solution to the homogeneous linear difference equation

$$\theta_{n+s} + \sum_{i=0}^{s-1} \alpha_i \theta_{n+i} = 0 \quad (12.69)$$

with constant coefficients α_i 's and the initial values

$$\begin{bmatrix} \theta_0 \\ \theta_{-1} \\ \vdots \\ \theta_{-s+2} \\ \theta_{-s+1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}. \quad (12.70)$$

Then the inhomogeneous equation

$$y_{n+s} + \sum_{i=0}^{s-1} \alpha_i y_{n+i} = \psi_{n+s} \quad (12.71)$$

with initial values $y_0, y_1, \dots, y_{s-1}$ is uniquely solved by

$$y_n = \sum_{i=0}^{s-1} \theta_{n-i} \tilde{y}_i + \sum_{i=s}^n \theta_{n-i} \psi_i \quad (12.72)$$

where

$$\begin{bmatrix} \tilde{y}_{s-1} \\ \tilde{y}_{s-2} \\ \tilde{y}_{s-3} \\ \vdots \\ \tilde{y}_1 \\ \tilde{y}_0 \end{bmatrix} = \begin{bmatrix} 1 & \theta_1 & \theta_2 & \cdots & \theta_{s-2} & \theta_{s-1} \\ & 1 & \theta_1 & \cdots & \theta_{s-3} & \theta_{s-2} \\ & & 1 & \cdots & \theta_{s-4} & \theta_{s-3} \\ & & & \ddots & \vdots & \vdots \\ & & & & 1 & \theta_1 \\ & & & & & 1 \end{bmatrix}^{-1} \begin{bmatrix} y_{s-1} \\ y_{s-2} \\ y_{s-3} \\ \vdots \\ y_1 \\ y_0 \end{bmatrix}. \quad (12.73)$$

Exercise 12.94. Prove Theorem 12.93 by induction.

12.4.5 Convergence

Definition 12.95. An LMM is said to be *convergent* if, for any fixed $T > 0$ and any given initial values

$$\forall i = 0, 1, \dots, s-1, \quad \mathbf{U}^i := \phi^i(\mathbf{u}(0), k)$$

satisfying

$$\forall i = 0, 1, \dots, s-1, \quad \lim_{k \rightarrow 0} \|\phi^i(\mathbf{u}(0), k) - \mathbf{u}(0)\| = 0, \quad (12.74)$$

its solution $\{\mathbf{U}^n \approx \mathbf{u}(t_n) : n = 0, 1, \dots, \frac{T}{k}, t_n = nk\}$ satisfies

$$\lim_{k \rightarrow 0} \max_{0 \leq n \leq \frac{T}{k}} \|\mathbf{U}^n - \mathbf{u}(t_n)\| = 0 \quad (12.75)$$

for any IVP with $\mathbf{f}(\mathbf{u}, t)$ continuous in t and Lipschitz continuous in $\mathbf{u}$.

Lemma 12.96. A convergent LMM is zero-stable.

Proof. Apply the convergent LMM to the trivial IVP

$$u'(t) = 0; \quad u(0) = 0 \quad (12.76)$$

and we have (12.53). Suppose the LMM is not zero-stable. Then Theorem 12.78 implies that the characteristic polynomial ρ either has a root ζ_1 with $|\zeta_1| > 1$ or has a multiple root ζ_2 with $|\zeta_2| = 1$. In the former case, we have

$$\sum_{j=0}^s \alpha_j \zeta_1^j = 0 \Rightarrow \zeta_1^m \sum_{j=0}^s \alpha_j \zeta_1^j = 0 \Rightarrow \sum_{j=0}^s \alpha_j Z^j \zeta_1^m = 0,$$

i.e., $U^n = \zeta_1^n$ is a solution of (12.53). In the latter case, we have $\rho(\zeta_2)\zeta_2^m = 0$ for any $m \in \mathbb{N}^+$. In addition, define

$$\chi(z) := (\rho(z)z^m)' = \left(\sum_{i=0}^s \alpha_i z^{m+i} \right)' = \sum_{i=0}^s \alpha_i (m+i) z^{m+i-1}$$

and we know from $\chi(\zeta_2) = 0$ that

$$\forall n \in \mathbb{N}, \quad V^n = n\zeta_2^{n-1}$$

is a solution of (12.53). For (12.76) and any fixed $T > 0$,

$$U_k(T) = \sqrt{k} \zeta_1^{T/k}, \quad V_k(T) = \frac{T}{\sqrt{k}} \zeta_2^{T/k-1}$$

satisfy condition (12.74) for initial values, but diverge as $k \rightarrow 0$, i.e. $n \rightarrow \infty$. This contradicts the convergence of the LMM and completes the proof. $\square$

Lemma 12.97. A convergent LMM is preconsistent.

Proof. The IVP $u'(t) = f(t) = 0$ with $u(0) = 1$ is solved by $u(T) = u(0) = 1$ for any $T > 0$. By Definition 12.45, the LMM reduces to $\sum_{j=0}^s \alpha_j U^{n-j} = 0$. Definition 12.95 gives

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N,$$

$$\forall i = 1, 2, \dots, n-s, n-s+1, \dots, n, \quad |U^i - 1| < \epsilon.$$

Then $\sum_{j=0}^s \alpha_j = \sum_{j=0}^s \alpha_j (1 - U^{n-j} + U^{n-j})$ and the linear difference equation $\sum_{j=0}^s \alpha_j U^{n-j} = 0$ yield

$$\left| \sum_{j=0}^s \alpha_j \right| \leq \sum_{j=0}^s |\alpha_j (1 - U^{n-j})| < \epsilon \sum_{j=0}^s |\alpha_j|,$$

which yields $\rho(1) = 0$. $\square$

Lemma 12.98. A convergent LMM is consistent.

Exercise 12.99. With the help of Lemma 12.97, prove Lemma 12.98 by an approach similar with that for Lemma 12.96, i.e., by considering the IVP

$$u'(t) = f(t) = 1, u(0) = 0. \quad (12.77)$$

Lemma 12.100. For an autonomous IVP, the one-step error of a consistent LMM satisfies

$$\|\mathcal{L}\mathbf{u}(t_n)\| \leq \sum_{j=0}^{s-1} \left(\frac{1}{2} (s-j)^2 |\alpha_j| + (s-j) |\beta_j| \right) LMk^2, \quad (12.78)$$

where L is the Lipschitz constant and M is an upper bound of $\|\mathbf{f}(\mathbf{u}(t))\|$ on $t \in [0, T]$.

Proof. By definition of the one-step error (12.19), we have

$$\begin{aligned}\mathcal{L}\mathbf{u}(t_n) &= \sum_{j=0}^s \alpha_j \mathbf{u}(t_{n+j}) - k \sum_{j=0}^s \beta_j \mathbf{u}'(t_{n+j}) \\ &= \sum_{j=0}^{s-1} \alpha_j \mathbf{u}(t_{n+j}) - \sum_{j=0}^{s-1} \alpha_j \mathbf{u}(t_{n+s}) \\ &\quad - k \sum_{j=0}^{s-1} ((j-s)\alpha_j - \beta_j) \mathbf{u}'(t_{n+s}) - k \sum_{j=0}^{s-1} \beta_j \mathbf{u}'(t_{n+j}) \\ &= \sum_{j=0}^{s-1} \alpha_j (\mathbf{u}(t_{n+j}) - \mathbf{u}(t_{n+s}) - (j-s)k\mathbf{u}'(t_{n+s})) \\ &\quad + k \sum_{j=0}^{s-1} \beta_j (\mathbf{u}'(t_{n+s}) - \mathbf{u}'(t_{n+j})),\end{aligned}$$

where the second step follows from the consistency condition (12.47), i.e.,

$$\begin{aligned}\alpha_s &= -\sum_{j=0}^{s-1} \alpha_j, \\ \beta_s &= \sum_{j=0}^s j\alpha_j - \sum_{j=0}^{s-1} \beta_j = \sum_{j=0}^{s-1} ((j-s)\alpha_j - \beta_j).\end{aligned}$$

Integration of the autonomous ODE $\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}(t))$ yields the identity

$$\begin{aligned}\mathbf{u}(t_{n+j}) - \mathbf{u}(t_{n+s}) - (j-s)k\mathbf{u}'(t_{n+s}) \\ = k \int_{s-j}^0 [\mathbf{f}(\mathbf{u}(t_{n+s} - \xi k)) - \mathbf{f}(\mathbf{u}(t_{n+s}))] d\xi,\end{aligned}$$

which, together with the Lipschitz condition, implies

$$\begin{aligned}\|\mathbf{u}(t_{n+j}) - \mathbf{u}(t_{n+s}) - (j-s)k\mathbf{u}'(t_{n+s})\| \\ \leq kL \int_0^{s-j} \|\mathbf{u}(t_{n+s} - \xi k) - \mathbf{u}(t_{n+s})\| d\xi \\ \leq \frac{1}{2}(s-j)^2 k^2 LM,\end{aligned}$$

where the second step follows from the mean value theorem and the condition of M being an upper bound of $\|\mathbf{f}(\mathbf{u}(t))\|$. Similarly, we have

$$\|\mathbf{f}(\mathbf{u}(t_{n+s})) - \mathbf{f}(\mathbf{u}(t_{n+j}))\| \leq LM(s-j)k.$$

Take a norm of $\mathcal{L}\mathbf{u}(t_n)$, apply the above two inequalities, and we have (12.78). $\square$

Lemma 12.101. For an autonomous IVP, the solution errors of a consistent LMM with $k < k_0$ and $k_0|\beta_s|L < 1$ satisfy

$$\left\| \mathbf{E}^{n+s} + \sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} \right\| \leq Ck \max_{i=0}^{s-1} \|\mathbf{E}^{n+i}\| + Dk^2, \quad (12.79)$$

where both C and D are positive constants.

Proof. By definitions of the LMM, its one-step errors, and its solution errors, we have

$$\begin{aligned}\mathbf{E}^{n+s} + \sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} \\ = \mathbf{U}^{n+s} - \mathbf{u}(t_{n+s}) + \sum_{i=0}^{s-1} \alpha_i (\mathbf{U}^{n+i} - \mathbf{u}(t_{n+i})) \\ = k \sum_{i=0}^s \beta_i (\mathbf{f}(\mathbf{U}^{n+i}) - \mathbf{f}(\mathbf{u}(t_{n+i}))) - \mathcal{L}\mathbf{u}(t_n),\end{aligned}$$

which yields

$$\begin{aligned}\left\| \mathbf{E}^{n+s} + \sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} \right\| \\ \leq \|\mathcal{L}\mathbf{u}(t_n)\| + k|\beta_s| \|\mathbf{f}(\mathbf{U}^{n+s}) - \mathbf{f}(\mathbf{u}(t_{n+s}))\| \\ + k \sum_{i=0}^{s-1} |\beta_i| \|\mathbf{f}(\mathbf{U}^{n+i}) - \mathbf{f}(\mathbf{u}(t_{n+i}))\| \\ \leq \|\mathcal{L}\mathbf{u}(t_n)\| + kL|\beta_s| \|\mathbf{E}^{n+s}\| + kL \sum_{i=0}^{s-1} |\beta_i| \|\mathbf{E}^{n+i}\| \\ \leq \|\mathcal{L}\mathbf{u}(t_n)\| + kL|\beta_s| \left\| \mathbf{E}^{n+s} + \sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} \right\| \\ + kL \sum_{i=0}^{s-1} |\alpha_i \beta_s| \|\mathbf{E}^{n+i}\| + kL \sum_{i=0}^{s-1} |\beta_i| \|\mathbf{E}^{n+i}\|,\end{aligned}$$

where the third step follows from the triangle inequality

$$\|\mathbf{E}^{n+s}\| \leq \left\| \mathbf{E}^{n+s} + \sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} \right\| + \sum_{i=0}^{s-1} \|\alpha_i \mathbf{E}^{n+i}\|.$$

Thus we have

$$\begin{aligned}(1 - kL|\beta_s|) \left\| \mathbf{E}^{n+s} + \sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} \right\| \\ \leq kL \sum_{i=0}^{s-1} (|\alpha_i \beta_s| + |\beta_i|) \|\mathbf{E}^{n+i}\| + \|\mathcal{L}\mathbf{u}(t_n)\|.\end{aligned}$$

For any $k < k_0 < \frac{1}{|\beta_s|L}$, dividing both sides by $(1 - kL|\beta_s|)$ and applying Lemma 12.100 yield (12.79). $\square$

Theorem 12.102. An LMM is convergent if and only if it is consistent and stable.

Proof. We only prove the sufficiency since the necessity has been stated in Lemmas 12.96 and 12.98. By Lemma 12.101, we have

$$\mathbf{E}^{n+s} = -\sum_{i=0}^{s-1} \alpha_i \mathbf{E}^{n+i} + \psi_{n+s},$$

where $\|\psi_{n+s}\| \leq Ck \max_{i=0}^{s-1} \|\mathbf{E}^{n+i}\| + Dk^2$ for any k sufficiently small. Then the zero-stability of the LMM and Theorem 12.93 imply the existence of bounded constants θ_i 's such that

$$\mathbf{E}^n = \sum_{i=0}^{s-1} \theta_{n-i} \widetilde{\mathbf{E}}^i + \sum_{i=s}^n \theta_{n-i} \psi_i,$$

where each $\widetilde{\mathbf{E}}^i$ is a linear combination of $\mathbf{E}^j$'s for $i, j = 0, 1, \dots, s-1$; see (12.73). Note that, in order to apply Theorem 12.93, we have shifted $\mathbf{E}^{n+i}$ to $\mathbf{E}^{n+i-s}$. It follows that

$$\|\mathbf{E}^n\| \leq \theta_m \sum_{i=0}^{s-1} \|\widetilde{\mathbf{E}}^i\| + \theta_m Cks \sum_{i=0}^{n-1} \|\mathbf{E}^i\| + \theta_m D(n-s+1)k^2,$$

where $\theta_m = \sup_{i=1}^n |\theta_i|$ and the factor s of the second summation is introduced to account for the fact that a local maximum value of $\|\mathbf{E}^{n-i}\|$ may appear in at most s adjacent terms. Construct a sequence (v_i) as an upper bound sequence of $(\|\mathbf{E}^i\|)$:

$$\begin{cases} v_0 = \theta_m \sum_{i=0}^{s-1} \|\widetilde{\mathbf{E}}^i\|; \\ v_1 = \theta_m Dk^2 + v_0; \\ \dots \\ v_n = \theta_m Cks \sum_{i=1}^{n-1} v_i + n\theta_m Dk^2 + v_0, \end{cases}$$

where $\lim_{k \rightarrow 0} v_0 = 0$ because Definition 12.95 implies $\lim_{k \rightarrow 0} \|\widetilde{\mathbf{E}}^i\| = 0$ for each $i = 0, 1, \dots, s-1$. It is straightforward to show that, for $n > 1$,

$$v_n + \frac{Dk}{Cs} = (1 + \theta_m Cks) \left(v_{n-1} + \frac{Dk}{Cs} \right),$$

which implies

$$\begin{aligned} v_n &= -\frac{Dk}{Cs} + (1 + \theta_m Cks)^{n-1} \left(v_1 + \frac{Dk}{Cs} \right) \\ &= (1 + \theta_m Cks)^{n-1} v_0 + [(1 + \theta_m Cks)^n - 1] \frac{Dk}{Cs} \\ &< \exp(\theta_m Csnk) v_0 + [\exp(\theta_m Csnk) - 1] \frac{Dk}{Cs}. \end{aligned}$$

For $n = T/k$, we have $\lim_{k \rightarrow 0} v_n = 0$. The proof is completed by the fact of $\|\mathbf{E}^n\| < v_n$ for each n . $\square$

Theorem 12.103. Consider an IVP of which $\mathbf{f}(\mathbf{u}, t)$ is p times continuously differentiable with respect to both t and $\mathbf{u}$. For a convergent LMM with consistency of order p and with its initial conditions satisfying

$$\forall i = 0, 1, \dots, s-1, \quad \|\mathbf{U}^i - \mathbf{u}(t_i)\| = O(k^p),$$

its numerical solution of the IVP satisfies

$$\|\mathbf{U}^{t/k} - \mathbf{u}(t)\| = O(k^p) \quad (12.80)$$

for all $t \in [0, T]$ and sufficiently small $k > 0$.

Proof. This proof is similar to that of Theorem 12.102. $\square$

Theorem 12.104. The s -step Adams and Nystrom formulas are stable for all $s \geq 1$. The s -step backward differentiation formulas are stable for $s = 1, 2, \dots, 6$, but unstable for $s \geq 7$.

Proof. See Hairer et al. [1993]. $\square$

Theorem 12.105 (The first Dahlquist's barrier). The order of accuracy p of a stable s -step LMM satisfies

$$p \leq \begin{cases} s & \text{if the LMM is explicit,} \\ s+1 & \text{else if } s \text{ is odd,} \\ s+2 & \text{else if } s \text{ is even.} \end{cases} \quad (12.81)$$

Proof. See Hairer et al. [1993]. $\square$

12.5 Stiff IVPs

Example 12.106. Consider the IVP

$$u'(t) = \lambda(u - \cos t) - \sin t, \quad u(0) = \eta. \quad (12.82)$$

By Duhamel's principle (12.10), the exact solution is

$$\begin{aligned} u_\eta(t) &= e^{\lambda t} \eta - \int_0^t e^{\lambda(t-\tau)} (\lambda \cos \tau + \sin \tau) d\tau \\ &= e^{\lambda t} \eta - \int_0^t \lambda e^{\lambda(t-\tau)} \cos \tau d\tau - \int_0^t e^{\lambda(t-\tau)} \sin \tau d\tau \\ &= e^{\lambda t} (\eta - 1) + \cos t, \end{aligned}$$

where the third equality follows from the integration-by-parts formula.

If $\eta = \cos(0) = 1$, then $u_1(t) = \cos t$ is the unique solution. If $\eta \neq 1$ and $\lambda < 0$, then the solution curve $u_\eta(t)$ decays exponentially to $u_1(t)$.

A negative λ with large magnitude has a dominant effect on nearby solutions of the ODE corresponding to different initial data; the plot in Figure 12.6 shows some solution curves with $\lambda = -100$.

For six values of k , the following table compares the results at $T = 1$ computed by the second-order Adams-Bashforth and the second-order BDF method. The results in Figure 12.6 indicate the curious effect that this property of the ODE has on numerical computations. To achieve a solution error $E(T) \leq \epsilon = 4 \times 10^{-5}$, the BDF2 method may use $k = 0.1$, the AB2 method has to use $k \leq 0.01$ while the time scale of the IVP is 1.

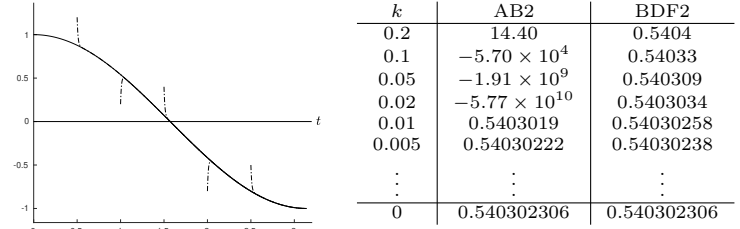


Figure 12.6: The example (12.82) of a linear nonautonomous stiff IVP.

12.5.1 The notion of stiffness

Definition 12.107. An IVP is said to be *stiff in an interval* if for some initial condition any numerical method with a bounded RAS is forced to use a time-step size that is unnecessarily and excessively smaller than the time scale of the true solution of the IVP.

Definition 12.108. For an IVP

$$\mathbf{u}'(t) = A\mathbf{u} + \mathbf{b}(t), \quad (12.83)$$

where $\mathbf{u}, \mathbf{b} \in \mathbb{R}^n$ and A is a constant, diagonalizable, $n \times n$ matrix with eigenvalues $\lambda_i \in \mathbb{C}, i = 1, 2, \dots, n$, its *stiffness ratio* is

$$\frac{\max_{\lambda \in \Lambda(A)} |\operatorname{Re} \lambda|}{\min_{\lambda \in \Lambda(A)} |\operatorname{Re} \lambda|}. \quad (12.84)$$

Example 12.109. Consider the linear IVP

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}' = \begin{pmatrix} -1000 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad t \in [0, 1] \quad (12.85)$$

with initial value $\mathbf{u}(0) = (1, 1)^\top$. Suppose we would like to solve (12.85) by a p -th order LMM with time-step size k so that $\|\mathbf{E}\|_\infty \leq \epsilon$, i.e.,

$$|U_1^N - e^{-1000}| \leq \epsilon, \quad |U_2^N - e^{-1}| \leq \epsilon.$$

For U_2^N to be sufficiently accurate, we need $k = O(\epsilon^{\frac{1}{p}})$. But for U_1^N to be sufficiently accurate, if the formula has a stability region of finite size like the Euler formula, the stability condition $|1 + k\lambda| \leq 1 + O(k)$ implies k should be smaller than $k_0 \approx 10^{-3}$. Most likely this is a much tighter restriction, e.g., $\epsilon = 10^{-4}$ and $p = 4$ give $(\epsilon)^{\frac{1}{p}} = 0.1$.

Example 12.110. Consider the nonlinear IVP

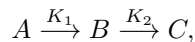
$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}' = \begin{pmatrix} -u_1 u_2 \\ \cos(u_1) - \exp(u_2) \end{pmatrix}. \quad (12.86)$$

The Jacobian matrix is

$$J = - \begin{pmatrix} u_2 & u_1 \\ \sin(u_1) & \exp(u_2) \end{pmatrix}.$$

Near a point t with $u_1(t) = 0$ and $u_2(t) \gg 1$, the matrix is diagonal with widely differing eigenvalues and the behavior will probably be stiff.

Example 12.111. Let A , B , and C represent chemical compounds and consider reactions of the form



where A is transformed into B with rate $K_1 > 0$ and B is simultaneously transformed into C with rate $K_2 > 0$. Denote by u_1 , u_2 , and u_3 the concentrations of A , B , and C , respectively. The dynamics is governed by

$$\mathbf{u}'(t) := \begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix} = \begin{bmatrix} -K_1 & 0 & 0 \\ K_1 & -K_2 & 0 \\ 0 & K_2 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}. \quad (12.87)$$

Note that the eigenvalues are $-K_1$, $-K_2$ and 0. With the assumption of $K_1 \neq K_2$, the general solution has the form

$$u_j(t) = c_{j1}e^{-K_1 t} + c_{j2}e^{-K_2 t} + c_{j3}.$$

Since A decays into B which decays into C , we expect that u_1 simply decays to 0 exponentially, and also that u_2 ultimately decays to 0 (although it may first grow if K_1 is larger than K_2), while u_3 grows and asymptotically approaches the value $u_1(0) + u_2(0) + u_3(0)$ as $t \rightarrow \infty$.

A typical solution for $K_1 = 3$ and $K_2 = 1$ with $\mathbf{u}(0) = [3, 4, 2]^T$ is shown in the top plot of Figure 12.7. For $K_1 \gg K_2$, e.g. $K_1 = 50$, $K_2 = 1$, we get the solution curves in the bottom left plot of Figure 12.7.

Suppose at time $t = 1$ we perturb the system by adding more of species A ; the solution curves are shown in the bottom right plot of Figure 12.7.

The sudden increment of A is rapidly converted into B and then slowly from B into C . The different time scales of these two procedures limit the time-step size.

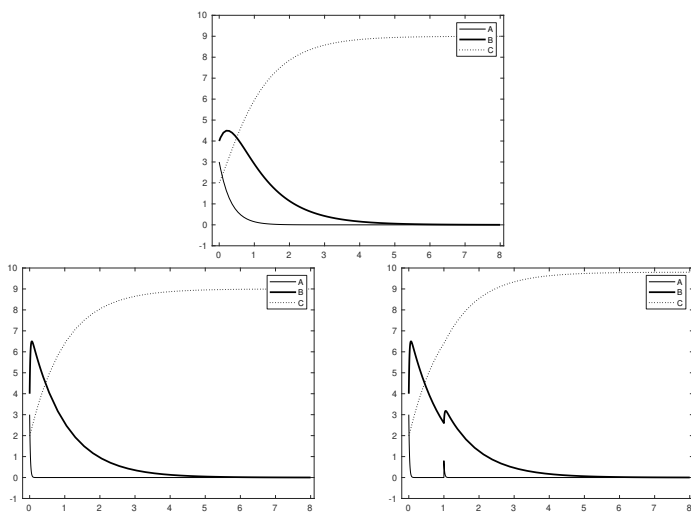


Figure 12.7: The example (12.87) of a nonlinear autonomous stiff IVP. An abscissa and an ordinate represent time t and concentration of elements of $\mathbf{u}(t)$, respectively.

12.5.2 A-stability

Definition 12.112. An LMM is *A-stable* iff its RAS $\mathcal{R}$ satisfies $\mathbb{C}^- \subseteq \mathcal{R}$, where

$$\mathbb{C}^- := \{z \in \mathbb{C} : \operatorname{Re} z \leq 0\}. \quad (12.88)$$

Example 12.113. Both the backward Euler method and the trapezoidal method are A-stable.

Theorem 12.114 (The second Dahlquist's barrier). The order of accuracy of an implicit A-stable LMM satisfies $p \leq 2$. An explicit LMM cannot be A-stable.

Definition 12.115. An IVP method is *A(α)-stable* if its region of absolute stability $\mathcal{R}$ satisfies $\mathbb{C}_\alpha^- \subseteq \mathcal{R}$, where

$$\mathbb{C}_\alpha^- := \{z \in \mathbb{C} : \pi - \alpha \leq \arg(z) \leq \pi + \alpha\}. \quad (12.89)$$

It is *A(0)-stable* if it is A(α)-stable for some $\alpha > 0$.

Example 12.116. As shown in Figure 12.5, the BDFs are A(α)-stable with $\alpha = 90^\circ$ for $p = 1, 2$ and $\alpha \approx 86^\circ, 73^\circ, 51^\circ$, and 17° for $p = 3, 4, 5, 6$ respectively. Note the large drop of α from $p = 5$ to $p = 6$.

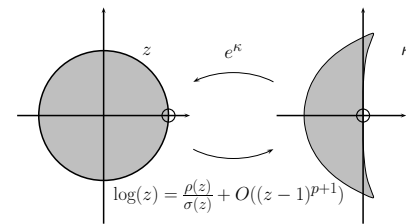


Figure 12.8: The analysis of LMMs is essentially the approximation of $\log z$ by a rational polynomial $\rho(z)/\sigma(z)$.

Summary 12.117. An LMM of the form

$$\mathbf{U}^{n+s} + \sum_{j=0}^{s-1} \alpha_j \mathbf{U}^{n+j} = k \sum_{j=0}^s \beta_j \mathbf{f}(\mathbf{U}^{n+j}, t_{n+j})$$

is completely characterized by the generating polynomials

$$\rho(z) = \sum_{j=0}^s \alpha_j z^j; \quad \sigma(z) = \sum_{j=0}^s \beta_j z^j.$$

By Theorem 12.70, an LMM with $\sigma(1) \neq 0$ is p th-order accurate if and only if

$$\frac{\rho(z)}{\sigma(z)} = \log z + O((z-1)^{p+1}).$$

In particular, any consistent LMM must have a root of $\rho(z)$ at $z = 1$, which is called the *principal root* of ρ .

By Theorem 12.78, zero stability is the condition that all simple roots of ρ (such as the principal root) satisfy $|z| \leq 1$ and all multiple roots satisfy $|z| < 1$. These conditions are explained by Remark ??, where the polynomial $p_Z^j = \prod_{i \neq j; i=1}^{s+1} (Z - \lambda_i)$ implies

$$\rho(Z) = \mathbf{0} \Leftrightarrow \forall j = 1, \dots, s+1, \quad Z p_Z^j U^0 = \lambda_j p_Z^j U^0.$$

For absolute stability, define $\kappa := k\lambda$ and $\pi_\kappa := \rho - \kappa\sigma$ and a similar argument for zero stability yields Theorem 12.81, which states that an LMM is absolutely stable for a given κ if and only if each simply root z of π_κ satisfies $|z| \leq 1 + O(k)$ and each multiple root z of π_κ satisfies $|z| < 1$.

For any LMM, it would be convenient to have its RAS so that a user can conveniently determine from it the stability of a chosen time step size k . The origin in the RAS corresponds to the principal root $z = 1$; see Figure 12.8.

Consistency is the condition that $\frac{\rho(z)}{\sigma(z)}$ matches $\log z$ to at least the second order at $z = 1$. When $\kappa = k\mathcal{D}$ approaches zero, $z = e^\kappa$ approaches the principal root $z = 1$. For this process to be stable, it must always satisfy $\operatorname{Re} z < 1$, which, due to the variable change $\kappa = \log z$, translates to $\kappa < 0$.

In summary, the study of crucial properties of LMMs have been completely reduced to the approximation of $\log z$ by the rational function $\frac{\rho(z)}{\sigma(z)}$. The stiff problems

12.6 Programming assignments

12.6.1 The context

The restricted three-body system is a model for studying the motion of an earth-moon satellite or of an asteroid close enough to the earth to be strongly influenced by the earth and the sun.

In this system, the two heavy bodies are regarded as revolving in fixed orbits about their common centre of mass while the small body is attracted by the heavy bodies but never affecting their motion. To further simplify the problem, we approximate orbits of the heavy bodies as circles so that we can work in a frame of reference that rotates with the two heavy bodies. In this frame, the two heavy bodies are considered as fixed in space with their rotation translated into a modification of the equations of gravitational motion. For simplicity, we scale the units to reduce a number of constants to one. We also scale the masses of the two heavy bodies to $1 - \mu$ and μ and their positions relative to the moving reference frame by the vectors μe_1 and $(\mu - 1)e_1$ respectively so that their center of mass is at the origin of coordinates.

Denote by u_1, u_2 and u_3 the scalar variables representing the position coordinates of the small body and by u_4, u_5 and u_6 the corresponding velocity coordinates. The equations of motion for $\mathbf{u}(t)$ are

$$\begin{cases} u'_1 = u_4, \\ u'_2 = u_5, \\ u'_3 = u_6, \\ u'_4 = 2u_5 + u_1 - \frac{\mu(u_1 + \mu - 1)}{(u_2^2 + u_3^2 + (u_1 + \mu - 1)^2)^{3/2}} \\ \quad - \frac{(1 - \mu)(u_1 + \mu)}{(u_2^2 + u_3^2 + (u_1 + \mu)^2)^{3/2}}, \\ u'_5 = -2u_4 + u_2 - \frac{\mu u_2}{(u_2^2 + u_3^2 + (u_1 + \mu - 1)^2)^{3/2}} \\ \quad - \frac{(1 - \mu)u_2}{(u_2^2 + u_3^2 + (u_1 + \mu)^2)^{3/2}}, \\ u'_6 = -\frac{\mu u_3}{(u_2^2 + u_3^2 + (u_1 + \mu - 1)^2)^{3/2}} \\ \quad - \frac{(1 - \mu)u_3}{(u_2^2 + u_3^2 + (u_1 + \mu)^2)^{3/2}}. \end{cases} \quad (12.90)$$

Planar motion is possible; that is, solutions which satisfy $u_3 = u_6 = 0$ at all times. One of these is shown in the

LHS plot of Figure 12.9, with the values of (u_1, u_2) plotted as the orbit evolves. The heavier mass is at the point $(-\mu, 0)$ and the lighter mass is at $(1 - \mu, 0)$, where $(0, 0)$ is marked 0 and $(1, 0)$ is marked 1. For this calculation the value of $\mu = 0.012277471$ was selected, corresponding to the earth-moon system. The initial values for this computation were

$$\mathbf{u}(0) = (0.994, 0, 0, 0, -2.0015851063790825224, 0) \quad (12.91)$$

and the period was $T_1 = 17.06521656015796$.

The RHS plot of Figure 12.9 illustrates another solution of (12.90) with a different initial condition

$$\mathbf{u}(0) = (0.879779227778, 0, 0, 0, -0.379677780949, 0) \quad (12.92)$$

and a different period $T_2 = 19.140540691377$.

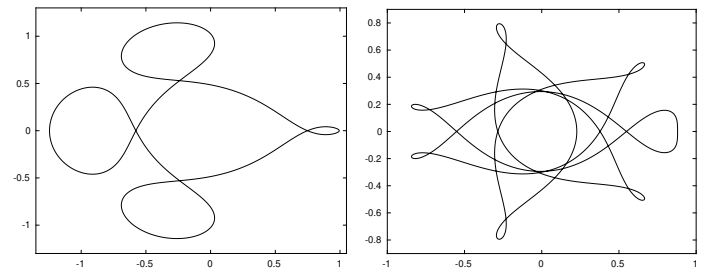


Figure 12.9: Planar orbits of the three-body ODE system (12.90) with initial conditions (12.91) and (12.92).

12.6.2 The assignments

Write a C++ package to implement

- Adams-Bashforth methods ($p = 1, 2, 3, 4$),
- Adams-Moulton methods ($p = 2, 3, 4, 5$),
- BDFs ($p = 1, 2, 3, 4$),

You must test each method against the restricted three-body system with (12.91) and (12.92) in Section 12.6.1 as the initial conditions. Each solution should be computed to T_i for $i = 1, 2$ with a stable time-step size. For (12.91), the solution is periodic and you should use the initial condition as the exact solution at T_1 for calculating the solution errors. For (12.92), you should calculate the difference between the results on two adjacent grids, set it as the computational error, and deduce the convergence rates. For each method and for each test case, you are supposed to

- (a) itemize all details of the test in an input file where the name and order of the method are parsed in the `main` program to generate an object of (a derived class of) the class `TimeIntegrator` by the singleton object `TimeIntegratorFactory`,
- (b) perform a sequence of grid refinement test where solution errors, convergence rates, and CPU time are reported to demonstrate convergence with a correct order of accuracy (this is the criterion of whether your test passes or fails!),
- (c) determine a time step size, as large as possible, with which the plot of the solutions is visually indistinguishable from the corresponding plot in Section 12.6.1.

Write a report to organize your results in a coherent story that relates your numerical results to the theory in this chapter.

Chapter 13

Runge–Kutta (RK) Methods for IVPs

13.1 One-step methods

Definition 13.1. A *one-step method* constructs numerical solutions of an IVP (12.3) at each time step $n = 0, 1, \dots$ by a formula of the form

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k\Phi(\mathbf{U}^n, t_n; k), \quad (13.1)$$

where $\Phi : \mathbb{R}^N \times [0, T] \times (0, +\infty) \rightarrow \mathbb{R}^N$ is the *increment function* given in terms of the RHS function $\mathbf{f} : \mathbb{R}^N \times [0, T] \rightarrow \mathbb{R}^N$ in (12.3).

13.1.1 Classical one-step methods

Example 13.2. For a scalar ODE $u' = f(u, t)$, the *Taylor series method* is obtained by replacing derivatives of $u(t)$ at t_n in the truncated Taylor expansion

$$U^{n+1} = U^n + ku'_n + \frac{k^2}{2}u''_n + \dots + \frac{k^p}{p!}u_n^{(p)} \quad (13.2)$$

with $f(u, t)$ and its derivatives evaluated at (U^n, t_n) ,

$$\begin{aligned} u'_n &= f, \\ u''_n &= f_u f + f_t, \\ u'''_n &= f_u^2 f + f_{uu} f^2 + f_u f_t + 2f_{tu} f + f_{tt}, \\ &\dots \end{aligned}$$

The one-step error of (13.2) is $O(k^{p+1})$. This method also applies to ODE systems.

Definition 13.3. The *improved Euler method* or the *explicit trapezoidal method* is a one-step method of the form

$$\begin{cases} \mathbf{y}_1 = \mathbf{f}(\mathbf{U}^n, t_n), \\ \mathbf{y}_2 = \mathbf{f}(\mathbf{U}^n + k\mathbf{y}_1, t_n + k), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + \frac{k}{2}(\mathbf{y}_1 + \mathbf{y}_2). \end{cases} \quad (13.3)$$

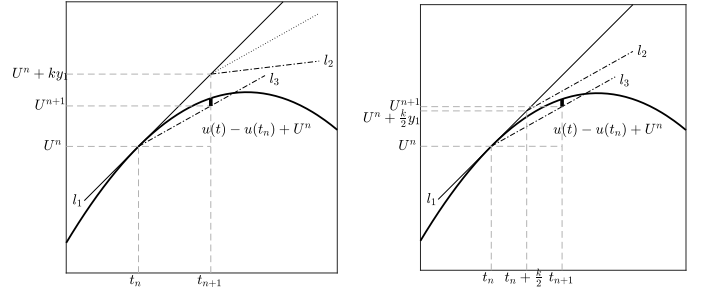


Figure 13.1: Geometric meanings of the improved Euler method (left) and the modified Euler method (right).

Definition 13.4. The *modified Euler method* or the *improved polygon method* or the *explicit midpoint method* is a one-step method of the form

$$\begin{cases} \mathbf{y}_1 = \mathbf{f}(\mathbf{U}^n, t_n), \\ \mathbf{y}_2 = \mathbf{f}(\mathbf{U}^n + \frac{k}{2}\mathbf{y}_1, t_n + \frac{k}{2}), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + k\mathbf{y}_2. \end{cases} \quad (13.4)$$

Exercise 13.5. Does the length of a short thick line segment in Figure 13.1 represent the one-step error in Definition 13.8? If so, prove it; otherwise derive an expression of the represented quantity.

Definition 13.6. The *TR-BDF2 method* is a one-step method of the form

$$\begin{cases} \mathbf{U}^* = \mathbf{U}^n + \frac{k}{4}(\mathbf{f}(\mathbf{U}^n, t_n) + \mathbf{f}(\mathbf{U}^*, t_n + \frac{k}{2})), \\ \mathbf{U}^{n+1} = \frac{1}{3}(4\mathbf{U}^* - \mathbf{U}^n + k\mathbf{f}(\mathbf{U}^{n+1}, t_{n+1})). \end{cases} \quad (13.5)$$

Exercise 13.7. Give a geometric interpretation of TR-BDF2 by drawing a figure similar to those for the improved Euler method and the modified Euler method.

13.1.2 Consistency and convergence

Definition 13.8. The *one-step error* of a one-step method (13.1) is

$$\mathcal{L}\mathbf{u}(t_n) := \mathbf{u}(t_{n+1}) - \mathbf{u}(t_n) - k\Phi(\mathbf{u}(t_n), t_n; k). \quad (13.6)$$

Definition 13.9. A one-step method is said to have *order of accuracy* p or to have *pth-order accuracy* or to be *pth-order accurate* if, as $k \rightarrow 0$,

$$\mathcal{L}\mathbf{u}(t_n) = \Theta(k^{p+1}). \quad (13.7)$$

Definition 13.10. A one-step method is *consistent* if

$$\lim_{k \rightarrow 0} \frac{1}{k} \mathcal{L}\mathbf{u}(t_n) = \mathbf{0}. \quad (13.8)$$

Exercise 13.11. Use recursive Taylor expansions to derive the k^3 term in the one-step error $\mathcal{L}\mathbf{u}(t_n)$ of the explicit midpoint method, verifying $\mathcal{L}\mathbf{u}(t_n) = \Theta(k^3)$, i.e., the explicit midpoint method is second-order accurate.

Theorem 13.12. A one-step method is consistent if and only if

$$\lim_{k \rightarrow 0} \Phi(\mathbf{u}, t; k) = \mathbf{f}(\mathbf{u}, t) \quad (13.9)$$

for any $(\mathbf{u}, t)$ in the domain of $\mathbf{f}$.

Proof. Definition 13.1 and a Taylor expansion of $\mathbf{u}(t_{n+1})$ at t_n yield

$$\frac{\mathcal{L}\mathbf{u}(t_n)}{k} = \mathbf{f}(\mathbf{u}(t_n), t_n) - \Phi(\mathbf{u}(t_n), t_n; k) + O(k).$$

The proof is completed by taking limit of the above equation in the asymptotic range of $k \rightarrow 0$, cf. Definition 13.10. $\square$

Corollary 13.13. The Euler method is consistent.

Proof. This follows from Theorem 13.12 and the fact that $\Phi(\mathbf{u}, t; 0) = \mathbf{f}(\mathbf{u}, t)$ for the Euler method. $\square$

Definition 13.14. A one-step method is *convergent* iff

$$\lim_{k \rightarrow 0; Nk=T} \mathbf{U}^N = \mathbf{u}(T), \quad (13.10)$$

i.e., its solution error tends to zero as $k \rightarrow 0$ for any $T > 0$ and for any initial condition $\mathbf{U}^0 = \mathbf{u}(0) + o(1)$.

Lemma 13.15. Let (ξ_n) be a sequence in $\mathbb{R}$ such that

$$|\xi_{n+1}| \leq (1 + C)|\xi_n| + D, \quad n \in \mathbb{N} \quad (13.11)$$

for some positive constants C and D . Then we have

$$|\xi_n| \leq e^{nC} |\xi_0| + \frac{D}{C} (e^{nC} - 1), \quad n \in \mathbb{N}. \quad (13.12)$$

Proof. The induction basis $n = 0$ clearly holds. Now suppose (13.12) holds for n , then for the inductive step, we have

$$\begin{aligned} |\xi_{n+1}| &\leq (1 + C)e^{nC} |\xi_0| + (1 + C) \frac{D}{C} (e^{nC} - 1) + D \\ &\leq e^{(n+1)C} |\xi_0| + \frac{D}{C} (e^{(n+1)C} - 1), \end{aligned}$$

where the first inequality follows from the induction hypothesis and the second from $1 + C \leq e^C$. Thus the estimate (13.12) holds for $n + 1$ as well. $\square$

Theorem 13.16. Suppose the increment function Φ that describes a one-step method is continuous (in $\mathbf{u}$, t , and k) and satisfies a Lipschitz condition

$$\|\Phi(\mathbf{u}, t; k) - \Phi(\mathbf{v}, t; k)\| \leq M \|\mathbf{u} - \mathbf{v}\| \quad (13.13)$$

for all $(\mathbf{u}, t)$ and $(\mathbf{v}, t)$ in the domain of $\mathbf{f}$ and for all sufficiently small k . Also suppose that the initial condition

satisfies $\|\mathbf{E}^0\| = O(k)$. Then the one-step method is convergent if and only if it is consistent. Furthermore, if the method has order of accuracy p , i.e., $\|\mathcal{L}\mathbf{u}(t_n)\| \leq Kk^{p+1}$, and the initial condition satisfies $\|\mathbf{E}^0\| = O(k^{p+1})$, then its solution error can be bounded as

$$\|\mathbf{E}^n\| \leq \frac{K}{M} (e^{MT} - 1) k^p. \quad (13.14)$$

Proof. For sufficiency, we assume that the one-step method is consistent and compute

$$\begin{aligned} \|\mathbf{E}^{n+1} - \mathbf{E}^n\| &= \|(\mathbf{U}^{n+1} - \mathbf{U}^n) - (\mathbf{u}(t_{n+1}) - \mathbf{u}(t_n))\| \\ &= \|k\Phi(\mathbf{U}^n, t_n; k) - (\mathbf{u}(t_{n+1}) - \mathbf{u}(t_n))\| \\ &= \|k\Phi(\mathbf{U}^n, t_n; k) - k\Phi(\mathbf{u}(t_n), t_n; k) - \mathcal{L}\mathbf{u}(t_n)\| \\ &\leq kM\|\mathbf{U}^n - \mathbf{u}(t_n)\| + kc(k), \end{aligned}$$

where the last step follows from the Lipschitz condition (13.13) and $\lim_{k \rightarrow 0} c(k) = \lim_{k \rightarrow 0} \frac{1}{k} \max \|\mathcal{L}\mathbf{u}(t)\| = 0$. Hence we have

$$\|\mathbf{E}^{n+1}\| \leq (1 + kM)\|\mathbf{E}^n\| + kc(k).$$

Lemma 13.15 with $C = kM$ and $D = kc(k)$ yields

$$\|\mathbf{E}^n\| \leq \|\mathbf{E}^0\| e^{MT} + \frac{c(k)}{M} (e^{MT} - 1),$$

which establishes the convergence since $\|\mathbf{E}^0\|$ and $c(k)$ both tend to 0 as $k \rightarrow 0$. In particular, (13.14) follows from this inequality and the condition of $c(k) \leq Kk^p$.

For necessity, we assume that the one-step method is convergent, i.e., the one-step method (13.1) converges to the solution of

$$\mathbf{u}'(t) = \mathbf{f}(\mathbf{u}, t), \quad \mathbf{u}(0) = \mathbf{u}_0,$$

for all final time $T > 0$. Consider

$$\mathbf{g}(\mathbf{u}, t) := \Phi(\mathbf{u}, t; 0)$$

and observe that by Theorem 13.12 the one-step method is consistent with the new IVP

$$\mathbf{u}'(t) = \mathbf{g}(\mathbf{u}, t), \quad \mathbf{u}(0) = \mathbf{u}_0.$$

Since we have already shown that consistency implies convergence, the one-step method also converges to this new IVP. Hence the solutions of the two IVPs coincide (why?) and we have $\mathbf{f}(\mathbf{u}(\tau), \tau) = \mathbf{g}(\mathbf{u}(\tau), \tau)$ for all $(\mathbf{u}(\tau), \tau)$ in the domain of $\mathbf{f}$. Then the continuity of Φ in k at $k = 0$ implies

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall k < \delta, \forall t \in [0, T],$$

$$\begin{aligned} &\|\Phi(\mathbf{u}, t; k) - \mathbf{f}(\mathbf{u}, t)\| \\ &\leq \|\Phi(\mathbf{u}, t; 0) - \mathbf{f}(\mathbf{u}, t)\| + \|\Phi(\mathbf{u}, t; k) - \Phi(\mathbf{u}, t; 0)\| < \epsilon, \end{aligned}$$

which implies uniform convergence of $\Phi(\mathbf{u}, t; k)$ to $\mathbf{f}$, cf. Definition D.168. Then the proof is completed by Theorem 13.12. $\square$

Corollary 13.17. Both the modified Euler method and the improved Euler method are convergent. If $\mathbf{f}$ in the IVP is twice continuously differentiable, then both methods are second-order accurate.

Proof. The increment function

$$\Phi(\mathbf{u}, t; k) = \mathbf{f}\left(\mathbf{u} + \frac{k}{2}\mathbf{f}(\mathbf{u}, t), t + \frac{k}{2}\right)$$

describing the modified Euler method (13.4) clearly satisfies the consistency condition (13.9) and hence by Theorem 13.16, it only remains to verify the Lipschitz condition of Φ . From the Lipschitz condition for $\mathbf{f}$ we obtain

$$\begin{aligned} & \|\Phi(\mathbf{u}, t; k) - \Phi(\mathbf{v}, t; k)\| \\ &= \left\| \mathbf{f}\left(\mathbf{u} + \frac{k}{2}\mathbf{f}(\mathbf{u}, t), t + \frac{k}{2}\right) - \mathbf{f}\left(\mathbf{v} + \frac{k}{2}\mathbf{f}(\mathbf{v}, t), t + \frac{k}{2}\right) \right\| \\ &\leq L\left(\|\mathbf{u} - \mathbf{v}\| + \frac{k}{2}\|\mathbf{f}(\mathbf{u}, t) - \mathbf{f}(\mathbf{v}, t)\|\right) \\ &\leq L\left(1 + \frac{kL}{2}\right)\|\mathbf{u} - \mathbf{v}\|, \end{aligned}$$

hence Φ also satisfies a Lipschitz condition.

If $\mathbf{f}$ is $\mathcal{C}^2$, then by Exercise 13.11, the one-step error of the modified Euler method satisfies $\|\mathcal{L}\mathbf{u}(t_n)\| \leq Kk^3$. Therefore the modified Euler method (13.4) has order of accuracy two by Theorem 13.16.

The same result concerning the improved Euler method (13.3) can be proved in a similar manner. $\square$

13.1.3 Absolute stability

Definition 13.18. The *stability function* of a one-step method is a function $R: \mathbb{C} \rightarrow \mathbb{C}$ that satisfies

$$U^{n+1} = R(z)U^n \quad (13.15)$$

for the test problem $u'(t) = \lambda u$ where $z := k\lambda$.

Example 13.19. The trapezoidal rule, when viewed as a one-step method, has its stability function as

$$R(z) = \frac{1 + \frac{1}{2}z}{1 - \frac{1}{2}z}, \quad (13.16)$$

which is also the root of the LMM stability polynomial in Example 12.85. Indeed, the stability function of the one-step method and the LMM stability polynomial have to coincide because they characterize the same concept from different viewpoints.

Exercise 13.20. Show that the TR-BDF2 method in (13.5) satisfies

$$R(z) = \frac{1 + \frac{5}{12}z}{1 - \frac{7}{12}z + \frac{1}{12}z^2}, \quad (13.17)$$

and $R(z) - e^z = O(z^3)$ as $z \rightarrow 0$.

Definition 13.21. The *RAS* of a one-step method is a subset of the complex plane

$$\mathcal{R} := \{z \in \mathbb{C} : |R(z)| \leq 1 + O(k)\}. \quad (13.18)$$

13.1.4 A-stability and L-stability

Definition 13.22. A one-step method is *A-stable* iff its RAS $\mathcal{R}$ satisfies $\mathbb{C}^- \subseteq \mathcal{R}$, cf. (12.88).

Example 13.23. We use the trapezoidal and backward Euler methods to solve the IVP (12.82) with $\lambda = -10^6$. By Example 12.113, both methods are A-stable. However, the following table shows different patterns of their errors at $T = 3$ with various values of k and the initial condition $u(0) = \eta$.

	k	Backward Euler	Trapezoidal
$\eta = 1$	0.2	9.7731e-08	4.7229e-10
	0.1	4.9223e-08	1.1772e-10
	0.05	2.4686e-08	2.9406e-11
$\eta = 1.5$	0.2	9.7731e-08	4.9985e-01
	0.1	4.9223e-08	4.9940e-01
	0.05	2.4686e-08	4.9761e-01

The stability functions are

$$\begin{aligned} R(z) &= \frac{1}{1-z} && \text{for backward Euler,} \\ R(z) &= \frac{2+z}{2-z} && \text{for trapezoidal.} \end{aligned}$$

Although the trapezoidal method is stable and the results stay bounded, but since

$$\forall k > 0, \quad \frac{1 + \frac{1}{2}k\lambda}{1 - \frac{1}{2}k\lambda} \rightarrow -1 \text{ as } \lambda \rightarrow -\infty,$$

the initial deviation from the smooth curve $\cos t$ is essentially negated in each time step. For $k = 0.1$, we have $\frac{2+k\lambda}{2-k\lambda} = -0.99996 \approx -1$ for trapezoidal method while for backward Euler method $(1 - k\lambda)^{-1} \approx 10^{-5}$. This leads to dramatically different results shown in Figure 13.2.

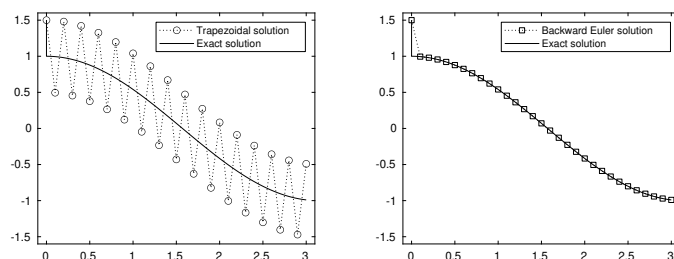


Figure 13.2: L-stability illustrated by Example 13.23 where results (left) of the second-order trapezoidal method without L-stability are much worse than those (right) of the first-order L-stable backward Euler method.

Definition 13.24. A one-step method is *L-stable* if it is A-stable and its stability function satisfies

$$\lim_{z \rightarrow \infty} |R(z)| = 0. \quad (13.19)$$

Exercise 13.25. Reproduce the results in Example 13.23 and explain in your own language why the first-order backward Euler method is superior to the second-order trapezoidal method.

13.2 Generic forms of RK methods

Definition 13.26. An *s-stage Runge-Kutta (RK) method* is a one-step method of the form

$$\begin{cases} \mathbf{y}_i = \mathbf{f}(\mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \mathbf{y}_j, t_n + c_i k), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + k \sum_{j=1}^s b_j \mathbf{y}_j, \end{cases} \quad (13.20)$$

where $i = 1, 2, \dots, s$ and the coefficients $a_{i,j}$, b_j , c_i are real.

Definition 13.27. The *Butcher tableau* is one way to organize the coefficients of an RK method as follows.

$$\begin{array}{c|ccc} c_1 & a_{1,1} & \cdots & a_{1,s} \\ \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s,1} & \cdots & a_{s,s} \\ \hline & b_1 & \cdots & b_s \end{array} \quad (13.21)$$

The matrix $A = (a_{i,j})_{i,j=1,\dots,s}$ is called the *RK matrix* while

$$\mathbf{b} = (b_1, b_2, \dots, b_s)^\top \text{ and } \mathbf{c} = (c_1, c_2, \dots, c_s)^\top$$

are called the *RK weights* and the *RK nodes*, respectively.

Notation 11. We follow Notation 3 in writing the matrix A in a Butcher tableau.

Exercise 13.28. Write down the Butcher tableaux of the modified Euler method, the improved Euler method, and Heun's third-order method in Definition 13.35.

Exercise 13.29. Verify that the RK method (13.20) can be rewritten as

$$\begin{cases} \xi_i = \mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \mathbf{f}(\xi_j, t_n + c_j k), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + k \sum_{j=1}^s b_j \mathbf{f}(\xi_j, t_n + c_j k), \end{cases} \quad (13.22)$$

where $i = 1, 2, \dots, s$.

Theorem 13.30. It suffices to only consider autonomous IVPs for the analysis of RK methods if we set

$$\forall i = 1, 2, \dots, s, \quad c_i = \sum_{j=1}^s a_{i,j}; \quad (13.23)$$

$$\sum_{i=1}^s b_i = 1. \quad (13.24)$$

Proof. A non-autonomous IVP can be converted to an equivalent autonomous IVP. Indeed, for the IVP

$$\mathbf{u}' = \mathbf{f}(\mathbf{u}, t), \quad \mathbf{u}(t_0) = \mathbf{u}_0, \quad (13.25)$$

define a new variable $\tilde{\mathbf{u}}(t) = \begin{pmatrix} \mathbf{u}(t) \\ t \end{pmatrix} \in \mathbb{R}^{N+1}$ and we have

$$\tilde{\mathbf{u}}' = \tilde{\mathbf{f}}(\tilde{\mathbf{u}}), \quad \tilde{\mathbf{u}}(t_0) = \begin{pmatrix} \mathbf{u}_0 \\ t_0 \end{pmatrix}, \quad (13.26)$$

$$\tilde{\mathbf{f}}(\tilde{\mathbf{u}}) := \begin{pmatrix} \mathbf{f}(\mathbf{u}, t) \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{f}((\tilde{u}_1, \dots, \tilde{u}_N)^\top, \tilde{u}_{N+1}) \\ 1 \end{pmatrix},$$

$(\tilde{u}_1, \dots, \tilde{u}_N)^\top = \mathbf{u}$, and $\tilde{u}_{N+1}$ is the time variable in the original formulation. The last equation of (13.26),

$$\frac{d}{dt} \tilde{u}_{N+1} = 1, \quad \tilde{u}_{N+1}(t_0) = t_0,$$

is solved by $\tilde{u}_{N+1}(t) = t$; the other equations are equivalent to the original system (13.25). Therefore, if (13.26) has a unique solution, (13.25) will have the same one.

For an RK method (13.20) satisfying (13.23) and (13.24), the solution for (13.26) is the same as that for (13.25). Indeed, for (13.25) consider its i th stage,

$$\mathbf{y}_i = \mathbf{f}(\mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \mathbf{y}_j, t_n + c_i k).$$

For the autonomous system (13.26), we have

$$\begin{aligned} \tilde{\mathbf{y}}_i &= \tilde{\mathbf{f}}\left(\tilde{\mathbf{U}}^n + k \sum_{j=1}^s a_{i,j} \tilde{\mathbf{y}}_j\right) \\ &= \begin{pmatrix} \mathbf{f}\left(\tilde{\mathbf{U}}_*^n + k \sum_{j=1}^s a_{i,j} \tilde{\mathbf{y}}_{j,*}, \tilde{U}_{N+1}^n + k \sum_{j=1}^s a_{i,j} \tilde{y}_{j,N+1}\right) \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{f}\left(\tilde{\mathbf{U}}_*^n + k \sum_{j=1}^s a_{i,j} \tilde{\mathbf{y}}_{j,*}, \tilde{U}_{N+1}^n + k \sum_{j=1}^s a_{i,j}\right) \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{f}\left(\mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \tilde{\mathbf{y}}_{j,*}, t_n + c_i k\right) \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{y}_i \\ 1 \end{pmatrix}, \end{aligned}$$

where $\tilde{\mathbf{U}}_*^n$ and $\tilde{\mathbf{y}}_{j,*}$ are respectively the first N components of $\tilde{\mathbf{U}}^n$ and $\tilde{\mathbf{y}}_j$, the third step follows from the $(N+1)$ th component of each $\tilde{\mathbf{y}}_i$ being 1, the fourth from $\tilde{\mathbf{U}}_*^n = \mathbf{U}^n$, $\tilde{U}_{N+1}^n = t_n$, and (13.23). Finally, (13.24) yields

$$\tilde{\mathbf{U}}_*^{n+1} = \tilde{\mathbf{U}}_*^n + k \sum_{j=1}^s b_j \tilde{\mathbf{y}}_{j,*} = \mathbf{U}^n + k \sum_{j=1}^s b_j \mathbf{y}_j = \mathbf{U}^{n+1};$$

$$\tilde{U}_{N+1}^{n+1} = \tilde{U}_{N+1}^n + k \sum_{j=1}^s b_j \tilde{y}_{j,N+1} = t_n + k \sum_{j=1}^s b_j = t_{n+1},$$

which complete the proof. $\square$

13.3 Explicit RK (ERK) methods

Definition 13.31. An s -stage explicit RK (ERK) method for solving the IVP (12.3) is an RK method of the form

$$\begin{cases} \mathbf{y}_i = \mathbf{f}(\mathbf{U}^n + k \sum_{j=1}^{i-1} a_{i,j} \mathbf{y}_j, t_n + c_i k), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + k \sum_{j=1}^s b_j \mathbf{y}_j, \end{cases} \quad (13.27)$$

where $a_{i,j} = 0$ for $i \leq j$ and

$$\forall i = 1, 2, \dots, s, \quad c_i = \sum_{j=1}^{i-1} a_{i,j}. \quad (13.28)$$

Example 13.32. The Butcher tableau of an s -stage ERK method is of the form

$$\begin{array}{c|ccccc} 0 & 0 & & & \\ c_2 & a_{2,1} & 0 & & \\ c_3 & a_{3,1} & a_{3,2} & 0 & \\ \vdots & \vdots & \vdots & \ddots & \ddots \\ c_s & a_{s,1} & a_{s,2} & \cdots & a_{s,s-1} & 0 \\ \hline & b_1 & b_2 & \cdots & b_{s-1} & b_s \end{array} \quad (13.29)$$

Lemma 13.33. For an ERK method, the condition (13.28) is equivalent to

$$\mathbf{y}_i(\mathbf{u}(t_n), t_n; k) = \mathbf{u}'(t_n + c_i k) + O(k^2), \quad (13.30)$$

for $i = 1, 2, \dots, s$.

Proof. We prove (13.30) by an induction on i . The induction basis of $i = 1$ holds because

$$\mathbf{y}_1(\mathbf{u}(t_n), t_n; k) = \mathbf{f}(\mathbf{u}(t_n), t_n) = \mathbf{u}'(t_n).$$

Suppose (13.30) holds for each $i = 1, 2, \dots, m$. Then for $m + 1$, we have

$$\begin{aligned} & \mathbf{y}_{m+1}(\mathbf{u}(t_n), t_n; k) \\ &= \mathbf{f}\left(\mathbf{u}(t_n) + k \sum_{j=1}^m a_{m+1,j} \mathbf{y}_j, t_n + c_{m+1}k\right) \\ &= \mathbf{f} + k \left(\mathbf{f}_{\mathbf{u}} \sum_{j=1}^m a_{m+1,j} \mathbf{y}_j + c_{m+1} \mathbf{f}_t \right) + O(k^2) \\ &= \mathbf{f} + k \left(\mathbf{f}_{\mathbf{u}} \sum_{j=1}^m a_{m+1,j} \mathbf{u}'(t_n + c_j k) + c_{m+1} \mathbf{f}_t \right) + O(k^2) \\ &= \mathbf{f} + k \left(\mathbf{f}_{\mathbf{u}} \mathbf{u}'(t_n) \sum_{j=1}^m a_{m+1,j} + c_{m+1} \mathbf{f}_t \right) + O(k^2), \end{aligned}$$

where all functions are evaluated at $(\mathbf{u}(t_n), t_n)$, the first step follows from the condition $a_{i,j} = 0$ for $i \leq j$, the second and fourth steps from Taylor expansions, and the third from the induction hypothesis. Taylor expansions also give

$$\begin{aligned} \mathbf{u}'(t_n + c_{m+1}k) &= \mathbf{u}'(t_n) + kc_{m+1} \mathbf{u}''(t_n) + O(k^2) \\ &= \mathbf{f} + kc_{m+1} (\mathbf{f}_{\mathbf{u}} \mathbf{u}'(t_n) + \mathbf{f}_t) + O(k^2). \end{aligned}$$

On one hand, the condition

$$(*) : \quad c_{m+1} = \sum_{j=1}^m a_{m+1,j}$$

yields (13.30) for $i = m + 1$; on the other hand, (13.30) for $i = m + 1$ implies $(*)$ since both c_{m+1} and $a_{m+1,j}$'s are independent on k . $\square$

Example 13.34. For the two-stage ERK method

$$\begin{cases} \mathbf{y}_1 = \mathbf{f}(\mathbf{U}^n, t_n), \\ \mathbf{y}_2 = \mathbf{f}(\mathbf{U}^n + ka_{2,1}\mathbf{y}_1, t_n + c_2k), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + k(b_1\mathbf{y}_1 + b_2\mathbf{y}_2), \end{cases} \quad (13.31)$$

the one-step error $\mathcal{L}\mathbf{u}(t_n)$ is

$$\begin{aligned} & \mathbf{u}(t_{n+1}) - \mathbf{u}(t_n) - kb_1\mathbf{f}(\mathbf{u}(t_n), t_n) \\ & \quad - kb_2\mathbf{f}(\mathbf{u}(t_n) + kc_2\mathbf{y}_1, t_n + c_2k) \\ &= \left(\mathbf{u} + k\mathbf{u}' + \frac{k^2}{2}\mathbf{u}'' + \frac{k^3}{6}\mathbf{u}''' + O(k^4) \right) - \mathbf{u} - kb_1\mathbf{u}' \\ & \quad - kb_2 \left[\mathbf{f} + kc_2(\mathbf{f}_{\mathbf{u}}\mathbf{f} + \mathbf{f}_t) + \frac{k^2c_2^2}{2}(\mathbf{f}^2\mathbf{f}_{\mathbf{uu}} + 2\mathbf{f}\mathbf{f}_{\mathbf{ut}} + \mathbf{f}_{\mathbf{tt}}) + O(k^3) \right] \\ &= k(1 - b_1 - b_2)\mathbf{u}' + k^2 \left(\frac{1}{2} - b_2c_2 \right) \mathbf{u}'' \\ & \quad + k^3 \left[\left(\frac{1}{6} - \frac{b_2c_2^2}{2} \right) \mathbf{u}''' + \frac{b_2c_2^2}{2} \mathbf{f}_{\mathbf{u}}\mathbf{u}'' \right] + O(k^4), \end{aligned}$$

where the term $\left(\frac{1}{6} - \frac{b_2c_2^2}{2} \right) \mathbf{u}''' + \frac{b_2c_2^2}{2} \mathbf{f}_{\mathbf{u}}\mathbf{u}''$ never vanishes.

To maximize the order of accuracy of (13.31), we choose

$$b_1 + b_2 = 1, \quad b_2c_2 = \frac{1}{2}$$

to obtain a family of two-stage, second-order ERK methods

$$\begin{array}{c|cc} 0 & 0 & \\ \alpha & \alpha & 0 \\ \hline & 1 - \frac{1}{2\alpha} & \frac{1}{2\alpha} \end{array}.$$

The choice of $\alpha = 1$ yields the explicit trapezoidal method (13.3), and that of $\alpha = \frac{1}{2}$ yields the explicit mid-point method (13.4). We have united the two methods under one derivation. Note that the above derivation is more general than Exercise 13.11.

Definition 13.35. Heun's third-order formula is an ERK method of the form

$$\begin{cases} \mathbf{y}_1 = \mathbf{f}(\mathbf{U}^n, t_n), \\ \mathbf{y}_2 = \mathbf{f}(\mathbf{U}^n + \frac{k}{3}\mathbf{y}_1, t_n + \frac{k}{3}), \\ \mathbf{y}_3 = \mathbf{f}(\mathbf{U}^n + \frac{2k}{3}\mathbf{y}_2, t_n + \frac{2k}{3}), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + \frac{k}{4}(\mathbf{y}_1 + 3\mathbf{y}_3). \end{cases} \quad (13.32)$$

Exercise 13.36. There are three one-parameter families of third-order three-stage ERK methods, one of which is

$$\begin{array}{c|ccc} 0 & 0 & & \\ \frac{2}{3} & \frac{2}{3} & 0 & \\ \frac{2}{3} & \frac{2}{3} - \frac{1}{4\alpha} & \frac{1}{4\alpha} & 0 \\ \hline & \frac{1}{4} & \frac{3}{4} - \alpha & \alpha \end{array},$$

where α is the free parameter. Derive the above family. Does Heun's third-order formula belong to this family?

Definition 13.37. The classical fourth-order RK method is an ERK method of the form

$$\begin{cases} \mathbf{y}_1 = \mathbf{f}(\mathbf{U}^n, t_n), \\ \mathbf{y}_2 = \mathbf{f}(\mathbf{U}^n + \frac{k}{2}\mathbf{y}_1, t_n + \frac{k}{2}), \\ \mathbf{y}_3 = \mathbf{f}(\mathbf{U}^n + \frac{k}{2}\mathbf{y}_2, t_n + \frac{k}{2}), \\ \mathbf{y}_4 = \mathbf{f}(\mathbf{U}^n + k\mathbf{y}_3, t_n + k), \\ \mathbf{U}^{n+1} = \mathbf{U}^n + \frac{k}{6}(\mathbf{y}_1 + 2\mathbf{y}_2 + 2\mathbf{y}_3 + \mathbf{y}_4). \end{cases} \quad (13.33)$$

Example 13.38. The Butcher tableau of the classical fourth-order RK method (13.33) is

$$\begin{array}{c|cccc} 0 & 0 & & & \\ \frac{1}{2} & \frac{1}{2} & 0 & & \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array} \quad (13.34)$$

13.4 Necessary order conditions

Theorem 13.39 (RK order conditions). To be p th-order accurate, an s -stage RK method (13.20) must satisfy the order conditions

$$\begin{cases} \forall l = 1, 2, \dots, p, \\ \forall m = 0, 1, \dots, p-l, \end{cases} \quad \mathbf{b}^\top A^m C^{l-1} \mathbf{1} = \frac{(l-1)!}{(l+m)!}, \quad (13.35)$$

where $C = \text{diag}(c_1, c_2, \dots, c_s)$ and $\mathbf{1} \in \mathbb{Z}^s$ is a column vector with all components equal to one.

Proof. For the scalar IVP

$$u' = u + t^{l-1}, \quad u(0) = 0, \quad (13.36)$$

where $l \in \mathbb{N}^+$, the first step of the RK method yields

$$\forall i = 1, 2, \dots, s, \quad y_i = k \sum_{j=1}^s a_{i,j} y_j + (c_i k)^{l-1},$$

which can be rewritten in matrix form as

$$(I - kA)\mathbf{y} = k^{l-1} C^{l-1} \mathbf{1},$$

where $\mathbf{y} = (y_1, y_2, \dots, y_s)^\top$. It follows that

$$\mathbf{y} = k^{l-1} (I - kA)^{-1} C^{l-1} \mathbf{1}$$

and we have $U^1 = U^0 + k\mathbf{b}^\top \mathbf{y} = k^l \mathbf{b}^\top (I - kA)^{-1} C^{l-1} \mathbf{1}$, i.e.,

$$U^1 = k^l \sum_{m=0}^{+\infty} k^m (\mathbf{b}^\top A^m C^{l-1} \mathbf{1}). \quad (13.37)$$

By Theorem 12.15, the exact solution of (13.36) is

$$u(t) = \int_0^t e^{t-\tau} \tau^{l-1} d\tau,$$

which, together with the chain rule or the Leibniz formula, implies

$$\begin{cases} u(0) = \dots = u^{(l-1)}(0) = 0, \\ \forall j \geq 0, \quad u^{(l+j)}(0) = (l-1)!. \end{cases}$$

The Taylor expansion of $u(t_1)$ at $t_0 = 0$ yields

$$u(t_1) = k^l \sum_{m=0}^{+\infty} k^m \frac{(l-1)!}{(l+m)!}. \quad (13.38)$$

Then the proof is completed by Definition 13.9, which implies that, in order for the RK method to be p th-order accurate, the series of U^1 and $u(t_1)$ in (13.37) and (13.38) must agree for the first $p - l + 1$ terms for each $l = 1, 2, \dots, p$. $\square$

Definition 13.40 (Pure quadrature order conditions). An s -stage RK method is said to be $B(r)$ if it satisfies

$$\forall l = 1, 2, \dots, r, \quad \mathbf{b}^\top C^{l-1} \mathbf{1} = \sum_{j=1}^s b_j c_j^{l-1} = \frac{1}{l}. \quad (13.39)$$

Definition 13.41. The *quadrature formula of an s -stage RK method* is the weighted quadrature formula

$$I_s(f) := k \sum_{j=1}^s b_j f(t_n + c_j k) \quad (13.40)$$

that approximates $I(f) := \int_{t_n}^{t_n+k} f(t) dt$.

Exercise 13.42. Show that the quadrature formula of a RK method is exact for all polynomials f of degree $< r$, i.e.,

$$\forall f \in \mathbb{P}_{r-1}, \quad I_s(f) = \int_{t_n}^{t_n+k} f(t) dt, \quad (13.41)$$

if and only if the RK method is $B(r)$.

Example 13.43. Exercise 13.42 justifies the name “pure quadrature order conditions” since (13.39) is the special case of $m = 0$ in (13.35) that corresponds to the RHS of an IVP depending only on time t .

If $m \neq 0$, the order conditions in (13.35) can be put as

$$\sum_{i,j_1,\dots,j_m} b_i a_{i,j_1} a_{j_1,j_2} \cdots a_{j_{m-1},j_m} c_{j_m}^{l-1} = \frac{(l-1)!}{(l+m)!}.$$

For $l = 1$, the order conditions in (13.35) reduce to

$$\forall m = 0, 1, \dots, p-1, \quad \mathbf{b}^\top A^m \mathbf{1} = \frac{1}{(m+1)!}. \quad (13.42)$$

In particular, $m = 0$ and $l = 1$ yield the condition (13.24) discussed in Theorem 13.30.

Corollary 13.44. An s -stage ERK method is at best s th-order accurate.

Proof. Since the RK matrix A of an s -stage ERK method is strictly lower triangular, A is nilpotent with $A^s = \mathbf{0}$. Suppose for the IVP (13.36) the order of accuracy of the ERK method is at least $s+1$. Then Definition 13.9, (13.37) and (13.38) yield

$$\mathcal{L}u(t_0) = u(t_1) - U^1 = o(k^{s+1}),$$

which is impossible because the coefficient of k^{s+1} of U^1 is $\mathbf{b}^\top A^s \mathbf{1} = 0$ whereas that of $u(t_1)$ is $\frac{1}{(s+1)!} \neq 0$. $\square$

Example 13.45. The classical fourth-order RK method (13.33) can be obtained as follows.

(a) Simpson’s rule (Definition 9.14) for quadrature is fourth-order accurate:

$$\int_0^1 f(t) dt \approx \frac{1}{6} f(0) + \frac{2}{3} f\left(\frac{1}{2}\right) + \frac{1}{6} f(1).$$

(b) Simpson’s rule has only three abscissas: 0, $\frac{1}{2}$, and 1. By Corollary 13.44, a fourth-order ERK method has at least four stages, so tentatively we settle for four stages with the abscissas as $c_i = 0, \frac{1}{2}, \frac{1}{2}, 1$.

(c) Simpson’s rule indicates the choice $b_1 = b_4 = \frac{1}{6}$. Set $l = 1$ in (13.39) and we have (13.24), i.e. $\sum_j b_j = 1$. A simple choice is then $b_2 = b_3 = \frac{1}{3}$.

(d) For $m = 3$ in (13.42), we have

$$\mathbf{b}^\top A^3 \mathbf{1} = b_4 a_{43} a_{32} a_{21} = \frac{1}{24},$$

which implies $a_{i+1,i} \neq 0$.

(e) Choose $a_{i+1,i}$ ’s to be the only nonzero elements of A and deduce $a_{i+1,i} = c_{i+1}$ for each $i = 1, 2, 3$ from (13.28).

The Butcher tableau (13.34) of the classical fourth-order RK method is now fully determined.

Theorem 13.46. ERK methods have the following limits on the attainable order of accuracy as a function of the number of stages:

number of stages	1	2	3	4	5	6	7	8	9	10
max order of accuracy	1	2	3	4	4	5	6	6	7	7

13.5 Implicit RK (IRK) methods

Definition 13.47. An *implicit RK (IRK) method* is an RK method with at least one $a_{i,j} \neq 0$ for $i \leq j$.

Definition 13.48. A *diagonal IRK (DIRK) method* is an IRK method with $a_{i,j} = 0$ whenever $i < j$. A *singly DIRK (SDIRK) method* is a DIRK method with $a_{1,1} = a_{2,2} = \cdots = a_{s,s} = \gamma \neq 0$.

Example 13.49. The TR-BDF2 method can be regarded as a second-order DIRK method since it can be expressed in the standard form of an IRK method:

$$\begin{cases} y_1 = \mathbf{f}(U^n, t_n) \\ y_2 = \mathbf{f}(U^*, t_n + \frac{k}{2}) \text{ where } U^* = U^n + \frac{k}{4}(y_1 + y_2), \\ y_3 = \mathbf{f}(U^n + \frac{k}{3}(y_1 + y_2 + y_3), t_n + k) \\ U^{n+1} = U^n + \frac{k}{3}(y_1 + y_2 + y_3), \end{cases}$$

which yields the Butcher tableau

$$\begin{array}{c|ccc} 0 & 0 & & \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} & \\ 1 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \hline & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{array}.$$

Example 13.50. It can be shown that, under the condition (13.23), a two-stage RK method (13.20) is third-order accurate if and only if it satisfies the order conditions (13.35) with $p = 3$. This fact is utilized to construct two-stage third-order IRK methods.

(13.23) and the order conditions in (13.35) yield

$$i = 1, 2: \quad c_i = \sum_{j=1}^2 a_{i,j}, \quad (13.43a)$$

$$l = 1, 2, 3: \quad \sum_{i=1}^2 b_i c_i^{l-1} = \frac{1}{l}, \quad (13.43b)$$

$$\sum_{i=1}^2 \sum_{j=1}^2 b_i a_{i,j} c_j = \frac{1}{6}, \quad (13.43c)$$

where (13.43a) is the same as (13.23) and the four equations in (13.43b,c) come from the six equations in (13.35) with $p = 3$. The eight equations in (13.23) with $s = 2$ and (13.35) with $p = 3$ are dependent in the sense that some equations can be deduced from others. For example, the cases $l = 1, m = 2$ and $l = 2, m = 1$ are the same, i.e.,

$$\mathbf{b}^\top A^2 \mathbf{1} = \mathbf{b}^\top A(A\mathbf{1}) = \mathbf{b}^\top A\mathbf{c} = \mathbf{b}^\top A\mathbf{c}\mathbf{1}.$$

By Definition 13.40 and Theorem 13.39, (13.43b) implies that the method is $B(3)$ and thus could have third-order accuracy. By Exercise 13.42, the quadrature formula of this RK method is exact for all polynomials in $\mathbb{P}_2$. By Theorem 9.24, the node polynomial is orthogonal to $\mathbb{P}_0$, i.e.,

$$\int_0^1 (x - c_1)(x - c_2)dx = 0,$$

which yields $c_2 = \frac{2-3c_1}{3-6c_1}$ ($c_1 \neq \frac{1}{2}$). By Lemma 9.8, we determine the weights from the nodes as

$$\begin{aligned} b_1 &= \int_0^1 \frac{c_2 - x}{c_2 - c_1} dx = \frac{c_2 - \frac{1}{2}}{c_2 - c_1}, \\ b_2 &= \int_0^1 \frac{c_1 - x}{c_1 - c_2} dx = \frac{c_1 - \frac{1}{2}}{c_1 - c_2}. \end{aligned}$$

In (13.43c), we insert (13.43a) and consider $a_{1,2}$ and c_1 as free parameters. This gives

$$a_{2,2} = \frac{\frac{1}{6} - b_1 a_{1,2}(c_2 - c_1) - \frac{1}{2}c_1}{b_2(c_2 - c_1)}.$$

For $a_{1,2} = 0$ we get a one-parameter family of DIRK methods of order 3. If we further require $a_{1,1} = a_{2,2} = \gamma$, then we obtain two two-stage third-order SDIRK methods with Butcher tableau given by

$$\begin{array}{c|cc} \gamma & \gamma & \\ 1 - \gamma & 1 - 2\gamma & \gamma \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}, \quad \gamma = \frac{3 \pm \sqrt{3}}{6}. \quad (13.44)$$

Definition 13.51. An *explicit DIRK (EDIRK) method* is a DIRK method with an explicit first stage, i.e., $a_{1,1} = 0$. An *explicit SDIRK (ESDIRK) method* is an EDIRK method with $a_{2,2} = a_{3,3} = \cdots = a_{s,s} = \gamma \neq 0$.

Example 13.52. A commonly used ESDIRK method is the following six-stage fourth-order ESDIRK method

$$\begin{array}{c|cccccc} 0 & 0 & & & & \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} & & & \\ \frac{83}{250} & \frac{8611}{62500} & \frac{-1743}{31250} & \frac{1}{4} & & \\ \frac{31}{50} & \frac{5012029}{34652500} & \frac{-654441}{2922500} & \frac{174375}{388108} & \frac{1}{4} & \\ \frac{17}{20} & \frac{15267082809}{155376265600} & \frac{-71443401}{120774400} & \frac{730878875}{902184768} & \frac{2285395}{8070912} & \frac{1}{4} \\ 1 & \frac{82889}{524892} & 0 & \frac{15625}{83664} & \frac{69875}{102672} & \frac{-2260}{8211} & \frac{1}{4} \\ & \frac{82889}{524892} & 0 & \frac{15625}{83664} & \frac{69875}{102672} & \frac{-2260}{8211} & \frac{1}{4} \end{array}. \quad (13.45)$$

Theorem 13.53. The maximal attainable order of an s -stage DIRK method cannot exceed $s + 1$.

13.6 Collocation methods

Definition 13.54. A numerical method for solving the IVP (12.3) is said to have the *dense output* property iff, in addition to the numerical results $U^{n+1} \approx \mathbf{u}(t_n + k)$, it provides cheap numerical approximations to $\mathbf{u}(t_n + \theta k)$ for the entire interval of the time step $\theta \in [0, 1]$.

Example 13.55. For fourth-order dense output, we can fit a cubic polynomial from the RK results $U^n, \mathbf{f}(U^n, t_n), U^{n+1}$, and $\mathbf{f}(U^{n+1}, t_{n+1})$ by solving Hermite interpolation problems as in Definition 6.43.

Definition 13.56. An s -stage *collocation method* is a numerical method for solving the IVP (12.3) in which we

- (i) choose s distinct *collocation parameters* $c_1, c_2, \dots, c_s$ (typically in $[0, 1]$),

(ii) seek s -degree *collocation polynomials* $\mathbf{p}$ satisfying

$$\begin{cases} \mathbf{p}(t_n) = \mathbf{U}^n, \\ \forall i = 1, 2, \dots, s, \quad \mathbf{p}'\left(t_n^{(i)}\right) = \mathbf{f}\left(\mathbf{p}(t_n^{(i)}), t_n^{(i)}\right), \end{cases} \quad (13.46)$$

where $t_n^{(i)} := t_n + c_i k$,

(iii) set $\mathbf{U}^{n+1} = \mathbf{p}(t_{n+1})$.

Corollary 13.57. A collocation method has dense output.

Proof. This follows from Definitions 13.56 and 13.54: we get the dense-output property as a byproduct of polynomial fitting, with (13.46) as the interpolation conditions. $\square$

Theorem 13.58. The collocation method in Definition 13.56 is an s -stage IRK method with

$$a_{i,j} = \int_0^{c_i} \ell_j(\tau) d\tau, \quad (13.47a)$$

$$b_j = \int_0^1 \ell_j(\tau) d\tau, \quad (13.47b)$$

where $i, j = 1, 2, \dots, s$ and $\ell_j(\tau)$ is the elementary Lagrange interpolation polynomial, i.e.,

$$\ell_j(\tau) = \prod_{i \neq j; i=1}^s \frac{\tau - c_i}{c_j - c_i}. \quad (13.48)$$

Proof. The Lagrange interpolation polynomial

$$\mathbf{r}(t_n + \tau k) := \sum_{j=1}^s \mathbf{p}'\left(t_n^{(j)}\right) \ell_j(\tau)$$

satisfies $\mathbf{r}(t_n^{(i)}) = \mathbf{p}'(t_n^{(i)})$ for each $i = 1, 2, \dots, s$. The coincidence of the polynomials $\mathbf{r}$ and $\mathbf{p}'$ at s points implies $\mathbf{r} \equiv \mathbf{p}'$ since their degrees are both $s - 1$. Then (13.46) yields

$$\mathbf{p}'(t_n + \tau k) = \sum_{j=1}^s \mathbf{p}'\left(t_n^{(j)}\right) \ell_j(\tau) = \sum_{j=1}^s \mathbf{f}\left(\mathbf{p}(t_n^{(j)}), t_n^{(j)}\right) \ell_j(\tau).$$

Integrate over $[t_n, t_n + ck]$, apply the fundamental theorem of calculus, notice $\mathbf{p}(t_n) = \mathbf{U}^n$, and we obtain

$$\begin{aligned} \mathbf{p}(t_n + ck) &= \mathbf{p}(t_n) + k \int_0^c \mathbf{p}'(t_n + \tau k) d\tau \\ &= \mathbf{U}^n + k \sum_{j=1}^s \mathbf{f}\left(\mathbf{p}(t_n^{(j)}), t_n^{(j)}\right) \int_0^c \ell_j(\tau) d\tau, \end{aligned}$$

which can be rewritten as

$$\mathbf{p}(t_n + ck) = \mathbf{U}^n + k \sum_{j=1}^s \mathbf{y}_j \int_0^c \ell_j(\tau) d\tau \quad (13.49)$$

if we set

$$\forall i = 1, 2, \dots, s, \quad \mathbf{y}_i := \mathbf{f}\left(\mathbf{p}(t_n^{(i)}), t_n^{(i)}\right). \quad (13.50)$$

Choose $c = c_i$ in (13.49), substitute the resulting equation into (13.50), apply (13.47a), and we have

$$\mathbf{y}_i = \mathbf{f}\left(\mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \mathbf{y}_j, t_n^{(i)}\right).$$

Choose $c = 1$ in (13.49), apply (13.47b), and we have

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k \sum_{j=1}^s b_j \mathbf{y}_j.$$

The above processes recover the formalism in (13.20) of an RK method. In general, we have $a_{i,j} \neq 0$ for $i \leq j$. Thus we conclude that the collocation method is an IRK method. $\square$

Exercise 13.59. Show that an s -stage collocation method is at least s th-order accurate.

Exercise 13.60. Prove that the collocation method viewed as an RK method satisfies (13.23) and (13.24).

Example 13.61. Not every IRK method originates in collocation. Let $s = 2, c_1 = 0, c_2 = \frac{2}{3}$, then the corresponding elementary Lagrange interpolation polynomials are

$$\ell_1(\tau) = \frac{\tau - \frac{2}{3}}{-\frac{2}{3}} = \frac{2 - 3\tau}{2}, \quad \ell_2(\tau) = \frac{3\tau}{2},$$

and (13.47) yields the IRK method with Butcher tableau

$$\begin{array}{c|cc} 0 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ \hline & \frac{1}{4} & \frac{3}{4} \end{array}.$$

By Definition 13.56, a collocation method is fully determined once the RK nodes c_i 's are given. Therefore, we deduce from the above Butcher tableau that the IRK method

$$\begin{array}{c|cc} 0 & \frac{1}{4} & -\frac{1}{4} \\ \frac{2}{3} & \frac{1}{4} & \frac{5}{12} \\ \hline & \frac{1}{4} & \frac{3}{4} \end{array}$$

is not a collocation method.

Exercise 13.62. Derive the three-stage IRK method that corresponds to the collocation points $c_1 = \frac{1}{4}, c_2 = \frac{1}{2}, c_3 = \frac{3}{4}$.

Definition 13.63. An s -stage RK method (13.20) is said to be $C(r)$ if

$$\begin{cases} \forall i = 1, 2, \dots, s, \\ \forall m = 1, 2, \dots, r, \end{cases} \quad \sum_{j=1}^s a_{i,j} c_j^{m-1} = \frac{c_i^m}{m}, \quad (13.51)$$

it is said to be $D(r)$ if

$$\begin{cases} \forall i = 1, 2, \dots, s, \\ \forall m = 1, 2, \dots, r, \end{cases} \quad \sum_{j=1}^s b_j c_j^{m-1} a_{j,i} = \frac{b_i}{m} (1 - c_i^m). \quad (13.52)$$

Theorem 13.64. Suppose an s -stage IRK method has order of accuracy $p \geq s$ and all its RK nodes are distinct. Then this IRK method is an s -stage collocation method if and only if $C(s)$ is true.

Proof. The necessity holds because, for each $i = 1, 2, \dots, s$,

$$\begin{aligned} \sum_{j=1}^s a_{i,j} c_j^{m-1} &= \sum_{j=1}^s c_j^{m-1} \int_0^{c_i} \ell_j(\tau) d\tau \\ &= \int_0^{c_i} \left(\sum_{j=1}^s c_j^{m-1} \ell_j(\tau) \right) d\tau \\ &= \int_0^{c_i} \tau^{m-1} d\tau = \frac{c_i^m}{m}, \end{aligned}$$

where the first equality follows from (13.47a) and the third from interpolating a polynomial of degree $m-1$ with the Lagrange formula, thanks to the fact of c_i 's being distinct.

For the sufficiency, for each $i = 1, 2, \dots, s$, the condition

$$\forall m = 1, 2, \dots, s, \quad \sum_{j=1}^s c_j^{m-1} a_{i,j} = \frac{c_i^m}{m}$$

in $C(s)$ can be rewritten as a Vandermonde linear system

$$V \begin{pmatrix} a_{i,1} \\ a_{i,2} \\ \vdots \\ a_{i,s} \end{pmatrix} := \begin{pmatrix} 1 & 1 & \cdots & 1 \\ c_1 & c_2 & \cdots & c_s \\ \vdots & \vdots & \ddots & \vdots \\ c_1^{s-1} & c_2^{s-1} & \cdots & c_s^{s-1} \end{pmatrix} \begin{pmatrix} a_{i,1} \\ a_{i,2} \\ \vdots \\ a_{i,s} \end{pmatrix} = \begin{pmatrix} \frac{c_i}{2} \\ \frac{c_i^2}{2} \\ \vdots \\ \frac{c_i^s}{s} \end{pmatrix}.$$

By Lemma 6.4 and the distinctness of c_i 's, the above Vandermonde linear system has a unique solution. Each element $a_{i,j}$ of the solution of the Vandermonde system is of the form (13.47a) because the linear map V is bijective and

$$\begin{aligned} \sum_{j=1}^s c_j^{m-1} a_{i,j} &= \frac{c_i^m}{m} = \int_0^{c_i} \tau^{m-1} d\tau \\ &= \int_0^{c_i} \left(\sum_{j=1}^s c_j^{m-1} \ell_j(\tau) \right) d\tau = \sum_{j=1}^s c_j^{m-1} \int_0^{c_i} \ell_j(\tau) d\tau. \end{aligned}$$

Since the IRK method has order at least s , by Exercise 13.42 the pure quadrature order conditions in (13.39) hold:

$$\forall m = 1, 2, \dots, s, \quad \sum_{j=1}^s c_j^{m-1} b_j = \frac{1}{m};$$

this can also be considered as a Vandermonde linear system, and its solution is given by b_i 's in (13.47b). The proof is then completed by Theorem 13.58. $\square$

Exercise 13.65. Show that $B(s+r)$, $C(s)$, and the nodes being distinctive imply $D(r)$ via multiplying the two vectors $u_j := \sum_{i=1}^s b_i c_i^{m-1} a_{i,j}$ and $v_j := \frac{1}{m} b_j (1 - c_j^m)$ by the Vandermonde matrix $V(c_1, c_2, \dots, c_s)$ in Definition 6.3.

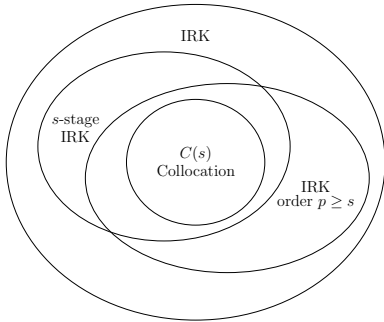


Figure 13.3: Relations of implicit RK methods and collocation methods.

Lemma 13.66. If an RK method satisfies $B(p)$, $C(\eta)$, and $D(\zeta)$ with $2\eta \geq p-1$ and $\zeta + \eta \geq p-1$, then the method has order of accuracy at least p .

Proof. See [Hairer et al., 1993, p. 208]. $\square$

Lemma 13.67. For each $r = 1, 2, \dots, s$, let $q_r \in \mathbb{P}_s$ be the polynomial of degree r that satisfies

$$\begin{cases} \forall p \in \mathbb{P}_{r-1}, & \langle q_r, p \rangle := \int_{t_n}^{t_n+k} q_r(x) p(x) dx = 0, \\ \langle q_r, t^r \rangle \neq 0. \end{cases} \quad (13.53)$$

If we design a weighted quadrature formula

$$I_s(f) := k \sum_{j=1}^s b_j f(\xi_j) \quad (13.54)$$

by selecting $\xi_j := t_n + c_j k$ as roots of the polynomial q_r and determine the weights b_j 's by the Vandermonde system

$$\forall m = 0, 1, \dots, s-1, \quad \sum_{j=1}^s \xi_j^m b_j = \frac{1}{k} \int_{t_n}^{t_n+k} \tau^m d\tau,$$

then the degree of exactness for $I_s(f)$ is $s+r-1$, i.e.,

$$\begin{cases} \forall f \in \mathbb{P}_{s+r-1}, & E_s(f) = \int_{t_n}^{t_n+k} f(t) dt - I_s(f) = 0; \\ \exists g \in \mathbb{P}_{s+r} \text{ s.t.} & E_s(g) \neq 0, \end{cases} \quad (13.55)$$

cf. Definition 9.6.

Proof. The weighted formula (13.54) has degree of exactness at least $s-1$ since for any pairwise distinct nodes $c_1, \dots, c_s$ we can interpolate f with a polynomial and derive a Newton-Cotes formula to approximate $\int_{t_n}^{t_n+k} f(t) dt$. Then the first statement of (13.55) follows from Theorem 9.24 with the weight function $\rho(x) = 1$. The proof is completed by setting $g(t) = t^r q_s(t) \in \mathbb{P}_{s+r}$ since

$$\begin{aligned} E_s(g) &= \int_{t_n}^{t_n+k} q_s(t) t^r dt - I_s(q_s t^r) \\ &= \langle q_s, t^r \rangle - k \sum_{j=1}^s b_j \xi_j^r q_s(\xi_j) \\ &= \langle q_s, t^r \rangle \neq 0, \end{aligned}$$

where the third step follows from $q_s(\xi_j) = 0$. $\square$

Theorem 13.68. An s -stage collocation method with its nodes c_j and weights b_j determined as those in (13.54) is at least $(s+r)$ th-order accurate for $r = 1, 2, \dots, s$.

Proof. By Lemma 13.66, it suffices to show that the collocation method is $B(s+r)$, $C(s)$, and $D(r)$.

It follows from Lemma 13.67 that the choices of c_j and b_j imply that the degree of exactness of the quadrature formula of the collocation method is $s+r-1$. Then by Exercise 13.42 the collocation method is $B(s+r)$.

By (13.53) and Definition 8.8, we can construct for the weight function $\rho \equiv 1$ a set of orthogonal polynomials, one for each degree $0, 1, \dots, r$ with q_r as the one with degree r .

By Theorem 8.12, the zeros of q_r are simple, hence the collocation parameters c_i 's are pairwise distinct. By Theorem 13.64 and Exercise 13.59, the method is $C(s)$.

The proof is then completed by Exercise 13.65. $\square$

Exercise 13.69. Determine the order of accuracy of the collocation method derived in Exercise 13.62.

Corollary 13.70. A collocation method with its RK nodes as zeros of q_s in (13.53) is $(2s)$ th-order accurate.

Proof. This follows from Theorem 13.68 with $r = s$. $\square$

Definition 13.71. A *Gauss-Legendre RK method* is an s -stage, $(2s)$ th-order collocation method based on Gauss-Legendre quadrature.

Lemma 13.72. The monic shifted Legendre polynomials in Definition 8.10 are

$$\tilde{P}_s(t) = \frac{(s!)^2}{(2s)!} \sum_{j=0}^s (-1)^{s-j} \binom{s}{j} \binom{s+j}{j} t^j. \quad (13.56)$$

Proof. Definition 8.10 and the binomial theorem yield

$$\begin{aligned} \frac{(2s)!}{(s!)^2} \tilde{P}_s(t) &= \frac{d^s}{dt^s} [(t^2 - t)^s] = \frac{d^s}{dt^s} \left[\sum_{j=0}^s \binom{s}{j} t^{2j} (-t)^{s-j} \right] \\ &= \sum_{j=0}^s (-1)^{s-j} \binom{s}{j} \frac{d^s}{dt^s} [t^{s+j}]. \end{aligned}$$

Then (13.56) follows from the identity $\binom{s+j}{j} = \frac{(s+j)!}{j!s!}$. $\square$

Example 13.73. By Lemma 13.72, we have

$$\tilde{P}_1(t) = t - \frac{1}{2}$$

and hence $c_1 = \frac{1}{2}$. Then Theorem 13.39 gives the one-stage second-order Gauss-Legendre RK method, which is also known as the *implicit midpoint method*:

$$\frac{\frac{1}{2}}{\frac{1}{2}} \Big| \frac{\frac{1}{2}}{1}. \quad (13.57)$$

Example 13.74. By Lemma 13.72, we have

$$\tilde{P}_2(t) = t^2 - t + \frac{1}{6}$$

and hence $c_1 = \frac{3-\sqrt{3}}{6}$ and $c_2 = \frac{3+\sqrt{3}}{6}$. Then (13.47) yields the 2-stage fourth-order Gauss-Legendre RK method as

$$\begin{array}{c|cc} \frac{3-\sqrt{3}}{6} & \frac{1}{4} & \frac{3-2\sqrt{3}}{12} \\ \frac{3+\sqrt{3}}{6} & \frac{3+2\sqrt{3}}{12} & \frac{1}{4} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}. \quad (13.58)$$

Example 13.75. By Lemma 13.72, we have

$$\tilde{P}_3(t) = t^3 - \frac{3}{2}t^2 + \frac{3}{5}t - \frac{1}{20}$$

and hence

$$c_1 = \frac{5-\sqrt{15}}{10}, \quad c_2 = \frac{1}{2}, \quad c_3 = \frac{5+\sqrt{15}}{10}.$$

Then (13.47) yields the 3-stage sixth-order Gauss-Legendre RK method as

$$\begin{array}{c|ccc} \frac{5-\sqrt{15}}{10} & \frac{5}{36} & \frac{2}{9} - \frac{\sqrt{15}}{15} & \frac{5}{36} - \frac{\sqrt{15}}{30} \\ \frac{1}{2} & \frac{5}{36} + \frac{\sqrt{15}}{24} & \frac{2}{9} & \frac{5}{36} - \frac{\sqrt{15}}{24} \\ \frac{5+\sqrt{15}}{10} & \frac{5}{36} + \frac{\sqrt{15}}{30} & \frac{2}{9} + \frac{\sqrt{15}}{15} & \frac{5}{36} \\ \hline & \frac{5}{18} & \frac{4}{9} & \frac{5}{18} \end{array}. \quad (13.59)$$

13.7 Analysis of RK methods

13.7.1 Consistency and convergence

Theorem 13.76. Consider an IVP (12.3) where the function $\mathbf{f} : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^N$ is continuous in t and Lipschitz continuous in $\mathbf{u}$ with Lipschitz constant L . The RK method (13.20) has a unique solution if its time step size k satisfies

$$k < \frac{1}{L\|A\|_\infty}, \quad (13.60)$$

where A is the RK matrix. Furthermore, $\mathbf{f}(\mathbf{u}, t) \in \mathcal{C}^p$ implies $\mathbf{y}_i(k) \in \mathcal{C}^p$.

Proof. Define $\mathbf{Y} := (\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_s)^\top$ and the first s equations in (13.20) state that $\mathbf{Y}$ is the fixed point $\mathbf{Y} = \mathbf{f}_k(\mathbf{Y})$ of the function $\mathbf{f}_k : \mathbb{R}^{sN} \rightarrow \mathbb{R}^{sN}$ given by

$$\mathbf{f}_k \left(\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_s \end{bmatrix} \right) = \begin{bmatrix} \mathbf{f} \left(\mathbf{U}^n + k \sum_{j=1}^s a_{1,j} \mathbf{y}_j, t_n + c_1 k \right) \\ \mathbf{f} \left(\mathbf{U}^n + k \sum_{j=1}^s a_{2,j} \mathbf{y}_j, t_n + c_2 k \right) \\ \vdots \\ \mathbf{f} \left(\mathbf{U}^n + k \sum_{j=1}^s a_{s,j} \mathbf{y}_j, t_n + c_s k \right) \end{bmatrix},$$

where the i th vector on the RHS is also denoted by $\mathbf{f}_k(\mathbf{Y})_i$. For the norm $\|\mathbf{Y}\| = \max_{1 \leq i \leq s} \|\mathbf{y}_i\|$, we have

$$\begin{aligned} \|\mathbf{f}_k(\mathbf{Y})_i - \mathbf{f}_k(\mathbf{Z})_i\| &\leq kL \left\| \sum_{j=1}^s a_{i,j} (\mathbf{y}_j - \mathbf{z}_j) \right\| \\ &\leq kL \sum_{j=1}^s |a_{i,j}| \|\mathbf{y}_j - \mathbf{z}_j\| \leq kL \sum_{j=1}^s |a_{i,j}| \|\mathbf{Y} - \mathbf{Z}\| \\ &\leq kL \|A\|_\infty \|\mathbf{Y} - \mathbf{Z}\|, \end{aligned}$$

where the first inequality follows from the Lipschitz condition of $\mathbf{f}$ and the second from the triangle inequality. Take the maximum over $i = 1, 2, \dots, s$ and we have

$$\|\mathbf{f}_k(\mathbf{Y}) - \mathbf{f}_k(\mathbf{Z})\| \leq kL \|A\|_\infty \|\mathbf{Y} - \mathbf{Z}\|.$$

By (13.60), $\mathbf{f}_k$ is a contractive mapping. Then the contractive mapping theorem D.120 ensures the existence and uniqueness of the solution.

For the last statement, we define $\mathbf{F} : \mathbb{R} \times \mathbb{R}^{sN} \rightarrow \mathbb{R}^{sN}$ as

$$\mathbf{F}(k, \mathbf{Y}) := \mathbf{Y} - \mathbf{f}_k(\mathbf{Y}),$$

and rewrite $\mathbf{Y} = \mathbf{f}_k(\mathbf{Y})$ as $\mathbf{F}(k, \mathbf{Y}) = \mathbf{0}$. For any $\mathbf{Y}_0 \in \mathbb{R}^{sN}$, we have $(0, \mathbf{Y}_0) \in \mathbb{R} \times \mathbb{R}^{sN}$ and $\mathbf{F}(0, \mathbf{Y}) = \mathbf{Y} - \mathbf{f}_0(\mathbf{U}^n, t_n)$ where $\mathbf{f}_0(\mathbf{U}^n, t_n)$ is independent of $\mathbf{Y}$. Thus the Jacobian matrix $\frac{\partial \mathbf{F}}{\partial \mathbf{Y}}(0, \mathbf{Y}_0) = I$ is invertible. Since $\mathbf{f}(\mathbf{u}, t) \in \mathcal{C}^p$ implies $\mathbf{F}(k, \mathbf{Y}) \in \mathcal{C}^p$, the rest follows from the implicit function theorem D.123. $\square$

Corollary 13.77. Consider an IVP (12.3) where the function $\mathbf{f} : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^N$ is continuous in t and Lipschitz continuous in $\mathbf{u}$ with Lipschitz constant L . The existence and uniqueness of the collocation polynomial $\mathbf{p}(t)$ in (13.46) are guaranteed if the time step size k satisfies (13.60) where A is the RK matrix of the collocation method.

Proof. This follows from Theorems 13.58 and 13.76. $\square$

Lemma 13.78. Consider an IVP (12.3) where the function $\mathbf{f} : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^N$ is continuous in t and Lipschitz continuous in $\mathbf{u}$ with Lipschitz constant L . The increment function Φ of an RK method with order of accuracy $p \geq 1$ satisfies the Lipschitz condition (13.13) with

$$M = \frac{L\|\mathbf{b}\|_1}{1 - kL\|A\|_\infty} \quad (13.61)$$

if its time step size k satisfies (13.60).

Proof. The increment function Φ that describes an RK method (13.20) is

$$\begin{aligned} \Phi &= \sum_{i=1}^s b_i \mathbf{y}_i, \\ \mathbf{y}_i(\mathbf{u}, t; k) &= \mathbf{f}\left(\mathbf{u} + k \sum_{j=1}^s a_{i,j} \mathbf{y}_j(\mathbf{u}, t; k), t + c_i k\right). \end{aligned}$$

The Lipschitz continuity of $\mathbf{f}$ yields

$$\begin{aligned} &\|\mathbf{y}_i(\mathbf{u}, t; k) - \mathbf{y}_i(\mathbf{v}, t; k)\| \\ &\leq L\|\mathbf{u} - \mathbf{v}\| + kL \left\| \sum_{j=1}^s a_{i,j} (\mathbf{y}_j(\mathbf{u}, t; k) - \mathbf{y}_j(\mathbf{v}, t; k)) \right\| \\ &\leq L\|\mathbf{u} - \mathbf{v}\| + kL \sum_{j=1}^s |a_{i,j}| \max_{1 \leq \ell \leq s} \|\mathbf{y}_\ell(\mathbf{u}, t; k) - \mathbf{y}_\ell(\mathbf{v}, t; k)\| \\ &\leq L\|\mathbf{u} - \mathbf{v}\| + kL\|A\|_\infty \max_{1 \leq \ell \leq s} \|\mathbf{y}_\ell(\mathbf{u}, t; k) - \mathbf{y}_\ell(\mathbf{v}, t; k)\| \end{aligned}$$

and taking maximum over $1 \leq i \leq s$ gives

$$\max_{1 \leq i \leq s} \|\mathbf{y}_i(\mathbf{u}, t; k) - \mathbf{y}_i(\mathbf{v}, t; k)\| \leq \frac{L}{1 - kL\|A\|_\infty} \|\mathbf{u} - \mathbf{v}\|.$$

The proof is then completed by

$$\begin{aligned} &\|\Phi(\mathbf{u}, t; k) - \Phi(\mathbf{v}, t; k)\| \\ &= \left\| \sum_{j=1}^s b_j (\mathbf{y}_j(\mathbf{u}, t; k) - \mathbf{y}_j(\mathbf{v}, t; k)) \right\| \\ &\leq \sum_{j=1}^s |b_j| \max_{1 \leq \ell \leq s} \|\mathbf{y}_\ell(\mathbf{u}, t; k) - \mathbf{y}_\ell(\mathbf{v}, t; k)\| \\ &\leq M\|\mathbf{u} - \mathbf{v}\|. \quad \square \end{aligned}$$

Theorem 13.79. An RK method with order of accuracy $p \geq 1$ and with initial error being $O(k^{p+1})$ is p th-order convergent for any IVP (12.3) with $\mathbf{f} \in \mathcal{C}^p$.

Proof. This follows directly from Lemma 13.78 and Theorem 13.16. $\square$

Lemma 13.80. The one-step error of the classical fourth-order RK method (13.33) is

$$\mathcal{L}\mathbf{u}(t_n) = O(k^5). \quad (13.62)$$

Exercise 13.81. Prove Lemma 13.80 for the scalar case.

Corollary 13.82. The classical fourth-order RK method (13.33) is convergent. Furthermore, it is fourth-order convergent for any IVP (12.3) with $\mathbf{f} \in \mathcal{C}^p$.

Proof. This follows directly from Lemma 13.80 and Theorem 13.79. $\square$

13.7.2 Absolute stability

Lemma 13.83. When applied to the scalar ODE

$$u'(t) = \lambda u(t),$$

the s -stage RK method (13.20) satisfies

$$U^{n+1} = [1 + k\lambda \mathbf{b}^\top (I - k\lambda A)^{-1} \mathbf{1}] U^n, \quad (13.63)$$

where the matrix $I - k\lambda A$ is assumed to be nonsingular and $\mathbf{1} \in \mathbb{Z}^s$ is a column vector with all components equal to one.

Proof. Apply the RK method (13.20) to the linear ODE $u' = \lambda u$ and we obtain

$$\forall i = 1, 2, \dots, s, \quad y_i = \lambda \left(U^n + k \sum_{j=1}^s a_{i,j} y_j \right).$$

Write $\mathbf{y} := [y_1, y_2, \dots, y_s]^\top$. Then $\mathbf{y} = \lambda U^n \mathbf{1} + k\lambda A \mathbf{y}$ and the exact solution of this linear algebraic system is

$$\mathbf{y} = \lambda U^n (I - k\lambda A)^{-1} \mathbf{1}.$$

Therefore we have

$$\begin{aligned} U^{n+1} &= U^n + k \sum_{j=1}^s b_j y_j = U^n + k\mathbf{b}^\top \mathbf{y} \\ &= [1 + k\lambda \mathbf{b}^\top (I - k\lambda A)^{-1} \mathbf{1}] U^n. \quad \square \end{aligned}$$

Notation 12. Denote the set of all rational polynomials by

$$\mathbb{P}_{m/n} := \left\{ \frac{P(z)}{Q(z)} : P(z) \in \mathbb{P}_m, Q(z) \in \mathbb{P}_n \right\}. \quad (13.64)$$

Lemma 13.84. When applied to the scalar IVP

$$u'(t) = \lambda u(t), \quad u(t_0) = u_0,$$

the s -stage RK method (13.20) yields

$$U^n = [R(k\lambda)]^n U^0, \quad (13.65)$$

where $R \in \mathbb{P}_{s/s}$. If the RK method is explicit, then $R \in \mathbb{P}_s$.

Proof. It follows from (13.63) that (13.65) holds with

$$R(z) = 1 + z\mathbf{b}^\top (I - zA)^{-1} \mathbf{1},$$

where $z := k\lambda \in \mathbb{C}$. It remains to verify $R \in \mathbb{P}_{s/s}$.

We represent the inverse of $I - zA$ by Corollary C.278,

$$(I - zA)^{-1} = \frac{\text{adj}(I - zA)}{\det(I - zA)},$$

where $\text{adj}(B)$ is the adjugate of the $s \times s$ matrix B as in Definition C.274. Since each entry of $I - zA$ is linear in z , each element of $\text{adj}(I - zA)$, as a determinant of an $(s-1) \times (s-1)$ matrix, is in $\mathbb{P}_{s-1}$. We thus deduce that $z\mathbf{b}^\top \text{adj}(I - zA)\mathbf{1}$ belongs to $\mathbb{P}_s$ and therefore $\det(I - zA) \in \mathbb{P}_s$ implies $R \in \mathbb{P}_{s/s}$.

Finally, if the method is explicit, then A is strictly lower triangular and $I - zA$ is, regardless of $z \in \mathbb{C}$, a lower triangular matrix with ones along the diagonal. Therefore $\det(I - zA) \equiv 1$ and R is a polynomial. $\square$

Corollary 13.85. The stability function of an s -stage RK method (13.20) is a rational polynomial $R(z) \in \mathbb{P}_{s/s}$:

$$R(z) = 1 + z\mathbf{b}^\top (I - zA)^{-1} \mathbf{1}, \quad z \in \mathbb{C}. \quad (13.66)$$

Proof. This follows immediately from Definition 13.18 and Lemmas 13.83 and 13.84. $\square$

Example 13.86. The stability function of the modified Euler method (13.4) is

$$\begin{aligned} R(z) &= 1 + z\mathbf{b}^\top(I - zA)^{-1}\mathbf{1} \\ &= 1 + z \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{z}{2} & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= 1 + z + \frac{z^2}{2}. \end{aligned}$$

Exercise 13.87. Show that the classical fourth-order RK method has its stability function as

$$R(z) = 1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \frac{1}{24}z^4. \quad (13.67)$$

Theorem 13.88. If the RK method (13.20) is at least p th-order accurate, then, as $z \rightarrow 0$,

$$R(z) = e^z + O(z^{p+1}). \quad (13.68)$$

Proof. For the scalar IVP $u' = \lambda u$, the one-step error of a p th- or higher-order RK method satisfies

$$O(k^{p+1}) = \mathcal{L}u(t_n) = u(t_{n+1}) - U^{n+1} = (e^{\lambda k} - R(\lambda k))e^{\lambda t_n},$$

where the first equality follows from Definition 13.9, the second from Definitions 13.8 and 13.26, and the third from Definition 13.18. The proof is completed by dividing both sides of the above equation by $e^{\lambda t_n}$. $\square$

Corollary 13.89. All s -stage, s th-order ERK methods have the same stability function

$$R(z) = \sum_{j=0}^s \frac{1}{j!} z^j, \quad z \in \mathbb{C}. \quad (13.69)$$

Proof. By Lemma 13.84, $R(z)$ is a polynomial of degree at most s . Theorem 13.88 states that $R(z) = e^z + O(z^{s+1})$ as $z \rightarrow 0$, which implies that $R(z)$ is a polynomial of degree at least s . These two conditions uniquely determine $R(z)$. $\square$

Corollary 13.90. All s -stage, s th-order ERK methods share the same RAS.

Proof. This follows immediately from Definition 13.21 and Corollary 13.89. $\square$

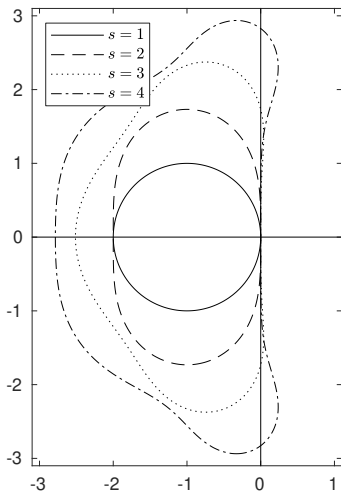


Figure 13.4: Bounded RASs of explicit RK methods with $s = 1, 2, 3, 4$.

Exercise 13.91. Define $S_s := \{z : |R_s(z)| \leq 1\}$ where $s = 1, 2, 3, 4$ and R_s is the stability function of the s -stage, s th-order ERK method. Show that

$$S_1 \subset S_2 \subset S_3. \quad (13.70)$$

Does this hold for ERK methods with a higher stage? Why?

13.7.3 I-stability and L-stability

Theorem 13.92. No ERK method is A-stable.

Proof. By Lemma 13.84, the stability function $R(z)$ of an ERK method is a polynomial, and hence

$$\lim_{z \rightarrow -\infty} |R(z)| = \infty,$$

which excludes A-stability. $\square$

Definition 13.93. An RK method is *I-stable* iff its stability function $R(z)$ satisfies

$$\forall y \in \mathbb{R}, |R(yi)| \leq 1. \quad (13.71)$$

Theorem 13.94 (Maximum modulus principle). Consider an analytic function $f : D \rightarrow \mathbb{C}$ over a path-connected nonempty open subset $D \subseteq \mathbb{C}$. If the modulus function $|f| : D \rightarrow \mathbb{R}^+ \cup \{0\}$ reaches its maximum on D , then f is a constant.

Proof. See [Freitag and Busam, 2009, page 129]. $\square$

Theorem 13.95. An RK method is A-stable if and only if it is I-stable and all poles of its stability function $R(z)$ have positive real parts.

Proof. The necessity follows directly from Definitions 13.22 and 13.93. As for the sufficiency, Corollary 13.85 yields

$$R(z) = \frac{P(z)}{Q(z)}, \quad P(z) = \sum_{i=0}^m p_i z^i, \quad Q(z) = \sum_{i=0}^n q_i z^i,$$

where $p_m \neq 0, q_n \neq 0$. Then Definition 13.93 implies $m < n$ or $m = n$, the latter of which yields $|p_m| \leq |q_n|$. Therefore the limit $\lim_{z \rightarrow \infty, z \in \mathbb{C}} |R(z)|$ exists and

$$\lim_{z \rightarrow \infty, z \in \mathbb{C}} |R(z)| \leq 1. \quad (13.72)$$

Since the poles of $R(z)$ have positive real parts, the rational function $R(z)$ is analytic in

$$\Omega_\rho := \{z \in \mathbb{C} : \operatorname{Re} z < 0 \text{ and } |z| < \rho\}$$

for any $\rho > 0$. Suppose the RK method were not A-stable. Then Definition 13.22 implies

$$\exists! Z \subset \mathbb{C}^-, Z \neq \emptyset \text{ s.t. } \forall z_0 \in Z, \operatorname{Re} z_0 < 0 \text{ and } |R(z_0)| > 1.$$

By (13.72) and the continuity of the function $z \mapsto |R(z)|$, there exists ρ_0 sufficiently large such that $Z \subset \Omega_{\rho_0}$ and

$$\forall z \in \partial\Omega_{\rho_0}, \quad |R(z)| \leq |R(z_m)|,$$

where $z_m \in Z$ is a point that maximizes $|R(z)|$; the existence of z_m is guaranteed by the continuity of $z \mapsto |R(z)|$ and the closure of Ω_{ρ_0} being compact. Therefore, the non-constant function $z \mapsto |R(z)|$ attains its maximum in Ω_{ρ_0} , which contradicts Theorem 13.94. $\square$

Example 13.96. The two-stage collocation method with $c_1 = \frac{1}{3}, c_2 = 1$ has its Butcher tableau as

$$\begin{array}{c|cc} \frac{1}{3} & \frac{5}{12} & -\frac{1}{12} \\ 1 & \frac{3}{4} & \frac{1}{4} \\ \hline & \frac{3}{4} & \frac{1}{4} \end{array}. \quad (13.73)$$

This method is A-stable since it satisfies the two conditions in Theorem 13.95. Its stability function is

$$R(z) = 1 + z\mathbf{b}^\top(I - zA)^{-1}\mathbf{1} = \frac{1 + \frac{1}{3}z}{1 - \frac{2}{3}z + \frac{1}{6}z^2}.$$

(i) I-stability (13.71) is equivalent to

$$\left|1 + \mathbf{i}\frac{1}{3}y\right|^2 \leq \left|1 - \frac{1}{6}y^2 - \mathbf{i}\frac{2}{3}y\right|^2, \quad \forall y \in \mathbb{R},$$

and hence to

$$1 + \frac{1}{9}y^2 \leq 1 + \frac{1}{9}y^2 + \frac{1}{36}y^4, \quad \forall y \in \mathbb{R}.$$

(ii) The poles of $R(z)$ are $2 \pm \mathbf{i}\sqrt{2}$, both residing at the open right half-plane.

This method is also L-stable in the sense of Definition 13.24, either by Exercise 13.97 or by Theorem 13.99.

Exercise 13.97. Prove that an A-stable RK method with its stability function as a rational polynomial $R(z) = \frac{P(z)}{Q(z)}$ is L-stable if and only if $\deg Q(z) > \deg P(z)$.

Definition 13.98. An RK method (13.20) is *stiffly accurate* if the last row of the RK matrix A is the same as the RK weight $\mathbf{b}^\top$, i.e.,

$$\forall j = 1, \dots, s, \quad a_{s,j} = b_j. \quad (13.74)$$

Theorem 13.99. If an A-stable RK method with a nonsingular RK matrix A is stiffly accurate, then it is L-stable.

Proof. By Definition 13.18 of the stability function $R(z)$ and the continuity of the matrix inverse function, we have

$$\begin{aligned} \lim_{z \rightarrow \infty} R(z) &= \lim_{z \rightarrow \infty} (1 + z\mathbf{b}^\top(I - zA)^{-1}\mathbf{1}) \\ &= 1 + \lim_{z \rightarrow \infty} \mathbf{b}^\top \left(\frac{1}{z}I - A\right)^{-1}\mathbf{1} \\ &= 1 - \mathbf{b}^\top A^{-1}\mathbf{1} \\ &= 1 - \mathbf{e}_s^\top \mathbf{1} = 0, \end{aligned}$$

where $\mathbf{e}_s = (0, \dots, 0, 1)^\top$ and the fourth equality holds because $A^\top \mathbf{e}_s = \mathbf{b}$ by (13.74). $\square$

Exercise 13.100. Show that if an A-stable RK method with a nonsingular RK matrix A satisfies

$$\forall i = 1, \dots, s, \quad a_{i,1} = b_1, \quad (13.75)$$

then it is L-stable.

Example 13.101. The trapezoidal method (12.16) with Butcher tableau

$$\begin{array}{c|cc} & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$$

is stiffly accurate. But Example 13.23 shows that it is not L-stable, which is caused by the fact that its RK matrix is singular. Similarly, the ESDIRK method in Example 13.52 is stiffly accurate, but is not L-stable.

Example 13.102. The 5th-order RK method embedded in the Dormand-Prince pair (13.93) is stiffly accurate. However, Theorem 13.92 dictates that it can never be A-stable, and thus cannot be L-stable either.

Example 13.103. The collocation method (13.73) is L-stable.

Example 13.104. The 3-stage 6th-order Gauss-Legendre RK method (13.59) is A-stable, but not L-stable since

$$\begin{aligned} \lim_{z \rightarrow \infty} R(z) &= \lim_{z \rightarrow \infty} (1 + z\mathbf{b}^\top(I - zA)^{-1}\mathbf{1}) \\ &= 1 + \lim_{z \rightarrow \infty} \mathbf{b}^\top \left(\frac{1}{z}I - A\right)^{-1}\mathbf{1} \\ &= 1 - \mathbf{b}^\top A^{-1}\mathbf{1} = -1, \end{aligned}$$

where the first step follows from Corollary 13.85, the third from the continuity of the matrix inverse function, and the last from (13.59).

Exercise 13.105. Show that the collocation method

$$\begin{array}{c|ccc} \frac{4-\sqrt{6}}{10} & \frac{88-7\sqrt{6}}{360} & \frac{296-169\sqrt{6}}{1800} & \frac{-2+3\sqrt{6}}{225} \\ \frac{4+\sqrt{6}}{10} & \frac{296+169\sqrt{6}}{1800} & \frac{88+7\sqrt{6}}{360} & \frac{-2-3\sqrt{6}}{225} \\ 1 & \frac{16-\sqrt{6}}{36} & \frac{16+\sqrt{6}}{36} & \frac{1}{9} \\ \hline & \frac{16-\sqrt{6}}{36} & \frac{16+\sqrt{6}}{36} & \frac{1}{9} \end{array} \quad (13.76)$$

is 5th-order accurate and L-stable.

13.7.4 Contractivity and B-stability

Definition 13.106. A function $\mathbf{f} : \mathbb{R}^N \times [0, +\infty) \rightarrow \mathbb{R}^N$ is said to satisfy a *one-sided Lipschitz condition* if

$$\forall t \geq 0, \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^N, \quad \langle \mathbf{u} - \mathbf{v}, \mathbf{f}(\mathbf{u}, t) - \mathbf{f}(\mathbf{v}, t) \rangle \leq \mu \|\mathbf{u} - \mathbf{v}\|^2, \quad (13.77)$$

where $\langle \cdot, \cdot \rangle$ is an inner product in $\mathbb{R}^N$, $\|\cdot\|$ the corresponding norm, and μ the *one-sided Lipschitz constant* of $\mathbf{f}$.

Lemma 13.107. If $\mathbf{f} : \mathbb{R}^N \times [0, +\infty) \rightarrow \mathbb{R}^N$ is Lipschitz continuous with Lipschitz constant L , then it satisfies the one-sided Lipschitz condition (13.77) with $\mu = L$.

Proof. The Cauchy-Schwarz inequality and the Lipschitz continuity of $\mathbf{f}$ yield

$$\begin{aligned} \langle \mathbf{u} - \mathbf{v}, \mathbf{f}(\mathbf{u}, t) - \mathbf{f}(\mathbf{v}, t) \rangle &\leq \|\mathbf{u} - \mathbf{v}\| \cdot \|\mathbf{f}(\mathbf{u}, t) - \mathbf{f}(\mathbf{v}, t)\| \leq L \|\mathbf{u} - \mathbf{v}\|^2. \end{aligned} \quad \square$$

Lemma 13.108. If $\mathbf{f} : \mathbb{R}^N \times [0, +\infty) \rightarrow \mathbb{R}^N$ is continuous and satisfies a one-sided Lipschitz condition (13.77), then any two solutions $\mathbf{u}(t)$ and $\mathbf{v}(t)$ of the ODE (12.1) satisfy

$$\|\mathbf{u}(t) - \mathbf{v}(t)\| \leq e^{\mu t} \|\mathbf{u}(0) - \mathbf{v}(0)\|, \quad (13.78)$$

where the norm is induced from an inner product.

Proof. Let $\phi(t) = \|\mathbf{u}(t) - \mathbf{v}(t)\|^2$, then

$$\begin{aligned}\phi'(t) &= \frac{d}{dt} \langle \mathbf{u}(t) - \mathbf{v}(t), \mathbf{u}(t) - \mathbf{v}(t) \rangle \\ &= 2 \langle \mathbf{u}(t) - \mathbf{v}(t), \mathbf{u}'(t) - \mathbf{v}'(t) \rangle \\ &= 2 \langle \mathbf{u}(t) - \mathbf{v}(t), \mathbf{f}(\mathbf{u}(t), t) - \mathbf{f}(\mathbf{v}(t), t) \rangle \leq 2\mu\phi(t),\end{aligned}$$

where the second step follows from the product rule and the symmetry of an inner product, the third from the ODE, and the last from (13.77). Therefore, we have

$$\phi(t) \leq e^{2\mu t} \phi(0),$$

the square root of which completes the proof. $\square$

Definition 13.109. An ODE system (12.1) is *contractive* or *monotone* if the RHS function $\mathbf{f}$ satisfies the one-sided Lipschitz condition (13.77) with $\mu = 0$, i.e.,

$$\forall t \geq 0, \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^N, \quad \langle \mathbf{u} - \mathbf{v}, \mathbf{f}(\mathbf{u}, t) - \mathbf{f}(\mathbf{v}, t) \rangle \leq 0. \quad (13.79)$$

Example 13.110. The linear constant-coefficient system

$$\mathbf{u}'(t) = A\mathbf{u}(t) + \mathbf{b}(t), \quad A \in \mathbb{R}^{N \times N}$$

is contractive if and only if the matrix A is negative semi-definite; see Definition 3.54.

Lemma 13.111. A contractive ODE system is *dissipative*: for any two solutions $\mathbf{u}(t)$ and $\mathbf{v}(t)$ of the ODE (12.1), the norm $\|\mathbf{u}(t) - \mathbf{v}(t)\|$ decreases monotonically for $t \geq 0$.

Proof. Following the same procedure as that in the proof of Lemma 13.108, we can show $\phi'(t) \leq 0$, and therefore $\|\mathbf{u}(t) - \mathbf{v}(t)\| = \sqrt{\phi(t)}$ is monotonically decreasing. $\square$

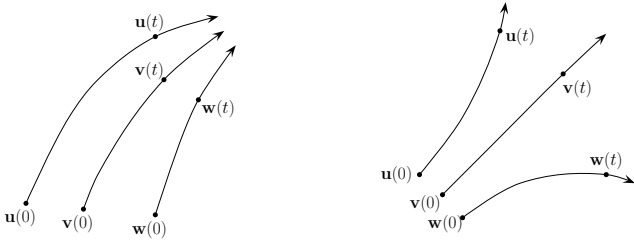


Figure 13.5: Illustrations of contractive and non-contractive ODEs.

Definition 13.112 (B-stability: Butcher [1975]). A one-step method is *B-stable* (or *contractive*) if, for any contractive ODE system, every pair of its numerical solutions $\mathbf{U}^n$ and $\mathbf{V}^n$ satisfy

$$\forall n = 0, 1, \dots, \quad \|\mathbf{U}^{n+1} - \mathbf{V}^{n+1}\| \leq \|\mathbf{U}^n - \mathbf{V}^n\|. \quad (13.80)$$

Example 13.113. For the backward Euler method (12.15), consider $\mathbf{e}^n := \mathbf{U}^n - \mathbf{V}^n$ for an arbitrary pair of numerical solutions $\mathbf{U}^n$ and $\mathbf{V}^n$. The Cauchy-Schwarz inequality and the contractivity in Definition 13.109 yield

$$\begin{aligned}\|\mathbf{e}^{n+1}\|^2 &= \langle \mathbf{e}^{n+1}, \mathbf{e}^{n+1} \rangle \\ &= \langle \mathbf{e}^n + k[\mathbf{f}(\mathbf{U}^{n+1}, t_{n+1}) - \mathbf{f}(\mathbf{V}^{n+1}, t_{n+1})], \mathbf{e}^{n+1} \rangle \\ &\leq \|\mathbf{e}^n\| \cdot \|\mathbf{e}^{n+1}\|,\end{aligned}$$

which implies $\|\mathbf{U}^{n+1} - \mathbf{V}^{n+1}\| \leq \|\mathbf{U}^n - \mathbf{V}^n\|$ and thus the backward Euler method is B-stable.

Exercise 13.114. Rewrite the implicit midpoint method

$$\mathbf{U}^{n+1} = \mathbf{U}^n + k\mathbf{f}\left(\frac{\mathbf{U}^n + \mathbf{U}^{n+1}}{2}, t_n + \frac{k}{2}\right)$$

in the standard form (13.20) and derive its Butcher tableau. Show that it is B-stable.

Example 13.115. The trapezoidal method (12.16) is not B-stable. To show this, consider a scalar autonomous ODE with its RHS function given by

$$f(u) = \begin{cases} -u & \text{if } u \geq 0, \\ -\frac{u}{\epsilon} & \text{if } u < 0, \end{cases}$$

where $\epsilon \in (0, \frac{1}{5})$ is a constant. It is easy to verify that f satisfies the contractivity condition (13.79). But if we apply the trapezoidal method to the ODE with initial conditions $U^0 = 0$ and $V^0 = -\epsilon$ and time step size $k = 1$, then

$$U^1 = 0, \quad V^1 = \frac{1 - 2\epsilon}{3},$$

and so $|U^1 - V^1| > |U^0 - V^0|$ contradicts (13.80).

Theorem 13.116. For any $s \in \mathbb{N}^+$, an s -stage Gauss-Legendre RK method is B-stable.

Proof. Let $\{\mathbf{U}^n\}$ and $\{\mathbf{V}^n\}$ be a pair of numerical solutions obtained by the Gauss-Legendre RK methods and denote by $\mathbf{p}_u(t)$ and $\mathbf{p}_v(t)$ the collocation polynomials for $\mathbf{U}^n$ and $\mathbf{V}^n$ in (13.46). Let $\phi(t) = \|\mathbf{p}_u(t) - \mathbf{p}_v(t)\|^2$, then at the collocation points $\xi_i = t_n + c_i k$ we have

$$\begin{aligned}\phi'(\xi_i) &= 2 \langle \mathbf{p}_u(\xi_i) - \mathbf{p}_v(\xi_i), \mathbf{p}'_u(\xi_i) - \mathbf{p}'_v(\xi_i) \rangle \\ &= 2 \langle \mathbf{p}_u(\xi_i) - \mathbf{p}_v(\xi_i), \mathbf{f}(\mathbf{p}_u(\xi_i), \xi_i) - \mathbf{f}(\mathbf{p}_v(\xi_i), \xi_i) \rangle \leq 0,\end{aligned}$$

where the second step follows from the definition (13.46) of the collocation method and the third from the contractivity condition (13.79). Therefore

$$\begin{aligned}\|\mathbf{U}^{n+1} - \mathbf{V}^{n+1}\|^2 &= \phi(t_n + k) = \phi(t_n) + \int_{t_n}^{t_n+k} \phi'(t) dt \\ &= \phi(t_n) + k \sum_{j=1}^s b_j \phi'(t_n + c_j k) \leq \phi(t_n) = \|\mathbf{U}^n - \mathbf{V}^n\|^2,\end{aligned}$$

where the second equality follows from the fundamental theorem of calculus, the third from Definition 13.71 that Gaussian quadrature integrates the polynomial $\phi'(t)$ (which is of degree $2s - 1$) exactly, and the fourth from the quadrature weights b_j being positive, cf. Lemma 9.30. $\square$

Theorem 13.117. B-stable one-step methods are A-stable.

Proof. It suffices to show that, for any contractive complex-valued scalar ODE

$$y' = \lambda y, \quad \lambda \in \mathbb{C} \text{ and } \operatorname{Re} \lambda \leq 0, \quad (13.81)$$

any B-stable method with any time step size $k > 0$ has a stability function $R : \mathbb{C} \rightarrow \mathbb{C}$ that satisfies $|R(k\lambda)| \leq 1$. After all, the conditions on λ and k ensure that the left half complex plane is covered by the set $\{k\lambda\}$. Let

$$y(t) = y_1(t) + \mathbf{i}y_2(t), \quad \lambda = \alpha + \mathbf{i}\beta.$$

Then (13.81) is equivalent to the real-valued ODE system

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad (13.82)$$

with $\alpha = \operatorname{Re} \lambda \leq 0$. By Definition 3.54, the constant-coefficient matrix in (13.82) is negative semi-definite, and hence (13.82) is contractive. Let $U^n = U_1^n + iU_2^n$ be a numerical solution of (13.81), then $\mathbf{U}^n = (U_1^n, U_2^n)^\top$ is a numerical solution of (13.82). Taking $\mathbf{V}^n \equiv \mathbf{0}$ in Definition 13.112 yields

$$\forall k > 0, \forall n = 0, 1, \dots, \quad \|\mathbf{U}^{n+1}\| \leq \|\mathbf{U}^n\|.$$

The fact $|U^n|^2 = |U_1^n|^2 + |U_2^n|^2 = \|\mathbf{U}^n\|^2$ implies

$$\forall k > 0, \forall n = 0, 1, \dots, \quad |U^{n+1}| \leq |U^n|.$$

Then the proof is completed by Definitions 13.18, 13.21, and 13.22. $\square$

13.7.5 Algebraic stability

Definition 13.118. An RK method is *algebraically stable* iff

(ABS-1) the RK weights $b_1, b_2, \dots, b_s$ are nonnegative,

(ABS-2) the symmetric matrix $M \in \mathbb{R}^{s \times s}$ with

$$m_{i,j} := b_i a_{i,j} + b_j a_{j,i} - b_i b_j \quad (13.83)$$

is positive semidefinite.

The matrix M is called the *algebraic stability matrix*.

Theorem 13.119. An algebraically stable RK method is B-stable and A-stable.

Proof. Apply the s -stage RK method (13.20) to the ODE (12.1) for two initial conditions $\mathbf{U}(0)$, $\mathbf{V}(0)$, and we have

$$\begin{aligned} \xi_i &= \mathbf{f} \left(\mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \xi_j, t_n + c_i k \right), \\ \mathbf{U}^{n+1} &= \mathbf{U}^n + k \sum_{j=1}^s b_j \xi_j; \\ \eta_i &= \mathbf{f} \left(\mathbf{V}^n + k \sum_{j=1}^s a_{i,j} \eta_j, t_n + c_i k \right), \\ \mathbf{V}^{n+1} &= \mathbf{V}^n + k \sum_{j=1}^s b_j \eta_j, \end{aligned}$$

where $i = 1, 2, \dots, s$. Define $\mathbf{e}^n := \mathbf{U}^n - \mathbf{V}^n$, $\mathbf{d}_i := \xi_i - \eta_i$ and we have

$$\begin{aligned} \|\mathbf{e}^{n+1}\|^2 &= \langle \mathbf{U}^{n+1} - \mathbf{V}^{n+1}, \mathbf{U}^{n+1} - \mathbf{V}^{n+1} \rangle \\ &= \langle \mathbf{e}^n + k \sum_{i=1}^s b_i \mathbf{d}_i, \mathbf{e}^n + k \sum_{j=1}^s b_j \mathbf{d}_j \rangle \\ &= \|\mathbf{e}^n\|^2 + 2k \langle \mathbf{e}^n, \sum_{i=1}^s b_i \mathbf{d}_i \rangle + k^2 \sum_{i=1}^s \sum_{j=1}^s b_i b_j \langle \mathbf{d}_i, \mathbf{d}_j \rangle. \end{aligned}$$

For B-stability it remains to show

$$2 \left\langle \mathbf{e}^n, \sum_{i=1}^s b_i \mathbf{d}_i \right\rangle + k \sum_{i=1}^s \sum_{j=1}^s b_i b_j \langle \mathbf{d}_i, \mathbf{d}_j \rangle \leq 0.$$

Define $\rho_i := \mathbf{U}^n + k \sum_{j=1}^s a_{i,j} \xi_j$, $\sigma_i := \mathbf{V}^n + k \sum_{j=1}^s a_{i,j} \eta_j$ and we have

$$\begin{aligned} \langle \mathbf{e}^n, \sum_{i=1}^s b_i \mathbf{d}_i \rangle &= \sum_{i=1}^s b_i \langle \mathbf{e}^n, \mathbf{d}_i \rangle \\ &= \sum_{i=1}^s b_i \left\langle \rho_i - \sigma_i - k \sum_{j=1}^s a_{i,j} \mathbf{d}_j, \mathbf{d}_i \right\rangle \\ &= \sum_{i=1}^s b_i \langle \rho_i - \sigma_i, \mathbf{d}_i \rangle - k \sum_{i=1}^s \sum_{j=1}^s b_i a_{i,j} \langle \mathbf{d}_i, \mathbf{d}_j \rangle. \end{aligned}$$

The ODE system being contractive and (ABS-1) yield

$$\begin{aligned} &\sum_{i=1}^s b_i \langle \rho_i - \sigma_i, \mathbf{d}_i \rangle \\ &= \sum_{i=1}^s b_i \langle \rho_i - \sigma_i, \mathbf{f}(\rho_i, t_n + c_i k) - \mathbf{f}(\sigma_i, t_n + c_i k) \rangle \leq 0, \end{aligned}$$

which further implies

$$\begin{aligned} \langle \mathbf{e}^n, \sum_{i=1}^s b_i \mathbf{d}_i \rangle &\leq -k \sum_{i=1}^s \sum_{j=1}^s b_i a_{i,j} \langle \mathbf{d}_i, \mathbf{d}_j \rangle, \\ \langle \mathbf{e}^n, \sum_{i=1}^s b_i \mathbf{d}_i \rangle &\leq -k \sum_{j=1}^s \sum_{i=1}^s b_j a_{j,i} \langle \mathbf{d}_j, \mathbf{d}_i \rangle. \end{aligned}$$

It follows that

$$\begin{aligned} &2 \langle \mathbf{e}^n, \sum_{i=1}^s b_i \mathbf{d}_i \rangle + k \sum_{i=1}^s \sum_{j=1}^s b_i b_j \langle \mathbf{d}_i, \mathbf{d}_j \rangle \\ &\leq -k \sum_{i=1}^s \sum_{j=1}^s (b_i a_{i,j} + b_j a_{j,i} - b_i b_j) \langle \mathbf{d}_i, \mathbf{d}_j \rangle \\ &= -k \sum_{i=1}^s \sum_{j=1}^s m_{i,j} \langle \mathbf{d}_i, \mathbf{d}_j \rangle. \end{aligned}$$

By (ABS-2), we can write $M = Q\Lambda Q^\top$, where Q is orthogonal, and Λ is a diagonal matrix with $\lambda_k = \Lambda_{k,k} \geq 0$. Then the RK method is B-stable because

$$\begin{aligned} &\sum_{i=1}^s \sum_{j=1}^s m_{i,j} \langle \mathbf{d}_i, \mathbf{d}_j \rangle \\ &= \sum_{i=1}^s \sum_{j=1}^s \left(\sum_{k=1}^s q_{i,k} \lambda_k q_{j,k} \right) \langle \mathbf{d}_i, \mathbf{d}_j \rangle \\ &= \sum_{k=1}^s \lambda_k \left\langle \sum_{i=1}^s q_{i,k} \mathbf{d}_i, \sum_{j=1}^s q_{j,k} \mathbf{d}_j \right\rangle \\ &= \sum_{k=1}^s \lambda_k \left\| \sum_{i=1}^s q_{i,k} \mathbf{d}_i \right\|^2 \geq 0. \end{aligned}$$

The A-stability follows from Theorem 13.117. $\square$

Example 13.120. A-stable RK methods may not be algebraically stable. For example, the three-stage method

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{5}{24} & \frac{1}{3} & -\frac{1}{24} \\ 1 & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \hline & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{array} \quad (13.84)$$

is fourth-order accurate and A-stable. However, its algebraic stability matrix

$$M = \frac{1}{36} \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

is not positive semidefinite.

Lemma 13.121. The algebraic stability matrix of an s -stage RK method is the zero matrix if

- (i) the RK nodes $c_1, c_2, \dots, c_s$ are pairwise distinct,
- (ii) it is both $B(2s)$ and $C(s)$ as in Definitions 13.40 and 13.63.

Proof. The Vandermonde matrix $V(c_1, c_2, \dots, c_s)$ in Definition 6.3 is nonsingular since c_i 's are pairwise distinct. Therefore $M = \mathbf{0}$ if and only if $\tilde{M} := V^T M V = \mathbf{0}$. This actually holds because $\forall i, j = 1, 2, \dots, s$, we have

$$\begin{aligned}\tilde{m}_{i,j} &= \sum_{k=1}^s \sum_{\ell=1}^s c_k^{i-1} m_{k,\ell} c_\ell^{j-1} \\ &= \sum_{k=1}^s \sum_{\ell=1}^s c_k^{i-1} (b_k a_{k,\ell} + b_\ell a_{\ell,k} - b_k b_\ell) c_\ell^{j-1} \\ &= \sum_{k=1}^s b_k c_k^{i-1} \sum_{\ell=1}^s a_{k,\ell} c_\ell^{j-1} + \sum_{\ell=1}^s b_\ell c_\ell^{j-1} \sum_{k=1}^s a_{\ell,k} c_k^{i-1} \\ &\quad - \sum_{k=1}^s b_k c_k^{i-1} \sum_{\ell=1}^s b_\ell c_\ell^{j-1} \\ &= \frac{1}{j} \sum_{k=1}^s b_k c_k^{i+j-1} + \frac{1}{i} \sum_{\ell=1}^s b_\ell c_\ell^{i+j-1} - \frac{1}{ij} \\ &= \left(\frac{1}{j} + \frac{1}{i}\right) \frac{1}{i+j} - \frac{1}{ij} = 0,\end{aligned}$$

where the second step follows from (ABS-2) in Definition (13.83) and the fourth and fifth steps from the conditions $B(2s)$ and $C(s)$. $\square$

Theorem 13.122. The Gauss-Legendre RK methods are algebraically stable for $s \geq 1$, and hence are also B-stable and A-stable.

Proof. We already know that the Gauss quadrature formula with s quadrature nodes has degree of exactness $2s - 1$ and that the quadrature weights b_i 's are positive. By Definition 13.118, Exercise 13.42, and Lemma 13.121, it suffices to show that the s -stage Gauss-Legendre RK method is $C(s)$:

$$\begin{aligned}\forall \ell = 1, \dots, s, \quad \sum_{j=1}^s a_{i,j} c_j^{\ell-1} &= \sum_{j=1}^s \left(\int_0^{c_i} \ell_j(\tau) d\tau \right) c_j^{\ell-1} \\ &= \int_0^{c_i} \left(\sum_{j=1}^s c_j^{\ell-1} \ell_j(\tau) \right) d\tau = \int_0^{c_i} \tau^{\ell-1} d\tau,\end{aligned}$$

where the first step follows from (13.47) and the third from the fact that the Lagrange interpolation polynomial of $\tau^{\ell-1}$ is itself. The proof is completed by Theorem 13.119. $\square$

13.8 Practical error estimation and step size control

Definition 13.123. The *Richardson extrapolation* is an algorithm to construct a more accurate solution from the increment function Φ of a p th-order one-step method (13.1).

(RET-1) Advance $\mathbf{u}_0$ by one step of size k :

$$\mathbf{U}^1 = \mathbf{u}_0 + k\Phi(\mathbf{u}_0, t_0; k).$$

(RET-2) Advance $\mathbf{u}_0$ by two consecutive steps of size $\frac{k}{2}$:

$$\begin{cases} \mathbf{U}^{\frac{1}{2}} &= \mathbf{u}_0 + \frac{k}{2}\Phi\left(\mathbf{u}_0, t_0; \frac{k}{2}\right), \\ \tilde{\mathbf{U}}^1 &= \mathbf{U}^{\frac{1}{2}} + \frac{k}{2}\Phi\left(\mathbf{U}^{\frac{1}{2}}, t_0 + \frac{k}{2}; \frac{k}{2}\right). \end{cases}$$

(RET-3) Set the more accurate solution to be

$$\hat{\mathbf{U}}^1 = \tilde{\mathbf{U}}^1 + \frac{1}{2^p - 1} (\tilde{\mathbf{U}}^1 - \mathbf{U}^1). \quad (13.85)$$

Lemma 13.124. The error of the solution $\hat{\mathbf{U}}^1$ given by the Richardson extrapolation for a consistent p th-order one-step method is

$$\hat{\mathbf{E}}^1 = \hat{\mathbf{U}}^1 - \mathbf{u}(t_0 + k) = O(k^{p+2}). \quad (13.86)$$

In addition, the error of the p th-order one-step method is

$$\mathbf{E}^1 = \mathbf{U}^1 - \mathbf{u}(t_0 + k) = \frac{2^p}{1 - 2^p} (\tilde{\mathbf{U}}^1 - \mathbf{U}^1) + O(k^{p+2}). \quad (13.87)$$

Proof. The p th-order accuracy of the one-step method gives

$$\mathbf{E}^1 = \tau(\mathbf{u}(t_0), t_0) k^{p+1} + O(k^{p+2}). \quad (13.88)$$

Similarly, we have

$$\mathbf{E}^{\frac{1}{2}} = \mathbf{U}^{\frac{1}{2}} - \mathbf{u}\left(t_0 + \frac{k}{2}\right) = \tau(\mathbf{u}(t_0), t_0) \left(\frac{k}{2}\right)^{p+1} + O(k^{p+2}).$$

By (RET-2), the error $\tilde{\mathbf{E}}^1 := \tilde{\mathbf{U}}^1 - \mathbf{u}(t_0 + k)$ includes $\mathbf{E}^{\frac{1}{2}}$ and the local error of the second substep:

$$\begin{aligned}\tilde{\mathbf{E}}^1 &= \mathbf{E}^{\frac{1}{2}} + \tau\left(\mathbf{u}\left(t_0 + \frac{k}{2}\right), t_0 + \frac{k}{2}\right) \left(\frac{k}{2}\right)^{p+1} + O(k^{p+2}) \\ &= 2\tau(\mathbf{u}(t_0), t_0) \left(\frac{k}{2}\right)^{p+1} + O(k^{p+2}),\end{aligned} \quad (13.89)$$

where the second step follows from the Taylor expansion of $\tau(\mathbf{u}(t_0 + \frac{k}{2}), t_0 + \frac{k}{2})$ at $(\mathbf{u}(t_0), t_0)$. Then (13.85), (13.88), and (13.89) yield

$$\hat{\mathbf{E}}^1 = \frac{2^p}{2^p - 1} \tilde{\mathbf{E}}^1 - \frac{1}{2^p - 1} \mathbf{E}^1 = O(k^{p+2}).$$

Similarly, (13.87) follows from (13.88) and (13.89). $\square$

Definition 13.125. An *embedded RK method* is a pair of RK methods of orders p and $\hat{p}$ that share the same stage computations:

$$\begin{array}{c|c} \mathbf{c} & A \\ \hline & \mathbf{b}^\top \end{array} \quad \text{and} \quad \begin{array}{c|c} \mathbf{c} & A \\ \hline & \hat{\mathbf{b}}^\top \end{array},$$

which, due to the same RK matrix A and the same RK node vector $\mathbf{c}$, often has the notation $p(\hat{p})$ and

$$\begin{array}{c|c} \mathbf{c} & A \\ \hline & \mathbf{b}^\top \\ \hline & \hat{\mathbf{b}}^\top \end{array}. \quad (13.90)$$

Example 13.126. The *Fehlberg 4(5) embedded RK method* has six stages and delivers a method of order 4 with an error estimate (or a method of order 5 without error estimates):

$$\begin{array}{c|cccccc} 0 & & & & & & \\ \hline \frac{1}{4} & \frac{1}{4} & & & & & \\ \frac{3}{8} & \frac{3}{32} & \frac{9}{32} & & & & \\ \hline \frac{12}{13} & \frac{1932}{2197} & -\frac{7200}{2197} & \frac{7296}{2197} & & & \\ \hline 1 & \frac{439}{216} & -8 & \frac{3680}{513} & -\frac{845}{4104} & & \\ \hline \frac{1}{2} & -\frac{8}{27} & 2 & -\frac{3544}{2565} & \frac{1859}{4104} & -\frac{11}{40} & \\ \hline & \frac{25}{216} & 0 & \frac{1408}{2565} & \frac{2197}{4104} & -\frac{1}{5} & 0 \\ \hline & \frac{16}{135} & 0 & \frac{6656}{12825} & \frac{28561}{56430} & -\frac{9}{50} & \frac{2}{55} \end{array}. \quad (13.91)$$

The somewhat unintuitive coefficients in (13.91) arise not only from satisfying the order conditions but also from minimizing the one-step error.

Definition 13.127. An RK method is said to have the *first-same-as-last (FSAL)* property iff it is stiffly accurate as in Definition 13.98 and

$$c_1 = 0, \quad c_s = 1. \quad (13.92)$$

Example 13.128. The best known general purpose ERK method is the *Dormand-Prince 5(4) embedded RK method*

0									
$\frac{1}{5}$	$\frac{1}{5}$								
$\frac{3}{10}$	$\frac{3}{40}$	$\frac{9}{40}$							
$\frac{4}{5}$	$\frac{44}{45}$	$-\frac{56}{15}$	$\frac{32}{9}$						
$\frac{8}{9}$	$\frac{19372}{6561}$	$-\frac{25360}{2187}$	$\frac{64448}{6561}$	$-\frac{212}{729}$					
1	$\frac{9017}{3168}$	$-\frac{355}{33}$	$\frac{46732}{5247}$	$\frac{49}{176}$	$-\frac{5103}{18656}$				
1	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$			
	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	0		
	$\frac{5179}{57600}$	0	$\frac{7571}{16695}$	$\frac{393}{640}$	$-\frac{92097}{339200}$	$\frac{187}{2100}$	$\frac{1}{40}$		

(13.93)

which has seven stages and is a local extrapolation method, i.e., the 5th-order method is used to advance the solution. (13.93) is also optimized to minimize the leading term of the one-step error.

Formula 13.129. To solve the IVP (12.3) of size N to meet the absolute tolerance $\mathbf{E}_{\text{abs}}$ and the relative tolerance $\mathbf{E}_{\text{rel}}$ of the one-step error, a $p(\hat{p})$ embedded RK method with *automatic step size control* advances $(\mathbf{U}^n, k)$ to $(\mathbf{U}^{n+1}, \tilde{k})$ as follows.

- Advance $\mathbf{U}^n$ to $(\mathbf{U}^{n+1}, \hat{\mathbf{U}}^{n+1})$ by the two methods in the embedded RK method.
- Compute an error indicator based on the max-norm or the 2-norm

$$E_{\text{ind}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{U}_i^{n+1} - U_i^{n+1}}{\varepsilon_i} \right)^2}, \quad (13.94)$$

where ε_i depends on the i th components of $\mathbf{U}^n$, $\mathbf{E}_{\text{abs}}$, and $\mathbf{E}_{\text{rel}}$:

$$\varepsilon_i := E_{\text{abs},i} + |U_i^n| E_{\text{rel},i}. \quad (13.95)$$

- Compute a new time step size $\tilde{k}$ by

$$\tilde{k} = k \min \left\{ \rho_{\max}, \max \left\{ \rho_{\min}, \rho (E_{\text{ind}})^{\frac{-1}{q+1}} \right\} \right\}, \quad (13.96)$$

where $q = \min(p, \hat{p})$, usually $\rho_{\max} \in [1.5, 5]$, $\rho = 0.8, 0.9, (0.25)^{\frac{1}{q+1}}$ or $(0.38)^{\frac{1}{q+1}}$, and $\rho_{\min} \in [0.2, \rho]$.

- If $E_{\text{ind}} \leq 1$, proceed to the next time step with $\mathbf{U}^{n+1}$, $\tilde{k}$, $\mathbf{E}_{\text{abs}}$, and $\mathbf{E}_{\text{rel}}$; otherwise reject $(\mathbf{U}^{n+1}, \hat{\mathbf{U}}^{n+1})$ and go back to step (a) with $\tilde{k}$ as the new initial step size.

Example 13.130. For the case $\hat{p} = p+1$ in Formula 13.129, we expect

$$\|\hat{\mathbf{U}}^{n+1} - \mathbf{U}^{n+1}\| \approx C k^{p+1}.$$

If we have $E_{\text{ind}} \leq 1$, i.e.,

$$\|\hat{\mathbf{U}}^{n+1} - \mathbf{U}^{n+1}\| \leq \|\epsilon\|,$$

we will increase the step size; otherwise we decrease it. The extent of increasing or decreasing is quantified as

$$\left(\frac{\tilde{k}}{k} \right)^{p+1} \approx \frac{\|\epsilon\|}{\|\hat{\mathbf{U}}^{n+1} - \mathbf{U}^{n+1}\|}$$

and the new step size can be obtained by one calculation

$$\tilde{k} = k \cdot (E_{\text{ind}})^{\frac{-1}{q+1}}, \quad (13.97)$$

where $q = \min(p, \hat{p})$. However, in practical computation k should not be allowed to increase nor to decrease too fast; (13.96) is designed to fulfill this purpose. As a price we pay for (13.96), the whole process in Formula 13.129 may be repeated multiple times until an acceptable step size is found. The values of the factor ρ are chosen such that the new time step size will be acceptable in the next time step with high probability. If $E_{\text{ind}} \leq 1$, the computed step is accepted and the solution is advanced with $\mathbf{U}^{n+1}$ and a new step is tried with $\tilde{k}$ as the step size; otherwise, the step is rejected and the computations are repeated with the new step size $\tilde{k}$.

13.9 Programming assignments

13.9.1 The assignments

Repeat the assignment in Section 12.6 by implementing in C++ methods as follows.

- the classical RK method,
- the ESDIRK method in Example 13.52,
- Gauss-Legendre RK methods ($s = 2, 3, 4, 5$),
- Fehlberg 4(5) embedded RK method,
- Dormand-Prince 5(4) embedded RK method.

Write a report to organize your results in a coherent story that relates your numerical results to the theory in this chapter. Your story must contain three plots of your solution to (12.91): the Euler method with 24000 steps of fixed size, the classical RK methods with 6000 steps of fixed size, and Dormand-Prince 5(4) embedded RK method with 100 steps of adaptive size. In addition, to achieve an error of 10^{-3} based on the max-norm of the solution error, which method is the winner in terms of total CPU time?

Chapter 14

Method of Lines (MOL)

14.1 MOL for the heat equation

Definition 14.1. A second-order, constant-coefficient, linear partial differential equation (PDE) of the form

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + F = 0 \quad (14.1)$$

is called a *parabolic PDE* if its coefficients satisfy

$$B^2 - 4AC = 0. \quad (14.2)$$

Definition 14.2. The *one-dimensional heat equation* is a parabolic PDE of the form

$$u_t = \nu u_{xx} \text{ in } \Omega := (0, 1) \times (0, T), \quad (14.3)$$

where $x \in (0, 1)$ is the spatial location, $t \in (0, T)$ the time and $\nu > 0$ the dynamic viscosity; the equation has to be supplemented with an *initial condition*

$$u(x, 0) = \eta(x), \text{ on } (0, 1) \times \{0\} \quad (14.4)$$

and appropriate boundary conditions at $\{0, 1\} \times (0, T)$.

Definition 14.3. An *initial-boundary value problem* is the problem of determining a solution to a differential equation with both initial conditions and boundary conditions.

Example 14.4. We derive the heat equation based on the principle of conservation. Let $u(x, t)$ be the *density* or *concentration* function (mass/length) for some substance, then the total mass within $[a, b]$ at time t is

$$M(t) := \int_a^b u(x, t) dx.$$

Let $F(x, t)$ denote the *flux* of material across the point x at time t (mass/time) and $\psi(x, t)$ the *source* or *sink* of the substance (mass/length/time). Since the total mass within $[a, b]$ changes due to the flux at the endpoints and the source (or sink) within $[a, b]$, we have

$$\begin{aligned} \frac{dM(t)}{dt} &= F(a, t) - F(b, t) + \int_a^b \psi(x, t) dx \\ &= -\kappa(a) \frac{\partial u(a, t)}{\partial x} + \kappa(b) \frac{\partial u(b, t)}{\partial x} + \int_a^b \psi(x, t) dx \\ &= \int_a^b \frac{\partial}{\partial x} \left(\kappa(x) \frac{\partial u(x, t)}{\partial x} \right) dx + \int_a^b \psi(x, t) dx, \end{aligned}$$

where the second step follows from Fick's law of diffusion and the third step from the second fundamental theorem of calculus (Theorem B.128). By definition of the total mass, we have

$$\frac{dM(t)}{dt} = \frac{d}{dt} \int_a^b u(x, t) dx = \int_a^b \frac{\partial u(x, t)}{\partial t} dx,$$

and therefore

$$\int_a^b \left[\frac{\partial u(x, t)}{\partial t} - \frac{\partial}{\partial x} \left(\kappa(x) \frac{\partial u(x, t)}{\partial x} \right) - \psi(x, t) \right] dx = 0.$$

Since the integral must be zero for all values of a and b , it follows that the integrand must be identically zero. This gives, finally, the differential equation

$$\frac{\partial u(x, t)}{\partial t} = \frac{\partial}{\partial x} \left(\kappa(x) \frac{\partial u(x, t)}{\partial x} \right) + \psi(x, t).$$

If the material is homogeneous, then $\kappa(x) = \kappa$ is independent of x and the above equation reduces to

$$\frac{\partial u(x, t)}{\partial t} = \kappa \frac{\partial^2 u(x, t)}{\partial x^2} + \psi(x, t).$$

Theorem 14.5. The exact solution to the heat equation (14.3) with Dirichlet conditions $g_0(t) = g_1(t) = 0$ is

$$u(x, t) = \sum_{j=0}^{\infty} \hat{u}_j(t) \sin(\pi j x), \quad (14.5)$$

where $\hat{u}_j(t) := \exp(-j^2 \pi^2 \nu t) \hat{u}_j(0)$ and $\hat{u}_j(0)$ is the coefficient of the Fourier mode $\sin(\pi j x)$ in the initial data $u(x, 0) = \sum_{j=0}^{\infty} \hat{u}_j(0) \sin(\pi j x)$, cf. (14.5).

Proof. It is straightforward to verify that (14.5) is indeed the solution of (14.3). $\square$

14.1.1 FTCS and Crank–Nicolson

Notation 13. As shown in the LHS of Figure 14.1, the space-time domain of the heat equation (14.3) can be discretized by the rectangular grids

$$x_i = ih, \quad t_n = nk, \quad (14.6)$$

$h = \frac{1}{m+1}$ is the uniform mesh spacing and k is the uniform time-step size. The unknowns U_i^n are located at nodes (x_i, t_n) .

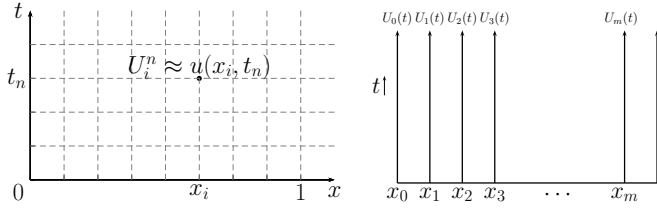


Figure 14.1: Spatiotemporal discretization (left) and lines in MOL (right).

Definition 14.6. The *method of lines* (MOL) is a technique for solving PDEs via

- (a) discretizing the spatial derivatives while leaving the time variable continuous;
- (b) solving the resulting ODEs with a numerical method designed for IVPs.

Example 14.7. Discretize the heat equation (14.3) in space at grid point x_i by

$$U_i'(t) = \frac{\nu}{h^2} (U_{i-1}(t) - 2U_i(t) + U_{i+1}(t)), \quad (14.7)$$

where $U_i(t) \approx u(x_i, t)$ for $i = 1, 2, \dots, m$.

For Dirichlet conditions

$$\begin{cases} u(0, t) = g_0(t), & \text{on } \{0\} \times (0, T); \\ u(1, t) = g_1(t), & \text{on } \{1\} \times (0, T), \end{cases} \quad (14.8)$$

this semi-discrete system (14.7) can be written as

$$\mathbf{U}'(t) = A\mathbf{U}(t) + \mathbf{g}(t), \quad (14.9)$$

where

$$A = \frac{\nu}{h^2} \begin{bmatrix} -2 & +1 & & & \\ +1 & -2 & +1 & & \\ & +1 & -2 & +1 & \\ & & \ddots & \ddots & \ddots \\ & & & +1 & -2 & +1 \\ & & & & +1 & -2 \end{bmatrix}, \quad (14.10)$$

$$\mathbf{U}(t) := \begin{bmatrix} U_1(t) \\ U_2(t) \\ U_3(t) \\ \vdots \\ U_{m-1}(t) \\ U_m(t) \end{bmatrix}, \quad \mathbf{g}(t) = \frac{\nu}{h^2} \begin{bmatrix} g_0(t) \\ 0 \\ 0 \\ \vdots \\ 0 \\ g_1(t) \end{bmatrix}. \quad (14.11)$$

By the negative definiteness of A , Definition 13.109, and Example 13.110, the ODE system (14.9) is dissipative.

Notation 14. The non-dimensional number

$$r := \frac{k\nu}{h^2} \quad (14.12)$$

is often used in numerically solving the heat equation.

Definition 14.8. The *FTCS* (forward in time, centered in space) method solves the heat equation (14.3) by

$$\frac{U_i^{n+1} - U_i^n}{k} = \frac{\nu}{h^2} (U_{i-1}^n - 2U_i^n + U_{i+1}^n), \quad (14.13)$$

or, equivalently

$$U_i^{n+1} = U_i^n + r(U_{i-1}^n - 2U_i^n + U_{i+1}^n). \quad (14.14)$$

Example 14.9. For homogeneous Dirichlet boundary conditions, the FTCS method can be written as

$$\mathbf{U}^{n+1} = (I + kA)\mathbf{U}^n, \quad (14.15)$$

where A is the matrix in (14.10) and

$$\mathbf{U}^n := \begin{bmatrix} U_1^n \\ U_2^n \\ \vdots \\ U_m^n \end{bmatrix}. \quad (14.16)$$

Definition 14.10. The *Crank–Nicolson method* solves the heat equation (14.3) by

$$\begin{aligned} \frac{U_i^{n+1} - U_i^n}{k} &= \frac{1}{2} (f(U_i^n, t_n) + f(U_i^{n+1}, t_{n+1})) \\ &= \frac{\nu}{2h^2} (U_{i-1}^n - 2U_i^n + U_{i+1}^n + U_{i-1}^{n+1} - 2U_i^{n+1} + U_{i+1}^{n+1}), \end{aligned} \quad (14.17)$$

or, equivalently

$$\begin{aligned} -rU_{i-1}^{n+1} + 2(1+r)U_i^{n+1} - rU_{i+1}^{n+1} \\ = rU_{i-1}^n + 2(1-r)U_i^n + rU_{i+1}^n. \end{aligned} \quad (14.18)$$

Exercise 14.11. Show that the matrix form of the Crank–Nicolson method for solving the heat equation (14.3) with Dirichlet conditions is

$$\left(I - \frac{k}{2}A\right)\mathbf{U}^{n+1} = \left(I + \frac{k}{2}A\right)\mathbf{U}^n + \mathbf{b}^n, \quad (14.19)$$

where

$$\mathbf{b}^n = \frac{r}{2} \begin{bmatrix} g_0(t_n) + g_0(t_{n+1}) \\ 0 \\ \vdots \\ 0 \\ g_1(t_n) + g_1(t_{n+1}) \end{bmatrix}.$$

Definition 14.12. For $\theta \in [0, 1]$, the θ -method solves the heat equation (14.3) by

$$\frac{U_i^{n+1} - U_i^n}{k} = \frac{\nu}{h^2} [\theta(U_{i-1}^{n+1} - 2U_i^{n+1} + U_{i+1}^{n+1}) + (1-\theta)(U_{i-1}^n - 2U_i^n + U_{i+1}^n)], \quad (14.20)$$

or, equivalently

$$\begin{aligned} -\theta r U_{i-1}^{n+1} + (1+2\theta r)U_i^{n+1} - \theta r U_{i+1}^{n+1} \\ = (1-\theta)r U_{i-1}^n + [1-2(1-\theta)r]U_i^n + (1-\theta)r U_{i+1}^n. \end{aligned} \quad (14.21)$$

Example 14.13. The θ -method with $\theta = 0$ is the FTCS method, that with $\theta = \frac{1}{2}$ is the Crank–Nicolson method, and that with $\theta = 1$ is the *BTCS* (backward in time and centered in space) of which the ODE solver is the backward Euler method.

14.1.2 Accuracy and consistency

Definition 14.14. The *local truncation error (LTE)* of an MOL for solving a PDE is the error τ of approximating the PDE caused by replacing continuous derivatives with finite difference formulas.

Definition 14.15. An MOL is said to be *consistent* if

$$\lim_{k,h \rightarrow 0} \tau(x, t) = 0. \quad (14.22)$$

Definition 14.16. An MOL is said to be *p th-order accurate in time* and *q th-order accurate in space* iff its LTE satisfies $\tau(x, t) = O(k^p + h^q)$.

Example 14.17. The LTE of the FTCS method in Definition 14.8 is

$$\begin{aligned} \tau(x, t) &= \frac{u(x, t+k) - u(x, t)}{k} \\ &\quad - \frac{\nu}{h^2} \left(u(x-h, t) - 2u(x, t) + u(x+h, t) \right) \\ &= \left(u_t + \frac{1}{2}ku_{tt} + \frac{1}{6}k^2u_{ttt} + \cdots \right) \\ &\quad - \nu \left(u_{xx} + \frac{1}{12}h^2u_{xxxx} + \cdots \right) \\ &= \left(\frac{1}{2}k\nu^2 - \frac{\nu}{12}h^2 \right) u_{xxxx} + O(k^2 + h^4), \end{aligned}$$

where the first step follows from Definition 14.14, the second from Taylor expansions and the last from $u_t = \nu u_{xx}$ and $u_{tt} = \nu u_{xxt} = \nu u_{txx} = \nu^2 u_{xxxx}$. By Definition 14.16, the FTCS method is second-order accurate in space and first-order accurate in time.

Lemma 14.18. The LTE of the θ -method is

$$\begin{aligned} \tau_i^{n+\frac{1}{2}} &= \left(\frac{1}{2} - \theta \right) k\nu u_{xxt} - \frac{1}{12}h^2\nu u_{xxxx} \\ &\quad + \frac{1}{12} \left(\frac{1}{2} - \theta \right) kh^2\nu u_{xxxxt} + O(k^2 + h^4). \end{aligned} \quad (14.23)$$

Proof. By Definition 14.14, we have

$$\begin{aligned} \tau_i^{n+\frac{1}{2}} &= \frac{u(x_i, t_n+k) - u(x_i, t_n)}{k} \\ &\quad - \frac{\nu}{h^2} \theta [u(x_i-h, t_n+k) - 2u(x_i, t_n+k) + u(x_i+h, t_n+k)] \\ &\quad - \frac{\nu}{h^2} (1-\theta) [u(x_i-h, t_n) - 2u(x_i, t_n) + u(x_i+h, t_n)], \end{aligned}$$

which, together with Taylor expansions at $(x_i, t_{n+\frac{1}{2}})$ and the equation $u_t = \nu u_{xx}$, yields (14.23). $\square$

Corollary 14.19. The Crank–Nicolson method is second-order accurate both in space and in time.

Proof. This follows from setting $\theta = \frac{1}{2}$ in Lemma 14.18. $\square$

Example 14.20. Set $\theta = \frac{1}{2} - \frac{h^2}{12k\nu}$ and we get a method whose LTE is $O(k^2 + h^4)$. Note that this requires $h^2 \leq 6k\nu$ to ensure $\theta \geq 0$, cf. Definition 14.12.

14.1.3 Absolute stability

Lemma 14.21. If A in (14.10) is treated as a linear operator on $\mathbb{R}^m$, its p th eigenvector $\mathbf{w}^p$ and the corresponding eigenvalue λ_p are

$$\lambda_p = -\frac{4\nu}{h^2} \sin^2 \left(\frac{p\pi h}{2} \right), \quad (14.24)$$

$$w_j^p = \sin(p\pi jh), \quad (14.25)$$

where $p, j = 1, 2, \dots, m$ and $h = \frac{1}{m+1}$. Furthermore, if A in (14.10) is considered as an operator on $\mathbb{C}^m$, its p th eigenvector would be $e^{ip\pi jh}$ while its p th eigenvalue remains the same as λ_p .

Proof. This follows directly from Lemma 10.25. $\square$

Example 14.22. For the FTCS method (14.15) to be absolutely stable, we must have $|1+k\lambda| \leq 1+O(k)$ for each eigenvalue in (14.24), which implies $-2 - O(k) \leq -4\nu k/h^2 \leq 0$. Since $k = o(k/h^2)$, we drop $O(k)$ from the inequality to have

$$k \leq \frac{h^2}{2\nu}, \quad (14.26)$$

which is equivalent to $r \leq \frac{1}{2}$.

Definition 14.23. An MOL is said to be *unconditionally stable (in the sense of absolute stability)* for a PDE if in solving the semi-discrete system of the PDE its ODE solver is absolutely stable for any $k > 0$.

Lemma 14.24. Suppose, for some $\alpha > 0$, the ODE solver of the MOL is $A(\alpha)$ -stable for the semi-discrete system that results from spatially discretizing the heat equation. Then the MOL is unconditionally stable for the heat equation.

Proof. For any $\alpha > 0$, the RAS of an $A(\alpha)$ -stable method contains the negative real axis. All eigenvalues of the semi-discrete system (14.9) are negative real numbers, hence $k\lambda$ is in the RAS for any $k > 0$. $\square$

Lemma 14.25. The θ -method (14.20) is unconditionally stable for $\theta \in [\frac{1}{2}, 1]$. For $\theta \in [0, \frac{1}{2})$, the time step size must satisfy $k \leq \frac{h^2}{2(1-2\theta)\nu}$ for the θ -method to be stable.

Exercise 14.26. Prove Lemma 14.25 via the stability function of one-step methods.

Corollary 14.27. The Crank–Nicolson method (14.18) is unconditionally stable for the heat equation.

Proof. This follows directly from Lemma 14.25. $\square$

14.1.4 Lax–Richtmyer stability

Definition 14.28. A set $\mathcal{S} \subseteq \mathbb{R}^m$ is *semianalytic* if there exist a finite number of analytic functions $g_i : \mathbb{R}^m \rightarrow \mathbb{R}$ such that $\mathcal{S}$ is in the universe of a finite Boolean algebra formed from the sets

$$\mathcal{X}_i = \{\mathbf{x} \in \mathbb{R}^m : g_i(\mathbf{x}) \leq 0\}. \quad (14.27)$$

The g_i 's are called the *generating functions* of $\mathcal{S}$. In particular, a semianalytic set is *semialgebraic* if all of its generating functions are polynomials.

Definition 14.29. A linear MOL of the form

$$\mathbf{U}^{n+1} = B_h(k)\mathbf{U}^n + \mathbf{b}_h^n(k) \quad (14.28)$$

is *Lax-Richtmyer stable* iff

$$\begin{aligned} \forall T > 0, \quad \exists h_0, k_0, C_T > 0 \text{ s.t.} \\ \forall k \in (0, k_0], \forall h \in (0, h_0], \forall n \in \mathbb{N}^+, \quad (14.29) \\ nk \leq T \implies \|B_h(k)^n\| \leq C_T, \end{aligned}$$

where B_h is the MOL iteration matrix for the grid with size h and the constant C_T depends neither on k nor on h .

More generally, the MOL (14.28) is *Lax-Richtmyer stable on the stability region $\mathcal{S}$* iff

$$\begin{aligned} \forall T > 0, \quad \exists C_T > 0 \text{ s.t. } \forall (k, h) \in \mathcal{S}, \forall n \in \mathbb{N}^+, \quad (14.30) \\ nk \leq T \implies \|B_h(k)^n\| \leq C_T, \end{aligned}$$

where $\mathcal{S} \subset \mathbb{R}^2$ is a semi-analytic set.

Definition 14.30. A linear MOL (14.28) is said to have *strong stability* if

$$\exists k_0 > 0 \text{ s.t. } \forall k \in (0, k_0], \forall h \in \mathbb{R}^+, \quad \|B_h(k)\|_2 \leq 1. \quad (14.31)$$

Corollary 14.31. The Crank–Nicolson method has strong stability with

$$B = \left(I - \frac{k}{2}A\right)^{-1} \left(I + \frac{k}{2}A\right). \quad (14.32)$$

Proof. Exercise 14.11 yields (14.32). The symmetry of A implies that of $I \pm \frac{k}{2}A$. The inverse matrix P of a symmetric matrix M is also symmetric since $(P^{-1})^\top = M^\top = M = P^{-1}$ and thus $(I - \frac{k}{2}A)^{-1}$ is also symmetric.

For $M := I - \frac{k}{2}A$ and $N = I + \frac{k}{2}A$, we have $MN = NM$. Then $\det M \neq 0$ implies $M^{-1}N = NM^{-1}$. i.e., M^{-1} commutes with N . Hence we have

$$B^\top = (M^{-1}N)^\top = N^\top(M^{-1})^\top = NM^{-1} = M^{-1}N = B.$$

Hence B is symmetric and Lemma C.306 implies

$$\|B\|_2 = \rho(B) = \max \left| \frac{1 + k\lambda_p/2}{1 - k\lambda_p/2} \right| \leq 1,$$

where the last step follows from (14.24). Then the proof is completed by Definition 14.30. $\square$

14.1.5 Convergence

Definition 14.32. The *solution error* of an MOL is

$$E_i^n = U_i^n - u(x_i, t_n), \quad (14.33)$$

where $u(x_i, t_n)$ is the exact solution of the PDE at the grid point (x_i, t_n) .

Definition 14.33. An MOL is *convergent* iff

$$\forall T > 0, \quad \lim_{k \rightarrow 0, h \rightarrow 0; kN=T} \|E^N\| = 0. \quad (14.34)$$

Lemma 14.34. Let $\{E_i\}_{i=1}^\infty$ be a sequence of Banach spaces. Define a new space E , a projection $P_j : E \rightarrow E_j$, and an embedding operator $I_j : E_j \rightarrow E$ as follows,

$$E := \bigoplus_{j=1}^\infty E_j = \left\{ (x_i)_{i \in \mathbb{N}^+} : x_i \in E_i, \sum_{i=1}^\infty \|x_i\| < +\infty \right\}, \quad (14.35)$$

$$\begin{cases} \forall (x_i) \in E, & P_j((x_i)) = x_j, \\ \forall x \in E_j, & I_j(x) = \bar{x}_j := (0, \dots, 0, x, 0, \dots). \end{cases} \quad (14.36)$$

Then we have

(a) $(E, \|\cdot\|)$ is a Banach space where

$$\forall (x_i)_{i \in \mathbb{N}^+} \in E, \quad \|(x_i)_{i \in \mathbb{N}^+}\| := \sum_{i=1}^\infty \|x_i\|; \quad (14.37)$$

(b) $\forall j \in \mathbb{N}^+$, both P_j and I_j are continuous linear maps, i.e., $P_j \in \mathcal{CL}(E, E_j)$, $I_j \in \mathcal{CL}(E_j, E)$, cf. Section E.2;

(c) $\|P_j\| = \|I_j\| = 1$ and $P_j I_j$ is the identity map on E_j ;

(d) $\forall T \in \mathcal{CL}(E_j, E_j)$, the linear operator $\bar{T} := I_j T P_j$ is a norm-preserving extension of T , i.e., $\bar{T} \in \mathcal{CL}(E, E)$ and

$$\forall n \in \mathbb{N}^+, \quad \|\bar{T}^n\| = \|T^n\|. \quad (14.38)$$

Proof. (a) follows from the condition of each E_j being a Banach space, the construction (14.35), and Definition E.25.

(b) follows from (14.36) and Theorem E.41.

(c) holds trivially from (14.36).

For (d), we consider an arbitrary element $(x_i) \in E$,

$$\begin{aligned} (*) : \quad \|\bar{T}^n((x_i))\| &= \|I_j T^n P_j((x_i))\| = \|I_j T^n(x_j)\| \\ &= \|(0, \dots, 0, T^n(x_j), 0, \dots, 0)\| = \|T^n(x_j)\|, \end{aligned}$$

where the first equality follows from (c), the second and the third from (14.36), and the fourth from (14.37). In (*), the range of x_j covers E_j while that of (x_i) covers E . Then the proof is completed by taking supremum of (*) and applying Lemma E.56. $\square$

Theorem 14.35 (Lax equivalence theorem). A consistent linear MOL (14.28) is convergent if and only if it is Lax–Richtmyer stable.

Proof. We first show the sufficiency. By Definition 14.14, the LTE of the numerical method (14.28) satisfies

$$(*) \quad \hat{\mathbf{U}}^{n+1} = B_h \hat{\mathbf{U}}^n + \mathbf{b}^n + k\tau^n,$$

where the dependence on k has been suppressed for clarity and $\hat{\mathbf{U}}^n$ is the exact solution,

$$\hat{\mathbf{U}}^n := \begin{bmatrix} u(x_1, t_n) \\ u(x_2, t_n) \\ \vdots \\ u(x_m, t_n) \end{bmatrix}, \quad \tau^n := \begin{bmatrix} \tau(x_1, t_n) \\ \tau(x_2, t_n) \\ \vdots \\ \tau(x_m, t_n) \end{bmatrix}.$$

Subtracting (*) from (14.28) gives the difference equation for the global error $E^n = \mathbf{U}^n - \hat{\mathbf{U}}^n$:

$$E^{n+1} = B_h E^n - k\tau^n,$$

and hence, by induction,

$$E^N = B_h^N E^0 - k \sum_{n=1}^N B_h^{N-n} \tau^{n-1},$$

from which we have

$$(**): \|E^N\| \leq \|B_h^N\| \|E^0\| + k \sum_{n=1}^N \|B_h^{N-n}\| \|\tau^{n-1}\|.$$

If the MOL is Lax–Richtmyer stable, we have, for $Nk \leq T$,

$$\|E^N\| \leq C_T \|E^0\| + kN \cdot C_T \max_{1 \leq n \leq N} \|\tau^{n-1}\|,$$

the RHS goes to 0 as $k \rightarrow 0$ and $h \rightarrow 0$.

As for the necessity, first we know from Definition 14.29 that it suffices to prove, for all sufficiently small $k, h \in \mathbb{R}^+$, the power boundedness of $B_h(k)$ for the homogeneous case, i.e., $\mathbf{b}_h(k) = \mathbf{0}$ in (14.28), since we have the same $B_h(k)$ for both homogeneous and non-homogeneous PDEs.

Second, we construct a space of all grid functions,

$$\mathcal{B} := \bigoplus_{m=1}^{+\infty} \mathbb{G}_{\frac{1}{m}}, \quad (14.39)$$

where “ $\bigoplus$ ” is given by (14.35) and $\mathbb{G}_{\frac{1}{m}}$ is the Banach space of grid functions on the grid $\mathbf{X}$ with size $h = \frac{1}{m}$,

$$\mathbb{G}_h := \{U : \mathbf{X} \rightarrow \mathbb{R} \mid \|U\|_q < +\infty\}, \quad (14.40)$$

and $\|\cdot\|_q$ is an L_q -norm in Definition 10.15. Since the PDE is linear, we can assume WLOG that the spatial domain is unit and each $h \in (0, 1]$ equals $\frac{1}{m}$ for some $m \in \mathbb{N}^+$. By Lemma 14.34, $(\mathcal{B}, \|\cdot\|)$ with $\|\cdot\|$ given in (14.37) is a Banach space.

Our main strategy is to deduce stability from convergence via the principle of uniform boundedness (Theorem E.83). More precisely, for the family of linear maps

$$\begin{aligned} \mathcal{T} &:= \{\overline{B_h(k)}^n : (h, k, n) \in I_{\mathcal{T}}\}, \\ I_{\mathcal{T}} &:= \{(h, k, n) : kn \leq T, h \in (0, h_0), k \in (0, k_0)\}, \end{aligned}$$

where $\overline{B_h(k)} : \mathcal{B} \rightarrow \mathcal{B}$ is the extension of $B_h(k)$ as defined in Lemma 14.34(d) and the constants h_0, k_0 come from Definition 14.33 (see below). We want to show

$$(\square) : \forall \eta \in \mathcal{B}, \exists M_{\eta} > 0 \text{ s.t. } \forall \overline{B_h(k)}^n \in \mathcal{T}, \|\overline{B_h(k)}^n(\eta)\| \leq M_{\eta}$$

so that the principle of uniform boundedness and Lemma 14.34(d) would imply stability.

Third, the convergence of the MOL and Definition 14.33 imply that we can pick some fixed $\epsilon > 0$ such that

$$\begin{aligned} \exists k_0, h_0 > 0 \text{ s.t. } \forall k \in (0, k_0], \forall h \in (0, h_0], \forall n \in \mathbb{N}^+, \\ t := kn \in [0, T] \implies \|E(t)\| = \|E^n\| < \epsilon. \end{aligned}$$

Finally, we prove $(\square)$ by contradiction. The negation of $(\square)$ yields

$$\begin{aligned} (\triangle) : \exists \eta \in \mathcal{B}, \exists (h_j, k_j, n_j)_{j \in \mathbb{N}} \subset I_{\mathcal{T}}, \text{ s.t. } \\ \begin{cases} \lim_{j \rightarrow \infty} n_j k_j = t' \in [0, T]; \\ \lim_{j \rightarrow \infty} \|\overline{B_{h_j}(k_j)}^{n_j}(\eta)\| = +\infty. \end{cases} \end{aligned}$$

By construction of $\overline{B_h(k)}$, we have

$$h_1 \neq h_2 \implies \overline{B_{h_1}(k)}(\overline{U_{h_2}^0}) = \mathbf{0} := (0, 0, \dots).$$

Hence $\mathcal{B} \setminus \overline{\mathbb{G}_{h_1}}$ is a subset of the null space of $\overline{B_{h_1}(k)}$, leading to $\overline{B_{h_j}(k_j)}^{n_j}(\eta) = B_{h_j}(k_j)^{n_j} U_{h_j}^0$ where $U_{h_j}^0$ is the component of η for grid size h_j . Let $u(t')$ be the exact solution at time $t = t'$. Then we have

$$\begin{aligned} \forall j \in \mathbb{N}^+, \quad \|\overline{B_{h_j}(k_j)}^{n_j}(\eta)\| &= \|\overline{B_{h_j}(k_j)}^{n_j} U_{h_j}^0\| \\ &\leq \|u(t')\| + \epsilon \leq C \|U_{h_j}^0\| + \epsilon \leq C \|\eta\| + \epsilon, \end{aligned}$$

where the second step follows from $\|E(t')\| < \epsilon$, the third from the well-posedness of the linear PDE, and the last from (14.37). This contradicts $(\triangle)$ and thus $(\square)$ holds. $\square$

Corollary 14.36. In solving the heat equation (14.3), the θ -method is convergent for any $\theta \in [0, 1]$.

Proof. For the θ -method, we have

$$\begin{aligned} B &= (I - \theta k A)^{-1} [I + (1 - \theta) k A], \\ \rho(B) &= \max \left| \frac{1 + (1 - \theta) k \lambda_p}{1 - \theta k \lambda_p} \right| \leq 1, \end{aligned}$$

where λ_p is given in (14.24) and the inequality holds for any k that is sufficiently small. The rest of the proof follows from Theorem 14.35 and Lemma 14.18. $\square$

14.1.6 Discrete maximum principle

Theorem 14.37 (Maximum principle of the heat equation). Suppose $u(x, t)$ satisfies the heat equation (14.3) in $\Omega := (0, 1) \times (0, T)$ and is continuous in $\overline{\Omega} := [0, 1] \times [0, T]$. Then both the maximum and the minimum of $u(x, t)$ over $\overline{\Omega}$ are assumed either initially at $t = 0$ or on the boundary $x = 0$ or $x = 1$. More precisely, define

$$\Gamma_{\Omega} := \{(x, t) \in \overline{\Omega} : t = 0 \text{ or } x = 0 \text{ or } x = 1\} \quad (14.41)$$

and we have

$$\max_{(x,t) \in \overline{\Omega}} \{u(x, t)\} = \max_{(x,t) \in \Gamma_{\Omega}} \{u(x, t)\}, \quad (14.42a)$$

$$\min_{(x,t) \in \overline{\Omega}} \{u(x, t)\} = \min_{(x,t) \in \Gamma_{\Omega}} \{u(x, t)\}. \quad (14.42b)$$

Theorem 14.38 (Discrete maximum principle). For the heat equation (14.3) in Ω , a θ -method with $\theta \in [0, 1]$ and $2r(1 - \theta) \leq 1$ satisfies

$$\min \mathcal{U}_{\Gamma} \leq U_j^n \leq \max \mathcal{U}_{\Gamma} \quad (14.43)$$

where U_j^n is a solution of the θ -method at $(x_j, t_n) \in \Omega$ and the set of values of U at the boundary points is

$$\mathcal{U}_{\Gamma} := \{U_0^n : n = 0, 1, \dots, N\} \cup \{U_{m+1}^n : n = 0, 1, \dots, N\} \cup \{U_j^0 : j = 0, 1, \dots, m+1\}.$$

Proof. Suppose there exists an internal point $U_j^{n+1} \notin \Gamma_\Omega$, cf. (14.41), such that U_j^{n+1} is a *global* maximum and satisfies $U_j^{n+1} > \max \mathcal{U}_\Gamma$. Then for U_j^{n+1} we write (14.21) as

$$\begin{aligned} (1 + 2\theta r)U_j^{n+1} &= \theta r(U_{j-1}^{n+1} + U_{j+1}^{n+1}) \\ &\quad + (1 - \theta)r(U_{j-1}^n + U_{j+1}^n) \quad (14.44) \\ &\quad + [1 - 2(1 - \theta)r]U_j^n, \end{aligned}$$

where all coefficients of the five values of U on the RHS are nonnegative and sum to $1 + 2\theta r$. Let U^* be the *local* maximum of the five values of U in the stencil of (x_j, t_{n+1}) on the RHS of (14.44). Then we have

$$\begin{aligned} (1 + 2\theta r)U_j^{n+1} &\leq \theta r(U^* + U^*) + (1 - \theta)r(U^* + U^*) \\ &\quad + [1 - 2(1 - \theta)r]U^* = (1 + 2\theta r)U^*, \end{aligned}$$

i.e., $U_j^{n+1} \leq U^*$. But since U_j^{n+1} is the global maximum, we also have $U_j^{n+1} \geq U^*$, and thus $U^* = U_j^{n+1}$. However, in order for the inequality “ $\leq$ ” to be “ $=$ ” so that $U^* = U_j^{n+1}$ holds, we must have

$$U_{j-1}^{n+1} = U_{j+1}^{n+1} = U_{j-1}^n = U_{j+1}^n = U_j^n = U^* = U_j^{n+1}.$$

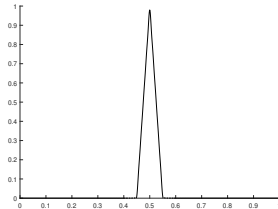
Now that each of the five values of U is also the global maximum, repeat the above arguments and we know that all points in the stencils of the five locations must also have the global maximum of U . Since the initial condition is always a Dirichlet condition and the discrete grid is always connected along the positive direction of the time axis, the values of U must be the same for all interior points and all boundary points. This contradicts the starting assumption.

The first inequality in (14.43) can be proven by similar arguments. $\square$

Example 14.39. Consider the model problem (14.3) with $\nu = 1$, the boundary conditions $u(0, t) = u(1, t) = 0$, and the initial condition

$$u(x, 0) = \varphi(x) = \begin{cases} 20(x - \frac{9}{20}) & \text{if } x \in [\frac{9}{20}, \frac{1}{2}); \\ -20(x - \frac{11}{20}) & \text{if } x \in [\frac{1}{2}, \frac{11}{20}); \\ 0 & \text{otherwise} \end{cases} \quad (14.45)$$

shown below.



The solution of the above problem is

$$u(t, x) = \sum_{k=1}^{\infty} A_k e^{-k^2 \pi^2 t} \sin(k\pi x) \quad (14.46)$$

where

$$\begin{aligned} A_k &= 2 \int_0^1 \varphi(\xi) \sin(k\pi \xi) d\xi \\ &= \frac{40}{k^2 \pi^2} \left[-\sin\left(\frac{9}{20}k\pi\right) + 2\sin\left(\frac{1}{2}k\pi\right) - \sin\left(\frac{11}{20}k\pi\right) \right]. \end{aligned}$$

In the rest of this example we fix $h = \frac{1}{20}$. The three plots in the first line of Figure 14.2 represent the exact solution (14.46) with $r = 1$ at $t = k$, $t = 2k$ and $t = 10k$, respectively. Results of other methods are also plotted in Figure 14.2.

For $r = 1$, Crank–Nicolson preserves the discrete maximum principle of the heat equation; but for $r = 2$, results of Crank–Nicolson violates the discrete maximum principle. This illustrates Theorem 14.38. The oscillation in the results of Crank–Nicolson after one time step is probably due to the fact that it is not L-stable. When we switch to the L-stable methods, either the BTCS or the collocation method in Example 13.96, the oscillations disappear and the monotonicity is preserved.

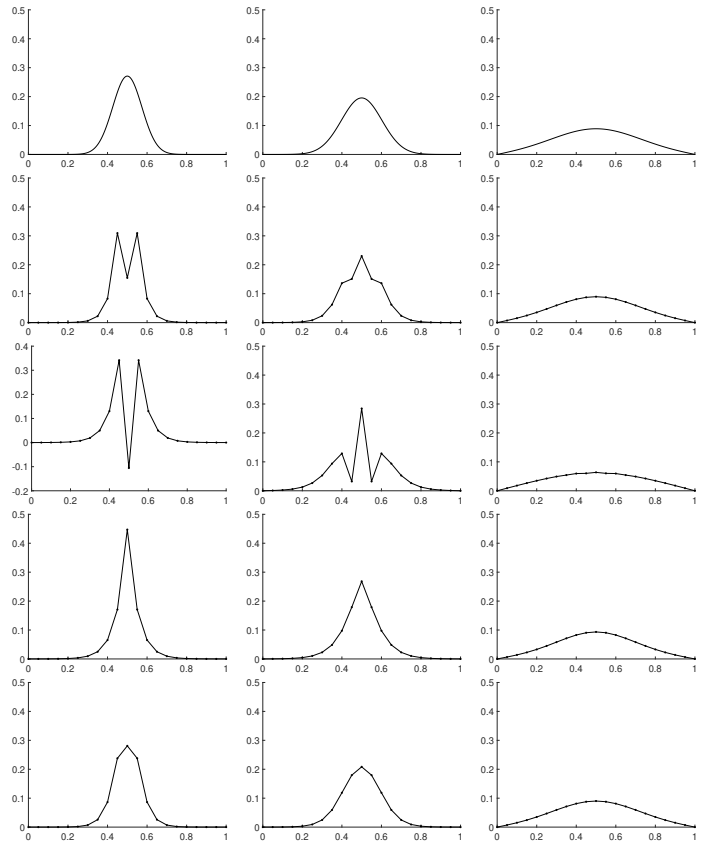


Figure 14.2: Illustrating discrete maximum principle and L-stability via results of different methods for solving the heat equation (14.3) with $\nu = 1$, $u(0, t) = u(1, t) = 0$, and the initial condition (14.45). The first line contains the exact solutions, the second line those of the Crank–Nicolson method with $r = 1$, the third line those of the Crank–Nicolson method with $r = 2$, the fourth line those of the BTCS method with $r = 1$, and the last line those of the collocation method in Example 13.96 with $r = 1$. The first, second, and third columns correspond to solutions at $t = k$, $2k$, and $10k$, respectively.

14.1.7 Von Neumann stability

Definition 14.40. An MOL for a first-order equation (F.44) is *von Neumann stable* iff

$$\begin{aligned} &\forall T > 0, \exists h_0, k_0, C_T > 0, \exists S \in \mathbb{N} \text{ s.t.} \\ &\forall k \in (0, k_0], \forall h \in (0, h_0], \forall n \in \mathbb{N}^+, \\ &nk \leq T \implies \|\mathbf{U}^n\|_{L^2}^2 \leq C_T \sum_{i=0}^S \|\mathbf{U}^i\|_{L^2}^2, \end{aligned} \quad (14.47)$$

where C_T is a constant depending only on T and $\|\mathbf{U}^n\|_{L^2}$ is the 2-norm of the grid function $\mathbf{U}^n$ in Definition 10.15.

More specifically, the MOL is *von Neumann stable under the constraint* $\mathbf{g}(k, h) \leq \mathbf{0}$ iff (14.47) holds with its third line replaced with

$$\mathbf{g}(k, h) \leq \mathbf{0}, \quad nk \leq T \implies \|\mathbf{U}^n\|_{L^2}^2 \leq C_T \sum_{i=0}^S \|\mathbf{U}^i\|_{L^2}^2,$$

where $\mathbf{g}(k, h) \leq \mathbf{0}$ means

$$\forall i = 1, 2, \dots, m, \quad g_i(k, h) \leq 0 \text{ or } g_i(k, h) < 0$$

and each $g_i(k, h)$ is an analytic function.

Lemma 14.41. Define $h\mathbb{Z} := \{hj : j \in \mathbb{Z}\}$ and

$$L^p(h\mathbb{Z}) := \left\{ g : h\mathbb{Z} \rightarrow \mathbb{R} \left| \sum_{j \in \mathbb{Z}} |g(jh)|^p < \infty \right. \right\}.$$

The Fourier transform of a function $\mathbf{U} \in L^1(h\mathbb{Z})$ is the continuous function $\hat{U} : [-\frac{\pi}{h}, \frac{\pi}{h}] \rightarrow \mathbb{C}$ given by

$$\hat{U}(\xi) = (\mathcal{F}U)(\xi) := \frac{1}{\sqrt{2\pi}} \sum_{m \in \mathbb{Z}} e^{-imh\xi} U(m)h \quad (14.48)$$

while its inverse Fourier transform is given by

$$U(m) = \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{+\frac{\pi}{h}} e^{imh\xi} \hat{U}(\xi) d\xi. \quad (14.49)$$

Proof. This follows from Definition F.19 and a change of variable. $\square$

Exercise 14.42. Show that any grid function in $L^1(h\mathbb{Z})$ can be recovered by a Fourier transform followed by an inverse Fourier transform.

Theorem 14.43. Parseval's equality holds for grid functions in terms of the L_2 -norm in Definition 10.15, i.e.,

$$\forall U \in L^1(h\mathbb{Z}), \quad \|U\|_{L_2} = \|\hat{U}\|_2. \quad (14.50)$$

Proof. Lemma 14.41 yields

$$\begin{aligned} &\|\hat{U}(\xi)\|_2^2 = \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} |\hat{U}(\xi)|^2 d\xi \\ &= \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \overline{\hat{U}(\xi)} \frac{1}{\sqrt{2\pi}} \sum_{m \in \mathbb{Z}} e^{-imh\xi} U(m)h d\xi \\ &= \frac{h}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \overline{\hat{U}(\xi)} \sum_{m \in \mathbb{Z}} e^{imh\xi} U(m)h d\xi \\ &= \frac{h}{\sqrt{2\pi}} \sum_{m \in \mathbb{Z}} U(m) \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \overline{\hat{U}(\xi)} e^{imh\xi} d\xi \\ &= h \sum_{m \in \mathbb{Z}} U(m) \overline{\hat{U}(m)} = h \sum_{m \in \mathbb{Z}} |U(m)|^2 = \|U\|_{L_2}^2. \quad \square \end{aligned}$$

Example 14.44. Consider the grid function

$$U(m) = \begin{cases} 1 & \text{if } |x_m| < 1, \\ \frac{1}{2} & \text{if } |x_m| = 1, \\ 0 & \text{if } |x_m| > 1, \end{cases}$$

where the grid spacing is $h = M^{-1}$ with $M \in \mathbb{N}^+$ and a grid point is $x_m = hm$. By (14.48), its Fourier transform is

$$\begin{aligned} \hat{U}(\xi) &= \frac{h}{2\sqrt{2\pi}} \left(e^{-iMh\xi} + e^{iMh\xi} + 2 \sum_{m=-M}^{M-1} e^{-imh\xi} \right) \\ &= \frac{h}{\sqrt{2\pi}} \cos(\xi) + \frac{h}{\sqrt{2\pi}} \frac{\sin((M-\frac{1}{2})h\xi)}{\sin(\frac{1}{2}h\xi)} \\ &= \frac{h}{\sqrt{2\pi}} \cos(\xi) + \frac{h}{\sqrt{2\pi}} \frac{\sin(Mh\xi) \cos(\frac{1}{2}h\xi)}{\sin(\frac{1}{2}h\xi)} \\ &\quad - \frac{h}{\sqrt{2\pi}} \frac{\cos(Mh\xi) \sin(\frac{1}{2}h\xi)}{\sin(\frac{1}{2}h\xi)} \\ &= \frac{h}{\sqrt{2\pi}} \sin(\xi) \cot\left(\frac{1}{2}h\xi\right). \end{aligned}$$

By Parseval's relation in Theorem 14.43, we must have

$$\begin{aligned} 2 - \frac{1}{2}h &= 2h \left(\frac{1}{2}\right)^2 + h \sum_{m=-M}^{M-1} 1 \\ &= \frac{h^2}{2\pi} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \sin^2(\xi) \cot^2\left(\frac{1}{2}h\xi\right) d\xi, \end{aligned}$$

which can be verified by directly evaluating the integral.

Example 14.45. For any constant $\alpha \in \mathbb{R}^+$, the Fourier transform of the grid function $U(m) = e^{-\alpha|m|h}$ is given by

$$\begin{aligned} \hat{U}(\xi) &= \frac{1}{\sqrt{2\pi}} \sum_{m \in \mathbb{Z}} e^{-imh\xi} e^{-\alpha|m|h} \\ &= \frac{h}{\sqrt{2\pi}} \left(1 + \sum_{m \in \mathbb{Z} \setminus \{0\}} e^{-imh\xi} e^{-\alpha|m|h} \right) \\ &= \frac{h}{\sqrt{2\pi}} \left(1 + \frac{e^{-(\alpha-i\xi)h}}{1-e^{-(\alpha-i\xi)h}} + \frac{e^{-(\alpha+i\xi)h}}{1-e^{-(\alpha+i\xi)h}} \right) \\ &= \frac{h}{\sqrt{2\pi}} \frac{1-e^{-2\alpha h}}{1-2e^{-\alpha h} \cos(h\xi) + e^{-2\alpha h}}. \end{aligned}$$

Then we have

$$\begin{aligned} \|\mathbf{U}\|_h^2 &= h \frac{1+e^{-2\alpha h}}{1-e^{-2\alpha h}} \\ &= \frac{h^2}{2\pi} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \left(\frac{1-e^{-2\alpha h}}{1-2e^{-\alpha h} \cos(h\xi) + e^{-2\alpha h}} \right)^2 d\xi = \|\hat{U}\|_2^2, \end{aligned}$$

which verifies Parseval's relation in Theorem 14.43.

Corollary 14.46. An MOL for a first-order equation (F.44) is von Neumann stable if and only if

$$\begin{aligned} &\forall T > 0, \exists h_0, k_0, C_T > 0, \exists S \in \mathbb{N} \text{ s.t.} \\ &\forall k \in (0, k_0], \forall h \in (0, h_0], \forall n \in \mathbb{N}^+, \\ &nk \leq T \implies \|\hat{U}^n\|_2 \leq C_T \sum_{i=0}^S \|\hat{U}^i\|_2, \end{aligned} \quad (14.51)$$

where C_T is a constant depending only on T and $\hat{U}^n$ is the Fourier transform of the grid function $\mathbf{U}^n$ advanced by the MOL.

Proof. This follows directly from Definition 14.40 and Theorem 14.43. $\square$

Example 14.47. Consider the FTCS method by the von Neumann analysis. Plug (14.49) into (14.14) and we have

$$\begin{aligned} U_j^{n+1} &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{+\frac{\pi}{h}} \chi_j(h\xi) \hat{U}^n(\xi) d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{+\frac{\pi}{h}} e^{ijh\xi} \hat{U}^{n+1}(\xi) d\xi, \end{aligned}$$

where $\chi_j(h\xi) := re^{i(j-1)h\xi} + re^{i(j+1)h\xi} + (1-2r)e^{ijh\xi}$ characterizes the FTCS method. Since the Fourier transform is unique, we have

$$\hat{U}^{n+1}(\xi) = g(h\xi)\hat{U}^n(\xi), \quad (14.52)$$

where $g(h\xi)$ is the *amplification factor*,

$$g(h\xi) = 1 - 4r \sin^2\left(\frac{\xi h}{2}\right).$$

To guarantee $|g(h\xi)| \leq 1$, we take $1 - \frac{4\nu k}{h^2} \geq -1$, which implies (14.26), i.e. $k \leq \frac{h^2}{2\nu}$. Therefore, the FTCS method is von Neumann stable under the constraint $k - \frac{h^2}{2\nu} \leq 0$, but not so for $k = \Theta(h)$.

Theorem 14.48. An MOL with a one-step time integrator and constant coefficients is von Neumann stable if and only if

$$\begin{aligned} \exists C, k_0, h_0 \in \mathbb{R}^+ \text{ s.t. } \forall h\xi \in [-\pi, \pi], \forall k \in (0, k_0], \forall h \in (0, h_0], \\ |g(h\xi, k, h)| \leq 1 + Ck, \end{aligned} \quad (14.53)$$

where the constant C is independent of ξ , k , and h .

Proof. Parseval's equality in Theorem 14.43 and the definition of g in (14.52) yield

$$\|\mathbf{U}^n\|_{L^2}^2 = \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} |g(h\xi, k, h)|^{2n} |\hat{U}^0(\xi)|^2 d\xi.$$

If $|g(h\xi, k, h)| \leq 1 + Ck$, we have

$$\begin{aligned} \|\mathbf{U}^n\|_{L^2}^2 &\leq \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} (1 + Ck)^{2n} |\hat{U}^0(\xi)|^2 d\xi \\ &= (1 + Ck)^{2n} \|\mathbf{U}^0\|_{L^2}^2. \end{aligned}$$

Then $n \leq \frac{T}{k}$ yields

$$(1 + Ck)^n \leq (1 + Ck)^{\frac{T}{k}} \leq e^{CT}.$$

Hence $\|\mathbf{U}^n\|_{L^2} \leq e^{CT} \|\mathbf{U}^0\|_{L^2}$ and the scheme is stable.

We now prove that if the inequality in (14.53) does not hold for any $C \in \mathbb{R}^+$, then the scheme is not stable. To do this we show that we can achieve any amount of growth in the solution, that is, we show that the stability inequality in Definition 14.40 cannot hold.

Suppose for any $C \in \mathbb{R}^+$ there are constants $h_0, k_0 \in \mathbb{R}^+$ and an interval $[\theta_1, \theta_2]$ such that $\xi h \in [\theta_1, \theta_2]$, $h \in (0, h_0]$, and $k \in (0, k_0]$ imply $|g(\xi h, k, h)| \geq 1 + Ck$. Then we construct a function U_m^0 as

$$\hat{U}^0(\xi) = \begin{cases} 0 & \text{if } h\xi \notin [\theta_1, \theta_2], \\ \sqrt{h(\theta_2 - \theta_1)^{-1}} & \text{if } h\xi \in [\theta_1, \theta_2]. \end{cases}$$

Then $\|\mathbf{U}^0\|_{L^2} = \|\hat{U}^0\|_2 = 1$ and

$$\begin{aligned} \|\mathbf{U}^n\|_{L^2}^2 &= \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} |g(h\xi, k, h)|^{2n} |\hat{U}^0(\xi)|^2 d\xi \\ &= \int_{\frac{\theta_1}{h}}^{\frac{\theta_2}{h}} |g(h\xi, k, h)|^{2n} \frac{h}{\theta_2 - \theta_1} d\xi \\ &\geq (1 + Ck)^{2n} \geq \frac{1}{2} e^{2TC} \|\mathbf{U}^0\|_{L^2}^2, \end{aligned}$$

for n near $\frac{T}{k}$. Hence the scheme is unstable. $\square$

Exercise 14.49. Prove Lemma 14.25 via Von Neumann analysis. What can you say after comparing this proof with that for Exercise 14.26?

14.2 MOL for advection equations

Definition 14.50. A second-order, constant-coefficient, linear partial differential equation (PDE) of the form

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + F = 0 \quad (14.54)$$

is called a *hyperbolic PDE* if its coefficients satisfy

$$B^2 - 4AC > 0. \quad (14.55)$$

Definition 14.51. The *one-dimensional wave equation* is a hyperbolic PDE of the form

$$u_{tt} = a^2 u_{xx}, \quad (14.56)$$

where $a > 0$ is the *wave speed*.

Definition 14.52. The *one-dimensional advection equation* is

$$u_t = -au_x \text{ in } \Omega := (0, 1) \times (0, T), \quad (14.57)$$

where $x \in (0, 1)$ is the spatial location and $t \in (0, T)$ the time; (14.57) is supplemented with an *initial condition*

$$u(x, 0) = \eta(x), \text{ on } (0, 1) \times \{0\} \quad (14.58)$$

and appropriate boundary conditions at either $\{0\} \times (0, T)$ or $\{1\} \times (0, T)$, depending on the sign of a .

Theorem 14.53. The exact solution of the Cauchy problem (14.57) is

$$u(x, t) = \eta(x - at). \quad (14.59)$$

Proof. It is straightforward to verify that

$$u_t + au_x = -a\eta'(x - at) + a\eta'(x - at) = 0. \quad \square$$

Lemma 14.54. The hyperbolic equation

$$u_t + au_x = f(x, t) - bu, \quad u(x, 0) = u_0(x) \quad (14.60)$$

with $a, b \in \mathbb{R}^+$ is solved by

$$u(x, t) = u_0(x - at)e^{-bt} + \int_0^t f(x - a(t - s), s)e^{-b(t-s)} ds. \quad (14.61)$$

Proof. It is straightforward to verify that (14.61) is indeed the solution of (14.60). We give a derivation below. For the following coordinate transformation and its inverse,

$$\begin{aligned} \tau &= t, & \xi &= x - at, \\ t &= \tau, & x &= \xi + a\tau, \end{aligned}$$

we write $\tilde{u}(\xi, \tau) = u(x, t)$ to indicate that $\tilde{u}$ is expressed in the new coordinates (ξ, τ) . Then the chain rule yields

$$\frac{\partial \tilde{u}}{\partial \tau} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{\partial x}{\partial \tau} = \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x},$$

which gives an ODE on $\tilde{u}$:

$$\frac{d\tilde{u}}{d\tau} = -b\tilde{u} + f(\xi + a\tau, \tau). \quad (14.62)$$

By Duhamel's principle (Theorem 12.15), we have

$$\tilde{u}(\xi, \tau) = u_0(\xi)e^{-b\tau} + \int_0^\tau f(\xi + a\sigma, \sigma)e^{-b(\tau-\sigma)} d\sigma.$$

Then the inverse transformation yields (14.61). $\square$

Lemma 14.55. The hyperbolic equation

$$u_t + a(x, t)u_x = f(x, t, u), \quad u(x, 0) = u_0(x) \quad (14.63)$$

is equivalent to the ODE system

$$\frac{d}{d\tau} \begin{bmatrix} x \\ \tilde{u} \end{bmatrix} = \begin{bmatrix} a(x(\tau), \tau) \\ f(x(\tau), \tau, \tilde{u}) \end{bmatrix}, \quad \begin{bmatrix} x(0) \\ \tilde{u}(\xi, 0) \end{bmatrix} = \begin{bmatrix} \xi \\ u_0(\xi) \end{bmatrix}, \quad (14.64)$$

where $\tau = t$ and $\tilde{u} = \tilde{u}(\xi, \tau)$.

Proof. For the new frame (ξ, τ) , we select ξ as the initial value of the ODE

$$\frac{dx}{d\tau} = a(x(\tau), \tau),$$

i.e., $x(0) = \xi$. Then the chain rule and the coordinate transformation $\tilde{u}(\xi, \tau) = u(x, t)$ yield

$$\frac{\partial \tilde{u}}{\partial \tau} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{\partial x}{\partial \tau} = \frac{\partial u}{\partial t} + a(x(\tau), \tau) \frac{\partial u}{\partial x}.$$

Then (14.63) gives another ODE

$$\frac{d\tilde{u}}{d\tau} = f(x(\tau), \tau, \tilde{u}),$$

which completes the proof. $\square$

Example 14.56. By Lemma 14.55, the PDE

$$u_t + xu_x = \alpha u, \quad u(x, 0) = \begin{cases} 1 & \text{if } x \in [0, 1]; \\ 0 & \text{otherwise} \end{cases} \quad (14.65)$$

with $-\alpha \in \mathbb{R}^+$ is solved by

$$u(x, t) = \begin{cases} e^{\alpha t} & \text{if } x \in [0, e^t]; \\ 0 & \text{otherwise.} \end{cases}$$

Definition 14.57. A system of PDEs

$$\mathbf{u}_t + A(x, t)\mathbf{u}_x + B(x, t)\mathbf{u} = F(x, t) \quad (14.66)$$

with $\mathbf{u}(x, 0) = \mathbf{u}_0(x)$ is *hyperbolic* if there is a matrix function $P(x, t)$ such that $P(x, t)AP(x, t)^{-1} = \Lambda(x, t)$ is diagonal with real eigenvalues and the matrix norms of $P(x, t)$ and $P(x, t)^{-1}$ are bounded for all $(x, t) \in \mathbb{R} \times [0, +\infty)$.

Example 14.58. The Euler equations are

$$\frac{\partial}{\partial t} \begin{bmatrix} p \\ u \end{bmatrix} + \begin{bmatrix} 0 & \kappa_0 \\ \frac{1}{\rho_0} & 0 \end{bmatrix} \frac{\partial}{\partial x} \begin{bmatrix} p \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (14.67)$$

The equation for the pressure p can be further written as

$$p_{tt} = a^2 p_{xx} \text{ with } a = \pm \sqrt{\kappa_0 / \rho_0}.$$

14.2.1 Classical MOLs

Example 14.59. Denote by $U_j(t) \approx u(x_j, t)$ the computed value of u at (x_j, t) and discretize the advection equation (14.57) in space at grid point x_j by

$$U'_j(t) = -\frac{a}{2h} (U_{j+1}(t) - U_{j-1}(t)). \quad (14.68)$$

For periodic boundary conditions we have

$$u(0, t) = u(1, t) \quad (14.69)$$

and thus $j = 1, 2, \dots, m+1$ in (14.68). In particular, the discretizations of (14.57) at $j = 1$ and $j = m+1$ are

$$U'_1(t) = -\frac{a}{2h} (U_2(t) - U_{m+1}(t)), \quad (14.70)$$

$$U'_{m+1}(t) = -\frac{a}{2h} (U_1(t) - U_m(t)). \quad (14.71)$$

Then the semi-discrete system can be written as

$$\mathbf{U}'(t) = A\mathbf{U}(t), \quad (14.72)$$

where $\mathbf{U}(t) = [U_1(t), U_2(t), \dots, U_{m+1}(t)]^\top$ and

$$A = -\frac{a}{2h} \begin{bmatrix} 0 & 1 & & & -1 \\ -1 & 0 & 1 & & \\ & -1 & 0 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 0 & 1 \\ 1 & & & & -1 & 0 \end{bmatrix}. \quad (14.73)$$

Lemma 14.60. The p th eigenvalue of A in (14.72) is

$$\lambda_p = -\frac{ia}{h} \sin(2\pi ph) \quad (14.74)$$

where $p = 1, 2, \dots, m+1$ and the corresponding eigenvector $\mathbf{w}^p$ has components

$$\forall j = 1, 2, \dots, m+1, \quad w_j^p = e^{2\pi i p j h}. \quad (14.75)$$

Proof. For $j = 2, 3, \dots, m$, we have

$$\begin{aligned} (A\mathbf{w}^p)_j &= -\frac{a}{2h} (w_{j+1}^p - w_{j-1}^p) \\ &= -\frac{a}{2h} e^{2\pi i p j h} (e^{2\pi i p h} - e^{-2\pi i p h}) \\ &= -\frac{ia}{h} \sin(2\pi ph) e^{2\pi i p j h} = \lambda_p w_j^p. \end{aligned} \quad (14.76)$$

By the periodicity and $h = \frac{1}{m+1}$, we have, for $j = 1$,

$$\begin{aligned} (A\mathbf{w}^p)_1 &= -\frac{a}{2h} (w_2^p - w_{m+1}^p) \\ &= -\frac{a}{2h} e^{2\pi i p h} (e^{2\pi i p h} - e^{2\pi i p m h}) \\ &= -\frac{a}{2h} e^{2\pi i p h} (e^{2\pi i p h} - e^{-2\pi i p h}) \\ &= -\frac{ia}{h} \sin(2\pi ph) e^{2\pi i p h} = \lambda_p w_1^p, \end{aligned} \quad (14.77)$$

where the third equality follows from $e^{2\pi i p m h} = e^{2\pi i p (1-h)}$ and $e^{2\pi i p} = 1$. Similar result holds for $j = m+1$. $\square$

The FTCS method

Notation 15. The *Courant number* for the advection equation (14.57) is

$$\mu := \frac{ak}{h}, \quad (14.78)$$

where k and h are respectively the uniform time-step size and the uniform grid size in an MOL.

Definition 14.61. The FTCS method for the advection equation (14.57) is

$$U_j^{n+1} = U_j^n - \frac{\mu}{2} (U_{j+1}^n - U_{j-1}^n), \quad (14.79)$$

or in matrix form,

$$\mathbf{U}^{n+1} = (I + kA)\mathbf{U}^n. \quad (14.80)$$

Corollary 14.62. The FTCS method for the advection equation (14.57) is unconditionally unstable with respect to absolute stability in Definition 14.23.

Proof. The RAS of the forward Euler's method is

$$\mathcal{R} = \{k\lambda_p \in \mathbb{C} : |1 + k\lambda_p| \leq 1 + O(k)\}.$$

For (14.80), we have

$$z_p := k\lambda_p = -i\mu \sin(2\pi ph),$$

which lies on the imaginary axis between $-i\mu$ and $i\mu$. Hence

$$\forall k = \Theta(h), \exists p \in \mathbb{N}^+ \text{ s.t. } |z_p| > \frac{\mu}{2} \text{ and } z_p = -i|z_p| \notin \mathcal{R},$$

where the Courant number $\mu = O(1)$. The instability for $k = \Theta(h)$ is illustrated in Figure 14.3. $\square$

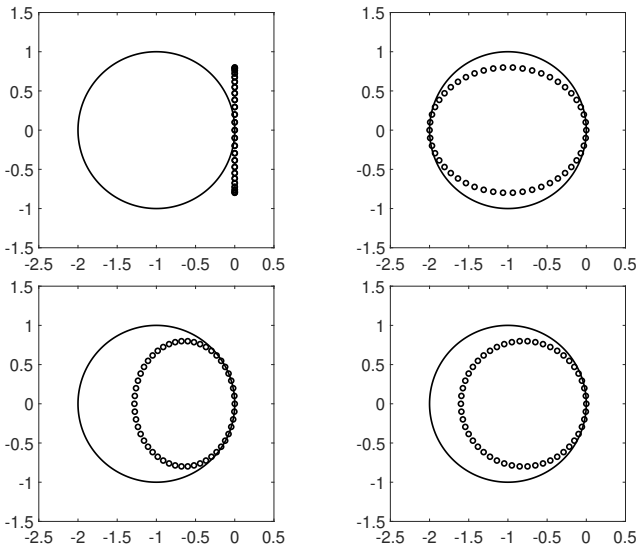


Figure 14.3: Stability of various methods for solving the advection equation (14.57) with $a = 1$, $h = \frac{1}{50}$, and the Courant number $\mu = \frac{ak}{h} = 0.8$. The top left and top right plots are for FTCS and Lax-Friedrichs, respectively, while the bottom left and bottom right plots are for Lax-Wendroff with $\epsilon = 0.008$ and the upwind method with $\epsilon = 0.01$, respectively. The circle represents the boundary of the bounded RAS of the forward Euler method while the hollow dots values of $z_p = k\lambda_p$ where λ_p is given by (14.74) for FTCS and by (14.86) otherwise.

Lemma 14.63. The FTCS method for the advection equation is Lax–Richtmyer stable for $k = O(h^2)$, cf. Definition 14.29.

Proof. Since λ_p is purely imaginary, we have

$$|1 + k\lambda_p|^2 = 1 + k \frac{k}{h^2} a^2 \sin^2(2\pi ph) \leq 1 + k\alpha,$$

for $\alpha = \frac{k}{h^2} a^2 = O(1)$. Since A is skew-symmetric, A is a normal operator as in Definition C.231. By the spectral theorem C.244, A has a diagonal matrix with respect to some orthonormal basis, so does $I + kA$. Thus we have

$$\|(I + kA)^n\|_2 \leq (\|I + kA\|_2^n)^{\frac{n}{2}} \leq (1 + k\alpha)^{\frac{n}{2}} \leq e^{\frac{\alpha}{2}T},$$

where the second inequality follows from the fact that A and $I + kA$ have the same eigenvectors. This shows the uniform boundedness of the iteration matrix needed for Lax–Richtmyer stability. $\square$

The leapfrog method

Definition 14.64. The *leapfrog method* for the advection equation (14.57) is

$$\frac{U_j^{n+1} - U_j^{n-1}}{2k} = -\frac{a}{2h} (U_{j+1}^n - U_{j-1}^n),$$

or, equivalently

$$U_j^{n+1} = U_j^{n-1} - \mu (U_{j+1}^n - U_{j-1}^n). \quad (14.81)$$

Lemma 14.65. The leapfrog method in Definition 14.65 is stable for $|\mu| \leq 1$.

Proof. By Example 12.86, the RAS of the midpoint method in Definition 12.21 is $\{z \in \mathbb{C} : \operatorname{Re}(z) = 0, |z| \leq 1\}$, which contains $z_p = k\lambda_p = -i\mu \sin(2\pi ph)$ for all p if $|\mu| \leq 1$. $\square$

Lax-Friedrichs

Definition 14.66. The *Lax-Friedrichs method* for the advection equation (14.57) is

$$U_j^{n+1} = \frac{1}{2} (U_{j+1}^n + U_{j-1}^n) - \frac{\mu}{2} (U_{j+1}^n - U_{j-1}^n). \quad (14.82)$$

Lemma 14.67. The advection-diffusion equation

$$u_t + au_x = \epsilon u_{xx} \quad (14.83)$$

with $\epsilon \in \mathbb{R}^+$ and the periodic boundary condition is discretized in space into the IVP system

$$\mathbf{U}'(t) = A_\epsilon \mathbf{U}(t) := (A + L_\epsilon) \mathbf{U}(t), \quad (14.84)$$

where A is given in (14.73) and

$$L_\epsilon = \frac{\epsilon}{h^2} \begin{bmatrix} -2 & 1 & & & 1 \\ 1 & -2 & 1 & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & -2 & 1 \\ 1 & & & & 1 & -2 \end{bmatrix}. \quad (14.85)$$

The p th eigenvalue of A_ϵ is

$$\lambda_p = -\frac{\mathbf{i}a}{h} \sin(2\pi ph) - \frac{2\epsilon}{h^2} [1 - \cos(2\pi ph)] \quad (14.86)$$

where $p = 1, 2, \dots, m+1$ while the corresponding eigenvector $\mathbf{w}^p$ has components

$$\forall j = 1, 2, \dots, m+1, \quad w_j^p = e^{2\pi i p j h}. \quad (14.87)$$

Proof. The conclusions follow from Lemma 14.60 and straightforward calculations. $\square$

Lemma 14.68. The Lax-Friedrichs method can be regarded as the MOL obtained by applying the forward Euler to the semi-discrete system (14.84) with $\epsilon = \frac{h^2}{2k}$.

Proof. The Lax-Friedrichs method can be rewritten as

$$U_j^{n+1} = U_j^n - \frac{\mu}{2} (U_{j+1}^n - U_{j-1}^n) + \frac{1}{2} (U_{j-1}^n - 2U_j^n + U_{j+1}^n),$$

which is equivalent to the standard MOL discretization of the advection-diffusion equation (14.83), i.e.,

$$\frac{U_j^{n+1} - U_j^n}{k} + a \left(\frac{U_{j+1}^n - U_{j-1}^n}{2h} \right) = \epsilon \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{h^2}.$$

$\square$

Theorem 14.69. The Lax-Friedrichs method (14.82) is convergent provided that $|\mu| \leq 1$.

Proof. By Lemma 14.68, we have

$$z_p = k\lambda_p = -\mathbf{i}\mu \sin(2\pi ph) - \frac{2k\epsilon}{h^2} [1 - \cos(2\pi ph)],$$

thus z_p 's lie on an ellipse centered at $\frac{-2k\epsilon}{h^2} = -1$ with semi-axes $(\frac{2k\epsilon}{h^2}, \mu) = (1, \mu)$. If $|\mu| \leq 1$, then this ellipse lies entirely inside the absolute region of stability of the forward Euler's method. By the Lax equivalence theorem 14.35, the Lax-Friedrichs method is convergent provided that $|\mu| \leq 1$. $\square$

Lax-Wendroff

Definition 14.70. The *Lax-Wendroff method* for the advection equation (14.57) is

$$U_j^{n+1} = U_j^n - \frac{\mu}{2} (U_{j+1}^n - U_{j-1}^n) + \frac{\mu^2}{2} (U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (14.88)$$

Lemma 14.71. The Lax-Wendroff method (14.88) is second-order accurate both in space and in time.

Proof. We calculate the LTE as

$$\begin{aligned} \tau(x, t) &= \frac{u(x, t+k) - u(x, t)}{k} + a \frac{u(x+h, t) - u(x-h, t)}{2h} \\ &\quad - \frac{ka^2}{2} \frac{u(x+h, t) - 2u(x, t) + u(x-h, t)}{h^2} \\ &= u_t(x, t) + \frac{k}{2} u_{tt}(x, t) + au_x(x, t) - \frac{ka^2}{2} u_{xx}(x, t) \\ &\quad + O(k^2 + h^2) \\ &= O(k^2 + h^2), \end{aligned}$$

where the first step follows from Definition 14.14, the second from Taylor expansions and the last from $u_t = -au_x$ and $u_{tt} = -au_{tx} = a^2 u_{xx}$. $\square$

Lemma 14.72. The Lax-Wendroff method (14.88) can be considered as the MOL obtained by applying the forward Euler to the semi-discrete system (14.84) with $\epsilon = \frac{1}{2}ka^2$.

Proof. The proof is similar to that for Lemma 14.68. $\square$

Theorem 14.73. The Lax-Wendroff method (14.88) is convergent provided $|\mu| \leq 1$.

Proof. By Lemma 14.72, we have

$$z_p = k\lambda_p = -\mathbf{i}\mu \sin(2\pi ph) + \mu^2 [\cos(2\pi ph) - 1].$$

These values all lie on an ellipse centered at $-\mu^2$ with semi-axes $(\mu^2, |\mu|)$. If $|\mu| \leq 1$, all of these values lie inside the stability region of the forward Euler's method, as shown in the bottom left plot of Figure 14.3. Thus the stability of the Lax-Wendroff method is ensured and the rest follows from the Lax equivalence theorem 14.35. $\square$

The upwind method

Definition 14.74. The *upwind method* for the advection equation (14.57) is

$$U_j^{n+1} = \begin{cases} U_j^n - \mu (U_j^n - U_{j-1}^n) & \text{if } a \geq 0; \\ U_j^n - \mu (U_{j+1}^n - U_j^n) & \text{if } a < 0. \end{cases} \quad (14.89)$$

Lemma 14.75. The upwind method (14.89) can be considered as the MOL obtained by applying the forward Euler to the semi-discrete system (14.84) with $\epsilon = \frac{h}{2}|a|$.

Proof. We only prove the case of $a > 0$. Then the upwind method can be rewritten as

$$U_j^{n+1} = U_j^n - \frac{\mu}{2} (U_{j+1}^n - U_{j-1}^n) + \frac{\mu}{2} (U_{j+1}^n - 2U_j^n + U_{j-1}^n),$$

which is the forward Euler's method applied to (14.84) with $\epsilon = \frac{ah}{2}$. $\square$

Theorem 14.76. For $a > 0$, the upwind method is convergent if and only if $\mu \leq 1$; for $a < 0$, the upwind method is convergent if and only if $\mu \geq -1$.

Proof. We only prove the case of $a > 0$. Lemmas 14.75 and 14.67 yield

$$z_p = k\lambda_p = -\mathbf{i}\mu \sin(2\pi ph) + \mu [\cos(2\pi ph) - 1].$$

These values all lie on a circle centered at $-\mu$ with radius μ . As shown in the bottom right plot of Figure 14.3, if $\mu \leq 1$, then all of these values lie inside the RAS of the forward Euler's method, thus ensuring the stability of the upwind method. For any choice of k, h satisfying $\mu > 1$, z_p would lie outside of the RAS and hence be unstable. $\square$

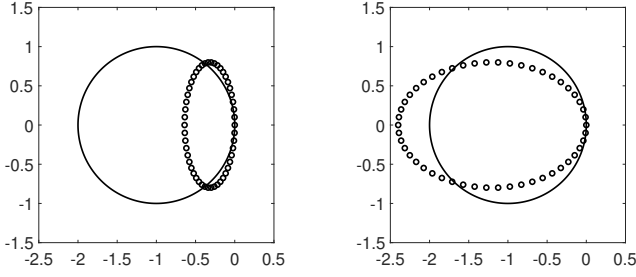


Figure 14.4: Out-of-range values of ϵ may lead to instability for the perturbed IVP system (14.84) with $a = 1$. The left and the right plots correspond to $\epsilon = 0.004$ and $\epsilon = 0.015$, respectively.

Corollary 14.77. The upwind method is equivalent to characteristic tracing followed by a linear interpolation.

Proof. As shown in Figure 14.5, for $\mu = 1$ the upwind method (14.89) reduces to

$$U_j^{n+1} = U_j^n - U_j^n + U_{j-1}^n = U_{j-1}^n,$$

which yields the exact solution by simply shifting the solution.

For $\mu < 1$, trace the characteristic and we have

$$u(x_j, t + k) = u(x_j - ak, t).$$

A linear interpolation of $u(x_j - ak, t)$ yields

$$u(x_j - ak, t) = \mu U_{j-1}^n + (1 - \mu) U_j^n + O(h^2),$$

which leads to the upwind method

$$U_j^{n+1} = \mu U_{j-1}^n + (1 - \mu) U_j^n = U_j^n - \mu (U_j^n - U_{j-1}^n). \quad \square$$

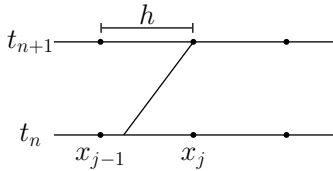


Figure 14.5: The upwind method is equivalent to characteristic tracing followed by a linear interpolation.

The Beam-Warming method

Definition 14.78. The *Beam-Warming method* solves the advection equation (14.57) by

$$U_j^{n+1} = U_j^n - \frac{\mu}{2} (3U_j^n - 4U_{j-1}^n + U_{j-2}^n) + \frac{\mu^2}{2} (U_j^n - 2U_{j-1}^n + U_{j-2}^n) \quad \text{if } a \geq 0; \quad (14.90a)$$

$$U_j^{n+1} = U_j^n - \frac{\mu}{2} (-3U_j^n + 4U_{j+1}^n - U_{j+2}^n) + \frac{\mu^2}{2} (U_j^n - 2U_{j+1}^n + U_{j+2}^n) \quad \text{if } a < 0. \quad (14.90b)$$

Exercise 14.79. Show that the Beam-Warming method is second-order accurate both in time and in space.

Exercise 14.80. Show that the Beam-Warming methods (14.90a) and (14.90b) are stable for $\mu \in [0, 2]$ and $\mu \in [-2, 0]$, respectively. Reproduce the plots in Figure 14.6.

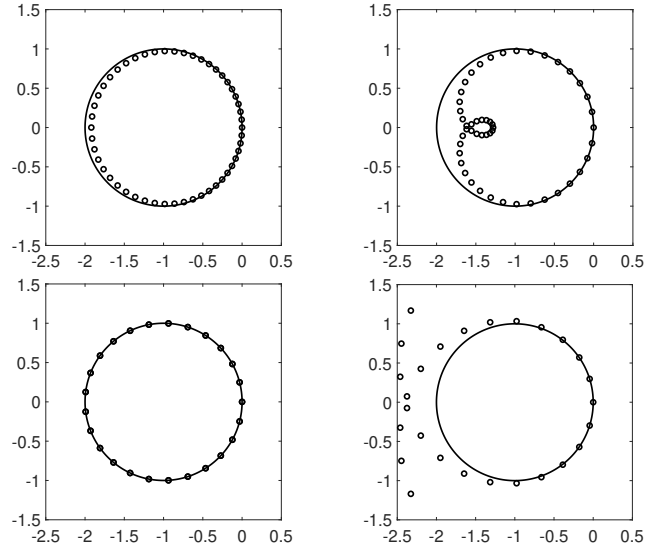


Figure 14.6: Absolute stability of the Beam-Warming method with $\mu = 0.8, 1.6, 2$, and 2.4 for the top left, top right, bottom left, and bottom right plots, respectively.

14.2.2 The CFL condition

Definition 14.81. For the advection equation (14.57), the *domain of dependence* of a point $(X, T) \in \Omega$ is

$$\mathcal{D}_{\text{ADV}}(X, T) = \{X - aT\}. \quad (14.91)$$

Definition 14.82. The *numerical domain of dependence* of a grid point (x_j, t_n) is the set of all grid points x_i such that U_i^0 at (x_i, t_0) affects U_j^n .

$$\mathcal{D}_N(x_j, t_n) = \{x_i : U_i^0 \text{ affects } U_j^n\}. \quad (14.92)$$

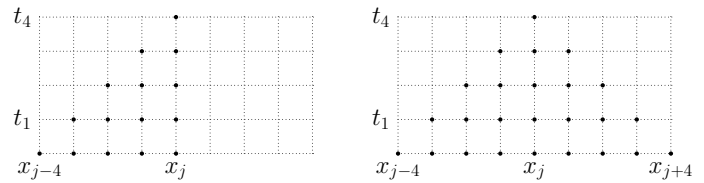


Figure 14.7: Examples of numerical domain of dependence. Solid dots on the left represent the numerical domain of dependence of a grid point (x_j, t_4) for the upwind method with $a > 0$ and $\mu \notin \{0, 1\}$ while solid dots on the right that of a grid point (x_j, t_4) for the Lax-Wendroff method with $\mu \notin \{0, 1, -1\}$.

Exercise 14.83. Plot the numerical domains of dependence of the grid point (x_j, t_3) for the upwind method with $a < 0$ and $\mu = 0, -1, -2$.

Exercise 14.84. Plot the numerical domains of dependence of the grid point (x_j, t_3) for the Lax-Wendroff method with $\mu = +1, -1$.

Theorem 14.85 (Courant-Friedrichs-Lewy). A numerical method can be convergent only if its numerical domain of dependence contains the domain of dependence of the PDE, at least in the limit of $k, h \rightarrow 0$.

Proof. It suffices to say that if some point p in the domain of dependence is not contained in the numerical domain of dependence, then we have no control over the value of p in the numerical method. Consequently, the numerical method cannot converge. $\square$

Example 14.86. For the advection equation with $a = 1$, any choice of $k > h$ results in instability. For $k = 3h$, the numerical domain of dependence for U_j^3 , as shown in the LHS of Figure 14.7, would be $\{x_j, x_{j-1}, x_{j-2}, x_{j-3}\}$ while the domain of dependence is the singleton set $\{x_j - 9h\}$. The former does not contain the latter and thus this choice of k will result in divergence.

Example 14.87. The heat equation

$$\begin{cases} u_t = \nu u_{xx} \\ u(x, 0) = \sqrt{\frac{\beta}{\pi}} e^{-\beta(x-\bar{x})^2}, \end{cases} \quad (14.93)$$

has its exact solution as

$$u(x, t) = \frac{1}{\sqrt{4\pi\nu t + \frac{\pi}{\beta}}} e^{-(x-\bar{x})^2/(4\nu t + \frac{1}{\beta})}. \quad (14.94)$$

The domain of dependence is the whole line, i.e.,

$$\mathcal{D}_{\text{DIFF}}(X, T) = (-\infty, +\infty), \quad (14.95)$$

because an initial point source $\lim_{\beta \rightarrow \infty} u(x, 0) = \delta(x - \bar{x})$ instantaneously affects each point on the real line:

$$\lim_{\beta \rightarrow \infty} u(x, t) = \frac{1}{\sqrt{4\pi\nu t}} e^{-\frac{(x-\bar{x})^2}{4\nu t}}.$$

This is very much an artifact of the mathematical model rather than the true physics.

14.2.3 Modified equations

Example 14.88. For the advection equation

$$u_t + u_x = 0 \quad (14.96)$$

with initial condition

$$u(x, 0) = \eta(x) = \exp(-20(x-2)^2) + \exp(-(x-5)^2), \quad (14.97)$$

the exact solution at $t = T$ is simply the initial data shifted by T . We solve this problem with $h = 0.05$ to $T = 17$ using the leapfrog method, the Lax-Friedrichs method, the Lax-Wendroff method, the upwind method, and the Beam-Warming method. The final results with $k = 0.8h$ and $k = h$ are shown in Figure 14.8, where the results invite a number of questions:

(a) Why are the solutions of Lax-Friedrichs and upwind much smoother than those of the other three methods?

(b) What caused the ripples in the solutions of the three methods in the right column?

(c) Why does the numerical solution of the leapfrog method contain more oscillations than that of the Lax-Wendroff method?

(d) For the Lax-Wendroff method, why do the ripples of numerical solutions lag behind the true crest?

(e) For the Beam-Warming method, why do the ripples of numerical solutions move ahead of the true crest?

(f) Why are numerical results with $k = h$ much better than those with $k = 0.8h$?

These questions concern the physics behind the different features of the results of different methods; they can be answered by the modified equations.

Definition 14.89. The *modified equation of an MOL* for solving a PDE (the original equation) is another PDE obtained from the formula of the MOL by

(1) replacing U_j^n in the MOL formula with a smooth function $v(x_j, t_n)$,

(2) dividing the equation by k , moving all terms to one side, expanding all terms in Taylor series at (x_j, t_n) , and expressing the second-order time derivative in terms of spatial derivatives,

(3) neglecting potentially high-order terms.

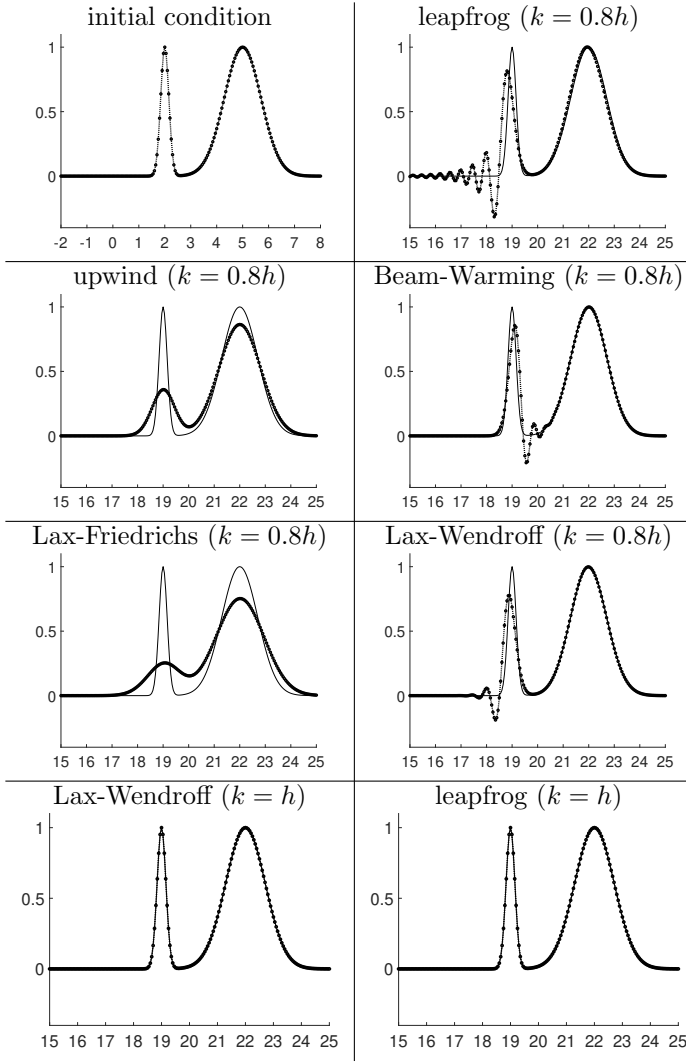


Figure 14.8: Accuracy of various methods in solving the advection equation (14.96) with initial condition (14.97) to $T = 17$ with $h = 0.05$. The thin solid line represents the exact solution while the circles connected by dotted lines the numerical results.

Example 14.90. Consider the upwind method for solving the advection equation

$$U_j^{n+1} = U_j^n - \mu (U_j^n - U_{j-1}^n).$$

The modified equation can be derived as follows.

(1) Replace U_j^n with $v(x_j, t_n)$ and we have

$$v(x, t + k) = v(x, t) - \mu (v(x, t) - v(x - h, t)).$$

(2) Expand all terms in Taylor series at (x, t) in a way similar to the calculation of the LTE.

$$\begin{aligned} 0 &= \frac{v(x, t + k) - v(x, t)}{k} + \frac{a}{h} (v(x, t) - v(x - h, t)) \\ &= \left(v_t + \frac{1}{2} k v_{tt} + \frac{1}{6} k^2 v_{ttt} + \cdots \right) \\ &\quad + a \left(v_x - \frac{1}{2} h v_{xx} + \frac{1}{6} h^2 v_{xxx} + \cdots \right), \end{aligned}$$

and thus

$$v_t + av_x = \frac{1}{2} (ahv_{xx} - kv_{tt}) - \frac{1}{6} (ah^2v_{xxx} + k^2v_{ttt}) + \cdots,$$

differentiating with respect to t and x gives

$$\begin{aligned} v_{tt} &= -av_{xt} + \frac{1}{2} (ahv_{xxt} - kv_{ttt}) + \cdots, \\ v_{tx} &= -av_{xx} + \frac{1}{2} (ahv_{xxx} - kv_{xtt}) + \cdots. \end{aligned}$$

Combining these gives

$$v_{tt} = a^2v_{xx} + O(h + k).$$

Therefore we have

$$v_t + av_x = \frac{1}{2} ah(1 - \mu)v_{xx} + O(h^2 + k^2),$$

(3) Neglect the high-order terms and we have the modified equation as

$$v_t + av_x = \frac{1}{2} ah(1 - \mu)v_{xx} := \beta v_{xx}, \quad (14.98)$$

which is satisfied better by the grid function than the advection equation $v_t + av_x = 0$.

Example 14.91. The modified equation of the Lax-Wendroff method for the advection equation is

$$v_t + av_x + \frac{ah^2}{6} (1 - \mu^2) v_{xxx} = 0, \quad (14.99)$$

which can be derived as follows.

First, replace U_j^n with $v(x, t)$ and we have

$$\begin{aligned} &\frac{v(x, t + k) - v(x, t)}{k} + \frac{a}{2h} (v(x + h, t) - v(x - h, t)) \\ &= \frac{a^2k}{2h^2} (v(x + h, t) - 2v(x, t) + v(x - h, t)). \end{aligned}$$

Second, we expand all terms in Taylor series about (x, t) to derive the modified equation as (for brevity, omit the dependence on (x, t))

$$v_t + \frac{1}{2} kv_{tt} + \frac{1}{6} k^2 v_{ttt} + a \left(v_x + \frac{h^2}{6} v_{xxx} \right) = \frac{k}{2} a^2 v_{xx} + O(k^3),$$

where we have assumed $h = O(k)$. It follows that

$$v_t + av_x = -\frac{1}{2} k (v_{tt} - a^2 v_{xx}) - \frac{1}{6} k^2 v_{ttt} - \frac{ah^2}{6} v_{xxx} + O(k^3).$$

Differentiate the above equation and we have

$$\begin{aligned} v_{tt} &= -av_{xt} - \frac{1}{2} kv_{ttt} + \frac{k}{2} a^2 v_{xxt} + O(k^2), \\ v_{xt} &= -av_{xx} - \frac{k}{2} v_{xtt} + \frac{k}{2} a^2 v_{xxx} + O(k^2), \\ v_{xxt} &= -av_{xxx} + O(k), \\ v_{xtt} &= -av_{xxt} + O(k) = a^2 v_{xxx} + O(k), \\ v_{ttt} &= -av_{xtt} + O(k) = -a^3 v_{xxx} + O(k). \end{aligned}$$

Combining the above equations gives

$$v_{tt} = a^2 v_{xx} + \frac{ka}{2}(v_{xtt} - a^2 v_{xxx}) + \frac{ka^3}{2}v_{xxx} - \frac{ka^3}{2}v_{xxx} + O(k^2) \\ = a^2 v_{xx} + O(k^2).$$

Hence we have

$$v_t + av_x = \frac{k^2}{6}a^3 v_{xxx} - \frac{ah^2}{6}v_{xxx} + O(k^3),$$

which implies (14.99).

Example 14.92. By Lemma F.38, the solution to the modified equation (14.99) is

$$v(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{v}(\xi) e^{i\xi(x - C_p t)} d\xi.$$

For Lax-Wendroff, (14.99) and Example F.52 yield

$$C_p(\xi) = a - \frac{ah^2}{6}(1 - \mu^2)\xi^2, \\ C_g(\xi) = a - \frac{ah^2}{2}(1 - \mu^2)\xi^2.$$

First, the phase velocity $C_p \neq a$ for $\mu \neq 1$, and its value depends on ξ ; this answers Question (b) of Example 14.88. For $\mu \neq 1$, both C_p and C_g have a magnitude smaller than $|a|$, hence numerical oscillations lag behind the true wave crest; this answers Question (d) of Example 14.88.

Exercise 14.93. Show that the modified equation of the leapfrog method is also (14.99). However, if one more term of higher-order derivative had been retained, the modified equation of the leapfrog method would have been

$$v_t + av_x + \frac{ah^2}{6}(1 - \mu^2)v_{xxx} = \epsilon_f v_{xxxxx} \quad (14.100)$$

while that of the Lax-Wendroff method would have been

$$v_t + av_x + \frac{ah^2}{6}(1 - \mu^2)v_{xxx} = \epsilon_w v_{xxxx}. \quad (14.101)$$

Exercise 14.94. Show that the modified equation of the Beam-Warming method applied to the advection equation (14.57) with $a \geq 0$ is

$$v_t + av_x + \frac{ah^2}{6}(-2 + 3\mu - \mu^2)v_{xxx} = 0. \quad (14.102)$$

Thus we have

$$C_p(\xi) = a + \frac{ah^2}{6}(\mu - 1)(\mu - 2)\xi^2, \\ C_g(\xi) = a + \frac{ah^2}{2}(\mu - 1)(\mu - 2)\xi^2.$$

How do these facts answer Question (e) of Example 14.88?

Exercise 14.95. What if $\mu = 1$? Discuss this case for both Lax-Wendroff and leapfrog methods to answer Question (f) of Example 14.88.

14.2.4 Von Neumann analysis

Example 14.96. For the advection equation (14.57) with $a \geq 0$, the von Neumann analysis of the upwind method yields its amplification factor as

$$g(\xi h) = (1 - \mu) + \mu e^{-i\xi h}. \quad (14.103)$$

Indeed, substituting $U_j^n = [g(\xi h)]^n e^{i\xi jh}$ into the upwind method

$$U_j^{n+1} = U_j^n - \mu(U_j^n - U_{j-1}^n)$$

gives

$$g(\xi h) = 1 - \mu(1 - e^{-i\xi h}) = (1 - \mu) + \mu e^{-i\xi h}.$$

Therefore

$$|g(\xi h)|^2 = (1 - \mu + \mu \cos(\xi h))^2 + \mu^2 \sin^2(\xi h) \\ = 1 + 2\mu(\mu - 1)(1 - \cos(\xi h)),$$

and hence the method is stable ($|g(\xi h)| \leq 1$) provided $\mu \leq 1$.

Exercise 14.97. Apply the von Neumann analysis to the Lax-Friedrichs method to derive its amplification factor as

$$g(\xi h) = \cos(\xi h) - \mu i \sin(\xi h). \quad (14.104)$$

For which values of μ would the method be stable?

Exercise 14.98. Apply the von Neumann analysis to the Lax-Wendroff method to derive its amplification factor as

$$g(\xi h) = 1 - 2\mu^2 \sin^2 \frac{\xi h}{2} - i\mu \sin(\xi h). \quad (14.105)$$

For which values of μ would the method be stable?

Example 14.99. When performing the analysis of modified equations, we typically neglect some higher-order terms of ξh in deriving the group velocity and the phase velocity. For ξh sufficiently small, the modified equation would be a reasonable model. However, for large ξh the terms we have neglected may play an equally important role. In this case it might be better to use an approach similar to von Neumann analysis by setting

$$v(x_j, t_n) := e^{i(\xi x_j - \omega t_n)}. \quad (14.106)$$

For the leapfrog method, this form yields

$$\sin(\omega k) = \mu \sin(\xi h), \quad (14.107)$$

which yields the group velocity as

$$\frac{d\omega}{d\xi} = \pm \frac{a \cos(\xi h)}{\sqrt{1 - \mu^2 \sin^2(\xi h)}}, \quad (14.108)$$

where the $\pm$ follows from the multivalued dispersion relation (14.107). For high-frequency modes satisfying $\xi h \approx \pi$, the group velocity may have a sign different from that of a .

14.3 Programming assignments

14.3.1 MOL for the heat equation

Write subroutines to solve the test problem in Example 14.39. More specifically, you need to

- (a) reproduce all plots in Example 14.39,
- (b) rerun your subroutine for BTCS and the collocation method in Example 13.96 (coupled with centered difference in space) with $r = \frac{1}{2h}$, show their results after 1, 2, and 10 steps,

- (c) write a subroutine for FTCS and show its results after 1, 2, and 10 steps with $r = \frac{1}{2}, 1$.
- (d) write a subroutine for the 1-stage Gauss-Legendre RK method in Example 13.73 (coupled with centered difference in space) and show its results after 1, 2, and 10 steps with $r = 1, \frac{1}{2h}$.
- (e) collect the results, compare them, and explain your understanding in your own words in a report.

14.3.2 MOL for the advection equation

Reproduce all results in Example 14.88.

Chapter 15

Finite Volume (FV) Methods

Notation 16. The rectangular problem domain Ω is discretized into a collection of square control volumes or cells, each indexed by a multi-index $\mathbf{i} \in \mathbb{Z}^D$ and its region denoted by

$$\mathcal{C}_{\mathbf{i}} = [\mathbf{x}_O + \mathbf{i}h, \mathbf{x}_O + (\mathbf{i} + \mathbf{1})h], \quad (15.1)$$

where $\mathbf{x}_O \in \mathbb{R}^D$ is some fixed origin of the coordinates, h the uniform mesh spacing, and $\mathbf{1} \in \mathbb{Z}^D$ the multi-index with all its components equal to one. The region of the higher face of $\mathcal{C}_{\mathbf{i}}$ in dimension d is denoted by

$$\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = [\mathbf{x}_O + (\mathbf{i} + \mathbf{e}^d)h, \mathbf{x}_O + (\mathbf{i} + \mathbf{1})h], \quad (15.2)$$

where $\mathbf{e}^d \in \mathbb{Z}^D$ a multi-index with 1 as its d -th component and 0 otherwise.

Exercise 15.1. Denote the interval of cell $\mathbf{i}$ along the d th dimension by

$$I_{\mathbf{i}}^d := x_{O,d} + [i_d h, (i_d + 1)h], \quad (15.3)$$

where $x_{O,d}$ is the d th component of $\mathbf{x}_O$ in Notation 16. Use (15.1) and (15.2) to show that the region of cell $\mathbf{i}$ is the Cartesian product of its lower face in the d th dimension and the interval $I_{\mathbf{i}}^d$, i.e.,

$$\mathcal{C}_{\mathbf{i}} = \mathcal{F}_{\mathbf{i}-\frac{1}{2}\mathbf{e}^d} \times I_{\mathbf{i}}^d. \quad (15.4)$$

15.1 The FV formulation

15.1.1 Averaged values

Definition 15.2. The *cell average* of a function $\phi : \mathbb{R}^D \rightarrow \mathbb{R}$ over the cell $\mathcal{C}_{\mathbf{i}}$ is a function $\mathbb{Z}^D \rightarrow \mathbb{R}$ given by

$$\langle \phi \rangle_{\mathbf{i}} = \frac{1}{h^D} \int_{\mathcal{C}_{\mathbf{i}}} \phi(\mathbf{x}) \, d\mathbf{x}. \quad (15.5)$$

Definition 15.3. The *face average* of a function $\phi : \mathbb{R}^D \rightarrow \mathbb{R}$ over the face $\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}$ is a function $\mathbb{Z}^D \rightarrow \mathbb{R}$ given by

$$\langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = \frac{1}{h^{D-1}} \int_{\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}} \phi(\mathbf{x}) \, d\mathbf{x}. \quad (15.6)$$

Notation 17. The three different types of quantities, viz. point values, cell averages, and face averages, are strictly distinguished in finite volume discretization. As shown in Figure 15.1, a symbol without the averaging operator $\langle \cdot \rangle$ denotes a point value; otherwise it is either a cell-averaged value if the subscript is an integer multi-index, or a face-averaged value if the subscript is a fractional multi-index.

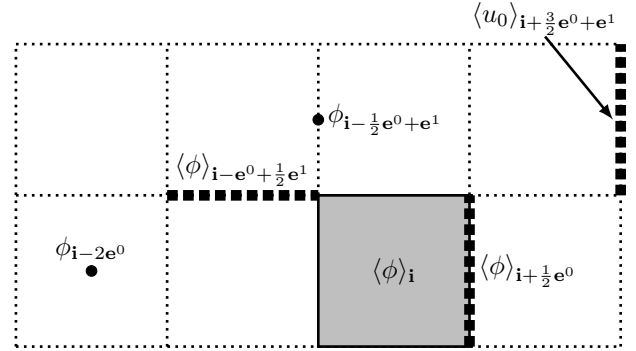


Figure 15.1: Point values, cell-averaged values, and face-averaged values in finite volume discretizations. $\phi_{\mathbf{i}-2\mathbf{e}^0}$, and $\phi_{\mathbf{i}-\frac{1}{2}\mathbf{e}^0+\mathbf{e}^1}$ are point values; $\langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^0}$, $\langle \phi \rangle_{\mathbf{i}-\mathbf{e}^0+\frac{1}{2}\mathbf{e}^1}$, and $\langle u_0 \rangle_{\mathbf{i}+\frac{3}{2}\mathbf{e}^0+\mathbf{e}^1}$ are face-averaged; $\langle \phi \rangle_{\mathbf{i}}$ is cell-averaged. Horizontal and vertical bold dotted lines represent the averaging processes over a vertical cell face and a horizontal cell face, respectively. Light gray area represents averaging over a cell.

Lemma 15.4. Point values can be converted to face averages and cell averages to the fourth-order accuracy via

$$\langle \phi \rangle_{\mathbf{i}} = \phi_{\mathbf{i}} + \frac{h^2}{24} \sum_{d=1}^D \frac{\partial^2 \phi(\mathbf{x})}{\partial x_d^2} \bigg|_{\mathbf{i}} + O(h^4), \quad (15.7)$$

$$\langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = \phi_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + \frac{h^2}{24} \sum_{d' \neq d} \frac{\partial^2 \phi(\mathbf{x})}{\partial x_{d'}^2} \bigg|_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + O(h^4). \quad (15.8)$$

Proof. The above identities follow from Taylor expansions of the integrands in (15.5) and (15.6). $\square$

Theorem 15.5. Cell averages can be converted to face averages to the fourth-order accuracy via

$$\langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = \frac{7}{12} \left(\langle \phi \rangle_{\mathbf{i}} + \langle \phi \rangle_{\mathbf{i}+\mathbf{e}^d} \right) - \frac{1}{12} \left(\langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} + \langle \phi \rangle_{\mathbf{i}+2\mathbf{e}^d} \right) + O(h^4), \quad (15.9)$$

$$\left\langle \frac{\partial \phi}{\partial x_d} \right\rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = \frac{15}{12h} \left(\langle \phi \rangle_{\mathbf{i}+\mathbf{e}^d} - \langle \phi \rangle_{\mathbf{i}} \right) - \frac{1}{12h} \left(\langle \phi \rangle_{\mathbf{i}+2\mathbf{e}^d} - \langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} \right) + O(h^4). \quad (15.10)$$

Proof. We prove (15.9) and (15.10) via the first fundamental theorem of calculus (Theorem B.127). Define an indefinite integral $\Phi^d : \mathbb{R}^D \rightarrow \mathbb{R}$ of ϕ along the d th axis as

$$\Phi^d(\mathbf{x}) := \int_{\xi}^{x_d} \phi(x_1, \dots, x_{d-1}, \eta, x_{d+1}, \dots, x_D) d\eta, \quad (15.11)$$

where the lower limit $\xi \in \mathbb{R}$ is fixed. Theorem B.127 implies

$$\phi(\mathbf{x}) = \frac{\partial \Phi^d}{\partial x_d}. \quad (15.12)$$

For the first dimension, define a function $\delta_{\mathbf{i}} : \mathbb{R}^{D-1} \rightarrow \mathbb{R}$ to represent the integral of ϕ over the interval $I_{\mathbf{i}}^1$ in (15.3):

$$\begin{aligned} \varphi_j &:= \Phi^1(x_{O,1} + jh, x_2, \dots, x_D); \\ \delta_{\mathbf{i}}(x_2, \dots, x_D) &:= \varphi_{i_1+1} - \varphi_{i_1}. \end{aligned} \quad (15.13)$$

Taylor expansions of φ_{i_1+2} , φ_{i_1} , φ_{i_1+3} , φ_{i_1-1} in the variable x_1 at $\bar{x} := x_{O,1} + (i_1 + 1)h$ yield

$$\begin{aligned} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ 2 & 4 & 8 & 16 \\ -2 & 4 & -8 & 16 \end{bmatrix} \begin{bmatrix} h & & & \\ & \frac{h^2}{2} & & \\ & & \frac{h^3}{6} & \\ & & & \frac{h^4}{24} \end{bmatrix} \begin{bmatrix} \frac{\partial \Phi^1}{\partial x_1} \\ \frac{\partial^2 \Phi^1}{\partial x_1^2} \\ \frac{\partial^3 \Phi^1}{\partial x_1^3} \\ \frac{\partial^4 \Phi^1}{\partial x_1^4} \end{bmatrix}_{x_1=\bar{x}} \\ &= \begin{bmatrix} \delta_{\mathbf{i}+\mathbf{e}^1} \\ -\delta_{\mathbf{i}} \\ \delta_{\mathbf{i}+\mathbf{e}^1} + \delta_{\mathbf{i}+2\mathbf{e}^1} \\ -\delta_{\mathbf{i}-\mathbf{e}^1} - \delta_{\mathbf{i}} \end{bmatrix} + O(h^5). \end{aligned} \quad (15.14)$$

Then (15.12) yields

$$\begin{bmatrix} h\phi \\ h^2 \frac{\partial \phi}{\partial x_1} \\ h^3 \frac{\partial^2 \phi}{\partial x_1^2} \\ h^4 \frac{\partial^3 \phi}{\partial x_1^3} \end{bmatrix}_{x_1=\bar{x}} = \mathbf{C} \begin{bmatrix} \delta_{\mathbf{i}+\mathbf{e}^1} \\ -\delta_{\mathbf{i}} \\ \delta_{\mathbf{i}+\mathbf{e}^1} + \delta_{\mathbf{i}+2\mathbf{e}^1} \\ -\delta_{\mathbf{i}-\mathbf{e}^1} - \delta_{\mathbf{i}} \end{bmatrix} + O(h^5), \quad (15.15)$$

where

$$\mathbf{C} = \begin{bmatrix} \frac{2}{3} & -\frac{2}{3} & -\frac{1}{12} & \frac{1}{12} \\ \frac{4}{3} & \frac{4}{3} & -\frac{1}{12} & -\frac{1}{12} \\ -1 & 1 & \frac{1}{2} & -\frac{1}{2} \\ -4 & -4 & 1 & 1 \end{bmatrix}.$$

Construct an auxiliary matrix

$$\mathbf{M} = \begin{bmatrix} 1 & & & \\ & -1 & & \\ -1 & & 1 & \\ & 1 & & -1 \end{bmatrix}, \quad (15.16)$$

add $\mathbf{M}^{-1}\mathbf{M}$ into the middle of the RHS of (15.15), and we have

$$\begin{bmatrix} \phi \\ h \frac{\partial \phi}{\partial x_1} \\ h^2 \frac{\partial^2 \phi}{\partial x_1^2} \\ h^3 \frac{\partial^3 \phi}{\partial x_1^3} \end{bmatrix}_{x_1=\bar{x}} = \frac{1}{h} \mathbf{T}^{(4)} \begin{bmatrix} \delta_{\mathbf{i}+\mathbf{e}^1} \\ \delta_{\mathbf{i}} \\ \delta_{\mathbf{i}+2\mathbf{e}^1} \\ \delta_{\mathbf{i}-\mathbf{e}^1} \end{bmatrix} + O(h^4), \quad (15.17)$$

where the fourth-order interpolation matrix $\mathbf{T}^{(4)}$ is

$$\mathbf{T}^{(4)} = \begin{bmatrix} \frac{7}{12} & \frac{7}{12} & -\frac{1}{12} & -\frac{1}{12} \\ \frac{5}{4} & -\frac{5}{4} & -\frac{1}{12} & \frac{1}{12} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -3 & 3 & 1 & -1 \end{bmatrix}. \quad (15.18)$$

Integrating the first row of (15.17) over $\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}$ yields (15.9) with $d = 1$, because (15.4) implies that, for any $j \in \mathbb{Z}$,

$$\begin{aligned} & \frac{1}{h^D} \int_{\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^1}} \delta_{\mathbf{i}+j\mathbf{e}^1} \\ &= \frac{1}{h^D} \int_{\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^1}} \int_{(i_1+j)h}^{(i_1+j+1)h} \phi(\eta, x_2, \dots, x_D) d\eta dx_2 \cdots dx_D \\ &= \frac{1}{h^D} \int_{\mathcal{C}_{\mathbf{i}+j\mathbf{e}^1}} \phi = \langle \phi \rangle_{\mathbf{i}+j\mathbf{e}^1}. \end{aligned}$$

The proof of (15.9) is completed by a repetition of the above arguments on the first dimension to all other dimensions. (15.10) can be proved by repeating the above procedures to obtain a fifth-order counterpart of (15.17) and then integrating its second equation; see Exercise 15.6. $\square$

Exercise 15.6. Repeat the procedures in the above proof to generate the fifth-order interpolation matrix

$$\mathbf{T}^{(5)} = \begin{bmatrix} \frac{47}{60} & \frac{9}{20} & -\frac{13}{60} & -\frac{1}{20} & \frac{1}{30} \\ \frac{5}{4} & -\frac{5}{4} & -\frac{1}{12} & \frac{1}{12} & 0 \\ -2 & \frac{1}{2} & \frac{3}{2} & \frac{1}{4} & -\frac{1}{4} \\ -3 & 3 & 1 & -1 & 0 \\ 6 & -4 & -4 & 1 & 1 \end{bmatrix}, \quad (15.19)$$

where the additional column is associated with $\langle \phi \rangle_{\mathbf{i}+3}$. Explain why the formulas of the fourth order and the fifth order coincide for $\frac{\partial \phi}{\partial x}$.

15.1.2 Discrete operators

Definition 15.7. The *discrete gradient*, the *discrete divergence*, and the *discrete Laplacian* are defined as

$$\mathbf{G}_d \langle \phi \rangle_{\mathbf{i}} = \frac{1}{12h} \left(-\langle \phi \rangle_{\mathbf{i}+2\mathbf{e}^d} + 8\langle \phi \rangle_{\mathbf{i}+\mathbf{e}^d} - 8\langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} + \langle \phi \rangle_{\mathbf{i}-2\mathbf{e}^d} \right), \quad (15.20)$$

$$\mathbf{D} \langle \mathbf{u} \rangle_{\mathbf{i}} = \frac{1}{12h} \sum_d \left(-\langle u_d \rangle_{\mathbf{i}+2\mathbf{e}^d} + 8\langle u_d \rangle_{\mathbf{i}+\mathbf{e}^d} - 8\langle u_d \rangle_{\mathbf{i}-\mathbf{e}^d} + \langle u_d \rangle_{\mathbf{i}-2\mathbf{e}^d} \right), \quad (15.21)$$

$$\mathbf{L} \langle \phi \rangle_{\mathbf{i}} = \frac{1}{12h^2} \sum_d \left(-\langle \phi \rangle_{\mathbf{i}+2\mathbf{e}^d} + 16\langle \phi \rangle_{\mathbf{i}+\mathbf{e}^d} - 30\langle \phi \rangle_{\mathbf{i}} + 16\langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} - \langle \phi \rangle_{\mathbf{i}-2\mathbf{e}^d} \right). \quad (15.22)$$

The discrete divergence operator also acts on tensor averages,

$$\mathbf{D} \langle \mathbf{u}\mathbf{u} \rangle_{\mathbf{i}} = \frac{1}{h} \sum_d \left(\mathbf{F} \langle u_d, \mathbf{u} \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} - \mathbf{F} \langle u_d, \mathbf{u} \rangle_{\mathbf{i}-\frac{1}{2}\mathbf{e}^d} \right), \quad (15.23)$$

where the discrete face average of the product of two scalar functions is

$$\begin{aligned} \mathbf{F} \langle \phi, \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} &= \langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \langle \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \\ &+ \frac{h^2}{12} \sum_{d' \neq d} (\mathbf{G}_{d'}^\perp \phi)_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} (\mathbf{G}_{d'}^\perp \psi)_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}, \end{aligned} \quad (15.24)$$

and $\mathbf{G}_{d'}^\perp$ is the discrete gradient operator in the transverse directions,

$$(\mathbf{G}_{d'}^\perp \phi)_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = \frac{1}{2h} \left(\langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d+\mathbf{e}^{d'}} - \langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d-\mathbf{e}^{d'}} \right). \quad (15.25)$$

Lemma 15.8. The operators in Definition 15.7 are fourth-order accurate, i.e.,

$$\mathbf{G}_d \langle \phi \rangle_{\mathbf{i}} = \frac{1}{h^D} \int_{C_i} \frac{\partial \phi}{\partial x_d} + O(h^4), \quad (15.26a)$$

$$\mathbf{D} \langle \mathbf{u} \rangle_{\mathbf{i}} = \frac{1}{h^D} \int_{C_i} \nabla \cdot \mathbf{u} + O(h^4), \quad (15.26b)$$

$$\mathbf{L} \langle \phi \rangle_{\mathbf{i}} = \frac{1}{h^D} \int_{C_i} \nabla^2 \phi + O(h^4), \quad (15.26c)$$

$$\mathbf{D} \langle \mathbf{u} \rangle_{\mathbf{i}} = O(h^4) \Rightarrow \mathbf{D} \langle \mathbf{u}\mathbf{u} \rangle_{\mathbf{i}} = \frac{1}{h^D} \int_{C_i} (\mathbf{u} \cdot \nabla) \mathbf{u} + O(h^4), \quad (15.26d)$$

$$\mathbf{F} \langle \phi, \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} = \frac{1}{h^{D-1}} \int_{\mathcal{F}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}} \phi \psi + O(h^4). \quad (15.26e)$$

Proof. (15.26a) follows from (15.9), the second fundamental theorem of calculus, and the fact that $\frac{\partial}{\partial x_d}$ commutes with $\langle \cdot \rangle$. (15.26b) follows from the divergence theorem and (15.9).

(15.26c) follows from the divergence theorem and (15.10). The rest of the proof concerns (15.26d) and (15.26e).

Denote the cell center of C_i by $\mathbf{x}_i = (\mathbf{i} + \frac{1}{2}\mathbf{1})h$, and the face centers by $\mathbf{x}_{i \pm \frac{1}{2}\mathbf{e}^d} = \mathbf{x}_i \pm \frac{h}{2}\mathbf{e}^d$. Let $\mathbf{x}_c = \mathbf{x}_{i+\frac{1}{2}\mathbf{e}^d}$ be the center of $\mathcal{F}_{i+\frac{1}{2}\mathbf{e}^d}$. Then the Taylor series of a function ϕ about $\mathbf{x}_c$ can be expressed in multi-index notation as

$$\begin{aligned} \phi(\mathbf{x}) &= \sum_{|\mathbf{j}| \leq 3} \frac{1}{\mathbf{j}!} (\mathbf{x} - \mathbf{x}_c)^{\mathbf{j}} \phi^{(\mathbf{j})}(\mathbf{x}_c) + O(h^4) \\ &= \sum_{|\mathbf{j}| \leq 3} \frac{1}{\mathbf{j}!} \boldsymbol{\eta}^{\mathbf{j}} \phi^{(\mathbf{j})}(\mathbf{x}_c) + O(h^4), \end{aligned} \quad (15.27)$$

where $\boldsymbol{\eta} = \mathbf{x} - \mathbf{x}_c$, so that $\eta_d = 0$ and $|\boldsymbol{\eta}| \approx O(h)$ on $\mathcal{F}_{i+\frac{1}{2}\mathbf{e}^d}$.

Then the product of two functions $\phi, \psi : \mathbb{R}^D \rightarrow \mathbb{R}$ is

$$\begin{aligned} \phi(\mathbf{x})\psi(\mathbf{x}) &= \left(\sum_{|\mathbf{j}| \leq 3} \frac{1}{\mathbf{j}!} \boldsymbol{\eta}^{\mathbf{j}} \phi^{(\mathbf{j})}(\mathbf{x}_c) \right) \left(\sum_{|\mathbf{k}| \leq 3} \frac{1}{\mathbf{k}!} \boldsymbol{\eta}^{\mathbf{k}} \psi^{(\mathbf{k})}(\mathbf{x}_c) \right) \\ &+ O(h^4) \\ &= \sum_{\mathbf{k}: |\mathbf{k}| \leq 3} \frac{1}{\mathbf{k}!} \boldsymbol{\eta}^{\mathbf{k}} \sum_{\mathbf{j}: \mathbf{j} \leq \mathbf{k}} \binom{\mathbf{k}}{\mathbf{j}} \phi^{(\mathbf{j})}(\mathbf{x}_c) \psi^{(\mathbf{k}-\mathbf{j})}(\mathbf{x}_c) \\ &+ O(h^4), \end{aligned}$$

where the second step follows from a variable substitution. Then the average over $\mathcal{F}_{i+\frac{1}{2}\mathbf{e}^d}$ (dropping indices on $\mathcal{F}$ and evaluation at $\mathbf{x}_c$) is

$$\begin{aligned} &\frac{1}{h^{D-1}} \int_{\mathcal{F}} \phi \psi \, d\mathbf{x} \\ &= \frac{1}{h^{D-1}} \int_{\mathcal{F}} \sum_{\mathbf{k}: |\mathbf{k}| \leq 3} \frac{1}{\mathbf{k}!} \boldsymbol{\eta}^{\mathbf{k}} \sum_{\mathbf{j}: \mathbf{j} \leq \mathbf{k}} \binom{\mathbf{k}}{\mathbf{j}} \phi^{(\mathbf{j})} \psi^{(\mathbf{k}-\mathbf{j})} \, d\mathbf{x} + O(h^4) \\ &= \sum_{\mathbf{k}: |\mathbf{k}| \leq 3} \frac{1}{\mathbf{k}!} \left(\frac{1}{h^{D-1}} \int_{\mathcal{F}} \boldsymbol{\eta}^{\mathbf{k}} \, d\mathbf{x} \right) \sum_{\mathbf{j}: \mathbf{j} \leq \mathbf{k}} \binom{\mathbf{k}}{\mathbf{j}} \phi^{(\mathbf{j})} \psi^{(\mathbf{k}-\mathbf{j})} + O(h^4). \end{aligned}$$

If $k_d \neq 0$ or $\mathbf{k}$ is odd in any component, then $\int_{\mathcal{F}} \boldsymbol{\eta}^{\mathbf{k}} \, d\mathbf{x} = 0$. Hence, the only nonzero terms come from the two cases $\mathbf{k} = \mathbf{0}, \mathbf{j} = \mathbf{0}$ and $\mathbf{k} = 2\mathbf{e}^{d'}, \mathbf{j} = \mathbf{0}, \mathbf{e}^{d'}, 2\mathbf{e}^{d'}$ with $d' \neq d$. Thus,

$$\begin{aligned} &\frac{1}{h^{D-1}} \int_{\mathcal{F}} \phi \psi \, d\mathbf{x} \\ &= \phi \psi + \frac{h^2}{24} \sum_{d' \neq d} \left(\phi^{(2\mathbf{e}^{d'})} \psi + \psi^{(2\mathbf{e}^{d'})} \phi \right) \\ &+ \frac{h^2}{12} \sum_{d' \neq d} \left(\phi^{(\mathbf{e}^{d'})} \psi^{(\mathbf{e}^{d'})} \right) + O(h^4) \\ &= \left(\phi + \frac{h^2}{24} \sum_{d' \neq d} \phi^{(2\mathbf{e}^{d'})} \right) \left(\psi + \frac{h^2}{24} \sum_{d' \neq d} \psi^{(2\mathbf{e}^{d'})} \right) \\ &+ \frac{h^2}{12} \sum_{d' \neq d} \left(\phi^{(\mathbf{e}^{d'})} \psi^{(\mathbf{e}^{d'})} \right) + O(h^4) \\ &= \langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \langle \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + \frac{h^2}{12} \sum_{d' \neq d} \left(\phi^{(\mathbf{e}^{d'})} \psi^{(\mathbf{e}^{d'})} \right) + O(h^4), \end{aligned}$$

where we have used (15.8) to convert the first two terms in parentheses to face averages. The last term representing the product of transverse gradients can be approximated with

$$\begin{aligned} \mathbf{G}_{d'}^\perp \phi|_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} &= \frac{1}{2h} \left(\langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d+\mathbf{e}^{d'}} - \langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d-\mathbf{e}^{d'}} \right) \\ &= \frac{\partial \phi}{\partial x_{d'}} \Big|_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + O(h^2), \end{aligned}$$

leading to $O(h^4)$ overall for the average flux formula:

$$\begin{aligned} \langle \phi \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} &= \langle \phi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \langle \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \\ &+ \frac{h^2}{12} \sum_{d' \neq d} \left(\mathbf{G}_{d'}^\perp \phi|_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \mathbf{G}_{d'}^\perp \psi|_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} \right) \\ &+ C_4(\mathbf{x}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d})h^4 + O(h^5). \end{aligned} \quad (15.28)$$

Furthermore,

$$\begin{aligned} \frac{1}{h^D} \int_{C_i} (\mathbf{u} \cdot \nabla) \mathbf{u} &= \frac{1}{h^D} \int_{C_i} \nabla \cdot (\mathbf{u}\mathbf{u}) - \frac{1}{h^D} \int_{C_i} (\nabla \cdot \mathbf{u}) \mathbf{u} \\ &= \frac{1}{h} \sum_d \mathbf{e}^d \cdot \left(\langle \mathbf{u}\mathbf{u} \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} - \langle \mathbf{u}\mathbf{u} \rangle_{\mathbf{i}-\frac{1}{2}\mathbf{e}^d} \right) + O(h^4), \end{aligned}$$

where we have applied the chain rule, the divergence theorem, (15.26b), and the given condition $\mathbf{D}\langle \mathbf{u} \rangle_{\mathbf{i}} = O(h^4)$. Then (15.26d) follows from (15.23) and (15.26e). Another necessary condition is the cancellation from the symmetry of the difference stencils, i.e.,

$$C_4(\mathbf{x}_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}) - C_4(\mathbf{x}_{\mathbf{i}-\frac{1}{2}\mathbf{e}^d}) = O(h). \quad \square$$

15.1.3 Ghost cells

Definition 15.9. *Ghost cells* are convenience devices for evaluating FD or FV discrete operators at cells near non-periodic domain boundaries.

Example 15.10. Two layers of ghost cells are used to enforce boundary conditions for fourth-order FV methods. We set the values of ghost cells by extrapolating those of the interior cells, with the boundary conditions incorporated in the extrapolation formulas. Referring to Figure 15.2, different boundary conditions entail different cell-averaged values for the cells $\mathbf{i} + \mathbf{e}^d$ and $\mathbf{i} + 2\mathbf{e}^d$. In particular, Dirichlet boundary conditions g are fulfilled to fourth-order accuracy by filling ghost cells with the following formulas:

$$\begin{aligned} \langle \phi \rangle_{\mathbf{i}+\mathbf{e}^d} &= \frac{1}{3} (-13 \langle \phi \rangle_{\mathbf{i}} + 5 \langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} - \langle \phi \rangle_{\mathbf{i}-2\mathbf{e}^d}) \\ &+ 4 \langle g \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + O(h^4); \end{aligned} \quad (15.29a)$$

$$\begin{aligned} \langle \phi \rangle_{\mathbf{i}+2\mathbf{e}^d} &= \frac{1}{3} (-70 \langle \phi \rangle_{\mathbf{i}} + 32 \langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} - 7 \langle \phi \rangle_{\mathbf{i}-2\mathbf{e}^d}) \\ &+ 16 \langle g \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + O(h^4). \end{aligned} \quad (15.29b)$$

Neumann boundary conditions are fulfilled to the fifth

order by

$$\begin{aligned} \langle \psi \rangle_{\mathbf{i}+\mathbf{e}^d} &= \frac{1}{10} (5 \langle \psi \rangle_{\mathbf{i}} + 9 \langle \psi \rangle_{\mathbf{i}-\mathbf{e}^d} - 5 \langle \psi \rangle_{\mathbf{i}-2\mathbf{e}^d} + \langle \psi \rangle_{\mathbf{i}-3\mathbf{e}^d}) \\ &+ \frac{6}{5} h \left\langle \frac{\partial \psi}{\partial n} \right\rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + O(h^5), \end{aligned} \quad (15.30a)$$

$$\begin{aligned} \langle \psi \rangle_{\mathbf{i}+2\mathbf{e}^d} &= \frac{1}{10} (-75 \langle \psi \rangle_{\mathbf{i}} + 145 \langle \psi \rangle_{\mathbf{i}-\mathbf{e}^d} - 75 \langle \psi \rangle_{\mathbf{i}-2\mathbf{e}^d} \\ &+ 15 \langle \psi \rangle_{\mathbf{i}-3\mathbf{e}^d}) + 6h \left\langle \frac{\partial \psi}{\partial n} \right\rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} + O(h^5), \end{aligned} \quad (15.30b)$$

where $\left\langle \frac{\partial \psi}{\partial n} \right\rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}$ is the Neumann condition for ψ .

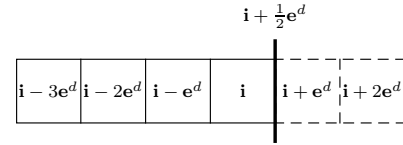


Figure 15.2: Ghost cells in FV discretization. The domain boundary is represented by the thick line, cells inside the domain by squares with solid lines, and ghost cells by squares with dashed lines.

Exercise 15.11. When no boundary conditions are known, a scalar ψ can be smoothly extended to fill a ghost cell abutting the boundary. Derive the following formulas

$$\begin{aligned} \langle \psi \rangle_{\mathbf{i}+\mathbf{e}^d} &= 5 \langle \psi \rangle_{\mathbf{i}} - 10 \langle \psi \rangle_{\mathbf{i}-\mathbf{e}^d} + 10 \langle \psi \rangle_{\mathbf{i}-2\mathbf{e}^d} - 5 \langle \psi \rangle_{\mathbf{i}-3\mathbf{e}^d} \\ &+ \langle \psi \rangle_{\mathbf{i}-4\mathbf{e}^d} + O(h^5); \end{aligned} \quad (15.31)$$

and those for its faced-averaged value at the boundary,

$$\begin{aligned} \langle \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} &= \frac{1}{60} (137 \langle \psi \rangle_{\mathbf{i}} - 163 \langle \psi \rangle_{\mathbf{i}-\mathbf{e}^d} + 137 \langle \psi \rangle_{\mathbf{i}-2\mathbf{e}^d} \\ &- 63 \langle \psi \rangle_{\mathbf{i}-3\mathbf{e}^d} + 12 \langle \psi \rangle_{\mathbf{i}-4\mathbf{e}^d}) + O(h^5). \end{aligned} \quad (15.32)$$

Lemma 15.12. Face-averaged derivatives can be calculated from known boundary conditions and interior cell averages:

$$\begin{aligned} \left\langle \frac{\partial \psi}{\partial n} \right\rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} &= \frac{1}{72h} (-415 \langle \psi \rangle_{\mathbf{i}} + 161 \langle \psi \rangle_{\mathbf{i}-\mathbf{e}^d} - 55 \langle \psi \rangle_{\mathbf{i}-2\mathbf{e}^d} \\ &+ 9 \langle \psi \rangle_{\mathbf{i}-3\mathbf{e}^d} + 300 \langle \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}) + O(h^4), \end{aligned} \quad (15.33a)$$

$$\begin{aligned} \left\langle \frac{\partial^2 \psi}{\partial n^2} \right\rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d} &= \frac{1}{48h^2} (-755 \langle \psi \rangle_{\mathbf{i}} + 493 \langle \psi \rangle_{\mathbf{i}-\mathbf{e}^d} - 191 \langle \psi \rangle_{\mathbf{i}-2\mathbf{e}^d} \\ &+ 33 \langle \psi \rangle_{\mathbf{i}-3\mathbf{e}^d} + 420 \langle \psi \rangle_{\mathbf{i}+\frac{1}{2}\mathbf{e}^d}) + O(h^3). \end{aligned} \quad (15.33b)$$

Algorithm 15.13. As shown in Figure 15.3, the discrete convection $\mathbf{D}(\mathbf{u}\mathbf{u})$ are evaluated as follows:

(Cnv-1) fill the ghost cells of cell-averaged velocity $\langle \mathbf{u} \rangle_{\mathbf{i}}$ using (15.29),

(Cnv-2) convert $\langle \mathbf{u} \rangle_{\mathbf{i}}$ to face-averaged *normal* velocity $\langle u_d \rangle_{\mathbf{i} \pm \frac{1}{2}\mathbf{e}^d}$ for each dimension d using (15.9),

- (Cnv-3) convert $\langle \mathbf{u} \rangle_{\mathbf{i}}$ to face-averaged velocity $\langle \mathbf{u} \rangle_{\mathbf{i} \pm \frac{1}{2} \mathbf{e}^d}$ for each component of the velocity and each dimension d using (15.9).
- (Cnv-4) calculate the discrete velocity product $\mathbf{F} \langle u_d, \mathbf{u} \rangle$ using (15.24) and (15.25).
- (Cnv-5) compute $\mathbf{D} \langle \mathbf{u} \mathbf{u} \rangle$ using (15.23).

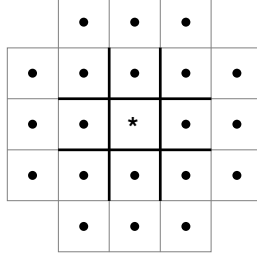


Figure 15.3: The stencil of the discrete convection operator in Algorithm 15.13. The star symbol represents the cell for which the convection operator is discretized while the solid dots cells involved in the calculation. Face averaged normal velocities are calculated on the solid lines.

15.2 FV methods for BVPs

Definition 15.14. A *domain* is a connected and bounded regular open subset $\Omega \subset \mathbb{R}^D$ with $D = 2, 3$.

Algorithm 15.15. To solve the Neumann BVP of the form

$$\begin{aligned} \Delta \phi &= \nabla \cdot \mathbf{u} \quad \text{in } \Omega, \\ \mathbf{n} \cdot \nabla \phi &= \mathbf{n} \cdot \mathbf{u} \quad \text{on } \partial\Omega, \end{aligned} \quad (15.34)$$

a fourth-order FV approximation is

$$\mathbf{L}_H \langle \phi \rangle = \mathbf{D}_H \langle \mathbf{u} \rangle, \quad (15.35)$$

where $\mathbf{L}_H, \mathbf{D}_H$ are the same as $\mathbf{L}, \mathbf{D}$ defined in (15.22) and (15.21), except that the fluxes at the wall boundaries are set to zero. More precisely, $\mathbf{D}_H \langle \mathbf{u} \rangle$ is calculated as follows,

- (DvH-1) smoothly extend $\langle u_n \rangle$ to the ghost cells abutting the domain by (15.31),
- (DvH-2) convert $\langle u_n \rangle$ to face averages using (15.9),
- (DvH-3) zero out the face averages on the wall boundaries,
- (DvH-4) sum up the face-averaged normal velocities for each interior cell.

$\mathbf{L}_H \langle \phi \rangle$ can also be calculated by first converting cell averages of $\nabla \phi$ to face averages and then apply the above steps, but this is not amenable to a multigrid solver. To facilitate the smoothing operations, $\mathbf{L}_H \langle \phi \rangle$ is evaluated by (15.22), with ghost cells that immediately abut the wall filled by (15.31) and those away from the wall by

$$\begin{aligned} \langle \phi \rangle_{\mathbf{i}+2\mathbf{e}^d} &= 60\langle \phi \rangle_{\mathbf{i}} - 149\langle \phi \rangle_{\mathbf{i}-\mathbf{e}^d} + 150\langle \phi \rangle_{\mathbf{i}-2\mathbf{e}^d} - 75\langle \phi \rangle_{\mathbf{i}-3\mathbf{e}^d} \\ &\quad + 15\langle \phi \rangle_{\mathbf{i}-4\mathbf{e}^d} + O(h^5) \end{aligned} \quad (15.36)$$

so that the flux $\left\langle \frac{\partial \phi}{\partial x_d} \right\rangle_{\mathbf{i} + \frac{1}{2} \mathbf{e}^d}$ resulting from (15.10) vanishes.

If $\mathbf{G} \langle \phi \rangle$ is needed, the ghost cells of $\langle \phi \rangle$ are filled by the formulas in (15.30), where the Neumann boundary value $\left\langle \frac{\partial \phi}{\partial x_d} \right\rangle_{\mathbf{i} + \frac{1}{2} \mathbf{e}^d}$ is obtained by extrapolating cell-averaged normal velocity $\langle u_n \rangle$'s to the wall boundary using (15.32).

15.3 FV for the advection-diffusion equation

Definition 15.16. The *advection-diffusion equation* is a PDE of the form

$$\frac{\partial \phi}{\partial t} = -\nabla \cdot (\mathbf{u} \phi) + \nu \Delta \phi + f, \quad (15.37)$$

where the constant diffusivity ν , the velocity field $\mathbf{u} : \mathbb{R}^D \times \mathbb{R} \rightarrow \mathbb{R}^D$, and the forcing term $f : \mathbb{R}^D \times \mathbb{R} \rightarrow \mathbb{R}$ are known *a priori*.

Example 15.17. To generate a semi-discrete system of the advection-diffusion equation in the FV formulation, we average (15.37) over a control volume $\mathcal{C}_{\mathbf{i}}$ and apply the divergence theorem to obtain a system of ODEs:

$$\frac{d \langle \phi \rangle_{\mathbf{i}}}{dt} = L_{\text{adv}}(\langle \phi \rangle, t)_{\mathbf{i}} + L_{\text{diff}}(\langle \phi \rangle)_{\mathbf{i}} + \langle f \rangle_{\mathbf{i}}, \quad (15.38)$$

where

$$\begin{aligned} L_{\text{adv}}(\langle \phi \rangle, t)_{\mathbf{i}} &= -\frac{1}{h} \sum_{d=1}^D \left(\langle u_d \phi \rangle_{\mathbf{i} + \frac{1}{2} \mathbf{e}^d} - \langle u_d \phi \rangle_{\mathbf{i} - \frac{1}{2} \mathbf{e}^d} \right), \\ L_{\text{diff}}(\langle \phi \rangle)_{\mathbf{i}} &= \nu \langle \Delta \phi \rangle_{\mathbf{i}} \\ &= \frac{\nu}{h} \sum_{d=1}^D \left(\left\langle \frac{\partial \phi}{\partial x_d} \right\rangle_{\mathbf{i} + \frac{1}{2} \mathbf{e}^d} - \left\langle \frac{\partial \phi}{\partial x_d} \right\rangle_{\mathbf{i} - \frac{1}{2} \mathbf{e}^d} \right). \end{aligned}$$

Definition 15.18. An *ERK-ESDIRK IMplicit-EXplicit (IMEX) Runge-Kutta scheme* for solving an ODE

$$\frac{d\phi}{dt} = \mathbf{X}^{[E]}(\phi, t) + \mathbf{X}^{[I]}(\phi) \quad (15.39)$$

consists of steps as follows:

$$\phi^{(1)} = \phi^n \approx \phi(t^n), \quad (15.40a)$$

$$\forall s = 2, 3, \dots, n_s,$$

$$(I - k\gamma \mathbf{X}^{[I]}) \phi^{(s)} = \phi^n + k \sum_{j=1}^{s-1} a_{s,j}^{[E]} \mathbf{X}^{[E]}(\phi^{(j)}, t^{(j)}) \quad (15.40b)$$

$$\begin{aligned} &+ k \sum_{j=1}^{s-1} a_{s,j}^{[I]} \mathbf{X}^{[I]} \phi^{(j)}, \\ \phi^{n+1} &= \phi^n + k \sum_{s=1}^{n_s} b_s^{[E]} \mathbf{X}^{[E]}(\phi^{(s)}, t^{(s)}) \quad (15.40c) \\ &+ k \sum_{s=1}^{n_s} b_s^{[I]} \mathbf{X}^{[I]} \phi^{(s)}, \end{aligned}$$

where the superscript (s) denotes an intermediate stage, $t^{(s)} = t^n + c_s k$ the time of that stage, n_s the number of stages, and $A, \mathbf{b}, \mathbf{c}$ standard coefficients of the Butcher tableau.

Algorithm 15.19. A fourth-order FV method for solving the advection-diffusion equation on periodic domains is obtained by directly applying the ERK-ESDIRK IMEX algorithm (15.40) to the ODE system (15.38) with

$$\mathbf{X}^{[E]} = L_{\text{adv}} + \langle f \rangle, \quad \mathbf{X}^{[I]} = L_{\text{diff}}. \quad (15.41)$$

Example 15.20. Kennedy and Carpenter [2003] studied a group of implicit-explicit Runge-Kutta schemes from third- to fifth-order accurate with the following form:

$$= \begin{array}{c|cccccc} \mathbf{c}^{[E]} & A^{[E]} & & & & \\ \hline & (\mathbf{b}^{[E]})^\top & & & & \\ \hline & (\hat{\mathbf{b}}^{[E]})^\top & & & & \\ \hline 0 & 0 & & & & \\ 2\gamma & 2\gamma & 0 & & & \\ c_3 & a_{31}^{[E]} & a_{32}^{[E]} & 0 & & \\ \vdots & \vdots & \vdots & \vdots & \ddots & \\ c_{s-1} & a_{s-1,1}^{[E]} & a_{s-1,2}^{[E]} & a_{s-1,3}^{[E]} & \cdots & 0 \\ 1 & a_{s,1}^{[E]} & a_{s,2}^{[E]} & a_{s,3}^{[E]} & \cdots & a_{s,s-1}^{[E]} & 0 \\ \hline & b_1 & b_2 & b_3 & \cdots & b_{s-1} & \gamma \\ \hline & \hat{b}_1 & \hat{b}_2 & \hat{b}_3 & \cdots & \hat{b}_{s-1} & \hat{b}_s \end{array}, \quad (15.42)$$

$$= \begin{array}{c|cccccc} \mathbf{c}^{[I]} & A^{[I]} & & & & \\ \hline & (\mathbf{b}^{[I]})^\top & & & & \\ \hline & (\hat{\mathbf{b}}^{[I]})^\top & & & & \\ \hline 0 & 0 & & & & \\ 2\gamma & \gamma & \gamma & & & \\ c_3 & a_{31}^{[I]} & a_{32}^{[I]} & \gamma & & \\ \vdots & \vdots & \vdots & \vdots & \ddots & \\ c_{s-1} & a_{s-1,1}^{[I]} & a_{s-1,2}^{[I]} & a_{s-1,3}^{[I]} & \cdots & \gamma \\ 1 & b_1 & b_2 & b_3 & \cdots & b_{s-1} & \gamma \\ \hline & b_1 & b_2 & b_3 & \cdots & b_{s-1} & \gamma \\ \hline & \hat{b}_1 & \hat{b}_2 & \hat{b}_3 & \cdots & \hat{b}_{s-1} & \hat{b}_s \end{array}, \quad (15.43)$$

where coefficients in $\hat{\mathbf{b}}$ are useful for error estimation.

The coefficients of ARK4(3)6L[2]SA, a particular IMEX scheme, are, in decimal form, $\gamma = 0.25$, $\mathbf{c}^{[E]} = \mathbf{c}^{[I]} = \mathbf{c}$, $\mathbf{b}^{[E]} = \mathbf{b}^{[I]} = \mathbf{b}$ where

$$\mathbf{c} = (0.0, 0.5, 0.332, 0.62, 0.85, 1.0)^\top$$

$$\begin{aligned} a_{31}^{[I]} &= 0.137776, \\ a_{32}^{[I]} &= -0.055776, \\ a_{41}^{[I]} &= 0.14463686602698217, \\ a_{42}^{[I]} &= -0.22393190761334475, \\ a_{43}^{[I]} &= 0.4492950415863626, \\ a_{51}^{[I]} &= 0.09825878328356477, \\ a_{52}^{[I]} &= -0.5915442428196704, \\ a_{53}^{[I]} &= 0.8101210538282996, \\ a_{54}^{[I]} &= 0.283164405707806, \end{aligned}$$

$$\begin{aligned} a_{31}^{[E]} &= 0.221776, \\ a_{32}^{[E]} &= 0.110224, \\ a_{41}^{[E]} &= -0.04884659515311857, \\ a_{42}^{[E]} &= -0.17772065232640102, \\ a_{43}^{[E]} &= 0.8465672474795197, \\ a_{51}^{[E]} &= -0.15541685842491548, \\ a_{52}^{[E]} &= -0.3567050098221991, \\ a_{53}^{[E]} &= 1.0587258798684427, \\ a_{54}^{[E]} &= 0.30339598837867193, \\ a_{61}^{[E]} &= 0.2014243506726763, \\ a_{62}^{[E]} &= 0.008742057842904185, \\ a_{63}^{[E]} &= 0.15993995707168115, \\ a_{64}^{[E]} &= 0.4038290605220775, \\ a_{65}^{[E]} &= 0.22606457389066084 \end{aligned}$$

$$\begin{aligned} b_1 &= 0.15791629516167136, \\ b_2 &= 0., \\ b_3 &= 0.18675894052400077, \\ b_4 &= 0.6805652953093346, \\ b_5 &= -0.27524053099500667, \end{aligned}$$

Lemma 15.21. The stability function of the IMEX Runge-Kutta methods in Definition 15.18 is

$$R(\bar{\lambda}^d + i\bar{\lambda}^a) = \frac{\det(I - \bar{\lambda}^d A^{[I]} - i\bar{\lambda}^a A^{[E]} + (\bar{\lambda}^d + i\bar{\lambda}^a) \mathbf{1} \otimes \mathbf{b}^\top)}{\det(I - \bar{\lambda}^d A^{[I]} - i\bar{\lambda}^a A^{[E]})}, \quad (15.44)$$

where $\bar{\lambda}^d = \lambda^d k$, $\bar{\lambda}^a = \lambda^a k$. The vector $\mathbf{b}$ and the matrices $A^{[E]}$ and $A^{[I]}$ are the coefficients in (15.42) and (15.43).

Proof. See Calvo et al. [2001]. $\square$

Lemma 15.22. For a periodic domain, the ODE system (15.38) with constant velocity $\mathbf{u}$ and $\langle f \rangle = 0$ can be converted to a system of decoupled ODEs of the form

$$\frac{dy}{dt} = \lambda y = (\lambda^d + i\lambda^a) y, \quad (15.45)$$

where $\lambda^d, i\lambda^a$ are the eigenvalues of the diffusion and advection operators:

$$\begin{aligned} \lambda^d &= -4 \frac{\nu}{h^2} \sum_{d=1}^D \sin^2 \frac{\theta_d}{2} \left(1 + \frac{1}{3} \sin^2 \frac{\theta_d}{2}\right), \\ \lambda^a &= -\frac{1}{h} \sum_{d=1}^D u_d \sin \theta_d \left(1 + \frac{2}{3} \sin^2 \frac{\theta_d}{2}\right), \end{aligned} \quad (15.46)$$

with $\theta_d = \xi_d h_d \in (0, 2\pi)$. Here ξ_d and h_d are the wavenumber and grid size in the d th dimension; we assume that a single Fourier mode of the cell average $\langle \phi \rangle_i$ is of the form $y(t)e^{i\xi \cdot \mathbf{x}_i}$.

Exercise 15.23. Prove Lemma 15.22.

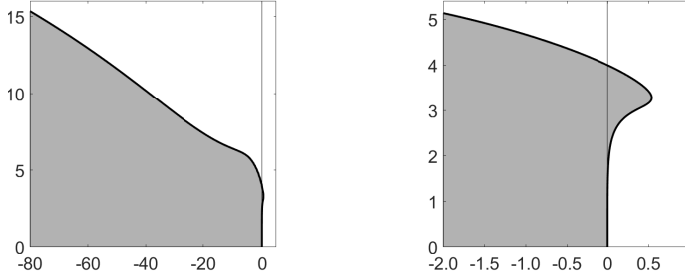


Figure 15.4: Stability regions of the MOL for solving the advection-diffusion equation (15.37), with Definition 15.7 as its spatial FV discretization and with ARK4(3)6L[2]SA in Example 15.19 as its time integrator. The left plot shows the stability region in the range $(\bar{\lambda}_d, \bar{\lambda}_a) \in [-80, 0] \times [0, 15]$ and the right is a zoom-in of the left near the origin.

Exercise 15.24. Reproduce the stability region $|R(z)| < 1$ in Figure 15.4. It is clear from these plots that

- the maximum stable Courant number increases as diffusion becomes stronger,
- in the absence of diffusion the scaled advection eigenvalue should be less than 4.

Deduce from the second plot and (15.46) that the range of stable Courant numbers for ARK4(3)6L[2]SA is

$$\mu \leq \frac{2.91}{D}. \quad (15.47)$$

Theorem 15.25. The FV method in Algorithm 15.19 is convergent with fourth-order accuracy.

Proof. This follows from Lemma 15.8, Lemma 15.22, and Theorem 14.35. $\square$

Appendix A

Sets and Logic

A.1 The algebra of sets

Definition A.1. A *set* $\mathcal{S}$ is a collection of *distinct* things,

$$\mathcal{S} = \{x : \text{the conditions that } x \text{ satisfies}\}. \quad (\text{A.1})$$

In particular, the *universe* is the set of all elements.

Definition A.2 (Membership). An object x in the universe is an *element* of a set $\mathcal{S}$, written $x \in \mathcal{S}$, if x belongs to $\mathcal{S}$; otherwise we write $x \notin \mathcal{S}$.

Definition A.3. The *empty set* is the set to which no elements in the universe belong.

Notation 18. $\mathbb{R}, \mathbb{Z}, \mathbb{N}, \mathbb{Q}, \mathbb{C}$ denote the sets of real numbers, integers, natural numbers, rational numbers and complex numbers, respectively. $\mathbb{R}^+, \mathbb{Z}^+, \mathbb{N}^+, \mathbb{Q}^+$ the sets of positive such numbers. In particular, $\mathbb{N}$ contains the number zero while $\mathbb{N}^+$ does not.

Definition A.4. $\mathcal{S}$ is a *subset* of $\mathcal{U}$, written $\mathcal{S} \subseteq \mathcal{U}$, if and only if (iff) $x \in \mathcal{S}$ implies $x \in \mathcal{U}$. $\mathcal{S}$ is a *proper subset* of $\mathcal{U}$, written $\mathcal{S} \subset \mathcal{U}$, if $\mathcal{S} \subseteq \mathcal{U}$ and there exists $x \in \mathcal{U}$ such that $x \notin \mathcal{S}$.

Definition A.5. The *power set* of a set A , written $\mathcal{P}(A)$, is the set of all subsets of A .

Definition A.6. Two sets A and B are *equal* if they contain the same elements, i.e., both $A \subseteq B$ and $B \subseteq A$ hold.

Definition A.7. The *complement* of a set B in A is the set

$$A \setminus B := \{x : x \in A \text{ and } x \notin B\}. \quad (\text{A.2})$$

In particular, the *complement* of a set $\mathcal{S}$, written $\mathcal{S}^c$, is the complement of $\mathcal{S}$ in the universe.

Definition A.8. The *intersection* of two sets is

$$A \cap B := \{x : x \in A \text{ and } x \in B\}. \quad (\text{A.3})$$

Definition A.9. The *union* of two sets is

$$A \cup B := \{x : x \in A \text{ or } x \in B\}. \quad (\text{A.4})$$

Theorem A.10 (Commutative laws). For any sets A and B , we have

$$A \cap B = B \cap A; \quad A \cup B = B \cup A. \quad (\text{A.5})$$

Theorem A.11 (Absorption laws). For any sets A and B , we have

$$A \cap (A \cup B) = A; \quad A \cup (A \cap B) = A. \quad (\text{A.6})$$

Theorem A.12 (Associative laws). For any sets A , B , and C , we have

$$A \cap (B \cap C) = (A \cap B) \cap C; \quad A \cup (B \cup C) = (A \cup B) \cup C. \quad (\text{A.7})$$

Theorem A.13 (Distributive laws). For any sets A , B , and C , we have

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \quad (\text{A.8})$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C). \quad (\text{A.9})$$

Theorem A.14 (DeMorgan's laws). For any sets A , B , and C , we have

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C), \quad (\text{A.10})$$

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C). \quad (\text{A.11})$$

A.2 First-order logic

Definition A.15 (Statements of first-order logic). A *universal statement* is a logical statement of the form

$$\mathbf{U} = (\forall x \in \mathcal{S}, A(x)). \quad (\text{A.12})$$

An *existential statement* has the form

$$\mathbf{E} = (\exists x \in \mathcal{S}, \text{ s.t. } A(x)), \quad (\text{A.13})$$

where $\forall$ (“for each”) and $\exists$ (“there exists”) are the *quantifiers*, $\mathcal{S}$ is a set, “s.t.” means “such that,” and $A(x)$ is the *formula*.

A statement of *implication/conditional* has the form

$$\mathbf{A} \implies \mathbf{B}. \quad (\text{A.14})$$

Example A.16. Universal and existential statements:

$$\forall x \in [2, +\infty), x > 1;$$

$$\forall x \in \mathbb{R}^+, x > 1;$$

$$\exists p, q \in \mathbb{Z}, \text{ s.t. } p/q = \sqrt{2};$$

$$\exists p, q \in \mathbb{Z}, \text{ s.t. } \sqrt{p} = \sqrt{q} + 1.$$

Definition A.17. *Uniqueness quantification* or *unique existential quantification*, written $\exists!$ or $\exists_{=1}$, indicates that exactly one object with a certain property exists.

Exercise A.18. Express the logical statement $\exists!x$, s.t. $A(x)$ with $\exists$, $\forall$, and $\iff$.

Definition A.19. A *universal-existential statement* is a logical statement of the form

$$U_E = (\forall x \in \mathcal{S}, \exists y \in \mathcal{T} \text{ s.t. } A(x, y)). \quad (\text{A.15})$$

An *existential-universal statement* has the form

$$E_U = (\exists y \in \mathcal{T}, \text{ s.t. } \forall x \in \mathcal{S}, A(x, y)). \quad (\text{A.16})$$

Example A.20. True or false:

$\forall x \in [2, +\infty), \exists y \in \mathbb{Z}^+ \text{ s.t. } x^y < 10^5$;

$\exists y \in \mathbb{R} \text{ s.t. } \forall x \in [2, +\infty), x > y$;

$\exists y \in \mathbb{R} \text{ s.t. } \forall x \in [2, +\infty), x < y$.

Example A.21 (Translating an English statement into a logical statement). Goldbach's conjecture states *every even natural number greater than 2 is the sum of two primes*. Let $\mathbb{P} \subset \mathbb{N}^+$ denote the set of prime numbers. Then Goldbach's conjecture is $\forall a \in 2\mathbb{N}^+ + 2, \exists p, q \in \mathbb{P}, \text{ s.t. } a = p + q$.

Theorem A.22. The existential-universal statement implies the corresponding universal-existential statement, but not vice versa.

Example A.23 (Translating a logical statement to an English statement). Let $\mathcal{S}$ be the set of all human beings.

$U_E = (\forall p \in \mathcal{S}, \exists q \in \mathcal{S} \text{ s.t. } q \text{ is } p\text{'s mom.})$

$E_U = (\exists q \in \mathcal{S} \text{ s.t. } \forall p \in \mathcal{S}, q \text{ is } p\text{'s mom.})$

U_E is probably true, but E_U is certainly false.

If E_U were true, then U_E would be true. Why?

Axiom A.24 (First-order negation of logical statements). The negations of the statements in Definition A.15 are

$$\neg U = (\exists x \in \mathcal{S}, \text{ s.t. } \neg A(x)). \quad (\text{A.17})$$

$$\neg E = (\forall x \in \mathcal{S}, \neg A(x)). \quad (\text{A.18})$$

Rule A.25. The negation of a more complicated logical statement abides by the following rules:

- switch the type of each quantifier until you reach the last formula without quantifiers;
- negate the last formula.

In particular, the negation of an implication formula $P \implies Q$, is $P \wedge \neg Q$.

Example A.26 (The negation of Goldbach's conjecture). $\exists a \in 2\mathbb{N}^+ + 2 \text{ s.t. } \forall p, q \in \mathbb{P}, a \neq p + q$.

Exercise A.27. Negate the logical statement in Definition B.106.

Axiom A.28 (Contraposition). A conditional statement is logically equivalent to its contrapositive.

$$(A \implies B) \iff (\neg B \implies \neg A) \quad (\text{A.19})$$

Example A.29. "If Jack is a man, then Jack is a human being," is equivalent to "If Jack is not a human being, then Jack is not a man."

Exercise A.30. Draw an Euler diagram of subsets to illustrate Example A.29.

Exercise A.31. Rewrite each of the following statements and its *negation* into *logical statements* using symbols, quantifiers, and formulas.

- The only even prime is 2.
- Multiplication of integers is associative.
- Goldbach's conjecture has at most a finite number of counterexamples.

A.3 Relations

Definition A.32. The *Cartesian product* $\mathcal{X} \times \mathcal{Y}$ between two sets $\mathcal{X}$ and $\mathcal{Y}$ is the set of all possible ordered pairs with first element from $\mathcal{X}$ and second element from $\mathcal{Y}$:

$$\mathcal{X} \times \mathcal{Y} = \{(x, y) \mid x \in \mathcal{X}, y \in \mathcal{Y}\}. \quad (\text{A.20})$$

Axiom A.33 (Fundamental principle of counting). Consider a task that consists of a sequence of k independent steps. Let n_i denote the number of different choices for the i -th step, the total number of distinct ways to complete the task is

$$\prod_{i=1}^k n_i = n_1 n_2 \cdots n_k. \quad (\text{A.21})$$

Example A.34. Let A, E, D be the set of appetizers, main entrees, desserts in a restaurant. $A \times E \times D$ is the set of possible dinner combos. If $\#A = 10$, $\#E = 5$, $\#D = 6$, $\#(A \times E \times D) = 300$.

Definition A.35. A *binary relation* between two sets $\mathcal{X}$ and $\mathcal{Y}$ is an ordered triple $(\mathcal{X}, \mathcal{Y}, \mathcal{G})$ where $\mathcal{G} \subseteq \mathcal{X} \times \mathcal{Y}$; it is a *binary relation on $\mathcal{X}$* if $\mathcal{X} = \mathcal{Y}$. The statement $(x, y) \in R$ reads " x is R -related to y " and is denoted by xRy or $R(x, y)$.

Definition A.36. A binary relation R on $\mathcal{X}$ is said to be

- *reflexive* if $\forall x \in \mathcal{X}, xRx$;
- *transitive* if $\forall a, b, c \in \mathcal{X}, aRb \text{ and } bRc \implies aRc$;
- *symmetric* if $\forall a, b \in \mathcal{X}, aRb \implies bRa$.

A relation that satisfies all the above three properties is called an *equivalence relation*.

Example A.37. An equivalence relation on a set $\mathcal{A}$ is also denoted by " $\sim$ " and we have, $\forall a, b, c \in \mathcal{A}$,

- $a \sim a$ (reflexivity);
- $a \sim b$ implies $b \sim a$ (symmetry);
- $a \sim b$ and $b \sim c$ imply $a \sim c$ (transitivity).

A.4 Functions

Definition A.38. A *function/map/mapping* f from $\mathcal{X}$ to $\mathcal{Y}$, written $f : \mathcal{X} \rightarrow \mathcal{Y}$ or $\mathcal{X} \mapsto \mathcal{Y}$, is a subset of the Cartesian product $\mathcal{X} \times \mathcal{Y}$ satisfying that $\forall x \in \mathcal{X}$, there is exactly one $y \in \mathcal{Y}$ s.t. $(x, y) \in \mathcal{X} \times \mathcal{Y}$. $\mathcal{X}$ and $\mathcal{Y}$ are the *domain* and *codomain* of f , respectively.

Definition A.39. A *binary function* or a *binary operation* on a set $\mathcal{S}$ is a map $\mathcal{S} \times \mathcal{S} \rightarrow \mathcal{S}$.

Definition A.40. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is said to be *injective* or *one-to-one* iff

$$\forall x_1 \in \mathcal{X}, \forall x_2 \in \mathcal{X}, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2). \quad (\text{A.22})$$

It is *surjective* or *onto* iff

$$\forall y \in \mathcal{Y}, \exists x \in \mathcal{X}, \text{ s.t. } y = f(x). \quad (\text{A.23})$$

It is *bijective* iff it is both injective and surjective.

Definition A.41. The *preimage* of a set $U \subset Y$ (or the *fiber over* U) under a function $f : X \rightarrow Y$ is

$$f^{-1}(U) := \{x \in X : f(x) \in U\}. \quad (\text{A.24})$$

Lemma A.42. For $f : X \rightarrow Y$, the operation f^{-1} preserves inclusions, unions, intersections, and differences of sets,

$$B_0 \subseteq B_1 \implies f^{-1}(B_0) \subseteq f^{-1}(B_1), \quad (\text{A.25a})$$

$$f^{-1}(B_0 \cup B_1) = f^{-1}(B_0) \cup f^{-1}(B_1), \quad (\text{A.25b})$$

$$f^{-1}(B_0 \cap B_1) = f^{-1}(B_0) \cap f^{-1}(B_1), \quad (\text{A.25c})$$

$$f^{-1}(B_0 \setminus B_1) = f^{-1}(B_0) \setminus f^{-1}(B_1), \quad (\text{A.25d})$$

while f only preserves inclusions and unions,

$$A_0 \subseteq A_1 \implies f(A_0) \subseteq f(A_1), \quad (\text{A.26a})$$

$$f(A_0 \cup A_1) = f(A_0) \cup f(A_1), \quad (\text{A.26b})$$

$$f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1), \quad (\text{A.26c})$$

$$f(A_0 \setminus A_1) \supseteq f(A_0) \setminus f(A_1), \quad (\text{A.26d})$$

where the equalities in the last two equations hold if f is injective.

Proof. We only prove the third equation in (A.26). Suppose $y \in f(A_0 \cap A_1)$. Then there exists $x \in A_0 \cap A_1$ such that $f(x) = y$. $x \in A_0$ implies $y \in f(A_0)$ while $x \in A_1$ implies $y \in f(A_1)$. Hence $y \in f(A_0) \cap f(A_1)$. The above argument can be reversed if f is injective. $\square$

Example A.43. For $f(x) = x^2$, $A_0 = [-1, 0]$, $A_1 = [0, 1]$, we have $\{0\} = f(A_0 \cap A_1) \subset f(A_0) \cap f(A_1) = A_1$.

Lemma A.44. For a map $f : X \rightarrow Y$, $A \subseteq X$, and $B \subseteq Y$, we have

$$A \subseteq f^{-1}(f(A)), \quad (\text{A.27})$$

$$f(f^{-1}(B)) \subseteq B, \quad (\text{A.28})$$

where the first inclusion is an equality if f is injective and the second is an equality if f is surjective or $B \subseteq f(X)$.

Proof. By (A.24), $a \in A$ implies $a \in f^{-1}(f(A))$. Conversely, $a \in f^{-1}(f(A))$ implies $f(a) \in f(A)$. f being injective dictates $a \in A$.

By (A.24), $b \in f(f^{-1}(B))$ implies $b \in B$. Furthermore, if f is surjective or $B \subseteq f(X)$, then for any $b \in B$ we have $f^{-1}(\{b\}) \neq \emptyset$ and thus

$$b \in f(f^{-1}(\{b\})) \subseteq f(f^{-1}(B)). \quad \square$$

A.5 Orderings

Definition A.45. A binary relation “ $\leq$ ” on some set $\mathcal{S}$ is a *total order* or *linear order* on $\mathcal{S}$ iff, $\forall a, b, c \in \mathcal{S}$,

- $a \leq b$ and $b \leq a$ imply $a = b$ (antisymmetry);
- $a \leq b$ and $b \leq c$ imply $a \leq c$ (transitivity);
- $a \leq b$ or $b \leq a$ (totality).

A set equipped with a total order is a *chain* or *totally ordered set*.

Example A.46. “less or equal” is a total order on the real numbers.

Example A.47. The dictionary order is a total order on the English letters of the alphabet.

Example A.48. The *lexicographical order* is a total order on the Cartesian product of a family of totally ordered sets.

Example A.49. Sort your book in lexicographical order and save a lot of time: $\log_{26} N \ll N$.

Definition A.50 (Maximum and minimum). Consider $\mathcal{S} \subseteq \mathbb{R}$, $\mathcal{S} \neq \emptyset$. If $\exists s_m \in \mathcal{S}$ s.t. $\forall x \in \mathcal{S}$, $x \leq s_m$, then s_m is the *maximum* of $\mathcal{S}$ and denoted by $\max \mathcal{S}$. If $\exists s_m \in \mathcal{S}$ s.t. $\forall x \in \mathcal{S}$, $x \geq s_m$, then s_m is the *minimum* of $\mathcal{S}$ and denoted by $\min \mathcal{S}$.

Definition A.51 (Upper and lower bounds). Consider $\mathcal{S} \subseteq \mathbb{R}$, $\mathcal{S} \neq \emptyset$. a is an *upper bound* of $\mathcal{S} \subseteq \mathbb{R}$ if $\forall x \in \mathcal{S}$, $x \leq a$; then the set $\mathcal{S}$ is said to be *bounded above*. a is a *lower bound* of $\mathcal{S}$ if $\forall x \in \mathcal{S}$, $x \geq a$; then the set $\mathcal{S}$ is said to be *bounded below*. $\mathcal{S}$ is *bounded* if it is bounded above and bounded below.

Definition A.52 (Supremum and infimum). Consider a nonempty set $\mathcal{S} \subseteq \mathbb{R}$. If $\mathcal{S}$ is bounded above and $\mathcal{S}$ has a least upper bound then we call it the *supremum* of $\mathcal{S}$ and denote it by $\sup \mathcal{S}$. If $\mathcal{S}$ is bounded below and $\mathcal{S}$ has a greatest lower bound, then we call it the *infimum* of $\mathcal{S}$ and denote it by $\inf \mathcal{S}$.

Example A.53. If a set $\mathcal{S} \subset \mathbb{R}$ has a maximum, we have $\max \mathcal{S} = \sup \mathcal{S}$.

Example A.54. $\sup[a, b] = \sup[a, b) = \sup(a, b] = \sup(a, b)$.

Theorem A.55 (Existence and uniqueness of least upper bound). Every nonempty subset of $\mathbb{R}$ that is bounded above has exactly one least upper bound.

Corollary A.56. Every nonempty subset of $\mathbb{R}$ that is bounded below has a greatest lower bound.

Definition A.57. A *well-order* or well-ordering or well-order relation on a set $\mathcal{S}$ is a total ordering on $\mathcal{S}$ such that every non-empty subset of $\mathcal{S}$ has a least element in this ordering. The set $\mathcal{S}$ together with the ordering is called a *well-ordered set* (woset).

Example A.58. $(\mathbb{N}, \leq)$ is a woset.

Definition A.59. A binary relation “ $\leq$ ” on some set $\mathcal{S}$ is a *partial order* on $\mathcal{S}$ iff, $\forall a, b, c \in \mathcal{S}$, antisymmetry, transitivity, and reflexivity ($a \leq a$) hold. A set equipped with a partial order is called a *poset*.

Example A.60. The set of subsets of a set $\mathcal{S}$ ordered by inclusion “ $\subseteq$.”

Example A.61. The natural numbers equipped with the relation of divisibility.

Example A.62. The set of stuff you will put on your body every morning with the time ordered: undershorts, pants, belt, shirt, tie, jacket, socks, shoes, watch.

Example A.63. Inheritance (“is-a” relation) is a partial order. $A \rightarrow B$ reads “ B is a special type of A ”.

Example A.64. Composition (“has-a” relation) is also a partial order. $A \rightsquigarrow B$ reads “ B has an instance/object of A .”

Example A.65. Implication “ $\implies$ ” is a partial order on the set of logical statements.

Example A.66. The set of definitions, axioms, propositions, theorems, lemmas, etc., is a poset with inheritance, composition, and implication. It is helpful to relate them with these partial orderings.

Definition A.67. An *upper bound* of a subset W of a poset M is an element $u \in M$ such that $x \leq u$ for each $x \in W$. A *maximal element* of a poset M is an $m \in M$ such that

$$\forall x \in M, x \geq m \implies x = m. \quad (\text{A.29})$$

Axiom A.68 (Zorn’s lemma). For a nonempty poset M , if every chain in M has an upper bound, then M has at least one maximal element.

Lemma A.69 (The Union Lemma). Let X be a set and $\mathcal{C}$ be a collection of subsets of X . Assume that for each $x \in X$, there is a set A_x in $\mathcal{C}$ such that $x \in A_x$. Then $\cup_{x \in X} A_x = X$.

A.6 Cardinality of sets

Definition A.70. A set $\mathcal{S}$ is *countably infinite* iff there exists a bijective function $f: \mathcal{S} \rightarrow \mathbb{N}^+$ that maps $\mathcal{S}$ to $\mathbb{N}^+$. A set is *countable* if it is either finite or countably infinite; it is *uncountable* if it is not countable.

Exercise A.71. Are the integers countable? Are the rationals countable? Are the real numbers countable?

Definition A.72. The *Cantor set* is a subset of $\mathbb{R}$ given by $C := \cap_{n=0}^{+\infty} C_n$ where $C_0 = [0, 1]$ and each C_{n+1} is obtained by deleting from C_n the open middle third of each closed interval.

Theorem A.73. Let $(E_n)_{n \in \mathbb{N}}$ be a sequence of countable sets. Then $S := \cup_{n \in \mathbb{N}} E_n$ is countable.

Theorem A.74. Let A be the set of all sequences whose elements are either 0 or 1. Then A is uncountable.

Appendix B

Basic Analysis

B.1 Sequences and limits

Definition B.1. A *sequence* is a function on $\mathbb{N}$.

Definition B.2. The *supremum* and *infimum* of a sequence $(a_n)_{n=m}^{\infty} \subset \mathbb{R}$ are respectively

$$\sup(a_n)_{n=m}^{\infty} := \sup\{a_n : n \geq m\}, \quad (\text{B.1})$$

$$\inf(a_n)_{n=m}^{\infty} := \inf\{a_n : n \geq m\}, \quad (\text{B.2})$$

where $\sup$ and $\inf$ on the RHS denotes the supremum and the infimum of a set in Definition A.52.

B.1.1 Convergence in $\mathbb{R}$

Definition B.3 (Limit of a sequence). A sequence (a_n) in $\mathbb{R}$ is said to *converge in $\mathbb{R}$* iff

$$\exists L \in \mathbb{R} \text{ s.t. } \forall \epsilon > 0, \exists N \in \mathbb{N}, \text{ s.t. } \forall n > N, |a_n - L| < \epsilon, \quad (\text{B.3})$$

where L is called the *limit* of (a_n) , written $\lim_{n \rightarrow \infty} a_n = L$ or $a_n \rightarrow L$ as $n \rightarrow \infty$, and (a_n) is said to *converge* to L . The sequence (a_n) is said to *diverge* iff it does not converge.

Example B.4 (A story of π). A famous estimation of π in ancient China is given by Zu, ChongZhi 1500 years ago,

$$\pi \approx \frac{355}{113} \approx 3.14159292.$$

In modern mathematics, we approximate π with a sequence for increasing accuracy, e.g.

$$\pi \approx 3.141592653589793 \dots \quad (\text{B.4})$$

As of March 2019, we human beings have more than 31 trillion digits of π . However, real world applications never use even a small fraction of the 31 trillion digits:

- If you want to build a fence over your backyard swimming pool, several digits of π is probably enough;
- in NASA, calculations involving π use 15 digits for Guidance Navigation and Control;
- if you want to compute the circumference of the entire universe to the accuracy of less than the diameter of a hydrogen atom, you need only 39 decimal places of π .

On one hand, computational mathematics is judged by a metric that is different from that of pure mathematics; this may cause a huge gap between what needs to be done and what has been done. On the other hand, a computational mathematician cannot assume that a fixed accuracy is good enough for all applications. In the approximation a number or a function, she must develop theory and algorithms to provide the user the choice of an ever-increasing amount of accuracy, so long as the user is willing to invest an increasing amount of computational resources. This is one of the main motivations of infinite sequence and series.

Lemma B.5. Let (a_n) be a sequence. Then

$$\lim_{n \rightarrow \infty} a_n = 0 \iff \left| \lim_{n \rightarrow \infty} a_n \right| = 0 \iff \lim_{n \rightarrow \infty} |a_n| = 0.$$

Proof. It follows from (B.3) with $L = 0$. $\square$

Lemma B.6. A convergent sequence has a unique limit.

Proof. Suppose a sequence (a_n) that converges to L also converges to $L_0 \neq L$. For $\epsilon = \frac{|L-L_0|}{2} > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - L| < \epsilon$ and $|a_n - L_0| < \epsilon$. It follows that

$$\begin{aligned} |L - L_0| &= |L - a_N + a_N - L_0| \\ &\leq |a_N - L| + |a_N - L_0| < 2\epsilon = |L - L_0|, \end{aligned}$$

which is a contradiction. $\square$

Lemma B.7. Any subsequence of a convergent sequence converges to the same limit of the sequence.

Proof. Suppose a sequence (a_n) converges to L . Then for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - L| < \epsilon$ for all $n \geq N$. Let (a_{n_j}) be a subsequence of (a_n) . Then there exists $j_0 \in \mathbb{N}$ such that $n_{j_0} \geq N$ and $|a_{n_j} - L| < \epsilon$ for all $j \geq j_0$, which implies the convergence of (a_{n_j}) to L . $\square$

Definition B.8. A sequence (a_n) is *Cauchy* if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } m, n > N \implies |a_n - a_m| < \epsilon. \quad (\text{B.5})$$

Lemma B.9. Every convergent sequence in $\mathbb{R}$ is Cauchy.

Proof. Since (a_n) converges to some $L \in \mathbb{R}$, for any given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$ we have $|a_n - L| < \frac{\epsilon}{2}$. It follows that

$$\begin{aligned} \forall n, m > N, |a_n - a_m| &= |a_n - L + L - a_m| \\ &\leq |a_n - L| + |a_m - L| < \epsilon. \quad \square \end{aligned}$$

Lemma B.10. If a Cauchy sequence contains a convergent subsequence, then the entire sequence converges to the same limit.

Proof. Suppose (a_n) is a Cauchy sequence and (a_{n_j}) is a subsequence converging to some $L \in \mathbb{R}$. It follows that

$$\begin{aligned} \forall \epsilon > 0, \exists n_0 \text{ s.t. } \forall m, n \geq n_0, |a_m - a_n| &< \frac{\epsilon}{2}; \\ \forall \epsilon > 0, \exists j_0 \text{ s.t. } \forall j \geq j_0, |a_{n_j} - L| &< \frac{\epsilon}{2}. \end{aligned}$$

Set $N = \max\{n_0, n_{j_0}\}$ and we have

$$\forall \epsilon > 0, \forall n \geq N, |a_n - L| \leq |a_n - a_N| + |a_N - L| < \epsilon,$$

which completes the proof. $\square$

Lemma B.11. Every Cauchy sequence is bounded.

Proof. Suppose (a_n) is a Cauchy sequence. For $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$ we have $|a_n - a_m| \leq 1$. It follows that

$$\forall n \geq N, |a_n| \leq |a_N| + |a_n - a_N| \leq |a_N| + 1.$$

Let $M := \max\{|a_1|, |a_2|, \dots, |a_{N-1}|, |a_N| + 1\}$ and we have $|a_n| \leq M$ for all $n \in \mathbb{N}$. $\square$

Definition B.12. A sequence $(a_n)_{n=m}^{+\infty} \subset \mathbb{R}$ is

- *monotonically increasing* iff $a_n \leq a_{n+1}$ for each n ;
- *monotonically decreasing* iff $a_n \geq a_{n+1}$ for each n .

$(a_n)_{n=m}^{+\infty}$ is a *monotone sequence* iff it is monotonically increasing or monotonically decreasing.

Lemma B.13. Every real sequence has a monotone subsequence.

Proof. Define $A := \{a_m : \forall n > m, a_n < a_m\}$. Clearly $A \subseteq (a_n)$.

If $\#A = +\infty$, we can choose infinitely many elements in A . Hence we take $1 \leq n_1 < n_2 < \dots$ s.t. $\forall j \geq 1, a_{n_j} \in A$, and then we get a subsequence (a_{n_j}) , which is monotonically decreasing by the definition of A .

If $\#A < +\infty$, then there exists some $N \in \mathbb{N}$ such that for any $n > N$ we have $a_n \notin A$. Thus

$$\forall n > N, \exists m > n, \text{ s.t. } a_n \leq a_m.$$

Hence for $n_1 > N$, there exists $n_2 > n_1$ s.t. $a_{n_1} \leq a_{n_2}$. For n_2 , there exists $n_3 > n_2$ s.t. $a_{n_2} \leq a_{n_3}$. By an easy induction, for n_j , there exists $n_{j+1} > n_j$ s.t. $a_{n_j} \leq a_{n_{j+1}}$. Hence we have constructed a monotonically increasing subsequence (a_{n_j}) . $\square$

Theorem B.14. A bounded monotone sequence is convergent.

Proof. Let (a_n) be a bounded monotone sequence. WLOG, we assume that (a_n) is monotonically increasing. Denote $L := \sup(a_n)_{n=1}^{\infty}$, which is finite because of the boundedness of (a_n) and Definition (B.1). Then we have

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } a_N > L - \epsilon.$$

Hence the monotonicity of (a_n) yields

$$\begin{aligned} \forall n \geq N, L - \epsilon &< a_N \leq a_n \leq L \\ \implies \forall n \geq N, |a_n - L| &< \epsilon, \end{aligned}$$

which implies the convergence of (a_n) to L . $\square$

Theorem B.15 (Bolzano-Weierstrass). Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Proof. Let (a_n) be a bounded sequence in $\mathbb{R}$. By Lemma B.13, (a_n) has a monotone subsequence (a_{n_j}) , which is also bounded. Then Theorem B.14 completes the proof. $\square$

Theorem B.16. Every Cauchy sequence in $\mathbb{R}$ converges to a limit in $\mathbb{R}$.

Proof. By Lemma B.11, the Cauchy sequence (a_n) is bounded. Theorem B.15 implies that $(a_n)_{n \in \mathbb{N}}$ has a convergent subsequence $(a_{n_j})_{j \in \mathbb{N}}$. Then Lemma B.10 completes the proof. $\square$

Theorem B.17 (Completeness of $\mathbb{R}$). A sequence of real numbers is Cauchy if and only if it is convergent.

Proof. This is a summary of Lemma B.9 and Theorem B.16. $\square$

B.1.2 Convergence in $\overline{\mathbb{R}}$

Definition B.18. The *extended real number system* is the set $\overline{\mathbb{R}}$ that consists of the real numbers and two symbols,

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}, \quad (\text{B.6})$$

such that the following rules hold:

$$x \in \mathbb{R} \implies \begin{cases} -\infty < x < +\infty; \\ x \pm \infty = \pm\infty; \\ \frac{x}{+\infty} = \frac{x}{-\infty} = 0, \end{cases} \quad (\text{B.7a})$$

$$x > 0 \implies x \cdot (\pm\infty) = \pm\infty, \quad (\text{B.7b})$$

$$x < 0 \implies x \cdot (\pm\infty) = \mp\infty. \quad (\text{B.7c})$$

Definition B.19. An extended real number $x \in \overline{\mathbb{R}}$ is *finite* iff $x \in \mathbb{R}$ and is *infinite* otherwise.

Definition B.20. A sequence $(a_n) \subset \mathbb{R}$ *diverges to $+\infty$ or $-\infty$* , written $\lim_{n \rightarrow +\infty} a_n = \pm\infty$, iff

$$\forall M > 0, \exists N \in \mathbb{N}, \text{ s.t. } \forall n > N, a_n > M \text{ or } a_n < -M. \quad (\text{B.8})$$

Definition B.21. A sequence $(a_n) \subset \mathbb{R}$ *converges in $\overline{\mathbb{R}}$* , written $\lim_{n \rightarrow \infty} a_n = L \in \overline{\mathbb{R}}$, iff (a_n) converges in $\mathbb{R}$ or diverges to $\pm\infty$.

Theorem B.22. Any monotone sequence in $\overline{\mathbb{R}}$ converges in $\overline{\mathbb{R}}$.

Proof. Let (a_n) be a monotone sequence. WLOG, we assume that (a_n) is monotonically increasing. Let $A := \sup(a_n)_{n=1}^\infty \in \overline{\mathbb{R}}$. If $A = -\infty$, then $(a_n)_{n=1}^\infty = (-\infty)_{n=1}^\infty$, which is a constant sequence and of course diverges to $-\infty$. If $A \in \mathbb{R}$, then (a_n) converges to A by the proof of Theorem B.14. If $A = +\infty$, then for any $M > 0$, there exists $N \in \mathbb{N}$ such that $a_N > M$. Hence for all $n \geq N$, we have $a_n \geq a_N > M$. Definition B.20 implies that (a_n) diverges to $+\infty$. $\square$

Corollary B.23. Let $(a_n)_{n=1}^\infty$ and $(b_n)_{n=1}^\infty$ be monotonically increasing and decreasing sequences in $\overline{\mathbb{R}}$, respectively. Then we have

$$\lim_{n \rightarrow \infty} a_n = \sup(a_n)_{n=1}^\infty, \quad (\text{B.9})$$

$$\lim_{n \rightarrow \infty} b_n = \inf(b_n)_{n=1}^\infty. \quad (\text{B.10})$$

Proof. This follows from the proof of Theorem B.22. $\square$

Theorem B.24. Consider $b_{ij} \geq 0$ in $\overline{\mathbb{R}}$ where $i, j \in \mathbb{N}$. If b_{ij} is monotonically increasing in i for each j and is monotonically increasing in j for each i , then we have

$$\lim_{i \rightarrow \infty} \lim_{j \rightarrow \infty} b_{ij} = \lim_{j \rightarrow \infty} \lim_{i \rightarrow \infty} b_{ij} \quad (\text{B.11})$$

with all the indicated limits existing in $\overline{\mathbb{R}}$.

Proof. Define $a_i := \lim_{j \rightarrow \infty} b_{ij}$ and $c_j := \lim_{i \rightarrow \infty} b_{ij}$. Since b_{ij} is monotonically increasing in j for each i , Corollary B.23 implies $a_i = \sup(b_{ij})_{j=0}^\infty$ for each $i \geq 0$. Similarly we have $c_i = \sup(b_{ij})_{i=0}^\infty$ for each $j \geq 0$. For $i_1 < i_2$, since

$$\forall j \geq 0, b_{i_1, j} \leq b_{i_2, j},$$

we have

$$a_{i_1} = \sup(b_{i_1, j})_{j=0}^\infty \leq \sup(b_{i_2, j})_{j=0}^\infty = a_{i_2}.$$

It follows that $(a_i)_{i=0}^\infty$ is monotonically increasing, and we have $\lim_{i \rightarrow \infty} a_i = \sup(a_i)_{i=0}^\infty$ from Corollary B.23. Similar arguments show that $\lim_{j \rightarrow \infty} c_j = \sup(c_j)_{j=0}^\infty$. Hence

$$\lim_{i \rightarrow \infty} \lim_{j \rightarrow \infty} b_{ij} = \lim_{i \rightarrow \infty} a_i = \sup(a_i)_{i=0}^\infty \geq \sup(b_{ij})_{i=0}^\infty = c_j$$

holds for all $j \geq 0$, which implies that

$$\lim_{i \rightarrow \infty} \lim_{j \rightarrow \infty} b_{ij} \geq \sup(c_j)_{j=0}^\infty = \lim_{j \rightarrow \infty} c_j = \lim_{j \rightarrow \infty} \lim_{i \rightarrow \infty} b_{ij}.$$

Similarly, we can also prove that

$$\lim_{i \rightarrow \infty} \lim_{j \rightarrow \infty} b_{ij} \leq \lim_{j \rightarrow \infty} \lim_{i \rightarrow \infty} b_{ij},$$

hence (B.11) holds. $\square$

Theorem B.25. Suppose (a_n) converges to $a \in \overline{\mathbb{R}}$ and $(b_n) \subset \mathbb{R}$ converges to $b \in \mathbb{R}$. Then

- (a) $(a_n \pm b_n)$ converges to $a \pm b$;
- (b) If $a \in \mathbb{R}$, then $(a_n b_n)$ converges to ab ;
- (c) $(b \neq 0 \text{ and } \forall n, b_n \neq 0) \implies \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}$.

Proof. (a) If $a \in \mathbb{R}$, the triangle inequality

$$|a_n \pm b_n - (a \pm b)| \leq |a_n - a| + |b_n - b|$$

yields $\lim_{n \rightarrow \infty} (a_n \pm b_n) = a \pm b$. If $a = \pm\infty$, (B.7a) yields

$$\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} (a_n) = a = a \pm b.$$

(b) By Lemmas B.11 and B.9, there exists $M \in \mathbb{R}^+$ such that $|b_n| \leq M$ for all n . The triangle inequality yields

$$\begin{aligned} |a_n b_n - ab| &\leq |a_n b_n - ab_n| + |ab_n - ab| \\ &\leq M|a_n - a| + |a||b_n - b|, \end{aligned}$$

which implies the convergence of $(a_n b_n)$ to ab .

(c) Since $b \neq 0$, there exists $N \in \mathbb{N}$ and $L > 0$ such that $|b_n| \geq L$ for all $n \geq N$. WLOG we assume that $|b_n| \geq L$ for all n . The triangle inequality yields

$$\begin{aligned} \left| \frac{a_n}{b_n} - \frac{a}{b} \right| &\leq \frac{1}{L|b|} |a_n b - ab_n| \leq \frac{1}{L|b|} (|a_n b - ab| + |ab - ab_n|) \\ &= \frac{1}{L} |a_n - a| + \frac{|a|}{L|b|} |b_n - b|, \end{aligned}$$

which implies the convergence of $(\frac{a_n}{b_n})$ to $\frac{a}{b}$. $\square$

Corollary B.26. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence. Then $\lim_{n \rightarrow \infty} a_n = L$ if and only if $\lim_{n \rightarrow \infty} |a_n - L| = 0$.

Proof. By Theorem B.25(a) and Lemma B.5,

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n = L &\iff \lim_{n \rightarrow \infty} (a_n - L) = 0 \\ &\iff \lim_{n \rightarrow \infty} |a_n - L| = 0. \end{aligned} \quad \square$$

B.1.3 $\limsup$, $\liminf$, and the asymptotic notation

Definition B.27. Let $\epsilon > 0$ be a real number. Two real numbers x, y are said to be ϵ -close iff $|x - y| \leq \epsilon$.

Definition B.28. A real number x is said to be ϵ -adherent to a sequence $(a_n)_{n=m}^\infty \subset \mathbb{R}$ iff there exists an $n \geq m$ such that a_n is ϵ -close to x . x is *continually* ϵ -adherent to $(a_n)_{n=m}^\infty$ iff it is ϵ -adherent to $(a_n)_{n=N}^\infty$ for every $N \geq m$.

Definition B.29. A real number x is a *limit point* or *adherent point* of a sequence $(a_n)_{n=m}^\infty \subset \mathbb{R}$ iff it is continually ϵ -adherent to $(a_n)_{n=m}^\infty$ for every $\epsilon > 0$.

Definition B.30. The *limit superior* of a sequence $(a_n)_{n=m}^\infty \subset \mathbb{R}$ is

$$\limsup_{n \rightarrow \infty} a_n := \inf(a_N^+)_{N=m}^\infty, \quad (\text{B.12})$$

where $a_N^+ = \sup(a_n)_{n=N}^\infty$. The *limit inferior* of $(a_n)_{n=m}^\infty$ is

$$\liminf_{n \rightarrow \infty} a_n := \sup(a_N^-)_{N=m}^\infty, \quad (\text{B.13})$$

where $a_N^- = \inf(a_n)_{n=N}^\infty$.

Example B.31. Let $(a_n)_{n=m}^{\infty}$ be the sequence

$$1.1, -1.01, 1.001, -1.0001, 1.00001, \dots$$

Then $(a_n^+)_{n=N}^{\infty}$ is the sequence

$$1.1, 1.001, 1.001, 1.00001, 1.00001, \dots$$

and $(a_n^-)_{n=N}^{\infty}$ is the sequence

$$-1.01, -1.01, -1.0001, -1.0001, -1.000001, -1.000001, \dots$$

Hence we have

$$\limsup_{n \rightarrow \infty} a_n = 1, \quad \liminf_{n \rightarrow \infty} a_n = -1.$$

Lemma B.32 (Properties of $\limsup$ and $\liminf$). The limit superior $L^+ = \limsup_{n \rightarrow \infty} a_n$ and the limit inferior $L^- = \liminf_{n \rightarrow \infty} a_n$ of a sequence $(a_n)_{n=m}^{\infty} \subset \mathbb{R}$ satisfy

(a) For every $x > L^+$, elements of the sequence are eventually less than x :

$$\forall x > L^+, \exists N \geq m \text{ s.t. } \forall n \geq N, a_n < x.$$

Similarly, for every $x < L^-$, elements of the sequence are eventually greater than x :

$$\forall x < L^-, \exists N \geq m \text{ s.t. } \forall n \geq N, a_n > x;$$

(b) For every $x < L^+$, there are an infinite number of elements in the sequence that are greater than x :

$$\forall x < L^+, \forall N \geq m, \exists n \geq N \text{ s.t. } a_n > x.$$

Similarly, for every $x > L^-$, there are an infinite number of elements in the sequence that are less than x :

$$\forall x > L^-, \forall N \geq m, \exists n \geq N \text{ s.t. } a_n < x;$$

(c) $\inf(a_n)_{n=m}^{\infty} \leq L^- \leq L^+ \leq \sup(a_n)_{n=m}^{\infty}$;

(d) Any limit point c of $(a_n)_{n=m}^{\infty}$ satisfies $L^- \leq c \leq L^+$;

(e) If L^+ (or L^-) is finite, then it is a limit point of $(a_n)_{n=m}^{\infty}$;

(f) $\lim_{n \rightarrow \infty} a_n = c$ if and only if $L^+ = L^- = c$.

Proof. (a) WLOG, we only prove the first statement. This statement clearly holds if $L^+ = +\infty$.

If $L^+ = -\infty$, then for every $x > -\infty$, there exists $N \geq m$ such that $a_N^+ < x$ by Definition A.52 and $L^+ = \inf(a_n^+)_{n=N}^{\infty}$. Then $a_N^+ = \sup(a_n)_{n=N}^{\infty}$ implies $a_n < x$ for all $n \geq N$.

If $L^+ \in \mathbb{R}$, define $\epsilon := \frac{x-L^+}{2} > 0$ for any $x > L^+$. Since $L^+ = \limsup a_n = \inf(a_n^+)_{n=N}^{\infty}$, there exists $N \geq m$ such that $a_N^+ < L^+ + \epsilon = \frac{x+L^+}{2} < x$. Then $a_N^+ = \sup(a_n)_{n=N}^{\infty}$ implies $a_n < x$ for all $n \geq N$.

(b) WLOG, we only prove the first statement. This statement clearly holds if $L^+ = -\infty$.

If $L^+ = +\infty$, then $a_N^+ = +\infty$ for all $N \geq m$ by the definition of L^+ . Hence for any $x < +\infty$ and $N \geq m$, there exists $n \geq N$ such that $a_n > x$ by the definition of a_N^+ .

If $L^+ \in \mathbb{R}$, for every $x < L^+$, we have $x < a_N^+$ for all $N \geq m$ by the definition of L^+ . Let $\epsilon = \frac{a_N^+ - x}{2} > 0$. Since $a_N^+ = \sup(a_n)_{n=N}^{\infty}$, there exists $n \geq N$ such that $a_n > a_N^+ - \epsilon = \frac{a_N^+ + x}{2} > x$.

(c) By definitions of L^+ and L^- , it follows that

$$L^+ = \inf(a_n^+)_{n=m}^{\infty} \leq a_m^+ = \sup(a_n)_{n=m}^{\infty},$$

$$L^- = \sup(a_n^-)_{n=m}^{\infty} \geq a_m^- = \inf(a_n)_{n=m}^{\infty}.$$

Hence it suffices to show that $L^- \leq L^+$. For any $k \in \mathbb{N}$,

$$\begin{aligned} a_N^+ \geq a_{N+k} &\implies \inf(a_n^+)_{n=N}^{\infty} \geq \inf(a_{N+k})_{n=N}^{\infty} \\ &\implies L^+ \geq \inf(a_n)_{n=N+k}^{\infty} = a_{m+k}^-. \end{aligned}$$

Hence $a_N^- \leq L^+$ for all $N \geq m$ and

$$L^- = \sup(a_n^-)_{n=m}^{\infty} \leq L^+.$$

(d) Suppose that $c > L^+$. By the proof of (a), there exists $N \geq m$ such that $a_n < \frac{L^+ + c}{2}$ for all $n \geq N$. Let $\epsilon = \frac{c-L^+}{2} > 0$. Then

$$(*) : \forall n \geq N, c - a_n > c - \frac{1}{2}(L^+ + c) = \frac{1}{2}(c - L^+) = \epsilon.$$

However, c is a limit point of $(a_n)_{n=m}^{\infty}$ implies there exists $n \geq N$ such that $|a_n - c| < \epsilon$, which contradicts $(*)$ and hence $c \leq L^+$. Similar arguments yield $c \geq L^-$.

(e) If L^+ is finite, by the definition of L^+ , for any $\epsilon > 0$, there exists $n_0 \geq m$ such that $L^+ \leq a_{n_0}^+ < L^+ + \epsilon$. For every $N \geq m$, since $a_{n_0}^+ \geq L^+ > L^+ - \epsilon$, there exists $n \geq \max\{N, n_0\}$ such that $a_n > L^+ - \epsilon$. Hence $L^+ - \epsilon < a_n \leq a_{n_0}^+ < L^+ + \epsilon$, i.e., a_n is ϵ -close to L^+ , which implies that L^+ is continually ϵ -adherent to $(a_n)_{n=m}^{\infty}$. Since $\epsilon > 0$ is arbitrary, we conclude that L^+ is a limit point of $(a_n)_{n=m}^{\infty}$. The case of L^- can be shown similarly.

(f) For the necessity, if $c = +\infty$, then for any $M > 0$, there exists $N_0 \geq m$ such that $a_n > M$ for all $n \geq N_0$. Hence

$$L^- = \sup(a_n^-)_{n=m}^{\infty} \geq a_{N_0}^- = \inf(a_n)_{n=N_0}^{\infty} \geq M.$$

Take $M \rightarrow +\infty$ and we have $L^- = +\infty$, which implies $L^+ = +\infty$ because of (c). Hence $L^+ = L^- = c = +\infty$. The case for $c = -\infty$ can be shown similarly.

If $c \in \mathbb{R}$, then $\lim_{n \rightarrow \infty} a_n = c$ implies that c is a limit point of $(a_n)_{n=m}^{\infty}$, hence $L^- \leq c \leq L^+$ from (d). Now for any $\epsilon > 0$, there exists $N_0 \geq m$ such that $|a_n - c| < \epsilon$ for all $n \geq N_0$. Hence

$$L^+ = \inf(a_n^+)_{n=m}^{\infty} \leq a_{N_0}^+ = \sup(a_n)_{n=N_0}^{\infty} \leq c + \epsilon.$$

Similarly we can get $L^- \geq c - \epsilon$. Take $\epsilon \rightarrow 0^+$ and we have $L^+ \leq c \leq L^-$, which shows that $L^+ = L^- = c$.

For the sufficiency, if $c = +\infty$, suppose that (a_n) does not diverge to $+\infty$, i.e.,

$$\exists M_0 > 0, \text{ s.t. } \forall N \geq m, \exists n_0 \geq N \text{ s.t. } a_{n_0} \leq M_0.$$

Then we can take $m \leq n_1 \leq n_2 \leq \dots$, s.t. $a_{n_j} \leq M_0$ for all $j \geq 1$. Then for any $N \geq m$, there exists $j_0 \geq 1$ such that $n_{j_0} \geq N$ and

$$a_N^- = \inf(a_n)_{n=N}^{\infty} \leq a_{n_{j_0}} \leq M_0.$$

Hence $L^- = \sup(a_N^-)_{N=m}^\infty \leq M_0 < +\infty$, which contradicts $L^- = c = +\infty$. The case of $c = -\infty$ can be shown similarly.

If $c \in \mathbb{R}$, suppose that (a_n) does not converge to c , i.e.,

$$\exists \epsilon_0 > 0, \text{ s.t. } \forall N \geq m, \exists n_0 \geq N \text{ s.t. } |a_{n_0} - c| \geq \epsilon_0.$$

Then we can take $m \leq n_1 \leq n_2 \leq \dots$, s.t. $|a_{n_j} - c| \geq \epsilon_0$ for all $j \geq 1$. Define the index sets

$$J_1 := \{j \geq 1 : a_{n_j} - c \geq \epsilon_0\}, \quad J_2 := \{j \geq 1 : a_{n_j} - c \leq -\epsilon_0\}.$$

Then one of them must be infinite. WLOG we assume J_1 is infinite. Then for any $N \geq m$, there exists $j_0 \in J_1$ such that $n_{j_0} \geq N$ and

$$a_N^+ = \sup(a_n)_{n=N}^\infty \geq a_{n_{j_0}} \geq \epsilon_0 + c.$$

Hence $L^+ = \inf(a_N^+)_{N=m}^\infty \geq \epsilon_0 + c > c$, which contradicts $L^+ = c$. $\square$

Lemma B.33. Suppose both the limit superior and limit inferior exist for a sequence $(a_n)_{n=m}^\infty$. Then

$$\forall M > m, \begin{cases} \limsup(a_n)_{n=m}^\infty = \limsup(a_n)_{n=M}^\infty, \\ \liminf(a_n)_{n=m}^\infty = \liminf(a_n)_{n=M}^\infty. \end{cases} \quad (\text{B.14})$$

Proof. This follows from Definition B.30. $\square$

Lemma B.34. Suppose $(a_n)_{n=m}^\infty \subset \mathbb{R}$ and $(b_n)_{n=m}^\infty \subset \mathbb{R}$ satisfy

$$\exists M \in \mathbb{N} \text{ s.t. } \forall n \geq M, a_n \leq b_n.$$

Then $\limsup a_n \leq \limsup b_n$ and $\liminf a_n \leq \liminf b_n$.

Proof. We only prove the first inequality since the second can be shown similarly.

$$\forall n \geq M, a_n \leq b_n$$

$$\implies \sup(a_n)_{n=M}^\infty \leq \sup(b_n)_{n=M}^\infty \iff (a_N^+)_{N=M}^\infty \leq (b_N^+)_{N=M}^\infty$$

$$\implies \inf(a_N^+)_{N=M}^\infty \leq \inf(b_N^+)_{N=M}^\infty$$

$$\iff \inf(a_N^+)_{N=m}^\infty \leq \inf(b_N^+)_{N=m}^\infty \iff \limsup a_n \leq \limsup b_n$$

where the second equivalence follows from Lemma B.33. $\square$

Lemma B.35. Let $(a_n)_{n=m}^\infty$ and $(b_n)_{n=m}^\infty$ be two convergent sequences of real numbers that satisfy

$$\exists M \in \mathbb{N} \text{ s.t. } \forall n \geq M, a_n \leq b_n.$$

Then we have $\lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n$.

Proof. Lemmas B.32(f) and B.34 yield

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} \sup_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} b_n. \quad \square$$

Theorem B.36 (Squeeze test or the sandwich theorem). Let $(a_n)_{n=m}^\infty$, $(b_n)_{n=m}^\infty$, and $(c_n)_{n=m}^\infty$ be sequences of real numbers that satisfy

$$\exists M \in \mathbb{N} \text{ s.t. } \forall n \geq M, a_n \leq b_n \leq c_n.$$

Suppose $(a_n)_{n=m}^\infty$ and $(c_n)_{n=m}^\infty$ both converge to the same limit L . Then $(b_n)_{n=m}^\infty$ also converges to L .

Proof. We have

$$\forall n \geq M, a_n \leq b_n \leq c_n$$

$$\implies \limsup a_n \leq \limsup b_n \leq \limsup c_n$$

$$\implies L \leq \limsup b_n \leq L$$

$$\implies \limsup b_n = L,$$

where the first step follows from Lemma B.34 and the second step from Lemma B.32(f). Similarly we have $\liminf b_n = L$. By Lemma B.32(f) again, we have $\lim_{n \rightarrow \infty} b_n = L$. $\square$

Notation 19 (Asymptotic notation). For $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, $f : \mathbb{R} \rightarrow \mathbb{R}$, and $a \in [0, +\infty]$, we write

$$f(x) = O(g(x)) \text{ as } x \rightarrow a$$

iff

$$\limsup_{x \rightarrow a} \frac{|f(x)|}{g(x)} < \infty.$$

In particular, we have

$$f(x) = o(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{|f(x)|}{g(x)} = 0.$$

We also write

$$f(x) = \Theta(g(x)) \text{ as } x \rightarrow a$$

iff

$$0 < \limsup_{x \rightarrow a} \frac{|f(x)|}{g(x)} < \infty.$$

When there is no danger of ambiguity, we often write $f(x) = O(g(x))$, $f(x) = o(g(x))$, and $f(x) = \Theta(g(x))$ without the conditional $x \rightarrow a$ in the above equations.

The notations $\mathbf{f} = O(g(k))$ and $\mathbf{f} = \Theta(g(k))$ for a function $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{R}^n$ mean that each component f_i of $\mathbf{f}$ satisfies $f_i = O(g(k))$ and $f_i = \Theta(g(k))$, respectively.

B.1.4 Series

Definition B.37 (Finite series). Let m, n be integers and let $(a_i)_{i=m}^n$ be a finite sequence of real numbers. The *finite series* or *finite sum* associated with the sequence $(a_i)_{i=m}^n$ is the number $\sum_{i=m}^n a_i$ given by the recursive formula

$$\sum_{i=m}^n a_i := \begin{cases} 0 & \text{if } n < m; \\ a_n + \sum_{i=m}^{n-1} a_i & \text{otherwise.} \end{cases} \quad (\text{B.15})$$

Definition B.38. A (formal) *infinite series* associated with an infinite sequence (a_n) is the expression $\sum_{n=0}^\infty a_n$.

Definition B.39. The n th term of the *sequence of partial sums* $(S_n)_{n=0}^\infty$ associated with a formal infinite series $\sum_{i=0}^\infty a_i$ is the sum of the sequence (a_i) from a_0 to a_n

$$S_n = \sum_{i=0}^n a_i. \quad (\text{B.16})$$

Definition B.40. A formal infinite series is said to be *convergent* and *converge* to L if its sequence of partial sums converges to some limit L . In this case we write $L = \sum_{n=0}^\infty a_n$ and call L the *sum of the infinite series*.

Lemma B.41. A convergent infinite series $\sum_{n=0}^{\infty} a_n$ of real numbers must have $\lim_{n \rightarrow \infty} a_n = 0$.

Proof. Assume that the series converges to $L \in \mathbb{R}$. For S_n in (B.16), we have $a_n = S_n - S_{n-1}$ for all n . Then Theorem B.25(a) yields $\lim_{n \rightarrow \infty} a_n = L - L = 0$. $\square$

Definition B.42. A formal infinite series is said to *diverge* or be *divergent* if its sequence of partial sums does not converge.

Lemma B.43 (Cauchy criterion). An infinite series $\sum_{n=0}^{\infty} a_n$ of real numbers is convergent if and only if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall p, q \geq N, \left| \sum_{n=p}^q a_n \right| \leq \epsilon. \quad (\text{B.17})$$

Proof. $\sum_{n=0}^{\infty} a_n$ is convergent if and only if its sequence (S_n) of partial sums in Definition B.39 is convergent, if and only if (S_n) is Cauchy by Theorem B.17, if and only if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall p, q \geq N, |S_q - S_{p-1}| = \left| \sum_{n=p}^q a_n \right| \leq \epsilon. \quad \square$$

Theorem B.44 (Comparison test). Let $\sum a_n$ be a series where $a_n \geq 0$ for all n .

- (i) If $\sum a_n$ converges and $|b_n| \leq a_n$ for all n , then $\sum b_n$ converges;
- (ii) If $\sum a_n = +\infty$ and $b_n \geq a_n$ for all n , then $\sum b_n = +\infty$.

Proof. For any $p, q \in \mathbb{N}$, we have

$$\left| \sum_{n=p}^q b_n \right| \leq \sum_{n=p}^q |b_n| \leq \sum_{n=p}^q a_n = \left| \sum_{n=p}^q a_n \right|.$$

Then (i) follows from Lemma B.43.

By Definition B.20 and the condition $\sum_{j=1}^n b_j \geq \sum_{j=1}^n a_j$ for all $n \in \mathbb{N}^+$, the divergence of $(\sum_{j=1}^n a_j)_{n=1}^{\infty}$ to $+\infty$ implies the divergence of $(\sum_{j=1}^n b_j)_{n=1}^{\infty}$ to $+\infty$. $\square$

Theorem B.45. The *geometric series* $\sum_{n=0}^{\infty} x^n$ diverges to $+\infty$ if $x \geq 1$ and converges to $\frac{1}{1-x}$ if $x \in [0, 1)$.

Proof. For $x = 1$, the geometric series $\sum_0^{\infty} x^n = 1 + 1 + \dots = \infty$. The case of $x > 1$ follows from the comparison test. For $x \in [0, 1)$, $\sum_{j=0}^n x^j = \frac{1-x^{n+1}}{1-x}$ and the conclusion follows. $\square$

Definition B.46. An infinite series $\sum_{n=0}^{\infty} a_n$ is *absolutely convergent* iff the series $\sum_{n=0}^{\infty} |a_n|$ is convergent.

Lemma B.47. An absolutely convergent infinite series is convergent.

Proof. Since $|a_n| \geq 0$, $\sum |a_n|$ converges and $|a_n| \leq |a_n|$ for all n , the proof is completed by Theorem B.44(i). $\square$

Theorem B.48 (Ratio test). A series $\sum a_n$ of nonzero real numbers

- (i) converges absolutely if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$;
- (ii) diverges if $\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$.

Otherwise, this test gives no information about the convergence of $\sum a_n$.

Proof. Write $\alpha := \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$. Choose $\epsilon > 0$ such that $\alpha + \epsilon \in (0, 1)$. Then Lemma B.32(a) implies

$$\exists N \in \mathbb{N}^+ \text{ s.t. } \forall n \geq N, \left| \frac{a_{n+1}}{a_n} \right| < \alpha + \epsilon$$

and hence

$$\forall n \geq N, |a_n| = \left| \frac{a_n}{a_{n-1}} \cdot \frac{a_{n-1}}{a_{n-2}} \cdots \frac{a_{N+1}}{a_N} \cdot a_N \right| < |a_N|(\alpha + \epsilon)^{n-N}.$$

Then (i) follows from Theorem B.44 by comparing $(|a_n|)$ with the geometric series. (ii) can be shown similarly. $\square$

Theorem B.49 (Root test). A series $\sum a_n$ of real numbers

- (i) converges absolutely if $\alpha < 1$;
- (ii) diverges if $\alpha > 1$,

where $\alpha = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}$. Otherwise, this test gives no information about the convergence of $\sum a_n$.

Proof. Since $\alpha < 1$, we can choose $\epsilon > 0$ so that $\alpha + \epsilon \in (0, 1)$. Lemma B.32(a) gives

$$\exists N \in \mathbb{N}^+ \text{ s.t. } \forall n \geq N, |a_n|^{\frac{1}{n}} < \alpha + \epsilon.$$

Hence $|a_n| < (\alpha + \epsilon)^n$ for all $n \geq N$. The comparison test with the geometric series yields (i).

We prove (ii) by contradiction. Suppose $\sum a_n$ converges. Then Lemma B.41 yields $\lim a_n = 0$. By Lemma B.32(b), there exists a subsequence a_{n_j} such that $|a_{n_j}|^{\frac{1}{n_j}} > 1$, i.e., $|a_{n_j}| > 1$ for all j . But Lemma B.7 implies $\lim_{j \rightarrow \infty} a_{n_j} = \lim_{n \rightarrow \infty} a_n = 0$, which contradicts $|a_{n_j}| > 1$ for all j . $\square$

Theorem B.50 (Integral test). Let $f : [0, \infty) \rightarrow \mathbb{R}^+ \cup \{0\}$ be a monotonically decreasing function, i.e., $x \geq y$ implies $f(x) \leq f(y)$. Then the series $\sum_{n=0}^{\infty} f(n)$ is convergent if and only if $\sup_{N \in \mathbb{N}^+} \int_0^N f$ is finite.

Proof. Since f is monotonically decreasing and non-negative, the integral mean value theorem B.124 yields

$$\forall n \in \mathbb{N}^+, f(n) \leq \int_{n-1}^n f \leq f(n-1),$$

which implies that

$$(*) : \quad \forall N \in \mathbb{N}^+, \sum_{n=1}^N f(n) \leq \int_0^N f \leq \sum_{n=0}^{N-1} f(n).$$

For necessity, suppose $\sum_{n=0}^{\infty} f(n) = L < +\infty$. Then

$$\forall N \in \mathbb{N}^+, \int_0^N f \leq \sum_{n=0}^{N-1} f(n) \leq \sum_{n=0}^{\infty} f(n) = L,$$

where the first inequality follows from $(*)$ and the second from $f \geq 0$.

For sufficiency, there exists $M > 0$ such that $\int_0^N f \leq M$ for all $N \in \mathbb{N}^+$. Then we have

$$\begin{aligned} \forall N \in \mathbb{N}^+, \sum_{n=0}^N f(n) &= \sum_{n=1}^N f(n) + f(0) \\ &\leq \int_0^N f + f(0) \leq M + f(0), \end{aligned}$$

where the first inequality follows from (*). By Definition B.12 and $f \geq 0$, the sequence $(\sum_{n=0}^N f(n))_{N \in \mathbb{N}^+}$ is monotonically decreasing. The rest follows from Theorem B.14. $\square$

B.2 Scalar functions $\mathbb{R} \rightarrow \mathbb{R}$

Definition B.51. A *scalar function* is a function whose range is a subset of $\mathbb{R}$.

Definition B.52 (Limit of a scalar function). Consider a function $f : I \rightarrow \mathbb{R}$ with $I = I(c, r) = (c - r, c) \cup (c, c + r)$, where $c \in \mathbb{R}$ and $r \in \mathbb{R}^+$. The *limit* of $f(x)$ exists as x approaches c , written $\lim_{x \rightarrow c} f(x) = L$, iff

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall x \in I(c, \delta), |f(x) - L| < \epsilon. \quad (\text{B.18})$$

Lemma B.53. Let $f : I(c, r) \rightarrow \mathbb{R}$ be a scalar function. Then

$$\lim_{x \rightarrow c} f(x) = 0 \iff \left| \lim_{x \rightarrow c} f(x) \right| = 0 \iff \lim_{x \rightarrow c} |f(x)| = 0.$$

Proof. It follows from (B.18) with $L = 0$. $\square$

Lemma B.54. Let f be a scalar function. Then the limit of $f(x)$ at c is unique.

Proof. The proof is similar to that of Lemma B.6. $\square$

Example B.55. We show $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2}$ as follows. If $\epsilon \geq \frac{1}{2}$, choose $\delta = 1$. Then $x \in (1, 3)$ implies $|\frac{1}{x} - \frac{1}{2}| < \frac{1}{2}$ since $\frac{1}{x} - \frac{1}{2}$ is a monotonically decreasing function with its supremum at $x = 1$.

If $\epsilon \in (0, \frac{1}{2})$, choose $\delta = \epsilon$. Then $x \in (2 - \epsilon, 2 + \epsilon) \subset (\frac{3}{2}, \frac{5}{2})$. Hence $|\frac{1}{x} - \frac{1}{2}| = \frac{|2-x|}{|2x|} < |2-x| < \epsilon$. The proof is completed by Definition B.52.

Theorem B.56. Let f, g be scalar functions. Suppose $\lim_{x \rightarrow c} f(x) = L_1$ and $\lim_{x \rightarrow c} g(x) = L_2$. Then

- (a) $\lim_{x \rightarrow c} (f \pm g)(x) = L_1 \pm L_2$;
- (b) $\lim_{x \rightarrow c} (fg)(x) = L_1 L_2$;
- (c) $(L_2 \neq 0 \text{ and } \forall x, g(x) \neq 0) \implies \lim_{x \rightarrow c} (\frac{f}{g})(x) = \frac{L_1}{L_2}$.

Proof. It follows from Definition B.52 and the proof of Theorem B.25. $\square$

Theorem B.57. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then $\lim_{x \rightarrow c} f(x) = L$ if and only if for any sequence $(a_n)_{n \in \mathbb{N}^+}$ satisfying $a_n \neq c$ for all n and $\lim_{n \rightarrow \infty} a_n = c$, we have $\lim_{n \rightarrow \infty} f(a_n) = L$.

Proof. For the necessity, Definition B.52 implies that

$$(*) : \quad \forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall 0 < |x - c| < \delta, |f(x) - L| < \epsilon.$$

Since $\lim_{n \rightarrow \infty} a_n = c$, there exists $N \in \mathbb{N}^+$ such that $|a_n - c| < \delta$ for all $n > N$. Then $a_n \neq c$ and (*) imply $|f(a_n) - L| < \epsilon$ for all $n > N$, which yields the convergence of $(f(a_n))$ to L .

For the sufficiency, suppose that $\lim_{x \rightarrow c} f(x) = L$ does not hold. Then we have

$$(**) : \quad \exists \epsilon_0 > 0, \text{ s.t. } \forall n \in \mathbb{N}^+, \exists a_n, 0 < |a_n - c| < \frac{1}{n},$$

$$\text{ s.t. } |f(a_n) - L| \geq \epsilon_0.$$

The squeeze test implies that $\lim_{n \rightarrow \infty} |a_n - c| = 0$, which, by Corollary B.26, is equivalent to $\lim_{n \rightarrow \infty} a_n = c$. Then we must have $\lim_{n \rightarrow \infty} f(a_n) = L$, which contradicts (**). $\square$

Example B.58. We show that $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist as follows. Suppose that $\lim_{x \rightarrow 0} f(x)$ exists, where $f(x) := \sin \frac{1}{x}, x \in \mathbb{R} \setminus \{0\}$. Then we define two sequences $(a_n), (b_n)$, where $a_n := (\frac{\pi}{2} + 2n\pi)^{-1}$, $b_n := (2n\pi)^{-1}$, $n \geq 1$. Since $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = 0$, Theorem B.57 implies that $\lim_{n \rightarrow \infty} f(a_n) = \lim_{n \rightarrow \infty} f(b_n)$. It contradicts the fact

$$\begin{aligned} \forall n \geq 1, \quad f(a_n) &= \sin \frac{1}{a_n} = \sin(\frac{\pi}{2} + 2n\pi) = 1, \\ f(b_n) &= \sin \frac{1}{b_n} = \sin(2n\pi) = 0. \end{aligned}$$

Theorem B.59 (Cauchy criterion for limit of a scalar function). Let $f : I \rightarrow \mathbb{R}$ with $I = I(c, r) = (c - r, c) \cup (c, c + r)$. Then $\lim_{x \rightarrow c} f(x)$ exists if and only if

$$\begin{aligned} \forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ \forall x, y \in I(c, \delta), |f(x) - f(y)| < \epsilon. \end{aligned} \quad (\text{B.19})$$

Proof. For the necessity, let $L := \lim_{x \rightarrow c} f(x)$. Definition B.52 implies

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall x \in I(c, \delta), |f(x) - L| < \frac{\epsilon}{2}.$$

Hence

$$\begin{aligned} \forall x, y \in I(c, \delta), \\ |f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

For the sufficiency, for any sequence (a_n) satisfying $a_n \neq c$ and $\lim a_n = c$, we have

$$\begin{aligned} \exists N \in \mathbb{N}^+, \text{ s.t. } \forall n > N, 0 < |a_n - c| < \delta \\ \implies \exists N \in \mathbb{N}^+, \text{ s.t. } \forall m, n > N, a_m, a_n \in I(c, \delta) \\ \implies \exists N \in \mathbb{N}^+, \text{ s.t. } |f(a_m) - f(a_n)| < \epsilon, \forall m, n > N. \end{aligned}$$

Hence Definition B.8 yields that $(f(a_n))$ is Cauchy. Then Theorems B.16 and B.57 complete the proof. $\square$

Definition B.60 (One sided limit of a scalar function). Consider a function $f : I \rightarrow \mathbb{R}$ with $I = (c - r, c) \cup (c, c + r)$. The *left-sided limit* or *left-hand limit* of $f(x)$ exists as x approaches c from the left side of c , written $\lim_{x \rightarrow c^-} f(x) = L$, iff

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall x \in (c - r, c), |f(x) - L| < \epsilon. \quad (\text{B.20})$$

When this limit exists, we usually denote it by $f(c^-)$, i.e.,

$$f(c^-) = \lim_{x \rightarrow c^-} f(x).$$

Similarly, $f(c^+)$, the *right-sided limit* or *right-hand limit* of $f(x)$ at c , can be also defined.

Theorem B.61. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then $\lim_{x \rightarrow c} f(x)$ exists if and only if $f(c^-) = f(c^+)$.

Proof. It follows from Definitions B.52 and B.60. $\square$

Theorem B.62. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then $f(c^-) = L$ (or $f(c^+) = L$) if and only if for any sequence $(a_n)_{n \in \mathbb{N}^+}$ satisfying $a_n < c$ (or $a_n > c$) for all n and $\lim_{n \rightarrow \infty} a_n = c$, we have $\lim_{n \rightarrow \infty} f(a_n) = L$.

Proof. It follows from the proof of Theorem B.57. $\square$

Theorem B.63 (Cauchy criterion for one sided limit of a scalar function). Let $f : I \rightarrow \mathbb{R}$ with $I = (c - r, c)$ (or $I = (c, c + r)$). Then $f(c^-)$ (or $f(c^+)$) exists if and only if

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ & \forall x, y \in (c - \delta, c) \text{ (or } (c, c + \delta)), |f(x) - f(y)| < \epsilon. \end{aligned} \quad (\text{B.21})$$

Proof. It follows from the proof of Theorem B.59. $\square$

B.2.1 Continuous scalar functions

Definition B.64. $f : \mathbb{R} \rightarrow \mathbb{R}$ is *continuous* at c iff

$$\lim_{x \rightarrow c} f(x) = f(c). \quad (\text{B.22})$$

Theorem B.65. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and $c \in \mathbb{R}$. Then f is continuous at c if and only if for any sequence $(a_n)_{n \in \mathbb{N}^+}$ satisfying $a_n \neq c$ for all n and $\lim_{n \rightarrow \infty} a_n = c$, the sequence $(f(a_n))_{n \in \mathbb{N}^+}$ is convergent to $f(c)$.

Proof. It follows from Definition B.64 and Theorem B.57 directly. $\square$

Definition B.66. $f : \mathbb{R} \rightarrow \mathbb{R}$ is *left-continuous* at c iff

$$f(c^-) = \lim_{x \rightarrow c^-} f(x) = f(c). \quad (\text{B.23})$$

f is *right-continuous* at c iff

$$f(c^+) = \lim_{x \rightarrow c^+} f(x) = f(c). \quad (\text{B.24})$$

Theorem B.67. $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at c if and only if $f(c^-) = f(c^+) = f(c)$.

Proof. It follows from Definition B.64, Definition B.66 and Theorem B.57. $\square$

Theorem B.68. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$. Then f is left-continuous (or right-continuous) at c if and only if for any sequence $(a_n)_{n \in \mathbb{N}^+}$ satisfying $a_n < c$ (or $a_n > c$) for all n and $\lim_{n \rightarrow \infty} a_n = c$, the sequence $(f(a_n))_{n \in \mathbb{N}^+}$ is convergent to $f(c)$.

Proof. It follows from Definition B.66 and Theorem B.62. $\square$

Theorem B.69. Let f and g be scalar functions. Suppose f and g are both continuous at x . Then $f \pm g$, fg are also continuous at x . If $g(x) \neq 0$, then $\frac{f}{g}$ is continuous at x .

Proof. It follows from Theorem B.56 and Definition B.64. $\square$

Definition B.70. A scalar function f is *continuous on* (a, b) , written $f \in \mathcal{C}(a, b)$, if (B.22) holds for all $x \in (a, b)$. f is *continuous on* $[a, b]$, written $f \in \mathcal{C}[a, b]$, if $f \in \mathcal{C}(a, b)$, f is right-continuous at a and f is left-continuous at b .

Theorem B.71. A scalar function f is bounded on $[a, b]$ if $f \in \mathcal{C}[a, b]$.

Proof. By Definition B.70, $-f \in \mathcal{C}[a, b]$. Hence it suffices to show that f is bounded above. We prove this by contradiction. Suppose that

$$(*) : \quad \forall n \in \mathbb{N}^+, \exists x_n \in [a, b], \text{ s.t. } f(x_n) > n.$$

Then the Bolzano-Weierstrass theorem ensures that the sequence (x_n) has a subsequence (x_{n_k}) which converges to $c \in \mathbb{R}$. Moreover, $a \leq x_{n_k} \leq b$ and Lemma B.35 imply that $a \leq c \leq b$. By $(*)$, there are only finitely many terms in (x_{n_k}) equal to c . Hence we may assume that $x_{n_k} \neq c$ for all $k \in \mathbb{N}^+$. Then Theorem B.65 yields

$$(**) : \quad \lim_{k \rightarrow \infty} f(x_{n_k}) = f(c).$$

However, we have $f(x_{n_k}) > n_k \geq k$ for all k . Then

$$\exists k_0 > |f(c)| + 1, \text{ s.t.}$$

$$\forall k > k_0, f(x_{n_k}) > k > k_0 > |f(c)| + 1 \geq f(c) + 1.$$

Hence Lemma B.35 yields

$$\lim_{k \rightarrow \infty} f(x_{n_k}) \geq f(c) + 1 > f(c),$$

which contradicts $(**)$. $\square$

Theorem B.72 (Extreme values). A continuous function $f : [a, b] \rightarrow \mathbb{R}$ attains its maximum at some point $x_{\max} \in [a, b]$ and its minimum at some point $x_{\min} \in [a, b]$.

Proof. WLOG, we only prove the first statement. Define $S := \{f(x) : x \in [a, b]\}$, which is a nonempty set. Theorem B.71 yields the boundedness of S , and then Theorem A.55 yields that S has exactly one least upper bound, denoted by M . Then we have

$$M = \sup S = \sup_{x \in [a, b]} f(x) < +\infty.$$

Hence Definition A.52 yields

$$\forall n \in \mathbb{N}^+, \exists x_n \in [a, b], \text{ s.t. } M - \frac{1}{n} < f(x_n) \leq M.$$

By similar arguments in the proof of Theorem B.71, we may assume $\lim_{n \rightarrow \infty} x_n = c \in [a, b]$ and $x_n \neq c$ for all n . Then the squeeze test and Theorem B.65 yield

$$f(c) = \lim_{n \rightarrow \infty} f(x_n) = M.$$

Hence f attains its maximum at $x_{\max} = c \in [a, b]$. $\square$

Theorem B.73 (Zero value). Let $f \in \mathcal{C}[a, b]$. If $f(a)f(b) < 0$, then there exists some $c \in (a, b)$ such that $f(c) = 0$.

Proof. WLOG, we assume that $f(a) < 0 < f(b)$. Define

$$S := \{x \in [a, b] : f(x) \leq 0\},$$

which is a nonempty bounded set since $a \in S \subset [a, b]$. Then Theorem A.55 yields that S has exactly one least upper bound, denoted by c . Since $a \in S$, we have $a \leq \sup S = c$; since $S \subset [a, b]$, b is an upper bound of S and hence $b \geq \sup S = c$. In summary, $c = \sup S \in [a, b]$. Then Definition A.52 yields

$$\forall n \in \mathbb{N}^+, \exists x_n \in S, \text{ s.t. } c - \frac{1}{n} < x_n \leq c.$$

The squeeze test implies $\lim_{n \rightarrow \infty} x_n = c$, and Theorem B.65 implies $\lim_{n \rightarrow \infty} f(x_n) = f(c)$. Since $x_n \in S$, we have $f(x_n) \leq 0$ for all n and then Lemma B.35 yields $f(c) \leq 0$. In addition, $f(b) > 0$ forces that $c < b$. By similar arguments, we have

$$\forall n \in \mathbb{N}, \exists y_n \in \left(c, c + \frac{1}{n}\right) \cap [a, b], \text{ s.t. } y_n \notin S, \text{ i.e., } f(y_n) > 0$$

and

$$f(c) = \lim_{n \rightarrow \infty} f(y_n) \geq 0.$$

Hence $f(c) = 0$, which completes the proof. $\square$

Theorem B.74 (Intermediate value). A scalar function $f \in \mathcal{C}[a, b]$ satisfies

$$\forall y \in [m, M], \exists \xi \in [a, b], \text{ s.t. } y = f(\xi) \quad (\text{B.25})$$

where $m = \inf_{x \in [a, b]} f(x)$ and $M = \sup_{x \in [a, b]} f(x)$.

Proof. If $m = M$, then f is constant and hence (B.25) holds clearly. If $m < M$, then by Theorem B.72, there exists $c, d \in [a, b]$ such that $f(c) = m, f(d) = M$. WLOG we assume $c < d$. If $y = m$ or $y = M$, then taking $\xi = c$ or $\xi = d$ completes the proof. Now let $y \in (m, M)$ and define $g(x) := f(x) - y$. Then $g \in \mathcal{C}[a, b]$ by Definition B.70, and we have $g(c) < 0, g(d) > 0$. Theorem B.73 implies that there exists some $\xi \in (c, d) \subset [a, b]$ such that $g(\xi) = 0$, i.e., $y = f(\xi)$. $\square$

Definition B.75. Let $I = (a, b)$. A function $f : I \rightarrow \mathbb{R}$ is *uniformly continuous* on I iff

$$\begin{aligned} &\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ &\forall x, y \in I, |x - y| < \delta \implies |f(x) - f(y)| < \epsilon. \end{aligned} \quad (\text{B.26})$$

Exercise B.76. Show that, on $(a, +\infty)$, $f(x) = \frac{1}{x}$ is uniformly continuous if $a > 0$ and is not so if $a = 0$.

Exercise B.77. On $(a, +\infty)$, $f(x) = \frac{1}{x^2}$ is uniformly continuous if $a > 0$ and is not so if $a = 0$.

Theorem B.78. Uniform continuity implies continuity but the converse is not true.

Proof. Let $f : I \rightarrow \mathbb{R}$ be uniformly continuous function on $I = (a, b)$. Then for each $x \in I$, (B.26) implies that

$$\begin{aligned} &\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ &\forall y \in I, 0 < |y - x| < \delta \implies |f(y) - f(x)| < \epsilon. \end{aligned}$$

Hence Definition B.52 gives that $\lim_{y \rightarrow x} f(y) = f(x)$, and then Definition B.64 shows that f is continuous at x . Since x is arbitrary, we conclude $f \in \mathcal{C}(a, b)$ by Definition B.70.

Now let $g(x) := \frac{1}{x}, x > 0$. Similar arguments in Example B.55 yield that $g \in \mathcal{C}(0, +\infty)$. However, Example B.76 shows that g is not uniformly continuous on $(0, +\infty)$. $\square$

Lemma B.79. $f : \mathbb{R} \rightarrow \mathbb{R}$ is uniformly continuous on (a, b) iff it can be extended to a continuous function $\tilde{f}$ on $[a, b]$.

Proof. First, we show that $f(a^+)$ and $f(b^-)$ exist. (B.26) implies that

$$\begin{aligned} &\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ &\forall x, y \in (a, a + \delta) \implies |f(x) - f(y)| < \epsilon. \end{aligned}$$

Then Theorem B.63 yields the existence of $f(a^+)$. The existence of $f(b^-)$ can be shown similarly.

Second, we define $\tilde{f} : [a, b] \rightarrow \mathbb{R}$ by

$$\tilde{f}(x) = \begin{cases} f(a^+), & x = a, \\ f(x), & a < x < b, \\ f(b^-), & x = b. \end{cases}$$

By Theorem B.78, $f \in \mathcal{C}(a, b)$ and hence $\tilde{f} \in \mathcal{C}(a, b)$. By the definition of $\tilde{f}$,

$$\tilde{f}(a) = f(a^+) = \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} \tilde{f}(x),$$

which implies the right-continuity of $\tilde{f}$ at a , cf. Definition B.66. The left-continuity of $\tilde{f}$ at b can be proved similarly. Then Definition B.70 yields that $\tilde{f} \in \mathcal{C}[a, b]$. $\square$

Theorem B.80 (Uniform continuity). A continuous function $f : [a, b] \rightarrow \mathbb{R}$ is uniformly continuous.

Proof. We prove the statement by contradiction. Negating the condition of uniform continuity, i.e., (B.26) yields

$$\begin{aligned} &\exists \epsilon_0 > 0, \text{ s.t. } \forall n \in \mathbb{N}^+, \exists x_n, y_n \in [a, b], |x_n - y_n| < \frac{1}{n} \\ &\implies |f(x_n) - f(y_n)| \geq \epsilon_0. \end{aligned}$$

By the Bolzano-Weierstrass theorem, $(x_n)_{n \in \mathbb{N}^+}$ has a subsequence $(x_{n_k})_{k \in \mathbb{N}^+}$ which converges to some $c \in \mathbb{R}$. Hence we have

$$\begin{aligned} (*) : & |x_{n_k} - y_{n_k}| < \frac{1}{n_k} \leq \frac{1}{k}, \\ (**) : & |f(x_{n_k}) - f(y_{n_k})| \geq \epsilon_0. \end{aligned}$$

Then $(*)$ gives that

$$0 \leq |y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| \leq \frac{1}{k} + |x_{n_k} - c|, k \in \mathbb{N}^+.$$

Since Theorem B.25 and Corollary B.26 imply

$$\lim_{k \rightarrow \infty} \left(\frac{1}{k} + |x_{n_k} - c| \right) = \lim_{k \rightarrow \infty} \frac{1}{k} + \lim_{k \rightarrow \infty} |x_{n_k} - c| = 0,$$

the squeeze test gives that $\lim_{k \rightarrow \infty} |y_{n_k} - c| = 0$ or $\lim_{k \rightarrow \infty} y_{n_k} = c$. Then we have

$$\begin{aligned} 0 &= |f(c) - f(c)| \\ &= \left| \lim_{k \rightarrow \infty} f(x_{n_k}) - \lim_{k \rightarrow \infty} f(y_{n_k}) \right| \\ &= \lim_{k \rightarrow \infty} |f(x_{n_k}) - f(y_{n_k})| \geq \epsilon_0 > 0, \end{aligned}$$

where the second step follows from the continuity of f and Theorem B.65, the third step from Theorem B.25 and Lemma B.5, and the fourth step from (**) and Lemma B.35. This contradiction shows that f must be uniformly continuous on $[a, b]$. $\square$

B.2.2 Differentiation of scalar functions

Definition B.81. The *derivative* of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at x is the limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \quad (\text{B.27})$$

If the limit exists, f is *differentiable* at x .

Definition B.82. The *left derivative* of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at x is the limit

$$f'(x^-) = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}. \quad (\text{B.28})$$

The *right derivative* of f at x , denoted by $f'(x^+)$, can be defined similarly.

Example B.83. For the power function $f(x) = x^\alpha$, $\alpha \neq 0$, we have $f' = \alpha x^{\alpha-1}$ due to Newton's generalized binomial theorem,

$$(x+h)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^{\alpha-n} h^n.$$

Lemma B.84. If a scalar function f is differentiable at x , then f is continuous at x .

Proof. Definition B.81 and Theorem B.56 yield

$$\begin{aligned} \lim_{h \rightarrow 0} (f(x+h) - f(x)) &= \lim_{h \rightarrow 0} h \cdot \frac{f(x+h) - f(x)}{h} \\ &= 0 \cdot f'(x) = 0. \end{aligned}$$

Then Definition B.64 completes the proof. $\square$

Theorem B.85. Let f and g be scalar functions. Suppose f and g are both differentiable at x . Then $f \pm g$, fg are also differentiable at x . If $g(x) \neq 0$, then $\frac{f}{g}$ is differentiable at x . To be specific, we have

$$(a) \quad (f \pm g)'(x) = f'(x) \pm g'(x);$$

$$(b) \quad (fg)'(x) = f'(x)g(x) + f(x)g'(x);$$

$$(c) \quad g(x) \neq 0 \implies \left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}.$$

Proof. Since f, g are differentiable at x , then

$$\begin{aligned} &\frac{(f \pm g)(x+h) - (f \pm g)(x)}{h} \\ &= \frac{f(x+h) - f(x)}{h} \pm \frac{g(x+h) - g(x)}{h}, \end{aligned}$$

which, together with Theorem B.56 and Definition B.81, implies (a). From Lemma B.84, g is continuous at x , i.e., $\lim_{h \rightarrow 0} g(x+h) = g(x)$. Then

$$\begin{aligned} &\frac{fg(x+h) - fg(x)}{h} \\ &= \frac{f(x+h) - f(x)}{h} g(x+h) + f(x) \frac{g(x+h) - g(x)}{h} \end{aligned}$$

implies (b). If $g(x) \neq 0$, the continuity of g at x implies that

$$\exists \delta > 0, \text{ s.t. } \forall |h| < \delta, g(x+h) \neq 0.$$

Then

$$\begin{aligned} \forall 0 < |h| < \delta, \quad \frac{1}{h} \left(\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} \right) &= \frac{1}{g(x+h)g(x)} \cdot \\ &\left(\frac{f(x+h) - f(x)}{h} g(x) - f(x) \frac{g(x+h) - g(x)}{h} \right) \end{aligned}$$

implies (c). $\square$

Example B.86. Consider the derivative of $\tan x$. Theorem B.85(c) implies

$$\begin{aligned} (\tan x)' &= \left(\frac{\sin x}{\cos x} \right)' = \frac{(\sin x)' \cos x - \sin x (\cos x)'}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}. \end{aligned}$$

Similarly, we have

$$\begin{aligned} (\cot x)' &= \left(\frac{\cos x}{\sin x} \right)' = \frac{(\cos x)' \sin x - \cos x (\sin x)'}{\sin^2 x} \\ &= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{1}{\sin^2 x}. \end{aligned}$$

Lemma B.87. Let f and g be scalar functions. If g is continuous at t_0 and f is continuous at $x_0 := g(t_0)$, then $f \circ g$ is continuous at t_0 .

Proof. It follows from Definition B.64. $\square$

Theorem B.88 (Chain rule for scalar functions). Let f and g be scalar functions. If g is differentiable at t_0 and f is differentiable at $x_0 := g(t_0)$, then $f \circ g$ is differentiable at t_0 , and

$$(f \circ g)'(t_0) = f'(g(t_0))g'(t_0). \quad (\text{B.29})$$

Proof. Define

$$h(x) := \begin{cases} \frac{f(x) - f(x_0)}{x - x_0}, & x \neq x_0, \\ f'(x_0), & x = x_0. \end{cases}$$

By Definitions B.81 and B.64, h is continuous at x_0 . Since

$$\begin{aligned} g(t) &\neq g(t_0) = x_0 \\ \Rightarrow \frac{f(g(t)) - f(g(t_0))}{t - t_0} &= h(g(t)) \cdot \frac{g(t) - g(t_0)}{t - t_0}, \\ g(t) &= g(t_0) = x_0 \\ \Rightarrow \frac{f(g(t)) - f(g(t_0))}{t - t_0} &= 0 = h(g(t)) \cdot \frac{g(t) - g(t_0)}{t - t_0}, \end{aligned}$$

we have

$$\forall t \neq t_0, \frac{f(g(t)) - f(g(t_0))}{t - t_0} = h(g(t)) \cdot \frac{g(t) - g(t_0)}{t - t_0}.$$

Taking $t \rightarrow t_0$ and using Theorem B.56 yield

$$\lim_{t \rightarrow t_0} \frac{f \circ g(t) - f \circ g(t_0)}{t - t_0} = h(g(t_0))g'(t_0) = f'(g(t_0))g'(t_0),$$

where $\lim_{t \rightarrow t_0} h(g(t)) = h(g(t_0))$ has been applied by using Lemma B.87. Then Definition B.81 completes the proof. $\square$

Definition B.89. A function $f(x)$ is k times *continuously differentiable* on (a, b) iff $f^{(k)}(x)$ exists on (a, b) and is itself continuous. The set or space of all such functions on (a, b) is denoted by $\mathcal{C}^k(a, b)$. In comparison, $\mathcal{C}^k[a, b]$ is the space of functions f for which $f^{(k)}(x)$ is bounded and uniformly continuous on (a, b) .

Theorem B.90 (Fermat). If $f : (a, b) \rightarrow \mathbb{R}$ assumes its maximum or minimum at $x_0 \in (a, b)$ and f is differentiable at x_0 , then $f'(x_0) = 0$.

Proof. WLOG we suppose $f'(x_0) > 0$. Then we have

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} > 0.$$

The definition of a limit implies

$$\exists \delta > 0 \text{ s.t. } a < x_0 - \delta < x_0 + \delta < b,$$

which, together with $|x - x_0| < \delta$, implies $\frac{f(x) - f(x_0)}{x - x_0} > 0$. This is a contradiction to $f(x_0)$ being a maximum when we choose $x \in (x_0, x_0 + \delta)$. $\square$

Theorem B.91 (Rolle's). If a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

- (i) $f \in \mathcal{C}[a, b]$ and f' exists on (a, b) ;
- (ii) $f(a) = f(b)$,

then $\exists x \in (a, b)$ s.t. $f'(x) = 0$.

Proof. By Theorem B.74, all values between $\sup f$ and $\inf f$ will be assumed. If $f(a) = f(b) = \sup f = \inf f$, then f is a constant on $[a, b]$ and thus the conclusion holds. Otherwise, Theorem B.90 completes the proof. $\square$

Theorem B.92 (Mean value). If $f \in \mathcal{C}[a, b]$ and if f' exists on (a, b) , then $\exists \xi \in (a, b)$ s.t. $f(b) - f(a) = f'(\xi)(b - a)$.

Proof. Construct a linear function $L : [a, b] \rightarrow \mathbb{R}$ such that $L(a) = f(a)$, $L(b) = f(b)$, then $\forall x \in (a, b)$, we have $L'(x) = \frac{f(b) - f(a)}{b - a}$. Consider $g(x) = f(x) - L(x)$ on $[a, b]$. Then $g \in \mathcal{C}[a, b]$, g' exists on (a, b) , and $g(a) = g(b) = 0$. By Theorem B.91, $\exists \xi \in [a, b]$ such that $g'(\xi) = 0$, which completes the proof. $\square$

B.2.3 Taylor series

Definition B.93. A *power series* centered at c is a series of the form

$$p(x) = \sum_{n=0}^{\infty} a_n(x - c)^n, \quad (\text{B.30})$$

where a_n 's are the *coefficients*. The *interval of convergence* is the set of values of x for which the series converges:

$$I_c(p) = \{x \mid p(x) \text{ converges}\}. \quad (\text{B.31})$$

Definition B.94. If the derivatives $f^{(i)}(x)$ with $i = 1, 2, \dots, n$ exist for a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at $x = c$, then

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x - c)^k \quad (\text{B.32})$$

is called the n th *Taylor polynomial* for $f(x)$ at c .

In particular, the *linear approximation* for $f(x)$ at c is

$$T_1(x) = f(c) + f'(c)(x - c). \quad (\text{B.33})$$

Example B.95. If $f \in \mathcal{C}^\infty$, then $\forall n \in \mathbb{N}$, we have

$$T_n^{(m)}(x) = \begin{cases} \sum_{k=m}^n \frac{f^{(k)}(c)}{(k-m)!} (x - c)^{k-m}, & m \in \mathbb{N}, m \leq n; \\ 0, & m \in \mathbb{N}, m > n. \end{cases}$$

This can be proved by induction. In the inductive step, we regroup the summation into a constant term and another shifted summation.

Definition B.96. The *Taylor series* or *Taylor expansion* for $f(x)$ at c is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x - c)^k. \quad (\text{B.34})$$

Definition B.97. The *remainder* of the n th *Taylor polynomial* in approximating $f(x)$ is

$$E_n(x) = f(x) - T_n(x). \quad (\text{B.35})$$

Theorem B.98. Let T_n be the n th Taylor polynomial for $f(x)$ at c . Then

$$\forall x \in \mathbb{R}, \lim_{n \rightarrow \infty} E_n(x) = 0 \iff \lim_{n \rightarrow \infty} T_n(x) = f(x). \quad (\text{B.36})$$

Proof. It follows from (B.35) and Theorem B.25(a). $\square$

Lemma B.99. $\forall m = 0, 1, 2, \dots, n$, $E_n^{(m)}(c) = 0$.

Proof. This follows from Definition B.94 and Example B.95. $\square$

Theorem B.100 (Taylor's theorem with Lagrangian form). Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$. If $f \in \mathcal{C}^n[c - d, c + d]$ and $f^{(n+1)}(x)$ exists on $(c - d, c + d)$, then $\forall x \in [c - d, c + d]$, there exists some ξ between c and x such that

$$E_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - c)^{n+1}. \quad (\text{B.37})$$

Proof. Fix $x \neq c$, let M be the unique solution of

$$E_n(x) = f(x) - T_n(x) = \frac{M(x-c)^{n+1}}{(n+1)!}.$$

Consider the function

$$(*) : \quad g(t) := E_n(t) - \frac{M(t-c)^{n+1}}{(n+1)!}.$$

Clearly $g(x) = 0$. By Lemma B.99, $g^{(k)}(c) = 0$ for each $k = 0, 1, \dots, n$. Then Rolle's theorem implies that

$$\exists x_1 \in (c, x) \text{ s.t. } g'(x_1) = 0.$$

If $x < c$, change (c, x) above to (x, c) . Apply Rolle's theorem to $g'(t)$ on (c, x_1) and we have

$$\exists x_2 \in (c, x_1) \text{ s.t. } g^{(2)}(x_2) = 0.$$

Repeatedly using Rolle's theorem, we have

$$(**) : \quad \exists x_{n+1} \in (c, x_n) \text{ s.t. } g^{(n+1)}(x_{n+1}) = 0.$$

Since T_n is a polynomial of degree n , we have $T_n^{(n+1)}(t) = 0$, which, together with (**), (*), and Theorem B.85(a), yields

$$f^{(n+1)}(x_{n+1}) - M = 0.$$

The proof is completed by identifying ξ with x_{n+1} . $\square$

Example B.101. How many terms are needed to compute e^2 correctly to four decimal places?

The requirement of four decimal places means an accuracy of at least $\epsilon = 10^{-5}$. By Definition B.96, the Taylor series of e^x at $c = 0$ is $e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!}$. Theorem B.100 yields

$$\exists \xi \in [0, 2] \text{ s.t. } E_n(2) = \frac{e^\xi 2^{n+1}}{(n+1)!} \leq \frac{e^2 2^{n+1}}{(n+1)!}.$$

Then $\frac{e^2 2^{n+1}}{(n+1)!} \leq \epsilon$ yields $n \geq 12$, i.e., 13 terms.

Definition B.102. A scalar function $f : (a, b) \rightarrow \mathbb{R}$ is *analytic* iff it is $\mathcal{C}^\infty$ and at any $x_0 \in (a, b)$ the Taylor series $T(x) := \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ converges pointwise to $f(x)$ for any x in a neighborhood of x_0 .

Theorem B.103 (Taylor's theorem with integral remainder). For a scalar function $f \in \mathcal{C}^{k+1}(c-d, c+d)$, the remainder of the n th Taylor polynomial at c is

$$\forall x \in (c-d, c+d), \quad E_n(x) = \int_c^x \frac{(x-t)^{n+1}}{n!} f^{(n+1)}(t) dt. \quad (\text{B.38})$$

Proof. This follows from an induction on n with the formula of integral by parts. $\square$

B.2.4 Riemann integral

Definition B.104. A *partition* of an interval $I = [a, b]$ is a totally-ordered finite subset $P_n \subseteq I$ of the form

$$P_n(a, b) = \{a = x_0 < x_1 < \dots < x_n = b\}. \quad (\text{B.39})$$

The interval $I_i = [x_{i-1}, x_i]$ is the i th *subinterval* of the partition. The *norm* of the partition is the length of the longest subinterval,

$$h_n = h(P_n) = \max(x_i - x_{i-1}), \quad i = 1, 2, \dots, n. \quad (\text{B.40})$$

Definition B.105. The *Riemann sum* of $f : \mathbb{R} \rightarrow \mathbb{R}$ over a partition P_n is

$$S_n(f) = \sum_{i=1}^n f(x_i^*) (x_i - x_{i-1}), \quad (\text{B.41})$$

where $x_i^* \in I_i$ is a *sample point* of the i th subinterval. In particular, when f is bounded, by Theorem A.55, the *upper Riemann sum* and *lower Riemann sum* over P_n can be defined as

$$M_n(f) = \sum_{i=1}^n (x_i - x_{i-1}) \sup f(I_i), \quad (\text{B.42})$$

$$m_n(f) = \sum_{i=1}^n (x_i - x_{i-1}) \inf f(I_i). \quad (\text{B.43})$$

From this definition, $m_n(f) \leq S_n(f) \leq M_n(f)$ always holds.

Definition B.106. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *Riemann integrable* on $[a, b]$ iff

$$\begin{aligned} &\exists L \in \mathbb{R}, \text{ s.t. } \forall \epsilon > 0, \exists \delta > 0 \text{ s.t.} \\ &\forall P_n(a, b) \text{ with } h(P_n) < \delta, |S_n(f) - L| < \epsilon. \end{aligned} \quad (\text{B.44})$$

In this case we write $L = \int_a^b f(x) dx$ and call it the *Riemann integral* of f on $[a, b]$.

Example B.107. $f \equiv 1$ is Riemann integrable on $[a, b]$. Indeed, for any partition $P_n(a, b)$ and all x_i^* 's, we always have $S_n(f) = b - a$ by (B.41). Then Definition B.106 implies that $\int_a^b f(x) dx = b - a$.

Example B.108. The Dirichlet function

$$D(x) := \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases} \quad (\text{B.45})$$

is not Riemann integrable on $[a, b]$. To see this, we first negate the logical statement in (B.44) to get

$$\begin{aligned} &\forall L \in \mathbb{R}, \exists \epsilon > 0, \text{ s.t. } \forall \delta > 0 \\ &\exists P_n(a, b) \text{ with } h(P_n) < \delta, \text{ s.t. } |S_n(D) - L| \geq \epsilon. \end{aligned}$$

If $|L| < \frac{b-a}{2}$, we choose all x_i^* 's to be rational so that $D(x_i^*) \equiv 1$; then (B.41) yields $S_n(D) = b - a$. For $\epsilon = \frac{b-a}{4}$, the formula $|S_n(D) - L| \geq \epsilon$ clearly holds.

If $|L| \geq \frac{b-a}{2}$, we choose all x_i^* 's to be irrational so that $D(x_i^*) \equiv 0$; then (B.41) yields $S_n(D) = 0$. For $\epsilon = \frac{b-a}{4}$, the formula $|S_n(D) - L| \geq \epsilon$ clearly holds.

Theorem B.109. If f is integrable on $[a, b]$, then f is bounded on $[a, b]$.

Proof. Let $L := \int_a^b f(x)dx$. Then (B.44) implies that

$$\exists \delta > 0, \text{ s.t. } \forall P_n(a, b) \text{ with } h(P_n) < \delta, |S_n(f) - L| < 1.$$

Let x_1^* be arbitrary and x_{i-1}, x_i^* be fixed, $i = 2, 3, \dots, n$. Then the triangle inequality yields

$$\begin{aligned} \forall x \in [x_0, x_1], \quad & |f(x)|(x_1 - x_0) \\ & \leq S_n(f) + \left| \sum_{i=2}^n f(x_i^*)(x_i - x_{i-1}) \right| \\ & < |L| + 1 + \left| \sum_{i=2}^n f(x_i^*)(x_i - x_{i-1}) \right|, \end{aligned}$$

which implies the boundedness of f on $[x_0, x_1]$. By similar arguments, f is bounded on $[x_{i-1}, x_i]$, $i = 2, 3, \dots, n$. Hence f is bounded on $[a, b]$. $\square$

Definition B.110. Let P_n and $P_{n'}$ be two partitions of $[a, b]$. If $P_n \subset P_{n'}$ or $P_{n'} \supset P_n$, we say P_n is *coarser* than $P_{n'}$, or $P_{n'}$ is *finer* than P_n . For any $P_n, P_{n'}$, define a new partition

$$P_n \oplus P_{n'} := P_n \cup P_{n'} \setminus (P_n \cap P_{n'}), \quad (\text{B.46})$$

which is called the *sum* of two partitions. In particular, $P_n \oplus P_{n'}$ is always finer than P_n and $P_{n'}$.

Theorem B.111. Let P_n and $P_{n'}$ be two partitions of $[a, b]$. If f is integrable on $[a, b]$ and P_n is coarser than $P_{n'}$, then

$$m_n(f) \leq m_{n'}'(f) \leq m_n(f) + (n' - n)\omega h_n, \quad (\text{B.47})$$

$$M_n(f) \geq M_{n'}'(f) \geq M_n(f) - (n' - n)\omega h_n, \quad (\text{B.48})$$

where $M_n(f)$ and $m_n(f)$ are the upper Riemann sum and lower Riemann sum over P_n , $M_{n'}'(f)$ and $m_{n'}'(f)$ are the upper Riemann sum and lower Riemann sum over $P_{n'}$, $\omega := \sup f([a, b]) - \inf f([a, b])$ (which is finite because of Theorem B.109), and h_n is defined in (B.40).

Proof. By induction and the finiteness of n and n' , it only to consider the case on $n' = n + 1$. Let

$$\begin{aligned} x_0 &= a < x' < x_1 < x_2 < \dots < x_n = b, \\ P_n &= \{x_0, x_1, \dots, x_n\}, \quad P_{n'} = P_n \cup \{x'\}. \end{aligned}$$

On the one hand, we have

$$\begin{aligned} & (x_1 - x_0) \inf f([x_0, x_1]) \\ &= (x_1 - x') \inf f([x_0, x_1]) + (x' - x_0) \inf f([x_0, x_1]) \\ &\leq (x_1 - x') \inf f([x', x_1]) + (x' - x_0) \inf f([x_0, x']), \end{aligned}$$

which, together with (B.43), implies $m_n(f) \leq m_{n'}'(f)$. On the other hand, we have

$$\begin{aligned} & (x_1 - x') \inf f([x', x_1]) + (x' - x_0) \inf f([x_0, x']) \\ &\leq (x_1 - x') \sup f([x_0, x_1]) + (x' - x_0) \sup f([x_0, x_1]) \\ &= (x_1 - x_0)(\sup f([x_0, x_1])) \\ &\leq (x_1 - x_0)(\omega + \inf f([x_0, x_1])) \\ &\leq h_n \omega + (x_1 - x_0) \inf f([x_0, x_1]), \end{aligned}$$

which, together with (B.43), implies $m_{n'}'(f) \leq h_n \omega + m_n(f)$. Hence (B.47) holds. (B.48) can be proved similarly. $\square$

Theorem B.112. Let f be integrable on $[a, b]$. Then for any partitions $P_n, P_{n'}$ of $[a, b]$, we have

$$m_n(f) \leq M_{n'}'(f), \quad (\text{B.49})$$

where $m_n(f)$ and $M_{n'}'(f)$ are defined in Theorem B.111.

Proof. Define $\tilde{P}_k := P_n \oplus P_{n'}$. Let $\tilde{M}_k(f)$ and $\tilde{m}_k(f)$ be the upper Riemann sum and lower Riemann sum over $\tilde{P}_k$. Then Definition B.110, Theorem B.111 and Definition B.105 yield

$$m_n(f) \leq \tilde{m}_k(f) \leq \tilde{M}_k(f) \leq M_{n'}'(f). \quad \square$$

Corollary B.113. Let f be integrable on $[a, b]$. Define the sets

$$\begin{aligned} A &:= \{M_n(f) : P_n \text{ is a partition of } [a, b]\}, \\ B &:= \{m_n(f) : P_n \text{ is a partition of } [a, b]\}. \end{aligned}$$

Then A has a greatest lower bound and B has a least upper bound.

Proof. For any partition P_n of $[a, b]$, by comparing with the partition $\{a, b\}$, Theorem B.112 yields

$$\begin{aligned} M_n(f) &\geq (b - a) \inf f([a, b]), \\ m_n(f) &\leq (b - a) \sup f([a, b]), \end{aligned}$$

where $\sup f([a, b])$ and $\inf f([a, b])$ are finite because of Theorem B.109. Then Theorem A.55 and Corollary A.56 complete the proof. $\square$

Definition B.114. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is integrable on $[a, b]$, then the limit of the Riemann sum of f is called the *definite integral* of f on $[a, b]$:

$$\int_a^b f(x)dx = \lim_{h_n \rightarrow 0} S_n(f). \quad (\text{B.50})$$

Definition B.115. Let f be a bounded function on $[a, b]$. The *upper* and *lower* Riemann integrals of f are defined by

$$\int_a^b f(x)dx := \inf_{h_n \rightarrow 0} M_n(f), \quad (\text{B.51})$$

$$\int_a^b f(x)dx := \sup_{h_n \rightarrow 0} m_n(f), \quad (\text{B.52})$$

where the existence of $\inf_{h_n \rightarrow 0} M_n(f)$ and $\sup_{h_n \rightarrow 0} m_n(f)$ is guaranteed by Corollary B.113.

Theorem B.116 (Darboux). A bounded function f on $[a, b]$ satisfies

$$\lim_{h_n \rightarrow 0} M_n(f) = \int_a^b f(x)dx, \quad (\text{B.53})$$

$$\lim_{h_n \rightarrow 0} m_n(f) = \int_a^b f(x)dx. \quad (\text{B.54})$$

Proof. Denote $\bar{L} := \int_a^b f(x)dx$. (B.51) implies that

$$(*) : \quad \forall \epsilon > 0, \exists P'_{n'}(a, b), \text{ s.t. } M'_{n'}(f) < \bar{L} + \frac{\epsilon}{2}.$$

Let $\delta := \frac{\epsilon}{2n'\omega+1} > 0$, where ω is defined in Theorem B.111. We claim that

$$\forall P_n(a, b) \text{ with } h_n < \delta, M_n(f) - \bar{L} < \epsilon.$$

Indeed, for this P_n , define $\tilde{P}_k := P_n \oplus P'_{n'}$. Then Definition B.110 yields that P_n and $P'_{n'}$ are both coarser than $\tilde{P}_k$ and $k - n \leq n'$. By Theorem B.111, we have

$$\begin{aligned} (**): \quad M'_{n'}(f) &\geq \tilde{M}_k(f) \geq M_n(f) - (k - n)\omega h_n \\ &\geq M_n(f) - n'\omega\delta \\ &= M_n(f) - \frac{n'\omega\epsilon}{2n'\omega+1} > M_n(f) - \frac{\epsilon}{2}. \end{aligned}$$

Hence the claim follows from $(*)$ and $(**)$. Then (B.44) and the fact $M_n(f) \geq \bar{L}$ imply (B.53). (B.54) can be proved similarly. $\square$

Theorem B.117. For a bounded function $f : [a, b] \rightarrow \mathbb{R}$, the following conditions are equivalent:

- (a) f is Riemann integrable;
- (b) $\lim_{h_n \rightarrow 0} (M_n(f) - m_n(f)) = 0$;
- (c) $\int_a^b f(x)dx = \int_a^b f(x)dx$.

Proof. (a) $\implies$ (b): Let $L := \int_a^b f(x)dx$. Then (B.50) yields

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.}$$

$$\forall P_n(a, b) \text{ with } h_n < \delta, L - \frac{\epsilon}{4} < S_n(f) < L + \frac{\epsilon}{4}.$$

For any $P_n(a, b)$ as above, by the definition of $M_n(f)$ in (B.42), there exists a Riemann sum $S_n(f)$ such that $M_n(f) - S_n(f) < \frac{\epsilon}{4}$. Hence $M_n(f) < S_n(f) + \frac{\epsilon}{4} < L + \frac{\epsilon}{2}$. Similarly, we can get $m_n(f) > L - \frac{\epsilon}{2}$. Then we have

$$0 \leq M_n(f) - m_n(f) < \left(L + \frac{\epsilon}{2}\right) - \left(L - \frac{\epsilon}{2}\right) = \epsilon,$$

where the first step follows from Definition B.105. Hence (B.44) yields (b).

(b) $\implies$ (c): Definition B.115 and Theorem B.112 yield that

$$(*) : \quad \int_a^b f(x)dx \leq \int_a^b f(x)dx.$$

Since $\lim_{h_n \rightarrow 0} (M_n(f) - m_n(f)) = 0$, we have

$$\begin{aligned} \forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall P_n(a, b) \text{ with } h(P_n) < \delta, \\ |(M_n(f) - m_n(f)) - 0| &= M_n(f) - m_n(f) < \epsilon. \end{aligned}$$

Fix some P_n as above and we have

$$0 \leq \int_a^b f(x)dx - \int_a^b f(x)dx \leq M_n(f) - m_n(f) < \epsilon,$$

where the first step follows from $(*)$ and the second step follows from Definition B.115. Since $\epsilon > 0$ is arbitrary, we get that $\int_a^b f(x)dx = \int_a^b f(x)dx$.

(c) $\implies$ (a): Let $L := \int_a^b f(x)dx = \int_a^b f(x)dx$. Since $m_n(f) \leq S_n(f) \leq M_n(f)$ holds for any partition P_n of $[a, b]$, taking $h_n \rightarrow 0$ and using Theorem B.116 yield $\lim_{h_n \rightarrow 0} S_n(f) = L$. Then Definition B.106 implies (a). $\square$

Example B.118. In Example B.108, we have shown that $D(x)$ is not integrable on $[a, b]$. Now we prove it by another way.

By Definition B.105, for any partition $P_n(a, b)$, choosing all x_i^* 's to be irrational yields $m_n(D) = 0$, and choosing all x_i^* 's to be rational yields $M_n(D) = b - a$. Hence Definition B.115 yields that $\int_a^b D(x)dx = 0$ and $\int_a^b D(x)dx = b - a > 0$. Then Theorem B.117 implies that $D(x)$ is not integrable on $[a, b]$.

Theorem B.119. A scalar function f is integrable on $[a, b]$ if $f \in \mathcal{C}[a, b]$.

Proof. By Theorem B.71, f is bounded on $[a, b]$. Theorem B.80 yields that f is uniformly continuous on $[a, b]$, i.e.,

$$\begin{aligned} (*) : \quad \forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ \forall s, t \in [a, b], |s - t| < \delta \implies |f(s) - f(t)| < \frac{\epsilon}{b - a}. \end{aligned}$$

For any $P_n(a, b)$ with $h_n < \delta$, the extreme values theorem (Theorem B.72) implies

$$\begin{aligned} \exists s_i, t_i \in [x_{i-1}, x_i], \text{ s.t. } \forall i = 1, 2, \dots, n, \\ f(s_i) = \sup f([x_{i-1}, x_i]), \quad f(t_i) = \inf f([x_{i-1}, x_i]). \end{aligned}$$

Then we have

$$\begin{aligned} M_n(f) - m_n(f) &= \sum_{i=1}^n (f(s_i) - f(t_i))(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^n \frac{\epsilon}{b - a} \cdot (x_i - x_{i-1}) = \epsilon, \end{aligned}$$

where the first step follows from Definition B.105, the second follows from $h_n < \delta$ and $(*)$. Hence (B.44) yields $\lim_{h_n \rightarrow 0} (M_n(f) - m_n(f)) = 0$ and then Theorem B.117 completes the proof. $\square$

Definition B.120. A *monotonic* function is a function between ordered sets that either preserves or reverses the given order. In particular, $f : \mathbb{R} \rightarrow \mathbb{R}$ is *monotonically increasing* if $\forall x, y, x \leq y \implies f(x) \leq f(y)$; $f : \mathbb{R} \rightarrow \mathbb{R}$ is *monotonically decreasing* if $\forall x, y, x \leq y \implies f(x) \geq f(y)$.

Theorem B.121. A scalar function is integrable on $[a, b]$ if it is monotonic on $[a, b]$.

Proof. WLOG, we assume a function f is monotonically increasing on $[a, b]$. If $f(a) = f(b)$, then f is constant on $[a, b]$ and is integrable by Definition B.106. If $f(a) \neq f(b)$, then $f(b) > f(a)$. For any $\epsilon > 0$, set $\delta := \frac{\epsilon}{f(b) - f(a)}$. For any $P_n(a, b)$ with $h_n < \delta$, we have

$$\begin{aligned} M_n(f) - m_n(f) &= \sum_{i=1}^n (\sup f([x_{i-1}, x_i]) - \inf f([x_{i-1}, x_i]))(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^n (f(x_i) - f(x_{i-1}))(x_i - x_{i-1}) \\ &< \sum_{i=1}^n (f(x_i) - f(x_{i-1}))\delta \\ &= (f(b) - f(a))\delta = \epsilon, \end{aligned}$$

where the first step follows from Definition B.105, the second follows from the fact that f is monotonically increasing. Hence (B.44) yields $\lim_{h_n \rightarrow 0} (M_n(f) - m_n(f)) = 0$ and then Theorem B.117 completes the proof. $\square$

Exercise B.122. True or false: a bijective function is either order-preserving or order-reversing?

Theorem B.123. Let f, g be integrable functions on $[a, b]$, and $c \in \mathbb{R}$.

(a) $cf + g$ is integrable on $[a, b]$, and

$$\int_a^b (cf + g)dx = c \int_a^b fdx + \int_a^b gdx; \quad (\text{B.55})$$

(b) fg is integrable on $[a, b]$;

(c) If $f \leq g$ on $[a, b]$, then $\int_a^b fdx \leq \int_a^b gdx$.

Proof. (a) follows from Definitions B.105 and B.106. By Theorem B.109, let $A := \sup f([a, b])$, $B = \sup g([a, b])$. By Theorem B.117,

$$\begin{aligned} \forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall P_n(a, b) \text{ with } h_n < \delta, \\ M_n(f) - m_n(f) < \frac{\epsilon}{3B}, \quad M_n(g) - m_n(g) < \frac{\epsilon}{3A}. \end{aligned}$$

Then for any P_n as above and any $s_i, t_i \in [x_{i-1}, x_i]$, define

$$\begin{aligned} A_i &:= \sup f([x_{i-1}, x_i]) - \inf f([x_{i-1}, x_i]), \\ B_i &:= \sup g([x_{i-1}, x_i]) - \inf g([x_{i-1}, x_i]), \end{aligned}$$

and we have

$$\begin{aligned} & \sum_{i=1}^n |f(s_i)g(s_i) - f(t_i)g(t_i)|(x_i - x_{i-1}) \\ & \leq \sum_{i=1}^n (|f(s_i)||g(s_i) - g(t_i)| + |g(t_i)||f(s_i) - f(t_i)|)(x_i - x_{i-1}) \\ & \leq \sum_{i=1}^n (AB_i + BA_i)(x_i - x_{i-1}) \\ & = A(M_n(g) - m_n(g)) + B(M_n(f) - m_n(f)) \\ & < A \cdot \frac{\epsilon}{3A} + B \cdot \frac{\epsilon}{3B} = \frac{2}{3}\epsilon. \end{aligned}$$

Let s_i and t_i go to the supremum and infimum of fg on $[x_{i-1}, x_i]$, respectively. Then

$$M_n(fg) - m_n(fg) \leq \frac{2}{3}\epsilon < \epsilon,$$

which implies $\lim_{h_n \rightarrow 0} (M_n(fg) - m_n(fg)) = 0$ and hence Theorem B.117 yields (b). For (c), by (a), it suffices to show that

$$f \geq 0 \text{ on } [a, b] \implies \int_a^b fdx = 0.$$

By Definition B.105, $m_n(f) \geq 0$ for any partition $P_n(a, b)$. Taking $h_n \rightarrow 0$, together with Theorem B.116, implies $\int_a^b fdx \geq 0$. From the proof of Theorem B.117, we have $\int_a^b fdx = \int_a^b fdx \geq 0$. $\square$

B.2.5 Crucial theorems of calculus

Theorem B.124 (Integral mean value). Let $w : [a, b] \rightarrow \mathbb{R}^+$ be integrable on $[a, b]$. For $f \in \mathcal{C}[a, b]$, $\exists \xi \in [a, b]$ s.t.

$$\int_a^b w(x)f(x)dx = f(\xi) \int_a^b w(x)dx. \quad (\text{B.56})$$

Proof. By Theorem B.71, we denote $m := \inf_{x \in [a, b]} f(x)$, $M := \sup_{x \in [a, b]} f(x)$, and $I := \int_a^b w(x)dx$. By Theorem B.119, f is integrable on $[a, b]$. Then Theorem B.123 and $mw(x) \leq f(x)w(x) \leq Mw(x)$ yield

$$mI \leq \int_a^b w(x)f(x)dx \leq MI.$$

$w > 0$ implies $I \neq 0$, hence

$$m \leq \frac{1}{I} \int_a^b w(x)f(x)dx \leq M.$$

Applying Theorem B.74 completes the proof. $\square$

Theorem B.125. If f is integrable on $[a, b]$, then so is $|f|$. Moreover,

$$\left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx. \quad (\text{B.57})$$

Proof. We claim that

$$(*) : \forall a \leq s < t \leq b, \quad \begin{aligned} \sup |f([s, t])| - \inf |f([s, t])| \\ \leq \sup f([s, t]) - \inf f([s, t]). \end{aligned}$$

Indeed, when $f > 0$ on $[s, t]$, $|f| = f$ on $[s, t]$ and $(*)$ holds. When $f < 0$ on $[s, t]$, we have

$$\begin{aligned} & \sup |f([s, t])| - \inf |f([s, t])| \\ & = \sup(-f([s, t])) - \inf(-f([s, t])) \\ & = -\inf(f([s, t])) + \sup(f([s, t])) \end{aligned}$$

and $(*)$ holds. When f changes its sign on $[s, t]$, denote $M_0 := \sup f([s, t]) > 0$ and $m_0 := \inf f([s, t]) < 0$. Then $\sup |f([s, t])| = \max\{M_0, -m_0\}$, $\inf |f([s, t])| = 0$ and

$$\sup |f([s, t])| - \inf |f([s, t])| = \max\{M_0, -m_0\} < M_0 - m_0,$$

which also implies $(*)$. Hence the claim holds. Then taking $s = x_{i-1}$, $t = x_i$ in $(*)$, we have

$$\begin{aligned} & \sup |f([x_{i-1}, x_i])| - \inf |f([x_{i-1}, x_i])| \\ & \leq \sup f([x_{i-1}, x_i]) - \inf f([x_{i-1}, x_i]), \end{aligned}$$

which, together with Definition B.105, implies

$$0 \leq M_n(|f|) - m_n(|f|) \leq M_n(f) - m_n(f).$$

Hence taking $h_n \rightarrow 0$ and applying Theorem B.117 get the integrability of $|f|$. Note that $-|f(x)| \leq f(x) \leq |f(x)|$, we have

$$-\int_a^b |f(x)|dx \leq \int_a^b f(x)dx \leq \int_a^b |f(x)|dx$$

and then (B.57) holds. $\square$

Theorem B.126. Let $f : [a, b] \rightarrow \mathbb{R}$ and $c \in (a, b)$. Then f is integrable on $[a, b]$ if and only if f is integrable on both $[a, c]$ and $[c, b]$. Moreover, if f is integrable on $[a, b]$, then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx. \quad (\text{B.58})$$

Proof. For the necessity, Theorem B.117 yields

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.}$$

$$\forall \tilde{P}_k(a, b) \text{ with } h(\tilde{P}_k) < \delta, \quad \tilde{M}_k(f) - \tilde{m}_k(f) < \epsilon.$$

Then for any partition $P_n(a, c)$ with $h(P_n) < \delta$, there exists $\tilde{P}_k(a, b) \supset P_n(a, c)$ with $h(\tilde{P}_k) < \delta$ such that

$$M_n(f) - m_n(f) \leq \tilde{M}_k(f) - \tilde{m}_k(f) < \epsilon.$$

By Theorem B.117, f is integrable on $[a, c]$. Similar arguments can show f is integrable on $[c, b]$.

For the sufficiency, Theorem B.117 yields

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.}$$

$$\forall P_n(a, c) \text{ with } h(P_n) < \delta, \quad M_n(f) - m_n(f) < \frac{\epsilon}{3},$$

$$\forall P'_{n'}(c, b) \text{ with } h(P'_{n'}) < \delta, \quad M'_{n'}(f) - m'_{n'}(f) < \frac{\epsilon}{3}.$$

Then for any $\tilde{P}_k(a, b)$ with $h(\tilde{P}_k) < \delta$, if $c \in \tilde{P}_k$, we have

$$\begin{aligned} \tilde{M}_k(f) - \tilde{m}_k(f) &= (M_n(f) + M'_{n'}(f)) - (m_n(f) + m'_{n'}(f)) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} < \epsilon. \end{aligned}$$

If $c \notin \tilde{P}_k$, consider a new partition $\tilde{P}'_{k'} := \tilde{P}_k \cup \{c\}$. Theorem B.111 yields

$$\tilde{M}_k(f) - \tilde{m}_k(f) \leq \tilde{M}'_{k'}(f) - \tilde{m}'_{k'}(f) + 2\omega h(\tilde{P}_k).$$

Taking $\delta' := \min\{\delta, \frac{\epsilon}{6\omega}\}$ yields

$$\begin{aligned} &\forall \tilde{P}_k(a, b) \text{ with } h(\tilde{P}_k) < \delta', \\ &\quad \tilde{M}_k(f) - \tilde{m}_k(f) \\ &\leq \tilde{M}'_{k'}(f) - \tilde{m}'_{k'}(f) + \frac{\epsilon}{3} \\ &= (M_n(f) + M'_{n'}(f)) - (m_n(f) + m'_{n'}(f)) + \frac{\epsilon}{3} \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

In a word, we have $\lim_{h(\tilde{P}_k) \rightarrow 0} (\tilde{M}_k(f) - \tilde{m}_k(f)) = 0$. Hence Theorem B.117 shows that f is integrable on $[a, b]$.

If f is integrable on $[a, b]$, denote $L_1 := \int_a^c f(x)dx$, $L_2 := \int_c^b f(x)dx$. Then the triangle inequality

$$\begin{aligned} |\tilde{S}_k(f) - L_1 - L_2| &= |S_n(f) + S'_{n'}(f) - L_1 - L_2| \\ &\leq |S_n(f) - L_1| + |S'_{n'}(f) - L_2| \end{aligned}$$

and Definition B.106 yield $\int_a^b f(x)dx = L_1 + L_2$. $\square$

Notation 20. Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable. For convenience, we define

$$\int_a^a f(x)dx := 0, \quad \int_b^a f(x)dx := -\int_a^b f(x)dx.$$

By adding the two equations, the additivity of integrals can be preserved, i.e., (B.58) holds for any $a, b, c \in \mathbb{R}$.

Theorem B.127 (First fundamental theorem of calculus). Let $a < b$ be real numbers. For a continuous function $f : [a, b] \rightarrow \mathbb{R}$ that is Riemann integrable, define a function $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) := \int_a^x f(y)dy. \quad (\text{B.59})$$

Then F is differentiable and

$$\forall x_0 \in [a, b], \quad F'(x_0) = f(x_0). \quad (\text{B.60})$$

Proof. For any $x_0, x_0 + h \in [a, b]$, Notation 20 and Theorem B.126 yield

$$\begin{aligned} F(x_0 + h) - F(x_0) &= \int_a^{x_0+h} f(y)dy - \int_a^{x_0} f(y)dy \\ &= \int_a^{x_0+h} f(y)dy + \int_{x_0}^a f(y)dy = \int_{x_0}^{x_0+h} f(y)dy. \end{aligned}$$

Thus Theorem B.123 implies

$$\begin{aligned} \frac{F(x_0+h) - F(x_0)}{h} - f(x_0) &= \frac{1}{h} \int_{x_0}^{x_0+h} f(y)dy - \frac{1}{h} \int_{x_0}^{x_0+h} f(x_0)dy \\ &= \frac{1}{h} \int_{x_0}^{x_0+h} (f(y) - f(x_0))dy. \end{aligned}$$

Since f is continuous at x_0 , we have

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.}$$

$$\forall y \in [a, b] \text{ with } |y - x_0| < \delta, \quad |f(y) - f(x_0)| < \epsilon.$$

Then Theorem B.125 and $0 < |h| < \delta$ give

$$\begin{aligned} \left| \frac{F(x_0+h) - F(x_0)}{h} - f(x_0) \right| &\leq \frac{1}{|h|} \int_{x_0}^{x_0+|h|} |f(y) - f(x_0)|dy \\ &< \frac{1}{|h|} \int_{x_0}^{x_0+|h|} \epsilon dy = \epsilon. \end{aligned}$$

Hence Definition B.52 yields

$$\lim_{h \rightarrow 0} \frac{F(x_0 + h) - F(x_0)}{h} = f(x_0).$$

Then by Definition B.81, F is differentiable at x_0 and $F'(x_0) = f(x_0)$. Since x_0 is arbitrary, the proof is completed. $\square$

Theorem B.128 (Second fundamental theorem of calculus). Let $a < b$ be real numbers and let $f : [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function. If $F : [a, b] \rightarrow \mathbb{R}$ is the antiderivative of f , i.e. $F'(x) = f(x)$, then

$$\int_a^b f(x)dx = F(b) - F(a). \quad (\text{B.61})$$

Proof. By Theorem B.127, $\int_a^x f(y)dy$ is an antiderivative of f . Hence there exists a constant C such that

$$\int_a^x f(y)dy = F(x) + C.$$

Then taking $x = a$ yields $C = -F(a)$. Now taking $x = b$ completes the proof. $\square$

Theorem B.129 (Integration by parts). Let $a < b$ be real numbers and let $f, g \in \mathcal{C}^1[a, b]$. Then

$$\int_a^b f(x)g'(x)dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)dx. \quad (\text{B.62})$$

Proof. By Theorem B.85, fg is differentiable and

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x).$$

Then Theorems B.123 and B.128 yield

$$\begin{aligned} \int_a^b f(x)g'(x)dx &= \int_a^b [(f(x)g(x))' - f'(x)g(x)]dx \\ &= f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)dx. \quad \square \end{aligned}$$

Theorem B.130 (Integration by substitution). Let $a < b$ be real numbers, and let $f \in \mathcal{C}[a, b]$. If $\varphi : [\alpha, \beta] \rightarrow [a, b]$ is a $\mathcal{C}^1$ function such that $\varphi(\alpha) = a$ and $\varphi(\beta) = b$, then

$$\int_a^b f(x)dx = \int_\alpha^\beta f(\varphi(t))\varphi'(t)dt. \quad (\text{B.63})$$

Proof. By Theorem B.119, f is integrable on $[a, b]$. Let F be an antiderivative of f . Then Theorem B.128 yields

$$\int_a^b f(x)dx = F(b) - F(a).$$

On the other hand, the chain rule (Theorem B.88) implies

$$\frac{d}{dt}F(\varphi(t)) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t),$$

which shows that $F \circ \varphi$ is an antiderivative of $(f \circ \varphi)\varphi'$. Note that by Theorem B.69 and Lemma B.87, $(f \circ \varphi)\varphi'$ is continuous on $[\alpha, \beta]$ and hence, by Theorem B.119, integrable on $[\alpha, \beta]$. Therefore using Theorem B.128 again, we have

$$\begin{aligned} \int_\alpha^\beta f(\varphi(t))\varphi'(t)dt &= F(\varphi(\beta)) - F(\varphi(\alpha)) \\ &= F(b) - F(a) = \int_a^b f(x)dx. \quad \square \end{aligned}$$

Appendix C

Linear Algebra

C.1 Vector spaces

Definition C.1. A *field* $\mathbb{F}$ is a set together with two binary operations, usually called “addition” and “multiplication” and denoted by “+” and “*”, such that $\forall a, b, c \in \mathbb{F}$, the following axioms hold,

- commutativity: $a + b = b + a$, $ab = ba$;
- associativity: $a + (b + c) = (a + b) + c$, $a(bc) = (ab)c$;
- identity: $a + 0 = a$, $a1 = a$;
- invertibility: $a + (-a) = 0$, $aa^{-1} = 1$ ($a \neq 0$);
- distributivity: $a(b + c) = ab + ac$.

Definition C.2. A *vector space* or *linear space* over a field $\mathbb{F}$ is a set $\mathcal{V}$ together with two binary operations “+” and “ $\times$ ” respectively called vector addition and scalar multiplication that satisfy the following axioms:

- (VSA-1) commutativity
 $\forall \mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
- (VSA-2) associativity
 $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
- (VSA-3) compatibility
 $\forall \mathbf{u} \in \mathcal{V}$, $\forall a, b \in \mathbb{F}$, $(ab)\mathbf{u} = a(b\mathbf{u})$;
- (VSA-4) additive identity
 $\exists \mathbf{0} \in \mathcal{V}$, $\forall \mathbf{u} \in \mathcal{V}$, s.t. $\mathbf{u} + \mathbf{0} = \mathbf{u}$;
- (VSA-5) additive inverse
 $\forall \mathbf{u} \in \mathcal{V}$, $\exists \mathbf{v} \in \mathcal{V}$, s.t. $\mathbf{u} + \mathbf{v} = \mathbf{0}$;
- (VSA-6) multiplicative identity
 $\exists 1 \in \mathbb{F}$, s.t. $\forall \mathbf{u} \in \mathcal{V}$, $1\mathbf{u} = \mathbf{u}$;
- (VSA-7) distributive laws

$$\forall \mathbf{u}, \mathbf{v} \in \mathcal{V}, \forall a, b \in \mathbb{F}, \begin{cases} (a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}, \\ a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}. \end{cases}$$

The elements of $\mathcal{V}$ are called *vectors* and the elements of $\mathbb{F}$ are called *scalars*.

Definition C.3. A *real vector space* or a *complex vector space* is a vector space with $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, respectively.

Exercise C.4. Show that a complex vector space always induces another real vector space.

Example C.5. The simplest vector space is $\{0\}$. Another simple example of a vector space over a field $\mathbb{F}$ is $\mathbb{F}$ itself, equipped with its standard addition and multiplication.

Definition C.6. An *algebra* is a vector space $\mathcal{V}$ with an associative and distributive multiplication $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$,

$$\forall u, v, w \in \mathcal{V}, \forall \alpha \in \mathbb{F}, \begin{cases} u(vw) = (uv)w, \\ (u + v)w = uw + vw, \quad u(v + w) = uv + uw, \\ \alpha(uv) = u(\alpha v) = (\alpha u)v. \end{cases} \quad (\text{C.1})$$

The *multiplicative identity* is the element $e \in \mathcal{V}$ such that $\forall v \in \mathcal{V}$, $ev = v = ve$.

C.1.1 Subspaces

Definition C.7. A subset $\mathcal{U}$ of $\mathcal{V}$ is called a *subspace* of $\mathcal{V}$ if $\mathcal{U}$ is also a vector space when equipped with the same addition and scalar multiplication on $\mathcal{V}$.

Definition C.8. Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subsets of $\mathcal{V}$. The *sum* of $\mathcal{U}_1, \dots, \mathcal{U}_m$ is the set of all possible sums of elements of $\mathcal{U}_1, \dots, \mathcal{U}_m$:

$$\mathcal{U}_1 + \dots + \mathcal{U}_m := \left\{ \sum_{j=1}^m \mathbf{u}_j : \mathbf{u}_j \in \mathcal{U}_j \right\}. \quad (\text{C.2})$$

Example C.9. For $\mathcal{U} = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$ and $\mathcal{W} = \{(x, x, x, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$, we have

$$\mathcal{U} + \mathcal{W} = \{(x, x, z, y) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}.$$

Lemma C.10. Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \dots + \mathcal{U}_m$ is the smallest subspace of $\mathcal{V}$ that contains $\mathcal{U}_1, \dots, \mathcal{U}_m$.

Definition C.11. Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. The sum $\mathcal{U}_1 + \dots + \mathcal{U}_m$ is called a *direct sum* if each element in $\mathcal{U}_1 + \dots + \mathcal{U}_m$ can be written in only one way as a sum $\sum_{j=1}^m \mathbf{u}_j$ with $\mathbf{u}_j \in \mathcal{U}_j$ for each $j = 1, \dots, m$. In this case we write the direct sum as $\mathcal{U}_1 \oplus \dots \oplus \mathcal{U}_m$.

Exercise C.12. Show that $\mathcal{U}_1 + \mathcal{U}_2 + \mathcal{U}_3$ is not a direct sum:

$$\begin{aligned} \mathcal{U}_1 &= \{(x, y, 0) \in \mathbb{F}^3 : x, y \in \mathbb{F}\}, \\ \mathcal{U}_2 &= \{(0, 0, z) \in \mathbb{F}^3 : z \in \mathbb{F}\}, \\ \mathcal{U}_3 &= \{(0, y, y) \in \mathbb{F}^3 : y \in \mathbb{F}\}. \end{aligned}$$

Lemma C.13. Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \dots + \mathcal{U}_m$ is a direct sum if and only if the only way to write $\mathbf{0}$ as a sum $\sum_{j=1}^m \mathbf{u}_j$, where $\mathbf{u}_j \in \mathcal{U}_j$ for each $j = 1, \dots, m$, is by taking each $\mathbf{u}_j$ equal to $\mathbf{0}$.

Theorem C.14. Suppose $\mathcal{U}$ and $\mathcal{W}$ are subspaces of $\mathcal{V}$. Then $\mathcal{U} + \mathcal{W}$ is a direct sum if and only if $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$.

C.1.2 Span and linear independence

Definition C.15. A *list of length n* or *n -tuple* is an ordered collection of n elements (which might be numbers, other lists, or more abstract entities) separated by commas and surrounded by parentheses: $\mathbf{x} = (x_1, x_2, \dots, x_n)$.

Definition C.16. A vector space composed of all the n -tuples of a field $\mathbb{F}$ is known as a *coordinate space*, denoted by $\mathbb{F}^n$ ($n \in \mathbb{N}^+$).

Example C.17. The properties of forces or velocities in the real world can be captured by a coordinate space $\mathbb{R}^2$ or $\mathbb{R}^3$.

Example C.18. The set of continuous real-valued functions on the interval $[a, b]$ forms a real vector space.

Notation 21. For a set $\mathcal{S}$, define a vector space

$$\mathbb{F}^{\mathcal{S}} := \{f : \mathcal{S} \rightarrow \mathbb{F}\}.$$

$\mathbb{F}^n$ is a special case of $\mathbb{F}^{\mathcal{S}}$ because n can be regarded as the set $\{1, 2, \dots, n\}$ and each element in $\mathbb{F}^n$ can be considered as a function $\{1, 2, \dots, n\} \mapsto \mathbb{F}$.

Definition C.19. A *linear combination* of a list of vectors $\{\mathbf{v}_i\}$ is a vector of the form $\sum_i a_i \mathbf{v}_i$ where $a_i \in \mathbb{F}$.

Example C.20. $(17, -4, 2)$ is a linear combination of $(2, 1, -3), (1, -2, 4)$ because

$$(17, -4, 2) = 6(2, 1, -3) + 5(1, -2, 4).$$

Example C.21. $(17, -4, 5)$ is not a linear combination of $(2, 1, -3), (1, -2, 4)$ because there do not exist numbers a_1, a_2 such that

$$(17, -4, 5) = a_1(2, 1, -3) + a_2(1, -2, 4).$$

Solving from the first two equations yields $a_1 = 6, a_2 = 5$, but $5 \neq -3 \times 6 + 4 \times 5$.

Definition C.22. The *span* of a list of vectors $(\mathbf{v}_i)$ is the set of all linear combinations of $(\mathbf{v}_i)$,

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m) = \left\{ \sum_{i=1}^m a_i \mathbf{v}_i : a_i \in \mathbb{F} \right\}. \quad (\text{C.3})$$

In particular, the span of the empty set is $\{\mathbf{0}\}$. We say that $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ *spans* $\mathcal{V}$ if $\mathcal{V} = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$.

Example C.23.

$$(17, -4, 2) \in \text{span}((2, 1, -3), (1, -2, 4))$$

$$(17, -4, 5) \notin \text{span}((2, 1, -3), (1, -2, 4))$$

Definition C.24. A vector space $\mathcal{V}$ is called *finite dimensional* if some list of vectors span $\mathcal{V}$; otherwise it is *infinite dimensional*.

Notation 22. The set of all polynomials with coefficients in $\mathbb{F}$ and degree at most n is denoted by

$$\mathbb{P}_n(\mathbb{F}) = \left\{ p : \mathbb{F} \rightarrow \mathbb{F}; p(z) = \sum_{i=0}^n a_i z^i, a_i \in \mathbb{F} \right\} \quad (\text{C.4})$$

and the set of all polynomials with coefficients in $\mathbb{F}$ by $\mathbb{P}(\mathbb{F}) := \mathbb{P}_{+\infty}(\mathbb{F})$. When omitting the parentheses, we mean polynomials with real coefficients, i.e., $\mathbb{P}_n = \mathbb{P}_n(\mathbb{R})$ and $\mathbb{P} = \mathbb{P}(\mathbb{R})$. The additive identity $0 \in \mathbb{F}$ is considered as a polynomial of degree -1 in $\mathbb{P}_n(\mathbb{F})$ while any nonzero constant $a \in \mathbb{F} \setminus \{0\}$ a polynomial of degree 0.

Example C.25. $\mathbb{P}_n(\mathbb{F})$ is a finite-dimensional vector space for each non-negative integer m while $\mathbb{P}(\mathbb{F})$ is infinite-dimensional. Both are subspaces of $\mathbb{F}^{\mathbb{F}}$ for $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Definition C.26. A list of vectors $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ in $\mathcal{V}$ is called *linearly independent* iff

$$a_1 \mathbf{v}_1 + \dots + a_m \mathbf{v}_m = \mathbf{0} \implies a_1 = \dots = a_m = 0. \quad (\text{C.5})$$

Otherwise the list of vectors is called *linearly dependent*.

Example C.27. The empty list is declared to be linearly independent. A list of one vector $(\mathbf{v})$ is linearly independent iff $\mathbf{v} \neq \mathbf{0}$. A list of two vectors is linearly independent iff neither vector is a scalar multiple of the other.

Example C.28. The list $(1, z, \dots, z^m)$ is linearly independent in $\mathbb{P}_m(\mathbb{F})$ for each $m \in \mathbb{N}$.

Example C.29. $(2, 3, 1), (1, -1, 2)$, and $(7, 3, 8)$ is linearly dependent in $\mathbb{R}^3$ because

$$2(2, 3, 1) + 3(1, -1, 2) + (-1)(7, 3, 8) = (0, 0, 0).$$

Example C.30. Every list of vectors containing the $\mathbf{0}$ vector is linearly dependent.

Lemma C.31 (Linear dependence lemma). Suppose $V = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ is a linearly dependent list in $\mathcal{V}$. Then there exists $j \in \{1, 2, \dots, m\}$ such that

- $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{j-1})$;
- if the j th term is removed from V , the span of the remaining list equals $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$.

Lemma C.32. In a finite-dimensional vector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.

C.1.3 Bases

Definition C.33. A *basis* of a vector space $\mathcal{V}$ is a list of vectors in $\mathcal{V}$ that is linearly independent and spans $\mathcal{V}$.

Definition C.34. The *standard basis* of $\mathbb{F}^n$ is the list of vectors

$$(1, 0, \dots, 0)^\top, (0, 1, 0, \dots, 0)^\top, \dots, (0, \dots, 0, 1)^\top. \quad (\text{C.6})$$

Example C.35. $(z^0, z^1, \dots, z^m)$ is a basis of $\mathbb{P}_m(\mathbb{F})$ in (C.4).

Lemma C.36. A list of vectors $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ iff every vector $\mathbf{u} \in \mathcal{V}$ can be written uniquely as

$$\mathbf{u} = \sum_{i=1}^n a_i \mathbf{v}_i, \quad (\text{C.7})$$

where $a_i \in \mathbb{F}$.

Lemma C.37. Every spanning list in a vector space $\mathcal{V}$ can be reduced to a basis of $\mathcal{V}$.

Lemma C.38. Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of that vector space.

C.1.4 Dimension

Lemma C.39. Any two bases of a finite-dimensional vector space have the same length.

Proof. Suppose B_1 and B_2 are two bases of V . Then B_1 is linearly independent in V and B_2 spans V . By Lemma C.32, the length of B_1 is no greater than B_2 . The proof is completed by switching the roles of B_1 and B_2 . $\square$

Definition C.40. The *dimension* of a finite-dimensional vector space $\mathcal{V}$, denoted $\dim \mathcal{V}$, is the length of any basis of the vector space.

Lemma C.41. If $\mathcal{V}$ is finite-dimensional, then every spanning list of vectors in $\mathcal{V}$ with length $\dim \mathcal{V}$ is a basis of $\mathcal{V}$.

Lemma C.42. If $\mathcal{V}$ is finite-dimensional, then every linearly independent list of vectors in $\mathcal{V}$ with length $\dim \mathcal{V}$ is a basis of $\mathcal{V}$.

Theorem C.43 (Grassmann's formula or subspace dimension theorem). Two finite-dimensional subspaces U and W of a vector space satisfy

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W). \quad (\text{C.8})$$

Corollary C.44. Suppose U and W are two subspaces of a vector space V such that $m := \dim U + \dim W - n \geq 1$ where $n = \dim V \in \mathbb{N}^+$. Then the dimension of the subspace $U \cap W$ is at least m .

Proof. Theorem C.43 yields

$$\begin{aligned} \dim(U \cap W) &= \dim U + \dim W - \dim(U + W) \\ &\geq \dim U + \dim W - n = m. \quad \square \end{aligned}$$

C.2 Linear maps

Definition C.45. A *linear map* or *linear transformation* between two vector spaces $\mathcal{V}$ and $\mathcal{W}$ is a function $T : \mathcal{V} \rightarrow \mathcal{W}$ that satisfies

$$\begin{aligned} (\text{LNM-1}) \text{ additivity} \\ \forall \mathbf{u}, \mathbf{v} \in \mathcal{V}, T(\mathbf{u} + \mathbf{v}) = T\mathbf{u} + T\mathbf{v}; \end{aligned}$$

(LNM-2) homogeneity

$$\forall a \in \mathbb{F}, \forall \mathbf{v} \in \mathcal{V}, T(a\mathbf{v}) = a(T\mathbf{v}),$$

where $\mathbb{F}$ is a scalar field. In particular, a linear map $T : \mathcal{V} \rightarrow \mathcal{W}$ is called a (*linear*) *operator* if $\mathcal{W} = \mathcal{V}$.

Notation 23. The set of all linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted by $\mathcal{L}(\mathcal{V}, \mathcal{W})$. The set of all linear operators from $\mathcal{V}$ to itself is denoted by $\mathcal{L}(\mathcal{V})$.

Example C.46. The differentiation map $D \in \mathcal{L}(\mathbb{P}(\mathbb{R}))$ given by $Dp = p'$ is a linear map.

Example C.47. $\mathcal{L}(\mathbb{F}^n, \mathbb{F}^m) \simeq \mathbb{F}^{m \times n}$ is a vector space with the zero map $\mathbf{0}$ as the additive identity.

Lemma C.48. The set $\mathcal{L}(\mathcal{V}, \mathcal{W})$, equipped with scalar multiplication $(aT)\mathbf{v} = a(T\mathbf{v})$ and vector addition $(S + T)\mathbf{v} = S\mathbf{v} + T\mathbf{v}$, is a vector space.

Proof. The scalar field $\mathbb{F}$ of $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is the same as that of $\mathcal{V}$ and $\mathcal{W}$. So multiplicative identity is still 1, the same as that of $\mathbb{F}$. However, the additive identity is the zero map $\mathbf{0} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. $\square$

Definition C.49. The *identity map*, denoted I , is the function on a vector space that assigns to each element the same element:

$$I\mathbf{v} = \mathbf{v}. \quad (\text{C.9})$$

Definition C.50. A *complex linear functional* is a linear map $T : \mathcal{V} \rightarrow \mathbb{C}$ with $\mathbb{C}$ being the underlying field of $\mathcal{V}$. A *real linear functional* is a map $T : \mathcal{V} \rightarrow \mathbb{R}$ such that (LNM-1) and (LNM-2) in Definition C.45 hold for $\mathbb{F} = \mathbb{R}$.

Lemma C.51. Let V be a complex vector space and f a complex linear functional on V . Then the real part $\text{Re } f(x) = u(x)$ is related to f by

$$\forall x \in V, \quad f(x) = u(x) - \mathbf{i}u(\mathbf{i}x). \quad (\text{C.10})$$

Proof. Any $\alpha, \beta \in \mathbb{R}$ and $z = \alpha + \mathbf{i}\beta \in \mathbb{C}$ satisfy

$$z = \text{Re } z - \mathbf{i}\text{Re } (\mathbf{i}z).$$

Set $z = f(x)$ and we have

$$\begin{aligned} f(x) &= \text{Re } f(x) - \mathbf{i}\text{Re } (\mathbf{i}f(x)) \\ &= u(x) - \mathbf{i}\text{Re } (f(\mathbf{i}x)) = u(x) - \mathbf{i}u(\mathbf{i}x). \quad \square \end{aligned}$$

Lemma C.52. Let V be a complex vector space and $u : V \rightarrow \mathbb{R}$ a real linear functional on V . Then the function $f : V \rightarrow \mathbb{C}$ defined by (C.10) is a complex linear functional.

Proof. The additivity (LNM-1) of f follows from the additivity of u and (C.10). For any $c \in \mathbb{R}$, we have $f(cx) = cf(x)$ from (C.10). The rest follows from the additivity of f and

$$\begin{aligned} f(\mathbf{i}x) &= u(\mathbf{i}x) - \mathbf{i}u(\mathbf{i}^2x) \\ &= u(\mathbf{i}x) + \mathbf{i}u(x) = \mathbf{i}(u(x) - \mathbf{i}u(\mathbf{i}x)) = \mathbf{i}f(x). \quad \square \end{aligned}$$

C.2.1 Null spaces and ranges

Definition C.53. The *null space* or *kernel* of a linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is the subset of $\mathcal{V}$ consisting of those vectors that T maps to the additive identity $\mathbf{0}$:

$$\text{null } T = \{\mathbf{v} \in \mathcal{V} : T\mathbf{v} = \mathbf{0}\}. \quad (\text{C.11})$$

Example C.54. The null space of the differentiation map in Example C.46 is $\mathbb{R}$.

Theorem C.55. A linear map $T \in \mathcal{L}(V, W)$ is injective if and only if $\text{null } T = \{\mathbf{0}\}$.

Definition C.56. The *range* of a linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is the subset of $\mathcal{W}$ consisting of those vectors that are of the form $T\mathbf{v}$ for some $\mathbf{v} \in \mathcal{V}$:

$$\text{range } T = \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\}. \quad (\text{C.12})$$

Example C.57. The range of $A \in \mathbb{C}^{m \times n}$ is the span of its column vectors.

Theorem C.58. The range of a linear map $T \in \mathcal{L}(V, W)$ is a subspace of W .

Theorem C.59 (The counting theorem or the fundamental theorem of linear maps). If $\mathcal{V}$ is a finite-dimensional vector space and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $\text{range } T$ is a finite-dimensional subspace of $\mathcal{W}$ and

$$\dim \mathcal{V} = \dim \text{null } T + \dim \text{range } T. \quad (\text{C.13})$$

Theorem C.60. For an operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional vector space $\mathcal{V}$, the following are equivalent:

- (a) T is invertible;
- (b) T is injective;
- (c) T is surjective.

C.2.2 The matrix of a linear map

Definition C.61. The *matrix of a linear map* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ with respect to the bases $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ of $\mathcal{V}$ and $(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m)$ of $\mathcal{W}$, denoted by

$$M_T := M(T, (\mathbf{v}_1, \dots, \mathbf{v}_n), (\mathbf{w}_1, \dots, \mathbf{w}_m)), \quad (\text{C.14})$$

is the $m \times n$ matrix $A(T)$ whose entries $a_{i,j} \in \mathbb{F}$ satisfy the linear system

$$\forall j = 1, 2, \dots, n, \quad T\mathbf{v}_j = \sum_{i=1}^m a_{i,j} \mathbf{w}_i. \quad (\text{C.15})$$

Corollary C.62. The matrix M_T in (C.14) of a linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfies

$$T[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m] M_T. \quad (\text{C.16})$$

Proof. This follows directly from (C.15). $\square$

C.2.3 Duality

Dual vector spaces

Definition C.63. The *dual space* of a vector space V is the vector space of all linear functionals on V ,

$$V' = \mathcal{L}(V, \mathbb{F}). \quad (\text{C.17})$$

Definition C.64. For a basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ of a vector space V , its *dual basis* is the list $\varphi_1, \dots, \varphi_n$ where each $\varphi_j \in V'$ is

$$\varphi_j(\mathbf{v}_k) = \begin{cases} 1 & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases} \quad (\text{C.18})$$

Exercise C.65. Show that the dual basis is a basis of the dual space.

Lemma C.66. A finite-dimensional vector space V satisfies

$$\dim V' = \dim V. \quad (\text{C.19})$$

Proof. This follows from Definition C.63 and the identity $\dim \mathcal{L}(V, W) = \dim(V) \dim(W)$. $\square$

Definition C.67. The *double dual space* of a vector space V , denoted by V'' , is the dual space of V' .

Lemma C.68. The function $\Lambda : V \rightarrow V''$ defined as

$$\forall v \in V, \forall \varphi \in V', \quad (\Lambda v)(\varphi) = \varphi(v) \quad (\text{C.20})$$

is a linear bijection.

Proof. It is easily verified that Λ is a linear map. The rest follows from Definitions C.63, C.67, and Lemma C.66. $\square$

Dual linear maps

Definition C.69. The *dual map* of a linear map $T : V \rightarrow W$ is the linear map $T' : W' \rightarrow V'$ defined as

$$\forall \varphi \in W', \quad T'(\varphi) = \varphi \circ T. \quad (\text{C.21})$$

Exercise C.70. Under the dual map of the linear map D in Example C.46, what is the image of the linear functional $\varphi(p) = \int_0^1 p$ on $\mathbb{P}(\mathbb{R})$?

Theorem C.71. The matrix of T' is the transpose of the matrix of T .

Proof. Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, $(\varphi_1, \dots, \varphi_n)$, $(\mathbf{w}_1, \dots, \mathbf{w}_m)$, $(\psi_1, \dots, \psi_m)$, be bases of V , V' , W , W' , respectively. Denote by A and C the matrices of $T : V \rightarrow W$ and $T' : W' \rightarrow V'$, respectively. We have

$$\psi_j \circ T = T'(\psi_j) = \sum_{r=1}^m c_{r,j} \varphi_r.$$

By Corollary C.62, applying this equation to $\mathbf{v}_k$ yields

$$(\psi_j \circ T)(\mathbf{v}_k) = \sum_{r=1}^m c_{r,j} \varphi_r(\mathbf{v}_k) = c_{k,j}.$$

On the other hand, we have

$$\begin{aligned} (\psi_j \circ T)(\mathbf{v}_k) &= \psi_j(T\mathbf{v}_k) = \psi_j\left(\sum_{r=1}^m a_{r,k} \mathbf{w}_r\right) \\ &= \sum_{r=1}^m a_{r,k} \psi_j(\mathbf{w}_r) = a_{j,k}. \end{aligned} \quad \square$$

Definition C.72. The *double dual map* of a linear map $T : V \rightarrow W$ is the linear map $T'' : V'' \rightarrow W''$ defined as $T'' = (T')'$.

Theorem C.73. For $T \in \mathcal{L}(V)$ and Λ in (C.20), we have

$$T'' \circ \Lambda = \Lambda \circ T. \quad (\text{C.22})$$

Proof. Definition C.72 and equation (C.20) yields

$$\begin{aligned} \forall v \in V, \forall \varphi \in V', \\ (T'' \circ \Lambda)v\varphi &= ((T')'\Lambda v)\varphi = (\Lambda v \circ T')\varphi = \Lambda v(T'\varphi) \\ &= (T'\varphi)(v) = \varphi(Tv) = \Lambda(Tv)(\varphi) \\ &= (\Lambda \circ T)v\varphi, \end{aligned}$$

where the third step is natural since T' send V' to V' . $\square$

Corollary C.74. For $T \in \mathcal{L}(V)$ where V is finite-dimensional, the double dual map is

$$T'' = \Lambda \circ T \circ \Lambda^{-1}. \quad (\text{C.23})$$

Proof. This follows directly from Theorem C.73 and Lemma C.68. $\square$

The null space and range of the dual of a linear map

Definition C.75. For $U \subset V$, the *annihilator* of U , denoted U^0 , is defined by

$$U^0 := \{\varphi \in V' : \forall \mathbf{u} \in U, \varphi(\mathbf{u}) = 0\}. \quad (\text{C.24})$$

Exercise C.76. Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_5$ denote the standard basis of $V = \mathbb{R}^5$, and $\varphi_1, \varphi_2, \varphi_3, \varphi_4, \varphi_5$ its dual basis of V' . Suppose

$$U = \text{span}(\mathbf{e}_1, \mathbf{e}_2) = \{(x_1, x_2, 0, 0, 0) \in \mathbb{R}^5 : x_1, x_2 \in \mathbb{R}\}.$$

Show that $U^0 = \text{span}(\varphi_3, \varphi_4, \varphi_5)$.

Exercise C.77. Let $i : U \hookrightarrow V$ be an inclusion. Show that $\text{null } i' = U^0$.

Lemma C.78. Suppose $U \subset V$. Then U^0 is a subspace of V' .

Exercise C.79. Suppose V is finite-dimensional. Prove every linear map on a subspace of V can be extended to a linear map on V .

Lemma C.80. Suppose V is finite-dimensional and U is a subspace of V . Then

$$\dim U + \dim U^0 = \dim V. \quad (\text{C.25})$$

Proof. Apply Theorem C.59 to the dual of an inclusion $i' : V' \rightarrow U'$ and we have

$$\begin{aligned} \dim \text{range } i' + \dim \text{null } i' &= \dim V' \\ \implies \dim \text{range } i' + \dim U^0 &= \dim V, \end{aligned}$$

where the second line follows from Exercise C.77 and Lemma C.66. For any $\varphi \in U'$, Exercise C.79 states that $\varphi \in U'$ can be extended to $\psi \in V'$ such that $i'(\psi) = \varphi$. Hence i' is surjective and we have $U' = \text{range } i'$. The proof is then completed by Lemma C.66. $\square$

Lemma C.81. Any linear map $T \in \mathcal{L}(V, W)$ satisfies

$$\text{null } T' = (\text{range } T)^0. \quad (\text{C.26})$$

Proof. Definitions C.53, C.56, C.69, and C.75 yield

$$\begin{aligned} \varphi \in \text{null } T' &\iff 0 = T'(\varphi) = \varphi \circ T \\ &\iff \forall v \in V, \varphi(Tv) = 0 \\ &\iff \varphi(\text{range } T) = 0 \\ &\iff \varphi \in (\text{range } T)^0. \quad \square \end{aligned}$$

Lemma C.82. For finite-dimensional vector spaces V and W , any linear map $T \in \mathcal{L}(V, W)$ satisfies

$$\dim \text{null } T' = \dim \text{null } T + \dim W - \dim V. \quad (\text{C.27})$$

Proof. Lemma C.81 and Theorem C.59 yield

$$\begin{aligned} \dim \text{null } T' &= \dim(\text{range } T)^0 = \dim W - \dim(\text{range } T) \\ &= \dim W - \dim V + \dim(\text{null } T) \\ &= \dim \text{null } T + \dim W - \dim V. \quad \square \end{aligned}$$

Corollary C.83. For finite-dimensional vector spaces V and W , any linear map $T \in \mathcal{L}(V, W)$ is surjective if and only if T' is injective.

Proof. T is surjective $\iff W = \text{range } T \iff (\text{range } T)^0 = \{0\} \iff \text{null } T' = \{0\} \iff T'$ is injective. The second step follows from Lemma C.80 applied to W :

$$\dim W = \dim(\text{range } T) + \dim(\text{range } T)^0. \quad \square$$

Lemma C.84. For finite-dimensional vector spaces V and W , any linear map $T \in \mathcal{L}(V, W)$ satisfies

$$\dim \text{range } T' = \dim \text{range } T. \quad (\text{C.28})$$

Proof. Theorem C.59, Lemma C.81, and Lemma C.80 yield

$$\begin{aligned} \dim \text{range } T' &= \dim W - \dim \text{null } T' \\ &= \dim W - \dim(\text{range } T)^0 \\ &= \dim(\text{range } T). \quad \square \end{aligned}$$

Lemma C.85. For finite-dimensional vector spaces V and W , any linear map $T \in \mathcal{L}(V, W)$ satisfies

$$\text{range } T' = (\text{null } T)^0. \quad (\text{C.29})$$

Proof. Theorem C.59, Lemma C.81, and Lemma C.80 yield

$$\begin{aligned} \varphi \in \text{range } T' &\implies \exists \psi \in W' \text{ s.t. } T'(\psi) = \varphi \\ &\implies \forall v \in \text{null } T, \varphi(v) = \psi(Tv) = 0 \\ &\implies \varphi \in (\text{null } T)^0. \end{aligned}$$

The proof is completed by

$$\begin{aligned} \dim \text{range } T' &= \dim(\text{range } T) \\ &= \dim V - \dim \text{null } T \\ &= \dim(\text{null } T)^0. \quad \square \end{aligned}$$

Corollary C.86. For finite-dimensional vector spaces V and W , any linear map $T \in \mathcal{L}(V, W)$ is injective if and only if T' is surjective.

Proof. T is injective $\iff \text{null } T = \{0\} \iff (\text{null } T)^0 = V' \iff \text{range } T' = V' \iff T'$ is surjective. The second step follows from Lemmas C.80 and C.66, and the third step follows from Lemma C.85. $\square$

Matrix ranks

Definition C.87. For a matrix $A \in \mathbb{F}^{m \times n} : \mathbb{F}^n \rightarrow \mathbb{F}^m$, its *column space* (or range or image) consists of all linear combinations of its columns, its *row space* (or coimage) is the column space of $A^\top$, its *null space* (or kernel) is the null space of A as a linear operator, and the *left null space* (or cokernel) is the null space of $A^\top$.

Definition C.88. The *column rank* and *row rank* of a matrix $A \in \mathbb{F}^{m \times n}$ is the dimension of its column space and row space, respectively.

Lemma C.89. Let A_T denote the matrix of a linear operator $T \in \mathcal{L}(V, W)$. Then the column rank of A_T is the dimension of $\text{range } T$.

Proof. For $\mathbf{u} = \sum_i c_i \mathbf{v}_i$, Corollary C.62 yields

$$T\mathbf{u} = \sum_i c_i T\mathbf{v}_i = T[\mathbf{v}_1, \dots, \mathbf{v}_n]\mathbf{c} = [\mathbf{w}_1, \dots, \mathbf{w}_m]A_T\mathbf{c}.$$

Hence we have

$$\{T\mathbf{u} : \mathbf{c} \in \mathbb{F}^n\} = \{[\mathbf{w}_1, \dots, \mathbf{w}_m]A_T\mathbf{c} : \mathbf{c} \in \mathbb{F}^n\}.$$

The LHS is $\text{range } T$ while $\{A_T\mathbf{c} : \mathbf{c} \in \mathbb{F}^n\}$ is the column space of A_T . Since $(\mathbf{w}_1, \dots, \mathbf{w}_m)$ is a basis, by Definition C.88 the column rank of the matrix $[\mathbf{w}_1, \dots, \mathbf{w}_m]$ is m . Taking $\dim$ to both sides of the above equation yields the conclusion. Note that the RHS is a subspace of $\mathbb{F}^m$ (why?) and the dimension of it does not depend on the special choice of its basis, hence we can choose $(\mathbf{w}_1, \dots, \mathbf{w}_m)$ to be the standard basis and then $[\mathbf{w}_1, \dots, \mathbf{w}_m]$ is simply the identity matrix. $\square$

Theorem C.90. For any $A \in \mathbb{F}^{m \times n}$, its row rank equals its column rank.

Proof. Define a linear map $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ as $T\mathbf{x} = A\mathbf{x}$. Clearly, A is the matrix of T for the standard bases of $\mathbb{F}^n$ and $\mathbb{F}^m$. Then we have,

$$\begin{aligned} \text{column rank of } A &= \dim \text{range } T \\ &= \dim \text{range } T' \\ &= \text{column rank of the matrix of } T' \\ &= \text{column rank of } A^\top \\ &= \text{row rank of } A, \end{aligned}$$

where the first step follows from Lemma C.89, the second from Lemma C.84, the third from Lemma C.89, the fourth from Theorem C.71, and the last from the definition of matrix transpose and matrix products. $\square$

Definition C.91. The *rank* of a matrix is its column rank.

Theorem C.92 (Fundamental theorem of linear algebra). For a matrix $A \in \mathbb{R}^{m \times n} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, its column space and row space both have dimension $r \leq \min(m, n)$; its null space and left null space have dimensions $n - r$ and $m - r$, respectively. In addition, we have

$$\mathbb{R}^m = \text{range } A \oplus \text{null } A^\top, \quad (\text{C.30a})$$

$$\mathbb{R}^n = \text{range } A^\top \oplus \text{null } A, \quad (\text{C.30b})$$

where $\text{range } A \perp \text{null } A^\top$ and $\text{range } A^\top \perp \text{null } A$.

Proof. The first sentence is a rephrase of Theorem C.90 and follows from Theorem C.59. For the second sentence, we only prove (C.30b). $\mathbf{x} \in \text{null } A$ implies $\mathbf{x} \in \mathbb{R}^n$ and $A\mathbf{x} = \mathbf{0}$. The latter expands to

$$\begin{bmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \vdots \\ \mathbf{a}_m^\top \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

which implies that $\forall j = 1, 2, \dots, m$, $\mathbf{a}_j \perp \mathbf{x}$. Hence $\mathbf{x}$ is orthogonal to each basis vector of $\text{range } A^\top$. The rest of the proof follows from Lemma C.84, Theorem C.71, Theorem C.59. $\square$

C.3 Eigenvalues and eigenvectors

C.3.1 Invariant subspaces

Definition C.93. Under a linear operator $T \in \mathcal{L}(\mathcal{V})$, a subspace $\mathcal{U}$ of $\mathcal{V}$ is *invariant* if $\mathbf{u} \in \mathcal{U}$ implies $T\mathbf{u} \in \mathcal{U}$.

Example C.94. Under $T \in \mathcal{L}(\mathcal{V})$, each of the following subspaces of $\mathcal{V}$ is invariant: $\{\mathbf{0}\}$, $\mathcal{V}$, $\text{null } T$, and $\text{range } T$.

Definition C.95. A number $\lambda \in \mathbb{F}$ is called an *eigenvalue* of an operator $T \in \mathcal{L}(\mathcal{V})$ if there exists $\mathbf{v} \in \mathcal{V}$ such that $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$. Then the vector $\mathbf{v}$ is called an *eigenvector* of T corresponding to λ .

Lemma C.96. Suppose V is finite-dimensional. $\lambda \in V$ is an eigenvalue of $T \in \mathcal{L}(V)$ if and only if $T - \lambda I$ is not injective.

Proof. This follows directly from Definition C.95. $\square$

Example C.97. For each eigenvector $\mathbf{v}$ of $T \in \mathcal{L}(\mathcal{V})$, the subspace $\text{span}(\mathbf{v})$ is a one-dimensional invariant subspace of $\mathcal{V}$.

Lemma C.98. Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$ with corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_m$. Then $\mathbf{v}_1, \dots, \mathbf{v}_m$ is linearly independent.

Lemma C.99. Suppose $\mathcal{V}$ is finite-dimensional. Then each operator on $\mathcal{V}$ has at most $\dim \mathcal{V}$ distinct eigenvalues.

Definition C.100. Suppose $T \in \mathcal{L}(V)$ and U is an invariant subspace of V under T . The *restriction operator* $T|_U \in \mathcal{L}(U)$ is defined by

$$\forall \mathbf{u} \in U, \quad T|_U(\mathbf{u}) = T\mathbf{u}. \quad (\text{C.31})$$

C.3.2 Existence of eigenvalues

Notation 24. Suppose $T \in \mathcal{L}(\mathcal{V})$ and $p \in \mathbb{P}(\mathbb{F})$ is a polynomial given by

$$p(z) = a_0 + a_1z + \cdots + a_mz^m$$

for $z \in \mathbb{F}$. Then $p(T)$ is the operator given by

$$p(T) = a_0I + a_1T + \cdots + a_mT^m,$$

where $I = T^0$ is the identity operator.

Example C.101. Suppose $D \in \mathcal{L}(\mathbb{P}(\mathbb{R}))$ is the differentiation operator defined by $Dq = q'$ and p is the polynomial defined by $p(x) = 7 - 3x + 5x^2$. Then we have

$$p(D) = 7 - 3D + 5D^2, \quad (p(D))q = 7q - 3q' + 5q''.$$

Definition C.102. The *product polynomial* of two polynomials $p, q \in \mathbb{P}(\mathbb{F})$ is the polynomial defined by

$$\forall z \in \mathbb{F}, \quad (pq)(z) := p(z)q(z). \quad (\text{C.32})$$

Lemma C.103. Any $T \in \mathcal{L}(\mathcal{V})$ and $p, q \in \mathbb{P}(\mathbb{F})$ satisfy

$$(pq)(T) = p(T)q(T) = q(T)p(T). \quad (\text{C.33})$$

Theorem C.104 (Existence of eigenvalues). Every operator $T \in \mathcal{L}(V)$ on a finite-dimensional, nonzero, complex vector space V has an eigenvalue.

Proof. Write $n := \dim V$. For a nonzero $\mathbf{v} \in V$, the $n + 1$ vectors $(\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots, T^n\mathbf{v})$ must be linear dependent, i.e.,

$$\mathbf{0} = a_0\mathbf{v} + a_1T\mathbf{v} + \cdots + a_nT^n\mathbf{v} = (a_0 + a_1T + \cdots + a_nT^n)\mathbf{v}$$

implies that there exists $j \in [1, n]$ such that $a_j \neq 0$. By the fundamental theorem of algebra, the polynomial $\sum_{i=0}^n a_iT^i$ has m roots, say, $\lambda_1, \dots, \lambda_m$, and thus for some $c \in \mathbb{C} \setminus \{0\}$, we have

$$\mathbf{0} = c(T - \lambda_1I)(T - \lambda_2I) \cdots (T - \lambda_mI)\mathbf{v}.$$

Hence $T - \lambda_jI$ is not injective for some λ_j and the proof is completed by Lemma C.96. $\square$

C.3.3 Similarity transformations

Definition C.105. Two matrices $A, B \in \mathbb{C}^{n \times n}$ are *similar* if there exists a nonsingular matrix S such that

$$B = SAS^{-1}, \quad (\text{C.34})$$

and the map $A \mapsto SAS^{-1}$ is a *similarity transformation*.

Lemma C.106. Two similar matrices A and B have the same set of eigenvalues.

Proof. Let $(\lambda, \mathbf{u})$ be an eigenpair of A , i.e., $A\mathbf{u} = \lambda\mathbf{u}$. Then (C.34) yields

$$BS\mathbf{u} = SA\mathbf{u} = \lambda S\mathbf{u},$$

and thus $(\lambda, S\mathbf{u})$ is an eigenpair of B . $\square$

C.3.4 Upper-triangular matrices

Definition C.107. The *matrix* of a linear operator $T \in \mathcal{L}(\mathcal{V})$ is the matrix of the linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{V})$, cf. Definition C.61.

Definition C.108. A matrix A is *upper triangular* iff $i > j$ implies $a_{i,j} = 0$. Similarly, a matrix A is *lower triangular* iff $i < j$ implies $a_{i,j} = 0$.

Theorem C.109. Suppose $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v}_1, \dots, \mathbf{v}_n$ is a basis of $\mathcal{V}$. Then the following are equivalent:

- (a) the matrix of T with respect to $\mathbf{v}_1, \dots, \mathbf{v}_n$ is upper triangular;
- (b) $T\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for each $j = 1, \dots, n$;
- (c) $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ is invariant under T for each $j = 1, \dots, n$.

Theorem C.110 (Schur). Every linear operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional complex vector space $\mathcal{V}$ has an upper-triangular matrix with respect to some basis of $\mathcal{V}$.

Theorem C.111. Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with respect to some basis of $\mathcal{V}$. Then T is invertible if and only if all the entries on the diagonal of that upper-triangular matrix are nonzero.

Theorem C.112. Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with respect to some basis of $\mathcal{V}$. Then the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.

C.3.5 Eigenspaces and diagonal matrices

Definition C.113. A *diagonal entry* of a matrix is an entry of the matrix of which the row index equals the column index. The *diagonal* of a matrix consists of all diagonal entries of the matrix. A *diagonal matrix* is a matrix that is zero everywhere except possibly along the diagonal.

Definition C.114. The *eigenspace* of $T \in \mathcal{L}(\mathcal{V})$ corresponding to $\lambda \in \mathbb{F}$ is

$$E(\lambda, T) := \text{null}(T - \lambda I). \quad (\text{C.35})$$

Lemma C.115. Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional space $\mathcal{V}$. Then

$$E(\lambda_1, T) + \cdots + E(\lambda_m, T)$$

is a direct sum and

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim \mathcal{V}. \quad (\text{C.36})$$

Definition C.116. An operator $T \in \mathcal{L}(\mathcal{V})$ is *diagonalizable* if it has a diagonal matrix with respect to some basis of $\mathcal{V}$.

Theorem C.117 (Conditions of diagonalizability). Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional space $\mathcal{V}$. Then the following are equivalent:

- (a) T is diagonalizable;

- (b) $\mathcal{V}$ has a basis consisting of eigenvectors of T ;
- (c) there exist one-dimensional subspaces $U_1, \dots, U_n$ of $\mathcal{V}$, each invariant under T , such that $\mathcal{V} = U_1 \oplus \dots \oplus U_n$;
- (d) $\mathcal{V} = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)$;
- (e) $\dim \mathcal{V} = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T)$.

Corollary C.118. An operator $T \in \mathcal{L}(\mathcal{V})$ is diagonalizable if T has $\dim \mathcal{V}$ distinct eigenvalues.

C.4 Operators on complex vector spaces

C.4.1 Generalized eigenvectors

Lemma C.119. For a linear operator $T \in \mathcal{L}(V)$, we have

$$\{0\} = \text{null } T^0 \subseteq \text{null } T^1 \subseteq \dots \subseteq \text{null } T^k \subseteq \text{null } T^{k+1} \subseteq \dots \quad (\text{C.37})$$

Proof. Suppose $\mathbf{v} \in \text{null } T^k$ for $k \in \mathbb{N}$. Thus $T^k \mathbf{v} = 0$. Then $T^{k+1} \mathbf{v} = T T^k \mathbf{v} = T 0 = 0$ and therefore $\mathbf{v} \in \text{null } T^{k+1}$. $\square$

Lemma C.120. Suppose a linear operator $T \in \mathcal{L}(V)$ satisfies $\text{null } T^m = \text{null } T^{m+1}$. Then we have

$$\text{null } T^m = \text{null } T^{m+1} = \text{null } T^{m+2} = \dots \quad (\text{C.38})$$

Proof. By Lemma C.119, it suffices to show

$$\forall k \in \mathbb{N}^+, \quad \text{null } T^{m+k+1} \subseteq \text{null } T^{m+k},$$

which indeed holds because

$$\begin{aligned} \mathbf{v} \in \text{null } T^{m+k+1} &\implies T^{m+k+1} \mathbf{v} = 0 \implies T^{m+1}(T^k \mathbf{v}) = 0 \\ &\implies T^k \mathbf{v} \in \text{null } T^{m+1} = \text{null } T^m \\ &\implies T^m T^k \mathbf{v} = 0 \implies T^{m+k} \mathbf{v} = 0 \\ &\implies \mathbf{v} \in \text{null } T^{m+k}. \quad \square \end{aligned}$$

Lemma C.121. A linear operator $T \in \mathcal{L}(V)$ satisfies

$$\text{null } T^n = \text{null } T^{n+1} = \text{null } T^{n+2} = \dots, \quad (\text{C.39})$$

where $n = \dim V$.

Proof. By Lemma C.120, it suffices to show

$$\text{null } T^n = \text{null } T^{n+1}.$$

Suppose this is not true. Then Lemmas C.119 and C.120 yield

$$\{0\} = \text{null } T^0 \subset \text{null } T^1 \subset \dots \subset \text{null } T^n \subset \text{null } T^{n+1},$$

where the symbol “ $\subset$ ” means strict inclusion, cf. Definition A.4. At each strict inclusion in the above chain, the dimension of the space increases by at least 1, and thus $\dim \text{null } T^{n+1} > n$. But the dimension of any subspace of V cannot exceed that of V . $\square$

Theorem C.122. A linear operator $T \in \mathcal{L}(V)$ satisfies

$$V = \text{null } T^n \oplus \text{range } T^n \quad (\text{C.40})$$

where $n = \dim V$.

Proof. We first show $\text{null } T^n \cap \text{range } T^n = \{0\}$. Indeed, if $\mathbf{v} \in \text{null } T^n \cap \text{range } T^n$, then $T^n \mathbf{v} = 0$ and there exists $\mathbf{u}$ such that $\mathbf{v} = T^n \mathbf{u}$. Hence $T^{2n} \mathbf{u} = 0$. Lemma C.121 further implies $T^n \mathbf{u} = 0$ and thus $\mathbf{v} = 0$. By Theorem C.14, $\text{null } T^n + \text{range } T^n$ is a direct sum. Then (C.40) follows from

$$\begin{aligned} \dim(\text{null } T^n \oplus \text{range } T^n) &= \dim \text{null } T^n + \dim \text{range } T^n \\ &= \dim V, \end{aligned}$$

where the second step follows from the fundamental theorem of linear maps (Theorem C.59). $\square$

Example C.123. For the operator $T \in \mathcal{L}(\mathbb{C}^3)$ given by

$$T(z_1, z_2, z_3) = (4z_2, 0, 5z_3), \quad (\text{C.41})$$

$\text{null } T + \text{range } T$ is not a direct sum of $\mathbb{C}^3$ because

$$\begin{aligned} \text{null } T &= \{(z_1, 0, 0) : z_1 \in \mathbb{C}\}, \\ \text{range } T &= \{(z_1, 0, z_3) : z_1, z_3 \in \mathbb{C}\}. \end{aligned}$$

In contrast, $T^3(z_1, z_2, z_3) = (0, 0, 125z_3)$ and thus

$$\begin{aligned} \text{null } T^3 &= \{(z_1, z_2, 0) : z_1, z_2 \in \mathbb{C}\}, \\ \text{range } T^3 &= \{(0, 0, z_3) : z_3 \in \mathbb{C}\}, \\ \text{null } T^3 \oplus \text{range } T^3 &= \mathbb{C}^3. \end{aligned}$$

Definition C.124. A *generalized eigenvector* of a linear operator $T \in \mathcal{L}(V)$ corresponding to the eigenvalue λ of T is a nonzero vector $\mathbf{v} \in V$ satisfying

$$\exists j \in \mathbb{N}^+ \text{ s.t. } (T - \lambda I)^j \mathbf{v} = 0. \quad (\text{C.42})$$

Definition C.125. The *generalized eigenspace* of a linear operator $T \in \mathcal{L}(V)$ corresponding to the eigenvalue λ of T , denoted $G(\lambda, T)$, is the set of all generalized eigenvectors of T corresponding to λ along with the zero vector.

Lemma C.126. For generalized eigenspaces, we have

$$\forall T \in \mathcal{L}(V), \forall \lambda \in \mathbb{F}, \quad G(\lambda, T) = \text{null}(T - \lambda I)^{\dim V}. \quad (\text{C.43})$$

Proof. Suppose $\mathbf{v} \in \text{null}(T - \lambda I)^{\dim V}$. Then Definitions C.124 and C.125 imply $\mathbf{v} \in G(\lambda, T)$. Conversely, $\mathbf{v} \in G(\lambda, T)$ implies that $(T - \lambda I)^j \mathbf{v} = 0$ for some $j \in \mathbb{N}^+$. Then we have $\mathbf{v} \in \text{null}(T - \lambda I)^{\dim V}$ from Lemmas C.119 and C.121. $\square$

Definition C.127. The *multiplicity or algebraic multiplicity* of the eigenvalue λ of an operator $T \in \mathcal{L}(V)$ is the dimension of the corresponding generalized eigenspace,

$$m_a(\lambda) := \dim G(\lambda, T) = \dim \text{null}(T - \lambda I)^{\dim V}, \quad (\text{C.44})$$

while the *geometric multiplicity* of an eigenvalue λ of T is the dimension of the corresponding eigenspace,

$$m_g(\lambda) := \dim E(\lambda, T) = \dim \text{null}(T - \lambda I). \quad (\text{C.45})$$

The *index* of an eigenvalue λ of T is the smallest integer k for which

$$\text{null}(T - \lambda I)^k = \text{null}(T - \lambda I)^{k+1}. \quad (\text{C.46})$$

Corollary C.128. Geometric multiplicity and algebraic multiplicity satisfy

$$1 \leq m_g(\lambda) \leq m_a(\lambda). \quad (\text{C.47})$$

Proof. The case $j = 1$ in Definition C.124 implies that every eigenvector of T is a generalized eigenvector of T . Then the proof is completed by Lemma C.119 and Definition C.127. $\square$

Definition C.129. An eigenvalue λ of A is *defective* iff

$$m_g(\lambda) < m_a(\lambda). \quad (\text{C.48})$$

A is *defective* iff A has one or more defective eigenvalues.

Example C.130. Eigenvalues of the operator $T \in \mathcal{L}(\mathbb{C}^3)$ in (C.41) are 0 and 5, with the corresponding eigenspaces as

$$\begin{aligned} E(0, T) &= \{(z_1, 0, 0) : z_1 \in \mathbb{C}\}, \\ E(5, T) &= \{(0, 0, z_3) : z_3 \in \mathbb{C}\}. \end{aligned}$$

The generalized eigenspaces are deduced as follows,

$$\begin{aligned} T^3(z_1, z_2, z_3) &= (0, 0, 125z_3), \\ G(0, T) &= \{(z_1, z_2, 0) : z_1, z_2 \in \mathbb{C}\}, \\ (T - 5I)^3(z_1, z_2, z_3) &= (-125z_1 + 300z_2, -125z_2, 0), \\ G(5, T) = E(5, T) &= \{(0, 0, z_3) : z_3 \in \mathbb{C}\}. \end{aligned}$$

Since $m_g(5) = 1 = m_a(5)$ and $m_g(0) = 1 < m_a(0) = 2$, the eigenvalue 5 is not defective but the eigenvalue 0 is. Hence the operator T is defective.

Lemma C.131. Generalized eigenvectors of distinct eigenvalues of an operator $T \in \mathcal{L}(V)$ are linearly independent.

Proof. Let $(\lambda_i, \mathbf{v}_i)$ be a generalized eigenpair of T . Define

$$\mathbf{w} := (T - \lambda_1 I)^k \mathbf{v}_1,$$

where $k \in \mathbb{N}$ is the largest integer such that $\mathbf{w} \neq \mathbf{0}$. By Definition C.124, $(T - \lambda_1 I)\mathbf{w} = \mathbf{0}$ and thus $\mathbf{w}$ is an eigenvector of T . Also, $(T - \lambda I)\mathbf{w} = (T - \lambda_1 I + \lambda_1 I - \lambda I)\mathbf{w} = (\lambda_1 - \lambda)\mathbf{w}$ yields

$$(*) : \quad \forall j \in \mathbb{N}^+, \forall \lambda \in \mathbb{C}, \quad (T - \lambda I)^j \mathbf{w} = (\lambda_1 - \lambda)^j \mathbf{w}.$$

Write $n := \dim V$ and define a polynomial of T by

$$p_1(T) = (T - \lambda_1 I)^k (T - \lambda_2 I)^n \cdots (T - \lambda_m I)^n;$$

By Lemma C.103, the factors of $p_1(T)$ commute.

Suppose $\mathbf{0} = \sum_{i=1}^m a_i \mathbf{v}_i$ where each $a_i \in \mathbb{C}$. Then the application of $p(T)$ to this equation yields $a_1 = 0$ because of the distinctness of the eigenvalues and

$$\begin{aligned} \mathbf{0} &= a_1 (T - \lambda_1 I)^k (T - \lambda_2 I)^n \cdots (T - \lambda_m I)^n \mathbf{v}_1 \\ &= a_1 (T - \lambda_2 I)^n \cdots (T - \lambda_m I)^n \mathbf{w} \\ &= a_1 (\lambda_1 - \lambda_2)^n \cdots (\lambda_1 - \lambda_m)^n \mathbf{w}, \end{aligned}$$

where the first equality follows from Lemma C.126, the second from the definition of $\mathbf{w}$, and the third from (*).

Similarly, the above arguments applied to the other generalized eigenpairs yield $a_2 = \cdots = a_m = 0$, which implies the linear independence of generalized eigenvectors. $\square$

C.4.2 Nilpotent operators

Definition C.132. An operator $N \in \mathcal{L}(V)$ is *nilpotent* iff $N^k = \mathbf{0}$ for some $k \in \mathbb{N}^+$.

Example C.133. The differentiation operator on the vector space of polynomials of degree at most m is nilpotent. The operator $N \in \mathcal{L}(\mathbb{F}^3)$ given by $N(x, y, z) = (y, z, 0)$ is nilpotent because $N^3 = \mathbf{0}$.

Lemma C.134. A nilpotent operator $N \in \mathcal{L}(V)$ satisfies

$$N^{\dim V} = \mathbf{0}. \quad (\text{C.49})$$

Proof. Definitions C.132 and C.125 imply $G(0, N) = V$. The rest follows from Lemma C.126. $\square$

Lemma C.135. Any nilpotent operator $N \in \mathcal{L}(V)$ has a strictly upper-triangular matrix M , i.e., $\forall i \geq j, M_{i,j} = 0$.

Proof. Write $n := \dim V$. Choose a basis of $\text{null } N$, extend it to a basis of $\text{null } N^2, \dots, \text{null } N^n$, and we have a matrix U whose columns are the vectors of this basis:

$$U = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n].$$

Corollary C.62 states that

$$N[\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n] = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n]M.$$

The fact of $\mathbf{u}_1 \in \text{null } N$ dictates that all entries in the first column of M must be zero. Let $\mathbf{u}_j$ be the first basis vector in $\text{null } N^2$ but not in $\text{null } N$. Then $N\mathbf{u}_j \in \text{null } N$, i.e., $N\mathbf{u}_j$ must be a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{j-1}$. Hence the first $j-1$ entries in the j th column of M could be nonzero while all other entries must be zero. Proceeding in this fashion completes the proof. $\square$

Lemma C.136. $I + N$ has a square root for any nilpotent operator $N \in \mathcal{L}(V)$.

Proof. By Definition C.132, $N^m = \mathbf{0}$ for some $m \in \mathbb{N}^+$. Construct $M := I + \sum_{i=1}^{m-1} a_i N^i$ and we claim that the coefficients a_i 's can be uniquely determined so that $M^2 = I + N$. Indeed, $M^2 = I + N$ gives $\left(I + \sum_{i=1}^{m-1} a_i N^i\right)^2 = I + N$, which is equivalent to

$$2a_1 = 1, \quad 2a_2 + a_1^2 = 0, \quad 2a_3 + 2a_1a_2 = 0, \quad \dots,$$

where $a_1 = \frac{1}{2}$ and each a_j can be uniquely determined from $a_1, a_2, \dots, a_{j-1}$. $\square$

C.4.3 Operator decomposition

Lemma C.137. Suppose $p(T)$ is a polynomial of an operator $T \in \mathcal{L}(V)$. Then both $\text{null } p(T)$ and $\text{range } p(T)$ are invariant under T .

Proof. $\mathbf{v} \in \text{null } p(T)$ implies $p(T)\mathbf{v} = \mathbf{0}$ and

$$p(T)(T\mathbf{v}) = T(p(T)\mathbf{v}) = T\mathbf{0} = \mathbf{0}.$$

Hence $T\mathbf{v} \in \text{null } p(T)$, and, by Definition C.93, $\text{null } p(T)$ is invariant under T . Similarly,

$$\mathbf{v} \in \text{range } p(T) \implies \exists \mathbf{u} \in V \text{ s.t. } p(T)\mathbf{u} = \mathbf{v}$$

and thus

$$T\mathbf{v} = Tp(T)\mathbf{u} = p(T)(T\mathbf{u}).$$

Since $T\mathbf{u} \in V$, we have $T\mathbf{v} \in \text{range } p(T)$ and, by Definition C.93, $\text{range } p(T)$ is invariant under T . $\square$

Theorem C.138 (Decomposing operators on complex vector spaces). Suppose V is a complex vector space, $T \in \mathcal{L}(V)$, and $\lambda_1, \lambda_2, \dots, \lambda_m$ are all the distinct eigenvalues of T . Then

(DOC-1) $V = G(\lambda_1, T) \oplus \dots \oplus G(\lambda_m, T)$;

(DOC-2) each $G(\lambda_j, T)$ is invariant under T ;

(DOC-3) each $(T - \lambda_j I)|_{G(\lambda_j, T)}$ is nilpotent.

Proof. Write $n := \dim V$. Lemma C.126 states that for each $j = 1, 2, \dots, m$ we have $G(\lambda_j, T) = \text{null}(T - \lambda_j I)^n$. Then (DOC-3) follows from Definitions C.132, C.124, and C.125 while (DOC-2) follows from Lemma C.137 by choosing $p(z) = (z - \lambda_j)^n$.

(DOC-1) can be proven by an induction on n . The induction basis of $n = 1$ clearly holds. As the induction hypothesis, (DOC-1) holds on all vector spaces of which the dimension is smaller than n .

By Theorem C.104, T has an eigenvalue λ_1 . The application of Lemma C.122 to $T - \lambda_1 I$ yields

$$(*) : V = G(\lambda_1, T) \oplus U,$$

where $U = \text{range}(T - \lambda_1 I)^n$ and Lemma C.126 states $G(\lambda, T) = \text{null}(T - \lambda I)^n$. If $U = \{\mathbf{0}\}$, (DOC-1) holds trivially; otherwise Lemma C.137 implies that U is invariant under T , furnishing the restriction operator $T|_U \in \mathcal{L}(U)$ in (C.31). By (*) and Theorem C.104, $T|_U$ has distinct eigenvalues $\lambda_2, \dots, \lambda_m$, each of which is different from λ_1 . Since $\dim G(\lambda_1, T) \geq 1$, we have $\dim U < n$ and thus we can apply the induction hypothesis to U to obtain

$$(**) : U = G(\lambda_2, T|_U) \oplus \dots \oplus G(\lambda_m, T|_U).$$

Combining (**) with (*) completes the proof if we can show

$$\forall j = 2, \dots, m, \quad G(\lambda_j, T|_U) = G(\lambda_j, T).$$

Indeed, apply $\cap G(\lambda_j, T)$ to both sides of (*) and we have

$$\begin{aligned} G(\lambda_j, T) &= (G(\lambda_1, T) \cap G(\lambda_j, T)) \oplus (U \cap G(\lambda_j, T)) \\ &= G(\lambda_j, T|_U), \end{aligned}$$

where the second step follows from Lemma C.131 and the identity $G(\lambda_j, T|_U) = G(\lambda_j, T) \cap U$. $\square$

Example C.139. For the operator $T \in \mathcal{L}(\mathbb{C}^3)$ in (C.41), T does not have enough eigenvectors to span $\mathbb{C}^3$. Example C.130 shows that

$$\begin{aligned} E(0, T) \oplus E(5, T) &\subset \mathbb{C}^3, \\ G(0, T) \oplus G(5, T) &= \mathbb{C}^3. \end{aligned}$$

Corollary C.140. Suppose V is a complex vector space and $T \in \mathcal{L}(V)$. Then there is a basis of V that consists of generalized eigenvectors of T .

Proof. This follows from (DOC-1) in Theorem C.138. $\square$

Definition C.141. A *block diagonal matrix* A is a square matrix of the form

$$\begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix}, \quad (\text{C.50})$$

where $A_1, \dots, A_m$ are square matrices along the diagonal and all other entries of A is 0.

Theorem C.142. Suppose V is a complex vector space, $T \in \mathcal{L}(V)$, and $\lambda_1, \lambda_2, \dots, \lambda_m$ are all the distinct eigenvalues of T with multiplicities $d_1, d_2, \dots, d_m$. Then there is a basis of V with respect to which T has a block diagonal form (C.50) where each A_j is a d_j -by- d_j upper-triangular matrix of the form

$$\begin{bmatrix} \lambda_j & \cdots & * \\ & \ddots & \vdots \\ & & \lambda_j \end{bmatrix}. \quad (\text{C.51})$$

Proof. By Theorem C.138, each $(T - \lambda_j I)|_{G(\lambda_j, T)}$ is nilpotent. By Lemma C.135, we can choose a basis of $G(\lambda_j, T)$ such that the matrix of $(T - \lambda_j I)|_{G(\lambda_j, T)}$ is strictly upper triangular. Then the form of (C.51) follows from

$$T|_{G(\lambda_j, T)} = (T - \lambda_j I)|_{G(\lambda_j, T)} + \lambda_j I|_{G(\lambda_j, T)}.$$

The rest follows from (DOC-1) in Theorem C.138. $\square$

Example C.143. For $T \in \mathcal{L}(\mathbb{F}^3)$ given by

$$T(x, y, z) = (6x + 3y + 4z, 6y + 2z, 7z), \quad (\text{C.52})$$

the matrix of T with respect to the standard basis is

$$T_S = \begin{pmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{pmatrix}.$$

It is readily verified that the eigenvalues of T are 6 and 7, with the corresponding generalized eigenspaces as

$$\begin{aligned} G(6, T) &= \text{span}\{(1, 0, 0), (0, 1, 0)\}; \\ G(7, T) &= \text{span}\{(10, 2, 1)\}. \end{aligned}$$

The matrix of T with respect to the basis

$$\{\mathbf{v}_1 = (1, 0, 0), \mathbf{v}_2 = (0, 1, 0), \mathbf{v}_3 = (10, 2, 1)\}$$

is of the block diagonal form:

$$T_B = \begin{bmatrix} \begin{bmatrix} 6 & 3 \\ 0 & 6 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 7 \end{bmatrix} & [7] \end{bmatrix}.$$

By Corollary C.62, we also have

$$T_S[\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]T_B.$$

Corollary C.144. Any operator $T \in \mathcal{L}(V)$ on a complex vector space V can be decomposed as $T = \Lambda + N$ where Λ is diagonalizable, N is nilpotent, and $\Lambda N = N\Lambda$.

Proof. By Theorem C.142, T has a block diagonal form where the diagonal matrices are $T_1, T_2, \dots, T_p$ and each T_j can be decomposed as $T_j = N_j + \Lambda_j$ with $N_j = T_j - \lambda_j I_j$ and $\Lambda_j = \lambda_j I_j$. Clearly N_j is nilpotent and Λ_j is diagonalizable. Also, any matrix commutes with the identity matrix, hence we have $N_j \lambda_j I_j = \lambda_j I_j N_j$. The rest follows from the block diagonal form of T . $\square$

Example C.145. If p is a polynomial of degree k , then $p(a+x)$ can be expressed as a Taylor series,

$$p(a+x) = \sum_{i=0}^k \frac{p^{(i)}(a)}{i!} x^i.$$

The formula is an algebraic identity and can be generalized to an operator $T = \Lambda + N$ with the help of $\Lambda N = N\Lambda$,

$$p(\Lambda + N) = \sum_{i=0}^k \frac{p^{(i)}(\Lambda)}{i!} N^i = \sum_{i=0}^m \frac{p^{(i)}(\Lambda)}{i!} N^i, \quad (\text{C.53})$$

where the nilpotent operator N satisfies $N^{m+1} = \mathbf{0}$.

Theorem C.146. Any invertible operator $T \in \mathcal{L}(V)$ on a complex vector space V has a square root.

Proof. Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be all distinct eigenvalues of T . By (DOC-3) in Theorem C.138, for each $j = 1, 2, \dots, m$ there exists a nilpotent operator $N_j \in \mathcal{L}(G(\lambda_j, T))$ such that $T|_{G(\lambda_j, T)} = \lambda_j I + N_j$. Since T is invertible, $\lambda_j \neq 0$ and

$$\forall j = 1, 2, \dots, m, \quad T|_{G(\lambda_j, T)} = \lambda_j \left(I + \frac{N_j}{\lambda_j} \right).$$

By Lemma C.136 and the condition of $\mathbb{F} = \mathbb{C}$, there exists an operator $R_j \in \mathcal{L}(G(\lambda_j, T))$ such that $R_j^2 = T|_{G(\lambda_j, T)}$.

By (DOC-1) in Theorem C.138, any vector $\mathbf{v} \in V$ can be uniquely expressed in the form $\mathbf{v} = \sum_{i=1}^m \mathbf{u}_i$ where $\mathbf{u}_i \in G(\lambda_i, T)$. Then it is straightforward to verify that the operator $R \in \mathcal{L}(V)$ given below satisfies $R^2 = T$:

$$R(\mathbf{v}) = \sum_{j=1}^m R_j \mathbf{u}_j. \quad \square$$

C.4.4 Minimal and characteristic polynomials

Definition C.147. A *monic polynomial* is a polynomial whose highest-degree coefficient equals 1.

Lemma C.148. For any $T \in \mathcal{L}(V)$ over a finite-dimensional vector space V , there exists a unique monic polynomial p of the smallest degree such that $p(T) = \mathbf{0}$.

Proof. Denote $n := \dim V$. Since the dimension of $\mathcal{L}(V)$ is n^2 , the list $I, T, T^2, \dots, T^m$ is linearly dependent for any $m \geq n^2$. Set m to the smallest integer such that the list is linearly dependent. Then by the linear dependence lemma

C.31, T^m can be expressed as a linear combination of the other terms on the list, which leads to a monic polynomial p satisfying $p(T) = \mathbf{0}$.

Suppose another monic polynomial q of degree m satisfies $q(T) = \mathbf{0}$. Then $(p-q)(T) = \mathbf{0}$ and m being the smallest integer for the linear dependence of the list dictate that $p-q = \mathbf{0}$ and thus p is unique. $\square$

Definition C.149. The minimal polynomial of an operator $T \in \mathcal{L}(V)$ is the unique monic polynomial p of the smallest degree satisfying $p(T) = \mathbf{0}$.

Example C.150. The minimal polynomial of any projection operator P defined as $P^2 = P$ is $p(z) = z^2 - z$.

Example C.151. For the operator $T \in \mathcal{L}(\mathbb{C}^5)$ whose matrix is

$$\begin{bmatrix} 0 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

its minimal polynomial is $p(z) = z^5 - 6z + 3$.

Definition C.152. The *characteristic polynomial* of an operator $T \in \mathcal{L}(V)$ on a complex vector space V is the polynomial

$$q(z) := (z - \lambda_1)^{m_1} \cdots (z - \lambda_n)^{m_n}, \quad (\text{C.54})$$

where λ_i 's are the n distinct eigenvalues of T and m_i 's their corresponding algebraic multiplicities.

Corollary C.153. The degree of the characteristic polynomial of $T \in \mathcal{L}(V)$ is $\dim V$.

Proof. This follows from Theorems C.138 and C.14. $\square$

Theorem C.154 (Cayley–Hamilton). The characteristic polynomial q of an operator $T \in \mathcal{L}(V)$ satisfies $q(T) = \mathbf{0}$.

Proof. Let $\lambda_1, \dots, \lambda_n$ be the distinct eigenvalues of T and $m_1, \dots, m_n$ their corresponding algebraic multiplicities, which, by Definition C.127, are the dimensions of the generalized eigenspaces. By Definitions C.125 and C.124, $(T - \lambda_i I)|_{G(\lambda_i, T)}$ is nilpotent for each $i = 1, 2, \dots, n$. Lemma C.134 implies

$$(*) : \quad (T - \lambda_i I)^{m_i}|_{G(\lambda_i, T)} = \mathbf{0}.$$

By Theorem C.138, any vector in V is a sum of vectors in $G(\lambda_1, T), \dots, G(\lambda_n, T)$. Thus it suffices to show that $q(T)|_{G(\lambda_i, T)} = \mathbf{0}$ for each i . Indeed, for any fixed $i \in \{1, 2, \dots, n\}$, we have

$$\begin{aligned} q(T) &= (T - \lambda_1 I)^{m_1} \cdots (T - \lambda_i I)^{m_i} \cdots (T - \lambda_n I)^{m_n} \\ &= (T - \lambda_1 I)^{m_1} \cdots (T - \lambda_n I)^{m_n} (T - \lambda_i I)^{m_i}, \end{aligned}$$

where the first step follows from (C.54) and the second from the identity $(T - \lambda_i I)(T - \lambda_j I) = (T - \lambda_j I)(T - \lambda_i I)$. Then $(*)$ completes the proof. $\square$

Lemma C.155. For $T \in \mathcal{L}(V)$, a polynomial $q(T) = \mathbf{0}$ if and only if $q(T)$ is a polynomial multiple of the minimal polynomial p of T .

Proof. The sufficiency is trivial and we only prove the necessity. Suppose $q(T) = \mathbf{0}$. Divide q by p and we have $q = ps + r$ where r is the remainder polynomial satisfying $\deg r < \deg p$. Then we have

$$\mathbf{0} = q(T) = p(T)s(T) + r(T) = r(T),$$

which implies $q(T) = p(T)s(T)$ and this completes the proof. $\square$

Corollary C.156. For $T \in \mathcal{L}(V)$, its characteristic polynomial is a multiple of its minimal polynomial.

Proof. This follows directly from Lemma C.155 and the Cayley-Hamilton theorem C.154. $\square$

Theorem C.157 (Fundamental theorem of algebra). If $p \in \mathcal{P}(\mathbb{C})$ is a nonconstant polynomial, then p has a unique factorization (except for the order of the factors) of the form

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m), \quad (\text{C.55})$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{C}$.

Lemma C.158. The set of roots of the minimal polynomial of an operator $T \in \mathcal{L}(V)$ equals the set of eigenvalues of T .

Proof. Denote by $p(z) := z^m + \sum_{i=0}^{m-1} a_i z^i$ the minimal polynomial of T . Suppose λ is a root of p . By Theorem C.157, we have $p(z) = (z - \lambda)q(z)$ where $q(z)$ is another monic polynomial. Then $p(T) = \mathbf{0}$ implies $(T - \lambda I)q(T)\mathbf{v} = \mathbf{0}$ for any $\mathbf{v} \in V$. Since the degree of $q(z)$ is $m - 1$, there exists at least one $\mathbf{u} \in V$ such that $q(T)\mathbf{u} \neq \mathbf{0}$. Hence λ is an eigenvalue of T .

Conversely, suppose λ is an eigenvalue of T . Then there exists a nonzero $\mathbf{u} \in V$ such that $T\mathbf{u} = \lambda\mathbf{u}$. Hence $T^i\mathbf{u} = \lambda^i\mathbf{u}$ for any $i \in \mathbb{N}^+$ and we have $\mathbf{0} = p(T)\mathbf{u} = p(\lambda)\mathbf{u}$. Therefore, λ is a root of p . $\square$

Example C.159. For the operator $T \in \mathcal{L}(\mathbb{C}^3)$ in Example C.143, both its minimal and characteristic polynomials are $p(z) = (z - 6)^2(z - 7)$.

Example C.160. The characteristic polynomial of the linear operator $T(z_1, z_2, z_3) := (6z_1, 6z_2, 7z_3)$ is $p(z) = (z - 6)^2(z - 7)$ while its minimal polynomial is $q(z) = (z - 6)(z - 7)$.

Example C.161. By numerically computing the eigenvalues of the operator T in Example C.151, we know that all eigenvalues of T have multiplicity 1. Thus the minimal polynomial of T is the same as its characteristic polynomial.

C.4.5 The Jordan canonical form

Definition C.162. A *Jordan block* of order k has the form

$$J(\lambda, k) = \lambda I_k + S_k, \quad (\text{C.56})$$

where

$$(S_k)_{i,j} = \begin{cases} 1, & i = j - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example C.163. The Jordan blocks of orders 1, 2, and 3 are

$$J(\lambda, 1) = \lambda, \quad J(\lambda, 2) = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}, \quad J(\lambda, 3) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}.$$

Example C.164. The nilpotent operator

$$N(x, y, z) = (0, x, y)$$

can be exploited to construct a basis of $\mathbb{F}^3$: $(N^2\mathbf{v}, N\mathbf{v}, \mathbf{v})$ where $\mathbf{v} = (1, 0, 0)$. With respect to this basis, the matrix of N is the Jordan block $J(0, 3)$.

Example C.165. The nilpotent operator

$$N(z_1, z_2, z_3, z_4, z_5, z_6) = (0, z_1, z_2, 0, z_4, 0)$$

can be exploited to construct a basis of $\mathbb{F}^6$:

$$(N^2\mathbf{v}_1, N\mathbf{v}_1, \mathbf{v}_1, N\mathbf{v}_2, \mathbf{v}_2, \mathbf{v}_3)$$

where

$$\begin{aligned} \mathbf{v}_1 &= (1, 0, 0, 0, 0, 0), \\ \mathbf{v}_2 &= (0, 0, 0, 1, 0, 0), \\ \mathbf{v}_3 &= (0, 0, 0, 0, 0, 1). \end{aligned}$$

With respect to this basis, the matrix of N is the block diagonal matrix

$$\begin{pmatrix} J(0, 3) & & \\ & J(0, 2) & \\ & & J(0, 1) \end{pmatrix}.$$

Lemma C.166. For a nilpotent operator $N \in \mathcal{L}(V)$, there exist $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in V$ and $m_1, m_2, \dots, m_n \in \mathbb{N}$ such that

- (a) $N^{m_1}\mathbf{v}_1, \dots, N\mathbf{v}_1, \mathbf{v}_1, \dots, N^{m_n}\mathbf{v}_n, \dots, N\mathbf{v}_n, \mathbf{v}_n$ form a basis of V ;
- (b) $N^{m_1+1}\mathbf{v}_1 = \dots = N^{m_n+1}\mathbf{v}_n = \mathbf{0}$.

Proof. If $\text{range } N = \{\mathbf{0}\}$, Theorem C.59 implies $\text{null } N = V$. Then $N = \mathbf{0}$ and both (a) and (b) hold trivially with each m_i as zero. Hereafter we prove this lemma by an induction on $\dim V$. By Lemma C.134, the induction basis for $\dim V = 1$ holds. Hereafter we assume that $\dim V > 1$ and (a) and (b) hold on all vector spaces of smaller dimensions.

Because N is nilpotent, N is not injective. By Theorem C.60, N is not surjective either. The range of N , a subspace of V (cf. Theorem C.58), has a smaller dimension than V . Thus we can apply our induction hypothesis to $\text{range } N$ to obtain that, for $N|_{\text{range } N} \in \mathcal{L}(\text{range } N)$, there exist vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \text{range } N$ and $m_1, m_2, \dots, m_n \in \mathbb{N}$ such that

$$(*) : N^{m_1}\mathbf{v}_1, \dots, N\mathbf{v}_1, \mathbf{v}_1, \dots, N^{m_n}\mathbf{v}_n, \dots, N\mathbf{v}_n, \mathbf{v}_n$$

is a basis of $\text{range } N$ and (b) holds for this basis. Then

$$\forall j = 1, \dots, n, \quad \exists \mathbf{u}_j \in V \text{ s.t. } \mathbf{v}_j = N\mathbf{u}_j.$$

Next, we claim that the following is an independent list of vectors in V ,

$$(\square) : N^{m_1+1}\mathbf{u}_1, \dots, N\mathbf{u}_1, \mathbf{u}_1, \dots, N^{m_n+1}\mathbf{u}_n, \dots, N\mathbf{u}_n, \mathbf{u}_n.$$

Indeed, suppose a linear combination of $(\square)$ equals $\mathbf{0}$. Then apply N to it and we get a linear combination of $(*)$ equal

to $\mathbf{0}$ and thus all the coefficients in the original linear combination must be 0 except for those of the vectors

$$N^{m_1+1}\mathbf{u}_1 = N^{m_1}\mathbf{v}_1, \dots, N^{m_n+1}\mathbf{u}_n = N^{m_n}\mathbf{v}_n.$$

Again, the linear independence of the list $(*)$ dictates that all coefficients of the above vectors must be 0.

By Lemma C.38, we can extend $(\square)$ to a basis of V ,

$$(\triangle): N^{m_1+1}\mathbf{u}_1, \dots, \mathbf{u}_1, \dots, N^{m_n+1}\mathbf{u}_n, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_p.$$

Since $N\mathbf{w}_j \in \text{range } N$, each $N\mathbf{w}_j$ is in the span of $(*)$, thus there exists $\mathbf{x}_j$ in the span of $(\square)$ such that $N\mathbf{w}_j = N\mathbf{x}_j$. Define $\mathbf{u}_{n+j} = \mathbf{w}_j - \mathbf{x}_j$ and we have $N\mathbf{u}_{n+j} = \mathbf{0}$. Therefore the following list of vectors satisfies (a) and (b):

$$N^{m_1+1}\mathbf{u}_1, \dots, \mathbf{u}_1, \dots, N^{m_n+1}\mathbf{u}_n, \dots, \mathbf{u}_n, \mathbf{u}_{n+1}, \dots, \mathbf{u}_{n+p}.$$

This completes the induction and the proof. $\square$

Definition C.167. A *Jordan basis for a linear operator* $T \in \mathcal{L}(V)$ is a basis of V with respect to which the matrix of T is a block diagonal matrix of the form

$$\begin{pmatrix} J(\lambda_1, k_1) & & & \\ & J(\lambda_2, k_2) & & \\ & & \ddots & \\ & & & J(\lambda_p, k_p) \end{pmatrix}, \quad (\text{C.57})$$

where the λ_j 's might not be distinct.

Theorem C.168 (Jordan). Any operator $T \in \mathcal{L}(V)$ on a complex vector space has a Jordan basis.

Proof. If T is a nilpotent operator N , we consider the basis given in Lemma C.166(a). For each j , N annihilates $N^{m_j}\mathbf{v}_j$ in the list $N^{m_j}\mathbf{v}_j, \dots, N\mathbf{v}_j, \mathbf{v}_j$ and sends another vector to its previous. Hence N has a block diagonal matrix where each matrix on the diagonal is the Jordan block $J(0, k_j)$. Thus the statement holds for nilpotent operators.

Otherwise let $\lambda_1, \lambda_2, \dots, \lambda_m$ be all distinct eigenvalues of T . By Theorem C.138, we have the decomposition

$$V = G(\lambda_1, T) \oplus \dots \oplus G(\lambda_m, T),$$

with each $N_j := (T - \lambda_j I)|_{G(\lambda_j, T)}$ being nilpotent. By the previous paragraph, the matrix of each N_j is a Jordan block. The combination of all these Jordan blocks is a Jordan basis for T . $\square$

Theorem C.169 (Jordan canonical form). Every matrix $A \in \mathbb{C}^{n \times n}$ has a similarity transformation

$$A = RJR^{-1}, \quad (\text{C.58})$$

where R is invertible and J is a block diagonal matrix of the form (C.57), and each $J(\lambda_i, k_i)$ is a Jordan block of order k_i , $\sum_{i=1}^s k_i = n$, and each k_i is no greater than the index of λ_i . Let m_a and m_g respectively denote the algebraic multiplicity and the geometric multiplicity of an eigenvalue λ of A . Then λ appears in m_g blocks and the sum of the orders of these blocks is m_a .

Proof. This is simply a restatement of Theorem C.168 using Definition C.127. $\square$

C.5 Inner product spaces

C.5.1 Inner products

Definition C.170. Denote by $\mathbb{F}$ the underlying field of a vector space $\mathcal{V}$. The *inner product* $\langle \mathbf{u}, \mathbf{v} \rangle$ on $\mathcal{V}$ is a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{F}$ that satisfies

- (IP-1) real positivity: $\forall \mathbf{v} \in \mathcal{V}, \langle \mathbf{v}, \mathbf{v} \rangle \geq 0$;
- (IP-2) definiteness: $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ iff $\mathbf{v} = \mathbf{0}$;
- (IP-3) additivity in the first slot:
 $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}, \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;
- (IP-4) homogeneity in the first slot:
 $\forall a \in \mathbb{F}, \forall \mathbf{v}, \mathbf{w} \in \mathcal{V}, \langle a\mathbf{v}, \mathbf{w} \rangle = a \langle \mathbf{v}, \mathbf{w} \rangle$;
- (IP-5) conjugate symmetry: $\forall \mathbf{v}, \mathbf{w} \in \mathcal{V}, \langle \mathbf{v}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{v} \rangle}$.

An *inner product space* is a vector space $\mathcal{V}$ equipped with an inner product on $\mathcal{V}$.

Corollary C.171. An inner product has additivity in the second slot, i.e. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.

Corollary C.172. An inner product has conjugate homogeneity in the second slot, i.e.

$$\forall a \in \mathbb{F}, \forall \mathbf{v}, \mathbf{w} \in \mathcal{V}, \quad \langle \mathbf{v}, a\mathbf{w} \rangle = \bar{a} \langle \mathbf{v}, \mathbf{w} \rangle. \quad (\text{C.59})$$

Exercise C.173. Prove Corollaries C.171 and C.172 from Definition C.170.

Definition C.174. The *Euclidean inner product* on $\mathbb{F}^n$ is

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i \overline{w_i}. \quad (\text{C.60})$$

C.5.2 Norms induced from inner products

Definition C.175. Let $\mathbb{F}$ be the underlying field of an inner product space $\mathcal{V}$. The *norm induced by an inner product* on $\mathcal{V}$ is a function $\mathcal{V} \rightarrow \mathbb{F}$:

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}. \quad (\text{C.61})$$

Definition C.176 (Vector p -norm). For a vector $\mathbf{v} \in \mathbb{F}^n$, its ℓ_∞ norm or the *infinity norm* or the *maximum norm* is

$$\|\mathbf{v}\|_\infty = \max_i |v_i|; \quad (\text{C.62})$$

for $p \in [1, \infty)$, its ℓ_p norm is

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{\frac{1}{p}}, \quad (\text{C.63})$$

which is also called the *taxicab norm* or the *Manhattan norm* for $p = 1$ and the *Euclidean norm* or the *2-norm* for $p = 2$.

Lemma C.177. Any $\mathbf{v} \in \mathbb{F}^n$ satisfies $\|\mathbf{v}\|_2^2 \leq \|\mathbf{v}\|_1 \|\mathbf{v}\|_\infty$.

Proof. This follows from the proof of the Hölder inequality in Theorem F.4 by setting $p = 1$, $q = \infty$, and $\mathbf{x} = \mathbf{y} = \mathbf{v}$. $\square$

Theorem C.178 (Equivalence of norms). Any two norms $\|\cdot\|_N$ and $\|\cdot\|_M$ on a finite dimensional vector space $\mathcal{V} = \mathbb{C}^n$ satisfy

$$\exists c_1, c_2 \in \mathbb{R}^+, \text{ s.t. } \forall \mathbf{x} \in \mathcal{V}, c_1 \|\mathbf{x}\|_M \leq \|\mathbf{x}\|_N \leq c_2 \|\mathbf{x}\|_M. \quad (\text{C.64})$$

Definition C.179. The angle between two vectors $\mathbf{v}, \mathbf{w}$ in an inner product space with $\mathbb{F} = \mathbb{R}$ is the number $\theta \in [0, \pi]$,

$$\theta = \arccos \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{\|\mathbf{v}\| \|\mathbf{w}\|}. \quad (\text{C.65})$$

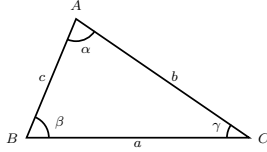


Figure C.1: The law of cosines.

Theorem C.180 (The law of cosines). A triangle in Figure C.1 satisfies

$$c^2 = a^2 + b^2 - 2ab \cos \gamma. \quad (\text{C.66})$$

Proof. The dot product of AB to $AB = CB - CA$ yields

$$c^2 = \langle AB, CB \rangle - \langle AB, CA \rangle.$$

The dot products of CB and CA to $AB = CB - CA$ yield

$$\begin{aligned} \langle CB, AB \rangle &= a^2 - \langle CB, CA \rangle; \\ -\langle CA, AB \rangle &= -\langle CA, CB \rangle + b^2. \end{aligned}$$

The proof is completed by adding up all three equations and applying (C.65). $\square$

Theorem C.181 (The law of cosines: abstract version). Any induced norm on a real vector space satisfies

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle. \quad (\text{C.67})$$

Proof. Definitions C.175 and C.170 and $\mathbb{F} = \mathbb{R}$ yield

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|^2 &= \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle. \end{aligned} \quad \square$$

C.5.3 Norms and induced inner-products

Definition C.182. A function $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{F}$ is a *norm* for a vector space $\mathcal{V}$ iff it satisfies

(NRM-1) real positivity: $\forall \mathbf{v} \in \mathcal{V}, \|\mathbf{v}\| \geq 0$;

(NRM-2) point separation: $\|\mathbf{v}\| = 0 \implies \mathbf{v} = \mathbf{0}$.

(NRM-3) absolute homogeneity:

$$\forall a \in \mathbb{F}, \forall \mathbf{v} \in \mathcal{V}, \|a\mathbf{v}\| = |a| \|\mathbf{v}\|;$$

(NRM-4) triangle inequality:

$$\forall \mathbf{u}, \mathbf{v} \in \mathcal{V}, \|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

The function $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{F}$ is called a *semi-norm* iff it satisfies (NRM-1,3,4).

Definition C.183. A *normed vector space* (or simply a *normed space*) is a vector space $\mathcal{V}$ equipped with a norm on $\mathcal{V}$.

Example C.184. For a vector norm $\|\cdot\|_p$ in Definition C.176, $(\mathbb{R}^n, \|\cdot\|_p)$ is a normed space.

Exercise C.185. Prove the backward triangle inequality of a norm $\|\cdot\|$, i.e.,

$$\forall u, v \in V, \quad |||u|| - ||v||| \leq \|u - v\|. \quad (\text{C.68})$$

Exercise C.186. Explain how (NRM-1,2,3,4) relate to the geometric meaning of the norm of vectors in $\mathbb{R}^3$.

Lemma C.187. The norm induced by an inner product is a norm as in Definition C.182.

Proof. The induced norm as in (C.61) satisfies (NRM-1,2) trivially. For (NRM-3),

$$\|a\mathbf{v}\|^2 = \langle a\mathbf{v}, a\mathbf{v} \rangle = a \langle \mathbf{v}, a\mathbf{v} \rangle = a\bar{a} \langle \mathbf{v}, \mathbf{v} \rangle = |a|^2 \|\mathbf{v}\|^2.$$

To prove (NRM-4), we have

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} \\ &\leq \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle + 2|\langle \mathbf{u}, \mathbf{v} \rangle| \\ &\leq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2, \end{aligned}$$

where the second step follows from (IP-5) and the fourth step from Cauchy-Schwarz inequality. $\square$

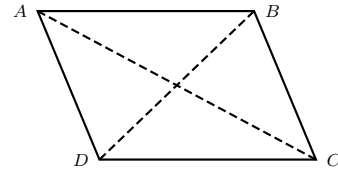


Figure C.2: The law of parallelograms.

Theorem C.188 (The parallelogram law). The sum of squares of the lengths of the four sides of a parallelogram equals the sum of squares of the two diagonals. More precisely, the parallelogram in Figure C.2 satisfies

$$(AB)^2 + (BC)^2 + (CD)^2 + (DA)^2 = (AC)^2 + (BD)^2. \quad (\text{C.69})$$

Proof. Apply the law of cosines to the two diagonals, add the two equations, and we obtain (C.69). $\square$

Theorem C.189 (The parallelogram law: abstract version). Any induced norm (C.61) satisfies

$$2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2 = \|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2. \quad (\text{C.70})$$

Proof. Replace $\mathbf{v}$ in (C.67) with $-\mathbf{v}$ and we have

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle.$$

(C.70) follows from adding the above equation to (C.67). $\square$

Exercise C.190. For vector norms in Definition C.176, show that the parallelogram law (C.70) holds if and only if $p = 2$.

Theorem C.191. The induced norm (C.61) holds for some inner product $\langle \cdot, \cdot \rangle$ if and only if the parallelogram law (C.70) holds for every pair of $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

Exercise C.192. Prove Theorem C.191.

Example C.193. By Theorem C.191 and Exercise C.190, none of the ℓ_1 and ℓ_∞ norms has a corresponding inner product.

C.5.4 Orthonormal bases

Definition C.194. Two vectors $\mathbf{u}, \mathbf{v}$ are called *orthogonal* if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, i.e., their inner product is the additive identity of the underlying field.

Example C.195. An inner product on the vector space of continuous real-valued functions on the interval $[-1, 1]$ is

$$\langle f, g \rangle = \int_{-1}^{+1} f(x)g(x)dx.$$

f and g are said to be orthogonal if the integral is zero.

Theorem C.196 (Pythagorean). If $\mathbf{u}, \mathbf{v}$ are orthogonal, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof. This follows from (C.67) and Definition C.194. $\square$

Theorem C.197 (Cauchy–Schwarz inequality).

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|, \quad (\text{C.71})$$

where the equality holds iff one of $\mathbf{u}, \mathbf{v}$ is a scalar multiple of the other.

Proof. For any complex number λ , (IP-1) implies

$$\begin{aligned} \langle \mathbf{u} + \lambda \mathbf{v}, \mathbf{u} + \lambda \mathbf{v} \rangle &\geq 0 \\ \implies \langle \mathbf{u}, \mathbf{u} \rangle + \lambda \langle \mathbf{v}, \mathbf{u} \rangle + \bar{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle + \lambda \bar{\lambda} \langle \mathbf{v}, \mathbf{v} \rangle &\geq 0. \end{aligned}$$

If $\mathbf{v} = \mathbf{0}$, (C.71) clearly holds. Otherwise (C.71) follows from substituting $\lambda = -\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}$ into the above equation. $\square$

Exercise C.198. To explain the choice of λ in the proof of Theorem C.197, what is the geometric meaning of (C.71) in the plane? When will the equality hold?

Example C.199. If $x_i, y_i \in \mathbb{R}$, then for any $n \in \mathbb{N}^+$

$$\left| \sum_{i=1}^n x_i y_i \right|^2 \leq \sum_{j=1}^n x_j^2 \sum_{k=1}^n y_k^2.$$

Example C.200. If $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous, then

$$\left| \int_a^b f(x)g(x)dx \right|^2 \leq \left(\int_a^b f^2(x)dx \right) \left(\int_a^b g^2(x)dx \right)$$

Definition C.201. A list of vectors $(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m)$ is called *orthonormal* if the vectors in it are pairwise orthogonal and each vector has norm 1, i.e.

$$\begin{cases} \forall i = 1, 2, \dots, m, & \|\mathbf{e}_i\| = 1; \\ \forall i \neq j, & \langle \mathbf{e}_i, \mathbf{e}_j \rangle = 0. \end{cases} \quad (\text{C.72})$$

Definition C.202. An *orthonormal basis* of an inner-product space $\mathcal{V}$ is an orthonormal list of vectors in $\mathcal{V}$ that is also a basis of $\mathcal{V}$.

Theorem C.203. If $(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ is an orthonormal basis of $\mathcal{V}$, then

$$\forall \mathbf{v} \in \mathcal{V}, \quad \mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i, \quad (\text{C.73a})$$

$$\|\mathbf{v}\|^2 = \sum_{i=1}^n |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2. \quad (\text{C.73b})$$

Lemma C.204. Every finite-dimensional inner-product space has an orthonormal basis.

Theorem C.205 (Schur). Every linear operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional complex vector space $\mathcal{V}$ has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. This follows from Theorem C.110, Lemma C.204 and the Gram-Schmidt process. $\square$

Definition C.206. A *linear functional* on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$, or, it is an element of $\mathcal{L}(\mathcal{V}, \mathbb{F})$.

Theorem C.207 (Riesz representation theorem). If $\mathcal{V}$ is a finite-dimensional vector space, then

$$\forall \varphi \in \mathcal{V}', \exists! \mathbf{u} \in \mathcal{V} \text{ s.t. } \forall \mathbf{v} \in \mathcal{V}, \quad \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle. \quad (\text{C.74})$$

Proof. Let $(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.

$$\begin{aligned} \varphi(\mathbf{v}) &= \varphi \left(\sum_{i=1}^n \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i \right) = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{e}_i \rangle \varphi(\mathbf{e}_i) \\ &= \sum_{i=1}^n \left\langle \mathbf{v}, \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i \right\rangle = \left\langle \mathbf{v}, \sum_{i=1}^n \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i \right\rangle, \end{aligned}$$

where the last two steps follow from Corollaries C.171 and C.172.

As for the uniqueness, suppose that $\exists \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{V}$ s.t. $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle$. Then for each $\mathbf{v} \in \mathcal{V}$,

$$0 = \langle \mathbf{v}, \mathbf{u}_1 \rangle - \langle \mathbf{v}, \mathbf{u}_2 \rangle = \langle \mathbf{v}, \mathbf{u}_1 - \mathbf{u}_2 \rangle.$$

Taking $\mathbf{v} = \mathbf{u}_1 - \mathbf{u}_2$ shows that $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{0}$. $\square$

C.6 Operators on inner-product spaces

C.6.1 Adjoint and self-adjoint operators

Definition C.208. The *adjoint of a linear map* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ between inner-product spaces is a function $T^* : \mathcal{W} \rightarrow \mathcal{V}$ that satisfies

$$\forall \mathbf{w} \in \mathcal{W}, \forall \mathbf{v} \in \mathcal{V}, \quad \langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T^*\mathbf{w} \rangle. \quad (\text{C.75})$$

Example C.209. Define a linear operator $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$,

$$T(x_1, x_2, x_3) = (x_2 + 3x_3, 2x_1).$$

Then $T^*(y_1, y_2) = (2y_2, y_1, 3y_1)$ because

$$\begin{aligned} \langle (x_1, x_2, x_3), T^*(y_1, y_2) \rangle &= \langle T(x_1, x_2, x_3), (y_1, y_2) \rangle \\ &= \langle (x_2 + 3x_3, 2x_1), (y_1, y_2) \rangle \\ &= x_2y_1 + 3x_3y_1 + 2x_1y_2 \\ &= \langle (x_1, x_2, x_3), (2y_2, y_1, 3y_1) \rangle. \end{aligned}$$

Lemma C.210. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T^* \in \mathcal{L}(\mathcal{W}, \mathcal{V})$.

Proof. Use Definition C.45. $\square$

Theorem C.211. The adjoint of a linear map has the following properties.

- (ADJ-1) additivity:
 $\forall S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}), \quad (S + T)^* = S^* + T^*;$
- (ADJ-2) conjugate homogeneity:
 $\forall T \in \mathcal{L}(\mathcal{V}, \mathcal{W}), \forall a \in \mathbb{F}, \quad (aT)^* = \bar{a}T^*;$
- (ADJ-3) adjoint of adjoint:
 $\forall T \in \mathcal{L}(\mathcal{V}, \mathcal{W}), \quad (T^*)^* = T;$
- (ADJ-4) identity: $I^* = I;$
- (ADJ-5) products: let $\mathcal{U}$ be an inner-product space,
 $\forall T \in \mathcal{L}(\mathcal{V}, \mathcal{W}), \forall S \in \mathcal{L}(\mathcal{W}, \mathcal{U}), \quad (ST)^* = T^*S^*.$

Proof. Use Definitions C.208 and C.170. $\square$

Lemma C.212. $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and T^* satisfy

- (a) $\text{null } T^* = (\text{range } T)^\perp;$
- (b) $\text{range } T^* = (\text{null } T)^\perp;$
- (c) $\text{null } T = (\text{range } T^*)^\perp;$
- (d) $\text{range } T = (\text{null } T^*)^\perp.$

Definition C.213. The *conjugate transpose*, or *Hermitian transpose*, or *Hermitian conjugate*, or *adjoint matrix*, of a matrix $A \in \mathbb{C}^{m \times n}$ is the matrix $A^* \in \mathbb{C}^{n \times m}$ defined by

$$(A^*)_{ij} = \overline{a_{ji}}, \quad (\text{C.76})$$

where $\overline{a_{ji}}$ denotes the complex conjugate of the entry a_{ji} .

Exercise C.214. Show that the conjugate transpose is an adjoint operator in $\mathcal{L}(\mathcal{V}, \mathcal{W})$ with $\mathcal{V} = \mathbb{C}^n$ and $\mathcal{W} = \mathbb{C}^m$.

Definition C.215. A matrix $U \in \mathbb{C}^{n \times n}$ is *unitary* iff $U^*U = I$. A matrix $U \in \mathbb{R}^{n \times n}$ is *orthogonal* iff $U^T U = I$.

Theorem C.216. A matrix $U \in \mathbb{C}^{n \times n}$ is unitary if and only if its columns form an orthonormal basis for $\mathbb{C}^n$.

Proof. This follows from considering the (i, j) th element of U^*U and applying $U^*U = I$ in Definition C.215. $\square$

Corollary C.217. A unitary matrix U preserves norms and inner products. More precisely, we have

$$\forall \mathbf{v}, \mathbf{w} \in \mathbb{C}^n, \quad \langle U\mathbf{v}, U\mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle.$$

Proof. This follows from Definitions C.208 and C.215. $\square$

Theorem C.218. Every unitary matrix $U \in \mathbb{C}^{2 \times 2}$ with $\det U = 1$ is of the form

$$U = \begin{bmatrix} a & b \\ -\bar{b} & \bar{a} \end{bmatrix}, \quad (\text{C.77})$$

where $|a|^2 + |b|^2 = 1$.

Proof. Let

$$U = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then Theorem C.216 and the condition $\det U = 1$ yield

$$\begin{aligned} a\bar{b} + c\bar{d} &= 0, \\ ad - cb &= 1. \end{aligned}$$

In other words, the linear system

$$\begin{bmatrix} \bar{b} & \bar{d} \\ d & -b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

has solution $x = a, y = c$. Furthermore, Theorem C.216 and the form of U yield $|b|^2 + |d|^2 = 1$. Hence the solution $x = a, y = c$ is unique and we have $a = \bar{d}$ and $c = -\bar{b}$, which completes the proof. $\square$

Theorem C.219. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of $\mathcal{V}$ and $f_1, \dots, f_m$ is an orthonormal basis of $\mathcal{W}$. Then

$$M(T^*, (f_1, \dots, f_m), (e_1, \dots, e_n))$$

is the conjugate transpose of

$$M(T, (e_1, \dots, e_n), (f_1, \dots, f_m)).$$

Proof. By Corollary C.62, we have

$$T[e_1, \dots, e_n] = [f_1, \dots, f_m]M_T.$$

The orthonormality of the two bases and Definition C.174 further imply

$$M_T = \begin{bmatrix} \langle Te_1, f_1 \rangle & \langle Te_2, f_1 \rangle & \cdots & \langle Te_n, f_1 \rangle \\ \langle Te_1, f_2 \rangle & \langle Te_2, f_2 \rangle & \cdots & \langle Te_n, f_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle Te_1, f_m \rangle & \langle Te_2, f_m \rangle & \cdots & \langle Te_n, f_m \rangle \end{bmatrix}.$$

The proof is completed by repeating the above derivation for T^* and then applying Definitions C.170 and C.208. $\square$

Lemma C.220. A linear operator $T \in \mathcal{L}(\mathcal{V})$ on a complex inner product space $\mathcal{V}$ satisfies $T = \mathbf{0}$ if

$$\forall \mathbf{v} \in \mathcal{V}, \quad \langle T\mathbf{v}, \mathbf{v} \rangle = 0. \quad (\text{C.78})$$

Proof. By Definition C.170 and (C.78), we have, $\forall \mathbf{u}, \mathbf{w} \in \mathcal{V}$,

$$\begin{aligned} \langle T\mathbf{u}, \mathbf{w} \rangle &= \frac{1}{4} \langle T(\mathbf{u} + \mathbf{w}), \mathbf{u} + \mathbf{w} \rangle - \frac{1}{4} \langle T(\mathbf{u} - \mathbf{w}), \mathbf{u} - \mathbf{w} \rangle \\ &\quad + \frac{\mathbf{i}}{4} \langle T(\mathbf{u} + \mathbf{i}\mathbf{w}), \mathbf{u} + \mathbf{i}\mathbf{w} \rangle - \frac{\mathbf{i}}{4} \langle T(\mathbf{u} - \mathbf{i}\mathbf{w}), \mathbf{u} - \mathbf{i}\mathbf{w} \rangle \\ &= 0. \end{aligned}$$

Setting $\mathbf{w} = T\mathbf{u}$ completes the proof. $\square$

Definition C.221. An operator $T \in \mathcal{L}(\mathcal{V})$ is *self-adjoint* iff $T = T^*$, i.e.,

$$\forall \mathbf{v}, \mathbf{w} \in \mathcal{V}, \quad \langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T\mathbf{w} \rangle. \quad (\text{C.79})$$

Definition C.222. A matrix $A \in \mathbb{C}^{n \times n}$ is *Hermitian* if it equals its adjoint A^* , i.e. $A = A^*$; A is *skew Hermitian* or *anti-Hermitian* if $A = -A^*$.

Lemma C.223. Any skew Hermitian matrix S satisfies

$$\forall \mathbf{x} \in \mathbb{C}^n, \quad \operatorname{Re} \langle S\mathbf{x}, \mathbf{x} \rangle = 0. \quad (\text{C.80})$$

Exercise C.224. Prove Lemma C.223.

Corollary C.225. Any eigenvalue of a skew Hermitian matrix is either 0 or purely imaginary.

Proof. This follows directly from Lemma C.223. $\square$

Lemma C.226. Every eigenvalue of a self-adjoint operator T is real.

Proof. Let $(\lambda, \mathbf{u})$ be an eigenpair of T . We have

$$\lambda \|\mathbf{u}\|^2 = \langle \lambda \mathbf{u}, \mathbf{u} \rangle = \langle T\mathbf{u}, \mathbf{u} \rangle = \langle \mathbf{u}, T\mathbf{u} \rangle = \langle \mathbf{u}, \lambda \mathbf{u} \rangle = \bar{\lambda} \|\mathbf{u}\|^2,$$

where the third step follows from Definition C.221. Then $\mathbf{u} \neq \mathbf{0}$ implies $\lambda = \bar{\lambda}$. $\square$

Theorem C.227. Suppose $\mathcal{V}$ is a complex inner product space and $T \in \mathcal{L}(\mathcal{V})$. Then T is self-adjoint if and only if

$$\forall \mathbf{v} \in \mathcal{V}, \quad \langle T\mathbf{v}, \mathbf{v} \rangle \in \mathbb{R}. \quad (\text{C.81})$$

Proof. By Definitions C.170, C.208, and C.221, we have

$$\begin{aligned} \langle T\mathbf{v}, \mathbf{v} \rangle - \overline{\langle T\mathbf{v}, \mathbf{v} \rangle} &= \langle T\mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{v}, T\mathbf{v} \rangle \\ &= \langle T\mathbf{v}, \mathbf{v} \rangle - \langle T^*\mathbf{v}, \mathbf{v} \rangle = \langle (T - T^*)\mathbf{v}, \mathbf{v} \rangle. \end{aligned}$$

Then Lemma C.220 completes the proof. $\square$

Lemma C.228. Suppose $\mathcal{V}$ is a real inner product space and $T \in \mathcal{L}(\mathcal{V})$. If T is self-adjoint and satisfies

$$\forall \mathbf{v} \in \mathcal{V}, \quad \langle T\mathbf{v}, \mathbf{v} \rangle = 0, \quad (\text{C.82})$$

then $T = \mathbf{0}$.

Proof. By the self-adjointness and the underlying field being real, we have $\langle T\mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{w}, T\mathbf{u} \rangle = \langle T\mathbf{u}, \mathbf{w} \rangle$, which, together with Definition C.170, implies

$$\langle T\mathbf{u}, \mathbf{w} \rangle = \frac{1}{4} \langle T(\mathbf{u} + \mathbf{w}), \mathbf{u} + \mathbf{w} \rangle - \frac{1}{4} \langle T(\mathbf{u} - \mathbf{w}), \mathbf{u} - \mathbf{w} \rangle.$$

Setting $\mathbf{w} = T\mathbf{u}$ completes the proof. $\square$

Definition C.229. The *Hermitian part* and the *skew Hermitian part* of a square matrix $A \in \mathbb{C}^{n \times n}$ are respectively the matrices H and $\mathbf{i}S$, where

$$\begin{aligned} H &= \frac{1}{2}(A + A^*), \\ S &= \frac{1}{2\mathbf{i}}(A - A^*), \end{aligned} \quad (\text{C.83})$$

such that $A = H + \mathbf{i}S$.

Lemma C.230. The Hermitian part H of a matrix $A \in \mathbb{R}^{n \times n}$ satisfies

$$\forall \mathbf{u} \in \mathbb{R}^n, \quad \langle H\mathbf{u}, \mathbf{u} \rangle = \langle A\mathbf{u}, \mathbf{u} \rangle. \quad (\text{C.84})$$

Proof. It is readily verified that both H and S are Hermitian and the matrix $\mathbf{i}S$ is skew Hermitian. As normal matrices, both H and $\mathbf{i}S$ have eigenvectors that span $\mathbb{C}^n$ and thus they also span $\mathbb{R}^n$. Then the rest of the proof follows from Corollary C.225. $\square$

C.6.2 Normal operators

Definition C.231. An operator $T \in \mathcal{L}(\mathcal{V})$ is *normal* iff $TT^* = T^*T$. In particular, a *normal matrix* A is a square matrix that commutes with its conjugate transpose, cf. Definition C.213.

Corollary C.232. Every self-adjoint operator is normal.

Lemma C.233. Both Hermitian and skew Hermitian matrices are normal matrices.

Lemma C.234. If a normal matrix is triangular, then it is a diagonal matrix.

Exercise C.235. Prove Lemma C.234.

Definition C.236. Two matrices A and B are *unitarily similar* if there exists a unitary matrix S such that (C.34) holds.

Theorem C.237. A matrix is normal if and only if it is unitarily similar to a diagonal matrix.

Proof. It is straightforward to verify the sufficiency from Definition C.231. As for the necessity, we express the matrix in the Schur form $A = QRQ^*$. Definition C.231 yields

$$QR^*RQ^* = QRR^*Q^*,$$

which implies $R^*R = RR^*$, i.e. the triangular matrix R is normal. The proof is then completed by Lemma C.234. $\square$

Corollary C.238. A normal matrix with real eigenvalues is Hermitian.

Proof. By Theorem C.237, we can write $A = QDQ^*$ where the diagonal matrix D has all eigenvalues of A . If D is real, then $D^* = D$ and thus $A^* = A$. $\square$

Corollary C.239. Any two eigenvectors of a normal matrix associated with two distinct eigenvalues are orthogonal.

Proof. This follows directly from Theorem C.237. $\square$

Lemma C.240. $T \in \mathcal{L}(\mathcal{V})$ is normal if and only if

$$\forall \mathbf{v} \in \mathcal{V}, \quad \|T\mathbf{v}\| = \|T^*\mathbf{v}\|. \quad (\text{C.85})$$

Proof. By Lemma C.228 and Definition C.208, we have

$$\begin{aligned} T^*T = TT^* &\iff \forall \mathbf{v} \in \mathcal{V}, \langle (T^*T - TT^*)\mathbf{v}, \mathbf{v} \rangle = 0 \\ &\iff \forall \mathbf{v} \in \mathcal{V}, \langle T^*T\mathbf{v}, \mathbf{v} \rangle = \langle TT^*\mathbf{v}, \mathbf{v} \rangle \\ &\iff \forall \mathbf{v} \in \mathcal{V}, \|T\mathbf{v}\|^2 = \|T^*\mathbf{v}\|^2. \end{aligned}$$

The positivity of a norm completes the proof. $\square$

Lemma C.241. $T \in \mathcal{L}(\mathcal{V})$ is normal if and only if each eigenvector of T is also an eigenvector of T^* .

Proof. If T is normal, so is $T - \lambda I$. By Lemma C.240, an eigenpair $(\lambda, \mathbf{u})$ of T satisfies

$$0 = \|(T - \lambda I)\mathbf{u}\| = \|(T - \lambda I)^*\mathbf{u}\| = \|(T^* - \bar{\lambda}I)\mathbf{u}\|,$$

and thus $\mathbf{u}$ is also an eigenvector of T^* .

Conversely, suppose each eigenvector $\mathbf{u}$ of T is also an eigenvector of T^* . Then the above equation implies that the corresponding eigenvalue of T^* is the conjugate of that of T . It suffices to prove that these eigenvectors form a basis of V , because then we have

$$TU = U\Lambda, \quad T^*U = U\Lambda^*,$$

where U is the matrix of these eigenvectors. Thus

$$TT^*U = TU\Lambda^* = U\Lambda\Lambda^* = U\Lambda^*\Lambda = T^*U\Lambda = T^*TU$$

and we have $TT^* = T^*T$ because U is nonsingular. By Theorem C.117, it suffices to show that T is diagonalizable. By Theorem C.168, we only need to show that for any eigenpair $(\lambda, \mathbf{u})$ of T ,

$$(A - \lambda I)^2\mathbf{u} = \mathbf{0} \implies (A - \lambda I)\mathbf{u} = \mathbf{0},$$

because this condition will annihilate all Jordan blocks of size greater than 1. Define $\mathbf{v} = (A - \lambda I)\mathbf{u}$ and we have

$$\begin{aligned} \langle A\mathbf{u}, \mathbf{v} \rangle &= \langle \mathbf{u}, A^*\mathbf{v} \rangle = \langle \mathbf{u}, \bar{\lambda}\mathbf{v} \rangle = \langle \lambda\mathbf{u}, \mathbf{v} \rangle, \\ \langle A\mathbf{u}, \mathbf{v} \rangle &= \langle \mathbf{v} + \lambda\mathbf{u}, \mathbf{v} \rangle = \|\mathbf{v}\|^2 + \langle \lambda\mathbf{u}, \mathbf{v} \rangle, \end{aligned}$$

which imply $\mathbf{v} = \mathbf{0}$, i.e., $(A - \lambda I)\mathbf{u} = \mathbf{0}$. $\square$

Theorem C.242. For a linear operator $T \in \mathcal{L}(\mathcal{V})$ on a two-dimensional real inner product space $\mathcal{V}$, the following are equivalent:

- (a) T is normal but not self-adjoint.
- (b) The matrix of T with respect to every orthonormal basis of $\mathcal{V}$ has the form

$$M(T) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \quad (\text{C.86})$$

where $b \neq 0$.

Proof. (b) $\implies$ (a) trivially holds, so we only prove (a) $\implies$ (b). Let (e_1, e_2) be an orthonormal basis of $\mathcal{V}$ and set

$$M(T, (e_1, e_2)) = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

By Definition C.231, we have

$$\begin{aligned} \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{bmatrix}. \end{aligned}$$

$b^2 = c^2$ and the condition of T being not self-adjoint further yields $c = -b \neq 0$, which, together with $ab + cd = ac + bd$, yields $a = d$. $\square$

Corollary C.243. Any Hermitian matrix is unitarily similar to a real diagonal matrix. Conversely, any matrix unitarily similar to a real diagonal matrix is a Hermitian matrix.

Proof. The first statement follows directly from Theorem C.237, Lemma C.233, and Lemma C.226. The second follows directly from Definition C.222. $\square$

C.6.3 The spectral theorems

Theorem C.244 (Complex spectral). For a linear operator $T \in \mathcal{L}(\mathcal{V})$ with $\mathbb{F} = \mathbb{C}$, the following are equivalent:

- (a) T is normal;
- (b) $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T ;
- (c) T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Corollary C.245. A normal operator T whose eigenvalues are real is self-adjoint.

Proof. By Theorem C.244, we write $T = V\Lambda V^{-1}$ and thus $T^* = V\Lambda^*V^{-1}$. Then all entries in Λ being real implies $T = T^*$. $\square$

Theorem C.246 (Real spectral). For a linear operator $T \in \mathcal{L}(\mathcal{V})$ with $\mathbb{F} = \mathbb{R}$, the following are equivalent:

- (a) T is self-adjoint;
- (b) $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T ;
- (c) T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

C.6.4 Isometries

Definition C.247. An operator $S \in \mathcal{L}(\mathcal{V})$ is called a (linear) *isometry* iff

$$\forall \mathbf{v} \in \mathcal{V}, \quad \|S\mathbf{v}\| = \|\mathbf{v}\|. \quad (\text{C.87})$$

Theorem C.248. An operator $S \in \mathcal{L}(\mathcal{V})$ on a real inner product space is an isometry if and only if there exists an orthonormal basis of $\mathcal{V}$ with respect to which S has a block diagonal matrix such that each block on the diagonal is a 1-by-1 matrix containing 1 or -1 , or, is a 2-by-2 matrix of the form

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (\text{C.88})$$

where $\theta \in (0, \pi)$.

Corollary C.249. For an operator $S \in \mathcal{L}(\mathcal{V})$ on a two-dimensional real inner product space, the following are equivalent:

- (a) S is an isometry;
- (b) S is either an identity or a reflection or a rotation.

C.6.5 Singular value decomposition

Definition C.250. An operator $T \in \mathcal{L}(V)$ is *positive semi-definite* iff

$$\forall \mathbf{v} \in V, \quad \langle T\mathbf{v}, \mathbf{v} \rangle \geq 0 \quad (\text{C.89})$$

and is *positive definite* iff

$$\forall \mathbf{v} \in V \setminus \{\mathbf{0}\}, \quad \langle T\mathbf{v}, \mathbf{v} \rangle > 0. \quad (\text{C.90})$$

An operator is *positive* if it is self-adjoint and positive semi-definite.

Corollary C.251. For any linear operator $f \in \mathcal{L}(\mathcal{V})$, both $f^* \circ f$ and $f \circ f^*$ are positive.

Proof. By Definition C.221, $f^* \circ f$ is self-adjoint since

$$\langle (f^* \circ f)\mathbf{u}, \mathbf{v} \rangle = \langle f\mathbf{u}, f\mathbf{v} \rangle = \langle \mathbf{u}, (f \circ f^*)\mathbf{v} \rangle.$$

Suppose $(\lambda, \mathbf{u})$ is an eigenpair of $(f^* \circ f)$. Then we have

$$\begin{aligned} \lambda \langle \mathbf{u}, \mathbf{u} \rangle &= \langle (f^* \circ f)\mathbf{u}, \mathbf{u} \rangle = \langle f\mathbf{u}, f\mathbf{u} \rangle \\ \Rightarrow \lambda &= \frac{\langle f\mathbf{u}, f\mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \geq 0. \end{aligned}$$

Similar arguments apply to $f \circ f^*$. $\square$

Definition C.252. The *singular values* of a linear map $f : \mathbb{C}^n \rightarrow \mathbb{C}^m$ are the non-negative square roots of the eigenvalues of $f^* \circ f$, usually sorted in non-increasing order as

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_n = 0,$$

where r is the rank of f .

Theorem C.253. For any matrix $A \in \mathbb{C}^{m \times n}$ with rank r , there exist orthonormal bases $\mathbb{C}^n = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ and $\mathbb{C}^m = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ such that

$$\forall j = 1, 2, \dots, r, \quad \begin{cases} A\mathbf{u}_j = \sigma_j \mathbf{v}_j; \\ A^* \mathbf{v}_j = \sigma_j \mathbf{u}_j, \end{cases} \quad (\text{C.91a})$$

$$\begin{cases} \forall j = r+1, r+2, \dots, n, & A\mathbf{u}_j = \mathbf{0}; \\ \forall j = r+1, r+2, \dots, m, & A^* \mathbf{v}_j = \mathbf{0}, \end{cases} \quad (\text{C.91b})$$

where σ_j 's are the singular values of A in Definition C.252.

Proof. The matrix $A^*A \in \mathbb{C}^{n \times n}$ is self-adjoint and thus normal. By Theorem C.244, $\mathbb{C}^n$ has an orthonormal basis that are also eigenvectors of A^*A ; choose them to be $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$. By Definition C.252 we have

$$A^*A\mathbf{u}_j = \sigma_j^2 \mathbf{u}_j.$$

Then we choose

$$\forall j = 1, 2, \dots, r, \quad \mathbf{v}_j := \frac{1}{\sigma_j} A\mathbf{u}_j,$$

which implies

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle A\mathbf{u}_i, A\mathbf{u}_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle \mathbf{u}_i, A^*A\mathbf{u}_j \rangle = \frac{\sigma_j}{\sigma_i} \langle \mathbf{u}_i, \mathbf{u}_j \rangle.$$

Then the orthonormality of $\mathbf{u}_j$'s implies that the first r $\mathbf{v}_j$'s are orthonormal and if $r < m$ we can extend them by the Gram-Schmidt process to arrive at an orthonormal basis of $\mathbb{C}^m$. Therefore (C.91a) holds.

The first line of (C.91b) follows from $\text{null } A = \text{null}(A^*A)$, which is implied by

$$\|A\mathbf{u}\|^2 = \langle A\mathbf{u}, A\mathbf{u} \rangle = \langle \mathbf{u}, A^*A\mathbf{u} \rangle.$$

The rank of A^* is r . If $r = m$, the second line of (C.91b) holds vacuously. Otherwise $r < m$. The fundamental theorem of linear algebra (Theorem C.92) implies that A^* has rank $m - r$, which completes the proof. $\square$

Definition C.254. The *singular value decomposition* (SVD) of a rectangular matrix $A \in \mathbb{C}^{m \times n}$ is the factorization $A = V\Sigma U^*$ where Σ is a diagonal matrix with its diagonal entries as the singular values in Definition C.252 and V and U are unitary matrices whose columns are respectively the vectors $\mathbf{v}_j$'s and $\mathbf{u}_j$'s specified in Theorem C.253, which are also called the *left singular vectors* and the *right singular vectors* of A , respectively. We also refer to the sequence of triples $(\sigma_j, \mathbf{u}_j, \mathbf{v}_j)$ as the *singular system* of A .

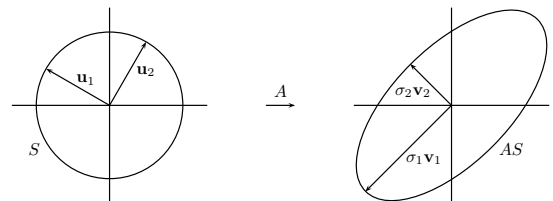


Figure C.3: Geometric interpretation of SVD.

Theorem C.255 (Solvability of a linear system). A linear system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{C}^{m \times n}$ admits a solution if and only if $\mathbf{b}$ is in the range of A , or equivalently, $\mathbf{b}$ is perpendicular to the null space of the adjoint of A , i.e.,

$$\forall \mathbf{z} \in \{\mathbf{y} \in \mathbb{C}^m : A^* \mathbf{y} = \mathbf{0}\}, \quad \langle \mathbf{b}, \mathbf{z} \rangle = 0, \quad (\text{C.92})$$

in which case a particular solution of the linear system is

$$\mathbf{x}^\oplus := \sum_{j=1}^r \frac{1}{\sigma_j} \langle \mathbf{b}, \mathbf{v}_j \rangle \mathbf{u}_j, \quad (\text{C.93})$$

where r is the rank of A and $(\sigma_j, \mathbf{u}_j, \mathbf{v}_j)$ is the singular system of A , cf. Definition C.254.

Proof. Suppose the linear system has a solution $\mathbf{x}$. Then

$$\langle \mathbf{b}, \mathbf{z} \rangle = \langle A\mathbf{x}, \mathbf{z} \rangle = \langle \mathbf{x}, A^* \mathbf{z} \rangle = 0.$$

As for the sufficiency, suppose (C.92) holds. Then we have

$$\mathbf{b} = \sum_{j=1}^r \langle \mathbf{b}, \mathbf{v}_j \rangle \mathbf{v}_j = \sum_{j=1}^r \frac{1}{\sigma_j} \langle \mathbf{b}, \mathbf{v}_j \rangle A \mathbf{u}_j = A \mathbf{x}^\oplus,$$

where the first step follows from (C.91b), the second from (C.91a), and the last from (C.93). $\square$

C.7 Functions of a single matrix

Roadmap C.256. Scalar-valued functions of a single (usually square) matrix often represent fundamental geometric or algebraic properties of the linear transformation the matrix represents. Apart from the condition number of a matrix discussed in Section 2.4, the five concepts covered in Sections C.7.1 to C.7.4 are probably the most critical scalar functions of a matrix. Sections C.7.5 and C.7.6 are two useful matrix functions for numerical analysis.

C.7.1 Trace and determinant

Definition C.257. The *trace* of a matrix A , denoted by $\text{Trace } A$, is the sum of the diagonal entries of A .

Lemma C.258. The trace of a matrix is the sum of its eigenvalues, each of which is repeated according to its multiplicity.

Example C.259. The signed volume of a parallelogram spanned by two vectors $\mathbf{v}_1 = (a, b)^\top$ and $\mathbf{v}_2 = (c, d)^\top$ is

$$\delta(\mathbf{v}_1, \mathbf{v}_2) = ad - bc = \langle \mathbf{v}_1^\perp, \mathbf{v}_2 \rangle. \quad (\text{C.94})$$

Indeed, high school geometry gives

$$\delta = |\mathbf{v}_1| |\mathbf{v}_2| \sin \alpha = |\mathbf{v}_1^\perp| |\mathbf{v}_2| \cos \left(\frac{\pi}{2} - \alpha \right) = \langle \mathbf{v}_1^\perp, \mathbf{v}_2 \rangle$$

where $|\cdot|$ denotes the Euclidean norm, α the angle between the two vectors, and $\mathbf{v}_1^\perp = (-b, a)^\top$ the vector obtained from $\mathbf{v}_1$ by a counterclockwise rotation of $\frac{\pi}{2}$.

Definition C.260. The *signed volume* of a parallelotope spanned by n vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$ is a function $\delta : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ that satisfies

(SVP-1) $\delta(I) = 1$;

(SVP-2) $\delta(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = 0$ if $\mathbf{v}_i = \mathbf{v}_j$ for some $i \neq j$;

(SVP-3) δ is linear, i.e., $\forall j = 1, \dots, n, \forall c \in \mathbb{R}$,

$$\begin{aligned} & \delta(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v} + c\mathbf{w}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_n) \\ &= \delta(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_n) \\ &+ c\delta(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{w}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_n). \end{aligned} \quad (\text{C.95})$$

Lemma C.261. Adding a multiple of one vector to another does not change the signed volume.

Proof. This follows directly from (SVP-2,3). $\square$

Lemma C.262. If the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly dependent, then $\delta(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = 0$.

Proof. WLOG, we assume $\mathbf{v}_1 = \sum_{i=2}^n c_i \mathbf{v}_i$. Then the result follows from (SVP-2,3). $\square$

Lemma C.263. The signed volume δ is alternating, i.e.,

$$\delta(\mathbf{v}_1, \dots, \mathbf{v}_i, \dots, \mathbf{v}_j, \dots, \mathbf{v}_n) = -\delta(\mathbf{v}_1, \dots, \mathbf{v}_j, \dots, \mathbf{v}_i, \dots, \mathbf{v}_n). \quad (\text{C.96})$$

Exercise C.264. Prove Lemma C.263 using (SVP-2,3).

Definition C.265. A *permutation* of a set A is a bijective function $\sigma : A \rightarrow A$.

Definition C.266. The *symmetric group* of order n , written S_n , is the set of all permutations of a set A of n elements.

Definition C.267 (Levi-Civita). Let σ be a permutation of $A := \{1, 2, \dots, n\}$ and let s denote the number of pairs of integers (j, k) with $1 \leq j < k \leq n$ such that j appears after k in the list $(m_1, \dots, m_n)$ given by $m_i = \sigma(i)$. The *Levi-Civita symbol* $\epsilon_{m_1 m_2 \dots m_n}$ or the *sign of the permutation* σ , written $\text{sgn}(\sigma)$, is 1 if s is even and -1 if s is odd.

Lemma C.268. Let M_σ denote the matrix of a permutation $\sigma : E \rightarrow E$ where E is the set of standard basis vectors in (C.6). Then we have $\delta(M_\sigma) = \text{sgn}(\sigma)$.

Proof. There is a one-to-one correspondence between the vectors in the matrix

$$M_\sigma = [e_{\sigma(1)}, e_{\sigma(2)}, \dots, e_{\sigma(n)}]$$

and the scalars in the one-line notation

$$(\sigma(1) \ \sigma(2) \ \dots \ \sigma(n)).$$

A sequence of transpositions taking σ to the identity map also takes M_σ to the identity matrix. By Lemma C.263, each transposition yields a multiplication factor -1 . Definition C.267 and (SVP-1) give $\delta(M_\sigma) = \text{sgn}(\sigma)\delta(I) = \text{sgn}(\sigma)$. $\square$

Definition C.269 (Leibniz formula of determinants). The *determinant* of a square matrix $A \in \mathbb{R}^{n \times n}$ is

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{\sigma(i), i}, \quad (\text{C.97})$$

where the sum is over the symmetric group S_n and $a_{\sigma(i), i}$ is the element of A at the $\sigma(i)$ th row and the i th column.

Lemma C.270. The determinant of a matrix is the product of its eigenvalues, each of which is repeated according to its multiplicity.

Theorem C.271. The signed volume function satisfying (SVP-1,2,3) in Definition C.260 is unique and is the same as the determinant in (C.97).

Proof. Let the parallelepiped be spanned by the column vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. We have

$$\begin{aligned} \delta \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{bmatrix} &= \sum_{i_1=1}^n v_{i_1 1} \delta \begin{bmatrix} | & v_{12} & \cdots & v_{1n} \\ e_{i_1} & v_{22} & \cdots & v_{2n} \\ | & \vdots & \ddots & \vdots \\ | & v_{n2} & \cdots & v_{nn} \end{bmatrix} \\ &= \sum_{i_1, i_2=1}^n v_{i_1 1} v_{i_2 2} \delta \begin{bmatrix} | & | & v_{13} & \cdots & v_{1n} \\ e_{i_1} & e_{i_2} & v_{23} & \cdots & v_{2n} \\ | & | & \vdots & \ddots & \vdots \\ | & | & v_{n2} & \cdots & v_{nn} \end{bmatrix}; \end{aligned}$$

a continuation in this manner gives

$$\begin{aligned} &\delta(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \\ &= \sum_{i_1, i_2, \dots, i_n=1}^n v_{i_1 1} v_{i_2 2} \cdots v_{i_n n} \delta \begin{bmatrix} | & | & \cdots & | \\ e_{i_1} & e_{i_2} & \cdots & e_{i_n} \\ | & | & \cdots & | \end{bmatrix} \\ &= \sum_{\sigma \in S_n} v_{\sigma(1),1} v_{\sigma(2),2} \cdots v_{\sigma(n),n} \delta \begin{bmatrix} | & | & \cdots & | \\ e_{\sigma(1)} & e_{\sigma(2)} & \cdots & e_{\sigma(n)} \\ | & | & \cdots & | \end{bmatrix} \\ &= \sum_{\sigma \in S_n} v_{\sigma(1),1} v_{\sigma(2),2} \cdots v_{\sigma(n),n} \operatorname{sgn}(\sigma) \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n v_{\sigma(i),i}, \end{aligned}$$

where the first four steps follow from (SVP-3), the sixth step from Lemma C.268, and the fifth step from (SVP-2). The signed volume $\delta(\cdot)$ is zero for any $i_j = i_k$ and hence the only nonzero terms are those of which $(i_1, i_2, \dots, i_n)$ is a permutation of $(1, 2, \dots, n)$. $\square$

Exercise C.272. Use the formula in (C.97) to show that $\det A = \det A^\top$.

Definition C.273. The (i, j) th *cofactor* of $A \in \mathbb{R}^{n \times n}$ is

$$C_{ij} := (-1)^{i+j} M_{ij}, \quad (\text{C.98})$$

where M_{ij} is the (i, j) *minor* of a matrix A , i.e. the determinant of the $(n-1) \times (n-1)$ matrix that results from deleting the i -th row and the j -th column of A .

Definition C.274. The *cofactor matrix* or the *matrix of cofactors* or the *comatrix* of A is the matrix formed by all of the cofactors of A , i.e.,

$$C := \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}. \quad (\text{C.99})$$

The *adjugate* of a matrix A , written $\operatorname{adj}(A)$, is the transpose of the cofactor matrix of A .

Theorem C.275 (Laplace formula of determinants). Given fixed indices $i, j \in \{1, 2, \dots, n\}$, the determinant of an n -by- n matrix A is given by

$$\det A = \sum_{j'=1}^n a_{ij'} C_{ij'} = \sum_{i'=1}^n a_{i'j} C_{i'j}, \quad (\text{C.100})$$

where $C_{ij'}$ is the (i, j') th cofactor of A in (C.98).

Exercise C.276. Prove Theorem C.275 by induction.

Corollary C.277. The product of a square matrix A and its adjugate is a diagonal matrix with $\det A$ on the diagonals, i.e., $A \operatorname{adj}(A) = (\det A)I$.

Corollary C.278. A square invertible matrix A has $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$.

Theorem C.279 (Cramer's rule). A linear system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{n \times n}$ and $\det A \neq 0$ has the unique solution

$$\forall i = 1, \dots, n, \quad x_i = \frac{\det A_i}{\det A}, \quad (\text{C.101})$$

where A_i is the matrix formed by replacing the i th column of A by $\mathbf{b}$.

Theorem C.280. The characteristic polynomial $p_A(z)$ of a linear operator $T \in \mathcal{L}(V)$ equals $\det(zI - M_T)$ where M_T is the matrix of T .

Proof. See [Axler, 2015, p. 309]. $\square$

C.7.2 Matrix norms

Definition C.281. The *matrix norm* of $A \in \mathbb{F}^{m \times n}$ induced from a vector p -norm with $p \in [1, \infty]$ is given by

$$\|A\|_p := \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|A\mathbf{x}\|_p}{\|\mathbf{x}\|_p}, \quad (\text{C.102})$$

where $\|\cdot\|_p$ on the RHS is given by Definition C.176. In particular, $\|A\|_2$ is also called the *spectral norm* of A .

Lemma C.282. Two induced matrix norms are given by

$$\begin{cases} \|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|, \\ \|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|. \end{cases} \quad (\text{C.103})$$

Lemma C.283. The induced matrix 1-norm and ∞ -norm satisfy

$$\|AB\|_1 \leq \|A\|_1 \|B\|_1, \quad \|AB\|_\infty \leq \|A\|_\infty \|B\|_\infty. \quad (\text{C.104})$$

Roadmap C.284. In Definition C.182, the concept of norms is defined with respect to a vector space $\mathcal{V}$, which is either $\mathbb{R}^n$ or $\mathbb{C}^n$ in lower-level classes such as linear algebra, but $\mathcal{V}$ could be a vector space of functions in upper-level classes such as functional analysis. To transit smoothly from linear algebra to functional analysis, we rephrase the induced matrix 2-norm as an operator norm. A careful study of this operator norm and Hilbert–Schmidt norm serves as a stepping stone to the more abstract discussion of norms in Sections E.1.3 and E.2. Then the equivalence of these two norms in Theorem C.303 is utilized to define the matrix exponential in Section C.7.6.

Notation 25. The length of a Euclidean vector $\mathbf{x} \in \mathbb{F}^n$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ is denoted by $|\mathbf{x}| := \|\mathbf{x}\|_2$ where $\|\cdot\|_2$ is the Euclidean norm in (C.63).

Lemma C.285. The length-scaling factor of a linear map $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ upon any vector is bounded, i.e.,

$$\forall T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m), \exists M \geq 0 \text{ s.t. } \forall \mathbf{x} \in \mathbb{F}^n, |T\mathbf{x}| \leq M|\mathbf{x}|, \quad (\text{C.105})$$

Proof. Since $\mathbf{x} = \sum_j x_j \mathbf{e}_j$ and T is a linear map, we have

$$\begin{aligned} |T\mathbf{x}| &= \left| \sum_j x_j T\mathbf{e}_j \right| \leq \sum_j |x_j T\mathbf{e}_j| = \sum_j |x_j| |T\mathbf{e}_j| \\ &\leq |\mathbf{x}| \sum_j |T\mathbf{e}_j|, \end{aligned}$$

where the second step follows from the triangle inequality in Definition C.182, the third step from the absolute homogeneity in Definition C.182, and the last step from $|x_j| \leq |\mathbf{x}|$. The proof is completed by setting $M := \sum_j |T\mathbf{e}_j|$. $\square$

Definition C.286. A function $\mathbf{f} : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is *uniformly continuous* on $U \subset \mathbb{F}^m$ iff

$$\begin{aligned} \forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ \forall \mathbf{x}, \mathbf{y} \in U, |\mathbf{x} - \mathbf{y}| < \delta \implies |\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})| < \epsilon. \end{aligned} \quad (\text{C.106})$$

Corollary C.287. Any linear map $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ is uniformly continuous on $\mathbb{F}^n$.

Proof. For any $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, we have

$$|T\mathbf{x} - T\mathbf{y}| = |T(\mathbf{x} - \mathbf{y})| \leq M|\mathbf{x} - \mathbf{y}|,$$

where the second step follows from Lemma C.285. The rest follows from setting $\delta = \frac{\epsilon}{M}$ in Definition C.286. $\square$

Definition C.288. The *operator norm* of a linear map $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ is the non-negative number

$$\|T\| := \inf \{M \geq 0 : \forall \mathbf{x} \in \mathbb{F}^n, |T\mathbf{x}| \leq M|\mathbf{x}|\}. \quad (\text{C.107})$$

Corollary C.289. $\forall \mathbf{x} \in \mathbb{F}^n, |T\mathbf{x}| \leq \|T\||\mathbf{x}|$.

Proof. For any given T , Lemma C.285 implies that the set

$$\mathcal{M} := \{M \geq 0 : \forall \mathbf{x} \in \mathbb{F}^n, |T\mathbf{x}| \leq M|\mathbf{x}|\}$$

is nonempty and bounded from below. For any sequence $\{M_n\}_{n=1}^\infty$ in $\mathcal{M}$ that converges to c , we can take the limit to the sequence of inequalities $|T\mathbf{x}| \leq M_n|\mathbf{x}|$ to obtain $|T\mathbf{x}| \leq c|\mathbf{x}|$. Hence $c \in \mathcal{M}$ and $\mathcal{M}$ is closed. It follows that the infimum of $\mathcal{M}$ is contained in $\mathcal{M}$. $\square$

Exercise C.290. Verify that (C.107) is indeed a norm in the sense of Definition C.182.

Corollary C.291. $\|T\| = \sup_{|\mathbf{x}| \leq 1} |T\mathbf{x}| = \sup_{|\mathbf{x}|=1} |T\mathbf{x}|$.

Proof. Since T is a linear map and $|\cdot|$ is a norm, we have

$$|T(c\mathbf{x})| = |cT\mathbf{x}| = |c||T\mathbf{x}|.$$

Hence the inequality $|T\mathbf{x}| \leq M|\mathbf{x}|$ in (C.107) holds for all $\mathbf{x} \neq 0$ if and only if it holds for all $\mathbf{x}$ with $|\mathbf{x}| \in (0, 1]$, if and only if it holds for all $\mathbf{x}$ with $|\mathbf{x}| = 1$. $\square$

Corollary C.292. The composition of two linear maps $S \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ and $T \in \mathcal{L}(\mathbb{F}^m, \mathbb{F}^k)$ satisfies

$$\|TS\| \leq \|T\|\|S\|. \quad (\text{C.108})$$

Proof. By Corollary C.289, we have

$$|(TS)(\mathbf{x})| = |T(S\mathbf{x})| \leq \|T\||S\mathbf{x}| \leq \|T\|\|S\||\mathbf{x}|.$$

Taking supremum of the above for $|\mathbf{x}| \leq 1$ and applying Corollary C.291 yield (C.108). $\square$

Corollary C.293. The identity function $I \in \mathcal{L}(\mathbb{F}^n)$ satisfies $\|I\| = 1$.

Proof. This follows directly from (C.107). $\square$

Exercise C.294. Verify that the space $\mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ becomes a metric space if we define the metric as $d(T, S) = \|T - S\|$.

Lemma C.295. For $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^m)$, suppose for each standard basis vectors $\mathbf{e}_j$ in Definition C.34 we have $T\mathbf{e}_j \in \mathbb{R}^m$. Then T carries $\mathbb{R}^n$ into $\mathbb{R}^m$ and $\|T\|$ is consistently defined in the sense that

$$\|T\| = \sup_{\mathbf{x} \in \mathbb{R}^n; |\mathbf{x}| \leq 1} |T\mathbf{x}| = \sup_{\mathbf{z} \in \mathbb{C}^n; |\mathbf{z}| \leq 1} |T\mathbf{z}|. \quad (\text{C.109})$$

Proof. First, $T(\mathbb{R}^n) \subset \mathbb{R}^m$ is trivial because the matrix of T is a real matrix. Thus, $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. From this fact, we can define an operator norm of T as

$$(*) : \quad \|T\| = \sup_{\mathbf{x} \in \mathbb{R}^n; |\mathbf{x}| \leq 1} |T\mathbf{x}|.$$

On the one hand, we have

$$\sup_{\mathbf{x} \in \mathbb{R}^n; |\mathbf{x}| \leq 1} |T\mathbf{x}| \leq \sup_{\mathbf{z} \in \mathbb{C}^n; |\mathbf{z}| \leq 1} |T\mathbf{z}|$$

since $\mathbb{R}^n \subset \mathbb{C}^n$. On the other hand, $\forall \mathbf{z} \in \mathbb{C}^n, |\mathbf{z}| \leq 1$, $\exists \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, s.t. $\mathbf{z} = \mathbf{x} + i\mathbf{y}$ and $|\mathbf{x}|^2 + |\mathbf{y}|^2 \leq 1$, and we have

$$\begin{aligned} |T\mathbf{z}| &= |T\mathbf{x} + iT\mathbf{y}| = \sqrt{|T\mathbf{x}|^2 + |T\mathbf{y}|^2} \\ &\leq \sqrt{\|T\|^2|\mathbf{x}|^2 + \|T\|^2|\mathbf{y}|^2} \\ &= \|T\|\sqrt{|\mathbf{x}|^2 + |\mathbf{y}|^2} \leq \|T\| = \sup_{\mathbf{x} \in \mathbb{R}^n; |\mathbf{x}| \leq 1} |T\mathbf{x}|, \end{aligned}$$

where the third step follows from Corollary C.289, the fifth step from $|\mathbf{x}|^2 + |\mathbf{y}|^2 \leq 1$, and the last step from (*). Thus we have

$$\sup_{\mathbf{x} \in \mathbb{R}^n; |\mathbf{x}| \leq 1} |T\mathbf{x}| \geq \sup_{\mathbf{z} \in \mathbb{C}^n; |\mathbf{z}| \leq 1} |T\mathbf{z}|,$$

which completes the proof. $\square$

Definition C.296. The *Hilbert–Schmidt norm* or the *Frobenius norm* of a linear map $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ is the non-negative number

$$|T| := \left(\sum_{j=1}^n |T\mathbf{e}_j|^2 \right)^{\frac{1}{2}} \quad (\text{C.110})$$

where $\mathbf{e}_1, \dots, \mathbf{e}_n$ is the standard basis in Definition C.34.

Exercise C.297. Verify that (C.110) is indeed a norm in the sense of Definition C.182.

Corollary C.298. The matrix A of $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ satisfies

$$|A| = \left(\sum_{i,j} |a_{ij}|^2 \right)^{\frac{1}{2}} \quad (\text{C.111})$$

Proof. Since A is a linear map, we rewrite (C.110) as

$$|A|^2 = |A\mathbf{e}_1|^2 + |A\mathbf{e}_2|^2 + \cdots + |A\mathbf{e}_n|^2.$$

The proof is completed by $|\cdot|$ being the Euclidean norm. $\square$

Corollary C.299. $\forall \mathbf{x} \in \mathbb{F}^n$, $|T\mathbf{x}| \leq |T||\mathbf{x}|$.

Proof. We have

$$\begin{aligned} |T\mathbf{x}| &= \left| \sum_j x_j T\mathbf{e}_j \right| \leq \sum_j |x_j| |T\mathbf{e}_j| \\ &\leq \left(\sum_j |x_j|^2 \right)^{\frac{1}{2}} \left(\sum_j |T\mathbf{e}_j|^2 \right)^{\frac{1}{2}} = |T||\mathbf{x}|, \end{aligned}$$

where the first inequality follows from (NRM-3,4) in Definition C.182, the second inequality from the Cauchy–Schwarz inequality (C.71), and the last step from Definition C.296. $\square$

Corollary C.300. The composition of two linear maps $S \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ and $T \in \mathcal{L}(\mathbb{F}^m, \mathbb{F}^k)$ satisfies

$$|TS| \leq |T||S|. \quad (\text{C.112})$$

Exercise C.301. Prove Corollary C.300.

Corollary C.302. The identity function $I \in \mathcal{L}(\mathbb{F}^n)$ satisfies $|I| = \sqrt{n}$.

Proof. This follows directly from (C.110). $\square$

Theorem C.303. The operator norm and the Hilbert–Schmidt norm on $\mathcal{L}(\mathbb{F}^n)$ are related by

$$\|T\| \leq |T| \leq \sqrt{n}\|T\|. \quad (\text{C.113})$$

Proof. Take supremum of Corollary C.299, apply Corollary C.291, and we have $\|T\| \leq |T|$. Then $|T| \leq \sqrt{n}\|T\|$ is given by

$$|T|^2 = \sum_j |T\mathbf{e}_j|^2 \leq \sum_j \|T\|^2 |\mathbf{e}_j|^2 = n\|T\|^2,$$

where the inequality follows from Corollary C.289. $\square$

Lemma C.304. For $A, B \in \mathcal{L}(\mathbb{R}^n)$, if A is invertible and $\|B - A\| \cdot \|A^{-1}\| < 1$, then B is also invertible.

Proof. Write $\alpha := \frac{1}{\|A^{-1}\|}$, $\beta := \|B - A\|$, and we have $\alpha > \beta$. Then, $\forall \mathbf{x} \in \mathbb{R}^n$,

$$\begin{aligned} \alpha|\mathbf{x}| &= \alpha|A^{-1}A\mathbf{x}| \leq \alpha\|A^{-1}\|\|A\mathbf{x}\| = |(A - B + B)\mathbf{x}| \\ &\leq \|B - A\||\mathbf{x}| + |B\mathbf{x}| = \beta|\mathbf{x}| + |B\mathbf{x}|, \end{aligned}$$

which implies $0 \leq (\alpha - \beta)|\mathbf{x}| \leq |B\mathbf{x}|$. Theorem C.55 implies that B is injective and Theorem C.60 further implies that B is invertible. $\square$

C.7.3 The spectral radius

Definition C.305. The *spectral radius* of a linear operator $T \in \mathcal{L}(\mathcal{V})$, written $\rho(T)$, is the maximum magnitude of its eigenvalues.

Lemma C.306. A square matrix A satisfies

$$\|A\|_2 = \sqrt{\rho(A^*A)} = \sigma_1(A). \quad (\text{C.114})$$

Proof. This follows from Theorems C.246 and C.253 and Definitions C.305, C.281, and C.252:

$$\begin{aligned} \|A\mathbf{x}\|_2^2 &= \|V\Sigma U^*\mathbf{x}\|_2^2 = \mathbf{x}^*U\Sigma^*V^*V\Sigma U^*\mathbf{x} \\ &= \sum_{i=1}^n |\sigma_i|^2 |x_i|^2 \leq \sigma_1^2 \|\mathbf{x}\|_2^2, \end{aligned} \quad (\text{C.115})$$

where the equality holds if $\mathbf{x} = (1, 0, \dots, 0)^\top$. $\square$

Theorem C.307 (Gelfand’s formula). For any matrix norm, we have

$$\lim_{n \rightarrow \infty} \|A^n\|^{\frac{1}{n}} = \rho(A). \quad (\text{C.116})$$

Proof. Consider an arbitrary $\epsilon > 0$. For $A_b = \frac{1}{\rho(A) + \epsilon}A$, Theorem C.326 yields

$$\begin{aligned} \rho(A_b) < 1 &\implies \lim_{n \rightarrow \infty} A_b^n = \mathbf{0} \\ &\implies \exists N_b \in \mathbb{N} \text{ s.t. } \forall m > N_b, \|A_b^m\| < 1 \\ &\implies \exists N_b \in \mathbb{N} \text{ s.t. } \forall m > N_b, \|A^m\|^{\frac{1}{m}} < \rho(A) + \epsilon. \end{aligned}$$

Similarly, for $A_u = \frac{1}{\rho(A) - \epsilon}A$, we have

$$\begin{aligned} \rho(A_u) > 1 &\implies \exists N_u \in \mathbb{N} \text{ s.t. } \forall m > N_u, \|A_u^m\| > 1 \\ &\implies \exists N_u \in \mathbb{N} \text{ s.t. } \forall m > N_u, \|A^m\|^{\frac{1}{m}} > \rho(A) - \epsilon. \end{aligned}$$

In summary, set $N = \max(N_b, N_u)$ and we have

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m > N, \left| \|A^m\|^{\frac{1}{m}} - \rho(A) \right| < \epsilon,$$

which, by Definition B.3, is equivalent to (C.116). $\square$

Lemma C.308. For any square matrix A and any matrix norm $\|\cdot\|$ induced from a vector norm, we have

$$\rho(A) \leq \|A\|. \quad (\text{C.117})$$

Proof. For each eigenvalue λ there exists an eigenvector $\mathbf{x}_0$ that is normalized with respect to the given vector norm. By Definition C.281, we have

$$\|A\| = \sup_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| \geq \|A\mathbf{x}_0\| = |\lambda| \|\mathbf{x}_0\| = |\lambda|. \quad \square$$

Corollary C.309. A square matrix A satisfies

$$\|A\|_2^2 \leq \|A\|_1 \|A\|_\infty. \quad (\text{C.118})$$

Proof. Lemmas C.306, C.308, C.283, and C.282 yield

$$\|A\|_2^2 = \rho(A^*A) \leq \|A^*A\|_\infty \leq \|A^*\|_\infty \|A\|_\infty = \|A\|_1 \|A\|_\infty. \quad \square$$

Theorem C.310. The spectral radius of a matrix $A \in \mathbb{C}^{n \times n}$ is the infimum of all vector-induced matrix norms; in other words, for any $\epsilon > 0$, there exists some vector-induced matrix norm $\|\cdot\|$ satisfying $\rho(A) \leq \|A\| \leq \rho(A) + \epsilon$.

Proof. The first inequality is given by Lemma C.308 and it remains to show that, for any $\epsilon > 0$, there exists some vector-induced matrix norm $\|\cdot\|$ satisfying $\|A\| \leq \rho(A) + \epsilon$.

By Theorem C.110 or C.169, there exists a non-singular matrix R such that

$$RAR^{-1} = J = \Lambda + U,$$

where Λ is a diagonal matrix whose diagonal entries are precisely the eigenvalues of A and the entries of U satisfy $u_{ij} = 0$ whenever $j \leq i$.

For $\delta \in (0, 1)$, we construct a diagonal matrix

$$D(\delta) := \begin{bmatrix} 1 & & & \\ & \delta^{-1} & & \\ & & \ddots & \\ & & & \delta^{1-n} \end{bmatrix},$$

which is clearly nonsingular. Define $C := DJD^{-1} = \Lambda + E$, where entries of $E = DUD^{-1}$ satisfy

$$e_{ij} = \begin{cases} 0 & \text{if } j \leq i; \\ u_{ij}\delta^{j-i} & \text{otherwise.} \end{cases}$$

Consequently, we have

$$C^*C = (\Lambda^* + E^*)(\Lambda + E) = \Lambda^*\Lambda + \mathcal{M}(\delta), \quad (\text{C.119})$$

where each entry of $\mathcal{M}(\delta)$ is $O(\delta)$, i.e., there exists a constant $K > 0$ that only depends on A such that each entry of $\mathcal{M}(\delta)$ is within the range $[-K\delta, K\delta]$.

Since $A = R^{-1}D^{-1}CDR$ and DR is non-singular, we define a vector norm $\|\cdot\|$ by

$$\|\mathbf{x}\| := \|DR\mathbf{x}\|_2 = (\mathbf{x}^*R^*D^*DR\mathbf{x})^{\frac{1}{2}}, \quad (\text{C.120})$$

where $*$ denotes the conjugate transpose in Definition C.213. Then the matrix norm induced by (C.120) is

$$\|A\| := \max_{\|\mathbf{y}\|=1} \|A\mathbf{y}\|. \quad (\text{C.121})$$

For a vector $\mathbf{y}$ with $\|\mathbf{y}\| = 1$, (C.120) implies that the vector $\mathbf{z} := DR\mathbf{y}$ satisfies $\|\mathbf{z}\|_2 = 1$ and $\mathcal{M} = O(\delta)$ implies

$$\begin{aligned} |\mathbf{z}^*\mathcal{M}(\delta)\mathbf{z}| &= |\sum_{i=1}^n \sum_{j=1}^n \bar{z}_i m_{ij} z_j| \\ &\leq \sum_{i=1}^n \sum_{j=1}^n |\bar{z}_i| |m_{ij}| |z_j| \leq k\delta (\sum_{i=1}^n |\bar{z}_i|) (\sum_{j=1}^n |z_j|) \\ &= |\langle \mathbf{y}, \mathbf{1} \rangle|^2 \leq \langle \mathbf{y}, \mathbf{y} \rangle \langle \mathbf{1}, \mathbf{1} \rangle = (\sum_{i=1}^n |z_i|^2) (\sum_{j=1}^n 1) = n\mathbf{z}^*\mathbf{z}, \end{aligned}$$

where $\mathbf{y} = (|z_1|, |z_2|, \dots, |z_n|)^T$, $\mathbf{1} \in \mathbb{R}^n$ is the vector with all its components as 1, and the last inequality follows from the Cauchy–Schwarz inequality (C.71). Then we have

$$\begin{aligned} \|A\mathbf{y}\|^2 &= \|DR A\mathbf{y}\|_2^2 = \|CD R\mathbf{y}\|_2^2 = \|C\mathbf{z}\|_2^2 = \mathbf{z}^* C^* C \mathbf{z} \\ &= \mathbf{z}^* \Lambda^* \Lambda \mathbf{z} + \mathbf{z}^* \mathcal{M}(\delta) \mathbf{z} \leq (\rho(A)^2 + n\delta) \|\mathbf{z}\|_2^2 \\ &= \rho(A)^2 + nK\delta. \end{aligned}$$

Consequently,

$$\|A\| \leq \sqrt{\rho(A)^2 + n^2 K \delta} \leq \rho(A) + \sqrt{nK\delta}.$$

To sum up, $\forall \epsilon > 0$, we choose $\delta = \min(\frac{1}{2}, \frac{\epsilon^2}{nK})$ so that, for the matrix norm in (C.121), $\|A\| \leq \rho(A) + \epsilon$ holds. $\square$

C.7.4 The numerical radius

Definition C.311. Let V be a finite-dimensional vector space with $\mathbb{F}$ as its scalar field. The *Rayleigh quotient* of a linear operator $T \in \mathcal{L}(V)$ is a function $\mu_T : V \setminus \{\mathbf{0}\} \rightarrow \mathbb{F}$ given by

$$\mu_T(\mathbf{x}) = \frac{\langle T\mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle}. \quad (\text{C.122})$$

Definition C.312. The *numerical range* or *field of values* of a linear operator T is the range of its Rayleigh quotients.

Example C.313. $\mu_A(\mathbf{x}) = \lambda$ for each eigenpair $(\lambda, \mathbf{x})$ of A .

Theorem C.314. The field of values of a normal operator A is the convex hull of its spectrum.

Proof. By Theorem C.237, A is unitarily similar to a diagonal matrix Λ , i.e., $A = Q\Lambda Q^*$ with $Q = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n]$ satisfying $Q^*Q = I$. Then each $(\lambda_i, \mathbf{q}_i)$ is an eigenpair of A . Consequently, any vector $\mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$ can be expressed as $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{q}_i$, for which we have

$$\mu_A(\mathbf{x}) = \frac{\langle \sum_{i=1}^n x_i \lambda_i \mathbf{q}_i, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} = \frac{\sum_{i=1}^n |x_i|^2 \lambda_i}{\sum_{i=1}^n |x_i|^2} = \sum_{i=1}^n \beta_i \lambda_i,$$

where $\beta_i = \frac{|x_i|^2}{\sum_{j=1}^n |x_j|^2}$. The proof is completed by the facts that $\sum_{i=1}^n \beta_i = 1$ and $\beta_i \geq 0$ for each i . $\square$

Theorem C.315 (Hausdorff). The field of values of an arbitrary matrix is a convex set that contains the convex hull of its spectrum.

Definition C.316. The *numerical radius* of a square matrix $A \in \mathbb{C}^{n \times n}$ is the radius of the smallest disk that contains its field of values, i.e.,

$$\nu(A) := \sup_{\mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\}} |\mu_A(\mathbf{x})|. \quad (\text{C.123})$$

Lemma C.317. Any square matrix $A \in \mathbb{C}^{n \times n}$ satisfies

$$\rho(A) \leq \nu(A) \leq \|A\|_2, \quad (\text{C.124})$$

where the equalities hold if A is normal.

Proof. For the first inequality, let $|\lambda| = \rho(A)$ and $\mathbf{u}$ be the corresponding eigenvector with unit length. Then

$$\nu(A) = \max_{\|\mathbf{x}\|_2=1} |\langle A\mathbf{x}, \mathbf{x} \rangle| \geq |\langle A\mathbf{u}, \mathbf{u} \rangle| = |\lambda| = \rho(A).$$

As for the second inequality, we have

$$\nu(A) = \max_{\|\mathbf{x}\|_2=1} |\langle A\mathbf{x}, \mathbf{x} \rangle| \leq \max_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2 \|\mathbf{x}\|_2 = \|A\|_2,$$

where the second step follows from the Cauchy–Schwarz inequality (Theorem C.197). $\square$

Exercise C.318. Is ν in (C.123) a norm on $\mathbb{C}^{n \times n}$? Prove your conclusion.

Lemma C.319. The field of values of a Hermitian matrix $A \in \mathbb{C}^{n \times n}$ is a compact interval in $\mathbb{R}$, where the Rayleigh quotient function μ_A attains its maximum and minimum.

Proof. By Lemma C.233, A is a normal operator on $V = \mathbb{C}^n$. Then Theorem C.314 and Lemma C.226 imply that the field of values of A is the convex hull of n points in $\mathbb{R}$. It follows from the extreme value theorem B.72 that both the maximum and the minimum of μ_A are attained on $\mu_A(V)$. $\square$

Theorem C.320 (Rayleigh). The extremum eigenvalues of a Hermitian matrix A can be characterized as

$$\begin{aligned}\lambda_{\max}(A) &= \max_{\mathbf{x} \neq \mathbf{0}} \mu_A(\mathbf{x}), \\ \lambda_{\min}(A) &= \min_{\mathbf{x} \neq \mathbf{0}} \mu_A(\mathbf{x}).\end{aligned}\quad (\text{C.125})$$

Proof. These identities follow directly from Lemma C.319 and Theorem C.314. $\square$

Theorem C.321 (Courant-Fisher min-max principle). The spectrum of a Hermitian matrix $A \in \mathbb{C}^{n \times n}$ is characterized by

$$\begin{aligned}\lambda_k &= \min_{\dim(S)=n-k+1} \max_{\mathbf{x} \in S \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}), \\ \lambda_k &= \max_{\dim(S)=k} \min_{\mathbf{x} \in S \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}),\end{aligned}\quad (\text{C.126})$$

where S is a subspace of $\mathbb{C}^n$ and λ_k is the k th eigenvalue of A in non-increasing order, i.e., $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.

Proof. By Theorem C.237, we have $A = U\Lambda U^*$ where the columns $\mathbf{u}_i$'s of the unitary matrix U are orthonormal eigenvectors and the diagonal entries λ_i 's of Λ are the corresponding eigenvalues.

Define $R := \text{span}(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k)$ and let S be any subspace of $\mathbb{C}^n$ with $\dim S = n - k + 1$. By Corollary C.44, the dimension of $R \cap S$ is at least 1. Then we have

$$\begin{aligned}\max_{\mathbf{x} \in S \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}) &\geq \max_{\mathbf{x} \in S \cap R \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}) \geq \min_{\mathbf{x} \in S \cap R \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}) \\ &\geq \min_{\mathbf{x} \in R \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}) = \lambda_k,\end{aligned}$$

where the existence of global extrema follows from Lemma C.319 and the last step follows from Theorem C.320. The minimum of the above inequality yields

$$\lambda_k \leq \min_{\dim(S)=n-k+1} \max_{\mathbf{x} \in S \setminus \{\mathbf{0}\}} \mu_A(\mathbf{x}),$$

where the " $\leq$ " reduces to " $=$ " because we can choose

$$S = \text{span}(\mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n). \quad \square$$

Exercise C.322. Prove the second equality in (C.126).

Theorem C.323. Any eigenvalue λ_j of a square matrix A is bounded as

$$\begin{aligned}\lambda_{\min}(H) &\leq \text{Re}(\lambda_j) \leq \lambda_{\max}(H), \\ \lambda_{\min}(S) &\leq \text{Im}(\lambda_j) \leq \lambda_{\max}(S),\end{aligned}\quad (\text{C.127})$$

where H, S are defined in (C.83).

Proof. Any eigenpair $(\lambda_j, \mathbf{u}_j)$ of A with $\|\mathbf{u}_j\| = 1$ satisfies

$$\lambda_j = \langle A\mathbf{u}_j, \mathbf{u}_j \rangle = \langle H\mathbf{u}_j, \mathbf{u}_j \rangle + i \langle S\mathbf{u}_j, \mathbf{u}_j \rangle.$$

The rest of the proof follows from Lemma C.319 and the Rayleigh theorem C.320. $\square$

Example C.324. The estimation of the range of eigenvalues of a matrix A from its field of values via Theorem C.323 may be inaccurate. For example, eigenvalues of the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 10^4 & 1 \end{bmatrix} \quad (\text{C.128})$$

are -99 and 101 , but those of H are $1 \pm \frac{1}{2}(10^4 + 1)$ and those of iS are $\pm \frac{i}{2}(10^4 - 1)$.

Lemma C.325. The numerical radius and the 2-norm of a square matrix A satisfy

$$\frac{1}{2}\|A\|_2 \leq \nu(A) \leq \|A\|_2. \quad (\text{C.129})$$

Proof. We only prove the first inequality since the second holds from Lemma C.317. The identity

$$\begin{aligned}4 \langle A\mathbf{x}, \mathbf{y} \rangle &= \langle A(\mathbf{x} + \mathbf{y}), (\mathbf{x} + \mathbf{y}) \rangle - \langle A(\mathbf{x} - \mathbf{y}), (\mathbf{x} - \mathbf{y}) \rangle \\ &\quad + i \langle A(\mathbf{x} + i\mathbf{y}), (\mathbf{x} + i\mathbf{y}) \rangle - i \langle A(\mathbf{x} - i\mathbf{y}), (\mathbf{x} - i\mathbf{y}) \rangle\end{aligned}$$

implies

$$\begin{aligned}4 |\langle A\mathbf{x}, \mathbf{y} \rangle| &\leq \nu(A)(\|\mathbf{x} + \mathbf{y}\|_2^2 + \|\mathbf{x} - \mathbf{y}\|_2^2) \\ &\quad + \nu(A)(\|\mathbf{x} + i\mathbf{y}\|_2^2 + \|\mathbf{x} - i\mathbf{y}\|_2^2) \\ &= 4\nu(A)(\|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2),\end{aligned}$$

where the last step follows from the parallelogram law. The choices of $\|\mathbf{x}\|_2 = 1$ and $\mathbf{y} = \frac{A\mathbf{x}}{\|A\mathbf{x}\|_2}$ yield

$$\|A\|_2 = \max_{\|\mathbf{x}\|_2=1} \langle A\mathbf{x}, \mathbf{y} \rangle \leq \nu(A)(\|\mathbf{x}\|_2 + \|\mathbf{y}\|_2) = 2\nu(A). \quad \square$$

C.7.5 Matrix powers

Theorem C.326. Let A be a square matrix of finite size. The sequence $(A^n)_{n \in \mathbb{N}}$ converges entry-by-entry to the zero matrix if and only if the spectral radius $\rho(A) < 1$.

Proof. For necessity, let $\mathbf{x}$ be a unit eigenvector associated with an eigenvalue λ of maximum modulus. Then we have

$$A^n \mathbf{x} = \lambda^n \mathbf{x} \implies \lim_{n \rightarrow \infty} |\lambda^n| = \lim_{n \rightarrow \infty} \|A^n \mathbf{x}\|_2 = 0$$

from $\lim_{n \rightarrow \infty} A^n = \mathbf{0}$. This implies $\rho(A) = |\lambda| < 1$.

For sufficiency, we start with the Jordan canonical form (C.58). To prove $\lim_{n \rightarrow \infty} A^n = \mathbf{0}$, it suffices to show that $\lim_{n \rightarrow \infty} J^n = \mathbf{0}$. Since J^n has the same block form as that of J , it suffices to show that each Jordan block $J_i = \lambda_i I + S_i$ satisfies $\lim_{n \rightarrow \infty} J_i^n = \mathbf{0}$. Denote by ℓ_i the index of the nilpotent matrix S_i , i.e., $\forall m \geq \ell_i, S_i^m = \mathbf{0}$. Therefore,

$$J_i^n = \sum_{j=0}^{\ell_i-1} \frac{n!}{j!(n-j)!} \lambda_i^{n-j} S_i^j.$$

The triangle inequality and the condition $\rho(A) < 1$ yield

$$\begin{aligned}\lim_{n \rightarrow \infty} \|J_i^n\| &= \lim_{n \rightarrow \infty} \left\| \sum_{j=0}^{\ell_i-1} \frac{n!}{j!(n-j)!} \lambda_i^{n-j} S_i^j \right\| \\ &\leq \lim_{n \rightarrow \infty} \sum_{j=0}^{\ell_i-1} \frac{n!}{j!(n-j)!} |\lambda_i|^{n-j} \|S_i^j\| = 0,\end{aligned}$$

where the last equality follows from $|\lambda_i| < 1$ and the fact that there are only a finite number of terms in the summation. $\square$

Lemma C.327. Let A be a square matrix of finite dimension. If $\rho(A) < 1$, then $I - A$ is nonsingular.

Exercise C.328. Prove Lemma C.327.

Theorem C.329. Let A be a square matrix of finite size. The series $\sum_{k=0}^{+\infty} A^k$ converges if and only if $\rho(A) < 1$. Also,

$$\sum_{k=0}^{+\infty} A^k = (I - A)^{-1}. \quad (\text{C.130})$$

Proof. In finite-dimensional normed spaces, a Cauchy sequence is equivalent to a convergent sequence. If $\sum_{k=0}^{+\infty} A^k$ converges, the sequence $(\sum_{k=0}^n A^k)_{n \in \mathbb{N}^+}$ is Cauchy and $\lim_{n \rightarrow \infty} \|A^n\| = 0$. Then Theorem C.326 yields $\rho(A) < 1$.

Conversely, $\rho(A) < 1$ and Lemma C.327 imply that $I - A$ is invertible. The equality

$$I - A^{k+1} = (I - A)(I + A + \dots + A^k)$$

implies

$$(I - A)^{-1} \lim_{n \rightarrow \infty} (I - A^{n+1}) = \lim_{n \rightarrow \infty} \sum_{k=0}^n A^k.$$

Then Theorem C.326 yields (C.130). $\square$

C.7.6 The matrix exponential

Definition C.330. The *matrix exponential* e^A of a complex matrix $A \in \mathbb{C}^{n \times n}$ is

$$e^A := \sum_{N=0}^{\infty} \frac{1}{N!} A^N = I + A + \frac{1}{2!} A^2 + \dots \quad (\text{C.131})$$

Lemma C.331. The series in (C.131) is *entry-by-entry* convergent.

Proof. Apply the Hilbert–Schmidt norm in Definition C.296 to (C.131) and we have

$$|e^A| = \left| \sum_{N=0}^{\infty} \frac{1}{N!} A^N \right| \leq \sum_{N=0}^{\infty} \frac{1}{N!} |A^N| \leq \sum_{N=0}^{\infty} \frac{1}{N!} |A|^N,$$

where the second inequality follows from Corollary C.300. Since the size of A is fixed, $|A|$ is a bounded nonnegative number. Then the ratio test (Theorem B.48) implies the convergence of the last series in the Hilbert–Schmidt norm, which, by (C.111), is equivalent to the entry-by-entry convergence. $\square$

Theorem C.332. Interpreted as a map $f : \mathbb{R}^{2n^2} \rightarrow \mathbb{R}^{2n^2}$, the matrix exponential $A \mapsto e^A$ is $\mathcal{C}^\infty$. In other words, every partial derivative of f to any order is entry-by-entry uniformly convergent to some continuous function.

Proof. Denote by E_j , $j = 1, \dots, 2n^2$, an n -by- n complex matrix that has 1 or $\mathbf{i}$ in one entry and 0 in all other entries. By Definition C.360, the partial derivative of f in the direction of E_j is

$$\frac{\partial f}{\partial E_j}(A) = \left. \frac{d}{dt} f(A + tE_j) \right|_{t=0}.$$

By (C.131) and Definition B.38, the sequence associated with the series $f(A)$ is $(\frac{1}{N!} A^N)_{N \in \mathbb{N}}$. Hence $\frac{\partial f}{\partial E_j}(A)$ is the sum of derivatives of all terms in the sequence; by the derivative operation as in Theorem B.85, each derivative is of the form

$$\frac{1}{N!} \sum_{i=1}^N g_1(A) \cdots g_{i-1}(A) \cdot \left. \frac{d}{dt} g_i(A + tE_j) \right|_{t=0} \cdot g_{i+1}(A) \cdots g_N(A),$$

where each g_i is A . Taking further partial derivatives preserves this general form, except that g_i is either A or E_j and that the number of products to be summed up is increased.

The k th-order partial derivative of the N th term $\frac{1}{N!} A^N$ in the sequence is a sum of N^k products, each product consisting of N terms and each term is either A or E_j satisfying

$$\max(|A|, |E_j|) \leq M := \max(|A|, 1).$$

Hence we have, for any fixed $k \in \mathbb{N}$,

$$\left| \frac{\partial^k}{\partial E_{j_1} \cdots \partial E_{j_k}} \left(\frac{1}{N!} A^N \right) \right| \leq \frac{N^k M^N}{N!}.$$

By the ratio test and Theorem E.38, the series $\frac{\partial^k f}{\partial E_{j_1} \cdots \partial E_{j_k}}(A)$ uniformly converges entry-by-entry to some function. The rest of the proof follows from Theorem D.181 and Lemma C.331. $\square$

Example C.333. For the real skew-symmetric matrix

$$A = \begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix},$$

we have

$$e^A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Indeed, define

$$I_2 := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad J_2 := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and we have

$$\begin{aligned} A^{4n} &= \theta^{4n} I_2, \quad A^{4n+2} = -\theta^{4n+2} I_2, \\ A^{4n+1} &= \theta^{4n+1} J_2, \quad A^{4n+3} = -\theta^{4n+3} J_2. \end{aligned}$$

It follows that $e^A = \cos \theta I_2 + \sin \theta J_2$.

Theorem C.334. The exponential of a skew-Hermitian matrix is a unitary matrix. Conversely, any unitary matrix must be the exponential of some skew-Hermitian matrix.

Exercise C.335. Prove Theorem C.334.

Lemma C.336. If two matrices X and Y commute, then

$$e^X e^Y = e^{X+Y}. \quad (\text{C.132})$$

Proof. By rearranging double summations, we have

$$\begin{aligned} e^X e^Y &= \left(\sum_{r=0}^{\infty} \frac{1}{r!} X^r \right) \left(\sum_{s=0}^{\infty} \frac{1}{s!} Y^s \right) = \sum_{r,s \in \mathbb{N}} \frac{1}{r!s!} X^r Y^s \\ &= \sum_{N=0}^{\infty} \sum_{k=0}^N \frac{X^k Y^{N-k}}{k!(N-k)!} = \sum_{N=0}^{\infty} \frac{1}{N!} \sum_{k=0}^N \binom{N}{k} X^k Y^{N-k} \\ &= \sum_{N=0}^{\infty} \frac{1}{N!} (X+Y)^N = e^{X+Y}, \end{aligned}$$

where the commutativity of X and Y ensures the validity of the last two steps. $\square$

Example C.337. If two matrices X and Y do not commute, then Lemma C.336 does not hold, e.g.,

$$X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Corollary C.338. The matrix e^X is nonsingular for any $X \in \mathbb{C}^{n \times n}$.

Proof. This follows from setting $Y = -X$ in (C.132) and taking determinant of both sides. $\square$

Lemma C.339. $\frac{d}{dt}(e^{tX}) = X e^{tX}$.

Proof. By (C.131) and Theorem C.332, we have

$$\begin{aligned} \frac{d}{dt}(e^{tX}) &= \frac{d}{dt} \sum_{N=0}^{\infty} \frac{1}{N!} (tX)^N = X \sum_{N=1}^{\infty} \frac{1}{(N-1)!} (tX)^{N-1} \\ &= X e^{tX}. \end{aligned} \quad \square$$

Lemma C.340. For any nonsingular matrix W , we have

$$e^{W^{-1}XW} = W^{-1}e^X W. \quad (\text{C.133})$$

Proof. By (C.131), we have

$$\begin{aligned} e^{W^{-1}XW} &= \sum_{N=0}^{\infty} \frac{1}{N!} (W^{-1}XW)^N = \sum_{N=0}^{\infty} \frac{1}{N!} W^{-1}X^N W \\ &= W^{-1}e^X W. \end{aligned} \quad \square$$

Corollary C.341. For a diagonalizable matrix $A = RAR^{-1}$, we have

$$e^A = R e^A R^{-1}. \quad (\text{C.134})$$

Proof. This follows directly from Lemma C.340. $\square$

Lemma C.342. If $\lambda_1, \dots, \lambda_n$ are eigenvalues of $A \in \mathbb{C}^{n \times n}$, then $e^{\lambda_1}, \dots, e^{\lambda_n}$ are eigenvalues of e^A . Furthermore, if $\mathbf{u}$ is an eigenvector of A for λ_i , then $\mathbf{u}$ is an eigenvector of e^A for e^{λ_i} .

Proof. By the Schur Theorem C.205, there exist an invertible matrix P and an upper-triangular matrix T such that

$$A = P^{-1}TP.$$

Then Lemma C.340 yields

$$e^A = e^{P^{-1}TP} = P^{-1}e^T P,$$

where e^T , by Definition C.330, is an upper-triangular matrix with its diagonal entries as $e^{\lambda_1}, \dots, e^{\lambda_n}$. If $\mathbf{u}$ is an eigenvector of A for the eigenvalue λ , then $\mathbf{u}$ is an eigenvector of A^n for the eigenvalue λ^n , the rest follows from Definition C.330. $\square$

Theorem C.343. $\det e^X = e^{\text{Trace } X}$.

Exercise C.344. Prove Theorem C.343.

C.8 Vector functions $\mathbb{R}^n \rightarrow \mathbb{R}^m$

C.8.1 Differentiation

Lemma C.345. For $f : E \rightarrow \mathbb{R}$ with $E \subset \mathbb{R}$, $x_0 \in E$, and $L \in \mathbb{R}$, the following two statements are equivalent,

- (a) f is differentiable at x_0 and $f'(x_0) = L$;
- (b) $\lim_{x \rightarrow x_0, x \in E \setminus \{x_0\}} \frac{|f(x) - f(x_0) - L(x - x_0)|}{|x - x_0|} = 0$.

Exercise C.346. Prove Lemma C.345.

Definition C.347. The *limit* of a function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ exists at $\mathbf{x}_0 \in E$ iff

$$\begin{aligned} &\forall \epsilon > 0, \exists \delta > 0, \text{ s.t.} \\ &\forall \mathbf{x} \in E \text{ with } 0 < |\mathbf{x} - \mathbf{x}_0| < \delta, |\mathbf{f}(\mathbf{x}) - \mathbf{l}| < \epsilon; \end{aligned} \quad (\text{C.135})$$

then we also write $\lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} \mathbf{f}(\mathbf{x}) = \mathbf{l}$.

Lemma C.348. For $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ and $\mathbf{x}_0 \in E$, we have

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} \mathbf{f}(\mathbf{x}) = \mathbf{l} \iff \lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} |\mathbf{f}(\mathbf{x}) - \mathbf{l}| = 0.$$

Proof. This follows directly from Definition C.347. $\square$

Lemma C.349. The limit of $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ at any $\mathbf{x}_0 \in E$ is unique.

Proof. The proof is similar to that of Lemma B.6. $\square$

Lemma C.350. Suppose $\mathbf{f}, \mathbf{g} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ and $\mathbf{x}_0 \in E$ satisfy

$$\begin{cases} \lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} \mathbf{f}(\mathbf{x}) = \mathbf{0}, \\ \exists r > 0 \text{ s.t. } \forall \mathbf{x} \in E \text{ with } 0 < |\mathbf{x} - \mathbf{x}_0| < r, |\mathbf{g}(\mathbf{x})| \leq |\mathbf{f}(\mathbf{x})|. \end{cases}$$

Then the limit of $\mathbf{g}(\mathbf{x})$ exists at $\mathbf{x}_0$, and

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} \mathbf{g}(\mathbf{x}) = \mathbf{0}.$$

Proof. From (C.135), setting $\delta' = \min\{r, \delta\}$ yields

$$\forall \mathbf{x} \in E \text{ with } 0 < |\mathbf{x} - \mathbf{x}_0| < \delta', |\mathbf{g}(\mathbf{x})| \leq |\mathbf{f}(\mathbf{x})| < \epsilon.$$

Then Definition C.347 completes the proof. $\square$

Definition C.351. A function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ is *continuous* at $\mathbf{x}_0 \in E$ iff

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} \mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{x}_0). \quad (\text{C.136})$$

Definition C.352 (Total derivative). A function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ is (totally) differentiable at $\mathbf{x}_0 \in E$ iff there exists an $\mathbb{R}$ -linear map $L \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ such that

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0, \mathbf{x} \in E \setminus \{\mathbf{x}_0\}} \frac{|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)|}{|\mathbf{x} - \mathbf{x}_0|} = 0; \quad (\text{C.137})$$

then L is written $\mathbf{f}'(\mathbf{x}_0)$ and called the *total derivative* of $\mathbf{f}$.

Example C.353. For $\mathbf{f}, L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$\mathbf{f}(x, y) := (x^2, y^2), \quad L(x, y) := (2x, 4y), \quad (\text{C.138})$$

$\mathbf{f}$ is differentiable at $\mathbf{x}_0 := (1, 2)$ with $\mathbf{f}'(\mathbf{x}_0) = L(\mathbf{x}_0)$ since

$$\begin{aligned} & \lim_{\substack{(x,y) \rightarrow (1,2) \\ (x,y) \neq (1,2)}} \frac{|\mathbf{f}(x,y) - \mathbf{f}(1,2) - L((x,y) - (1,2))|}{|(x,y) - (1,2)|} \\ &= \lim_{\substack{(a,b) \rightarrow (0,0) \\ (a,b) \neq (0,0)}} \frac{|\mathbf{f}(1+a, 2+b) - \mathbf{f}(1,2) - L(a,b)|}{|(a,b)|} \\ &= \lim_{\substack{(a,b) \rightarrow (0,0) \\ (a,b) \neq (0,0)}} \frac{|((1+a)^2, (2+b)^2) - (1,4) - (2a, 4b)|}{|(a,b)|} \\ &= \lim_{\substack{(a,b) \rightarrow (0,0) \\ (a,b) \neq (0,0)}} \frac{|(a^2, b^2)|}{|(a,b)|} \leq \lim_{\substack{(a,b) \rightarrow (0,0) \\ (a,b) \neq (0,0)}} \left(\frac{|(a^2, 0)|}{|(a,b)|} + \frac{|(0, b^2)|}{|(a,b)|} \right) \\ &= \lim_{\substack{(a,b) \rightarrow (0,0) \\ (a,b) \neq (0,0)}} \sqrt{a^2 + b^2} = 0, \end{aligned}$$

where the inequality follows from Lemma C.350.

Lemma C.354. Suppose a function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subseteq \mathbb{R}^n$, is differentiable at an interior point $\mathbf{x}_0$ of E with derivative L_1 and also differentiable at $\mathbf{x}_0$ with derivative L_2 . Then $L_1 = L_2$.

Exercise C.355. Prove Lemma C.354.

Definition C.356 (Directional derivative). A function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subseteq \mathbb{R}^n$ is differentiable in the direction $\mathbf{v} \in \mathbb{R}^n$ at an interior point $\mathbf{x}_0$ of E iff the limit $\lim_{t \rightarrow 0; t \neq 0, \mathbf{x}_0 + t\mathbf{v} \in E} \frac{\mathbf{f}(\mathbf{x}_0 + t\mathbf{v}) - \mathbf{f}(\mathbf{x}_0)}{t}$ exists; then we write

$$D_{\mathbf{v}}\mathbf{f}(\mathbf{x}_0) := \lim_{t \rightarrow 0; t \neq 0, \mathbf{x}_0 + t\mathbf{v} \in E} \frac{\mathbf{f}(\mathbf{x}_0 + t\mathbf{v}) - \mathbf{f}(\mathbf{x}_0)}{t}. \quad (\text{C.139})$$

Example C.357. For $\mathbf{v} = (3, 4)$ and $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined in (C.138), we have $D_{\mathbf{v}}\mathbf{f}(1, 2) = (6, 16)$.

Example C.358. For $f : \mathbb{R} \rightarrow \mathbb{R}$, $D_{+1}f(x)$ is the right derivative of f at x (if it exists), and similarly $D_{-1}f(x)$ is the left derivative of f at x (if it exists), cf. Definition B.82.

Lemma C.359. If a function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subseteq \mathbb{R}^n$ is differentiable at an interior point $\mathbf{x}_0$ of E , then it is differentiable in any direction $\mathbf{v} \in \mathbb{R}^n$ at $\mathbf{x}_0$ and

$$D_{\mathbf{v}}\mathbf{f}(\mathbf{x}_0) = \mathbf{f}'(\mathbf{x}_0)\mathbf{v}. \quad (\text{C.140})$$

Proof. Since $\mathbf{f}$ is differentiable at $\mathbf{x}_0$, Definitions C.347 and C.352 yield

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall 0 < t < \frac{\delta}{|\mathbf{v}|} \text{ with } \mathbf{x}_0 + t\mathbf{v} \in E, \\ & \frac{|\mathbf{f}(\mathbf{x}_0 + t\mathbf{v}) - \mathbf{f}(\mathbf{x}_0) - \mathbf{f}'(\mathbf{x}_0)(\mathbf{x}_0 + t\mathbf{v} - \mathbf{x}_0)|}{|\mathbf{x}_0 + t\mathbf{v} - \mathbf{x}_0|} < \frac{\epsilon}{|\mathbf{v}|}, \end{aligned}$$

which simplifies to $\frac{|\mathbf{f}(\mathbf{x}_0 + t\mathbf{v}) - \mathbf{f}(\mathbf{x}_0) - t\mathbf{f}'(\mathbf{x}_0)\mathbf{v}|}{|t|} < \epsilon$. Then Definition C.347 gives

$$\lim_{t \rightarrow 0; t \neq 0, \mathbf{x}_0 + t\mathbf{v} \in E} \frac{\mathbf{f}(\mathbf{x}_0 + t\mathbf{v}) - \mathbf{f}(\mathbf{x}_0)}{t} = \mathbf{f}'(\mathbf{x}_0)\mathbf{v}$$

and the proof is completed by Definition C.356. $\square$

Definition C.360 (Partial derivative). The j th partial derivative of a function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subseteq \mathbb{R}^n$ at an interior point $\mathbf{x}_0$ of E is the directional derivative of $\mathbf{f}$ at $\mathbf{x}_0$ in the direction $\mathbf{e}_j$, the j th standard basis vector of $\mathbb{R}^n$, i.e.,

$$\frac{\partial \mathbf{f}}{\partial x_j}(\mathbf{x}_0) := D_{\mathbf{e}_j}\mathbf{f}(\mathbf{x}_0). \quad (\text{C.141})$$

Definition C.361. The *gradient* of a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is

$$\nabla f := \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)^\top. \quad (\text{C.142})$$

Corollary C.362. A differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies $D_{\mathbf{v}}f(\mathbf{x}_0) = \mathbf{v} \cdot \nabla f(\mathbf{x}_0)$.

Proof. Write $\mathbf{v} = (v_1, v_2, \dots, v_n)^\top = \sum_{j=1}^n v_j \mathbf{e}_j$. Then

$$\begin{aligned} D_{\mathbf{v}}f(\mathbf{x}_0) &= f'(\mathbf{x}_0)\mathbf{v} = f'(\mathbf{x}_0) \sum_{j=1}^n v_j \mathbf{e}_j = \sum_{j=1}^n v_j f'(\mathbf{x}_0) \mathbf{e}_j \\ &= \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0) = \mathbf{v} \cdot \nabla f(\mathbf{x}_0), \end{aligned} \quad (\text{C.143})$$

where the first step follows from Lemma C.359, the third step from $f'(\mathbf{x}_0)$ being a linear map, and the fourth step from (C.141) and (C.140). $\square$

Exercise C.363. Show that the existence of partial derivatives at $\mathbf{x}_0$ does not imply that the function is differentiable at $\mathbf{x}_0$ by considering the differentiability of the following function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at $(0, 0)$:

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0); \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Definition C.364. The *derivative matrix* or *differential matrix* or *Jacobian matrix* of a differentiable function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a $m \times n$ matrix,

$$D\mathbf{f} := \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}. \quad (\text{C.144})$$

Definition C.365. A differentiable function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ open is *continuously differentiable*, written $\mathbf{f} \in \mathcal{C}^1(E)$, iff the map $\mathbf{f}' : E \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous.

Lemma C.366. A function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ has a limit (or is continuous/differentiable) at $\mathbf{x}_0 \in E$ if and only if f_i has a limit (or is continuous/differentiable) at $\mathbf{x}_0$ for each $i = 1, 2, \dots, m$.

Proof. This follows from Lemma C.350, Definitions C.347, C.351, C.352 and

$$\forall j = 1, \dots, m, |f_j(\mathbf{x})| \leq |\mathbf{f}(\mathbf{x})| = \left(\sum_{i=1}^m |f_i(\mathbf{x})|^2 \right)^{\frac{1}{2}}. \quad \square$$

Theorem C.367. A function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ is differentiable at an interior point $\mathbf{x}_0$ of E if and only if

$$\exists L \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) \text{ s.t. } \forall \mathbf{h} \in \mathbb{R}^n, \quad \mathbf{f}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f}(\mathbf{x}_0) = L\mathbf{h} + \sum_{j=1}^n \mathbf{b}_j(\mathbf{h})h_j, \quad (\text{C.145})$$

where the function $\mathbf{b}_j : \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies

$$\forall j = 1, 2, \dots, n, \quad \lim_{\mathbf{h} \rightarrow \mathbf{0}} \mathbf{b}_j(\mathbf{h}) = \mathbf{0}; \quad (\text{C.146})$$

then $L = \mathbf{f}'(\mathbf{x}_0)$.

Proof. For necessity, we define

$$\begin{aligned} \forall \mathbf{h} \in \mathbb{R}^n, \quad \mathbf{E}(\mathbf{h}) &:= \mathbf{f}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f}(\mathbf{x}_0) - \mathbf{f}'(\mathbf{x}_0)\mathbf{h}, \\ \forall j = 1, 2, \dots, n, \quad \mathbf{b}_j(\mathbf{h}) &:= \frac{h_j}{|\mathbf{h}|^2} \mathbf{E}(\mathbf{h}). \end{aligned}$$

Then (C.145) follows from setting $L = \mathbf{f}'(\mathbf{x}_0)$ and (C.146) follows from $|\mathbf{b}_j(\mathbf{h})| = \frac{|h_j|}{|\mathbf{h}|} \cdot \frac{|\mathbf{E}(\mathbf{h})|}{|\mathbf{h}|} \leq \frac{|\mathbf{E}(\mathbf{h})|}{|\mathbf{h}|}$, (C.137), and Lemma C.350.

For sufficiency, we have

$$\begin{aligned} \frac{|\mathbf{f}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f}(\mathbf{x}_0) - L\mathbf{h}|}{|\mathbf{h}|} &= \frac{1}{|\mathbf{h}|} \left| \sum_{j=1}^n \mathbf{b}_j(\mathbf{h})h_j \right| \\ &\leq \sum_{j=1}^n |\mathbf{b}_j(\mathbf{h})| \frac{|h_j|}{|\mathbf{h}|} \leq \sum_{j=1}^n |\mathbf{b}_j(\mathbf{h})|, \end{aligned}$$

where the first step follows from (C.145) and the second from (NRM-3,4) in Definition C.182. Then the proof is completed by Lemma C.350 and Definition C.347. $\square$

Theorem C.368. A function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ with $E \subset \mathbb{R}^n$ being an open subset is continuously differentiable if and only if the partial derivatives $\frac{\partial f_i}{\partial x_j}$ exist and are continuous on E for each $i = 1, \dots, m$ and $j = 1, \dots, n$. In this case, the linear map $\mathbf{f}'(\mathbf{x}_0) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is given by

$$\mathbf{f}'(\mathbf{x}_0)\mathbf{v} = \sum_{j=1}^n v_j \frac{\partial \mathbf{f}}{\partial x_j}(\mathbf{x}_0). \quad (\text{C.147})$$

Proof. Let $\mathbf{x}_0 \in E$. For necessity, we fix $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$. Then the existence of $\frac{\partial f_i}{\partial x_j}(\mathbf{x}_0)$ is guaranteed by Lemma C.359 and Definitions C.360 and C.365. The Hilbert–Schmidt norm in Definition C.296 implies

$$\begin{aligned} \forall \mathbf{h} \in \mathbb{R}^n \text{ with } \mathbf{x}_0 + \mathbf{h} \in E, \\ \left| \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0 + \mathbf{h}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| \leq |\mathbf{f}'(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f}'(\mathbf{x}_0)|. \end{aligned}$$

Hence Definition C.351, the continuity of $\mathbf{f}'$ at $\mathbf{x}_0$, and Lemma C.350 imply the continuity of $\frac{\partial f_i}{\partial x_j}$ at $\mathbf{x}_0$.

For sufficiency, we first prove that $\mathbf{f}$ is differentiable at $\mathbf{x}_0$ by an induction on n . By Lemma C.366, we may assume that $m = 1$. Then the induction hypothesis ($n = 1$) follows from Definition B.81. Suppose that $\mathbf{f}$ is differentiable at $\mathbf{x}_0$ for $n - 1$. Then for n , we have

$$\begin{aligned} \forall \mathbf{h} \in \mathbb{R}^n \text{ with } \mathbf{x}_0 + \mathbf{h} \in E, \\ (\triangle) : \quad f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) = K_1 + K_2, \end{aligned}$$

where $\mathbf{x}_0 := (x_1, \dots, x_n)^\top$ and

$$\begin{aligned} K_1 &:= f(x_1 + h_1, \dots, x_{n-1} + h_{n-1}, x_n + h_n) - \\ &\quad f(x_1 + h_1, \dots, x_{n-1} + h_{n-1}, x_n), \\ K_2 &:= f(x_1 + h_1, \dots, x_{n-1} + h_{n-1}, x_n) - f(x_1, x_2, \dots, x_n). \end{aligned}$$

The mean value theorem (Theorem B.92) implies

$$(*) : \exists \theta \in (0, 1) \text{ s.t. } K_1 = \frac{\partial f}{\partial x_n}(\mathbf{x}_\theta)h_n = \frac{\partial f}{\partial x_n}(\mathbf{x}_0)h_n + b_n(\mathbf{h})h_n,$$

where $\mathbf{x}_\theta := (x_1 + h_1, \dots, x_{n-1} + h_{n-1}, x_n + \theta h_n)^\top$ and $b_n(\mathbf{h}) := \frac{\partial f}{\partial x_n}(\mathbf{x}_\theta) - \frac{\partial f}{\partial x_n}(\mathbf{x}_0)$. Then the continuity of $\frac{\partial f}{\partial x_n}$ at $\mathbf{x}_0$ and Definition C.351 imply $b_n(\mathbf{h}) \rightarrow 0$ as $\mathbf{h} \rightarrow \mathbf{0}$. Then the induction hypothesis, Definition C.352, and Theorem C.367 yield

$$(**) : \quad K_2 = \sum_{j=1}^{n-1} h_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0) + \sum_{j=1}^{n-1} b_j(\mathbf{h})h_j,$$

where $b_j(\mathbf{h}) = b_j(h_1, \dots, h_{n-1}) \rightarrow 0$ as $\mathbf{h} \rightarrow \mathbf{0}$ for each $j = 1, 2, \dots, n - 1$. Then the differentiability of f at $\mathbf{x}_0$ follows from (*), (**), ($\triangle$), and Theorem C.367.

The definition of the Hilbert–Schmidt norm gives

$\forall \mathbf{h} \in \mathbb{R}^n$ with $\mathbf{x}_0 + \mathbf{h} \in E$,

$$|\mathbf{f}'(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f}'(\mathbf{x}_0)|^2 = \sum_{i=1}^n \sum_{j=1}^m \left| \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0 + \mathbf{h}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right|^2.$$

The continuity of $\frac{\partial f_i}{\partial x_j}$ and Definition C.351 implies the continuity of $\mathbf{f}'$ at $\mathbf{x}_0$. Then $\mathbf{f}' \in \mathcal{C}^1(E)$ follows from Definition C.365 and $\mathbf{x}_0$ being arbitrary.

Finally, (C.147) follows from the second, the third, and the fourth equalities of (C.143). $\square$

Theorem C.369 (Chain rule for vector functions). Consider two functions $\mathbf{g} : E \rightarrow \mathbb{R}^p$ with $E \subset \mathbb{R}^n$ open and $\mathbf{f} : F \rightarrow \mathbb{R}^m$ with $F \supset \mathbf{g}(E)$ open. If $\mathbf{g}$ is differentiable at $\mathbf{x}_0 \in E$ and $\mathbf{f}$ is differentiable at $\mathbf{g}(\mathbf{x}_0)$, then $\mathbf{f} \circ \mathbf{g} : E \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{x}_0$ and

$$(\mathbf{f} \circ \mathbf{g})'(\mathbf{x}_0) = \mathbf{f}'(\mathbf{g}(\mathbf{x}_0))\mathbf{g}'(\mathbf{x}_0). \quad (\text{C.148})$$

Proof. Write $\mathbf{y}_0 := \mathbf{g}(\mathbf{x}_0) \in \mathbb{R}^p$, $A := \mathbf{f}'(\mathbf{y}_0) \in \mathcal{L}(\mathbb{R}^p, \mathbb{R}^m)$ and $B := \mathbf{g}'(\mathbf{x}_0) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^p)$. Definition C.352 yields

$$\begin{aligned} (*) : \quad &\forall \mathbf{h} \in \mathbb{R}^n \text{ with } \mathbf{x}_0 + \mathbf{h} \in E, \\ &\quad \mathbf{g}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{g}(\mathbf{x}_0) = (B + \mathbf{b}(\mathbf{h}))\mathbf{h}, \\ (**) : \quad &\forall \mathbf{k} \in \mathbb{R}^p \text{ with } \mathbf{y}_0 + \mathbf{k} \in F, \\ &\quad \mathbf{f}(\mathbf{y}_0 + \mathbf{k}) - \mathbf{f}(\mathbf{y}_0) = (A + \mathbf{a}(\mathbf{k}))\mathbf{k}, \end{aligned}$$

where

$$\begin{aligned} (\triangle) : \quad &\mathbf{a} : \mathbb{R}^p \rightarrow \mathbb{R}^m, \mathbf{a}(\mathbf{k}) \rightarrow \mathbf{0}(\mathbf{k} \rightarrow \mathbf{0}), \\ (\square) : \quad &\mathbf{b} : \mathbb{R}^n \rightarrow \mathbb{R}^p, \mathbf{b}(\mathbf{h}) \rightarrow \mathbf{0}(\mathbf{h} \rightarrow \mathbf{0}). \end{aligned}$$

For each $\mathbf{h}$, let $\mathbf{k} = \mathbf{g}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{g}(\mathbf{x}_0)$. Then (*) gives

$$(\diamond) : \quad |\mathbf{k}| \leq |B + \mathbf{b}(\mathbf{h})||\mathbf{h}| \leq (|B| + |\mathbf{b}(\mathbf{h})|)|\mathbf{h}|,$$

where the first step follows from Corollary C.289 and the second from the triangle inequality in Definition C.182. Hence

$$\begin{aligned} &|\mathbf{f} \circ \mathbf{g}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f} \circ \mathbf{g}(\mathbf{x}_0) - AB\mathbf{h}| \\ &= |\mathbf{f}(\mathbf{y}_0 + \mathbf{k}) - \mathbf{f}(\mathbf{y}_0) - AB\mathbf{h}| \\ &= |(A + \mathbf{a}(\mathbf{k}))\mathbf{k} - AB\mathbf{h}| = |A(\mathbf{k} - B\mathbf{h}) + \mathbf{a}(\mathbf{k})\mathbf{k}| \\ &= |A\mathbf{b}(\mathbf{h})\mathbf{h} + \mathbf{a}(\mathbf{k})\mathbf{k}| \\ &\leq |A||\mathbf{b}(\mathbf{h})||\mathbf{h}| + |\mathbf{a}(\mathbf{k})||\mathbf{k}| \\ &\leq |A||\mathbf{b}(\mathbf{h})||\mathbf{h}| + |\mathbf{a}(\mathbf{k})|(|B| + |\mathbf{b}(\mathbf{h})|)|\mathbf{h}| \\ &= (|A||\mathbf{b}(\mathbf{h})| + |\mathbf{a}(\mathbf{k})|(|B| + |\mathbf{b}(\mathbf{h})|))|\mathbf{h}|, \end{aligned}$$

where the first step follows from $\mathbf{y}_0 + \mathbf{k} = \mathbf{g}(\mathbf{x}_0 + \mathbf{h})$, the second from (**), the fourth from (*), the fifth from Corollary C.289 and the triangle inequality in Definition C.182, and the sixth from $(\diamond)$. Lemma C.350, $(\triangle)$, and $(\square)$ imply

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}, \mathbf{x}_0 + \mathbf{h} \in E} \frac{|\mathbf{f} \circ \mathbf{g}(\mathbf{x}_0 + \mathbf{h}) - \mathbf{f} \circ \mathbf{g}(\mathbf{x}_0) - AB\mathbf{h}|}{|\mathbf{h}|} = 0.$$

Then the proof is completed by Definition C.352. $\square$

Theorem C.370 (Vector mean value). Suppose a continuous function $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^m$ is differentiable on (a, b) . Then

$$\exists x \in (a, b) \text{ s.t. } |\mathbf{f}(b) - \mathbf{f}(a)| \leq (b - a)|\mathbf{f}'(x)|. \quad (\text{C.149})$$

Proof. Write $\mathbf{y} := \mathbf{f}(b) - \mathbf{f}(a)$ and define $\varphi : [a, b] \rightarrow \mathbb{R}$ by

$$\varphi(t) := \mathbf{y} \cdot \mathbf{f}(t).$$

The mean value theorem (Theorem B.92) yields

$$\exists x \in [a, b] \text{ s.t. } \varphi(b) - \varphi(a) = \varphi'(x)(b - a) = \mathbf{y} \cdot \mathbf{f}'(x)(b - a).$$

On the other hand, we have

$$\varphi(b) - \varphi(a) = \mathbf{y} \cdot [\mathbf{f}(b) - \mathbf{f}(a)] = |\mathbf{y}|^2.$$

The above two equations and the Cauchy–Schwarz inequality (Theorem C.197) imply

$$|\mathbf{y}|^2 = (b - a)|\mathbf{y} \cdot \mathbf{f}'(x)| \leq (b - a)|\mathbf{y}||\mathbf{f}'(x)|,$$

which yields (D.32). $\square$

C.8.2 Basics of curves and surfaces

Definition C.371. An (open) *curve* in the D -dimensional space $\mathbb{R}^D$ is (the image of) a continuous map $\gamma : (a, b) \rightarrow \mathbb{R}^D$ for some a, b with $-\infty \leq a < b \leq +\infty$, where γ is called the *parametrization* of the curve.

Definition C.372. A curve γ is *simple* if the map γ is injective.

Definition C.373. A *closed curve* is (the image of) a continuous map $\tilde{\gamma} : [0, 1] \rightarrow \mathbb{R}^D$ that satisfies $\tilde{\gamma}(0) = \tilde{\gamma}(1)$. If the restriction of $\tilde{\gamma}$ to $[0, 1)$ is further injective, then the closed curve is a *simple closed curve* or *Jordan curve*.

Definition C.374. A *parametrized differentiable curve* is a curve whose parametrization is differentiable.

Definition C.375. A curve $\alpha : (a, b) \rightarrow \mathbb{C}$ is *smooth* iff $\alpha(t)$ has derivatives of all orders for all $t \in (a, b)$.

Definition C.376. A curve $\alpha : (a, b) \rightarrow \mathbb{C}$ is *piecewise smooth* iff there is a partition $a = a_0 < a_1 < \dots < a_n = b$ such that each of the restrictions

$$\forall j = 0, 1, \dots, n-1, \quad \alpha_j := \alpha|_{[a_j, a_{j+1}]} \quad (\text{C.150})$$

is smooth.

Definition C.377. The *tangent vector* of a parametrized differentiable curve $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{R}^n$ at the point $p = \gamma(0)$ is its differential $d\gamma_p = \gamma'(0)$

$$\gamma' := \frac{d\gamma}{dt} \quad (\text{C.151})$$

where the differentiation operates on each of its n components. The *unit tangent vector* of γ , denoted by $\mathbf{t}$, is the normalization of its tangent vector.

Definition C.378. A *unit-speed curve* is a curve whose tangent vector has unit length at each of its points.

Definition C.379. A point $\gamma(t_0)$ is a *regular point* of a curve γ if $\mathbf{t}(t_0)$ exists and $\mathbf{t}(t_0) \neq \mathbf{0}$ holds; a curve is *regular* if all of its points are regular.

Definition C.380. The *signed unit normal* of a curve, denoted by $\mathbf{n}_s$, is the unit vector obtained by rotating its unit tangent vector counterclockwise by $\frac{\pi}{2}$.

Definition C.381. The *signed curvature* of a unit-speed curve γ is $\kappa_s := \gamma'' \cdot \mathbf{n}_s$.

Definition C.382. The *arc length* of a regular curve starting at the point $\gamma(t_0)$ is defined as

$$s_\gamma(t) = \int_{t_0}^t \|\gamma'(u)\|_2 du. \quad (\text{C.152})$$

Definition C.383. A map $X \mapsto Y$ is a *homeomorphism* if it is continuous and bijective and its inverse is also continuous; then the two sets X and Y are said to be *homeomorphic*.

Definition C.384. A curve $\tilde{\gamma} : (\tilde{a}, \tilde{b}) \rightarrow \mathbb{R}^n$ is a *reparametrization* of another curve $\gamma : (a, b) \rightarrow \mathbb{R}^n$ if there exists a homeomorphism $\phi : (\tilde{a}, \tilde{b}) \rightarrow (a, b)$ such that $\tilde{\gamma}(\tilde{t}) = \gamma(\phi(\tilde{t}))$ for each $\tilde{t} \in (\tilde{a}, \tilde{b})$.

Lemma C.385. A reparametrization of a regular curve is unit-speed if and only if it is based on the arc length.

Example C.386. The spiral $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$ given by

$$\gamma(t) = (e^t \cos t, e^t \sin t) \quad (\text{C.153})$$

is a curve. Its tangent vector is

$$\gamma'(t) = (e^t(\cos t - \sin t), e^t(\cos t + \sin t)) \quad (\text{C.154})$$

and thus the modulus of the tangent vector is

$$\|\gamma'(t)\|_2 = \sqrt{2}e^t.$$

By Definition C.382, the arc length of the spiral is

$$s(t) = \int_0^t e^\tau \sqrt{2} d\tau = \sqrt{2}(e^t - 1) \quad (\text{C.155})$$

and we have $t = \ln(\frac{s}{\sqrt{2}} + 1)$. According to Lemma C.385, the spiral can be expressed as a unit-speed curve

$$\gamma(s) = \left(\frac{s}{\sqrt{2}} + 1 \right) \left(\cos \left(\ln \left(\frac{s}{\sqrt{2}} + 1 \right) \right), \sin \left(\ln \left(\frac{s}{\sqrt{2}} + 1 \right) \right) \right). \quad (\text{C.156})$$

Despite of its complicated form, the parametrization of the spiral in (C.156) makes the curve unit-speed; this is a prominent advantage over the parametrization in (C.153).

Definition C.387. A *surface patch* is (the image of) a continuous map $(0, 1)^2 \rightarrow \mathbb{R}^3$.

Definition C.388 (Regular surface). A subset $\mathcal{S} \subset \mathbb{R}^3$ is called a *regular surface* iff for each $p \in \mathcal{S}$ there exist a neighborhood $V \subset \mathbb{R}^3$, an open set $U \subset \mathbb{R}^2$, and a surjective map $\mathbf{x} : U \rightarrow V \cap \mathcal{S}$ such that

(RSF-1) $\mathbf{x}$ is totally differentiable;

(RSF-2) $\mathbf{x}$ is a homeomorphism;

(RSF-3) For each $q \in U$, the differential $d\mathbf{x}_q : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is injective.

The neighborhood $V \cap \mathcal{S}$ of p in $\mathcal{S}$ is called a *coordinate neighborhood*, for which the map $\mathbf{x}$ is the *parametrization* or the *system of (local) coordinates*.

Exercise C.389. Show that the unit sphere

$$\mathbb{S}^2 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\} \quad (\text{C.157})$$

is a regular surface.

Definition C.390. A *tangent vector to a regular surface $\mathcal{S}$ at a point $p \in \mathcal{S}$* is the tangent vector $w = \alpha'(0)$ of a differentiable parametrized curve $\alpha : (-\epsilon, \epsilon) \rightarrow \mathcal{S}$ satisfying $\alpha(0) = p$.

Lemma C.391. Let $\mathbf{x} : U \subset \mathbb{R}^2 \rightarrow \mathcal{S}$ be a parametrization of a regular surface $\mathcal{S}$ and let $q \in U$. The vector space

$$d\mathbf{x}_q(\mathbb{R}^2) \subset \mathbb{R}^3 \quad (\text{C.158})$$

coincides with the set of tangent vectors to $\mathcal{S}$ at $\mathbf{x}(q)$.

Proof. For the curve $\beta := \mathbf{x}^{-1} \circ \alpha : (-\epsilon, \epsilon) \rightarrow U$, we have $\mathbf{x} \circ \beta = \alpha$. The chain rule in Theorem C.369 gives

$$d\mathbf{x} \circ d\beta = d\alpha, \quad (\text{C.159})$$

which evaluates at q to $\mathbf{w} = \alpha'(0) = d\mathbf{x}_q(\beta'(0))$ with $\beta'(0) \in \mathbb{R}^2$. Hence $\mathbf{w} \in d\mathbf{x}_q(\mathbb{R}^2)$.

On the other hand, for any given vector $\mathbf{w} = d\mathbf{x}_q(\mathbf{v})$, we have $\mathbf{v} \in \mathbb{R}^2$. Then $\mathbf{v}$ can be considered as the tangent vector of the curve $\gamma : (-\epsilon, \epsilon) \rightarrow U$ defined as

$$\gamma(t) = t\mathbf{v} + q. \quad (\text{C.160})$$

Hence the curve $\alpha := \mathbf{x} \circ \gamma$ has the signature $(-\epsilon, \epsilon) \rightarrow \mathcal{S}$. Consequently, $\alpha'(0) = d\mathbf{x}_q \circ \gamma'(0) = \mathbf{w}$, which shows that $\mathbf{w}$ is indeed a tangent vector to $\mathcal{S}$ at q . $\square$

Definition C.392. The *tangent plane to a regular surface $\mathcal{S}$ at $p \in \mathcal{S}$* is the vector subspace $d\mathbf{x}_p(\mathbb{R}^2)$ in (C.158).

Appendix D

Metric Spaces

Motivation D.1. Complex analysis and real analysis are respectively calculus in $\mathbb{C}$ and $\mathbb{R}^n$, which are both finite-dimensional vector spaces. Elements of these spaces are often regarded as passively being acted upon by functions. However, nothing stops us from grouping a family of functions into a space, e.g., $\mathcal{C}[a, b]$ is the vector spaces of continuous functions $[a, b] \rightarrow \mathbb{R}$. A function that acts on a space of functions is called an *operator*. It is then natural to call the study of these function spaces *functional analysis* and the study of linear operators on function spaces *operator theory*. This appendix serves as a foundation for functional analysis.

Functional analysis features frequent and deep utilization of topological and algebraic structures. Accordingly, a solid ground on topology and linear algebra may be helpful. On the other hand, the central concepts of basic analysis such as sequence, limit, convergence, continuity, differentiation, and integration are still the major concerns of functional analysis: they will be re-examined in the context of infinite-dimensional vector spaces.

As the most fundamental difference of functional analysis from basic analysis, elements in a function space have infinite degrees of freedom, as illustrated in Examples D.2 and D.3. Consequently, lemmas and theorems for finite-dimensional spaces may or may not hold for infinite-dimensional spaces. Indeed, it is the generalization of results in basic analysis to function spaces that motivated the study of functional analysis.

Example D.2. In classical mechanics, a physical quantity such as position, momentum, or energy is represented by a tuple of real numbers and thus the variable lies in $\mathbb{R}^n$ where n is a fixed positive integer. In comparison, a physical quantity in quantum mechanics is represented by an *observable*, a self-adjoint linear operator acting on some Hilbert space. An *eigenstate* of an observable is an eigenvector of the operator and a *quantum state* of an observable is a linear combination of the eigenstates, known as *quantum superposition*. The time evolution of a quantum state is governed by the *Schrodinger equation*. When an observable is measured by an experiment, the result will be an eigenvalue of the quantum state with probability given by the Born rule. After the measurement, the quantum state is postulated, in the nondegenerate case, to *collapse* to the eigenvector that corresponds to the observed eigenvalue. Due to the prob-

abilistic nature stemming from measurements, a quantum state cannot be described by a finite degree of freedoms and thus does not lie in any finite-dimensional spaces.

The aforementioned postulations of quantum mechanics come from the need to reconcile the contradiction of classical mechanics to real-world phenomena such as the double-slit experiment. Historically, quantum mechanics has been a major driving force behind the development of functional analysis, which is no exception to the fact that mathematics is developed for the bare necessity to understand the world.

Example D.3. A copper mining company mines in a mountain that has an estimated total amount of Q tonnes of copper. Let $x(t)$ denote the amount of copper removed during the period $[0, t]$, with $x(0) = 0$ and $x(T) = Q$. Assume x is a continuous function $[0, T] \rightarrow \mathbb{R}$ and the cost of extracting copper per unit tonne at time t is

$$c(t) = ax(t) + bx'(t), \quad (\text{D.1})$$

where $a, b \in \mathbb{R}^+$. What is the optimal mining operation $x(t)$ that minimizes the cost function

$$f(x) = \int_0^T (ax(t) + bx'(t))x'(t)dt?$$

In math terms, we would like to minimize $f : \mathcal{C}_Q^1[0, T] \rightarrow \mathbb{R}^+$ where $\mathcal{C}_Q^1[0, T]$ is the set of continuously differentiable functions $x : [0, T] \rightarrow \mathbb{R}$ satisfying $x(0) = 0$ and $x(T) = Q$.

In calculus, the minimizer x_* of a function $f \in C^2$ is usually found by the condition $f'(x_*) = 0$ and $f''(x_*) > 0$. However, the above problem does not fit into the usual framework of calculus, since x is not a number but a function that belongs to an infinite-dimensional function space. Solving this problem requires a number of techniques in functional analysis.

Definition D.4. A *metric* on a set $\mathcal{X}$ is a function $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, +\infty)$ that satisfies, for all $x, y, z \in \mathcal{X}$,

- (1) non-negativity: $d(x, y) \geq 0$;
- (2) identity of indiscernibles: $x = y \iff d(x, y) = 0$;
- (3) symmetry: $d(x, y) = d(y, x)$;
- (4) triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

A *pseudometric* is a function that satisfies (1), (3), and (4).

Example D.5. The function

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y, \\ 0 & \text{if } x = y \end{cases} \quad (\text{D.2})$$

is a metric called the *discrete metric*.

Exercise D.6. Let $\mathcal{X}$ be the set of all bounded and unbounded sequences of complex numbers. Show that the function d given by

$$\forall x = (\xi_j), \forall y = (\eta_j), \quad d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|\xi_j - \eta_j|}{1 + |\xi_j - \eta_j|} \quad (\text{D.3})$$

is a metric on $\mathcal{X}$.

Definition D.7. A *metric/pseudometric space* is an ordered pair $(\mathcal{X}, d)$ where $\mathcal{X}$ is a set and d is a metric/pseudometric on $\mathcal{X}$.

Example D.8. Denote by $\mathcal{C}[a, b]$ the set of bounded continuous functions $[a, b] \rightarrow \mathbb{R}$. The metric space

$$(\mathcal{C}[a, b], d) \text{ with } d(x, y) = \sup_{t \in [a, b]} |x(t) - y(t)| \quad (\text{D.4})$$

was heavily studied in the early history of functional analysis and serves as an excellent pedagogical device for many important concepts. Here d is called a *uniform metric*.

Notation 26. For a metric space $(\mathcal{X}, d_{\mathcal{X}})$, sometimes we use the notation

$$|u - v|_{\mathcal{X}} := d_{\mathcal{X}}(u, v). \quad (\text{D.5})$$

Definition D.9. In a metric space $(\mathcal{X}, d)$, an *open ball* $B_r(x)$ centered at $x \in \mathcal{X}$ with radius r is the point set

$$B_r(x) := \{y \in \mathcal{X} : d(x, y) < r\}. \quad (\text{D.6})$$

Sometimes we also write $B(x, r) := B_r(x)$.

Roadmap D.10. The storyline of this chapter is as follows. Section D.1 concerns complete metric spaces where Cauchy sequences in such a space $\mathcal{X}$ converges in $\mathcal{X}$. The contents of Section D.2 are motivated by a particular family of subsets $\{E : E \subset \mathcal{X}\}$ in $\mathcal{X}$ such that Cauchy sequences in E also converges in E . These sets are called closed sets, from which follows other important concepts such open sets, limit points, and topology. Section D.3 is a long chain of inheritance with various specialized concepts of continuous functions. The contents of Section D.4 are motivated by the quest for a special family of closed sets where significant theorems in calculus can be generalized. As the central result of this quest, the Ascoli–Arzela theorem D.158 gives a sufficient condition on how to improve closed subsets to compact subsets. In Section D.5, Cauchy sequences of functions are studied with the help of Theorem D.158, leading to the celebrated Stone–Weierstrass theorem.

D.1 Complete metric spaces

D.1.1 Completeness for Cauchy sequences

Definition D.11. A sequence $(x_n)_{n \in \mathbb{N}}$ in a metric space $\mathcal{X}$ is said to *converge* in $\mathcal{X}$ iff

$$\exists L \in \mathcal{X}, \forall \epsilon > 0, \exists N \in \mathbb{N}, \text{ s.t. } \forall n > N, |x_n - L|_{\mathcal{X}} < \epsilon, \quad (\text{D.7})$$

where $L := \lim_{n \rightarrow \infty} x_n$ is called the *limit* of the sequence.

Definition D.12. A *Cauchy sequence in a metric space* $(\mathcal{X}, d)$ is a sequence $(x_n) \subset \mathcal{X}$ satisfying

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } m, n > N \implies d(x_m, x_n) < \epsilon. \quad (\text{D.8})$$

Lemma D.13. Convergent sequences in a metric space are Cauchy.

Proof. By Definition D.11, the convergence of (x_i) to x gives

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, d(x_n, x) < \frac{\epsilon}{2}.$$

It follows that $\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m, n > N,$

$$d(x_n, x_m) < d(x_n, x) + d(x_m, x) < \epsilon. \quad \square$$

Definition D.14. A metric space $\mathcal{X}$ is *complete* iff every Cauchy sequence in $\mathcal{X}$ converges to an element in $\mathcal{X}$.

Example D.15 ($\mathbb{Q}$ is not complete). The sequence

$$x_1 = \frac{3}{2}; \quad \forall n > 1, \quad x_n = \frac{4 + 3x_{n-1}}{3 + 2x_{n-1}} \quad (\text{D.9})$$

is bounded below by $\sqrt{2}$ and is monotonically decreasing. By Theorem B.14, (x_n) is convergent in $\mathbb{R}$. However, although (x_n) is Cauchy in $\mathbb{Q}$, it is not convergent in $\mathbb{Q}$ because

$$L = \frac{4 + 3L}{3 + 2L} \implies L = \sqrt{2}.$$

Example D.16. The sequence $(\sum_{k=1}^n \frac{1}{k^k})_{n \in \mathbb{N}}$ is Cauchy, but we do not know yet whether the limit is rational or irrational.

Lemma D.17. The metric space (D.4) is complete.

Proof. By Definition D.12, a Cauchy sequence $(f_n) \subset \mathcal{C}[a, b]$ satisfies

$$(*) : \quad \forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m, n \geq N, \sup_{t \in [a, b]} |f_n(t) - f_m(t)| < \frac{\epsilon}{3}.$$

First, at any fixed $t \in [a, b]$, we can reduce the above Cauchy sequence to a number sequence $(f_n(t)) \subset \mathbb{R}$. The completeness of $\mathbb{R}$ (Theorem B.17) yields $\lim_{n \rightarrow \infty} f_n(t) \in \mathbb{R}$. This process furnishes a function $f : [a, b] \rightarrow \mathbb{R}$ given by

$$f(t) = \lim_{n \rightarrow \infty} f_n(t).$$

Second, we show $f \in \mathcal{C}[a, b]$, i.e., f is continuous. In $(*)$, set $m = N$, let $n \rightarrow \infty$, and we have

$$\forall t \in [a, b], \quad |f_N(t) - f(t)| < \frac{\epsilon}{3}.$$

The condition of $f_N \in \mathcal{C}[a, b]$ implies

$$\forall t \in [a, b], \forall \epsilon > 0, \exists \delta > 0 \text{ s.t.} \\ |t - \tau| < \delta \implies |f_N(t) - f_N(\tau)| < \frac{\epsilon}{3}.$$

The above two equations yield

$$\begin{aligned} \forall t \in [a, b], \forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |t - \tau| < \delta \implies \\ |f(t) - f(\tau)| &\leq |f(t) - f_N(t)| + |f_N(t) - f_N(\tau)| \\ &\quad + |f_N(\tau) - f(\tau)| \\ &\leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon, \end{aligned}$$

which shows that f is continuous at every $t \in [a, b]$.

Finally, we show that $(f_n)_{n \in \mathbb{N}^+}$ indeed converges to f in $\mathcal{C}[a, b]$. Fix $n > N$ in (*) and we have

$$\forall m > N, \forall t \in [a, b], |f_n(t) - f_m(t)| < \epsilon,$$

which implies

$$\forall t \in [a, b], |f_n(t) - f(t)| = \left| f_n(t) - \lim_{m \rightarrow \infty} f_m(t) \right| < \epsilon.$$

Hence $d(f_n, f) = \max_{t \in [a, b]} |f_n(t) - f(t)| < \epsilon$. In the above process, we could have fixed any $n > N$ at the outset to obtain the same result. Therefore we have

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, d(f_n, f) < \epsilon,$$

which, by Definition D.11, implies $\lim_{n \rightarrow \infty} f_n = f$. $\square$

Exercise D.18. The completeness depends on the metric. For the metric $d_1(x, y) = \int_a^b |x(t) - y(t)| dt$, show that the metric space $(\mathcal{C}[a, b], d_1)$ is not complete.

D.1.2 Dense subsets and separable spaces

Definition D.19. A subset A in a metric space $\mathcal{X}$ is *dense* in another subset $E \subseteq \mathcal{X}$ iff

$$\forall u \in E, \forall \epsilon > 0, \exists v \in A \text{ s.t. } |v - u|_{\mathcal{X}} < \epsilon. \quad (\text{D.10})$$

Example D.20. $\mathbb{Q}$ is a dense subset of $\mathbb{R}$ and $\mathbb{R}$ is the completion of $\mathbb{Q}$: each element of $\mathbb{R}$ is constructed as an equivalence class of Cauchy sequences of rational numbers.

Example D.21. The set $[0, 1] \setminus C$ is dense in $[0, 1]$ where $C = \bigcap_{n \in \mathbb{N}^+} C_n$ is the Cantor set in Definition A.72. Indeed, by construction C_n has 2^n subintervals, each of which is of length $\frac{1}{3^n}$. Thus we have

$$\forall x \in C, \forall \epsilon \in (0, 1), \exists y \in [0, 1] \setminus C \text{ s.t. } |x - y| < \epsilon,$$

where y is obtained by first calculating $N = \lceil \log \frac{1}{\epsilon} \rceil + 1$ and then choose any $y \in C_N \setminus C_{N+1}$.

Definition D.22. A subset M of a metric space $(\mathcal{X}, d)$ is *bounded* iff

$$\exists x \in \mathcal{X}, \exists r \in \mathbb{R}^+ \text{ s.t. } M \subseteq B_r(x). \quad (\text{D.11})$$

The *diameter* of a bounded set M is

$$\text{diam } M := \sup\{d(x, y) : x, y \in M\}. \quad (\text{D.12})$$

Definition D.23. The ϵ -net of a set M in a metric space $(\mathcal{X}, d)$ is a subset $N \subset \mathcal{X}$ satisfying

$$\forall x \in M, \exists y \in N \text{ s.t. } d(x, y) < \epsilon. \quad (\text{D.13})$$

In particular, N is a *finite ϵ -net* of M if N is a finite set.

Example D.24. For any $\epsilon > 0$, the set $\epsilon\mathbb{N}^+$ is a ϵ -net of $\mathbb{R}^+$ in $\mathbb{R}$.

Example D.25. For any $\epsilon \in (0, 1]$, the interval $M = (0, 1]$ in $\mathbb{R}$ has a finite ϵ -net $N := \{i\epsilon : i = 1, \dots, \lfloor \frac{1}{\epsilon} \rfloor\}$.

Definition D.26. A subset M of a metric space $(\mathcal{X}, d)$ is *totally bounded* or *precompact* iff it has a finite ϵ -net for every $\epsilon > 0$.

Example D.27. The interval $M = (0, 1]$ in Example D.25 is totally bounded in $\mathbb{R}$. Why?

Lemma D.28. A totally bounded set is bounded.

Proof. This follows from Definitions D.26 and D.22 and the fact that the union of bounded sets is bounded. $\square$

Example D.29. For the metric space $(\mathbb{Q}, d)$ with d as the discrete metric in (D.2), no finite ϵ -net with $\epsilon < 1$ exists for $(0, 1)$: each open ball only contains the center and thus no finite set of open balls can cover the infinite set $(0, 1)$.

Definition D.30. A set in a metric space is *separable* iff it has a countable dense subset.

Lemma D.31. A totally bounded metric space is separable.

Proof. By Definition D.26, we can construct, for each $n \in \mathbb{N}^+$, a sequence (N_n) of finite $\frac{1}{n}$ -net of the metric space $\mathcal{X}$. By Definition D.19, the union $\bigcup_{n \in \mathbb{N}^+} N_n$ is a countable dense subset of $\mathcal{X}$. $\square$

D.1.3 Completion via isometry

Definition D.32. An *isometry* is a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ such that

$$\forall u, v \in \mathcal{X}, d_{\mathcal{Y}}(f(u), f(v)) = d_{\mathcal{X}}(u, v). \quad (\text{D.14})$$

Definition D.33. A *global isometry* or *isometric isomorphism* or *congruence mapping* is a bijective isometry as in Definition D.32. Two metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ are *isomorphic* if there exists a global isometry between them.

Definition D.34. A subset $\mathcal{X} \subseteq \mathcal{Y}$ is a *subspace* of the metric space $(\mathcal{Y}, d_{\mathcal{Y}})$ iff $(\mathcal{X}, d_{\mathcal{Y}})$ is a metric space itself.

Definition D.35. An *isometric embedding of a metric space* $(\mathcal{X}, d_{\mathcal{X}})$ in another metric space $(\mathcal{Y}, d_{\mathcal{Y}})$ is a map $f : \mathcal{X} \rightarrow \mathcal{Y}$ such that the map $f' : \mathcal{X} \rightarrow f(\mathcal{X})$ obtained by restricting the codomain to $f(\mathcal{X})$ is a global isometry.

Definition D.36. The *completion of a metric space* $(\mathcal{X}, d)$ is the intersection of all complete metric spaces that contain $(\mathcal{X}, d)$ as a subspace.

Lemma D.37. Suppose that $(\mathcal{X}, d)$ is a dense subspace of a complete metric space $(\mathcal{X}_1, d_1)$ with $d_1|_{\mathcal{X} \times \mathcal{X}} = d$. Then $(\mathcal{X}_1, d_1)$ is the completion of $(\mathcal{X}, d)$.

Proof. By Definition D.36, it suffices to show that $(\mathcal{X}_1, d_1)$ is the smallest complete metric space that contains $(\mathcal{X}, d)$ as a subspace. Since $(\mathcal{X}, d)$ is dense in $(\mathcal{X}_1, d_1)$, we have

$$\forall \xi_1 \in \mathcal{X}_1, \exists (x_n) \subset \mathcal{X} \text{ s.t. } \lim_{n \rightarrow \infty} d_1(x_n, \xi_1) = 0.$$

If another complete metric space $(\mathcal{X}_2, d_2)$ also contains $(\mathcal{X}, d)$ as a subspace, we have

$$\exists \xi_2 \in \mathcal{X}_2 \text{ s.t. } \lim_{n \rightarrow \infty} d_2(x_n, \xi_2) = 0.$$

The above two equations furnish a map $f : \mathcal{X}_1 \rightarrow \mathcal{X}_2$ that sends ξ_1 to ξ_2 . For another Cauchy sequence $(y_n) \subset \mathcal{X}$ that converges to $\eta_1 \in \mathcal{X}_1$ and to $\eta_2 \in \mathcal{X}_2$, we have

$$d_2(\xi_2, \eta_2) = \lim_{n \rightarrow \infty} d_2(x_n, y_n) = \lim_{n \rightarrow \infty} d_1(x_n, y_n) = d_1(\xi_1, \eta_1),$$

which, by Definition D.34, implies that $(\mathcal{X}_1, d_1)$ is a subspace of $(\mathcal{X}_2, d_2)$. $\square$

Theorem D.38. Every metric space has a unique completion.

Proof. For an arbitrary metric space (X, d) , define an equivalence relation on the Cauchy sequences in X as

$$(*) : \quad \forall (x_n), (y_n) \subset X, (x_n) \sim (y_n) \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0$$

and denote by $\mathcal{X}_2$ the set of equivalence classes of Cauchy sequences in (X, d) that converges in X , i.e.,

$$(**) : \quad \mathcal{X}_2 := \{\xi_x \in \mathcal{X}_1 : (x, x, \dots, x, \dots) \in \xi_x \text{ and } x \in X\}.$$

In comparison to $\mathcal{X}_2$, denote by $\mathcal{X}_1$ the set of equivalence classes of Cauchy sequences in (X, d) that may or may not converge in X . Thus $\mathcal{X}_2 \subset \mathcal{X}_1$. By $(*)$, we can define a function $d_1 : \mathcal{X}_1 \times \mathcal{X}_1 \rightarrow [0, +\infty)$ by

$$\forall \xi, \eta \in \mathcal{X}_1, \quad d_1(\xi, \eta) = \lim_{n \rightarrow \infty} d(x_n, y_n),$$

where (x_n) and (y_n) are arbitrary Cauchy sequences in ξ and η , respectively.

It is straightforward to show

- (a) $(\mathcal{X}_1, d_1)$ is a metric space,
- (b) $\forall x \in X, f(x) = \xi_x$ is a global isometry between (X, d) and $(\mathcal{X}_2, d_1)$, where ξ_x is defined by $(**)$,
- (c) $\mathcal{X}_2$ is a dense subspace of $\mathcal{X}_1$.

(c) holds because, for any Cauchy sequence $(x_n) \subset X$, the sequence $(f(x_n)) \subset \mathcal{X}_2$ converges to the equivalence class of (x_n) in $\mathcal{X}_1$.

By Lemma D.37 and (b), $(\mathcal{X}_1, d_1)$ can be regarded as the completion of (X, d) if we can prove $(\mathcal{X}_1, d_1)$ is complete. Indeed, for any Cauchy sequence $(\xi_k) \subset \mathcal{X}_1$, we have

$$(\Delta) : \quad \begin{aligned} & \implies \forall k \in \mathbb{N}^+, \exists x_k \in X \text{ s.t. } d_1(\xi_k, f(x_k)) < \frac{1}{k} \\ & d(x_n, x_m) = d_1(f(x_n), f(x_m)) \\ & \leq d_1(f(x_n), \xi_n) + d_1(\xi_n, \xi_m) + d_1(\xi_m, f(x_m)) \\ & < \chi(m, n) := \frac{1}{n} + \frac{1}{m} + d_1(\xi_n, \xi_m), \end{aligned}$$

where the first line follows from (c), the second from (b), and the last from the first line. Since (ξ_n) is Cauchy, we have $\chi(m, n) \rightarrow 0$ as $m, n \rightarrow \infty$ and thus the sequence $(x_k) \subset X$ is Cauchy. Then the triangle inequality $d_1(\xi_n, \zeta) \leq d_1(\xi_n, f(x_n)) + d_1(f(x_n), \zeta)$ and the first line of (Δ) imply that $d_1(\xi_n, \zeta) \rightarrow 0$ as $n \rightarrow \infty$ and thus $\zeta \in \mathcal{X}_1$ with $(x_k) \in \zeta$ is the limit of the Cauchy sequence $(\xi_n) \subset \mathcal{X}_1$.

Finally, the uniqueness of the completion follows from Definition D.36. $\square$

Example D.39. By Theorem 5.38, the set of polynomials $\mathbb{P}[a, b]$ is a dense subset of the metric space (D.4), which is the completion of $\mathbb{P}[a, b]$.

D.1.4 Sequence spaces

Definition D.40. The ℓ^∞ sequence space is a metric space (ℓ^∞, d) , where ℓ^∞ is the set of all bounded sequences of complex numbers,

$$\ell^\infty := \left\{ (\xi_1, \xi_2, \dots) : \text{s.t. } \sup_i |\xi_i| < \infty \right\} \quad (\text{D.15})$$

and the metric is given by $d(x, y) = \sup_i |\xi_i - \eta_i|$, where both $x = (\xi_j)$ and $y = (\eta_j)$ are in ℓ^∞ .

Definition D.41. For a real number $p \geq 1$, the ℓ^p sequence space is the metric space (ℓ^p, d) with

$$\ell^p := \left\{ (\xi_j)_{j=1}^\infty : \xi_j \in \mathbb{C}; \sum_{j=1}^\infty |\xi_j|^p < \infty \right\}; \quad (\text{D.16})$$

$$d(x, y) = \left(\sum_{j=1}^\infty |\xi_j - \eta_j|^p \right)^{1/p}, \quad (\text{D.17})$$

where $x = (\xi_j)$ and $y = (\eta_j)$ are both in ℓ^p . In particular, the Hilbert sequence space ℓ^2 is the ℓ^p space with $p = 2$.

Exercise D.42. Show that the ℓ^p space in Definition D.41 is complete for $p \in [1, +\infty)$.

Notation 27. Denote by c_{00} the space of all sequences that are eventually 0, c_0 the space of null sequences, i.e., the space of all sequences that converge to 0, and c the space of all sequences that converge.

Lemma D.43. The sequence spaces in Notation 27 and Definitions D.40 and D.41 satisfy

$$c_{00} \subset \ell^1 \subset \ell^p \subset \ell^q \subset c_0 \subset c \subset \ell^\infty, \quad (\text{D.18})$$

where $1 < p < q < \infty$.

Proof. Notation 27 and Definition D.40 yield $c_{00} \subset c_0 \subset c \subset \ell^\infty$. For any $q \in [1, \infty)$, we have $\ell^q \subset c_0$ because

$$(a_n)_{n \in \mathbb{N}} \in \ell^q \implies \sum_{n \in \mathbb{N}^+} |a_n|^q < \infty \implies \lim_{n \rightarrow \infty} |a_n|^q = 0,$$

which also implies $c_{00} \subset \ell^q$. For $p, q \in [1, \infty)$ with $p < q$, consider $(a_n) \in \ell^p$ and we have

$$\begin{aligned} \sum_{n \in \mathbb{N}^+} |a_n|^p < \infty & \implies \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, |a_n| < 1 \\ \implies \sum_{n=N}^\infty |a_n|^q & < \sum_{n=N}^\infty |a_n|^p < \infty \implies (a_n) \in \ell^q, \end{aligned}$$

which gives $\ell^p \subset \ell^q$. $\square$

Exercise D.44. Show that the sequence space c_{00} in Notation 27 is a dense subset of ℓ^p in Definition D.41 with $p \in [1, \infty)$.

Lemma D.45. ℓ^∞ in Definition D.40 is not separable.

Proof. Suppose ℓ^∞ is separable. Then ℓ^∞ has a countable dense subset E . For the set A of sequences with each term being either 0 or 1, we have

$$\forall a, b \in A, \quad a \neq b \iff d(a, b) = 1.$$

It follows from E being dense in ℓ^∞ that

$$\forall a \in A, \exists x_{n(a)} \in E \text{ s.t. } x_{n(a)} \in B_{\frac{1}{2}}(a).$$

Because the open balls are pairwise distinct, the map $f : A \rightarrow \mathbb{N}$ given by $f(a) = n(a)$ is injective. However, the construction of A implies that A is uncountable because A has a one-to-one correspondence with all real numbers in $[0, 1)$ via binary expansion. Thus $f : A \rightarrow \mathbb{N}$ cannot be injective and this completes the proof. $\square$

Lemma D.46. ℓ^p in Definition D.41 is separable for all $p \in [1, \infty)$.

Proof. Consider the set $E \subset c_{00}$ of all finitely supported sequences with rational terms. We claim that E is a countable dense subset of ℓ^p . By Definition D.41, we have

$$\forall \epsilon > 0, \forall x = (\xi_i) \in \ell^p, \exists N \in \mathbb{N} \text{ s.t. } \sum_{i=N}^{\infty} |\xi_i|^p < \frac{\epsilon}{2}.$$

Then the proof is completed by

$$\forall i = 1, \dots, N, \exists \eta_i \in \mathbb{Q} \text{ s.t. } |\xi_i - \eta_i|^p < \frac{\epsilon}{2N}. \quad \square$$

D.2 Basic metric topology

D.2.1 Open sets and closed sets

Definition D.47. A point $p \in E$ is an *interior point* of $E \subset \mathcal{X}$ iff there exists $r > 0$ such that $B_r(p) \subset E$. The *interior of a set* E is the set of its interior points.

Definition D.48. A subset U of a metric space is *open* iff every point $x \in U$ is an interior point of U , i.e.,

$$\forall x \in U, \exists r > 0 \text{ s.t. } B_r(x) \subset U.$$

Example D.49. By Lemma C.304, the set of invertible linear operators is open in $\mathcal{L}(\mathbb{R}^n)$.

Definition D.50. A subset E of a metric space $\mathcal{X}$ is *closed* iff its complement $E^c := \mathcal{X} \setminus E$ is open.

Exercise D.51. Show that $\emptyset$ and $\mathcal{X}$ are both open and closed.

Lemma D.52. The intersection of two open sets is open.

Proof. The statement holds trivially If the two open sets U and V are disjoint. Otherwise for any $x \in U \cap V$ we have

$$\exists r_1 > 0, r_2 > 0 \text{ s.t. } B_{r_1}(x) \subset U \text{ and } B_{r_2}(x) \subset V.$$

$\partial B_{r_1}(x)$, the boundary of $B_{r_1}(x)$, is bounded since it is a subset of a bounded set. Thus there exists $r \in \mathbb{R}^+$ less than both $\inf_{y \in \partial B_{r_1}(x)} d(x, y)$ and $\inf_{z \in \partial B_{r_2}(x)} d(x, z)$ such that $B_r(x) \subset U \cap V$. Thus $U \cap V$ is open. $\square$

Lemma D.53. The union of two closed sets is closed.

Proof. This follows from Definition D.50, Lemma D.52, and the DeMorgan's law (A.11). $\square$

Exercise D.54. If a closed set F in a metric space $\mathcal{X}$ does not contain any nonempty open set, then $\mathcal{X} \setminus F$ is dense in $\mathcal{X}$.

D.2.2 Limit points and closure

Definition D.55. Let $(\mathcal{X}, d)$ be a metric space. A point $x_0 \in \mathcal{X}$ is a *point of closure* of $E \subset \mathcal{X}$ or a *closure point* or a *contact point* iff

$$\forall r > 0, E \cap B_r(x_0) \neq \emptyset. \quad (\text{D.19})$$

Example D.56. Any point in the set

$$K = \left\{ \frac{1}{n} : n \in \mathbb{Z}^+ \right\} \quad (\text{D.20})$$

is a closure point of K , so is 0.

Definition D.57. In a metric space $\mathcal{X}$, a point x is an *accumulation point* or a *limit point* of $A \subseteq \mathcal{X}$ iff

$$\forall r > 0, (B_r(x) \setminus \{x\}) \cap A \neq \emptyset. \quad (\text{D.21})$$

Lemma D.58. For any limit point x of $E \subseteq \mathcal{X}$, there exists a sequence in $E \setminus \{x\}$ that converges to x in $\mathcal{X}$.

Proof. By Definition D.57, we have

$$\forall n \in \mathbb{N}^+, \exists p_n \in E \text{ s.t. } p_n \neq x \text{ and } d(p_n, x) < \frac{1}{n}.$$

Then Definition D.11 implies the convergence of the sequence (p_n) to x in $\mathcal{X}$. $\square$

Example D.59. Every point in an open set U is a limit point of U .

Example D.60. The only limit point of the set K in (D.20) is 0.

Example D.61. Each number in $\mathbb{R}$ is a limit point of $\mathbb{Q}$.

Lemma D.62. In a metric space $\mathcal{X}$, a set $A \subseteq \mathcal{X}$ is dense in another set $E \subseteq \mathcal{X}$ if and only if each point of E is a point in A or a limit point of A or both.

Proof. This follows from Definitions D.19 and D.57. $\square$

Definition D.63. A point x in a metric space $\mathcal{X}$ is an *isolated point* of $A \subseteq \mathcal{X}$ iff

$$\exists r > 0 \text{ s.t. } (B_r(x) \setminus \{x\}) \cap A = \emptyset. \quad (\text{D.22})$$

Definition D.64. A subset E of a metric space is *perfect* iff E is closed and every point of E is a limit point of E .

Lemma D.65. A subset of a metric space is closed if and only if it contains all of its limit points.

Proof. Suppose E is closed in $\mathcal{X}$. Then any point $p \in \mathcal{X} \setminus E$ has an open ball $B_r(p) \subseteq \mathcal{X} \setminus E$ for some $r > 0$. By Definition D.57, p cannot be a limit point of E . Therefore all limit points of E must belong to E itself.

Conversely, suppose the limit of any convergent sequence $(x_n) \subset E \subset \mathcal{X}$ is in E . Then E^c is open because

$$\forall p \in E^c, \exists r > 0 \text{ s.t. } B_r(p) \subset E^c;$$

otherwise the negation of the above statement would imply that p be a limit point of E . The proof is finished by Definition D.50. $\square$

Corollary D.66. A closed subset of a complete metric space $\mathcal{X}$ form a complete subspace of $\mathcal{X}$.

Proof. This follows directly from Lemma D.65 and Definitions D.14 and D.34. $\square$

Definition D.67. In a metric space, the *closure* of a subset A , written $\overline{A}$, is the union of A with the set of all limit points of A .

Lemma D.68. The closure of a subset $A \subset \mathcal{X}$ is closed.

Proof. By Lemma D.65, it suffices to show that any limit point x of $\overline{A}$ is in $\overline{A}$. Suppose $x \notin \overline{A}$. Then x is neither a point in A nor a limit point of A . Hence we have

$$(*) : \exists r > 0 \text{ s.t. } B_r(x) \cap A = \emptyset.$$

Since x is a limit point of $\overline{A}$, there exists $y \in \overline{A}$ such that $y \in B_r(x)$. Hence there exists $r_0 \in (0, r)$ such that $B_{r_0}(y) \subset B_r(x)$, which, together with (*), implies that y is not a point of A but a limit point of A . Therefore, $B_{r_0}(y)$ must have a point of A and this contradicts (*). $\square$

Corollary D.69. $\overline{A}$ is the smallest closed set containing A , i.e., the intersection of all closed sets containing A .

Proof. By Lemma D.68, $\overline{A}$ is a closed set that contains A . Now consider an arbitrary closed set E that contains A . Lemma D.65 implies that E must contain all limit points of A . Then Definition D.67 implies $\overline{A} \subseteq E$. $\square$

Corollary D.70. A subset A of a metric space is closed if and only if $A = \overline{A}$.

Proof. This follows from Lemma D.68 and Corollary D.69. $\square$

Corollary D.71. For any subset $A \subset \mathcal{X}$, we have $\overline{\overline{A}} = \overline{A}$.

Proof. This follows from Corollary D.70. $\square$

Corollary D.72. The subset relation is preserved by taking closure, i.e., $A \subset E$ implies $\overline{A} \subset \overline{E}$.

Proof. Definition D.67 implies $A \subset \overline{A}$ and thus $A \subset \overline{E}$. By Corollaries D.70 and D.71, $\overline{E}$ is closed. Then $\overline{A} \subset \overline{E}$ follows from Corollary D.69. $\square$

D.2.3 Closed sets and convergence

Lemma D.73. Any convergent sequence $(x_n) \subset \mathcal{X}$ with $\lim_n x_n = x$ either has countably infinite images whose (single) limit point is x , or, is eventually equal to x .

Proof. The number of images of the function of the sequence is either finite or countably infinite. Suppose it is infinite. Then removing all terms equal to x from (x_n) yields a subsequence (x_{n_k}) which is still infinite. Definition D.11 implies

$$\forall r > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall k > N, x_{n_k} \in B_r(x).$$

Since $x_{n_k} \neq x$, Definition D.57 dictates that x be a limit point of (x_n) .

Suppose the sequence has a finite number of images $\{p_1, \dots, p_m\}$. Then there exists some p_j such that $x_{n_k} = p_j$ for an infinite number of k . As a subsequence of a convergent sequence, (x_{n_k}) must converge to the same limit x . The rest follows from the finite number of possible values of x_n . $\square$

Lemma D.74. A subset E of a metric space $\mathcal{X}$ is closed if and only if any sequence $(x_n) \subset E$ that converges in $\mathcal{X}$ also converges in E .

Proof. Suppose E is closed and the sequence $(x_n) \subset E$ converges in $\mathcal{X}$. By Lemma D.73, (x_n) has either countably infinite images whose singleton limit point x is in $\mathcal{X}$ or finite images, one of which is x . In both cases, the limit x is in E , cf. Lemma D.65. Conversely, suppose the sequence (x_n) that converges in $\mathcal{X}$ has $\lim x_n = x \in E$. By Lemma D.58, there is a subsequence $(x_{n_k}) \subset (E \setminus \{x\})$ that converges to x . Thus E is closed because the limit point x is in E . $\square$

D.2.4 The Baire category theorem

Definition D.75. A subset A of topological space X is *dense* iff $\overline{A} = X$, i.e., the closure of A in X equals X .

Definition D.76. A subset of a topological space is *nowhere dense* if the interior of its closure is an empty set.

Example D.77. The Cantor set in Definition A.72 is nowhere dense in the metric space $([0, 1], |\cdot|)$.

Definition D.78. A subset of a topological space X is *meager* or *of the first category* in X iff it is a countable union of nowhere dense subsets of X ; otherwise it is *nonmeager* or *of the second category* in X .

Definition D.79. A *Baire space* is a topological space $\mathcal{X}$ in which every countable intersection of open dense sets is dense in $\mathcal{X}$.

Theorem D.80 (Baire category). In a complete metric space $\mathcal{X}$, the intersection of countably many open dense sets is dense in $\mathcal{X}$.

Proof. The statement holds trivially if $\mathcal{X} = \emptyset$. Otherwise, by Definition D.19, it suffices to show that for any open set V and any sequence $(W_n)_{n \in \mathbb{N}}$ of open dense sets in $\mathcal{X}$, there exists $x \in \bigcap_{n \in \mathbb{N}} U_n$ where $U_n := W_n \cap V$ such that $x \in V$.

We prove by induction that

$$\forall n \in \mathbb{N}^+, \exists x_n \in U_n, r_n > 0 \text{ s.t. } \begin{cases} r_n < \frac{1}{n}; \\ \overline{B_{r_n}(x_n)} \subset \bigcap_{i=1}^n U_i; \\ \overline{B_{r_n}(x_n)} \subset B_{r_{n-1}}(x_{n-1}), \end{cases}$$

where $B_{r_0}(x_0) := U_0$. The induction basis holds since U_1 is open and nonempty. For the inductive hypothesis, we assume that the above equation holds for some $n \in \mathbb{N}^+$. Since U_{n+1} is dense and open, Lemma D.52 implies $E_n := \overline{B_{r_n}(x_n)} \cap U_{n+1}$ is also open and nonempty. Thus we have

$$\exists x_{n+1} \in E_n, \exists r_{n+1} < \frac{1}{n+1} \text{ s.t. } \overline{B_{r_{n+1}}(x_{n+1})} \subset E_n,$$

which completes the inductive step.

It is straightforward to show that (x_n) is Cauchy and converges to x in the complete metric space of $\mathcal{X}$. Then the proof is completed by the fact that $x \in \bigcap_{i=0}^{\infty} U_i$. $\square$

Corollary D.81. A complete metric space $\mathcal{X}$ is not the union of countably many nowhere dense sets (U_n) .

Proof. We prove a stronger statement that $\mathcal{X} \neq \bigcup_{n \in \mathbb{N}} \overline{U_n}$. Suppose $\mathcal{X}$ is the union of countably many closed nowhere dense sets. On the one hand, the complement of this union is the empty set. On the other hand, DeMorgan's law (A.11) and Theorem D.80 dictate that this complement is nonempty. This contradiction completes the proof. $\square$

D.3 Continuous functions

D.3.1 Pointwise continuity

Definition D.82. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ is *continuous* at $x \in \mathcal{X}$ iff any sequence with $\lim_{n \rightarrow \infty} x_n = x$ implies the existence of $\lim_{n \rightarrow \infty} f(x_n)$, i.e.,

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x_n \in \mathcal{X}, \quad (D.23) \\ d_{\mathcal{X}}(x, x_n) < \delta \implies d_{\mathcal{Y}}(f(x), f(x_n)) < \epsilon.$$

Lemma D.83. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ is continuous at $x \in \mathcal{X}$ iff

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } f(B_{\delta}(x)) \subset B_{\epsilon}(f(x)). \quad (D.24)$$

Proof. This follows directly from Definition D.82. $\square$

Definition D.84. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ is (pointwise) *continuous* iff it is continuous at all $x \in \mathcal{X}$, i.e.,

$$\forall \epsilon > 0, \forall x \in \mathcal{X}, \exists \delta > 0 \text{ s.t. } \forall y \in \mathcal{X}, \quad (D.25) \\ d_{\mathcal{X}}(x, y) < \delta \implies d_{\mathcal{Y}}(f(x), f(y)) < \epsilon$$

Theorem D.85 (Continuous extension at a limit point).

Consider a continuous function $f : E \rightarrow \mathcal{Y}$ that maps a subset E of a metric space $\mathcal{X}$ to a complete metric space $\mathcal{Y}$. Suppose, at a limit point $x \in \mathcal{X}$ of E with $x \notin E$, the sequence $(f(x_n)) \subset \mathcal{Y}$ is Cauchy for any Cauchy sequence $(x_n) \subset E$ that converges to x . Then f can be uniquely extended to a continuous function $F : E \cup \{x\} \rightarrow \mathcal{Y}$ such that $F|_E = f$.

Proof. By the completeness of $\mathcal{Y}$, $(f(x_n)) \subset \mathcal{Y}$ converges to some point $y \in \mathcal{Y}$. Let (x_n) and $(\hat{x}_n)$ be a Cauchy sequence. Suppose $(f(x_n))$ and $(f(\hat{x}_n))$ converge respectively to y and $\hat{y}$ with $y \neq \hat{y}$. Then the new sequence $(\bar{x}_n)$ constructed by interlacing (x_n) and $(\hat{x}_n)$ is another Cauchy sequence that converges to x . Hence $(f(\bar{x}_n))$ is also a Cauchy sequence, but has two subsequences that converges to two different limits, which contradicts Lemma B.10. Hence, for any Cauchy sequences $(x_n) \subset E$ that converges to x , $(f(x_n))$ converges to a unique value which we define as $F(x)$. The continuity of F follows from the above construction and Definition D.82. $\square$

Theorem D.86. A map $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces is (pointwise) continuous if and only if the preimage $f^{-1}(V)$ of any open set V in $\mathcal{Y}$ is open in $\mathcal{X}$.

Proof. For necessity, first note the empty set is open in $\mathcal{X}$ because the condition in Definition D.48 holds vacuously; this corresponds to the special case $f^{-1}(V) = \emptyset$. Now consider $x \in f^{-1}(V)$. (A.26) yields $f(x) \in V$. Since V is open, Definition D.48 implies that $\exists \epsilon > 0$ s.t. $B_{\epsilon}(f(x)) \subset V$. By Lemma D.83, the continuity of f implies

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } f(B_{\delta}(x)) \subset B_{\epsilon}(f(x)) \subset V,$$

which, together with Lemma A.42, yields $B_{\delta}(x) \subset f^{-1}(V)$. Thus $f^{-1}(V)$ is open.

For sufficiency, consider $B_{\epsilon}(f(x))$ for any given x and $\epsilon > 0$. Since $B_{\epsilon}(f(x))$ is open, $E := f^{-1}(B_{\epsilon}(f(x)))$ is also open in $\mathcal{X}$. Hence $x \in E$ implies $\exists \delta > 0$ s.t. $B_{\delta}(x) \subset E$, which further implies

$$f(B_{\delta}(x)) \subset f(E) \subseteq B_{\epsilon}(f(x)),$$

where the last step follows from Lemma A.44. Then the proof is completed by Lemma D.83. $\square$

Theorem D.87. For a map $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces, the following conditions are equivalent:

- (a) $f^{-1}(V)$ is open in $\mathcal{X}$ for any open set V in $\mathcal{Y}$.
- (b) $\forall A \subseteq \mathcal{X}, f(\overline{A}) \subseteq \overline{f(A)}$.
- (c) $\forall E \subseteq \mathcal{Y}, \overline{f^{-1}(E)} \subseteq f^{-1}(\overline{E})$,
- (d) $f^{-1}(V)$ is closed in $\mathcal{X}$ for any closed set V in $\mathcal{Y}$,
- (e) f is continuous.

Proof. By Theorem D.86, (a) and (e) are equivalent.

(a) $\implies$ (b): $y \in f(\overline{A})$ and (A.25) imply $f^{-1}(y) \subset \overline{A}$. If $f^{-1}(y) \subset A$, (A.26) gives $y \in f(A) \subseteq \overline{f(A)}$. Otherwise consider any limit point $x \in f^{-1}(y)$ of A satisfying $x \notin A$. By Lemma D.58, there exists a sequence $(x_n) \subset A$ converging to x . Then (a), Theorem D.86, and Definition D.82 imply the convergence of the sequence $(f(x_n)) \subset f(A)$ to $y = f(x)$. Thus y is a limit point of $f(A)$.

(b) $\implies$ (c): for $E \subset \mathcal{Y}$, write $A := f^{-1}(E)$. Then (A.28) yields $\overline{f(A)} = \overline{f(f^{-1}(E))} \subseteq E$ and Corollary D.72 further implies $\overline{f(A)} \subseteq \overline{E}$. (A.27) and (b) gives

$$\overline{f^{-1}(E)} = \overline{A} \subseteq f^{-1}(\overline{f(A)}) \subseteq f^{-1}(\overline{E}).$$

(c) $\implies$ (d): for any closed set $V \subset \mathcal{Y}$, (c) and Corollary D.70 imply $\overline{f^{-1}(V)} \subseteq f^{-1}(\overline{V}) = f^{-1}(V)$. On the other hand, $f^{-1}(V) \subseteq \overline{f^{-1}(V)}$ always holds. Hence we have $f^{-1}(V) = \overline{f^{-1}(V)}$ and (d) follows from Corollary D.70.

(d) $\implies$ (a): any open set U in $\mathcal{Y}$ can be expressed as $U = \mathcal{Y} \setminus V$ for some close set V , then (A.25) gives

$$f^{-1}(U) = f^{-1}(\mathcal{Y} \setminus V) = f^{-1}(\mathcal{Y}) \setminus f^{-1}(V) = \mathcal{X} \setminus f^{-1}(V),$$

which is open since $f^{-1}(V)$ is closed. $\square$

Lemma D.88. If $f : \mathcal{X} \rightarrow \mathcal{Y}$ and $g : \mathcal{Y} \rightarrow \mathcal{Z}$ are continuous functions between metric spaces, then the composition map $g \circ f : \mathcal{X} \rightarrow \mathcal{Z}$ is continuous.

Proof. This follows from Definition D.82. $\square$

Lemma D.89. For continuous functions $f, g : \mathcal{X} \rightarrow \mathbb{F}^n$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, the functions cf , $f + g$, $f - g$, and $|f|$ are continuous for any $c \in \mathbb{F}$. In particular,

- (a) $\frac{1}{f}$ is continuous for $n = 1$;
- (b) both $\max\{f, g\}$ and $\min\{f, g\}$ are continuous for $n = 1$ and $\mathbb{F} = \mathbb{R}$;
- (c) $\bar{f}$, the complex conjugate of f , is continuous for $n = 1$ and $\mathbb{F} = \mathbb{C}$.

Proof. These follows from Lemma D.88 and Definition D.82. $\square$

Lemma D.90. For the set $\mathcal{B}(\mathbb{R}^n)^{-1} \subset \mathcal{L}(\mathbb{R}^n)$ of invertible linear operators equipped with the operator norm in Definition C.288, the inverse function $f : \mathcal{B}(\mathbb{R}^n)^{-1} \rightarrow \mathcal{B}(\mathbb{R}^n)^{-1}$ given by $f(A) = A^{-1}$ is continuous in the sense of Definition D.82.

Proof. It suffices to show that f is continuous at an arbitrary $A \in \mathcal{B}(\mathbb{R}^n)^{-1}$, i.e.,

$$(*) : \quad \|B^{-1} - A^{-1}\| \rightarrow 0 \text{ as } \|B - A\| \rightarrow 0.$$

Write $\alpha := \|A^{-1}\|$ and we have

$$\|B^{-1}\| - \|A^{-1}\| \leq \|B^{-1} - A^{-1}\| \leq \|B^{-1}\| \|A - B\| \|A^{-1}\|,$$

where the last step follows from Corollary C.300 and the identity $B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1}$. Consequently, we have $\|B^{-1}\| \leq \frac{\alpha}{1 - \alpha\|A - B\|}$ and its substitution into the last inequality above yields

$$\|B^{-1} - A^{-1}\| \leq \frac{\alpha^2 \|A - B\|}{1 - \alpha\|A - B\|},$$

which implies $(*)$ since $g(t) = \frac{\alpha^2 t}{1 - \alpha t}$ is monotonically increasing on $\mathbb{R}^+$. $\square$

D.3.2 Separation properties of metric spaces

Definition D.91. In a metric space $(\mathcal{X}, d)$, the *distance between a nonempty set $A \subset \mathcal{X}$ and a point $x \in \mathcal{X}$* is

$$\rho(x, A) = \inf\{d(x, a) : a \in A\}. \quad (\text{D.26})$$

Lemma D.92. For any subset A of a metric space $\mathcal{X}$, the function $f : \mathcal{X} \rightarrow \mathbb{R}$ given by $f(x) := \rho(x, A)$ in (D.26) is continuous.

Proof. Definition D.91 and the triangle inequality yield

$$\forall x, y \in \mathcal{X}, \forall z \in A, \rho(x, A) \leq d(x, z) \leq d(x, y) + d(y, z).$$

Taking infimum over $z \in A$ gives $\rho(x, A) \leq d(x, y) + \rho(y, A)$. The symmetry of x and y yields $\rho(y, A) \leq d(x, y) + \rho(x, A)$. These two inequalities imply $|\rho(x, A) - \rho(y, A)| \leq d(x, y)$. Then the proof is completed by Definition D.82 from setting $\delta = \epsilon$. $\square$

Corollary D.93. For any point y of a metric space $(\mathcal{X}, d)$, the function $f : \mathcal{X} \rightarrow \mathbb{R}$ given by $f(x) := d(x, y)$ is continuous.

Proof. This follows from Lemma D.92 with $A = \{y\}$. $\square$

Corollary D.94. In a metric space $(\mathcal{X}, d)$, the closed ball $\{x \in \mathcal{X} : d(x, y) \leq r\}$ is a closed set for any $y \in \mathcal{X}$ and $r \in \mathbb{R}^+$.

Proof. By Corollary D.93, the function $f(x) := d(x, y)$ is continuous. Then Theorem D.87 dictates that the closed ball, as the preimage $f^{-1}([0, r])$, be closed. $\square$

Lemma D.95. Any nonempty subset A of a metric space $(\mathcal{X}, d)$ satisfies $\bar{A} = \{x \in \mathcal{X} : \rho(x, A) = 0\}$.

Proof. By Lemma D.92, the function $f(x) := \rho(x, A)$ is continuous. By Theorem D.87, the preimage $f^{-1}(\{0\})$ is closed. Then Corollary D.69 implies

$$\bar{A} \subseteq \{x \in \mathcal{X} : \rho(x, A) = 0\}.$$

For the reverse inclusion, suppose there exists $x \in f^{-1}(\{0\})$ such that $x \notin \bar{A}$. Then x is neither a point in A nor a limit point of A . Thus Definition D.57 implies that there exists $r > 0$ such that $B_r(x) \cap A = \emptyset$. Hence

$$\forall y \in A, d(x, y) \geq r \implies \rho(x, A) \geq r > 0,$$

which contradicts $x \in f^{-1}(\{0\})$. $\square$

Lemma D.96. For any two disjoint closed sets E and F in a metric space $\mathcal{X}$, there exists a continuous function $f : \mathcal{X} \rightarrow [0, 1]$ such that $f(E) = \{0\}$ and $f(F) = \{1\}$.

Proof. Corollary D.70 implies $E = \bar{E}$ and $F = \bar{F}$. By the disjointness of $\bar{E}$ and $\bar{F}$, we have $\rho(x, E) + \rho(x, F) > 0$ for any $x \in \mathcal{X}$. By Lemmas D.88, D.89, and D.92, the function $f(x) := \frac{\rho(x, E)}{\rho(x, E) + \rho(x, F)}$ is continuous. Then the proof is completed by Lemma D.95. $\square$

Corollary D.97. A metric space $\mathcal{X}$ is *normal*, i.e., for any two disjoint closed subsets E, F of $\mathcal{X}$, there exist disjoint open sets U, V such that $E \subset U$ and $F \subset V$.

Proof. Utilizing the function f in the proof of Lemma D.96, we define $U := f^{-1}((-\infty, \frac{1}{2}))$ and $V := f^{-1}((\frac{1}{2}, +\infty))$. Then Lemma D.96 implies $E \subset U$ and $F \subset V$. The proof is completed by Theorem D.87. $\square$

Corollary D.98. A metric space $\mathcal{X}$ is *T1*, i.e., every one point subset of $\mathcal{X}$ is closed.

Proof. For any singleton set $\{p\} \subset \mathcal{X}$, the only converging sequence in $\mathcal{X}$ is the constant sequence (p) . The proof is completed by Lemma D.74. $\square$

Corollary D.99. A metric space $\mathcal{X}$ is *regular*, i.e., for any closed set E of $\mathcal{X}$ and any point $x \in \mathcal{X}$ with $x \notin E$ there exist disjoint open sets U, V such that $E \subset U$ and $x \in V$.

Proof. This follows from Corollaries D.97 and D.98. $\square$

Corollary D.100. A metric space $\mathcal{X}$ is *Hausdorff* or *T2*, i.e., for any two disjoint points x, y in $\mathcal{X}$, there exist disjoint open sets U, V such that $x \in U$ and $y \in V$.

Proof. This follows from Corollaries D.99 and D.98. $\square$

D.3.3 Homeomorphisms

Definition D.101. A map $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces is *open* if its image of any open set is open; it is *closed* if its image of any closed set is closed.

Definition D.102. A *homeomorphism* $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces is a bijective map such that both f and f^{-1} are continuous. Two spaces $\mathcal{X}$ and $\mathcal{Y}$ are said to be *homeomorphic* if there exists a homeomorphism between them.

Lemma D.103. A homeomorphism is both open and closed.

Proof. This follows from Definition D.101 and Theorem D.87. $\square$

Definition D.104. A *topological property* of a metric space $\mathcal{X}$ is a property of $\mathcal{X}$ that can be expressed entirely in terms of the open sets of $\mathcal{X}$.

Corollary D.105. A homeomorphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ preserves all topological properties of $\mathcal{X}$.

Proof. This follows directly from Definition D.104 and Lemma D.103. $\square$

Definition D.106. A (*topological*) *embedding* of $\mathcal{X}$ in $\mathcal{Y}$ is an injective map $f : \mathcal{X} \rightarrow \mathcal{Y}$ such that the map $g : \mathcal{X} \rightarrow f(\mathcal{X})$ obtained by restricting the codomain of f to $f(\mathcal{X})$ is a homeomorphism.

D.3.4 Uniform continuity

Definition D.107. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ is *uniformly continuous* iff

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x, y \in \mathcal{X}, \quad d_{\mathcal{X}}(x, y) < \delta \implies d_{\mathcal{Y}}(f(x), f(y)) < \epsilon. \quad (\text{D.27})$$

Example D.108. Let d_1 and d_2 denote the metrics induced from the operator norm and the Hilbert-Schmidt norm in Definitions C.288 and C.296, respectively. By Theorem C.303 and Definition D.107, the identity map

$$I : (\mathcal{L}(\mathbb{F}^n, \mathbb{F}^m), d_1) \rightarrow (\mathcal{L}(\mathbb{F}^n, \mathbb{F}^m), d_2) \quad (\text{D.28})$$

is uniformly continuous and has a uniformly continuous inverse.

Theorem D.109 (Continuous extension). A uniformly continuous function $f : E \rightarrow \mathcal{Y}$ that maps a dense subset E of a metric space $\mathcal{X}$ to a complete metric space $\mathcal{Y}$ can be uniquely extended to a continuous function $F : \mathcal{X} \rightarrow \mathcal{Y}$ such that $F|_E = f$.

Proof. By Theorem D.85 and Lemma D.37, it suffices to show that the sequence $(f(x_n)) \subset \mathcal{Y}$ is Cauchy for any Cauchy sequence $(x_n) \subset E$ that converges to a limit point $x \in \mathcal{X}$ of E . The uniform continuity of f implies

$$(*) : \quad \begin{cases} \forall \epsilon > 0, \exists \delta > 0 \text{ s.t.} \\ \forall x, y \in E, d_{\mathcal{X}}(x, y) < \delta \implies d_{\mathcal{Y}}(f(x), f(y)) < \epsilon. \end{cases}$$

(x_n) being Cauchy and $(*)$ yield

$$\begin{aligned} & \exists N(\delta) \in \mathbb{N} \text{ s.t. } \forall m, n > N, d_{\mathcal{X}}(x_m, x_n) < \delta \\ & \implies \begin{cases} \forall \epsilon > 0, \exists M = N(\delta) \in \mathbb{N} \text{ s.t.} \\ \forall m, n > N, d_{\mathcal{Y}}(f(x_m), f(x_n)) < \epsilon. \end{cases} \end{aligned}$$

where ϵ, δ are the same as that in $(*)$. Hence $(f(x_n))$ is a Cauchy sequence. $\square$

Definition D.110. The function F in Theorem D.109 is called the continuous extension of $f : E \rightarrow \mathcal{Y}$ to the completion of E .

Definition D.111. The *Cantor function* $F : [0, 1] \rightarrow [0, 1]$ is given by two steps. First, we define $f : [0, 1] \setminus C \rightarrow [0, 1]$ as

- (a) $f([0, 1] \setminus C_1) = \frac{1}{2}$,
- (b) $f(I_{n,i}) = \frac{2i-1}{2^{n+1}}$ for each $i = 1, \dots, 2^n$ where $I_{n,i}$ is one of the 2^n closed intervals of C_n in ascending order.

Second, F is the continuous extension of f to $[0, 1]$.

Exercise D.112. Show that the Cantor function F is indeed well defined by proving the uniform continuity of f in Definition D.111. Prove that F is monotonically increasing. Plot the graph of F on $[0, 1] \setminus C_n$ for $n = 5$. What are the values of $F(0)$ and $F(1)$?

D.3.5 Equicontinuity

Definition D.113. Let $E \subset \mathcal{X}$ denote a subset of a metric space $\mathcal{X}$. A family of scalar functions $\mathcal{F} := \{f : E \rightarrow \mathbb{C}\}$ is *equicontinuous* on E iff

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall f \in \mathcal{F}, \forall x, y \in E, \quad d_{\mathcal{X}}(x, y) < \delta \implies |f(x) - f(y)| < \epsilon. \quad (\text{D.29})$$

D.3.6 Lipschitz continuity

Definition D.114. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ is *Lipschitz continuous* iff

$$\exists L > 0 \text{ s.t. } \forall x, y \in \mathcal{X}, \quad d_{\mathcal{Y}}(f(x), f(y)) < L d_{\mathcal{X}}(x, y). \quad (\text{D.30})$$

Exercise D.115. What is the connection between the logical statements in (D.27) and (D.30)?

Definition D.116. A subset $S \subset \mathbb{R}^n$ is *star-shaped* with respect to a point $\mathbf{p} \in S$ if for each $\mathbf{x} \in S$ the line segment from $\mathbf{p}$ to $\mathbf{x}$ lies in S .

Theorem D.117. For a continuously differentiable function $\mathbf{f} : S \rightarrow \mathbb{R}^m$ where $S \subset \mathbb{R}^n$ is star-shaped with respect to $\mathbf{p} = (p_1, p_2, \dots, p_n) \in S$, there exist continuous functions $\mathbf{g}_1(\mathbf{x}), \mathbf{g}_2(\mathbf{x}), \dots, \mathbf{g}_n(\mathbf{x})$ such that $\mathbf{g}_j(\mathbf{p}) = \frac{\partial \mathbf{f}}{\partial x_j}(\mathbf{p})$ and

$$\mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{p}) + \sum_{j=1}^n (x_j - p_j) \mathbf{g}_j(\mathbf{x}). \quad (\text{D.31})$$

Proof. Since S is star-shaped, for any given $\mathbf{y} \in S$ and $t \in [0, 1]$, $\mathbf{f}(\mathbf{x})$ is defined for $\mathbf{x} = \mathbf{p} + t(\mathbf{y} - \mathbf{p})$. Then the chain rule yields

$$\frac{d}{dt} \mathbf{f}(\mathbf{x}) = \sum_i \frac{\partial \mathbf{f}}{\partial x_i} \frac{dx_i}{dt} = \sum_i (y_i - p_i) \frac{\partial \mathbf{f}}{\partial x_i}(\mathbf{x}).$$

An integration with respect to t from 0 to 1 leads to

$$\begin{aligned} \mathbf{f}(\mathbf{y}) - \mathbf{f}(\mathbf{p}) &= \sum_i (y_i - p_i) \mathbf{g}_i(\mathbf{y}), \\ \mathbf{g}_i(\mathbf{y}) &= \int_0^1 \frac{\partial \mathbf{f}}{\partial x_i}(\mathbf{p} + t(\mathbf{y} - \mathbf{p})) dt, \end{aligned}$$

where the function $\mathbf{g}_i(\mathbf{p}) = \frac{\partial \mathbf{f}}{\partial x_i}(\mathbf{p})$. Then the continuity of $\mathbf{g}_i$ follows from the continuity of $\frac{\partial \mathbf{f}}{\partial x_i}$ and the inequality $|\int_0^1 f| \leq \int_0^1 |f|$ for any function f . $\square$

Theorem D.118. Consider a differentiable function $\mathbf{f} : E \rightarrow \mathbb{R}^m$ where $E \subset \mathbb{R}^n$ is a convex open set. Suppose there exists $L > 0$ such that $\forall \mathbf{x} \in E$, $\|\mathbf{f}'(\mathbf{x})\| \leq L$. Then

$$\forall \mathbf{a} \in E, \mathbf{b} \in E, \quad \|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\| \leq L \|\mathbf{b} - \mathbf{a}\|. \quad (\text{D.32})$$

Proof. Fix $\mathbf{a} \in E$, $\mathbf{b} \in E$, and define $\varphi : [0, 1] \rightarrow E$ by

$$\varphi(t) := (1 - t)\mathbf{a} + t\mathbf{b}.$$

Write $\mathbf{g}(t) := \mathbf{f}(\varphi(t))$ and we have $\mathbf{g}'(t) = \mathbf{f}'(\varphi(t))\varphi'(t) = \mathbf{f}'(\varphi(t))(\mathbf{b} - \mathbf{a})$, which yields

$$\forall t \in [0, 1], \quad \|\mathbf{g}'(t)\| \leq \|\mathbf{f}'(\varphi(t))\| \|\mathbf{b} - \mathbf{a}\|.$$

Then (D.32) follows from the vector mean value theorem C.370. $\square$

D.3.7 The Banach fixed point theorem

Definition D.119. A function $f : \mathcal{X} \rightarrow \mathcal{X}$ on a metric space $(\mathcal{X}, d)$ is *contractive* iff

$$\exists \alpha \in (0, 1) \text{ s.t. } \forall x, y \in \mathcal{X}, \quad d(f(x), f(y)) \leq \alpha d(x, y). \quad (\text{D.33})$$

Theorem D.120 (Banach fixed point). A contractive mapping $f : \mathcal{X} \rightarrow \mathcal{X}$ on a complete metric space $(\mathcal{X}, d)$, as in Definition D.119, has a unique fixed point $x = f(x) \in \mathcal{X}$.

Proof. For any $x_0 \in \mathcal{X}$, construct a sequence of points by $x_n = f(x_{n-1})$. Definition D.119 yields $d(x_n, x_{n+1}) \leq \alpha^n d(x_0, x_1)$, which implies that, $\forall m \in \mathbb{N}^+$,

$$d(x_n, x_{n+m}) \leq \sum_{i=1}^m d(x_{n+i-1}, x_{n+i}) < \frac{\alpha^n}{1 - \alpha} d(x_0, x_1).$$

Thus the sequence (x_n) is Cauchy and converges to some x in the complete metric space $\mathcal{X}$. Take limit of $x_n = f(x_{n-1})$ and we have from the continuity of f that $x = f(x)$.

Suppose y is another fixed point of f . Then we have $d(x, y) = d(f(x), f(y)) \leq \alpha d(x, y)$, which implies $d(x, y) = 0$ and $x = y$. $\square$

Example D.121. Theorem D.120 can be used to prove the fundamental theorem of ODEs (Theorem 12.8).

D.3.8 The inverse function theorem and the implicit function theorem

Theorem D.122 (Inverse function). Consider a function $\mathbf{f} : E \rightarrow \mathbb{R}^n$ with $E \subset \mathbb{R}^n$ being an open subset. Suppose

- (i) $\mathbf{f} \in \mathcal{C}^1(E)$,
- (ii) $\exists \mathbf{a} \in E$ s. t. $\det |\mathbf{f}'(\mathbf{a})| \neq 0$.

Then we have

- (a) there exist open sets $U \subset E$ such that $\mathbf{a} \in U$, $\mathbf{f}$ is injective on U , and $V := \mathbf{f}(U)$ is open.
- (b) $\exists \mathbf{g} \in \mathcal{C}^1(V, U)$ s.t. $\forall \mathbf{x} \in U$, $\mathbf{g}(\mathbf{f}(\mathbf{x})) = \mathbf{x}$, and

$$\forall \mathbf{y} \in V, \quad \mathbf{g}'(\mathbf{y}) = [\mathbf{f}'(\mathbf{g}(\mathbf{y}))]^{-1}. \quad (\text{D.34})$$

Proof. Write $A := \mathbf{f}'(\mathbf{a})$ and $\lambda := \frac{1}{2\|\mathbf{A}^{-1}\|}$. By (i), there exists an open set $U \subset E$ such that

$$\forall \mathbf{x} \in U, \quad \|\mathbf{f}'(\mathbf{x}) - A\| < \lambda.$$

For any given $\mathbf{y} \in \mathbb{R}^n$, define a function $\varphi_{\mathbf{y}} : U \rightarrow \mathbb{R}^n$ by

$$(*) : \quad \varphi_{\mathbf{y}}(\mathbf{x}) := \mathbf{x} + A^{-1}(\mathbf{y} - \mathbf{f}(\mathbf{x})).$$

It follows that $\varphi'_{\mathbf{y}}(\mathbf{x}) := A^{-1}(A - \mathbf{f}'(\mathbf{x}))$ and consequently

$$\forall \mathbf{x} \in U, \quad \|\varphi'_{\mathbf{y}}(\mathbf{x})\| \leq \|A^{-1}\| \|A - \mathbf{f}'(\mathbf{x})\| < \frac{1}{2},$$

which, together with Theorem D.118, yields,

$$(**) : \quad \forall \mathbf{x}, \mathbf{z} \in \overline{U}, \quad |\varphi_{\mathbf{y}}(\mathbf{x}) - \varphi_{\mathbf{y}}(\mathbf{z})| \leq \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|.$$

By (*), $\mathbf{x}$ is a fixed point of $\varphi_{\mathbf{y}}(\mathbf{x})$ if and only of $\mathbf{y} = \mathbf{f}(\mathbf{x})$. By (**) and the proof of Theorem D.120, $\varphi_{\mathbf{y}}$ has at most one fixed point in U . Thus $\mathbf{f}$ is injective on U .

To show V is open, fix $\mathbf{y}_0 \in V$. There exist $\mathbf{x}_0 \in U$ and $r > 0$ such that $\mathbf{f}(\mathbf{x}_0) = \mathbf{y}_0$ and $\overline{B_r(\mathbf{x}_0)} \subset U$. Then we have, $\forall \mathbf{y} \in \overline{B_{\lambda r}(\mathbf{y}_0)}$,

$$\begin{aligned} & |\varphi_{\mathbf{y}}(\mathbf{x}_0) - \mathbf{x}_0| \leq \|A^{-1}\| \|\mathbf{y} - \mathbf{y}_0\| < \frac{r}{2} \\ \Rightarrow & |\varphi_{\mathbf{y}}(\mathbf{x}) - \mathbf{x}_0| \leq |\varphi_{\mathbf{y}}(\mathbf{x}) - \varphi_{\mathbf{y}}(\mathbf{x}_0)| + |\varphi_{\mathbf{y}}(\mathbf{x}_0) - \mathbf{x}_0| < r \\ \Rightarrow & \varphi_{\mathbf{y}}(\overline{B_r(\mathbf{x}_0)}) \subset \overline{B_r(\mathbf{x}_0)}. \end{aligned}$$

By Definition D.119, $\varphi_{\mathbf{y}}$ is a contraction on the complete metric space $\overline{B_r(\mathbf{x}_0)}$ and, by Theorem D.120, $\varphi_{\mathbf{y}}$ has a unique fixed point $\mathbf{x} \in U$ and $\mathbf{y} = \mathbf{f}(\mathbf{x}) \in V$. Thus $\overline{B_{\lambda r}(\mathbf{y}_0)} \subset \mathbf{f}(U) = V$, which implies that V is open. This completes the proof of (a).

To prove (b), we first note that, for any $\mathbf{y} \in V$ and $\mathbf{y} + \mathbf{k} \in V$, there exist $\mathbf{x} \in U$ and $\mathbf{x} + \mathbf{h} \in U$ such that $\mathbf{y} = \mathbf{f}(\mathbf{x})$ and $\mathbf{y} + \mathbf{k} = \mathbf{f}(\mathbf{x} + \mathbf{h})$. Then (*) yields

$$\varphi_{\mathbf{y}}(\mathbf{x} + \mathbf{h}) - \varphi_{\mathbf{y}}(\mathbf{x}) = \mathbf{h} + A^{-1}[\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x} + \mathbf{h})] = \mathbf{h} - A^{-1}\mathbf{k}.$$

(**) yields $|\mathbf{h} - A^{-1}\mathbf{k}| \leq \frac{1}{2}|\mathbf{h}|$ and thus $|A^{-1}\mathbf{k}| \geq \frac{1}{2}|\mathbf{h}|$, i.e., $|\mathbf{h}| \leq \frac{1}{\lambda}|\mathbf{k}|$. Write $T := [\mathbf{f}'(\mathbf{x})]^{-1}$ and we have $\mathbf{g}(\mathbf{y} + \mathbf{k}) - \mathbf{g}(\mathbf{y}) - T\mathbf{k} = -T[\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - \mathbf{f}'(\mathbf{x})\mathbf{h}]$, yielding

$$\frac{|\mathbf{g}(\mathbf{y} + \mathbf{k}) - \mathbf{g}(\mathbf{y}) - T\mathbf{k}|}{|\mathbf{k}|} \leq \frac{\|T\|}{\lambda} \frac{|\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - \mathbf{f}'(\mathbf{x})\mathbf{h}|}{|\mathbf{h}|}.$$

By (i) and Definition C.352, the RHS in the above equation goes to zero, which implies that the LHS must also go to zero; thus Definition C.352 dictates that $\mathbf{g}$ be differentiable at $\mathbf{y}$ and $\mathbf{g}'(\mathbf{y}) = T$, which is the same as (D.34).

By (D.34), $\mathbf{g}'$ can be considered as the composition of three continuous functions: $\mathbf{g}$ is continuous because it is differentiable, $\mathbf{f}'$ is continuous because of (i), and the inverse function is continuous because of Lemma D.90. Then Lemma D.88 dictates the continuity of $\mathbf{g}'$. $\square$

Theorem D.123 (Implicit function theorem). For a function $\mathbf{g} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$, suppose there exist $U \times V \subset \mathbb{R}^n \times \mathbb{R}^m$ and $(\mathbf{x}_0, \mathbf{y}_0) \in U \times V$ such that

- (i) $\mathbf{g}(\mathbf{x}_0, \mathbf{y}_0) = \mathbf{0}$;
- (ii) $\det \left| \frac{\partial \mathbf{g}}{\partial \mathbf{y}}(\mathbf{x}_0, \mathbf{y}_0) \right| \neq 0$;
- (iii) $\mathbf{g} \in \mathcal{C}^1(U \times V)$.

Then there exists an open set $U_0 \times V_0 \subset U \times V$ with $(\mathbf{x}_0, \mathbf{y}_0) \in U_0 \times V_0$ such that

- (a) $\forall \mathbf{x} \in U_0, \exists! \mathbf{y} \in V_0$ s.t. $\mathbf{g}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$;

the function $\mathbf{f} : U_0 \rightarrow V_0$ given by $\mathbf{f}(\mathbf{x}) := \mathbf{y}$ in (a) satisfies

- (b) $\mathbf{f} \in \mathcal{C}^1(U_0)$ and $\mathbf{f}' = -[\frac{\partial \mathbf{g}}{\partial \mathbf{y}}]^{-1}[\frac{\partial \mathbf{g}}{\partial \mathbf{x}}]$.

For $p > 1$, we have $\mathbf{f} \in \mathcal{C}^p(U_0)$ if $\mathbf{g} \in \mathcal{C}^p(U \times V)$.

Proof. Define $\varphi : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^{n+m}$ by $\varphi(\mathbf{x}, \mathbf{y}) := (\mathbf{x}, \mathbf{g}(\mathbf{x}, \mathbf{y}))$. (i) gives $\varphi(\mathbf{x}_0, \mathbf{y}_0) = (\mathbf{x}_0, \mathbf{0})$. Denote by $I_n \in \mathbb{R}^{n \times n}$ the identity matrix. By (ii), the following matrix is invertible:

$$\varphi'(\mathbf{x}_0, \mathbf{y}_0) = \begin{bmatrix} I_n & \mathbf{0} \\ \frac{\partial \mathbf{g}}{\partial \mathbf{x}}(\mathbf{x}_0, \mathbf{y}_0) & \frac{\partial \mathbf{g}}{\partial \mathbf{y}}(\mathbf{x}_0, \mathbf{y}_0) \end{bmatrix}.$$

By (iii) and the inverse function theorem D.122, there exists an open set $U_0 \times V_0 \subset U \times V$ such that $(\mathbf{x}_0, \mathbf{y}_0) \in U_0 \times V_0$, φ is injective on the open set $W_0 := \varphi(U_0 \times V_0)$, and φ has an inverse $\varphi^{-1} : W_0 \rightarrow U_0 \times V_0$. For any $\mathbf{x} \in U_0$, associate with it $\mathbf{y} \in V_0$ such that $(\mathbf{x}, \mathbf{y}) := \varphi^{-1}(\mathbf{x}, \mathbf{0})$ and thus

$$(*) : (\mathbf{x}, \mathbf{0}) = \varphi \varphi^{-1}(\mathbf{x}, \mathbf{0}) = \varphi(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, \mathbf{g}(\mathbf{x}, \mathbf{y})).$$

Then (a) follows from the uniqueness of $\mathbf{y}$ in (*), which in turn follows from φ being injective and

$$\begin{aligned} & \mathbf{g}(\mathbf{x}, \mathbf{y}_1) = \mathbf{g}(\mathbf{x}, \mathbf{y}_2) \\ \Rightarrow & \varphi(\mathbf{x}, \mathbf{y}_1) = (\mathbf{x}, \mathbf{g}(\mathbf{x}, \mathbf{y}_1)) = (\mathbf{x}, \mathbf{g}(\mathbf{x}, \mathbf{y}_2)) = \varphi(\mathbf{x}, \mathbf{y}_2). \end{aligned}$$

Theorem D.122 also gives $\varphi^{-1} \in \mathcal{C}^1(W_0)$ and $\mathbf{f} \in \mathcal{C}^1(U_0)$. The formula of $\mathbf{f}'$ follows from differentiating $\mathbf{g}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ with respect to $\mathbf{x}$ and applying the chain rule. The last statement follows from (b), (D.34), and the chain rule. $\square$

D.4 Sequential compactness

Definition D.124. A subset K of a metric space $(\mathcal{X}, d)$ is *sequentially compact* iff every sequence $(x_n)_{n \in \mathbb{N}^+} \subset K$ has a convergent subsequence that converges in K , i.e.,

$$\forall (x_n) \subset K, \exists k : \mathbb{N} \rightarrow \mathbb{N}, \exists L \in K \text{ s.t. } \lim_{n \rightarrow +\infty} d(x_{k(n)}, L) = 0, \quad (\text{D.35})$$

where the function k is monotonically increasing.

Lemma D.125. In a sequentially compact metric space $\mathcal{X}$, any Cauchy sequence converges in $\mathcal{X}$.

Proof. By Definition D.124, any Cauchy sequence $\{x_n\}$ has a subsequence $\{x_{n_j}\}$ that converges to some $x \in \mathcal{X}$. Thus

$$\begin{aligned} \forall \epsilon > 0, \exists n_0 \text{ s.t. } \forall m, n \geq n_0, & |x_m - x_n| \leq \frac{\epsilon}{2}; \\ \forall \epsilon > 0, \exists j_0 \text{ s.t. } \forall j \geq j_0, & |x_{n_j} - x| \leq \frac{\epsilon}{2}. \end{aligned}$$

Set $N = \max\{n_0, n_{j_0}\}$ and we have

$$\forall \epsilon > 0, \forall n \geq N, |x_n - x| \leq |x_n - x_N| + |x_N - x| \leq \epsilon,$$

which completes the proof. $\square$

Corollary D.126. Any sequentially compact metric space is complete.

Proof. This follows directly from Lemma D.125 and Definition D.14. $\square$

Lemma D.127. A sequentially compact subset K of a metric space $\mathcal{X}$ is closed and bounded.

Proof. Suppose K is sequentially compact but not bounded. Take an arbitrary point $O \in \mathcal{X}$ and we have

$$\forall n \in \mathbb{N}, \exists \mathbf{x}_n \in K \text{ s.t. } d(\mathbf{x}_n, O) \geq n.$$

Hence for any given point $y \in K$ we have $d(y, O) < \infty$, and thus no subsequence of $(\mathbf{x}_n)_{n \in \mathbb{N}}$ could possibly converge to y , which contradicts the sequential compactness of K .

For any convergent sequence $(\mathbf{x}_n)_{n \in \mathbb{N}} \subset K$, Definition D.124 implies that it has a convergent subsequence that converges in K . The uniqueness of limit (Lemma B.6) dictates that the two sequences converge to the same limit in K . Now that any convergent sequence converges to some limit point in K , Lemma D.65 implies that K is closed. $\square$

D.4.1 In finite-dimensional metric spaces

Example D.128. Any interval $[a, b]$ is sequentially compact in $\mathbb{R}$. Indeed, any sequence in $[a, b]$ is bounded, and by the Bolzano-Weierstrass theorem B.15 it has a convergent subsequence, of which the limit must be in $[a, b]$, thanks to the completeness of $\mathbb{R}$ in Theorem B.17.

Example D.129. (a, b) is not sequentially compact since the sequence $(a + \frac{b-a}{2^n})_{n \in \mathbb{N}^+}$ is contained in (a, b) , but its limit a is not contained in (a, b) .

Example D.130. $\mathbb{R}$ is not sequentially compact because the sequence $(n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ cannot have a convergent subsequence: the distance between any two terms on any subsequence is at least 1.

Lemma D.131. Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. We prove this statement by an induction on n . The induction basis is the Bolzano-Weierstrass theorem (Theorem B.15). Suppose the statement holds for $n \geq 1$. For a bounded sequence $(\mathbf{x}_m)_{m \in \mathbb{N}} \subset \mathbb{R}^{n+1}$, we split each $\mathbf{x}_m$ as $\mathbf{x}_m = (\alpha_m, \beta_m)$, where $\alpha_m \in \mathbb{R}^n$ and $\beta_m \in \mathbb{R}$. Since $\mathbf{x}_m$ is bounded and $\|\alpha_m\|_2 \leq \|\mathbf{x}_m\|_2$, α_m is also bounded. By the induction hypothesis, $(\alpha_m)_{m \in \mathbb{N}}$ has a convergent subsequence, say $(\alpha_{m_k})_{k \in \mathbb{N}}$, that converges to $\alpha \in \mathbb{R}^n$. Then $(\beta_{m_k})_{k \in \mathbb{N}}$ is bounded and by Theorem B.15 it has a convergent subsequence $(\beta_{m_{k_p}})_{p \in \mathbb{N}}$ that converges to $\beta \in \mathbb{R}$. Therefore we have

$$\lim_{p \rightarrow \infty} \mathbf{x}_{m_{k_p}} = \lim_{p \rightarrow \infty} (\alpha_{m_{k_p}}, \beta_{m_{k_p}}) = (\alpha, \beta) \in \mathbb{R}^{n+1},$$

which completes the proof. $\square$

Lemma D.132. A Cauchy sequence in $\mathbb{R}^n$ converges in $\mathbb{R}^n$.

Proof. By Lemma B.11 and n being finite, the Cauchy sequence is bounded and thus, by Lemma D.131, has a convergent subsequence. By arguments similar to those in the proof of Lemma D.125, it converges in $\mathbb{R}^n$. $\square$

Theorem D.133. A subset K of $\mathbb{R}^n$ is sequentially compact if and only if K is closed and bounded.

Proof. The necessity follows from Lemma D.127. As for the sufficiency, any sequence $(\mathbf{x}_n)_{n \in \mathbb{N}} \subset K$ is bounded because K is bounded. By Lemma D.131, $(\mathbf{x}_n)_{n \in \mathbb{N}}$ has a convergent subsequence. Because each term $\mathbf{x}_n \in K$ and K is closed, Lemma D.65 dictates that the limit point of $(\mathbf{x}_n)_{n \in \mathbb{N}}$ be also in K . Then Definition D.124 completes the proof. $\square$

Example D.134. The Cantor set in Definition A.72 is an intersection of closed set and thus it is closed. It is also bounded and thus by Theorem D.133 it is sequentially compact.

D.4.2 In infinite-dimensional metric spaces

Example D.135. The closed unit ball in $(\mathcal{C}[0, 1], \|\cdot\|_\infty)$

$$K := \{f \in \mathcal{C}[0, 1] : \|f\|_\infty \leq 1\} \quad (\text{D.36})$$

is closed and bounded, but not sequentially compact. Consider the hat function

$$B_n(x) = \begin{cases} \frac{x-a_n}{b_n-a_n} & x \in [a_n, b_n], \\ \frac{x-c_n}{b_n-c_n} & x \in [b_n, c_n], \\ 0 & \text{otherwise,} \end{cases} \quad (\text{D.37})$$

where $a_n = 1 - \frac{1}{2^n}$, $c_n = a_{n+1}$, and $b_n = \frac{a_n + c_n}{2}$. Then the sequence $(B_n)_{n \in \mathbb{N}}$ has no convergent subsequence.

Example D.136. The closed unit ball in ℓ^2 ,

$$K := \{\mathbf{x} \in \ell^2 : \|\mathbf{x}\|_2 \leq 1\}, \quad (\text{D.38})$$

is closed and bounded, but not sequentially compact. For

$$\mathbf{e}_n = (0, \dots, 0, 1, 0, \dots) \in K \subset \ell^2$$

where all terms are zero except that the n th term is 1, the sequence $(\mathbf{e}_n)_{n \in \mathbb{N}^+}$ has no convergent subsequence.

Lemma D.137. Any sequentially compact set is totally bounded.

Proof. Suppose M is sequentially compact but not totally bounded. By Definition D.26, there exists $\epsilon > 0$ such that M does not have any finite ϵ -net. Consequently, the induction of choosing any $x_{n+1} \in M \setminus \bigcup_{i=1}^n \mathcal{B}_\epsilon(x_i)$ from the arbitrary basis $x_1 \in M$ leads to an infinite sequence that has no convergent subsequence, which contradicts the sequential compactness of M . $\square$

Lemma D.138. In a complete metric space, a closed and totally bounded set M is sequentially compact.

Proof. We must find a convergent subsequence for any sequence (x_n) . By Definition D.26, we have

$$\exists y_1 \in M \text{ s.t. } (x_n^1) := \mathcal{B}_1(y_1) \cap (x_n) \text{ is an infinite sequence.}$$

With the above as the induction basis, the inductive step

$$\exists y_{m+1} \in \mathcal{B}_{\frac{1}{m}}(y_m) \text{ s.t. } (x_n^{m+1}) := \mathcal{B}_{\frac{1}{m+1}}(y_{m+1}) \cap (x_n^m)$$

yields a sequence $((x_n^m)_{m \in \mathbb{N}})_{n \in \mathbb{N}}$ of infinite sequences. Then we set $z_k := x_k^k$ to construct a diagonal sequence (z_k) , which is Cauchy because

$$\forall \epsilon > 0, \exists N > \frac{2}{\epsilon} \text{ s.t. } \forall j \in \mathbb{N} \\ d(z_N, z_{N+j}) \leq d(z_N, y_N) + d(z_{N+j}, y_N) < \frac{1}{N} + \frac{1}{N} < \epsilon,$$

where the penultimate inequality follows from y_N being the center of the open ball at the N th step of construction. Since the metric space is complete and M is closed, (z_k) converges in M . $\square$

Exercise D.139. Show that the *Hilbert cube* in the metric space ℓ^2 ,

$$C := \left\{ (x_n)_{n \in \mathbb{N}^+} : x_n \in \left[0, \frac{1}{n}\right] \right\}, \quad (\text{D.39})$$

is sequentially compact.

D.4.3 Generalizing the extreme value theorem B.72

Lemma D.140. Let X, Y be metric spaces and let K be a sequentially compact subset of X . If $f : X \rightarrow Y$ is continuous at each $x \in K$, then $f(K)$ is a sequentially compact subset of Y .

Proof. For a sequence $(y_n)_{n \in \mathbb{N}} \subset f(K)$, there exists for each $n \in \mathbb{N}$ an $x_n \in K$ such that $f(x_n) = y_n$. This defines a sequence $(x_n)_{n \in \mathbb{N}} \subset K$. Because K is sequentially compact, Definition D.124 implies the existence of a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ that converges to $L \in K$. Since f is continuous, Definition D.84 implies that $(y_n)_{n \in \mathbb{N}}$ converges to $f(L) \in f(K)$. $\square$

Theorem D.141 (Weierstrass). Suppose K is a nonempty sequentially compact subset of a metric space $\mathcal{X}$ and the function $f : \mathcal{X} \rightarrow \mathbb{R}$ is continuous at each $x \in K$. Then

$$\exists a, b \in K \text{ s.t. } \begin{cases} f(a) = \sup\{f(x) : x \in K\}, \\ f(b) = \inf\{f(x) : x \in K\}. \end{cases}$$

Proof. It suffices to only prove the first clause. By Lemma D.140, $f(K)$ is sequentially compact, and thus by Lemma D.127 $f(K)$ is bounded. $f(K)$ is also nonempty because K is nonempty. Then Theorem A.55 implies that $f(K) \subset \mathbb{R}$ must have a unique supremum

$$M := \sup\{f(x) : x \in K\} \in \mathbb{R},$$

and hence there exists a sequence $(x_n)_{n \in \mathbb{N}} \subset K$ satisfying $\lim_{n \rightarrow \infty} f(x_n) = M$. By Definition D.124, $(x_n)_{n \in \mathbb{N}}$ has a convergent subsequence $(x_{n_k})_{k \in \mathbb{N}}$ that converges to some $c \in K$. Then the continuity of f and Lemma B.10 yield

$$\lim_{k \rightarrow \infty} f(x_{n_k}) = f(c) = M. \quad \square$$

Example D.142. Since the set $K = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\|_2 = 1\}$ is sequentially compact in $\mathbb{R}^3$ and the function $\mathbf{x} \mapsto \sum_{j=1}^3 x_j$ is continuous, the optimization problem

$$\begin{cases} \text{minimize } \sum_{j=1}^3 x_j, \\ \text{subject to } \|\mathbf{x}\|_2 = 1, \end{cases}$$

has a minimizer.

D.4.4 Sequential compactness = compactness

Definition D.143. A cover of a subset K in a metric space $\mathcal{X}$ is a collection α of subsets in $\mathcal{X}$ such that $K \subseteq \bigcup \alpha$. A subcover of K is a subcollection of a cover that also covers K . An open cover is a cover in which each subset is open.

Example D.144. Consider K in (D.20) and $\mathcal{X} = K \cup \{0\}$. An open cover of K in $\mathbb{R}$ is $\{U_n : n \in \mathbb{N}^+\}$ where

$$U_n = \left(\frac{1}{n} - \epsilon_n, \frac{1}{n} + \epsilon_n \right), \quad \epsilon_n := \frac{1}{n(n+1)};$$

elements of this open cover are pairwise disjoint for all $n > 1$. An open cover of $\mathcal{X}$ in $\mathbb{R}$ is $\{U_n : n \in \mathbb{N}^+\} \cup (-\epsilon, \epsilon)$ with $\epsilon := \frac{1}{N}$ for some $N \in \mathbb{N}^+$.

Definition D.145. A subset K of a metric space $\mathcal{X}$ is compact iff every open cover of K contains a finite subcover.

Lemma D.146. A compact subset $M \subset \mathcal{X}$ is closed in $\mathcal{X}$.

Proof. For any $y \in \mathcal{X} \setminus M$, the open cover of M ,

$$\mathcal{B}_y := \{B_{\rho(x)}(x) : x \in M; \rho(x) := \tfrac{1}{2}d(x, y)\},$$

has a finite subcover $\mathcal{B}_n := \{B_{\rho(x_i)}(x_i) : i = 1, 2, \dots, n\}$. Set $\delta := \min_{i=1}^n \rho(x_i)$ and we have, $\forall z \in B_\delta(y)$, $\forall i = 1, 2, \dots, n$,

$$\begin{aligned} d(z, x_i) + d(z, y) &\geq d(x_i, y) \geq 2\delta \\ \implies d(z, x_i) &\geq d(y, x_i) - d(z, y) > \tfrac{1}{2}d(y, x_i), \end{aligned}$$

which implies $z \notin \mathcal{B}_n$ and consequently $B_\delta(y) \subset \mathcal{X} \setminus M$. Therefore $\mathcal{X} \setminus M$ is open and M is closed. $\square$

Theorem D.147. A subset K of a metric space $\mathcal{X}$ is compact if and only if it is sequentially compact.

Proof. Suppose K is compact but not sequentially compact. Then there exists a sequence (x_n) with no convergent subsequence. WLOG, we assume the elements in (x_n) are pairwise distinct. For each $n \in \mathbb{N}^+$ define $S_n := (x_n)_{n \in \mathbb{N}^+} \setminus \{x_n\}$. By Lemma D.65, each S_n is closed since it has no limit points. Thus $\mathcal{X} \setminus S_n$ is open and $\bigcup_{n \in \mathbb{N}^+} (\mathcal{X} \setminus S_n)$ is an open cover of K . The compactness of K dictates that

$$\exists N \in \mathbb{N}^+ \text{ s.t. } K \subset \bigcup_{n=1}^N (\mathcal{X} \setminus S_n) = \mathcal{X} \setminus \bigcap_{n=1}^N S_n,$$

where the last step follows from DeMorgan's law (Theorem A.14). The above equation is false since $x_{N+1} \in K$ but $x_{N+1} \notin (\mathcal{X} \setminus \bigcap_{n=1}^N S_n)$.

For sufficiency, suppose K is sequentially compact but not compact. By Lemma D.137, K is totally bounded. By Definition D.26, for each $n \in \mathbb{N}^+$, K has a finite $\frac{1}{n}$ -net $N_n = \{x_1^{(n)}, x_2^{(n)}, \dots, x_{m(n)}^{(n)}\}$ such that $K \subset \bigcup_{y \in N_n} B_{\frac{1}{n}}(y)$. Since there exists an open cover $\{V_i\}$ of K with no finite subcover of K , there exists at least one $y_n \in N_n$ such that $K \cap B_{\frac{1}{n}}(y_n)$ is not covered by any finite subcollection of $\{V_i\}$. By the sequentially compactness of K , the sequence (y_n) has a convergent subsequence $(y_{k(n)})$ that converges to $y_0 \in K$.

$$(*) : \quad \exists V_0 \in \{V_i\}, \exists \delta > 0 \text{ s.t. } B_\delta(y_0) \subset V_0.$$

Since $\lim_{n \rightarrow \infty} y_{k(n)} = y_0$, we have, for δ in $(*)$,

$$\exists N \in \mathbb{N} \text{ s.t. } n > N \implies d(y_{k(n)}, y_0) < \tfrac{\delta}{2} \text{ and } k(n) > \tfrac{2}{\delta},$$

which further implies,

$$\begin{aligned} \forall x \in B_{\frac{1}{k(n)}}(y_{k(n)}), \quad d(x, y_0) &\leq d(x, y_{k(n)}) + d(y_0, y_{k(n)}) \\ &< \tfrac{1}{k(n)} + \tfrac{\delta}{2} < \delta. \end{aligned}$$

Then $(*)$ yields $B_{\frac{1}{k(n)}}(y_{k(n)}) \subset V_0$, a contradiction. $\square$

D.4.5 Generalizing the uniform continuity theorem B.80

Lemma D.148 (Lebesgue number). Let $\mathcal{A}$ be an open cover of the metric space $(\mathcal{X}, d)$. If $\mathcal{X}$ is compact, we have

$$\exists \delta > 0 \text{ s.t. } \forall K \subset \mathcal{X}, \text{diam}(K) < \delta \implies \exists A \in \mathcal{A} \text{ s.t. } K \subseteq A. \quad (\text{D.40})$$

Proof. Since $\mathcal{X}$ is compact, it has a finite subcover $\{A_i : i = 1, 2, \dots, n\}$. If $\mathcal{X}$ is contained in a single A_i , we are done. Otherwise, for any $x \in \mathcal{X}$, define

$$f(x) := \frac{1}{n} \sum_{i=1}^n \rho(x, \mathcal{X} \setminus A_i),$$

where $\rho(x, A)$ is the distance from a point x to a set A in Definition D.91. Since $\mathcal{X} \setminus A_i \neq \emptyset$ and at least one of the $\rho(x, \mathcal{X} \setminus A_i)$'s is positive, we have $f(x) > 0$ for all $x \in \mathcal{X}$. By the generalized extreme value theorem D.141, $f(x)$ attains its infimum on $\mathcal{X}$, written $\delta := \inf_{x \in \mathcal{X}} f(x)$.

Consider $K \subset \mathcal{X}$ with $\text{diam}(K) < \delta$. For any $x_0 \in K$, we have $\delta \leq f(x_0) < \rho(x_0, \mathcal{X} \setminus A_m)$ where $\rho(x_0, \mathcal{X} \setminus A_i)$ attains its supremum for $i = m$. Therefore, $K \subset B_\delta(x_0) \subset A_m$. $\square$

Definition D.149. The number δ in (D.40) is called a *Lebesgue number* of the open cover $\mathcal{A}$ for $\mathcal{X}$.

Theorem D.150. A continuous map $f : \mathcal{X} \rightarrow \mathcal{Y}$ from a compact metric space $\mathcal{X}$ to another metric space $\mathcal{Y}$ is uniformly continuous.

Proof. For any given $\epsilon > 0$, $\{B_{\frac{\epsilon}{2}}(y) : y \in \mathcal{Y}\}$ is an open cover of $\mathcal{Y}$. By Theorem D.86, any $f^{-1}(B_{\frac{\epsilon}{2}}(y))$ is open in $\mathcal{X}$. All these preimages form an open cover $\mathcal{A}$ of $\mathcal{X}$; and denote by δ a Lebesgue number of $\mathcal{A}$ for $\mathcal{X}$. For any two points $x, z \in \mathcal{X}$ with $d_{\mathcal{X}}(x, z) < \delta$, Lemma D.148 dicates that $\{x, z\}$ be contained in some cover in $\mathcal{A}$, and thus $d_{\mathcal{Y}}(f(x), f(z)) < \epsilon$. The proof is then completed by Definition D.107. $\square$

D.4.6 Partition of unity in $\mathbb{R}^n$

Lemma D.151. For a compact set L of $\mathbb{R}^n$ and an open set U with $L \subset U$, there exists an open set V satisfying $L \subset V \subset \overline{V} \subset U$.

Proof. By the given conditions, L and $U^c = \mathbb{R}^n \setminus U$ are disjoint closed sets. By Corollary D.97, there exist disjoint open sets V and W such that $L \subset V$ and $U^c \subset W$. It follows that $L \subset V \subset \overline{V} \subset W^c \subset U$. $\square$

Lemma D.152. For any compact set K of $\mathbb{R}^n$ and any finite open cover $\{U_i : i = 1, \dots, m\}$ of K , there exists another open cover $\{V_i : i = 1, \dots, m\}$ of K such that $\overline{V_i}$ is compact and $\overline{V_i} \subset U_i$ for each i .

Proof. Since $\{U_1, \dots, U_m\}$ is an open cover of K , we have $K \subset \cup_{i=1}^m U_i$, which implies $K \setminus (\cup_{i=2}^m U_i) \subset U_1$. By Lemma D.151, there exists an open set V_1 such that $\overline{V_1}$ is compact and

$$K \setminus (\cup_{i=2}^m U_i) \subset V_1 \subset \overline{V_1} \subset U_1.$$

Thus $\{V_1, U_2, \dots, U_m\}$ is an open cover of K . The rest of the proof is a straightforward induction on i with the above arguments as the induction basis. $\square$

Theorem D.153. For any compact set K of $\mathbb{R}^n$ and any finite open cover $\{U_i : i = 1, \dots, m\}$ of K , there exist a set of functions $\{f_i : i = 1, \dots, m\}$ such that

$$(a) \quad \forall i = 1, \dots, m, f_i \in \mathcal{C}(\mathbb{R}^n, [0, 1]);$$

$$(b) \quad \forall i = 1, \dots, m, \text{supp}(f_i) \subset U_i;$$

$$(c) \quad \forall x \in K, \sum_{i=1}^m f_i(x) = 1,$$

where $\text{supp}(f_i)$ is the support of the function f_i in Definition 7.29.

Proof. By Lemma D.152, there exist two open covers of K , $\{W_i : i = 1, \dots, m\}$ and $\{V_i : i = 1, \dots, m\}$, such that

$$\forall i = 1, \dots, m, V_i \subset \overline{V_i} \subset W_i \subset \overline{W_i} \subset U_i.$$

By Lemma D.96, for each $i = 1, \dots, m$ there exists a continuous function $g_i : \mathbb{R}^n \rightarrow [0, 1]$ such that

$$g_i(x) := \begin{cases} 1 & \text{if } x \in \overline{V_i}; \\ 0 & \text{if } x \in \mathbb{R}^n \setminus W_i. \end{cases}$$

By Lemma D.89, the function defined by $g := \sum_{i=1}^m g_i$ is continuous. Then by Theorem D.87 and Corollary D.98, the set $g^{-1}(\{0\})$ is closed. By Lemma D.96, there exists a continuous function $h : \mathbb{R}^n \rightarrow [0, 1]$ such that

$$h(x) := \begin{cases} 1 & \text{if } x \in g^{-1}(\{0\}); \\ 0 & \text{if } x \in K, \end{cases}$$

where K and $g^{-1}(\{0\})$ are disjoint. By the above constructions, $g + h$ is everywhere positive. The proof is completed by defining $f_i := \frac{g_i}{g+h}$. $\square$

Definition D.154. A *partition of unity* of a metric space $\mathcal{X}$ is a family of functions $\{f_i : i \in \mathcal{I}\}$ such that

$$(a) \quad \forall i = 1, \dots, m, f_i \in \mathcal{C}(\mathcal{X}, [0, 1]);$$

$$(b) \quad \forall x \in \mathcal{X}, \exists r > 0 \text{ s.t. } \{f_i : i \in \mathcal{I}; f_i(B_r(x)) \neq \{0\}\} \text{ is a finite set};$$

$$(c) \quad \forall x \in \mathcal{X}, \sum_{i \in \mathcal{I}} f_i(x) = 1,$$

where the index set $\mathcal{I}$ may be finite or infinite. This partition of unity is *subordinate to an open cover* $\{U_i : i \in \mathcal{I}\}$ iff

$$\forall i \in \mathcal{I}, \exists j \in \mathcal{I} \text{ s.t. } \text{supp}(f_i) \subset U_j. \quad (\text{D.41})$$

Example D.155. The B-spline functions discussed in Section 7.2 form a partition of unity for any closed interval of $\mathbb{R}$, cf. Corollary 7.51, so do the Bernstein base polynomials, cf. Lemma 5.34.

D.4.7 The Ascoli–Arzelà theorem

Notation 28. $\mathcal{C}^k(X, Y)$ denotes the set of k -times continuously differentiable functions $X \rightarrow Y$. In particular, $\mathcal{C}^k(M)$ denotes the subset of $\mathcal{C}^k(X, Y)$ where $Y = \mathbb{R}$ or $Y = \mathbb{C}$ and $\mathcal{C}(X, Y) := \mathcal{C}^0(X, Y)$ contains all continuous functions from X to Y . When necessary, we also write $\mathcal{C}^k(M, \mathbb{R})$ or $\mathcal{C}^k(M, \mathbb{C})$ to indicate the codomain of the functions.

Definition D.156. For a subset E of a metric space $\mathcal{X}$, a subset F of $\mathcal{C}(E)$ is *uniformly bounded* iff

$$\exists M > 0 \text{ s.t. } \forall x \in E, \forall f \in F, |f(x)| \leq M. \quad (\text{D.42})$$

Lemma D.157. For a compact set M , the metric space $\mathcal{C}(M)$ with the uniform metric is complete.

Proof. The proof is similar to that of Lemma D.17, $\square$

Theorem D.158 (Ascoli–Arzelà). Denote by M a compact metric space and d the uniform metric. A closed set $\mathcal{F}$ in the metric space $(\mathcal{C}(M), d)$ is sequentially compact if and only if $\mathcal{F}$ is uniformly bounded and equicontinuous.

Proof. By Lemma D.157, $(\mathcal{C}(M), d)$ is complete. By Lemmas D.137 and D.138, the compactness of $\mathcal{F}$ is equivalent to total boundedness.

By Lemma D.28 and Definitions D.22 and D.156, total boundedness implies uniform boundedness. For necessity, we also need to show the equicontinuity of $\mathcal{F}$, i.e.,

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall f \in \mathcal{F}, \forall x, y \in M, \\ d_M(x, y) < \delta \implies |f(x) - f(y)| < \epsilon.$$

By Definition D.26, the $\frac{\epsilon}{3}$ -net of $\mathcal{F}$ is finite, say, $N_{\frac{\epsilon}{3}} = \{f_i : i = 1, 2, \dots, n\}$. By Theorem D.150 and the compactness of M , each function in $\mathcal{F}$ is uniformly continuous. Then, for any $f \in \mathcal{F}$ there exists $f_i \in N_{\frac{\epsilon}{3}}$ such that

$$\begin{aligned} & |f(x) - f(y)| \\ & \leq |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| \\ & < \frac{2}{3}\epsilon + |f_i(x) - f_i(y)| < \epsilon, \end{aligned}$$

where the first inequality follows from the triangle inequality, the second from the uniform metric and $f_i \in N_{\frac{\epsilon}{3}}$, and the last from the uniform continuity of f_i and the requirement of $d_M(x, y) < \delta(\frac{\epsilon}{3})$.

For sufficiency, given any $\epsilon > 0$, we need to construct a finite ϵ -net for $\mathcal{F}$ from the conditions of $\mathcal{F}$ being uniformly bounded and equicontinuous. By Definition D.113, we have

$$(*) : \forall \epsilon > 0, \exists \delta(\frac{\epsilon}{3}) > 0 \text{ s.t. } \forall f \in \mathcal{F}, \forall x, y \in M, \\ d_M(x, y) < \delta(\frac{\epsilon}{3}) \implies |f(x) - f(y)| < \frac{\epsilon}{3}.$$

Since M is compact, Lemma D.137 implies that M is totally bounded. For the above $\delta(\frac{\epsilon}{3})$, construct a finite δ -net for M ,

$$N_\delta := \{x_i : i = 1, 2, \dots, m\},$$

and define a function $T : \mathcal{F} \rightarrow \mathbb{R}^m$ by

$$\forall f \in \mathcal{F}, \quad Tf := [f(x_1), f(x_2), \dots, f(x_m)]^T.$$

Since $\mathcal{F}$ is uniformly bounded, we have

$$\begin{aligned} & \exists L > 0 \text{ s.t. } \sup_{f \in \mathcal{F}} |f(M)| < L \\ \implies & \forall f \in \mathcal{F}, \|Tf\|_2 = \left(\sum_{i=1}^m |f(x_i)|^2 \right)^{\frac{1}{2}} \leq \sqrt{m}L. \end{aligned}$$

Thus $T(\mathcal{F})$ is bounded in $\mathbb{R}^m$ and also totally bounded in $\mathbb{R}^m$. From the finite $\frac{\epsilon}{3}$ -net $N_T := \{Tf_i : i = 1, 2, \dots, n\}$ for $T(\mathcal{F})$, we define

$$N_{\mathcal{F}} := \{f_i : i = 1, 2, \dots, n\}$$

and claim that $\{f_i : i = 1, 2, \dots, n\}$ is a finite ϵ -net for $\mathcal{F}$. Indeed, for any $f \in \mathcal{F}$ there exists $f_i \in N_{\mathcal{F}}$ such that

$$\begin{aligned} & |f(x) - f_i(x)| \\ & \leq |f(x) - f(x_k)| + |f(x_k) - f_i(x_k)| + |f_i(x_k) - f_i(x)| \\ & < \frac{2}{3}\epsilon + |f(x_k) - f_i(x_k)| < \frac{2}{3}\epsilon + \|Tf_i - Tf\|_2 < \epsilon, \end{aligned}$$

where the first inequality follows from the triangle inequality with $x_k \in N_\delta$ and $x \in B_\delta(x_k)$, the second from (*), the third from the definition of T , and the fourth from N_T being the finite $\frac{\epsilon}{3}$ -net of $T(\mathcal{F})$. $\square$

Exercise D.159. Denote by $\Omega \subset \mathbb{R}^m$ a bounded open convex set. For $M_1, M_2 \in \mathbb{R}^+$, show that the set

$$\mathcal{F} := \{f \in \mathcal{C}^1(\bar{\Omega}) : \forall x \in \Omega, |f(x)| \leq M_1; \|\nabla f(x)\| \leq M_2\} \quad (\text{D.43})$$

is totally bounded and equicontinuous, but not sequentially compact.

D.5 Function sequences

D.5.1 Pointwise convergence

Definition D.160 (Limiting value of a function). Let $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ be metric spaces. Let E be a subset of $\mathcal{X}$ and $x_0 \in \mathcal{X}$ be a limit point of E . A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is said to *converge* to $L \in \mathcal{Y}$ as x converges to $x_0 \in E$, written $\lim_{x \rightarrow x_0; x \in E} f(x) = L$, iff

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x \in E, \\ & |x - x_0|_{\mathcal{X}} < \delta \implies |f(x) - L|_{\mathcal{Y}} < \epsilon. \end{aligned} \quad (\text{D.44})$$

Definition D.161 (Pointwise convergence). A sequence $(f_n)_{n \in \mathbb{N}^+}$ of functions between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ *converges pointwise* to a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ iff

$$\forall x \in \mathcal{X}, \quad \lim_{n \rightarrow \infty} f_n(x) = f(x), \quad (\text{D.45})$$

or, equivalently,

$$\forall \epsilon > 0, \forall x \in \mathcal{X}, \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n > N, |f_n(x) - f(x)|_{\mathcal{Y}} < \epsilon. \quad (\text{D.46})$$

Example D.162. The function (B.45) can be constructed as the limit of a double sequence,

$$f(x) := \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} (\cos m! \pi x)^{2n}, \quad (\text{D.47})$$

where $m, n \in \mathbb{N}$. Switching the order of the two limits would result in f being not well defined since the double limit would not even exist.

Exercise D.163. Show that the functions in (D.47) and (B.45) are identical.

Example D.164. Consider $f_n : [0, 1] \rightarrow \mathbb{R}$ defined by $f_n(x) := x^n$ and $f : [0, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) := \begin{cases} 1 & \text{if } x = 1; \\ 0 & \text{if } x \in [0, 1). \end{cases}$$

The functions f_n are continuous and converge pointwise to f , which is discontinuous. Hence pointwise convergence does not preserve continuity.

Example D.165. For functions in Example D.164, we have $\lim_{x \rightarrow 1; x \in [0, 1)} x^n = 1$ for all n and $\lim_{x \rightarrow 1; x \in [0, 1)} f(x) = 0$; it follows that

$$\lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0; x \in \mathcal{X}} f_n(x) \neq \lim_{x \rightarrow x_0; x \in \mathcal{X}} \lim_{n \rightarrow \infty} f_n(x).$$

Hence pointwise convergence does not preserve limits.

Example D.166. Consider the interval $[a, b] = [0, 1]$, and the function sequence $f_n : [a, b] \rightarrow \mathbb{R}$ given by

$$f_n(x) := \begin{cases} 2n & \text{if } x \in [\frac{1}{2n}, \frac{1}{n}]; \\ 0 & \text{otherwise.} \end{cases}$$

Then (f_n) converges pointwise to $f(x) = 0$. However, $\int_a^b f_n = 1$ for every n while $\int_a^b f = 0$. Hence

$$\lim_{n \rightarrow \infty} \int_a^b f_n \neq \int_a^b \lim_{n \rightarrow \infty} f_n.$$

Hence pointwise convergence does not preserve integral.

Example D.167. Pointwise convergence does not preserve boundedness. For example, the function sequence

$$f_n(x) = \begin{cases} \exp(x) & \text{if } \exp(x) \leq n; \\ n & \text{if } \exp(x) > n \end{cases} \quad (\text{D.48})$$

converges pointwise to $f(x) = \exp(x)$. Similarly, the function sequence

$$f_n(x) = \begin{cases} \frac{1}{x} & \text{if } x \geq \frac{1}{n}; \\ 0 & \text{if } x \in (0, \frac{1}{n}) \end{cases} \quad (\text{D.49})$$

converges pointwise to $f(x) = \frac{1}{x}$. As another example, the function sequence

$$f_n(x) = n \sin \frac{x}{n} \quad (\text{D.50})$$

converges pointwise to $f(x) = x$.

D.5.2 Uniform convergence

Definition D.168 (Uniform convergence). Let $(f_n)_{n \in \mathbb{N}^+}$ be a sequence of functions between metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$. We say that $(f_n)_{n \in \mathbb{N}^+}$ converges uniformly to another function $f : \mathcal{X} \rightarrow \mathcal{Y}$ on $\mathcal{X}$ iff

$$\forall \epsilon > 0, \exists N \in \mathbb{N}^+ \text{ s.t. } \forall x \in \mathcal{X}, \forall n > N, |f_n(x) - f(x)|_{\mathcal{Y}} < \epsilon. \quad (\text{D.51})$$

The sequence (f_n) is *locally uniformly convergent* to f iff for every point $x \in \mathcal{X}$ there is an $r > 0$ such that $(f_n|_{B_r(x) \cap \mathcal{X}})$ is uniformly convergent to f on $B_r(x) \cap \mathcal{X}$.

Example D.169. Consider $f_n : [0, 1] \rightarrow \mathbb{R}$ with $f_n(x) := \frac{x}{n}$. Then (f_n) converges uniformly to $f(x) = 0$.

Theorem D.170. Uniform convergence implies local uniform convergence, which further implies pointwise convergence.

Proof. This follows directly from (D.46), (D.51), and Theorem A.22. $\square$

Example D.171 (Uniform convergence of Taylor series). Consider the scalar function $f : \mathbb{R} \rightarrow \mathbb{R}$ and the sequence of its Taylor polynomial $(T_n)_{n \in \mathbb{N}^+}$ in Definition B.94. For any interval $I_r := (a - r, a + r)$, $(T_n)_{n \in \mathbb{N}^+}$ converges locally uniformly to $f|_{I_r}$ if r is less or equal to the radius of convergence of f at a .

Exercise D.172. If the radius of convergence of f in Example D.171 is $+\infty$, does $(T_n)_{n \in \mathbb{N}^+}$ converge to f uniformly or locally uniformly?

Criteria

Theorem D.173 (Cauchy criterion). A sequence of functions $(f_n)_{n \in \mathbb{N}^+}$ from a metric space $\mathcal{X}$ to another complete metric space $\mathcal{Y}$ converges uniformly if and only if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m, n > N, d_{\mathcal{Y}}(f_n, f_m) < \epsilon, \quad (\text{D.52})$$

where $d_{\mathcal{Y}}$ is the uniform metric.

Proof. By condition (D.52), the sequence is Cauchy and the necessity follows from Lemma D.13. As for the sufficiency, the proof is similar to the first and final steps of the proof of Lemma D.17. $\square$

Lemma D.174. Suppose a sequence of functions $(f_n)_{n \in \mathbb{N}^+}$ between metric spaces $\mathcal{X}$ and $\mathcal{Y}$ converges pointwise to f . Then $(f_n)_{n \in \mathbb{N}^+}$ converges uniformly to f if and only if $\lim_{n \rightarrow \infty} M_n = 0$ where $M_n := \sup_{x \in \mathcal{X}} |f_n(x) - f(x)|_{\mathcal{Y}}$.

Proof. This follows directly from Definition D.168. $\square$

Consequences

Lemma D.175. Consider a uniformly convergent sequence of functions $(f_n)_{n \in \mathbb{N}^+}$ from a subset E of a metric space $\mathcal{X}$ to a complete metric space $\mathcal{Y}$. If a limit point $x \in E$ satisfies

$$\forall n \in \mathbb{N}^+, \exists y_n \in \mathcal{Y} \text{ s.t. } \lim_{t \rightarrow x} f_n(t) = y_n. \quad (\text{D.53})$$

Then $(y_n)_{n \in \mathbb{N}^+}$ converges in $\mathcal{Y}$ and

$$\lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t) = \lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t). \quad (\text{D.54})$$

Proof. By the uniform convergence of (f_n) to f and the completeness of $\mathcal{Y}$, we have

$$\forall t \in E, \exists f(t) \in \mathcal{Y} \text{ s.t. } \lim_{n \rightarrow \infty} f_n(t) = f(t).$$

By (D.53), $\lim_{t \rightarrow x} f_n(t)$ exists. For the case of $x \notin E$, Theorem D.85 guarantees a unique continuous extension of f_n to $F_n : E \cup \{x\} \rightarrow \mathcal{Y}$ with $y_n := F_n(x)$. By the second step

in the proof of Lemma D.13, f can also be continuously extended to x . Hence the RHS of (D.54) is $f(x)$. By Lemma D.13, (f_n) is Cauchy:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall t \in E, \forall n, m > N, |f_n(t) - f_m(t)| < \epsilon.$$

The limit $t \rightarrow x$ of the above equation implies that (y_n) be Cauchy. The completeness of $\mathcal{Y}$ and the limit of $m \rightarrow \infty$ further imply $\lim_{n \rightarrow \infty} y_n = f(x)$. $\square$

Corollary D.176. Suppose a sequence of continuous functions $(f_n)_{n \in \mathbb{N}^+}$ defined on a subset E of a metric space $\mathcal{X}$ converges uniformly to f . Then f is continuous on E .

Proof. This follows directly from Lemma D.175 and Definition D.82. $\square$

Corollary D.177. Denote by $\mathcal{C}(\mathcal{X})$ the set of all bounded continuous functions $\mathcal{X} \mapsto \mathbb{C}$ where $\mathcal{X}$ is a metric space. Then the following is a complete metric space,

$$(\mathcal{C}(\mathcal{X}), d) \quad \text{where } d(f, g) = \sup_{t \in \mathcal{X}} |f(t) - g(t)|. \quad (\text{D.55})$$

Proof. This follows from Corollary D.176. $\square$

Theorem D.178. Suppose a sequence of scalar functions $(f_n)_{n \in \mathbb{N}^+}$ converges uniformly to $f : [a, b] \rightarrow \mathbb{R}$. If each f_n is Riemann integrable on $[a, b]$, then so is f . Furthermore,

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n. \quad (\text{D.56})$$

Proof. Define $\epsilon_n := \sup_{x \in [a, b]} |f_n(x) - f(x)|$. Lemma D.174 and the uniform convergence of (f_n) yield $\lim_{n \rightarrow \infty} \epsilon_n = 0$. Then we have

$$\begin{aligned} & \forall n \in \mathbb{N}^+, f_n - \epsilon_n \leq f \leq f_n + \epsilon_n. \\ \Rightarrow & \int_a^b (f_n - \epsilon_n) \leq \int_a^b f \leq \int_a^b (f_n + \epsilon_n). \\ \Rightarrow & \left| \int_a^b f - \int_a^b f_n \right| \leq 2\epsilon_n(b-a) \\ \Rightarrow & f \text{ is integrable on } [a, b], \end{aligned}$$

where the second line follows from Definition B.115, the third from the integral mean value theorem B.124, and the last from $\lim_{n \rightarrow \infty} \epsilon_n = 0$ and Theorem B.117. Now that f is integrable, the second line of the above equation implies

$$\left| \int_a^b f - \int_a^b f_n \right| \leq \epsilon_n(b-a),$$

which yields (D.56). $\square$

Example D.179. Consider the functions $f_n : [0, 2\pi] \rightarrow \mathbb{R}$ defined by $f_n(x) := n^{-\frac{1}{2}} \sin(nx)$. Define $f : [0, 2\pi] \rightarrow \mathbb{R}$ by $f(x) := 0$. We have

$$\forall \epsilon > 0, \exists N = \frac{1}{\epsilon^2} \text{ s.t. } n > N \Rightarrow |f_n(x) - f(x)| \leq n^{-\frac{1}{2}} < \epsilon.$$

Hence (f_n) converges uniformly to f . However,

$$f'_n(x) = n^{\frac{1}{2}} \cos(nx), \quad |f'_n(x) - f'(x)| = n^{\frac{1}{2}} |\cos(nx)|.$$

Thus f'_n does not converge pointwise to f' , letting alone uniform convergence. In other words, we have

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) \neq \lim_{n \rightarrow \infty} \frac{df_n(x)}{dx}.$$

Example D.180. Consider the functions $f_n : [-1, 1] \rightarrow \mathbb{R}$ defined by $f_n(x) := \sqrt{\frac{1}{n^2} + x^2}$. Define $f : [-1, 1] \rightarrow \mathbb{R}$ by $f(x) := |x|$. We have

$$\forall x \in [-1, 1], \quad |x| \leq f_n(x) \leq |x| + \frac{1}{n}.$$

Hence by the squeeze test, (f_n) converges uniformly to f . But f is not differentiable at $x = 0$.

Theorem D.181. For a sequence of differentiable scalar functions (f_n) , suppose that $(f_n(x_0))$ converges for some $x_0 \in [a, b]$ and (f'_n) converges uniformly on (a, b) . Then (f_n) converges uniformly on $[a, b]$ to a function $f : [a, b] \rightarrow \mathbb{R}$ and

$$\forall x \in [a, b], \quad f'(x) = \lim_{n \rightarrow \infty} f'_n(x). \quad (\text{D.57})$$

Proof. The convergence of $(f_n(x_0))$ and the uniform convergence of (f'_n) yield

$$\begin{aligned} & \forall \epsilon > 0, \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n, m > N, \\ (*) : & \begin{cases} |f_n(x_0) - f_m(x_0)| < \frac{\epsilon}{2}, \\ \forall t \in [a, b], |f'_n(t) - f'_m(t)| < \frac{\epsilon}{2(b-a)}. \end{cases} \end{aligned}$$

For the same m, n , and ϵ above, the second clause in $(*)$ and the application of the mean value theorem B.92 to $f_n - f_m$ yield, $\forall x, t \in [a, b]$,

$$(**) : |f_n(x) - f_m(x) - f_n(t) + f_m(t)| \leq \frac{\epsilon |x - t|}{2(b-a)} \leq \frac{\epsilon}{2},$$

which, together with the first clause in $(*)$ and the triangle inequality, gives

$$\forall \epsilon > 0, \exists N \in \mathbb{N}^+ \text{ s.t. } n, m > N \Rightarrow \sup_{x \in [a, b]} |f_n(x) - f_m(x)| < \epsilon.$$

Hence the sequence (f_n) is Cauchy with respect to the uniform metric and Theorem D.173 yields its uniform convergence to some function f given by $f(x) := \lim_{n \rightarrow \infty} f_n(x)$.

For any given point $x \in [a, b]$, construct scalar functions

$$\forall t \in [a, b], \quad \phi_n(t) := \frac{f_n(t) - f_n(x)}{t - x}, \quad \phi(t) := \frac{f(t) - f(x)}{t - x}.$$

The first inequality in $(*)$ implies that the sequence (ϕ_n) is Cauchy and Theorem D.173 further implies its uniform convergence. Apply Theorem D.175 to (ϕ_n) and we have

$$\lim_{t \rightarrow x} \phi(t) = \lim_{t \rightarrow x} \lim_{n \rightarrow \infty} \phi_n = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} \phi_n = \lim_{n \rightarrow \infty} f'_n(x),$$

which yields (D.57). $\square$

Exercise D.182. Give a different proof of Theorem D.181 with the strengthened condition of each f_n being *continuously* differentiable.

Exercise D.183. Show that the Weierstrass function

$$f(x) = \sum_{n=1}^{\infty} 4^{-n} \cos(32^n \pi x) \quad (\text{D.58})$$

is uniformly convergent to a continuous function, but it is nowhere differentiable.

Compactness $\implies$ equicontinuity

Theorem D.184. For a compact metric space K , a uniformly convergent sequence of functions $(f_n)_{n \in \mathbb{N}^+} \subset \mathcal{C}(K)$ is equicontinuous on K .

Proof. The uniform convergence implies

$$\forall \epsilon > 0, \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n \geq N, \forall x \in K, |f_n(x) - f_N(x)| < \frac{1}{3}\epsilon.$$

By Theorem D.150, each continuous function in (f_n) has

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall i = 1, 2, \dots, N, \\ d(x, y) < \delta \implies |f_i(x) - f_i(y)| < \frac{1}{3}\epsilon.$$

The above two equations yield

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall n \in \mathbb{N}^+, \forall x, y \in K, \\ d(x, y) < \delta \implies |f_n(x) - f_n(y)| \\ < |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \epsilon.$$

The proof is then completed by Definition D.113. $\square$

D.5.3 Pointwise and uniform boundedness

Definition D.185. Let E be a subset of a metric space $\mathcal{X}$. A sequence of complex-valued functions $(f_n : E \rightarrow \mathbb{C})$ is *pointwise bounded* on E if the sequence $(f_n(x))$ is bounded for any $x \in E$, i.e.,

$$\exists \phi : E \rightarrow \mathbb{R}^+, \text{ s.t. } \forall x \in E, \forall n \in \mathbb{N}, |f_n(x)| \leq \phi(x); \quad (\text{D.59})$$

it is *uniformly bounded* on E iff

$$\exists M \in \mathbb{R}^+, \text{ s.t. } \forall x \in E, \forall n \in \mathbb{N}, |f_n(x)| \leq M. \quad (\text{D.60})$$

Example D.186. The function sequence $(f_n) \subset \mathcal{C}[0, 1]$ given by

$$f_n(x) = \frac{x^2}{x^2 + (1 - nx)^2} \quad (\text{D.61})$$

is uniformly bounded because $|f_n(x)| \leq 1$. Although (f_n) converges pointwise to $\mathbf{0}$, no subsequence of (f_n) can converge uniformly because $f_n(\frac{1}{n}) = 1$. Therefore, a uniformly bounded sequence of functions needs not to have a uniformly convergent subsequence.

Lemma D.187. Any pointwise bounded function sequence $(f_n : E \rightarrow \mathbb{C})$ over a countable set E has a subsequence (f_{n_k}) such that $(f_{n_k}(x))$ converges for any $x \in E$.

Proof. Arrange the points of E in a sequence: $E = (x_n)$. For any fixed point x_i , the sequence $(f_n(x_i))_{n \in \mathbb{N}^+}$ is bounded and thus by the Bolzano-Weierstrass Theorem B.15 it has a convergent subsequence $(f_{i,k}(x_i))$. Construct an array of functions

$$\begin{array}{cccccc} S_1 : & f_{1,1} & f_{1,2} & f_{1,3} & f_{1,4} & \cdots \\ S_2 : & f_{2,1} & f_{2,2} & f_{2,3} & f_{2,4} & \cdots \\ S_3 : & f_{3,1} & f_{3,2} & f_{3,3} & f_{3,4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

that satisfies

$$(a) \quad \forall n = 2, 3, 4, \dots, S_n \subset S_{n-1};$$

$$(b) \quad \text{the ordering of functions in } S_n \text{ is the same as that in } S_{n-1},$$

which, together with the convergence of $(f_{i,k}(x_i))$, imply the convergence of the diagonal sequence $(f_{n,n}(x))$ for any $x \in E$. $\square$

Theorem D.188 (Ascoli). Suppose a scalar function sequence $(f_n) \subset \mathcal{C}(K)$ over a compact set K is pointwise bounded and equicontinuous. Then (f_n) is uniformly bounded on K and contains a uniformly convergent subsequence.

Proof. (f_n) being pointwise bounded implies

$$\exists \phi : K \rightarrow \mathbb{R}^+, \text{ s.t. } \forall x \in K, \forall n \in \mathbb{N}^+, |f_n(x)| < \phi(x).$$

The equicontinuity of (f_n) gives

$$\forall \epsilon > 0, \exists \delta(\frac{\epsilon}{3}) > 0 \text{ s.t. } \forall n \in \mathbb{N}^+, \forall x, y \in K, \\ d_K(x, y) < \delta \implies |f_n(x) - f_n(y)| < \frac{\epsilon}{3}.$$

For the above δ , Lemma D.137 and Definition D.26 imply that K have a finite δ -net $N_\delta := \{x_1, \dots, x_m\}$ satisfying

$$\forall x \in K, \exists x_s \in N_\delta \text{ s.t. } \forall n \in \mathbb{N}^+, \\ |f_n(x)| \leq |f_n(x) - f_n(x_s)| + |f_n(x_s)| \leq \frac{\epsilon}{3} + \max_{1 \leq i \leq m} |\phi(x_i)|,$$

where the first inequality follows from the triangle inequality with $x_s \in N_\delta$ and $x \in B_\delta(x_s)$ and the second from the equicontinuity and pointwise boundedness of (f_n) . Thus (f_n) is uniformly bounded on K .

By Theorem D.147, Lemma D.137, Lemma D.31, and Definition D.30, K has a countable dense subset E . Then Lemma D.187 implies that (f_n) has a subsequence, say (g_n) , that converges for any $x \in E$. It suffices to show the uniform convergence of (g_n) . Assume WLOG $N_\delta \subset E$ and we have

$$\exists N \in \mathbb{N}^+ \text{ s.t. } \forall n, m > N, \forall x \in K, |g_n(x) - g_m(x)| \\ \leq |g_n(x) - g_n(x_s)| + |g_n(x_s) - g_m(x_s)| + |g_m(x_s) - g_m(x)| \\ < \frac{2}{3}\epsilon + |g_n(x_s) - g_m(x_s)| < \epsilon,$$

where the first inequality follows from the triangle inequality with $x_s \in N_\delta$ and $x \in B_\delta(x_s)$, the second from the equicontinuity of (g_n) , and the last from the convergence of $(g_n(x))$ for any $x \in E$. Therefore (g_n) is a Cauchy sequence in K and the proof is completed by Lemma D.157. $\square$

Corollary D.189. For an equicontinuous sequence $(f_n) \subset \mathcal{C}(K)$ of scalar functions over a compact set K , pointwise boundedness is equivalent to uniform boundedness.

Proof. This follows directly from Theorem D.188 and Definition D.185. $\square$

D.5.4 The Stone–Weierstrass theorem

Definition D.190. An algebra of scalar functions is a algebra $\mathcal{A}$ (as in Definition C.6) closed under addition, multiplication, and scalar multiplication, i.e.,

$$\forall f, g \in \mathcal{A}, \forall c \in \mathbb{C}, f + g \in \mathcal{A}, fg \in \mathcal{A}, cf \in \mathcal{A}, \quad (\text{D.62})$$

where fg is the multiplication of function values in $\mathbb{C}$.

Definition D.191. An algebra of scalar functions is *uniformly closed* iff any sequence $(f_n) \subset \mathcal{A}$ that is uniformly convergent converges to some $f \in \mathcal{A}$.

Definition D.192. The *uniform closure* of an algebra $\mathcal{A}$ of scalar functions is the set of all functions that are limits of uniformly convergent sequences of functions in $\mathcal{A}$.

Example D.193. The Weierstrass approximation theorem 5.38 states that $\mathcal{C}([a, b], \mathbb{R})$ is the uniform closure of the set of polynomials on $[a, b]$.

Lemma D.194. The uniform closure $\mathcal{B}$ of any algebra $\mathcal{A}$ of bounded scalar functions is a uniformly closed algebra.

Proof. By Definition D.192, we have $\mathcal{A} \subset \mathcal{B}$ since any function sequence $(f_n = f)$ converges uniformly to f . For any $f \in \mathcal{B}$ and $g \in \mathcal{B}$, Definition D.192 implies the existence of uniformly convergent sequences $(f_n) \subset \mathcal{A}$ and $(g_n) \subset \mathcal{A}$ that converge to f and g , respectively. Since all functions are bounded, it is straightforward to show, $\forall c \in \mathbb{C}$,

$$\lim_{n \rightarrow \infty} (f_n + g_n) = f + g, \quad \lim_{n \rightarrow \infty} (f_n g_n) = fg, \quad \lim_{n \rightarrow \infty} (cf_n) = cf.$$

Then Definition D.192 implies $f + g \in \mathcal{B}$, $fg \in \mathcal{B}$, and $cf \in \mathcal{B}$. The proof is completed by Definitions D.190 and D.191. $\square$

Lemma D.195. For any interval $[-a, a]$ there exists a sequence (p_n) of real polynomials such that $p_n(0) = 0$ and (p_n) converges uniformly to $|x|$ on $[-a, a]$.

Proof. By the Weierstrass approximation theorem 5.38, there exists a sequence (q_n) of real polynomials that converges uniformly to $|x|$ on $[-a, a]$. The proof is completed by setting $p_n := q_n - q_n(0)$. $\square$

Definition D.196. A set $\mathcal{A}$ of scalar functions over E is said to *separate points* on E iff

$$\forall x_1, x_2 \in E, x_1 \neq x_2 \implies \exists f \in \mathcal{A} \text{ s.t. } f(x_1) \neq f(x_2).$$

Example D.197. The set of all even polynomials does not separate points on $[-1, 1]$.

Example D.198. The algebra of continuous complex-valued periodic functions on $[-\pi, \pi]$ does not separate points, neither does that of complex-valued trigonometric polynomials $\sum_{n=-N}^{+N} c_n e^{inx}$: every function f in question has $f(-\pi) = f(\pi)$.

Definition D.199. A set $\mathcal{A}$ of scalar functions on E is said to *vanish at no point* of E iff

$$\forall x \in E, \exists f \in \mathcal{A} \text{ s.t. } f(x) \neq 0.$$

Example D.200. The set of all polynomials separate points on $E := [-a, a]$ and vanishes at no point of E for any $a \in \mathbb{R}^+$.

Lemma D.201. Suppose an algebra $\mathcal{A}$ of scalar functions on E separates points on E and vanishes at no point of E . Then we have

$$\forall x, y \in E, \forall \xi, \eta \in \mathbb{F}, \exists f \in \mathcal{A} \text{ s.t. } f(x) = \xi, f(y) = \eta,$$

where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and $\xi = \eta$ if $x = y$.

Proof. By Definitions D.196 and D.199, there exist functions $g, h, k \in \mathcal{A}$ such that $g(x) \neq g(y)$, $h(x) \neq 0$, $k(y) \neq 0$. Define $u := gk - g(x)k$ and $v := gh - g(y)h$. Definition D.190 implies $u \in \mathcal{A}$, $v \in \mathcal{A}$, $u(x) = v(y) = 0$, $u(x) \neq 0$, and $v(x) \neq 0$. Then the proof is completed by the construction

$$f := \frac{\xi v}{v(x)} + \frac{\eta u}{u(y)}. \quad \square$$

Lemma D.202. Suppose an algebra $\mathcal{A}$ of continuous scalar functions on a compact set K separates points on K and vanishes at no point of K . For the uniform closure $\mathcal{B}$ of $\mathcal{A}$, we have

- (a) $f \in \mathcal{B} \implies |f| \in \mathcal{B}$;
- (b) $f, g \in \mathcal{B} \implies \max(f, g) \in \mathcal{B}$ and $\min(f, g) \in \mathcal{B}$,

where $h = \max(f, g)$ is defined as

$$h(x) := \begin{cases} f(x) & \text{if } f(x) \geq g(x); \\ g(x) & \text{otherwise} \end{cases} \quad (\text{D.63})$$

and $\min(f, g)$ is defined similarly.

Proof. By Definition D.141, $f \in \mathcal{B}$ and K being compact imply that f attains its maximum and minimum on K . Write $a := \max_{x \in K} |f(x)|$.

Consider any given $\epsilon > 0$. Lemma D.195 yields

$$\exists c_1, \dots, c_n \in \mathbb{F} \text{ s.t. } \forall y \in [-a, a], \left| \sum_{i=1}^n c_i y^i - |y| \right| < \epsilon.$$

By Lemma D.194, $\mathcal{B}$ is an algebra and thus the function $g := \sum_{i=1}^n c_i f^i$ is also in $\mathcal{B}$. Therefore, we have $\max_{x \in K} |g(x) - |f(x)|| < \epsilon$ and this proves (a). Then (b) follows from (a) and the identities

$$\begin{cases} \max(f, g) = \frac{f+g}{2} + \frac{|f-g|}{2}, \\ \min(f, g) = \frac{f+g}{2} - \frac{|f-g|}{2}. \end{cases} \quad \square$$

Theorem D.203 (Stone–Weierstrass). Suppose an algebra $\mathcal{A}$ of continuous real-valued functions on a compact set K separates points on K and vanishes at no point of K . Then the uniform closure $\mathcal{B}$ of $\mathcal{A}$ consists of all real continuous functions on K . In other words, $\mathcal{B} = \mathcal{C}(K, \mathbb{R})$ and $\mathcal{A}$ is dense in $\mathcal{B}$.

Proof. By Definition D.192, it suffices to prove

$$\forall f \in \mathcal{C}(K, \mathbb{R}), \forall \epsilon > 0, \exists h \in \mathcal{B} \text{ s.t. } \max_{t \in K} |h(t) - f(t)| < \epsilon,$$

which we show in two steps. In the first step, we prove

$$(*) : \forall x \in K, \exists g_x \in \mathcal{B} \text{ s.t. } \begin{cases} g_x(x) = f(x); \\ \forall t \in K, g_x(t) > f(t) - \epsilon. \end{cases}$$

Indeed, Lemma D.201 implies

$$\forall x, y \in K, \exists h_y \in \mathcal{B} \text{ s.t. } h_y(x) = f(x); h_y(y) = f(y).$$

By Theorem D.150, the continuity of h_y and the compactness of K yield the uniform continuity of h_y . Thus

$$\exists \delta_y > 0 \text{ s.t. } \forall u, v \in K, d(u, v) < \delta_y \implies |h_y(u) - h_y(v)| < \epsilon.$$

By Definition D.145, Theorem D.147, Lemma D.137, and Definition D.26, K has a finite $\delta_{\min}$ -net $\{y_1, \dots, y_n\}$ such that the set of open balls $\{B_{\delta_{\min}}(y_i) : i = 1, \dots, n\}$ cover K where $\delta_{\min} = \min(\delta_{y_1}, \dots, \delta_{y_n})$ and each δ_{y_i} satisfies the above equation. By Lemma D.202 and Theorem D.150,

$$(\triangle) : g_x := \max(h_{y_1}, h_{y_2}, \dots, h_{y_n})$$

satisfies $(*)$ and is uniformly continuous.

In the second step, $(*)$ and the continuity of g_x yield

$$\forall x \in K, \exists \delta_x > 0 \text{ s.t. } \forall t \in B_{\delta_x}(x), g_x(t) < f(t) + \epsilon.$$

By arguments similar to those in the first step, K has a finite η -net $\{x_1, \dots, x_m\}$ such that $K \subset \cup_{i=1}^m B_{\eta}(x_i)$ and

$$\forall i = 1, \dots, m, \forall u, v \in B_{\eta}(x_i), |g_{x_i}(u) - g_{x_i}(v)| < \epsilon.$$

Then the function $h : K \rightarrow \mathbb{R}$ given by

$$h := \min(g_{x_1}, g_{x_2}, \dots, g_{x_m})$$

satisfies $h(t) < f(t) + \epsilon$ for any $t \in K$. Meanwhile $(\triangle)$ implies $h(t) > f(t) - \epsilon$ for any $t \in K$. The combination of these two inequalities completes the proof. $\square$

Exercise D.204. Theorem D.203 cannot be generalized directly to the set of continuous complex-valued functions. Denote by $\mathbb{S}^1$ the unit circle in the complex plane and $\mathcal{A}$ the algebra of all functions of the form $f(e^{i\theta}) = \sum_{n=0}^N c_n e^{in\theta}$ where $\theta \in \mathbb{R}$. Show that $\mathcal{A}$ separates points on $\mathbb{S}^1$, vanishes at no points of $\mathbb{S}^1$, but there still exists continuous functions $\mathbb{S}^1 \rightarrow \mathbb{C}$ that do not belong to the uniform closure of $\mathcal{A}$.

Corollary D.205. Suppose an algebra $\mathcal{A}$ of continuous complex-valued functions on a compact set K separates points on K , vanishes at no point of K , and is closed under complex conjugation, $f(x) := \overline{f(x)}$. Then the uniform closure $\mathcal{B}$ of $\mathcal{A}$ consists of all complex continuous functions on K . In other words, $\mathcal{B} = \mathcal{C}(K, \mathbb{C})$ and $\mathcal{A}$ is dense in $\mathcal{B}$.

Proof. Denote by $\mathcal{A}_{\mathbb{R}}$ the set of real-valued functions in $\mathcal{A}$. For any $f = u + iv \in \mathcal{A}$, we have $u = \frac{1}{2}(f + \overline{f}) \in \mathcal{A}_{\mathbb{R}}$. For disjoint $x, y \in K$, Lemma D.201 implies that there exists $f \in \mathcal{A}$ such that $f(x) = 1 \neq 0 = f(y)$, i.e., $u(x) = 1 \neq 0 = u(y)$. By Definition D.196, $\mathcal{A}_{\mathbb{R}}$ separates points on K . Furthermore,

$$\forall x \in K, \exists g \in \mathcal{A}, \lambda \in \mathbb{C} \text{ s.t. } \lambda g(x) > 0.$$

Then the function $f = \lambda g = u + iv$ satisfies $u(x) > 0$ and $v(x) = 0$. By Definition D.199, $\mathcal{A}_{\mathbb{R}}$ vanishes at no points of K . Then Theorem D.203 dictates that every continuous real-valued function on K belong to $\mathcal{B}_{\mathbb{R}}$, the uniform closure of $\mathcal{A}_{\mathbb{R}}$.

Finally, we show $\mathcal{B} = \mathcal{C}(K, \mathbb{C})$. For any $f = u + iv \in \mathcal{C}(K, \mathbb{C})$, we have $u, v \in \mathcal{B}_{\mathbb{R}} \subset \mathcal{B}$. Then Definition D.190 implies that $f \in \mathcal{B}$. Conversely, $f \in \mathcal{A}$ implies $f \in \mathcal{C}(K, \mathbb{C})$. By Definitions D.124 and D.192, every uniformly convergent sequence $(f_n) \subset \mathcal{A}$ converges in $\mathcal{C}(K, \mathbb{C})$. $\square$

Example D.206. To fix the problem of complex trigonometric polynomials in Example D.198, we change their domain to $\mathbb{S}^1$ parameterized by $e^{i\theta}$ with $\theta \in [0, 2\pi)$. Then $\mathbb{S}^1$ is compact in $\mathbb{C}$. The algebra of complex trigonometric polynomials restricted to $\mathbb{S}^1$ separates points, vanishes at no points of $\mathbb{S}^1$, and thus by Corollary D.205 is dense in the space of complex-valued continuous periodic functions.

Definition D.207. A k -cell I^k is a compact set in $\mathbb{R}^k$ of the form

$$I^k = \{(x_1, \dots, x_k) : \forall i = 1, \dots, k, x_i \in [a_i, b_i]\}. \quad (\text{D.64})$$

Lemma D.208. Consider functions $f : I^k \rightarrow \mathbb{R}$ satisfying

$$\exists f_i \in [a_i, b_i] \rightarrow \mathbb{R} \text{ s.t. } f(\mathbf{x}) = \prod_{i=1}^k f_i(x_i). \quad (\text{D.65})$$

The set $\mathcal{A}$ of all finite sums of such functions form an algebra and is dense in $\mathcal{C}(I^k, \mathbb{R})$.

Proof. By Definitions D.190, D.196, and D.199, it is straightforward to verify that $\mathcal{A}$ is indeed an algebra of continuous real-valued functions, that $\mathcal{A}$ separates points in I^k , and that $\mathcal{A}$ vanishes at point of I^k . Then the proof is completed by the Stone–Weierstrass theorem D.203. $\square$

Corollary D.209. For any continuous function $f : I^m \times I^n \rightarrow \mathbb{R}$, we have

$$\int_{I^m} \int_{I^n} f = \int_{I^n} \int_{I^m} f, \quad (\text{D.66})$$

where $\int_{I^n}$ denotes the Riemann integral over the n -cell I^n .

Proof. This follows directly from Lemma D.208 and an iteration of integrals over one-dimensional intervals. $\square$

Appendix E

Banach Spaces

E.1 Normed spaces

E.1.1 As special metric spaces

Definition E.1. The *ambient metric space* of a normed space $(V, \|\cdot\|)$ is the metric space (V, d) where the metric d is given by $d(x, y) = \|x - y\|$.

Exercise E.2. Show that the ambient metric space is indeed a metric space.

Definition E.3. A sequence (u_n) in a normed space $(V, \|\cdot\|)$ is *convergent* to u iff (u_n) converges to u in the ambient complete metric space as in Definition D.11.

Lemma E.4. Any open ball in a normed space is a convex set, cf. Definitions 4.23 and D.9.

Proof. $\alpha \in [0, 1]$ and $\mathbf{x}, \mathbf{y} \in B_r(\mathbf{0})$ imply $\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in B_r(\mathbf{0})$ because

$$\begin{aligned} \|\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}\| &\leq \|\alpha\mathbf{x}\| + \|(1 - \alpha)\mathbf{y}\| \\ &\leq \alpha\|\mathbf{x}\| + (1 - \alpha)\|\mathbf{y}\| < \alpha r + (1 - \alpha)r = r, \end{aligned}$$

where we have applied properties of norms. $\square$

Example E.5. Consider the complementary case to Example C.184: is $(\mathbb{R}^n, \|\cdot\|_p)$ a normed space for $p \in (0, 1)$? By Lemma E.4, it is not a norm if any open ball is not convex. For $p = \frac{1}{2}$, the boundary of the open ball $\{\mathbf{x} : \|\mathbf{x}\|_{\frac{1}{2}} < 1\}$ with $\|\mathbf{x}\|_{\frac{1}{2}} := (\sqrt{x_1} + \sqrt{x_2})^2$ shown in Figure E.1 bounds a concave region and hence $\|\cdot\|_p$ is not a norm for $p \in (0, 1)$.

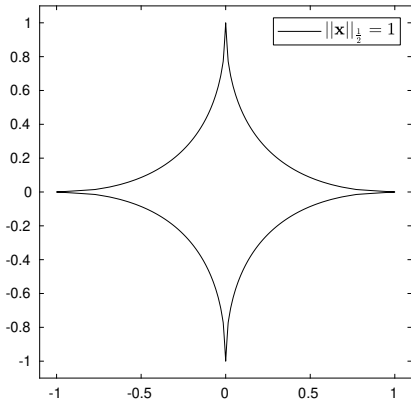


Figure E.1: $(\mathbb{R}^n, \|\cdot\|_p)$ with $p \in (0, 1)$ is not a normed space.

Definition E.6. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set. The *p-norm of a continuous scalar function* in the linear space $\mathcal{C}(\overline{\Omega})$ is

$$\forall v \in \mathcal{C}(\overline{\Omega}), \quad \|v\|_p := \left[\int_{\Omega} |v(\mathbf{x})|^p d\mathbf{x} \right]^{\frac{1}{p}} \quad (\text{E.1})$$

and the ∞ -norm or *maximum norm* is given by

$$\forall v \in \mathcal{C}(\overline{\Omega}), \quad \|v\|_{\infty} := \max_{\mathbf{x} \in \overline{\Omega}} |v(\mathbf{x})|. \quad (\text{E.2})$$

Example E.7. $(\mathcal{C}(\overline{\Omega}), \|\cdot\|_{\infty})$ in Definition E.6 is a normed space, so is $(\mathcal{C}(\overline{\Omega}), \|\cdot\|_p)$ for any $p \in [1, \infty)$.

Example E.8. For the ℓ^{∞} sequence space in (D.15),

$$\ell^{\infty} := \left\{ (a_n)_{n \in \mathbb{N}} : \sup_{n \in \mathbb{N}} |a_n| < \infty \right\},$$

define $\|\cdot\|_{\infty} : \ell^{\infty} \rightarrow \mathbb{R}^+ \cup \{0\}$ as

$$\|(a_n)_{n \in \mathbb{N}}\|_{\infty} = \sup_{n \in \mathbb{N}} |a_n|. \quad (\text{E.3})$$

Then $(\ell^{\infty}, \|\cdot\|_{\infty})$ is a normed space.

Example E.9. For the ℓ^p space in (D.16) with $p \in [1, \infty)$,

$$\ell^p := \left\{ (a_n)_{n \in \mathbb{N}} : a_n \in \mathbb{C}; \sum_{n \in \mathbb{N}} |a_n|^p < \infty \right\},$$

we have

$$\begin{aligned} &\left\{ \begin{array}{l} (a_n)_{n \in \mathbb{N}} \in \ell^p, (b_n)_{n \in \mathbb{N}} \in \ell^p \\ |a + b|^p \leq (|a| + |b|)^p \leq 2^p (\max(|a|, |b|))^p \leq 2^p (|a|^p + |b|^p) \end{array} \right. \\ &\implies (a_n)_{n \in \mathbb{N}} + (b_n)_{n \in \mathbb{N}} \in \ell^p, \end{aligned}$$

where the comparison test is applied.

Then $(\ell^p, \|\cdot\|_p)$ is a normed space where

$$\|(a_n)_{n \in \mathbb{N}}\|_p := \left(\sum_{n=1}^{\infty} |a_n|^p \right)^{\frac{1}{p}}. \quad (\text{E.4})$$

E.1.2 Continuous maps of normed spaces

Lemma E.10. The norm function $\|\cdot\|$ is continuous.

Proof. By Definition D.161, we have $\lim_{n \rightarrow \infty} \|u_n - u\| = 0$ from $\lim_{n \rightarrow \infty} u_n = u$. The rest of the proof follows from the backward triangle inequality (C.68). $\square$

Lemma E.11. The vector norms in Definition C.176 satisfy a monotonicity property:

$$1 \leq p \leq q \leq \infty \Rightarrow \forall \mathbf{x} \in \mathbb{R}^n \quad \|\mathbf{x}\|_q \leq \|\mathbf{x}\|_p. \quad (\text{E.5})$$

Proof. By Lemma E.10, for any $1 \leq p \leq \infty$, the function $f : \mathbb{R}^n \rightarrow \mathbb{R}^+ \cup \{0\}$ given by $f(p) := \|x\|_p$ is continuous. Hence it suffices to verify its monotonicity in $\mathbb{Q}$.

For $p, q \in \mathbb{Q}$, we have $p = \frac{i_1}{j_1}$ and $q = \frac{i_2}{j_2}$ for some $i_1, i_2, j_1, j_2 \in \mathbb{Z}$. We assume $p \leq q$, i.e., $i_1 j_2 \leq i_2 j_1$. Define $a_k = |x_k|^{\frac{1}{j_1 j_2}}$ and we only need to verify

$$\left(\sum_{k=1}^n a_k^{i_1 j_2} \right)^{i_2 j_1} \geq \left(\sum_{k=1}^n a_k^{i_2 j_1} \right)^{i_1 j_2}.$$

Assume $a_m = \max\{a_k\}$ and we have

$$\begin{aligned} \left(\sum_{k=1}^n \left(\frac{a_k}{a_m} \right)^{i_1 j_2} \right)^{i_2 j_1} &\geq \left(\sum_{k=1}^n \left(\frac{a_k}{a_m} \right)^{i_2 j_1} \right)^{i_1 j_2} \\ &\geq \left(\sum_{k=1}^n \left(\frac{a_k}{a_m} \right)^{i_2 j_1} \right)^{i_1 j_2}, \end{aligned}$$

where the first inequality follows from $\frac{a_k}{a_m} \leq 1$ and the second from $\sum_{k=1}^n \left(\frac{a_k}{a_m} \right)^{i_2 j_1} \geq \left(\frac{a_m}{a_m} \right)^{i_2 j_1} = 1$. Multiply both side with $a_m^{i_1 i_2 j_1 j_2}$ and we have (E.5). $\square$

Example E.12. The map $S : (\mathcal{C}[0, 1], \|\cdot\|_\infty) \rightarrow (\mathbb{R}, |\cdot|)$,

$$S(f) = \int_0^1 f^2(x) dx, \quad (\text{E.6})$$

is continuous. Indeed, for any $g \in \mathcal{C}[0, 1]$, we have

$$\begin{aligned} |S(f) - S(g)| &= \left| \int_0^1 f^2(x) dx - \int_0^1 g^2(x) dx \right| \\ &\leq \int_0^1 |f(x) - g(x)| |f(x) + g(x)| dx \\ &\leq \int_0^1 \|f - g\|_\infty (\|f - g\|_\infty + 2\|g\|_\infty) dx, \end{aligned}$$

which yields

$$\begin{aligned} \forall \epsilon > 0, \exists \delta = \min \left(1, \frac{\epsilon}{1 + 2\|g\|_\infty} \right) \text{ s.t. } \|f - g\|_\infty < \delta \Rightarrow \\ \|f - g\|_\infty (\|f - g\|_\infty + 2\|g\|_\infty) &< \|f - g\|_\infty (1 + 2\|g\|_\infty) < \epsilon \\ \Rightarrow |S(f) - S(g)| &< \epsilon. \end{aligned}$$

Exercise E.13. For $V = \mathcal{C}[0, 1]$ and $x_0 \in [0, 1]$, define $\ell_{x_0} : V \rightarrow \mathbb{R}$ as $\ell_{x_0}(v) := v(x_0)$. Show that ℓ_{x_0} is continuous on $\mathcal{C}[0, 1]$.

Example E.14. The differentiation map

$$\frac{d}{dt} : (\mathcal{C}^1[a, b], \|\cdot\|_\infty) \rightarrow (\mathcal{C}[a, b], \|\cdot\|_\infty)$$

is not continuous, but can be made continuous if we change the norm on $\mathcal{C}^1[a, b]$ to

$$\|f\|_{1,\infty} := \|f\|_\infty + \|f'\|_\infty. \quad (\text{E.7})$$

Indeed, for $f_n(t) = \frac{1}{\sqrt{n}} \cos(2\pi n t)$, we have

$$\forall n \in \mathbb{N}^+, \quad \|f'_n - \mathbf{0}'\|_\infty = 2\pi\sqrt{n} > 1,$$

yet $\|f_n - \mathbf{0}\|$ can be made arbitrarily small as $n \rightarrow \infty$. In contrast, $D : (\mathcal{C}^1[a, b], \|\cdot\|_{1,\infty}) \rightarrow (\mathcal{C}[a, b], \|\cdot\|_\infty)$ is continuous because

$$\begin{aligned} \forall \epsilon > 0, \exists \delta = \epsilon, \text{ s.t. } \forall f, g \in \mathcal{C}^1[0, 1], \|f - g\|_{1,\infty} < \delta \Rightarrow \\ \|Df - Dg\|_\infty = \|f' - g'\|_\infty &\leq \|f - g\|_{1,\infty} < \delta = \epsilon. \end{aligned}$$

Exercise E.15. Show that the arc length function $L : \mathcal{C}^1[0, 1] \rightarrow \mathbb{R}$,

$$L(f) := \int_0^1 \sqrt{1 + (f'(t))^2} dt, \quad (\text{E.8})$$

is not continuous if the norm of $\mathcal{C}^1[0, 1]$ is $\|\cdot\|_\infty$, whereas it is continuous if we equip $\mathcal{C}^1[0, 1]$ with (E.7).

Exercise E.16. Is the function $S : (c_{00}, \|\cdot\|_\infty) \rightarrow (\mathbb{R}, |\cdot|)$,

$$S((a_n)_{n \in \mathbb{N}}) = \sum_{n=1}^{\infty} a_n^2 \quad (\text{E.9})$$

continuous? Here c_{00} is defined in Notation 27.

E.1.3 Norm equivalence

Definition E.17. Two norms $\|\cdot\|_A$ and $\|\cdot\|_B$ on V are *equivalent*, written $\|\cdot\|_A \sim \|\cdot\|_B$, iff

$$\exists c_1, c_2 \in \mathbb{R}^+ \text{ s.t. } \forall v \in V, \quad c_1 \|v\|_A \leq \|v\|_B \leq c_2 \|v\|_A. \quad (\text{E.10})$$

Example E.18. The vector norms in Definition C.176 satisfy

$$\forall \mathbf{x} \in \mathbb{R}^n, \quad \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_p \leq \sqrt[p]{n} \|\mathbf{x}\|_\infty. \quad (\text{E.11})$$

Therefore all the ℓ_p norms are equivalent.

Example E.19. The operator norm and the Hilbert-Schmidt norm in Definitions C.288 and C.296 are equivalent.

Exercise E.20. Show that $\sim$ in Definition E.17 defines an equivalence relation on the set of all norms on V .

Lemma E.21. Show that two norms $\|\cdot\|_A$ and $\|\cdot\|_B$ on a linear space V are equivalent if and only if each sequence converging with respect to one norm also converges with respect to the other.

Proof. The necessity part follows directly from Definition E.17. For the sufficiency part, we start from

$$\lim_{n \rightarrow \infty} \|u_n - u\|_A = 0 \iff \lim_{n \rightarrow \infty} \|u_n - u\|_B = 0$$

and the backward triangle inequality (C.68) to obtain

$$\lim_{n \rightarrow \infty} \|u_n\|_A = \|u\|_A \iff \lim_{n \rightarrow \infty} \|u_n\|_B = \|u\|_B.$$

If $\|u\|_A = 0$, then u must be 0 and $\|u\|_B = 0$, in which case any $c_1, c_2 \in \mathbb{R}^+$ satisfy (E.10). Therefore, $\|u\|_A$ and $\|u\|_B$ must either simultaneously be zero or simultaneously be bounded.

Suppose the two norms are not equivalent. Then we have

$$\forall C > 0, \exists x \in V \text{ s.t. } \|x\|_B > C\|x\|_A.$$

For each $n \in \mathbb{N}^+$, we have one x_n satisfying

$$\|x_n\|_B > n^2\|x_n\|_A.$$

The sequence $\{y_n = \frac{1}{n\|x_n\|_A}x_n\}_{n=1}^\infty$ converges to 0 under the $\|\cdot\|_A$ norm, but diverges under the $\|\cdot\|_B$ norm:

$$\lim_{n \rightarrow \infty} \frac{\|x_n\|_B}{n\|x_n\|_A} > \lim_{n \rightarrow \infty} n = +\infty;$$

this contradicts the conclusion of the last paragraph. $\square$

Theorem E.22. All norms are equivalent on $\mathbb{R}^n$ or $\mathbb{C}^n$.

Proof. By Exercise E.20, the equivalence of norms is an equivalence relation on the set of all norms. Thus it suffices to prove that any norm is equivalent to the 2-norm. Let $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ be a basis of $\mathbb{F}^n$ where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$. Then any vector $\mathbf{x} \in \mathbb{R}^n$ can be expressed as $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$. Thus

$$\|\mathbf{x}\| = \left\| \sum_{i=1}^n x_i \mathbf{e}_i \right\| \leq \sum_{i=1}^n |x_i| \|\mathbf{e}_i\| \leq M \|\mathbf{x}\|_2,$$

where $M = \sqrt{\sum_{i=1}^n \|\mathbf{e}_i\|^2}$ and the last inequality follows from the Cauchy–Schwarz inequality (Theorem C.197). Set

$$K := \{\mathbf{y} \in \mathbb{F}^n : \|\mathbf{y}\|_2 = 1\}.$$

Since K is a sequentially compact set and the norm $\|\cdot\| : K \rightarrow \mathbb{R}$ is a continuous function (Lemma E.10), $\|\cdot\|$ must attain its minimum value m on K . Furthermore, $m > 0$ since $\mathbf{0} \notin K$. For $\mathbf{x} \neq \mathbf{0}$, we have $\mathbf{y} := \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \in K$ since $\|\mathbf{y}\|_2 = 1$. Then $\|\mathbf{y}\| \geq m$ implies $m\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|$. $\square$

Corollary E.23. Over a finite-dimensional space, any two norms are equivalent.

Proof. This follows from Theorem E.22 and the isomorphism of linear spaces. $\square$

Example E.24. In the normed space $V := \mathcal{C}[0, 1]$, consider a sequence of functions (u_n) given by

$$u_n(x) := \begin{cases} 1 - nx, & x \in [0, \frac{1}{n}]; \\ 0, & x \in (\frac{1}{n}, 1]. \end{cases}$$

For the p -norm in (E.1), we have $\|u_n\|_p = [n(p+1)]^{-\frac{1}{p}}$ and thus the sequence (u_n) converges to $u = 0$ in $(V, \|\cdot\|_p)$. However, for the ∞ -norm in (E.2), we have $\|u_n\|_\infty = 1$. By Definition E.17, these two norms can not be equivalent.

E.1.4 Completeness

Definition E.25 (Banach spaces). A *Banach space* is a *complete* normed space, i.e., a normed space V such that every Cauchy sequence in V converges in V .

Definition E.26. A *Hilbert space* is a Banach space equipped with an inner product.

Theorem E.27. $(\mathcal{C}[a, b], \|\cdot\|_\infty)$ is a Banach space.

Proof. It is straightforward to show that $\|\cdot\|_\infty$ is a norm. The rest of the proof is similar to that of Lemma D.17. $\square$

Example E.28. For $p \in [1, \infty)$, the space $(\mathcal{C}(\overline{\Omega}), \|\cdot\|_p)$ is not a Banach space. Consider $(u_n)_{n \in \mathbb{N}} \subset \mathcal{C}[0, 1]$ given by

$$u_n(x) = \begin{cases} 0, & x \in [0, \frac{1}{2} - \frac{1}{2n}]; \\ nx - \frac{n-1}{2}, & x \in [\frac{1}{2} - \frac{1}{2n}, \frac{1}{2} + \frac{1}{2n}]; \\ 1, & x \in [\frac{1}{2} + \frac{1}{2n}, 1]. \end{cases} \quad (\text{E.12})$$

$(u_n)_{n \in \mathbb{N}}$ is clearly Cauchy and we have

$$\lim_{n \rightarrow \infty} u_n = u(x) = \begin{cases} 0, & x \in [0, \frac{1}{2}); \\ 1, & x \in (\frac{1}{2}, 1]. \end{cases}$$

But $u(x)$ cannot be in $\mathcal{C}(\overline{\Omega})$ no matter how we define $u(\frac{1}{2})$.

Definition E.29. Two normed spaces X and Y are *isomorphic*, written $X \simeq Y$, iff there exists a bijective linear map $T \in \mathcal{L}(X, Y)$ such that

$$\forall v \in X, \quad \|Tv\|_Y = \|v\|_X. \quad (\text{E.13})$$

Then T is an (*isometric*) *isomorphism* between X and Y .

Theorem E.30. Every normed space V has a unique *completion* up to isometric isomorphism, i.e., there exists a unique Banach space W such that V is isomorphic to a dense subspace of W .

Proof. This follows from Theorem D.38 and Definitions E.1 and E.3. $\square$

Definition E.31. The Banach space W in Theorem E.30 is called the *completion of the normed space* V .

Exercise E.32. Show that the sequence space $(\ell^\infty, \|\cdot\|_\infty)$ is complete.

Exercise E.33. Define $\mathcal{C}_b[0, \infty)$ as the set of all functions f that are continuous on $[0, \infty)$ and satisfy

$$\|f\|_\infty := \sup_{x \geq 0} |f(x)| < \infty.$$

Show $\mathcal{C}_b[0, \infty)$ with this norm is complete.

Exercise E.34. Define $\mathcal{C}^\alpha[a, b]$ as the set of all functions $f \in \mathcal{C}[a, b]$ satisfying

$$M_\alpha(f) := \sup_{x, y \in [a, b]; x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha} < \infty.$$

Define $\|f\|_\alpha := \|f\|_\infty + M_\alpha(f)$. Show that $(\mathcal{C}^\alpha[a, b], \|\cdot\|_\alpha)$ is a Banach space.

E.1.5 Convergence of function series

Theorem E.35. In a Banach space, absolutely convergent series converge. More precisely, if $(x_n)_{n \in \mathbb{N}}$ is a sequence in a Banach space $(X, \|\cdot\|)$ such that $\sum_{n=1}^{\infty} \|x_n\|$ converges, then $\sum_{n=1}^{\infty} x_n$ converges in X . Furthermore,

$$\left\| \sum_{n=1}^{\infty} x_n \right\| \leq \sum_{n=1}^{\infty} \|x_n\|. \quad (\text{E.14})$$

Proof. Since X is Banach, it suffices to prove that the sequence $(s_n = \sum_{i=1}^n x_i)_{n \in \mathbb{N}}$ is Cauchy. Since the real sequence $(\sigma_n = \sum_{i=1}^n \|x_i\|)_{n \in \mathbb{N}}$ is Cauchy, we have

$$\forall \epsilon > 0, \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n > m > N, \sum_{i=m+1}^n \|x_i\| < \epsilon,$$

which implies that $(s_n = \sum_{i=1}^n x_i)_{n \in \mathbb{N}}$ is Cauchy:

$$\left\| \sum_{i=m+1}^n x_i \right\| \leq \sum_{i=m+1}^n \|x_i\| < \epsilon.$$

Set $L := \sum_{i=m+1}^n x_i$ and we have

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, \|s_n - L\| < \epsilon,$$

which implies

$$\|L\| \leq \|s_n - L\| + \|s_n\| < \epsilon + \sigma_n < \epsilon + \sum_{n=1}^{\infty} \|x_n\|,$$

where the second inequality follows from the triangle inequality and the third from n being a finite number. Then (E.14) holds because ϵ can be made arbitrarily small. $\square$

Example E.36. The series $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin(nx)$ converges in $(\mathcal{C}[0, 2\pi], \|\cdot\|_{\infty})$ since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges in $\mathbb{R}$. Hence $x \mapsto \sum_{n=1}^{\infty} \frac{1}{n^2} \sin(nx)$ defines a continuous function.

Exercise E.37. Prove the converse of Theorem E.35, i.e., a normed space X is complete if every absolutely convergent series converges in X .

Theorem E.38 (Weierstrass M-test). Suppose a sequence of functions $(f_n)_{n=1}^{\infty}$ defined on a complete normed space $\mathcal{X}$ satisfies

$$\forall x \in \mathcal{X}, \|f_n(x)\| \leq M_n. \quad (\text{E.15})$$

Then the convergence of $\sum_{i=1}^n M_i$ implies the uniform convergence of the series $\sum_{i=1}^n f_i$ to some $f \in \mathcal{X}$.

Proof. The convergence of $\sum_{i=1}^n M_i$ and (E.15) imply

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n, m > N, \sum_{i=m}^n \|f_i\| \leq \sum_{i=m}^n M_i < \epsilon.$$

The proof is then completed by Theorem E.35. $\square$

Exercise E.39. Prove Theorem E.38 via Theorem D.173.

E.2 The space $\mathcal{CL}(X, Y)$

Notation 29. For normed spaces X and Y , the set of all continuous linear transformations from X to Y is

$$\mathcal{CL}(X, Y) := \mathcal{C}(X, Y) \cap \mathcal{L}(X, Y), \quad (\text{E.16})$$

where $\mathcal{C}(X, Y)$ and $\mathcal{L}(X, Y)$ are given by Notation 28 and Notation 23, respectively. We also write $\mathcal{CL}(X)$ if $Y = X$,

E.2.1 Boundedness = continuity

Definition E.40. A linear map $T \in \mathcal{L}(X, Y)$ is *bounded* if it maps any bounded set in $\mathcal{X}$ to a bounded set in $\mathcal{Y}$, i.e.,

$$\exists M \in \mathbb{R}^+ \text{ s.t. } \forall x \in X, \|Tx\|_Y \leq M\|x\|_X. \quad (\text{E.17})$$

Theorem E.41. For any map $T \in \mathcal{L}(X, Y)$, the following statements are equivalent:

- (a) T is continuous,
- (b) T is continuous at $\mathbf{0}$,
- (c) T is bounded.

Proof. (a) $\Rightarrow$ (b) follows from Definition D.161. For (b) $\Rightarrow$ (c), the continuity of T at $\mathbf{0}$ implies

$$\text{for } \epsilon = 1, \exists \delta > 0 \text{ s.t. } \|x\| < \delta \implies \|Tx\| < 1.$$

Replacing x with $y = \frac{\delta}{2} \frac{x}{\|x\|}$ in the above inequalities yields $\|Tx\| \leq M\|x\|$ with $M = \frac{2}{\delta}$. Finally, (c) $\Rightarrow$ (a) follows from

$$\forall \epsilon > 0, \exists \delta = \frac{\epsilon}{M} \text{ s.t. } \|x - y\| < \delta \implies$$

$$\|Tx - Ty\| = \|T(x - y)\| \leq M\|x - y\| = \frac{\epsilon}{\delta}\|x - y\| < \epsilon. \quad \square$$

Example E.42. The left shift operator $L : \ell^2 \rightarrow \ell^2$ and right shift operator $R : \ell^2 \rightarrow \ell^2$,

$$L(a_1, a_2, a_3, \dots) = (a_2, a_3, \dots), \quad (\text{E.18})$$

$$R(a_1, a_2, a_3, \dots) = (0, a_1, a_2, \dots), \quad (\text{E.19})$$

are linear operators. Furthermore, $L, R \in \mathcal{CL}(\ell^2)$ because they are bounded:

$$\|L(a_n)_{n \in \mathbb{N}}\| \leq \|(a_n)_{n \in \mathbb{N}}\|, \quad \|R(a_n)_{n \in \mathbb{N}}\| = \|(a_n)_{n \in \mathbb{N}}\|.$$

Example E.43. The linear map $T : (\mathcal{C}[a, b], \|\cdot\|_{\infty}) \rightarrow \mathbb{R}$ given by $T(f) = \int_a^b f(t) dt$ is continuous because

$$|T(f)| = \left| \int_a^b f(t) dt \right| \leq \int_a^b \|f\|_{\infty} dt = (b - a)\|f\|_{\infty}.$$

By Definition D.82, T preserves convergent sequences:

$$\lim_{n \rightarrow \infty} f_n = f \implies \lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b f.$$

In other words, the continuity of T under $\|\cdot\|_{\infty}$ guarantees that T and $\lim_{n \rightarrow \infty}$ are commutative; see Section D.5.

Example E.44. The continuity of differentiation maps in Example E.14 can be determined by Theorem E.41. For $\|\cdot\|_{1, \infty}$, we have $\|D\mathbf{x}\|_{\infty} \leq \|\mathbf{x}\|_{1, \infty}$, and thus the operator $D : (\mathcal{C}[0, 1], \|\cdot\|_{1, \infty}) \rightarrow (\mathcal{C}[0, 1], \|\cdot\|_{\infty})$ is continuous.

In comparison, $D : (\mathcal{C}[0, 1], \|\cdot\|_{\infty}) \rightarrow (\mathcal{C}[0, 1], \|\cdot\|_{\infty})$ is not continuous: for $\mathbf{x}_n = t^n$, we have $\|\mathbf{x}_n\|_{\infty} = 1$ yet $\lim_{n \rightarrow \infty} \|\mathbf{x}'_n\|_{\infty} = \infty$.

Corollary E.45. $\mathcal{CL}(X, Y)$ is a subspace of $\mathcal{L}(X, Y)$.

Proof. Since $\mathcal{CL}(X, Y) \subset \mathcal{L}(X, Y)$, it suffices to verify that $\mathcal{CL}(X, Y)$ is a vector space. Any $T_1, T_2 \in \mathcal{CL}(X, Y)$ satisfy

$$\|(T_1 + T_2)\mathbf{x}\|_Y \leq \|T_1\mathbf{x}\|_Y + \|T_2\mathbf{x}\|_Y \leq (M_1 + M_2)\|\mathbf{x}\|_X,$$

which means $T_1 + T_2 \in \mathcal{CL}(X, Y)$ by Theorem E.41. For $T \in \mathcal{CL}(X, Y)$ and $a \in \mathbb{R}$, we have

$$\|aT\mathbf{x}\|_Y = |a|\|T\mathbf{x}\|_Y \leq |a|M\|\mathbf{x}\|_X,$$

which means $aT \in \mathcal{CL}(X, Y)$ by Theorem E.41. $\square$

Corollary E.46. For finite-dimensional normed spaces X and Y , we have $\mathcal{L}(X, Y) = \mathcal{CL}(X, Y)$.

Proof. Each linear transformation $T_A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ has a matrix $A \in \mathbb{R}^{m \times n}$ such that

$$\begin{aligned} \|T_A\mathbf{x}\|_2^2 &= \|A\mathbf{x}\|_2^2 = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij}x_j \right)^2 \\ &\leq \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} \right)^2 \left(\sum_{j=1}^n x_j^2 \right) = \|\mathbf{x}\|_2^2 \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2, \end{aligned}$$

where the inequality follows from the Cauchy–Schwarz inequality. The proof is completed by Theorem E.41, Theorem E.22, and the isomorphism of linear spaces. $\square$

Exercise E.47. For an infinite-dimensional matrix A satisfying $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij}^2 < \infty$, define $T_A : \ell^2 \rightarrow \ell^2$ by

$$\forall \mathbf{x} = (x_j)_{j \in \mathbb{N}} \in \ell^2, \quad T_A\mathbf{x} = A\mathbf{x} = \left(\sum_{j=1}^{\infty} a_{ij}x_j \right)_{i \in \mathbb{N}^+}. \quad (\text{E.20})$$

Prove $T_A \in \mathcal{CL}(\ell^2)$.

Exercise E.48. For $A, B \in \mathcal{C}[a, b]$ and

$$S := \{f \in \mathcal{C}^1[a, b] : f(a) = f(b) = 0\}, \quad (\text{E.21})$$

show that the map $L : (S, \|\cdot\|_{1,\infty}) \rightarrow \mathbb{R}$ given by

$$L(f) = \int_a^b [A(t)f(t) + B(t)f'(t)] dt$$

is a bounded linear transformation.

E.2.2 Consequences on the topology

Notation 30. For a vector space X and its subsets A, A_1, A_2 , we write

$$\begin{aligned} \forall \alpha \in \mathbb{R}, \quad \alpha A &:= \{\alpha a, a \in A\}; \\ \forall w \in X, \quad A + w &:= \{a + w, a \in A\}. \\ \forall A_1, A_2 \subset X, \quad A_1 + A_2 &:= \{a_1 + a_2 : a_1 \in A_1, a_2 \in A_2\}. \end{aligned}$$

Lemma E.49. The null space of any linear map $A \in \mathbb{R}^{m \times n}$ is closed in $\mathbb{R}^n$.

Proof. A is a continuous linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ and $\{\mathbf{0}\}$ is a closed set in $\mathbb{R}^m$. The rest follows from Definition C.53 and Theorems D.86 and D.87. $\square$

Theorem E.50. Every subspace of $\mathbb{R}^n$ is closed.

Proof. Since $\mathbb{R}^n$ is closed, it suffices to show that any subspace V with $\dim V = k < n$ is also closed. Construct an orthonormal basis $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n)$ of $\mathbb{R}^n$ via the Gram-Schmidt process in Definition 8.13. Then we have $\ker A = V$ for the matrix

$$A := \begin{bmatrix} \mathbf{u}_{k+1}^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix}.$$

The rest of the proof follows from Lemma E.49. $\square$

Lemma E.51. Let X and Y be normed spaces. A bounded linear map $T \in \mathcal{CL}(X, Y)$ is open in the sense of Definition D.101 if and only if the image of the unit open ball in X under T contains some open ball centered at $\mathbf{0}_Y$ in Y , i.e.,

$$\exists \delta > 0 \text{ s.t. } B(\mathbf{0}_Y, \delta) \subset T(B(\mathbf{0}_X, 1)). \quad (\text{E.22})$$

Proof. For necessity, T being an open map implies that the image $T(B(\mathbf{0}_X, 1))$ is open. The linearity of T implies $\mathbf{0}_Y \in T(B(\mathbf{0}_X, 1))$. Then Definition D.48 yields (E.22).

For sufficiency, let $U \subset X$ be open, we need to show

$$(*) : \quad \forall y_0 \in T(U), \exists r_Y > 0 \text{ s.t. } B(y_0, r_Y) \subset T(U).$$

$y_0 \in T(U)$ implies there exists $x_0 \in U$ such that $Tx_0 = y_0$. Since U is open, we have

$$(**) : \quad \exists r_X > 0 \text{ s.t. } B(x_0, r_X) \subset U.$$

Choose $r_Y = \delta r_X$ and we have

$$\begin{aligned} B(Tx_0, r_Y) &= Tx_0 + B(\mathbf{0}_Y, \delta r_X) \subset Tx_0 + TB(\mathbf{0}_X, r_X) \\ &= TB(x_0, \delta r_X) \subset T(U), \end{aligned}$$

where the second step follows from (E.22). $\square$

Theorem E.52 (Baire). Suppose $(F_n)_{n \in \mathbb{N}}$ is a sequence of closed sets in a Banach space X such that $X = \bigcup_{n \in \mathbb{N}} F_n$. Then there exists $n \in \mathbb{N}$ and a nonempty open set U such that $U \subset F_n$.

Proof. Suppose the conclusion does not hold, i.e.,

$$(*) : \quad \forall n \in \mathbb{N}, \forall U \subset X \text{ open}, U \not\subset F_n.$$

By Definition D.76, each F_n is nowhere dense. Then $(*)$ contradicts Corollary D.81. $\square$

Lemma E.53 (Unit open ball). Suppose X and Y are Banach spaces and $T \in \mathcal{CL}(X, Y)$ is surjective. Then the image $T(B_0)$ of the open ball $B_0 := B(\mathbf{0}_X, 1)$ contains an open ball about $\mathbf{0}_Y$.

Proof. Define $B_n := B(\mathbf{0}_X, \frac{1}{2^n})$ and we need to show that $\overline{T(B_1)}$ contains an open ball. Indeed,

$$\forall x \in X, \exists k > 2\|x\| \text{ s.t. } x \in kB_1$$

where we have applied Notation 30. Thus $X = \cup_{k \in \mathbb{N}} kB_1$. T being surjective implies

$$Y = T(X) = T\left(\bigcup_{k \in \mathbb{N}^+} kB_1\right) = \bigcup_{k \in \mathbb{N}^+} kT(B_1) = \bigcup_{k \in \mathbb{N}^+} \overline{kT(B_1)},$$

where the last step follows from Definition D.67 and the condition of Y being a Banach space. By Theorem E.52, there exists some $\overline{kT(B_1)}$ that contains a nonempty open ball, which implies that $\overline{T(B_1)}$ also contains an open ball, say, $B(y_0, \epsilon) \subset \overline{T(B_1)}$, which implies

$$\begin{aligned} B(\mathbf{0}_Y, \epsilon) &= B(y_0, \epsilon) - y_0 \subset \overline{T(B_1)} + \overline{T(B_1)} \\ &\subset \overline{T(B_1) + T(B_1)} = \overline{T(B_0)}, \end{aligned}$$

where we have applied Notation 30. To sum up the above arguments, we have

$$(*) : B(\mathbf{0}_Y, \epsilon) \subset \overline{T(B_0)}.$$

To complete the proof, we show

$$V_1 = B\left(\mathbf{0}_Y, \frac{\epsilon}{2}\right) \subset T(B_0),$$

where $V_n := B(\mathbf{0}_Y, \frac{\epsilon}{2^n})$. The linearity of T and $(*)$ imply

$$(\Delta) : \forall n \in \mathbb{N}, V_n \subset \overline{T(B_n)}.$$

For $y \in V_1$, we have $y \in \overline{T(B_1)}$. Since both T and $\|\cdot\|$ are continuous, the map $x \mapsto \|y - Tx\|$ is also continuous, and therefore

$$\exists x_1 \in B_1 \text{ s.t. } \|y - Tx_1\| < \frac{\epsilon}{4}.$$

By definition of V_n and (Δ) , $y - Tx_1 \in V_2 \subset \overline{T(B_2)}$. Thus

$$\exists x_2 \in B_2 \text{ s.t. } \|(y - Tx_1) - Tx_2\| < \frac{\epsilon}{8}.$$

Proceed inductively and we have

$$\forall k = 1, 2, \dots, n, \exists x_k \in B_k \text{ s.t. } \left\| y - T \sum_{k=1}^n x_k \right\| < \frac{\epsilon}{2^{n+1}}.$$

Take limit of the above and we have

$$(\square) : y = T \lim_{n \rightarrow \infty} \sum_{k=1}^n x_k.$$

For each k , $x_k \in B_k$ implies $\|x_k\| < \frac{1}{2^k}$. Hence

$$\sum_{k=1}^{\infty} \|x_k\| < \sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

The completeness of X and Theorem E.35 yield

$$\exists x \in X \text{ s.t. } x = \sum_{k=1}^{\infty} x_k, \quad \|x\| < 1.$$

Then $(\square)$ yields $y = Tx \in T(B_0)$. $\square$

Theorem E.54 (Banach-Schauder or open mapping). For Banach spaces X and Y , any surjective map $T \in \mathcal{CL}(X, Y)$ is open.

Proof. This follows from Lemmas E.51 and E.53. $\square$

Example E.55. The following function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = \begin{cases} x+1 & \text{if } x \in (-\infty, -1]; \\ 0 & \text{if } x \in (-1, +1); \\ x-1 & \text{if } x \in [+1, +\infty), \end{cases}$$

is surjective and continuous; but since $f((-1, 1)) = \{0\}$ is closed, f is not open. By the open mapping theorem, if a map between two Banach spaces is not open but surjective and continuous, then it must be nonlinear.

E.2.3 Operator norms

Lemma E.56. The operator norm $\|\cdot\| : \mathcal{CL}(X, Y) \rightarrow \mathbb{R}$ of an operator $T \in \mathcal{CL}(X, Y)$ given by

$$\|T\| = \sup S \text{ where } S := \{\|Tx\| : x \in X, \|x\| \leq 1\} \quad (\text{E.23})$$

is well defined, i.e., $\|T\|$ is a unique bounded real number.

Proof. By Theorem A.55, it suffices to show that S is a nonempty bounded subset of $\mathbb{R}$. S is nonempty because $\mathbf{0} \in X$ and $T\mathbf{0} = \mathbf{0}_Y$, imply $0 \in S$. The boundedness of S follows from Theorem E.41 (c) and $\|x\|_X \leq 1$. $\square$

Lemma E.57. Any $T \in \mathcal{CL}(X, Y)$ satisfies

$$(\forall x \in X, \|Tx\| \leq M\|x\|) \implies \|T\| \leq M. \quad (\text{E.24})$$

Proof. M is an upper bound of the set S in (E.23) while $\|T\|$ is the least upper bound of S . $\square$

Lemma E.58. $\forall T \in \mathcal{CL}(X, Y), \forall x \in X, \|Tx\| \leq \|T\|\|x\|$.

Proof. The statement holds trivially for $x = \mathbf{0}$. Otherwise for $y = \frac{x}{\|x\|}$ we have $\|Ty\| \in S$ where S is in (E.23). Hence

$$\|Ty\| \leq \|T\| \implies \|Tx\| \leq \|T\|\|x\|. \quad \square$$

Lemma E.59. $\forall S \in \mathcal{CL}(X, Y), \forall T \in \mathcal{CL}(Y, Z)$, we have $\|ST\| \leq \|S\|\|T\|$.

Proof. This follows from Lemmas E.57 and E.58. $\square$

Theorem E.60. $(\mathcal{CL}(X, Y), \|\cdot\|)$ is a normed space.

Proof. (NRM-1,2,3) are trivial, we only verify the triangular inequality. Indeed, for $T_1, T_2 \in \mathcal{CL}(X, Y)$, we have

$$\begin{aligned} \|T_1 + T_2\| &= \sup\{\|(T_1 + T_2)x\| : x \in X, \|x\| \leq 1\} \\ &\leq \sup\{\|T_1\|\|x\| + \|T_2\|\|x\| : x \in X, \|x\| \leq 1\} \\ &\leq \|T_1\| + \|T_2\|. \end{aligned} \quad \square$$

Lemma E.61. For a normed space X , the normed space $(\mathcal{CL}(X, Y), \|\cdot\|)$ is complete if and only if Y is a Banach space. $\square$

Proof. The necessity holds trivially as each $y \in Y$ can be considered as a constant map $X \rightarrow \{y\}$.

Let $(T_n)_{n \in \mathbb{N}^+}$ be a Cauchy sequence in $\mathcal{CL}(X, Y)$. For any $x \in X$, $(T_n x)_{n \in \mathbb{N}^+} \subset Y$ is Cauchy as Lemma E.58 yields

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \|x\|.$$

Since Y is complete, $(T_n x)_{n \in \mathbb{N}}$ converges to some $L(x) \in Y$. This defines a map $T(x) = L(x)$.

The second step is to show $T \in \mathcal{CL}(X, Y)$.

$$\begin{aligned} T(x + y) &= \lim_{n \rightarrow \infty} T_n(x + y) = \lim_{n \rightarrow \infty} T_n(x) + \lim_{n \rightarrow \infty} T_n(y) \\ &= T(x) + T(y). \end{aligned}$$

Since each map in $(T_n)_{n \in \mathbb{N}}$ is bounded, we can find $M = \sup_{n \in \mathbb{N}^+} \|T_n\|$. Because Y is Banach, $\forall \epsilon > 0, \forall x \in X, x \neq 0$, we can find $m \in \mathbb{N}$ s.t. $\|T x - T_m x\| \leq \epsilon \|x\|$. Then

$$\|T x\| \leq \|T x - T_m x\| + \|T_m x\| \leq \epsilon \|x\| + M \|x\|$$

and Theorem E.41 yield $T \in \mathcal{CL}(X, Y)$.

The third step is to show $\lim_{n \rightarrow \infty} T_n = T$. The sequence $(T_n)_{n \in \mathbb{N}^+}$ being Cauchy implies

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m, n \geq N, \|T_n x - T_m x\| \leq \epsilon \|x\|,$$

which implies

$$\forall x \text{ with } \|x\| \leq 1, \|T_n x - T x\| \leq \epsilon \|x\| \leq \epsilon.$$

Then Lemma E.56 yields $\|T_n - T\| \leq \epsilon$, which means $\lim_{n \rightarrow \infty} T_n = T$. $\square$

Corollary E.62. If X is a normed space over $\mathbb{R}$, then the dual space of X , written $X' = \mathcal{CL}(X, \mathbb{R})$, is a Banach space with the operator norm.

Proof. This follows directly from Lemma E.61. $\square$

Corollary E.63. If X is a Banach space, then $\mathcal{CL}(X)$ is a Banach space with the operator norm.

Proof. This follows directly from Lemma E.61. $\square$

Definition E.64. A *normed algebra* is an algebra V (as in Definition C.6) with a norm $\|\cdot\|$ satisfying

$$\forall u, v \in V, \quad \|uv\| \leq \|u\| \|v\|. \quad (\text{E.25})$$

A *Banach algebra* is a normed algebra that is complete.

E.2.4 Invertible operators

Lemma E.65. In a finite-dimensional vector space X , if two operators $T, S \in \mathcal{L}(X)$ satisfy $TS = I$, then $ST = I$.

Proof. $TS = I$ implies $\ker S = \{0\}$ because

$$Sx = 0 \Rightarrow TSx = 0 \Rightarrow x = 0.$$

Thus for any basis $(v_i)_{i=1}^n$ of X , $(Sv_i)_{i=1}^n$ is also a basis.

$$\forall x \in X, \exists (\beta_i)_{i=1}^n \text{ s.t. } x = \sum_{i=1}^n \beta_i Sv_i = S \sum_{i=1}^n \beta_i v_i.$$

It follows that

$$\forall x \in X, STx = STS \sum_{i=1}^n \beta_i v_i = S \sum_{i=1}^n \beta_i v_i = x,$$

which implies $ST = I$. $\square$

Example E.66. For the shift operators on ℓ^2 in Example E.42, we have $LR = I$ but $RL \neq I$,

$$RL(1, 0, 0, \dots) = (0, 0, 0, \dots).$$

Definition E.67. For vector spaces X and Y , a linear map $A \in \mathcal{L}(X, Y)$ is *invertible* iff there exists $B \in \mathcal{L}(Y, X)$ such that $AB = I \in \mathcal{L}(Y)$ and $BA = I \in \mathcal{L}(X)$. Then B is called the *inverse* of A .

Exercise E.68. Prove that the inverse of $A \in \mathcal{L}(X, Y)$ is unique if A is invertible.

Lemma E.69. For any vector spaces X and Y , if a linear map $A \in \mathcal{L}(X, Y)$ is invertible, then A is bijective.

Proof. A is injective because

$$Ax = Ay \Rightarrow A^{-1}Ax = A^{-1}Ay \Rightarrow x = y.$$

A is surjective because $\forall y \in Y, A^{-1}y \in X$ implies $y = Ax$ for some $x \in X$. $\square$

Lemma E.70. For two vector spaces X and Y , if a map $A \in \mathcal{L}(X, Y)$ is invertible, then its inverse A^{-1} is linear.

Proof. For any $x, y \in X$, set $z = A^{-1}(x + y)$ and we have

$$x + y = Az \Rightarrow A^{-1}x + A^{-1}y = z = A^{-1}(x + y).$$

Similarly, for any $\alpha \in \mathbb{F}$, set $z = A^{-1}(\alpha x)$ and we have

$$\begin{aligned} Az = \alpha x &\Rightarrow A \frac{z}{\alpha} = x \Rightarrow \frac{z}{\alpha} = A^{-1}x \\ \Rightarrow \alpha A^{-1}x &= z = A^{-1}(\alpha x). \end{aligned} \quad \square$$

Lemma E.71. For two vector spaces X and Y , a linear map $A \in \mathcal{L}(X, Y)$ is invertible provided that it is bijective.

Proof. For the bijective map A , define a map $B : Y \rightarrow X$ by

$$\forall v \in Y, \quad A(Bv) = v.$$

The existence and uniqueness of Bv are guaranteed by the surjectivity and injectivity of A . Therefore, $AB = I$. Furthermore, $BA = I$ follows from the injectivity of A and

$$\forall v \in X, \quad A(BAv) = (AB)Av = Av.$$

Finally, Lemma E.70 implies that B is a linear map. $\square$

Theorem E.72. For finite-dimensional normed spaces X and Y , a linear map $A \in \mathcal{CL}(X, Y)$ is invertible if and only if A is bijective, in which case we also have $A^{-1} \in \mathcal{CL}(Y, X)$.

Proof. This follows from Lemmas E.69, E.70, E.71, and Corollary E.46. $\square$

Example E.73. The map $A : c_{00} \rightarrow c_{00}$ given by

$$\forall (x_n)_{n \in \mathbb{N}} \in c_{00}, \quad A(x_1, x_2, x_3, \dots) = (x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots)$$

is linear, bijective, and continuous (since $\|A\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_\infty$). However, it is not invertible in $\mathcal{CL}(c_{00})$. Suppose it is and $B \in \mathcal{CL}(c_{00})$ is the inverse of A . Then for the sequences $\mathbf{e}_m := (0, \dots, 0, 1, 0, \dots)$ where all terms are 0 except that the m th term is 1, we have

$$1 = \|\mathbf{e}_m\|_\infty = \|BA\mathbf{e}_m\|_\infty \leq \|B\| \|A\mathbf{e}_m\|_\infty = \frac{\|B\|}{m}.$$

Hence $\forall m \in \mathbb{N}$, $\|B\| \geq m$ and this contradicts Lemma E.56.

Theorem E.74 (Banach). For Banach spaces X and Y , a map $T \in \mathcal{CL}(X, Y)$ is invertible with $T^{-1} \in \mathcal{CL}(Y, X)$ if and only if T is bijective.

Proof. The necessity follows from Lemma E.69. For sufficiency, the bijective map T induces a map $T^{-1} : Y \rightarrow X$,

$$\forall y = Tx \in Y, \quad T^{-1}(y) = x.$$

Since the bijectiveness of T guarantees that T^{-1} is well defined, T^{-1} is indeed an inverse of T . By Lemma E.70, T^{-1} is linear. It remains to show that T^{-1} is continuous. By the surjectivity of T and Theorem E.54, T is open. Hence $T(U)$ is open whenever U is open. Meanwhile we have

$$\begin{aligned} (T^{-1})^{-1}(U) &= \{y \in Y : T^{-1}y \in U\} \\ &= \{y \in Y : y \in T(U)\} = T(U). \end{aligned}$$

Thus T^{-1} is continuous by Theorem D.86. $\square$

Theorem E.75 (Closed graph). For two Banach spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$, a map $T \in \mathcal{L}(X, Y)$ is continuous if and only if its graph $\mathcal{G}(T) := \{(x, Tx) : x \in X\}$ is closed in $(X \times Y, \|\cdot\|_\infty)$ where

$$\forall (x, y) \in X \times Y, \quad \|(x, y)\|_\infty := \max(\|x\|_X, \|y\|_Y). \quad (\text{E.26})$$

Proof. For necessity we show that any convergent sequence $(x_n, Tx_n)_{n \in \mathbb{N}} \subset \mathcal{G}(T)$ converges in $\mathcal{G}(T)$ and then Lemma D.65 implies that $\mathcal{G}(T)$ is closed.

For sufficiency we define $p : \mathcal{G}(T) \rightarrow X$ by $p(x, Tx) = x$. It is straightforward to show that p is linear, continuous, and bijective. It can also be shown that $X \times Y$ is Banach, and thus $\mathcal{G}(T)$, a closed subset of $X \times Y$, is also Banach. By Theorem E.74, p is invertible and its inverse p^{-1} is also a bounded linear map. Then we have

$$\|Tx\|_Y \leq \|(x, Tx)\|_\infty = \|p^{-1}x\|_\infty \leq \|p^{-1}\| \|x\|_\infty,$$

and the proof is completed by Theorem E.41. $\square$

E.2.5 Neumann series

Theorem E.76 (Neumann series). Suppose X is a Banach space and $A \in \mathcal{CL}(X)$ has $\|A\| < 1$. Then we have

(NST-1) $I - A$ is invertible in $\mathcal{CL}(X)$,

(NST-2) $(I - A)^{-1} = I + A + \dots + A^n + \dots = \sum_{n=0}^{\infty} A^n$,

(NST-3) $\|(I - A)^{-1}\| \leq \frac{1}{1 - \|A\|}$.

Proof. Since X is a Banach space, Corollary E.63 states that $\mathcal{CL}(X)$ is also a Banach space. By Theorem E.35, the convergence of $\sum_{n=0}^{\infty} \|A\|^n$ implies that the sequence $(S_n)_{n \in \mathbb{N}}$ with

$$S_n = \sum_{k=0}^n A^k$$

converges to some $S \in \mathcal{CL}(X)$. It follows that

$$\begin{aligned} S_n A &= A S_n = \sum_{k=1}^{n+1} A^k = S_{n+1} - I \\ \Rightarrow \begin{cases} \|A S_n - A S\| \leq \|A\| \|S_n - S\|, \\ \|S_n A - S A\| \leq \|A\| \|S - S_n\|. \end{cases} \\ \Rightarrow S A &= A S = S - I \\ \Rightarrow (I - A) S &= I = S(I - A) \\ \Rightarrow (I - A)^{-1} &= S = \sum_{n=0}^{\infty} A^n, \end{aligned}$$

where the last step follows from Definition E.67. Finally, (NST-3) follows from

$$\|(I - A)^{-1}\| = \left\| \sum_{n=0}^{\infty} A^n \right\| \leq \sum_{n=0}^{\infty} \|A^n\| \leq \sum_{n=0}^{\infty} \|A\|^n = \frac{1}{1 - \|A\|}$$

where the first inequality follows from Theorem E.35 and the second inequality from Lemma E.59. $\square$

Theorem E.77. Suppose X is a Banach space. Then the exponential of $A \in \mathcal{CL}(X)$, defined as

$$e^A := \sum_{n=0}^{\infty} \frac{1}{n!} A^n, \quad (\text{E.27})$$

converges in $\mathcal{CL}(X)$.

Proof. By Lemma E.59, we have

$$\sum_{n=0}^{\infty} \left\| \frac{1}{n!} A^n \right\| \leq \sum_{n=0}^{\infty} \frac{1}{n!} \|A\|^n = e^{\|A\|}.$$

By the comparison test, $\sum_{n=0}^{\infty} \frac{1}{n!} A^n$ converges absolutely. The rest of the proof follows from Theorem E.35. $\square$

Lemma E.78. For a Banach space X , $A \in \mathcal{CL}(X)$ satisfies

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A. \quad (\text{E.28})$$

Lemma E.79. For a Banach space X , if $A, B \in \mathcal{CL}(X)$ commute, i.e. $AB = BA$, then

$$e^{A+B} = e^A e^B. \quad (\text{E.29})$$

Corollary E.80. For a Banach space X and $A \in \mathcal{CL}(X)$, e^A is always invertible with its inverse as e^{-A} .

Theorem E.81 (Existence and uniqueness of ODEs). For a Banach space X and $A \in \mathcal{CL}(X)$, the IVP $\frac{dx}{dt}(t) = Ax(t)$ with initial condition $x(0) = x_0 \in X$ has a unique solution $x(t) = e^{tA} x_0$ for $t \in \mathbb{R}$.

Proof. If $x(t)$ solves the IVP, then

$$\frac{d}{dt} (e^{-tA} x(t)) = e^{-tA} (-A) x(t) + e^{-tA} \frac{d}{dt} (x(t)) = 0,$$

which implies $e^{-tA} x(t) = x_0$ and thus $x(t) = e^{tA} x_0$. $\square$

E.2.6 Uniform boundedness

Lemma E.82. Suppose X is a normed space and a subset $A \subset X$ satisfies

- A is symmetric, i.e., $-A = A$;
- A is mid-point convex, i.e., $\forall x, y \in A, \frac{x+y}{2} \in A$,
- there exists a nonempty open set $U \subset A$.

Then there exists $\delta > 0$ such that $B(\mathbf{0}_X, \delta) \subset A$.

Proof. For $a \in X$ and $\alpha \neq 0$, the maps $x \mapsto x + a$ and $x \mapsto \alpha x$ are both continuous with continuous inverses. By Theorem D.86, U being open in X implies that its preimage $U + \{-a\}$ under $x \mapsto x + a$ is also open in X . Adopting Notation 30, we find that the set

$$U + (-A) := \cup_{a \in A} (U + \{-a\})$$

is open since it is a union of open sets. For $a \in U$, we have

$$\mathbf{0}_X = \frac{a - a}{2} \in \frac{U + (-A)}{2} \subset \frac{A + (-A)}{2} = \frac{A + A}{2} = A,$$

where the last two equalities follow from A being symmetric and mid-point convex, respectively. The proof is completed by Definition D.48 and $\frac{U+(-A)}{2}$ being open. $\square$

Theorem E.83 (Principle of uniform boundedness). Suppose X is a Banach space and Y is a normed space. For a family of maps $T_i \in \mathcal{CL}(X, Y)$, $i \in I$, “pointwise boundedness” implies “uniform boundedness,”

$$\forall x \in X, \sup_{i \in I} \|T_i x\| < +\infty \implies \sup_{i \in I} \|T_i\| < +\infty. \quad (\text{E.30})$$

Proof. For any given $n \in \mathbb{N}$, we define

$$F_n := \cap_{i \in I} \{x \in X : \|T_i x\| \leq n\} = \{x \in X : \sup_{i \in I} \|T_i x\| \leq n\}.$$

As intersection of closed sets, each F_n is closed. By pointwise boundedness, we have $X = \cup_{n \in \mathbb{N}} F_n$. The Baire theorem E.52 implies that there exists some F_n that contains a nonempty open subset. Since F_n is also symmetric and mid-point convex, Lemma E.82 implies that F_n contains an open ball $B(\mathbf{0}_X, \delta)$. Consequently, $x \in B(\mathbf{0}_X, \delta) \subset F_n$ implies $\|T_i x\| \leq n$ for all $i \in I$. Thus for any $y \in X$, there exists $x = \frac{\delta}{2} \frac{y}{\|y\|}$ such that

$$\forall i \in I, \|T_i x\| \leq n \implies \|T_i y\| \leq \frac{2n}{\delta} \|y\|,$$

and the proof is completed by Lemma E.57. $\square$

Theorem E.84 (Banach-Steinhaus). Suppose X and Y are Banach spaces. If a sequence $(T_n)_{n \in \mathbb{N}} \in \mathcal{CL}(X, Y)$ has

$$\forall x \in X, \lim_{n \rightarrow \infty} \|T_n x\| < \infty,$$

then the map $x \mapsto \lim_{n \rightarrow \infty} T_n x$ belongs to $\mathcal{CL}(X, Y)$.

Proof. Clearly the map $T(x) = \lim_{n \rightarrow \infty} T_n x$ is linear, it remains to show that it is continuous. Because the limit $\lim_{n \rightarrow \infty} T_n x$ exists, we have $\sup_{n \in \mathbb{N}} \|T_n x\| < \infty$ for all $x \in X$. Then Theorem E.83 implies $\sup_{n \in \mathbb{N}} \|T_n\| < \infty$. Hence we have

$$\exists M \in \mathbb{R}, \forall x \in X, \forall n \in \mathbb{N}, \|T_n x\| \leq M \|x\|,$$

the limit of which yields $\forall x \in X, \|Tx\| \leq M \|x\|$. Then the proof is completed by Theorem E.41. $\square$

Example E.85. A linear PDE can be written in the form

$$Tx = y \quad (\text{E.31})$$

where x is the unknown, T is a continuous linear operator, and y is a known vector incorporating initial and boundary conditions. If the PDE is well-posed, we can often assume that T is a bijection, hence by Theorem E.74 the inverse of T is a bounded linear operator and we write $x = T^{-1}y$. In numerically solving the PDE, we usually approximate y by a grid function y_n and approximate T^{-1} by a discrete operator T_n^{-1} . Suppose the numerical computation satisfies

$$\forall y \in \mathcal{C}^r(\bar{\Omega}), \lim_{n \rightarrow \infty} y_n = y, \sup_{n \rightarrow \infty} \|T_n^{-1} y_n\| < \infty.$$

Then Theorem E.84 yields $\sup_{n \in \mathbb{N}} \|T_n^{-1}\| < \infty$, which usually implies some form of numerical stability and thus the desired convergence $\lim_{n \rightarrow \infty} T_n^{-1} y_n = x$. The entire process of numerically solving (E.31) can be summarized as follows.

$$\begin{array}{ccc} y & \xrightarrow{T^{-1}} & x \\ \downarrow \text{sampling} & \text{The Banach-Steinhaus theorem} & \uparrow \\ (y_n) & \xrightarrow{T_n^{-1}} & (T_n^{-1} y_n) \end{array}$$

Appendix F

Fourier Analysis

F.1 L^p spaces

Definition F.1. A pair of *conjugate exponents* are two real numbers $p, q \in [1, \infty]$ satisfying

$$p + q = pq, \text{ i.e., } \frac{1}{p} + \frac{1}{q} = 1. \quad (\text{F.1})$$

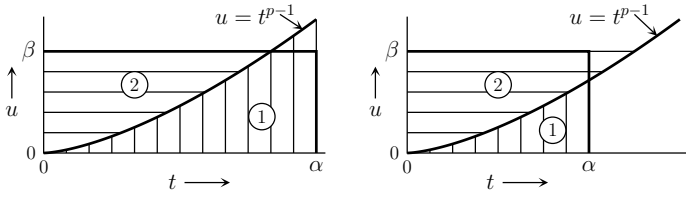


Figure F.1: The proof of the inequality (F.2) involving conjugate exponents.

Lemma F.2. Any two positive real numbers α, β satisfy

$$\alpha\beta \leq \frac{\alpha^p}{p} + \frac{\beta^q}{q}, \quad (\text{F.2})$$

where p and q are conjugate exponents and the equality holds if $\beta = \alpha^{p-1}$.

Proof. By (F.1), we have

$$u = t^{p-1} \Rightarrow t = u^{q-1}.$$

It follows from Figure F.1 that

$$\alpha\beta \leq \int_0^\alpha t^{p-1} dt + \int_0^\beta u^{q-1} du = \frac{\alpha^p}{p} + \frac{\beta^q}{q},$$

where the equality holds if $\beta = \alpha^{p-1}$ since $p = q(p-1)$. $\square$

Corollary F.3. A pair of conjugate exponents p, q satisfy

$$\forall a, b \in [0, +\infty), \quad a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{a}{p} + \frac{b}{q}. \quad (\text{F.3})$$

Proof. This follows directly from Lemma F.2. $\square$

Theorem F.4 (Hölder's inequality). For $n \in \mathbb{N}^+ \cup \{+\infty\}$, $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ and conjugate exponents $p, q \in [1, \infty]$, we have

$$\sum_{j=1}^n |x_j y_j| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q, \quad (\text{F.4})$$

where $\|\cdot\|_p$ is a vector norm in Definition C.176. For $p, q \in (1, \infty)$, the equality in (F.4) holds if

$$\exists c \in \mathbb{R} \text{ s.t. } \forall j = 1, \dots, n, \quad |x_j|^p = c |y_j|^q. \quad (\text{F.5})$$

Proof. If $\sum_{j=1}^n |x_j|^p = 0$ or $\sum_{j=1}^n |y_j|^q = 0$ or $p = \infty$ or $q = \infty$, then (F.4) holds trivially. Otherwise we define

$$a_i := \frac{|x_i|^p}{\sum_{j=1}^n |x_j|^p}, \quad b_i := \frac{|y_i|^q}{\sum_{j=1}^n |y_j|^q}.$$

It follows from (F.3) that

$$\frac{|x_i y_i|}{\left(\sum_{j=1}^n |x_j|^p\right)^{\frac{1}{p}} \left(\sum_{j=1}^n |y_j|^q\right)^{\frac{1}{q}}} \leq \frac{|x_i|^p}{p \sum_{j=1}^n |x_j|^p} + \frac{|y_i|^q}{q \sum_{j=1}^n |y_j|^q}.$$

Sum up all equations for $i = 1, \dots, n$ and we have

$$\frac{\sum_{j=1}^n |x_j y_j|}{\|\mathbf{x}\|_p \|\mathbf{y}\|_q} \leq \frac{1}{p} + \frac{1}{q} = 1,$$

which yields (F.4). Substitute (F.5) into (F.4) and we have the equality. $\square$

Example F.5. The Cauchy–Schwarz inequality in Theorem C.197 is a special case of the Hölder inequality (F.4) for $p = q = 2$.

Exercise F.6. Use Hölder's inequality (F.4) to verify the triangle inequality for a vector norm in Definition C.176.

Exercise F.7. Prove that (D.17) satisfies the triangle inequality and is indeed a metric.

(a) The Hölder inequality implies the *Minkowski inequality*, i.e. for any $p \geq 1$, $(\xi_j) \in \ell^p$, and $(\eta_j) \in \ell^p$,

$$\left(\sum_{j=1}^{\infty} |\xi_j + \eta_j|^p \right)^{\frac{1}{p}} \leq \left(\sum_{k=1}^{\infty} |\xi_k|^p \right)^{\frac{1}{p}} + \left(\sum_{m=1}^{\infty} |\eta_m|^p \right)^{\frac{1}{p}}. \quad (\text{F.6})$$

(b) The Minkowski inequality implies that the triangle inequality holds for (D.17).

Definition F.8. Let Ω be a domain in $\mathbb{R}^n$, i.e. a Lebesgue-measurable subset of $\mathbb{R}^n$ with nonempty interior. The L^p -norm of a Lebesgue-measurable function $f : \Omega \rightarrow \mathbb{R}$ is

$$\|f\|_{L^p(\Omega)} := \begin{cases} \left(\int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}} & \text{if } p \in [1, +\infty); \\ \text{ess sup}\{|f(x)| : x \in \Omega\} & \text{if } p = \infty, \end{cases} \quad (\text{F.7})$$

where “ dx ” denotes the Lebesgue measure, “ $\int_{\Omega}$ ” the Lebesgue integral, and “ess sup” the essential supremum:

$$\text{ess sup}\{|f(x)| : x \in \Omega\} := \inf_{\text{meas}(\Omega')=0} \sup_{x \in \Omega \setminus \Omega'} |f(x)|. \quad (\text{F.8})$$

The L^p -space or the Lebesgue space is the set of all Lebesgue-measurable functions on Ω with bounded L^p -norms, written

$$L^p(\Omega) := \{f : \|f\|_{L^p(\Omega)} < \infty\}. \quad (\text{F.9})$$

Notation 31. In this chapter we write $\|u\|_p := \|u\|_{L^p(\Omega)}$ when there is no danger of confusion of Ω ; in particular, we use the shorthand notation $\|u\| := \|u\|_{L^2}$ since the L^2 norm will be mostly used.

Theorem F.9 (Hölder’s Inequality). For $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$, $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$ imply $fg \in L^1(\Omega)$ and

$$\|fg\|_{L^1(\Omega)} \leq \|f\|_{L^p(\Omega)} \|g\|_{L^q(\Omega)}. \quad (\text{F.10})$$

Corollary F.10 (Schwarz’s Inequality). $f \in L^2(\Omega)$ and $g \in L^2(\Omega)$ imply $fg \in L^1(\Omega)$ and

$$\int_{\Omega} |fg| dx \leq \|f\|_{L^2(\Omega)} \|g\|_{L^2(\Omega)}. \quad (\text{F.11})$$

Theorem F.11 (Minkowski’s Inequality). For $p \in [1, \infty]$ and $f, g \in L^p(\Omega)$, we have

$$\|f + g\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)} + \|g\|_{L^p(\Omega)}. \quad (\text{F.12})$$

Theorem F.12. $L^p(\Omega)$ is a Banach space for any $p \in [1, \infty]$.

Theorem F.13 (Lebesgue’s Dominated Convergence). Let $\{f_n\}_{n=0}^{\infty}$ be a sequence of functions in $L^1(\Omega)$ such that

- (i) $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ a.e. in Ω ;
- (ii) $\exists g \in L^1(\Omega)$ s.t. $\forall n \in \mathbb{N}$, $|f_n(x)| \leq g(x)$ a.e. in Ω .

Then $f \in L^1(\Omega)$ and $\lim_{n \rightarrow \infty} \int_{\Omega} f_n dx = \int_{\Omega} f dx$.

Theorem F.14 (Fubini). Suppose $f \in L^1(\Omega_1 \times \Omega_2)$ where $\Omega_1 \subset \mathbb{R}$ and $\Omega_2 \subset \mathbb{R}$. Then

$$\begin{aligned} \int_{\Omega_1 \times \Omega_2} f(x, y) dx dy &= \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) dx \right) dy \\ &= \int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) dy \right) dx. \end{aligned}$$

Example F.15. By Definitions D.19 and D.30, $L^p(\Omega)$ is separable since the set of all polynomials with rational coefficients is countable and is dense in $L^p(\Omega)$.

F.2 Fourier transforms

Definition F.16. The Fourier transform of a function $u \in L^1(\mathbb{R})$ is the function $\hat{u} : \mathbb{R} \rightarrow \mathbb{C}$ given by

$$\hat{u}(\xi) = (\mathcal{F}u)(\xi) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\xi x} u(x) dx. \quad (\text{F.13})$$

Example F.17. The Fourier transform of $u \in L^1(\mathbb{R})$ is the function $\hat{u} : \mathbb{R} \rightarrow \mathbb{C}$ given by (F.13). Then $\hat{u} \in \mathcal{C}_b(\mathbb{R})$, i.e., $\hat{u}$ is a bounded continuous function, because

$$\begin{aligned} |\hat{u}(\xi)| &= \frac{1}{\sqrt{2\pi}} \left| \int_{-\infty}^{\infty} e^{-i\xi x} u(x) dx \right| \\ &\leq \int_{-\infty}^{\infty} |e^{-i\xi x}| |u(x)| dx = \int_{-\infty}^{\infty} |u(x)| dx. \end{aligned}$$

Therefore we have $\hat{\cdot} \in \mathcal{CL}(L^1(\mathbb{R}), \mathcal{C}_b(\mathbb{R}))$, i.e. the Fourier transform $\hat{\cdot} : (L^1(\mathbb{R}), \|\cdot\|_1) \rightarrow (\mathcal{C}_b(\mathbb{R}), \|\cdot\|_{\infty})$ is a bounded linear transformation.

Theorem F.18. If $u \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, then its Fourier transform (F.13) also belongs to $L^1 \cap L^2$, and u can be recovered from $\hat{u}$ by the inverse Fourier transform

$$u(x) = (\mathcal{F}^{-1}\hat{u})(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\xi x} \hat{u}(\xi) d\xi. \quad (\text{F.14})$$

Definition F.19. The Fourier transform of a grid function $U \in L^1(\mathbb{Z})$ is the function $\hat{U} : [-\pi, \pi] \rightarrow \mathbb{C}$ given by

$$\hat{U}(\xi) = (\mathcal{F}U)(\xi) := \frac{1}{\sqrt{2\pi}} \sum_{m \in \mathbb{Z}} e^{-im\xi} U(m) \quad (\text{F.15})$$

where $\xi \in [-\pi, \pi]$. Its inverse Fourier transform is given by

$$U(m) = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{+\pi} e^{im\xi} \hat{U}(\xi) d\xi. \quad (\text{F.16})$$

Theorem F.20. Let $u, v \in L^1$ have Fourier transforms $\hat{u} = \mathcal{F}u, \hat{v} = \mathcal{F}v$. Then

(a) Linearity. If $c \in \mathbb{R}$, then

$$\mathcal{F}\{u + v\}(\xi) = \hat{u}(\xi) + \hat{v}(\xi), \quad (\text{F.17})$$

$$\mathcal{F}\{cu\}(\xi) = c\hat{u}(\xi). \quad (\text{F.18})$$

(b) Translation. If $x_0 \in \mathbb{R}$, then

$$\mathcal{F}\{u(x + x_0)\}(\xi) = e^{ix_0\xi} \hat{u}(\xi). \quad (\text{F.19})$$

(c) Modulation. If $\xi_0 \in \mathbb{R}$, then

$$\mathcal{F}\{e^{i\xi_0 x} u(x)\}(\xi) = \hat{u}(\xi - \xi_0). \quad (\text{F.20})$$

(d) Dilation. If $c \in \mathbb{R}$ with $c \neq 0$, then

$$\mathcal{F}\{u(cx)\}(\xi) = \frac{1}{|c|} \hat{u}\left(\frac{\xi}{c}\right). \quad (\text{F.21})$$

(e) Conjugation.

$$\mathcal{F}\{\bar{u}\}(\xi) = \overline{\hat{u}(-\xi)}. \quad (\text{F.22})$$

(f) Differentiation. If $u_x \in L^2$, then

$$\mathcal{F}\{u_x\}(\xi) = i\xi \hat{u}(\xi). \quad (\text{F.23})$$

(g) Inversion.

$$\mathcal{F}^{-1}\{u\}(\xi) = \hat{u}(-\xi). \quad (\text{F.24})$$

Proof. Most conclusions follow directly from (F.13). We only show (F.23) by

$$\begin{aligned} \widehat{u_x}(\xi) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\xi x} u_x(x) dx \\ &= -\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (-i\xi) e^{-i\xi x} u(x) dx = i\xi \hat{u}(\xi), \end{aligned}$$

where we have applied integration by parts with the assumptions that $u(x)$ is smooth and vanishes at ∞ . $\square$

Definition F.21. A function $u(x)$ is *even*, *odd*, *real*, or *imaginary* if $\overline{u(x)} = u(-x)$, $u(x) = -u(-x)$, $u(x) = \overline{u(x)}$, or $u(x) = -\overline{u(x)}$, respectively; $u(x)$ is *Hermitian* or *skew-Hermitian* if $u(x) = \overline{u(-x)}$ or $u(x) = -\overline{u(-x)}$, respectively.

Theorem F.22. Denote by $\hat{u} = \mathcal{F}u$ the Fourier transform of $u \in L^1$. Then

- (a) $u(x)$ is even (odd) $\iff \hat{u}(\xi)$ is even (odd).
- (b) $u(x)$ is real (imaginary) $\iff \hat{u}(\xi)$ is Hermitian (skew-Hermitian) and therefore
- (c) $u(x)$ is real and even $\iff \hat{u}(\xi)$ is real and even.
- (d) $u(x)$ is real and odd $\iff \hat{u}(\xi)$ is imaginary and odd.
- (e) $u(x)$ is imaginary and even $\iff \hat{u}(\xi)$ is imaginary and even.
- (f) $u(x)$ is imaginary and odd $\iff \hat{u}(\xi)$ is real and odd.

Definition F.23. The *convolution* of two functions u, v is the function $u * v$ defined by

$$\begin{aligned} (u * v)(x) &= (v * u)(x) \\ &= \int_{-\infty}^{\infty} u(x-y)v(y)dy = \int_{-\infty}^{\infty} u(y)v(x-y)dy, \end{aligned}$$

assuming these integrals exist.

Theorem F.24. If $u \in L^1 \cap L^2$ and $v \in L^1$ (or vice versa), then $u * v \in L^1 \cap L^2$, and $\widehat{u * v}$ satisfies

$$\widehat{u * v}(\xi) = \sqrt{2\pi} \hat{u}(\xi) \hat{v}(\xi). \quad (\text{F.25})$$

Proof. Definition F.23 and Theorem F.14 yield

$$\begin{aligned} \int_{-\infty}^{\infty} |u * v(x)| dx &= \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} u(y)v(x-y) dy \right| dx \\ &\leq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |u(y)v(x-y)| dy dx \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} |u(y)v(x-y)| dx \right) dy \\ &= \int_{-\infty}^{\infty} \left(|u(y)| \int_{-\infty}^{\infty} |v(x-y)| dx \right) dy \\ &= \int_{-\infty}^{\infty} |u(y)| dy \int_{-\infty}^{\infty} |v(z)| dz < \infty, \end{aligned}$$

where the fifth step follows from the variable substitution $z = x - y$. We omit the proof of $u * v \in L^2$ and refer the

reader to a standard text on functional analysis. By Definition F.23, F.16, and Theorem F.14, we calculate

$$\begin{aligned} \widehat{u * v}(\xi) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} u(y)v(x-y) dy \right) e^{-i\xi x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(y)v(x-y) e^{-i\xi x} dx dy \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(y)v(z) e^{-i\xi(y+z)} dz dy \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(y) e^{-i\xi y} dy \int_{-\infty}^{\infty} v(z) e^{-i\xi z} dz \\ &= \sqrt{2\pi} \hat{u}(\xi) \hat{v}(\xi), \end{aligned}$$

where the third step follows from the variable substitution $z = x - y$. $\square$

Theorem F.25 (Parseval's equality or Plancherel theorem). The L^2 -norms of u and $\hat{u}$ are related by

$$\forall u \in L^1(\mathbb{R}) \cap L^2(\mathbb{R}), \quad \|\hat{u}\| = \|u\|. \quad (\text{F.26})$$

Proof. We calculate

$$\begin{aligned} \int |u(x)|^2 dx &= \int \overline{u(x)} u(x) dx = \left(\overline{u(x)} * u(-x) \right) (0) \\ &= \frac{1}{\sqrt{2\pi}} \int \overline{u(x)} * u(-x)(\xi) d\xi \\ &= \int \widehat{\overline{u(x)}}(\xi) \widehat{u(-x)}(\xi) d\xi \\ &= \int \widehat{\overline{u}}(-\xi) \hat{u}(-\xi) d\xi = \int |\hat{u}(\xi)|^2 d\xi, \end{aligned}$$

where in the first step $\overline{u(x)}$ denotes the conjugate of $u(x)$, the second step follows from Definition F.23, the third from (F.14), the fourth from Theorem F.24, the fifth from (F.21) and (F.22), and the last from the symmetry of the integral limits. $\square$

Example F.26 (B-splines). For the function

$$u(x) = \begin{cases} \sqrt{\frac{\pi}{2}} & \text{if } x \in [-1, 1]; \\ 0 & \text{otherwise,} \end{cases} \quad (\text{F.27})$$

we have $\|u\| = \sqrt{\pi}$. Then (F.13) gives

$$\hat{u}(\xi) = \frac{1}{2} \int_{-1}^1 e^{-i\xi x} dx = \frac{e^{-i\xi x}}{-2i\xi} \Big|_{-1}^1 = \frac{\sin \xi}{\xi}, \quad (\text{F.28})$$

where the function $\xi \mapsto \frac{\sin \xi}{\xi}$ is known as the *sinc function*. From the indispensable identity

$$\int_{-\infty}^{\infty} \frac{\sin^2 s}{s^2} ds = \pi, \quad (\text{F.29})$$

which can be derived by complex contour integration, we calculate $\|\hat{u}\| = \sqrt{\pi}$, which confirms (F.26).

By Definition F.23, it is readily verified that

$$(u * u)(x) = \begin{cases} \frac{\pi}{2}(2 - |x|) & \text{if } x \in [-2, 2]; \\ 0 & \text{otherwise,} \end{cases} \quad (\text{F.30})$$

and

$$(u * u * u)(x) = \begin{cases} \frac{3(\sqrt{2\pi})^3}{8} - \frac{(\sqrt{2\pi})^3}{8}x^2 & \text{if } |x| \leq 1; \\ \frac{(\sqrt{2\pi})^3}{16}(9 - 6|x| + x^2) & \text{if } 1 \leq |x| \leq 3; \\ 0 & \text{otherwise.} \end{cases}$$

By (F.25) and (F.28), their Fourier transforms are

$$\widehat{u * u}(\xi) = \sqrt{2\pi} \frac{\sin^2 \xi}{\xi^2}, \quad \widehat{u * u * u}(\xi) = 2\pi \frac{\sin^3 \xi}{\xi^3}; \quad (\text{F.31})$$

see Figure F.2. In general, a convolution $u_{(p)}$ of p copies of u has the Fourier transform

$$\widehat{u_{(p)}}(\xi) := \mathcal{F}\{u * u * \cdots * u\}(\xi) = (2\pi)^{\frac{p-1}{2}} \left(\frac{\sin \xi}{\xi} \right)^p.$$

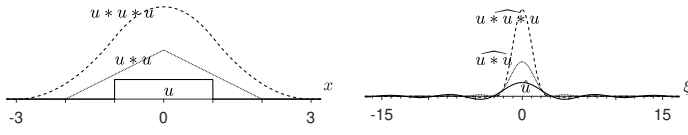


Figure F.2: Three splines (left) and their corresponding Fourier transforms.

Example F.27. The function

$$u(x) = \begin{cases} \frac{\pi}{2}, & \text{for } -2 \leq x < 0, \\ -\frac{\pi}{2}, & \text{for } 0 < x \leq 2, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{F.32})$$

has its Fourier transform as

$$\begin{aligned} \hat{u}(\xi) &= \frac{\sqrt{2\pi}}{4} \int_{-2}^0 e^{-i\xi x} dx - \frac{\sqrt{2\pi}}{4} \int_0^2 e^{-i\xi x} dx \\ &= \frac{\sqrt{2\pi}}{-4i\xi} (1 - e^{2i\xi} - e^{-2i\xi} + 1) \\ &= \frac{\sqrt{2\pi}}{4i\xi} (e^{i\xi} - e^{-i\xi})^2 = \sqrt{2\pi} \frac{i \sin^2 \xi}{\xi}. \end{aligned}$$

Hence $\hat{u}(\xi)$ is $i\xi$ times the Fourier transform (F.31) of the triangular hat function (F.30), which confirms (F.23).

Definition F.28. A function u defined on $\mathbb{R}$ is said to have *bounded variation* if there is a constant M such that for any finite m and any points $x_0 < x_1 < \cdots < x_m$,

$$\sum_{j=1}^m |u(x_j) - u(x_{j-1})| \leq M. \quad (\text{F.33})$$

Theorem F.29. Let u be a function in L^2 .

- (a) If u has $p-1$ continuous derivatives in L^2 for some $p \geq 0$, and a p th derivative in L^2 that has bounded variation, then

$$\hat{u}(\xi) = O(|\xi|^{-p-1}) \quad \text{as } |\xi| \rightarrow \infty. \quad (\text{F.34})$$

- (b) If u has infinitely many continuous derivatives in L^2 , then we have

$$\hat{u}(\xi) = O(|\xi|^{-M}) \quad \text{as } |\xi| \rightarrow \infty \text{ for all } M. \quad (\text{F.35})$$

The converse also holds.

- (c) If u can be extended to an entire function such that

$$\exists a > 0 \text{ s.t. } |u(z)| = O(e^{a|z|}) \text{ as } |z| \rightarrow \infty,$$

then $\hat{u}$ has compact support contained in $[-a, a]$, i.e.,

$$\hat{u}(\xi) = 0 \quad \text{for all } |\xi| > a. \quad (\text{F.36})$$

The converse also holds.

Example F.30. The function u in (F.27) satisfies condition (a) of Theorem F.29 with $p = 0$, so its Fourier transform should satisfy

$$|\hat{u}(\xi)| = O(|\xi|^{-1}),$$

as is verified by (F.28). On the other hand, suppose we interchange the role of u and $\hat{u}$ and apply the theorem again. The function $u(\xi) = \frac{\sin \xi}{\xi}$ is entire, and since $\sin(\xi) = \frac{1}{2i}(e^{i\xi} - e^{-i\xi})$, it satisfies

$$u(\xi) = O(e^{|\xi|}) \quad \text{as } |\xi| \rightarrow \infty$$

(with ξ now taking complex values). By part (c) of Theorem F.29, it follows that $u(x)$ must have compact support contained in $[-1, 1]$, as indeed it does.

Repeating the example for $u * u$, condition (a) now applies with $p = 1$, and the Fourier transform (F.31) is indeed of magnitude $O(|\xi|^{-2})$, as required. Interchanging u and $\hat{u}$, we note that $\frac{\sin^2 \xi}{\xi^2}$ is an entire function of magnitude $O(e^{2|\xi|})$ as $|\xi| \rightarrow \infty$, and $u * u$ has support contained in $[-2, 2]$.

F.3 Green's function of the heat equation in $(-\infty, +\infty)$

Lemma F.31. The Fourier transform of a Gaussian centered at the origin is another such Gaussian.

Proof. For $f(x) = e^{-x^2}$, we have

$$\hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} e^{-i\xi x} dx.$$

Differentiating with respect to ξ yields

$$\begin{aligned} \frac{d}{d\xi} \hat{f}(\xi) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} (-ix) e^{-i\xi x} dx \\ &= \frac{i}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{de^{-x^2}}{dx} e^{-i\xi x} dx \\ &= -\frac{\xi}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} e^{-i\xi x} dx \\ &= -\frac{\xi}{2} \hat{f}(\xi), \end{aligned}$$

where the third step follows from the integration by parts formula. The unique solution to this ODE is given by

$$\hat{f}(\xi) = c \cdot e^{-\frac{\xi^2}{4}},$$

where $c = \hat{f}(0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} dx = \frac{\sqrt{2}}{2}$. The proof is completed by the dilation property (F.21) of Fourier transform. In particular, the Fourier transform of a Gaussian with $a = 1$, $b = 0$, and c is another Gaussian with $a' = c$, $b' = 0$, and $c' = \frac{1}{c}$. $\square$

Lemma F.32. For any $u \in L^2$ satisfying

$$\forall n \in \mathbb{N}, \quad \lim_{x \rightarrow \pm\infty} u^{(n)}(x) = 0, \quad (\text{F.37})$$

we have

$$\widehat{\frac{d^2 u}{dx^2}} = -\xi^2 \hat{u}. \quad (\text{F.38})$$

Proof. Repeated application of (F.37) yields

$$\begin{aligned} \sqrt{2\pi} \cdot \widehat{\frac{d^2 u}{dx^2}} &= \int_{-\infty}^{+\infty} e^{-i\xi x} \frac{d^2 u}{dx^2} dx \\ &= e^{-i\xi x} \frac{du}{dx} \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{du}{dx} (-i\xi) e^{-i\xi x} dx \\ &= i\xi \int_{-\infty}^{+\infty} \frac{du}{dx} e^{-i\xi x} dx \\ &= i\xi (e^{-i\xi x} u) \Big|_{-\infty}^{+\infty} + (i\xi)^2 \int_{-\infty}^{+\infty} u e^{-i\xi x} dx \\ &= -\xi^2 \int_{-\infty}^{+\infty} u e^{-i\xi x} dx = -\xi^2 \sqrt{2\pi} \hat{u}, \end{aligned}$$

where the first and the last steps follow from Definition F.16, the second and the fourth from the integration by parts formula, and the third from (F.37). $\square$

Theorem F.33. The solution to the heat equation

$$u_t = \nu u_{xx} \text{ on } (-\infty, +\infty) \quad (\text{F.39})$$

with the initial condition $\eta(x) = e^{-\beta x^2}$ is

$$u(x, t) = \frac{1}{\sqrt{4\beta\nu t + 1}} e^{-\frac{x^2}{4\nu t + 1/\beta}}. \quad (\text{F.40})$$

Proof. By Lemma F.32, the Fourier transform of (F.39) leads to the ODE

$$\hat{u}_t(\xi, t) = -\nu \xi^2 \hat{u}(\xi, t),$$

the solution of which with the initial condition $\hat{u}(\xi, 0) = \hat{\eta}(\xi)$ yields $\hat{u}(\xi, t) = e^{-\nu \xi^2 t} \hat{\eta}(\xi)$. Then

$$\begin{aligned} u(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \hat{u}(\xi, t) e^{i\xi x} d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\nu \xi^2 t} \hat{\eta}(\xi) e^{i\xi x} d\xi \\ &= \frac{1}{2\sqrt{\pi\beta}} \int_{-\infty}^{+\infty} e^{-\xi^2(\nu t + \frac{1}{4\beta})} e^{i\xi x} d\xi. \end{aligned}$$

Define $C = \frac{1}{4\nu t + 1/\beta}$ and we have

$$\begin{aligned} u(x, t) &= \frac{1}{2\sqrt{\pi\beta}} \int_{-\infty}^{+\infty} e^{-\frac{\xi^2}{4C}} e^{i\xi x} d\xi = \frac{1}{2\sqrt{\pi\beta}} \sqrt{4\pi C} \cdot e^{-x^2 C} \\ &= \frac{1}{\sqrt{4\beta\nu t + 1}} e^{-\frac{x^2}{4\nu t + 1/\beta}}. \end{aligned}$$

As t increases this Gaussian becomes more spread out and the magnitude decreases. $\square$

Corollary F.34. A translation of the initial condition $\eta(x) = e^{-\beta(x-\bar{x})^2}$ of the heat equation (F.39) leads to a translation of the solution, i.e.,

$$u(x, t) = \frac{1}{\sqrt{4\beta\nu t + 1}} e^{-\frac{(x-\bar{x})^2}{4\nu t + 1/\beta}}. \quad (\text{F.41})$$

Corollary F.35. For the heat equation (F.39) with the initial condition as $\omega_\beta(x, 0; \bar{x}) = \sqrt{\frac{\beta}{\pi}} e^{-\beta(x-\bar{x})^2}$, its solution is

$$\omega_\beta(x, t; \bar{x}) = \frac{1}{\sqrt{4\pi\nu t + \pi/\beta}} e^{-\frac{(x-\bar{x})^2}{4\nu t + 1/\beta}}. \quad (\text{F.42})$$

Definition F.36. The *Green's function*

$$G(x, t; \bar{x}) := \lim_{\beta \rightarrow +\infty} \omega_\beta(x, t; \bar{x}) = \frac{1}{\sqrt{4\pi\nu t}} e^{-\frac{(x-\bar{x})^2}{4\nu t}} \quad (\text{F.43})$$

is the solution of the heat equation (F.39) with its initial condition as the Dirac delta function in Definition 8.41.

F.4 Fourier solutions of linear PDEs

Lemma F.37. For a linear PDE of the form

$$\frac{\partial u}{\partial t} + \sum_{n=1}^N a_n \frac{\partial^n u}{\partial x^n} = 0, \quad (\text{F.44})$$

the evolution of a single Fourier mode of wavenumber ξ satisfies the ODE

$$\frac{\partial \hat{u}}{\partial t}(\xi, t) + \sum_{n=1}^N a_n (i\xi)^n \hat{u}(\xi, t) = 0. \quad (\text{F.45})$$

Proof. Differentiations of the inverse Fourier transform (F.14) with respect to t and x yield, respectively,

$$\begin{aligned} \frac{\partial u(x, t)}{\partial t} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{u}_t(\xi, t) e^{i\xi x} d\xi, \\ \frac{\partial^n u(x, t)}{\partial x^n} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{u}(\xi, t) (i\xi)^n e^{i\xi x} d\xi. \end{aligned}$$

Plug these equations into (F.44) and we have (F.45). $\square$

Lemma F.38. The solution to the linear PDE (F.44) with initial condition $u(x, 0) = \eta(x)$ is

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\eta}(\xi) e^{i(\xi x - \omega t)} d\xi, \quad (\text{F.46})$$

where the frequency is

$$\omega := \sum_{n=1}^N a_n \xi^n i^{n-1}. \quad (\text{F.47})$$

Proof. Rewrite (F.45) as $\frac{\partial \hat{u}(\xi, t)}{\partial t} = -i\omega \hat{u}(\xi, t)$ and we have from Duhamel's principle (Theorem 12.15)

$$\hat{u}(\xi, t) = e^{-i\omega t} \hat{\eta}(\xi). \quad (\text{F.48})$$

Then the inverse Fourier transform yields

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{u}(\xi, t) e^{i\xi x} d\xi = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\eta}(\xi) e^{i(\xi x - \omega t)} d\xi. \quad \square$$

F.5 Dissipation and dispersion

Definition F.39. The *dispersion relation* of a PDE is the relation between the frequency ω and the wavenumber ξ ,

$$\omega = \omega(\xi). \quad (\text{F.49})$$

Example F.40. The dispersion relation of (F.44) is (F.47). In particular, the linear Korteweg-deVries (KdV) equation

$$\varphi_t + c_0\varphi_x + \nu\varphi_{xxx} = 0 \quad (\text{F.50})$$

is characterized by its dispersion relation $\omega = c_0\xi - \nu\xi^3$.

Example F.41. The beam equation

$$\varphi_{tt} + \gamma^2\varphi_{xxx} = 0 \quad (\text{F.51})$$

is characterized by its dispersion relation $\omega = \pm\gamma\xi^2\mathbf{i}$ since it is equivalent to $\varphi_t = \pm\gamma\varphi_{xx}$.

F.6 Phase velocity and group velocity

Definition F.42. For a wave function of the form

$$\varphi(x, t) = \varphi_0 e^{i\theta(x, t)} = \varphi_0 e^{-i\omega t} e^{i\xi x} e^{-i\theta_0}, \quad (\text{F.52})$$

φ_0 is called the *amplitude*, $e^{-i\omega t}$ the *oscillation*, $e^{i\xi x}$ the *propagation*, θ the *phase*, and the constant θ_0 the *phase offset*.

Definition F.43. The *phase velocity* of a monochromatic wave with wavenumber ξ is the speed that a point of constant phase propagates:

$$C_p(\xi) := \frac{\omega(\xi)}{\xi}. \quad (\text{F.53})$$

Example F.44. The speed of light is a phase velocity,

$$C_p = \frac{\lambda}{T} = \frac{\omega}{\xi},$$

where λ and T are the wavelength and the period, respectively.

Definition F.45. The system (F.44) is said to be *hyperbolic* if the PDE is hyperbolic; it is *dissipative* if ω is purely imaginary; it is *dispersive* if $\omega(\xi)$ is real and $\omega(\xi)/\xi$ is not a constant.

Definition F.46. The *group velocity* of a monochromatic wave with wavenumber ξ is

$$C_g(\xi) := \frac{d\omega(\xi)}{d\xi}. \quad (\text{F.54})$$

Example F.47. For the linear PDE (F.44), the phase velocity of a single Fourier mode is

$$C_p(\xi) = \sum_{n=1}^N a_n \xi^{n-1} \mathbf{i}^{n-1}$$

while the group velocity is

$$C_g(\xi) = \sum_{n=1}^N n a_n \xi^{n-1} \mathbf{i}^{n-1}.$$

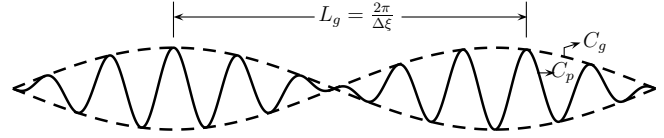


Figure F.3: Phase velocities and group velocities.

Lemma F.48. The group velocity and the phase velocity are related via

$$C_g(\xi) = \frac{C_p(\xi)}{1 - \xi \frac{dC_p}{d\xi}}. \quad (\text{F.55})$$

Proof. Definition F.46 yields

$$C_g(\xi) = \frac{d\omega}{d\xi} = \frac{1}{\frac{d\xi}{d\omega}} = \frac{1}{\frac{d}{d\omega} \left(\frac{\omega}{C_p} \right)},$$

which implies (F.55). $\square$

Example F.49. For the advection equation, we have $a_1 = a$ and $a_j = 0$ for all $j > 1$. Consequently, we have

$$\omega = a\xi, \quad C_p = a = C_g,$$

hence Lemma F.38 yields

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\eta}(\xi) e^{i\xi(x-at)} d\xi = \eta(x - at).$$

Thus all wave modes that constitute the initial data $\eta(x)$ move at the same phase speed, which is also the moving speed of energy.

Example F.50. For the heat equation $u_t = \nu u_{xx}$, we have $a_2 = -\nu < 0$, $a_1 = a_3 = a_4 = \dots = 0$, and thus

$$\omega = a_2 \xi^2 \mathbf{i}, \quad C_p = a_2 \xi \mathbf{i}, \quad C_g = 2a_2 \xi \mathbf{i}.$$

Lemma F.38 yields

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\eta}(\xi) e^{i\xi x} e^{a_2 \xi^2 t} d\xi,$$

where the term “ $e^{i\xi x}$ ” denotes the initial mode ξ that does not move while “ $e^{-\nu\xi^2 t}$ ” represents the exponential decay with respect to time.

Example F.51. For the dispersion equation

$$u_t = u_{xxx},$$

we have $a_3 = -1$, $a_1 = a_2 = a_4 = a_5 = \dots = 0$,

$$\omega = \xi^3, \quad C_p = \xi^2, \quad C_g = 3\xi^2.$$

Lemma F.38 yields

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\eta} e^{i\xi(x-\xi^2 t)} d\xi,$$

thus there is no damping, but different phases move with different speed ξ^2 .

Example F.52. For the equation

$$u_t + au_x + bu_{xxx} = 0,$$

we have $a_1 = a$, $a_3 = b$, $a_2 = a_4 = a_5 = \dots = 0$, and

$$\omega = a\xi - b\xi^3, \quad C_p = a - b\xi^2, \quad C_g = a - 3b\xi^2.$$

Lemma F.38 yields

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\eta}(\xi) e^{i\xi(x - (a - b\xi^2)t)} d\xi.$$

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